

Preface

DNA binding proteins are involved in a number of basic cellular processes including transcription, DNA replication, transposition, restriction, recombination, and DNA repair. The chapters in this volume cover a number of important methods, some old and some new, that are now widely used in the study of DNA binding proteins. These include purification and protein characterization, assays of protein–DNA binding and protein-induced DNA bending which can be used *in vitro* and *in vivo*, and biochemical and genetic methods for probing the structure, energy, and specificity of protein–DNA interactions. Previous volumes of this series that contain related methods include 65, 100, 130, 154, 155, and 170.

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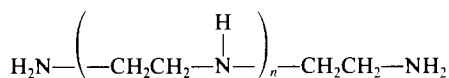
[1] Use of Polyethyleneimine in Purification of DNA-Binding Proteins

By RICHARD R. BURGESS

One of the important early steps in purifying a DNA-binding protein is to separate the protein of interest from cellular nucleic acids. A large variety of methods have been used, including precipitation with streptomycin sulfate,¹ polyethylene glycol,^{2,3} batch treatment of extracts with DEAE-cellulose,⁴ and treatment with DNase I.⁵ Perhaps the most effective method involves precipitation with polyethyleneimine.⁶⁻⁸ This chapter will focus on the use of polyethyleneimine in early stages of the purification of DNA-binding proteins. It should be noted, however, that the use of polyethyleneimine in protein purification is not restricted to DNA-binding proteins and has been used successfully for proteins from both prokaryotic and eukaryotic sources. The most comprehensive summary of the applications of polyethyleneimine is found in a review by Jendrisak.⁹

Properties of Polyethyleneimine (PEI)

Polyethyleneimine (PEI) is a product of polymerization of ethyleneimine to yield a basic linear polymer with the structure



Typically, n equals 700–2000 to give a molecular weight range of 30,000–90,000. Since the $\text{p}K_a$ value of the imino group is 10–11, PEI is a positively charged molecule in solutions of neutral pH. It has been produced for over 30 years in large quantities for uses such as a mordant for the dye industry by BASF (Charlotte, NC) under the trade name, Polymine P.

¹ M. J. Chamberlin and P. Berg, *Proc. Natl. Acad. Sci. U.S.A.* **48**, 81 (1962).

² C. Babinet, *Biochem. Biophys. Res. Commun.* **26**, 639 (1967).

³ B. Alberts and G. Herrick, this series, Vol. 21, [11] (1971).

⁴ W. F. Mangel, *Biochim. Biophys. Acta* **163**, 172 (1974).

⁵ R. R. Burgess, *J. Biol. Chem.* **244**, 6160 (1969).

⁶ W. Zillig, K. Zechel, and H. Halbwachs, *Hoppe-Seyler's Z. Physiol. Chem.* **351**, 221 (1970).

⁷ R. R. Burgess and J. J. Jendrisak, *Biochemistry* **14**, 4634 (1975).

⁸ J. J. Jendrisak and R. R. Burgess, *Biochemistry* **14**, 4639 (1975).

⁹ J. J. Jendrisak, in "Protein Purification: Micro to Macro" (R. R. Burgess, ed.), p. 75, Alan R. Liss, New York, 1987.

For precipitation of nucleic acids and some proteins, it has the desirable properties of being inexpensive, working well at slightly alkaline pH's where proteins are usually stable, and rapidly forming finely divided precipitates which may be centrifuged and resuspended very easily.

Principles of PEI Precipitation

PEI causes precipitation of nucleic acids and acidic proteins at low ionic strength by forming charge neutralization complexes and cross-bridges between the complexes. This leads to flocculation.¹⁰

This precipitation is basically a titration. As PEI is added, it becomes intimately involved in the networked complexes and is thus part of the precipitate. PEI precipitation thus differs fundamentally from salt or solvent precipitation. In ammonium sulfate precipitation, for example, the precipitant sequesters H₂O, decreases H₂O available to solvate the protein, and thus decreases the solubility of the protein, without itself being substantially precipitated. To illustrate this difference, assume that a given extract requires 50% saturated ammonium sulfate or 0.2% (w/v) PEI to achieve 90% precipitation of a given enzyme. If the extract is diluted 10-fold, it will take *higher* ammonium sulfate (about 56% saturation) but *lower* PEI concentration (0.02%) to achieve a similar 90% precipitation of the enzyme.

The fraction of the total protein precipitated by PEI will depend on the pH (at higher pH, more proteins will carry a negative charge and be precipitated) and on the ionic strength (increasing ionic strength will weaken PEI-protein interactions). At low ionic strength (0–0.1 M NaCl), DNA and all acidic proteins will be precipitated. At higher ionic strength (0.1–1 M NaCl), only DNA and highly negatively charged proteins will precipitate. At greater than 1 M NaCl, only nucleic acids will precipitate. DNA-PEI precipitates dissolve at 1.4 to 1.6 M NaCl. There are thus three basic strategies for using PEI in the purification of DNA binding proteins.

Strategy A: Precipitate with PEI at high ionic strength to remove nucleic acid. Almost no proteins will bind to DNA or to PEI in 1 M NaCl. Therefore one can adjust the salt concentration of an extract to 1 M NaCl and titrate with PEI to achieve precipitation of nucleic acids, leaving the proteins in the supernatant. This titration can be conveniently followed by determining the UV absorption spectrum of the supernatants⁹ and adding PEI until the supernatant shows an A_{280} value greater than the A_{260} , characteristic of protein solutions. While this strategy can be used, it is not common. One reason is that while nucleic acid is effectively removed,

¹⁰ D. J. Bell, M. Hoare, and P. Dunhill, in "Advances in Biochemical Engineering/Biotechnology" (A. Fiechler, ed.), p. 1. Springer-Verlag, Berlin, 1982.

no purification from other proteins is achieved, unlike strategies B or C (below). Another problem is that if more PEI is added than is needed to precipitate the nucleic acids, excess PEI will be present in the supernatant with the proteins. When the supernatant is then diluted or dialyzed to lower ionic strength, PEI–acidic protein complexes will precipitate. This excess PEI can be removed by ammonium sulfate precipitation as described below.

Strategy B: Precipitate with PEI at low to medium ionic strength to remove nucleic acids and some proteins, leaving enzyme of interest in supernatant. This is like strategy A but, because it is carried out at lower ionic strength, a substantial number of proteins will be precipitated. The enzyme of interest will remain in the supernatant and will not only be nucleic acid free but also will be free of the precipitated acidic proteins. This strategy works well for neutral or slightly basic DNA-binding proteins that do not bind to DNA under the ionic conditions of the PEI precipitation.

Strategy C: Precipitate with PEI at lower ionic strength to precipitate nucleic acids and some proteins and then elute desired enzyme out of pellet with higher ionic strength. This strategy is by far the most commonly used. Generally one chooses a pH of 7.5–7.9 and an ionic strength of 0.1 to 0.2 M NaCl and then performs a PEI titration test to determine the minimal amount of PEI solution that causes complete precipitation of the enzyme of interest. The pellet, which contains nucleic acids and some proteins, is resuspended in a predetermined low ionic strength buffer to wash out any trapped proteins but not the desired enzyme. The washed pellet is then resuspended in a buffer of predetermined higher ionic strength to elute the enzyme, leaving the nucleic acids in the precipitate. The precise amounts of PEI added and the ionic strength of the wash and elution buffers are determined empirically as described below. The eluate contains the enzyme of interest, some other proteins, and a certain amount of PEI that was originally coprecipitated with the eluted proteins. The PEI is generally removed by ammonium sulfate precipitation of the protein, leaving the PEI in the ammonium sulfate supernatant. The ammonium sulfate pellet is dissolved in buffer and subsequent purification steps are performed.

Methods and Procedures

Preparing Extracts

Extracts may be prepared in a variety of ways from bacterial,¹¹ animal and plant¹² cells, or tissues. Generally, the crude extract is centrifuged at

¹¹ M. Cull and C. S. McHenry, this series, Vol. 182, [12] (1990).

¹² P. Gegenheimer, this series, Vol. 182, [14] (1990).

10,000 *g* for 10–30 min to remove unbroken cells and cell debris. The resulting supernatant is referred to as extract.

Preparing Neutralized 5% (w/v) PEI Stock Solution

Concentrated PEI solutions, which are usually (but not always!) 50% (w/v), can be obtained from several commercial suppliers, including ICN Biomedicals (Irvine, CA), Aldrich (Milwaukee, WI), and Sigma (St. Louis, MO). One hundred milliliters of 50% (w/v) PEI is dissolved in 800 ml of deionized water. The solution is adjusted to pH 7.9 by adding concentrated HCl (approximately 38–40 ml) with stirring. After diluting to 1000 ml, the neutralized solution is filtered through one layer of Miracloth, Calbiochem, San Diego, CA (to remove slight particulate matter), resulting in a clear, slightly bluish solution. This 5% (w/v) or 10% (v/v) stock solution is stored refrigerated and appears to be stable indefinitely.

While PEI solutions are sometimes dialyzed to remove low-molecular-weight components, this does not seem to be necessary.

Performing PEI Test Titration of Escherichia coli Extract (Strategy C)

Materials

PEI stock solution, 5% (w/v), pH 7.9 (see above)

E. coli extract (about 12 mg protein/ml and 0.2 *M* NaCl, pH 7.9) prepared as described⁷

Procedure

1. Two-milliliter portions of extract are placed in 11 small centrifuge tubes and 5% (w/v) PEI stock solution is added to give final PEI concentrations of from 0 to 0.5% (w/v) in 0.05% (w/v) intervals.
2. Tubes are rapidly mixed after PEI addition, incubated for 5 min at 0°, mixed again, and centrifuged for 5 min.
3. Supernatants are assayed for protein concentration and for the presence of the enzyme of interest. This may be done by enzymatic assay, by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) analysis, or by immunologic assay.

In the example given in Figs. 1A and 2A,⁷ the *E. coli* RNA polymerase, measured by the presence of the high-molecular-weight β' and β subunits on Coomassie-stained SDS-PAGE, was found to be completely precipitated after addition of PEI to a final concentration of 0.175% (w/v) or 0.35 ml of PEI stock solution per 10 ml of extract. At this PEI concentration, all the nucleic acids and about 30% of the total protein precipitated.

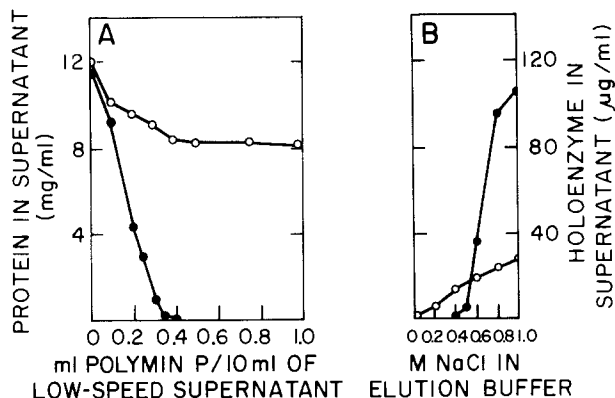


FIG. 1. PEI (Polymix P) precipitation curve and NaCl elution curve. (A) To 2-ml portions of the extract (low-speed supernatant), 5% (w/v) PEI was added in the indicated amounts and processed as described in Methods and Procedures. (B) An NaCl elution curve was prepared as described in Methods and Procedures. Supernatants were assayed for total protein (○) and for β' subunits of RNA polymerase on SDS-polyacrylamide gels shown in Fig. 2A and B (●). (Adapted from Ref. 7, with permission.)

Determining NaCl Elution Conditions (Strategy C)

Procedure

1. A 0.35-ml portion of 5% (w/v) PEI solution is added per 10 ml of extract as above, mixed, divided into 2-ml portions, centrifuged, and the supernatant discarded.
2. The identical pellets are resuspended for 10 min in 2 ml of TGED buffer (50 mM Tris-HCl, pH 7.9, 5% glycerol, 0.1 mM EDTA, 0.1 mM dithiothreitol) containing various concentrations of NaCl from 0 to 1 M and centrifuged.
3. Supernatants are again assayed for protein and enzyme.

As shown in Figs. 1B and 2B,⁷ the *E. coli* RNA polymerase remained in the pellet during a 0.5 M NaCl wash but was completely eluted by 1 M NaCl. When a larger scale preparation was carried out with 0.175% (w/v) PEI precipitation, a 0.5 M NaCl wash, and a 1 M NaCl elution, a complete removal of nucleic acids, a 15-fold purification, a 29-fold increase in specific activity, and a 90% recovery of RNA polymerase was achieved.

Removal of PEI from Eluted Protein

When protein is eluted from the PEI pellet, PEI is also released into solution. If the eluate is dialyzed or diluted to lower ionic strength, the free protein and PEI will again interact and precipitate. Therefore, one

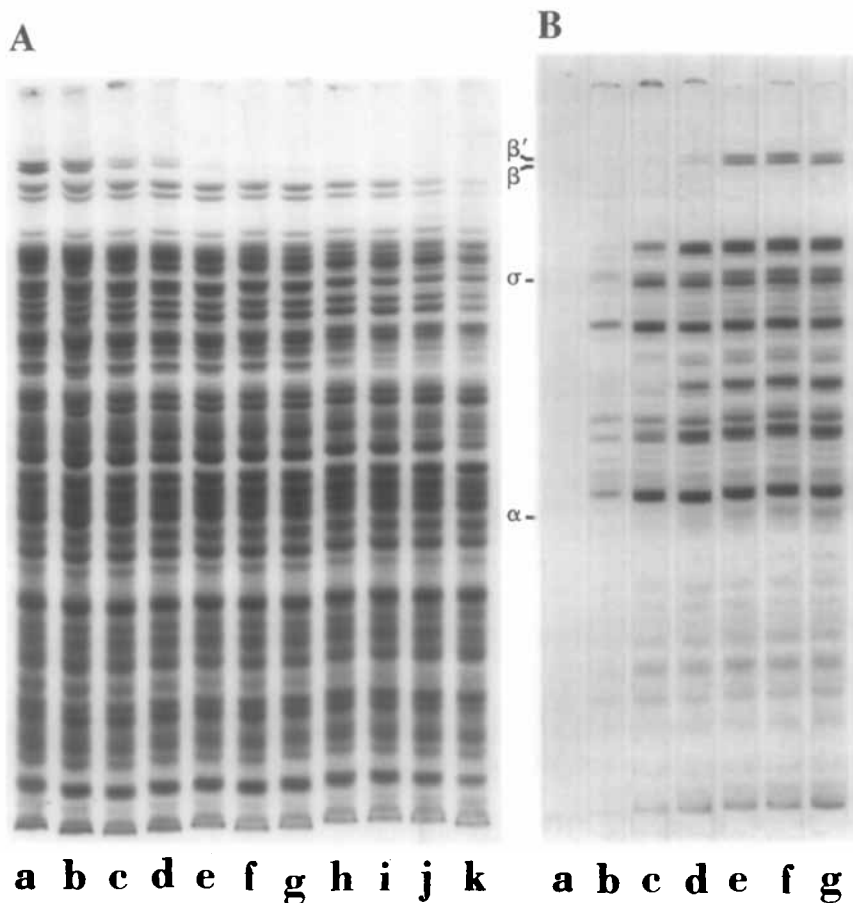


FIG. 2. SDS-polyacrylamide gel electrophoretic analysis of supernatants from the PEI precipitation curve and NaCl elution curve. Portions of supernatants were electrophoresed on an 8.75% acrylamide gel and stained with Coomassie Blue. The positions of the RNA polymerase subunits β' , β , σ , and α (with apparent molecular weights of 165,000, 155,000, 87,000, and 39,000, respectively) are indicated. (A) Portions of supernatants of a PEI precipitation curve containing the following final PEI concentrations were analyzed: (a) 0%, (b) 0.05%, (c) 0.1%, (d) 0.125%, (e) 0.15%, (f) 0.175%, (g) 0.2%, (h) 0.25%, (i) 0.375%, (j) 0.5%, (k) 0.6%. (B) Portions of supernatants of a NaCl elution curve from Fig. 1B containing the following concentrations of NaCl: (a) 0.0 M, (b) 0.2 M, (c) 0.4 M, (d) 0.6 M, (e) 0.8 M, (f) 1.0 M, (g) 1.2 M. (Adapted from Ref. 7, with permission.)

must remove the free PEI. This is most commonly done by adding enough ammonium sulfate to the eluate to precipitate the protein. At this high ionic strength (often 40–60% saturated ammonium sulfate), PEI will not bind to protein and will remain in the supernatant after the ammonium sulfate precipitate is pelleted. This procedure removes the vast majority of the PEI and is sufficient for most proteins. If a protein is especially sensitive to residual traces of PEI, then a further step is required. The easiest method is to thoroughly resuspend the ammonium sulfate pellet in 40–60% saturated ammonium sulfate (sufficient to keep the protein insoluble) and recentrifuge. This way, any PEI physically trapped in the pellet will be removed.

Another method to remove PEI involves passing the eluate, or the dissolved ammonium sulfate precipitate, through a phosphocellulose column at 1 M NaCl. The protein will not bind to the column at this high ionic strength, but the PEI binds very tightly and is completely removed.¹³

Final Precautions and Comments

This basic approach for nucleic acid removal and protein purification has been widely employed (see references in Ref. 9). A few final comments, which might improve the success of this method, follow.

It is wise to carry out test titrations regularly, especially if the methods for preparing the extract change. The amount of PEI needed to achieve precipitation of the enzyme of interest will decrease if cell breakage is poor because the resulting extract will be more dilute.

Many proteins that bind to DNA are also acidic proteins that will precipitate with PEI even in the absence of nucleic acids. These are best treated with strategy C. Some DNA-binding proteins are neutral or basic and would only be expected to precipitate with PEI if they are bound to DNA and come down in the PEI–DNA precipitate. Therefore, the ionic strength at which PEI precipitation is done will determine whether one should use strategy B (high enough ionic strength to keep the protein of interest from binding to DNA and thus not precipitated with PEI) or strategy C (low enough ionic strength to allow the protein to bind DNA and be PEI precipitated). The optimal conditions must be determined empirically.

Precipitation with PEI occurs rapidly. There is a need to mix well during addition of PEI to prevent local overprecipitation. On large-scale (1–4 liters) quantities of extract this mixing is conveniently done in a 1-gal Waring blender.

¹³ Z. F. Burton and D. Eisenberg, *Arch. Biochem. Biophys.* **205**, 478 (1980).

Occasionally, PEI is sold as a 25 (w/v) or 50% (v/v) solution diluted with an equal volume of H₂O to decrease viscosity and increase ease of dispensing. This can lead to an incorrect stock solution, so caution is advised. One way this can be detected is by observing the amount of HCl needed to titrate 100 ml of PEI concentrate to pH 7.9.

Acknowledgments

I would like to thank Mark Knuth for his helpful comments. This work was supported by NIH Grants CA07175, CA23076, and GM28575 and is dedicated to the outstanding research and training environment of McArdle Laboratory for Cancer Research on its 50th anniversary.

[2] Purification of Sequence-Specific Binding Proteins by DNA Affinity Chromatography

By JAMES T. KADONAGA

Many biological processes, such as transcription, replication, and recombination, involve the action of sequence-specific DNA-binding proteins. To characterize the biochemical properties of these factors, it is necessary to purify them to homogeneity. In addition, the purified factors could be used to raise antibodies as well as to isolate the genes encoding the proteins. It has generally been difficult, however, to isolate these DNA-binding factors by conventional chromatography because they typically constitute less than 0.01% of the total cellular protein. Fortunately, it is now possible to purify sequence-specific DNA-binding proteins by DNA affinity chromatography.

DNA affinity chromatography was originally developed with nonspecific DNAs, such as calf thymus DNA, attached to cellulose¹ and agarose² supports. More recently, a variety of procedures have been described for DNA affinity purification of sequence-specific DNA-binding proteins. The different DNA affinity resins that have been employed include plasmid DNA adsorbed to cellulose,³ biotinylated DNA fragments attached to agarose, cellulose, or magnetic beads by biotin-avidin or biotin-streptavi-

¹ B. Alberts and G. Herrick, this series, Vol. 21, p. 198.

² D. J. Arndt-Jovin, T. M. Jovin, W. Bähr, A.-M. Frischauf, and M. Marquardt, *Eur. J. Biochem.* **54**, 411 (1975).

³ P. J. Rosenfeld and T. J. Kelly, *J. Biol. Chem.* **261**, 1398 (1986).

din interactions,⁴⁻⁷ monomeric synthetic oligodeoxynucleotides covalently attached to agarose,^{8,9} a Teflon-based support on which synthetic oligonucleotides are synthesized,¹⁰ and multimeric synthetic oligodeoxynucleotides covalently coupled to agarose beads¹¹ or latex particles.¹² A sequence-specific DNA-binding protein was also purified by a preparative gel mobility shift procedure.¹³ It is likely that many, if not all, of these different techniques are effective in the purification of sequence-specific DNA-binding proteins.

This chapter describes the preparation and use of a sequence-specific DNA affinity resin that consists of multimerized synthetic oligodeoxynucleotides covalently attached to an agarose support.¹¹ This particular procedure has been successfully employed in the purification of at least 50 different sequence-specific DNA-binding proteins, a few of which are listed in Table I. A brief description of the procedure is as follows. Complementary chemically synthesized oligodeoxynucleotides that contain a recognition site for a sequence-specific DNA-binding protein are annealed and ligated to give oligomers. The DNA is then covalently coupled to agarose beads with cyanogen bromide to yield the affinity resin. A partially purified protein fraction is first incubated with nonspecific competitor DNA to which the desired protein has very low affinity, and the resulting mixture is subsequently applied to the affinity resin. The desired sequence-specific DNA-binding protein binds to the recognition sites in the affinity resin rather than to the competitor DNA in solution, while other proteins, including nonspecific DNA-binding proteins, flow through the resin. The sequence-specific factor is then eluted from the affinity resin with a salt gradient. A typical protein can be purified 500- to 1000-fold with 30% yield by two sequential affinity chromatography steps. In addition, the use of tandem affinity columns containing different protein-binding sites allows the simultane-

⁴ L. A. Chodosh, R. W. Carthew, and P. A. Sharp, *Mol. Cell. Biol.* **6**, 4723 (1986).

⁵ M. S. Kasher, D. Pintel, and D. C. Ward, *Mol. Cell. Biol.* **6**, 3117 (1986).

⁶ M. Leblond-Francillard, M. Dreyfus, and F. Rougeon, *Eur. J. Biochem.* **166**, 351 (1987).

⁷ O. S. Gabrielsen, E. Hornes, L. Korsnes, A. Ruet, and T. B. Øyen, *Nucleic Acids Res.* **17**, 6253 (1989).

⁸ C. Wu, S. Wilson, B. Walker, I. Dawid, T. Paisley, V. Zimarino, and H. Ueda, *Science* **238**, 1247 (1987).

⁹ R. Blanks and L. W. McLaughlin, *Nucleic Acids Res.* **16**, 10283 (1988).

¹⁰ C. H. Duncan and S. L. Cavalier, *Anal. Biochem.* **169**, 104 (1988).

¹¹ J. T. Kadonaga and R. Tjian, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 5889 (1986).

¹² H. Kawaguchi, A. Asai, Y. Ohtsuka, H. Watanabe, T. Wada, and H. Handa, *Nucleic Acids Res.* **17**, 6229 (1989).

¹³ I. Gander, R. Foeckler, L. Rogge, M. Meisterernst, R. Schneider, R. Mertz, F. Lottspeich, and E.-L. Winnacker, *Biochim. Biophys. Acta* **951**, 411 (1988).

TABLE I
SEQUENCE-SPECIFIC DNA-BINDING PROTEINS PURIFIED BY DNA AFFINITY TECHNIQUE

Factor	Source	Function	Refs.
Mammalian			
Sp1	HeLa cells	RNA polymerase II transcription	<i>a</i>
CTF/NFI	HeLa cells	RNA polymerase II transcription/ adenovirus DNA replication	<i>b</i>
AP-1	HeLa cells	RNA polymerase II transcription	<i>c,d</i>
AP-2	HeLa cells	RNA polymerase II transcription	<i>e,f</i>
NF- κ B	Namalwa cells	RNA polymerase II transcription	<i>g</i>
	HeLa cells	RNA polymerase II transcription	<i>h</i>
Pit-1/GHF-1/PUF-I	Bovine spleen	RNA polymerase II transcription	<i>i</i>
	GC2 cells	RNA polymerase II transcription	<i>j,k</i>
	GH ₃ cells	RNA polymerase II transcription	<i>l</i>
μ EBP-E	Mouse plasmacytoma cells	RNA polymerase II transcription	<i>m</i>
ATP	HeLa cells	RNA polymerase II transcription	<i>n</i>
SRF/f-EBP/CBF	HeLa cells	RNA polymerase II transcription	<i>o-q</i>
TEF-1	HeLa cells	RNA polymerase II transcription	<i>r</i>
OBP	HSV-1-infected Vero cells	DNA replication	<i>s</i>
UBF1	HeLa cells	RNA polymerase I transcription	<i>t</i>
TFIIIC2	HeLa cells	RNA polymerase III transcription	<i>u</i>
Avian			
COUP factor	Chicken oviducts	RNA polymerase II transcription	<i>v</i>
PAL	Chicken erythrocytes	RNA polymerase II transcription	<i>w</i>
<i>Drosophila melanogaster</i>			
GAGA	Kc cells, embryos	RNA polymerase II transcription	<i>x,y</i>
DTF-1	Embryos	RNA polymerase II transcription	<i>z</i>
Zeste	Kc cells	RNA polymerase II transcription/ transvection	<i>aa</i>
Transposase	MT Δ 2-3 cells	P element-mediated transposition	<i>bb</i>
<i>Saccharomyces cerevisiae</i>			
HSE-binding/HSTF	Strain BJ2168	RNA polymerase II transcription	<i>cc</i>
	Strain BJ926	RNA polymerase II transcription	<i>dd</i>
RAP-1/GRFI	Strain BJ2168	RNA polymerase II transcription/ DNA attachment to nuclear scaffold	<i>ee</i>
	Strain BJ926		<i>ff</i>
ABF-1/SBFI	Strain PEP4D	DNA replication	<i>gg</i>
CBP-1	Strain SK1	Centromere-binding protein	<i>hh</i>

^a M. R. Briggs, J. T. Kadonaga, S. P. Bell, and R. Tjian, *Science* **234**, 47 (1986).

^b K. A. Jones, J. T. Kadonaga, P. J. Rosenfeld, T. J. Kelly, and R. Tjian, *Cell (Cambridge, Mass.)* **48**, 79 (1987).

^c W. Lee, P. Mitchell, and R. Tjian, *Cell (Cambridge, Mass.)* **49**, 741 (1987).

^d P. Angel, M. Imagawa, R. Chiu, B. Stein, R. J. Imbra, H. J. Rahmsdorf, C. Jonat, P. Herrlich, and M. Karin, *Cell (Cambridge, Mass.)* **49**, 729 (1987).

^e P. J. Mitchell, C. Wang, and R. Tjian, *Cell (Cambridge, Mass.)* **50**, 847 (1987).

ous purification of multiple DNA-binding proteins from a single protein fraction. This technique is not only effective, but it is also simple and straightforward to perform. Preparation of the DNA affinity resin can be easily accomplished within 2 days, and a DNA affinity chromatography experiment can be carried out in about 4 hr. Table I is a partial list of proteins that have been purified by using this technique. It may serve as a helpful guide for the purification of related proteins, and, in particular, the listed references may provide important information on

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- ^f M. Imagawa, R. Chiu, and M. Karin, *Cell (Cambridge, Mass.)* **51**, 251 (1987).
^g K. Kawakami, C. Scheidereit, and R. G. Roeder, *Proc. Natl. Acad. U.S.A.* **85**, 4700 (1988).
^h P. A. Baeuerle and D. Baltimore, *Genes Dev.* **3**, 1689 (1989).
ⁱ M. J. Lenardo, A. Kuang, A. Gifford, and D. Baltimore, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8825 (1988).
^j H. J. Mangalam, V. R. Albert, H. A. Ingraham, M. Kapiloff, L. Wilson, C. Nelson, H. Elsholtz, and M. G. Rosenfeld, *Genes Dev.* **3**, 946 (1989).
^k J.-L. Castrillo, M. Bodner, and M. Karin, *Science* **243**, 814 (1989).
^l Z. Cao, E. A. Barron, and Z. D. Sharp, *Mol. Cell. Biol.* **8**, 5432 (1988).
^m C. L. Peterson, S. Eaton, and K. Calame, *Mol. Cell. Biol.* **8**, 4972 (1988).
ⁿ T. Hai, F. Liu, E. A. Allegretto, Marin, and M. R. Green, *Genes Dev.* **2**, 1216 (1988).
^o R. Treisman, *EMBO J.* **6**, 2711 (1987).
^p R. Prywes and R. G. Roeder, *Mol. Cell. Biol.* **7**, 3482 (1987).
^q L. M. Boxer, R. Prywes, R. G. Roeder, and L. Kedes, *Mol. Cell. Biol.* **9**, 515 (1989).
^r I. Davidson, J. H. Xiao, R. Rosales, A. Staub, and P. Chambon, *Cell (Cambridge, Mass.)* **54**, 931 (1988).
^s P. Elias and I. R. Lehman, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 2959 (1988).
^t S. P. Bell, R. M. Learned, H.-M. Jantzen, and R. Tjian, *Science* **241**, 1192 (1988).
^u S. K. Yoshinaga, N. D. L'Etoile, and A. J. Berk, *J. Biol. Chem.* **264**, 10726 (1989).
^v M. K. Bagchi, S. Y. Tsai, M.-J. Tsai, and B. W. O'Malley, *Mol. Cell. Biol.* **7**, 4151 (1987).
^w B. M. Emerson, J. M. Nickol, and T. C. Fong, *Cell (Cambridge, Mass.)* **57**, 1189 (1989).
^x M. D. Biggin and R. Tjian, *Cell (Cambridge, Mass.)* **53**, 699 (1988).
^y D. S. Gilmour, G. H. Thomas, and S. C. R. Elgin, *Science* **245**, 1487 (1989).
^z K. K. Perkins, G. M. Dailey, and R. Tjian, *Genes Dev.* **2**, 1615 (1988).
^{aa} M. D. Biggin, S. Bickel, M. Genson, V. Pirrotta, and R. Tjian, *Cell (Cambridge, Mass.)* **53**, 713 (1988).
^{bb} P. D. Kaufman, R. F. Doll, and D. C. Rio, *Cell (Cambridge, Mass.)* **59**, 359 (1989).
^{cc} P. K. Sorger and H. R. B. Pelham, *EMBO J.* **6**, 3035 (1987).
^{dd} G. Wiederrecht, D. J. Shuey, W. A. Kibbe, and C. S. Parker, *Cell (Cambridge, Mass.)* **48**, 507 (1987).
^{ee} J. F.-X. Hofmann, T. Laroche, A. H. Brand, and S. M. Gasser, *Cell (Cambridge, Mass.)* **57**, 725 (1989).
^{ff} A. R. Buchman, N. F. Lue, and R. D. Kornberg, *Mol. Cell. Biol.* **8**, 5086 (1988).
^{gg} K. S. Sweder, P. R. Rhode, and J. L. Campbell, *J. Biol. Chem.* **263**, 17270 (1988).
^{hh} M. Cai and R. W. Davis, *Mol. Cell. Biol.* **9**, 2544 (1989).

the preparation of partially purified protein fractions that are suitable for affinity chromatography.

General Approach to Purification of Sequence-Specific DNA-Binding Proteins

A strategy for the purification of sequence-specific DNA-binding proteins is as follows. First, the DNA sequence to which the factor binds is determined by techniques such as DNase I footprinting,¹⁴ methidiumpropyl EDTA-Fe(II) footprinting,¹⁵ and dimethyl sulfate methylation protection.¹⁶ If possible, several different binding sites should be surveyed to identify a high-affinity recognition sequence, which would be useful in the construction of the DNA affinity resin. Second, optimal conditions are established for binding of the factor. The effect of temperature, Mg(II) concentration, pH, and monovalent ions, such as chloride versus glutamate,¹⁷ should be examined. In addition, the sequence-specific binding of the factor should be tested in the presence of nonspecific DNAs, such as calf thymus DNA, poly(dI-dC), poly(dG-dC), and poly(dA-dT) (see Proper Use of Nonspecific Competitor DNAs, below). Third, the factor is partially purified to remove nucleases and other contaminants that might adversely affect the affinity resin (this will be discussed later). Finally, two or more DNA affinity resins that contain high-affinity recognition sites with different flanking DNA sequences are prepared. A control resin that does not contain any binding sites should also be made. If a protein is purified by passage through two different affinity resins, then contamination by proteins that bind fortuitously to flanking oligodeoxynucleotide sequences is minimized. The control resin is used to identify proteins that bind nonspecifically to DNA-agarose. The desired protein can be tentatively identified by using this approach because a species that is enriched by chromatography through the specific affinity resins, but is not enriched by chromatography through the control resin, is good candidate for either the sequence-specific DNA-binding protein or a molecule that is closely associated with the DNA-binding protein.

It is important to identify nonspecific DNA-binding proteins that may contaminate the protein preparations after DNA affinity chromatography. The use of a control DNA affinity resin that does not contain binding sites for the sequence-specific factor in parallel with the specific DNA affinity

¹⁴ D. Galas and A. Schmitz, *Nucleic Acids Res.* **5**, 3157 (1978).

¹⁵ M. W. Van Dyke and P. B. Dervan, *Nucleic Acids Res.* **11**, 5555 (1983).

¹⁶ U. Siebenlist and W. Gilbert, *Proc. Natl. Acad. Sci. U.S.A.* **77**, 122 (1980).

¹⁷ S. Leirimo, C. Harrison, D. S. Cayley, R. R. Burgess, and M. T. Record, Jr., *Biochemistry* **26**, 2095 (1987).

columns (using the same protein fraction as the starting material) should minimize misidentification of a nonspecific DNA-binding protein for the desired factor. In addition, the polypeptide that is responsible for the sequence-specific DNA-binding activity can be identified by recovery of protein from an SDS-PAGE gel and renaturation of the denatured polypeptide¹⁸ followed by a DNA-binding assay (preferably a DNase I footprinting assay). In HeLa cells, there are two common nonspecific DNA-binding proteins that often contaminate preparations of affinity-purified factors: poly(ADP-ribose) polymerase (NAD⁺ ADP-ribosyltransferase),^{19,20} which has an M_r of 116,000, and the Ku antigen,²¹ which consists of two polypeptides of M_r 70,000 and 80,000. Thus, when purifying DNA-binding factors from HeLa cells, it is prudent to be suspicious of polypeptides of M_r 70,000, 80,000, and 116,000.

Preparation of DNA Affinity Resin

Preliminary Considerations

A few practical considerations are worth noting before preparation of the DNA affinity resin (5-ml scale). First, oligodeoxynucleotides ranging in size from 14 to 51 bases have been used successfully. Since the coupling of the DNA to the CNBr-activated resin probably occurs via primary amino groups on unpaired bases, the oligodeoxynucleotides should have a 4-base overhang; I usually include 5'-GATC . . . at the 5' end of the oligodeoxynucleotides. Also, do not use 21- or 42-mers because, if the DNA sequence has a bend, the oligodeoxynucleotides may circularize on ligation. The amount of DNA that is coupled to the Sepharose is typically 80–90 $\mu\text{g/ml}$ resin, which corresponds to a protein-binding capacity of 7 nmol/ml resin if there is one recognition site per 20 base pairs (bp). The affinity resins are stable in the storage buffer at 4° for at least 1 year and can be reused >30 times without any detectable decrease in the protein-binding capacity.

Materials and Reagents

Synthetic oligodeoxynucleotides, 250 μg each (500 μg total DNA) of complementary oligodeoxynucleotides that contain the recognition sequence of the desired protein. Note the recommendations given in

¹⁸ D. A. Hager and R. R. Burgess, *Anal. Biochem.* **109**, 76 (1980).

¹⁹ K. Ueda and O. Hayaishi, *Annu. Rev. Biochem.* **54**, 73 (1985).

²⁰ E. Slattery, J. D. Dignam, T. Matsui, and R. G. Roeder, *J. Biol. Chem.* **258**, 5955 (1983).

²¹ T. Mimori, J. A. Hardin, and J. A. Steitz, *J. Biol. Chem.* **261**, 2774 (1986).

the previous section for the design of the oligodeoxynucleotides. The oligodeoxynucleotides should be purified by electrophoresis in a polyacrylamide-urea gel. To estimate concentration of single-stranded DNA, assume that $1 A_{260\text{ nm}} \text{ unit} = 40 \mu\text{g/ml DNA}$

10 $\times$ T4 polynucleotide kinase buffer: 500 mM Tris-HCl, pH 7.6, containing 100 mM MgCl_2 , 50 mM dithiothreitol (DTT), 1 mM spermidine, and 1 mM EDTA. Store at 4°. Add extra 10 mM DTT just before use

T4 polynucleotide kinase (#201; New England Biolabs, Boston, MA)

10 $\times$ Linker-kinase buffer: 660 mM Tris-HCl, pH 7.6, containing 100 mM MgCl_2 , 150 mM DTT, and 10 mM spermidine. Store at 4°. Add extra 10 mM DTT just before use

T4 DNA ligase (#202; New England Biolabs)

Miscellaneous solutions for 5' phosphorylation and ligation of the oligodeoxynucleotides for coupling to the Sepharose: TE buffer (10 mM Tris-HCl, pH 7.6, containing 0.1 mM EDTA); 10 mM ammonium acetate; 20 mM ATP (Na^+), pH 7.0; absolute ethanol; phenol equilibrated with TE; phenol-chloroform (1 : 1, v/v); chloroform-isoamyl alcohol (24 : 1, v/v); 3 M sodium acetate; 2-propanol

Sepharose CL-2B (17-0140-01; Pharmacia, Piscataway, NJ)

Cyanogen bromide (caution: *highly toxic*) (C9, 149-2; Aldrich)

Miscellaneous solutions required for coupling of the DNA to the Sepharose: 5 M NaOH; *N,N*-dimethylformamide (27,054-7; Aldrich); 10 mM potassium phosphate, pH 8.0 (500 ml); 1 M ethanolamine-HCl, pH 8.0 (200 ml); 1 M potassium phosphate, pH 8.0 (200 ml); 1 M KCl (200 ml)

Column storage buffer: 10 mM Tris-HCl, pH 7.6, containing 1 mM EDTA, 0.3 M NaCl, and 0.04% (w/v) sodium azide (freshly added)

Materials: Standard equipment for manipulation of DNA [microcentrifuges, water baths, Speedvac evaporator/concentrator (Savant, Hicksville, NY)], 60-ml coarse-sintered glass funnel; rotating wheel (multipurpose rotator, model 151; Scientific Industries, Inc.)

5'-Phosphorylation of Oligodeoxynucleotides

1. Combine 65 μl DNA (250 μg of *each* oligodeoxynucleotide in TE) with 10 μl of 10 $\times$ T4 polynucleotide kinase buffer. Incubate at 88° for 2 min; 65° for 10 min; 37° for 10 min; and room temperature for 5 min.

2. Add 15 μl of 20 mM ATP, pH 7.0, containing about 5 μCi [$\gamma\text{-}^{32}\text{P}$]ATP, and 10 μl of T4 polynucleotide kinase (100 units) to a final volume of 100 μl . Incubate the solution at 37° for 2 hr.

3. Add 50 μl of 10 M ammonium acetate and 100 μl water to a final

volume of 250 μ l. Heat at 65° for 15 min to inactivate the kinase. Let the sample cool to room temperature, and then add 750 μ l ethanol. Mix by inversion. Spin (microcentrifuge) for 15 min at room temperature to pellet DNA. Discard supernatant.

4. Add 225 μ l TE. Dissolve the pellet by vortexing. Then add 250 μ l phenol–chloroform (1 : 1, v/v). Vortex for 1 min. Spin 5 min to separate phases. Transfer the upper layer to a new tube. Then add 250 μ l chloroform–isoamyl alcohol (24 : 1, v/v). Vortex for 1 min. Spin 5 min to separate phases. Transfer the upper layer to a new tube.

5. Add 25 μ l of 3 M sodium acetate. Mix by vortexing. Then add 750 μ l ethanol. Mix by inversion. Spin 15 min to pellet DNA. Discard supernatant. Then add 800 μ l of 75% ethanol. Mix by vortexing. Spin 5 min at room temperature. Discard supernatant. Dry pellet in Speedvac.

Ligation of Oligodeoxynucleotides

1. Dissolve the 5'-phosphorylated DNA pellet from above in 65 μ l water. Then add 10 μ l of 10 $\times$ linker–kinase buffer to give a final volume of 75 μ l. Vortex thoroughly to ensure that the pellet is completely dissolved.

2. Add 20 μ l of 20 mM ATP, pH 7.0, and 5 μ l of T4 DNA ligase (30 Weiss units) to a final volume of 100 μ l. Incubate at 15° overnight.

Monitor the progress of ligation by agarose gel electrophoresis (use 0.5 μ l of ligation reaction per gel lane). The average length of the ligated oligodeoxynucleotides should be at least 10-mers. Depending on the oligodeoxynucleotides used, the optimal temperature for ligation will vary from 4 to 30°. Short oligodeoxynucleotides ($\leq$ 15-mers) tend to ligate better at lower temperatures (4 to 15°), whereas oligodeoxynucleotides that have a moderate degree of palindromic symmetry, which have a tendency to self-anneal, ligate better at higher temperatures (15 to 30°).

Also, 5'-phosphorylated oligodeoxynucleotides often do not ligate on the first attempt. Thus, if ligation does not occur, extract the DNA once with 1 : 1 (v/v) phenol–chloroform, extract the DNA once with chloroform, and ethanol precipitate (using sodium acetate as salt). Dissolve the DNA in 225 μ l TE, add 25 μ l 3 M sodium acetate, and reprecipitate with ethanol (750 μ l). Wash the 75% ethanol, dry in the Speedvac, and try the ligation again.

Preparation of Ligated DNA for Coupling to Sepharose CL-2B

1. Add 100 μ l phenol to the 100 μ l ligation reaction. Vortex 1 min. Spin 5 min at room temperature. Transfer upper layer to a new tube.

2. Add 100 μ l of chloroform–isoamyl alcohol (24 : 1, v/v). Vortex 1 min. Spin 5 min at room temperature. Transfer upper layer to a new tube.

3. Add 33 μ l 10 M ammonium acetate. Mix by vortexing.
4. Add 133 μ l 2-propanol. Mix by inversion. Incubate at -20° for 20 min. Spin 15 min to pellet DNA. Discard supernatant.
5. Add 225 μ l TE. Vortex to dissolve pellet. Add 25 μ l of 3 M sodium acetate. Mix by vortexing. Add 750 μ l ethanol. Mix by inversion. Spin 15 min to pellet DNA. Discard supernatant.
6. Wash DNA two times with 75% ethanol. Dry pellet in Speedvac.
7. Dissolve the DNA in 50 μ l glass-distilled water. Store at -20° . *Do not dissolve the DNA in TE; it will interfere with the coupling reaction.*

Coupling of DNA to Sepharose

The coupling of the DNA to the Sepharose CL-2B is similar to the procedure described by Arndt-Jovin *et al.*²

1. Wash 10 to 15 ml (settled bed volume) of Sepharose CL-2B extensively with glass-distilled water (500 ml) in a 60-ml coarse-sintered glass funnel.
2. Transfer the moist Sepharose to a 25-ml graduated cylinder and estimate a 10-ml settled bed volume of resin. Then add water to a 20-ml final volume. Transfer this slurry to a 150-ml glass beaker in a water bath equilibrated to 15° over a magnetic stirrer *in a fume hood*.
3. *In the fume hood*, measure 1.1 g of cyanogen bromide (**DANGER—TOXIC**) in a 25-ml Erlenmeyer flask with Parafilm covering the mouth of the flask. It is better to have slightly more than 1.1 g than slightly less than 1.1 g. Dissolve the cyanogen bromide in 2 ml of *N,N*-dimethylformamide. It will instantly dissolve. Add the cyanogen bromide solution dropwise over 1 min to the stirring slurry of Sepharose *in the fume hood*.
4. Immediately add NaOH as follows. Add 30 μ l of 5 M NaOH every 10 sec to the stirring mixture (at 15°) for 10 min until the total volume of NaOH added is 1.8 ml.
5. Immediately add 100 ml of ice-cold water to the beaker and pour the mixture into a 60-ml coarse-sintered glass funnel. *At this point, it is very important that the resin is not suction filtered into a dry cake.* If the resin is accidentally dried during suction filtration, it will be necessary to start the CNBr activation procedure again from the beginning.
6. Wash the resin four times with 100 ml ice-cold water ($\leq 4^{\circ}$) each time, followed by two washes with 100 ml ice-cold 10 mM potassium phosphate, pH 8.0.
7. Immediately transfer one-half (5 ml) of the activated resin to a 15-ml polypropylene screw cap tube and add about 2 ml of 10 mM potassium phosphate, pH 8.0, until the resin has the consistency of a thick slurry.

[The remaining half (5 ml) of the activated resin could also be used for coupling another sample of DNA.]

8. Immediately add the DNA (in 50 μ l water). Incubate on a rotating wheel overnight at room temperature.

9. Transfer the resin to a 60-ml coarse-sintered glass funnel, and wash twice with 100 ml water followed by a wash with 100 ml 1 M ethanolamine-hydrochloride, pH 8.0. Compare the level of radioactivity in the first few milliliters of the filtrate with the level of radioactivity in the washed resin to estimate the efficiency of incorporation of DNA to the resin. Usually, all detectable radioactivity is present only in the resin.

10. Transfer the resin to a 15-ml polypropylene screw cap tube and add 1 M ethanolamine hydrochloride, pH 8.0 until the mixture is a smooth slurry. Incubate the tube on a rotating wheel at room temperature for 4 to 6 hr (this step is to inactivate unreacted cyanogen bromide-activated Sepharose).

11. Wash the resin with the following solutions: 100 ml of 10 mM potassium phosphate, pH 8.0; 100 ml of 1 M potassium phosphate, pH 8.0; 100 ml of 1 M KCl; 100 ml water; and 100 ml of column storage buffer.

12. Store the resin at 4° (do not freeze the resin). The resin is stable for at least 1 year at 4°.

DNA Affinity Chromatography

Partial Purification of DNA-Binding Proteins before DNA Affinity Chromatography

The protein fraction that is to be subjected to affinity chromatography generally does not need to be extensively purified beyond the crude extract, although it is important that the fraction does not contain high levels of nucleases, which would destroy the affinity resin. The references listed in Table I may provide useful information for the preparation of crude extracts and partial purification of the DNA-binding factors. Typically, a crude nuclear extract is prepared, and the desired protein is partially purified by one column chromatography step and then applied to the affinity resins. For example, in the isolation of many sequence-specific transcription factors from HeLa cells, such as Sp1, CTF/NFI, AP-1, AP-2, AP-3, and AP-4 (for review, see Ref. 22), it is sufficient to prepare a standard nuclear extract²³ and then to subject the extract to Sephacryl S-300 gel filtration before affinity chromatography. Other chromatographic

²² P. J. Mitchell and R. Tjian, *Science* **245**, 371 (1989).

²³ J. D. Dignam, R. M. Lebovitz, and R. G. Roeder, *Nucleic Acids Res.* **11**, 1475 (1983).

resins that are commonly used to carry out partial purification of DNA-binding proteins include heparin-agarose, DNA-cellulose,¹ phosphocellulose (P11; Whatman, Clifton, NJ), DEAE-cellulose (DE-52; Whatman), Mono S (Pharmacia), and Mono Q (Pharmacia) columns. It is also possible to purify partially the DNA-binding factor by chromatography with a specific DNA-Sepharose resin that is constructed with oligodeoxynucleotides that *do not* contain binding sites for the desired protein (for an example, see Ref. 24). This approach is effective if the desired factor elutes from this column at a low salt concentration while high-affinity nonspecific DNA-binding proteins elute from the resin at a high salt concentration. If such a separation of the desired protein and high-affinity nonspecific DNA-binding proteins can be achieved, then it would not be necessary to add competitor DNA during subsequent chromatography with a DNA affinity resin that contains binding sites for the desired factor. Finally, a variety of eukaryotic sequence-specific transcription factors have been found to contain multiple O-linked *N*-acetylglucosamine monosaccharide residues.²⁵ It has been demonstrated that three of these factors, Sp1, CTF/NFI, and HNF1, can be purified in high yield by wheat germ agglutinin affinity chromatography followed by sequence-specific DNA affinity chromatography.^{26,27} Hence, it may be worthwhile to examine whether or not the desired DNA-binding factor can be partially purified by wheat germ agglutinin affinity chromatography.

Proper Use of Nonspecific Competitor DNAs

The proper use of nonspecific competitor DNAs is an important feature of successful purification of proteins. It is necessary to determine experimentally the appropriate amount and type of competitor DNA to add for each affinity chromatography experiment. DNA-binding assays (DNase I footprinting or gel mobility shift experiments) should be carried out with the starting material (the identical protein fraction that is to be applied to the affinity resin) in the presence of different types and amounts of competitor DNAs. Competitor DNAs that are commonly used include sonicated calf thymus DNA, poly(dI-dC), poly(dA-dT), and poly(dG-dC). Poly(dI-dC), poly(dA-dT), or poly(dG-dC) is dissolved to a final concentration of 10 $A_{260\text{ nm}}$ units in TE containing 100 mM NaCl, heated to 90°, and slowly cooled over 30 to 60 min to room temperature before use. The average length of the competitor DNA should be around 1 kilobase (kb). It is

²⁴ P. D. Kaufman, R. F. Doll, and D. C. Rio, *Cell (Cambridge, Mass.)* **59**, 359 (1989).

²⁵ S. P. Jackson and R. Tjian, *Cell (Cambridge, Mass.)* **55**, 125 (1988).

²⁶ S. P. Jackson and R. Tjian, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 1781 (1989).

²⁷ S. Lichtsteiner and U. Schibler, *Cell (Cambridge, Mass.)* **57**, 1179 (1989).

common for poly(dI-dC) to be sold in lengths greater than 9 kb. If the DNA is too long, it can be fragmented by sonication. Also, a mixture of different competitor DNAs can be used in a single experiment.

The appropriate amount and type of competitor DNA to use in affinity chromatography are estimated as follows. First, DNA-binding experiments are carried out using the partially purified protein fraction with varying levels of different types of competitor DNAs. In these studies, which can be viewed as small-scale DNA affinity experiments, the highest quantity of competitor DNA that does not adversely affect the binding of the protein is determined. The amounts of competitor DNA and protein fraction used in the DNA-binding studies are adjusted to full scale, and then the actual amount of competitor DNA that will be used for affinity chromatography is one-fifth of the quantity that was obtained by adjustment of the small-scale DNA-binding assay to full scale.

The method of estimation of competitor DNA is best explained by a simple example. For instance, the sequence-specific DNA binding of protein XX is strongly inhibited by either calf thymus DNA or poly(dA-dT), whereas it is weakly inhibited by poly(dI-dC). In a DNA-binding assay, the binding of XX (10 μ l of a protein fraction containing XX) is affected as follows: (1) no detectable inhibition by 1 μ g or less of poly(dI-dC); (2) weak inhibition by 2 μ g of poly(dI-dC); (3) strong inhibition by 4 μ g or more of poly(dI-dC). Thus, the highest level of competitor DNA that does not affect XX binding with 10 μ l of the protein fraction is 1 μ g of poly(dI-dC). If we have 10 ml of the protein fraction containing XX (which is $1000 \times 10 \mu$ l), a direct unmodified scale-up of competitor DNA would be $1000 \times 1 \mu$ g = 1000 μ g poly(dI-dC). However, use one-fifth of the direct scale-up amount, which in this case would be 1000μ g $\times 1/5$ = 200 μ g poly(dI-dC).

Materials and Reagents

Buffer Z: 25 mM HEPES, K⁺, pH 7.6, containing variable KCl (as indicated in the protocol), 12.5 mM MgCl₂, 1 mM DTT (freshly added just before use), 20% (v/v) glycerol, and 0.1% (v/v) Nonidet P-40 (NP-40). Adjust the pH of 1 $\times$ buffer Z with KOH. Do not make a 10 $\times$ buffer. Store at 4°. *Note:* Some preparations of HEPES buffer contain potent inhibitors of DNA-binding proteins. We typically use Sigma #H3375

Buffer Z^c: 25 mM HEPES, K⁺, pH 7.6, containing variable KCl (as indicated), 1 mM DTT (freshly added just before use), 20% (v/v) glycerol, and 0.1% (v/v) Nonidet P-40. See comments on buffer Z

TM buffer: 50 mM Tris-HCl, pH 7.9, containing variable KCl (as indicated), 12.5 mM MgCl₂, 1 mM DTT, 20% (v/v) glycerol, and 0.1%

(v/v) NP-40. Store at 4°. This buffer can be used as a substitute for buffer Z if inhibitors of DNA-binding proteins are present in the HEPES buffer

Column regeneration buffer: 10 mM Tris-HCl, pH 7.8, containing 1 mM EDTA, 2.5 M NaCl, and 1% (v/v) NP-40. Store at room temperature.

This solution will be cloudy and separate into two phases (NP-40 and aqueous) on storage. Just before use, mix by swirling and shaking

Column storage buffer: 10 mM Tris-HCl, pH 7.6, containing 1 mM EDTA, 0.3 M NaCl, and 0.04% (w/v) sodium azide (freshly added)

Chromatography columns: Bio-Rad (Richmond, CA) 2-ml disposable Poly-Prep columns (#731-1550)

Competitor DNAs: Sonicated calf thymus DNA; poly(dI-dC); poly(dA-dT); poly(dG-dC). The average length of the DNA should be about 1 kb. If the average length is greater than 1 kb, the DNA should be sonicated until it is sheared to the appropriate size

Affinity Chromatography

All operations are performed at 4°. This procedure is described with buffer Z, but in some instances, factors have been found to bind to DNA with higher affinity in buffer Z^c, which is identical to buffer Z except that it does not contain MgCl₂.

1. Equilibrate the DNA affinity resin (1 ml settled bed volume) in a Bio-Rad 2-ml Poly-Prep column with two washes of 10 ml of buffer Z + 0.1 M KCl.

2. The protein fraction (in TM + 0.1 M KCl, buffer Z + 0.1 M KCl, or buffer Z^c + 0.1 M KCl) is combined with nonspecific competitor DNA.

3. The protein-DNA mixture is allowed to stand for 10 min. Then centrifuge this solution at 10,000 rpm for 10 min in a SS-34 rotor to pellet insoluble protein-DNA complexes.

4. The columns are loaded at gravity flow (15 ml/hr/column). When purifying a large quantity of material, use multiple 1-ml columns.

5. After loading the starting material, wash each column four times with 2 ml of buffer Z + 0.1 M KCl.

6. Elute the protein from each column by successive addition of 1-ml portions of buffer Z containing 0.2 M KCl, 0.3 M KCl, 0.4 M KCl, 0.5 M KCl, 0.6 M KCl, 0.7 M KCl, 0.8 M KCl, and 0.9 M KCl, and three times with 1 ml of buffer Z containing 1 M KCl. Collect 1-ml fractions that correspond to the addition of the 1-ml portions of buffer.

7. Save small aliquots of each fraction (20 μ l), freeze all of the fractions and aliquots in liquid nitrogen, and store at -80°.

8. Regenerate the affinity resin as follows. At room temperature, stop

the flow of the column, and then add 5 ml of regeneration buffer. Stir the resin with a narrow siliconized glass rod to mix the resin with the regeneration buffer. Let the buffer flow out of the column. Stop the flow of the column again, and repeat the addition and mixing of the regeneration buffer with the resin. Then, equilibrate the resin with two washes of 15 ml of column storage buffer, and store at 4°.

The protein fractions should be assayed for the sequence-specific DNA-binding activity. If further purification is desired, the fractions that contain the activity are combined. Then the pooled fractions are either diluted (with buffer Z without KCl) or dialyzed (against buffer Z plus 0.1 *M* KCl) to 0.1 *M* KCl final concentration. The protein fraction is subsequently combined with nonspecific competitor DNA (the amount and type of which need to be determined again experimentally) and reapplied to a DNA affinity resin.

Suggestions for Use and Handling of Affinity-Purified DNA-Binding Proteins

1. Quick-freeze the protein in liquid nitrogen.
2. Store the protein at -70°.
3. Thaw the protein quickly in water that is either cold or at room temperature. The thawed protein should then be maintained at 4°.
4. Since the protein concentration of affinity-purified proteins is typically low (in the range of 5 to 50 $\mu\text{g/ml}$), the buffers should contain 0.01 to 0.1% (v/v) NP-40 (a nonionic detergent) to minimize loss of protein by adsorption to plastic or glass.
5. Minimize handling of protein, especially with regard to exposure to plastic and glass, to which the protein might be irreversibly adsorbed. The use of siliconized plastic test tubes and pipette tips will minimize loss of protein by adsorption to plastic, but it is usually not necessary to use siliconized plasticware.
6. Typically, 2 to 10 ng of purified protein is sufficient for a complete footprint (with ≈ 10 fmol of DNA probe). Do not add competitor DNA to footprinting reactions with purified protein as it will severely inhibit the binding of the factors.

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[3] Using Protein Affinity Chromatography to Probe Structure of Protein Machines

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Most proteins in cells function in large assemblies of proteins created by specific protein-protein interactions. Thus, for example, such diverse processes as DNA replication, RNA synthesis, transport vesicle budding and fusion, and the incorporation of specific proteins into membranes can be considered to be mediated by protein complexes that function as "protein machines." Many of the protein-protein interactions that play important roles in assembling and regulating these complex protein machines are too weak to allow copurification of the interacting species from cellular extracts. Such relatively weak binding is to be expected, since components that must bind to one another reversibly in the concentrated environment of the intracellular fluid will fall apart rapidly in a dilute extract. Since the study of processes mediated by protein machines requires the *in vitro* reassembly of active complexes, the subunit composition and enzymatic properties of which are the same as those found inside the cell, it is important to have methods for identifying and purifying all of the interacting proteins, starting with one member of a protein complex.

In this chapter we describe the use of protein affinity chromatography to characterize and isolate the interacting components of protein complexes. We summarize the techniques used to construct affinity matrices, the preparation of extracts from prokaryotic and eukaryotic sources, and methods for analyzing the binding fractions. This technique has been used successfully to study interactions between elements of the cytoskeleton,^{1,2} protein transport machinery,³ transcription complexes^{4,5} and the DNA replication apparatus.^{6,7} A chapter that focuses exclusively on microtubule

¹ D. G. Drubin, *Cell Motil. Cytoskeleton* **15**, 7 (1990).

² K. G. Miller, C. M. Field, B. M. Alberts, and D. R. Kellogg, this series, Vol. **196**, p. 303.

³ S. Tajima, L. Lauffer, V. L. Rath, and P. Walter, *J. Cell Biol.* **103**, 1167 (1986).

⁴ Z. F. Burton, M. Killeen, M. Sopta, L. G. Ortolan, and J. Greenblatt, *Mol. Cell. Biol.* **8**, 1602 (1988).

⁵ R. J. Horwitz, J. Li, and J. Greenblatt, *Cell (Cambridge, Mass.)* **51**, 631 (1987).

⁶ T. Formosa, R. L. Burke, and B. M. Alberts, *Proc. Natl. Acad. Sci. U.S.A.* **80**, 2442 (1983).

⁷ B. M. Alberts, J. Barry, P. Bedinger, T. Formosa, C. V. Jongeneel, and K. Kreuzer, *Cold Spring Harbor Symp. Quant. Biol.* **47**, 655 (1983).

and actin filament affinity columns has been published previously in this series.² Given the widespread occurrence and importance of protein–protein interactions in biology, protein affinity chromatography should be useful for studying nearly all biological processes.

Preparing Affinity Columns

Selecting Activated Matrix

Protein affinity chromatography allows sensitive detection of protein–protein interactions because of the high concentration of binding sites presented by the matrix. To achieve optimal sensitivity, it is important to choose a matrix that will couple a maximal amount of protein in a minimal volume without introducing potentially denaturing multiple cross-links to individual protein molecules.

A second consideration in choosing a matrix is the minimization of interactions between proteins in a cellular extract and the column matrix itself. Agarose is a good neutral chromatographic matrix, but the process of activating it to accept a protein ligand can alter the agarose, creating unacceptably high levels of nonspecific protein binding. For example, the common method of activating agarose by treating it at high pH with cyanogen bromide leaves the matrix covered with a variety of charged residues.⁸ These residues act as nonspecific ion exchangers, and sometimes bind such high levels of proteins themselves that weak interactions with the covalently bound protein substituent are obscured. Therefore, although results with cyanogen bromide-activated agarose have sometimes been acceptable,^{9–11} the nonspecific binding by such matrices is often a problem.

We have had our most consistent success in optimizing for these considerations by coupling proteins to the *N*-hydroxysuccinimide-activated agarose matrix sold commercially by Bio-Rad (Richmond, CA), Affi-Gel 10 (we have had less favorable experiences with the similarly constructed but cationic matrix Affi-Gel 15). This matrix has not been cyanogen bromide treated, and it contains the active moiety at the end of a neutral spacer arm coupled to an agarose matrix. The active groups are present at a high enough concentration to couple proteins to the 20 mg/ml level, yet leave a neutral matrix with low background binding characteristics.

⁸ M. Wilchek, T. Miron, and J. Kohn, this series, Vol. 104, p. 3.

⁹ J. Greenblatt and J. Li, *J. Mol. Biol.* **147**, 11 (1981).

¹⁰ J. Greenblatt and J. Li, *Cell (Cambridge, Mass.)* **24**, 421 (1981).

¹¹ M. Sopta, R. W. Carthrew, and J. Greenblatt, *J. Biol. Chem.* **260**, 10353 (1985).

In addition, the commercial availability allows good reproducibility and convenience.

Preparation of Protein Ligand

Ideally, the protein ligand that is used for affinity chromatography should be pure in order to ensure that every interacting protein that is detected is binding to the ligand rather than to a contaminant in the preparation. In practice, since true homogeneity of a protein preparation is not achievable, it is wise to develop independent criteria, either genetic^{9,12,13} or biochemical,^{2,4,11,14} that the proteins truly interact with each other. Optimally, one would like to be able to construct an appropriate control column that contains all of the contaminants, but none of the true ligand.¹⁵ This can sometimes be achieved by performing parallel protein purifications from two strains or cell lines that differ only in their content of the protein of interest.

Coupling Proteins to Activated Matrix

To restrict the coupling reaction to the protein of interest, protein samples should be buffered with reagents lacking primary amines or sulfhydryls that will react with the activated Affi-Gel 10 matrix. Tris buffers usually are not suitable, but *N*-2-hydroxyethylpiperazine-*N'*-2-ethanesulfonic acid (HEPES), 4-morpholinopropanesulfonic acid (MOPS), and bicarbonate buffers are. The protein sample may contain glycerol, MgCl₂, NaCl and other reagents as long as no primary amines or sulfhydryls are present. (Some exceptions can be made if a protein couples efficiently and rapidly. In this case, the high concentration of active coupling groups ensures that sufficient sites are available for covalent binding of both the protein and the buffer component.) Our standard protocol is to dialyze about 0.5 to 3 mg of protein in a volume of 1–2 ml against several changes of coupling buffer [25 mM HEPES, pH 7.5–7.9, 5–20% (w/v) glycerol, 50–500 mM NaCl], at 4° for 5–15 hr. This protein sample is then coupled to an appropriate amount (often about 0.5 ml) of washed, activated matrix. Glycerol is useful for stabilizing nearly all proteins; it is therefore routinely used at a concentration of 10 to 20% in all buffers. High levels of NaCl have been used for coupling proteins that tend to aggregate at low ionic strengths.

¹² T. C. Huffaker, M. A. Hoyt, and D. Botstein, *Annu. Rev. Genet.* **21**, 259 (1987).

¹³ D. G. Drubin, K. G. Miller, and D. Botstein, *J. Cell Biol.* **107**, 2551 (1988).

¹⁴ A. E. Adams, D. Botstein, and D. G. Drubin, *Science* **243**, 231 (1989).

¹⁵ K. F. Stringer, C. J. Ingles, and J. Greenblatt, *Nature (London)* **345**, 783 (1990).

The Affi-Gel 10 matrix is supplied as a slurry in 2-propanol and is stored at -20° . 2-Propanol must be removed before adding the protein sample. The washing and coupling can be done in a single tube; the matrix is quite sticky, so transferring should be minimized. We have found it convenient to wash activated matrix in 4–10 ml capped polycarbonate centrifuge tubes [e.g., Falcon 2059 or 2063 tubes], or in 1.5-ml microfuge tubes for smaller volumes. The matrix is equilibrated to 4° , shaken to produce a slurry, then aliquots are removed using an Eppendorf pipette with the end of the disposable tip cut off to avoid clogging. Each aliquot is added directly to about 5 vol of cold, distilled H_2O , then the matrix is collected by centrifugation (1000 g) for 2 min. The supernatant is removed, the gel is washed twice with 5 vol of cold coupling buffer, and finally the matrix is suspended in the dialyzed protein sample. The coupling proceeds with gentle agitation on a rotator at 4° for several hours to overnight.

Using polycarbonate tubes and collecting the matrix by centrifugation decreases losses of material and drying out of the matrix compared to the use of filtration. However, the matrix can also be washed by gentle filtration, first with cold water and then with cold coupling buffer, usually six washes in all. The gel is then transferred into a plastic tube of appropriate size, weighed, and suspended in the dialyzed protein sample (2–5 ml of protein solution per gram of gel). We have found that the coupling is efficient even when the filtration process has made the gel somewhat difficult to resuspend. Most of the coupling occurs within a few hours, so the reaction can be terminated after 3–4 hr if desired. However, if short coupling times are used, the matrix should be blocked by treatment with neutralized ethanolamine (60–200 mM) for at least 2 hr.

Measuring Coupling Efficiency

The efficiency of coupling can be monitored in one of three ways. If the protein is pure, one can perform a Bradford assay for protein on a sample of the dialyzed protein used for the coupling and compare this with the supernatant from the reaction. Alternatively, the absorbance at 280 nm of both fractions can be measured. If the absorbance method is used, the sample and background controls should contain 0.1 M HCl to remove absorbance contributed by the unprotonated form of the *N*-hydroxysuccinimide moiety that is released during the coupling reaction. If the protein ligand is not pure, it is best to compare aliquots of the dialyzed protein and the postcoupling supernatant by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) followed by staining with Co-

massie Brilliant Blue and densitometric determination of the amount of protein in the gel bands.

The Affi-Gel matrices contain about 15 $\mu\text{mol/ml}$ of active esters. Assuming random distribution, these active groups are spaced about 5 nm apart, which is about equal to the diameter of a typical globular protein molecule of moderate size. It is therefore possible that multiple cross-links to the same protein molecule will form. If this acts to "freeze" proteins in denatured configurations, the columns will have lower than optimal binding capacities. In practice this has not usually been a problem. It can be detected and circumvented by allowing the activated matrix to hydrolyze in the presence of the aqueous coupling buffer for various amounts of time before adding the ligand protein. To prevent this problem, some researchers recommend using coupling conditions that allow not more than 80% of the ligand protein to be bound to the matrix²; controlling the linkage of microtubules to the agarose matrix in this way can produce columns that have a capacity that is three to four times greater than more heavily coupled columns.^{9,16}

Constructing Affinity Column

The matrix is recovered by allowing it to settle for several minutes, removing a sample of the supernatant to check the coupling efficiency as described above, and then suspending the matrix in a suitable column buffer. Affinity matrices can be used to detect binding proteins using batch elution techniques; in this case, matrices are suspended in cell lysates and then collected by centrifugation, washed, and eluted by suspending the matrix in buffers of appropriate composition. However, greater resolution and more efficient washing can be obtained by continuous-flow column chromatography. A variety of column sizes, types, and geometries can be used. We describe below methods for preparing micro-, standard, and preparative scale columns.

Microscale Columns. For small volumes, columns (20–100 μl) can be poured in capillary tubes. This is accomplished by heating a 200- μl capillary tube [for example, Clay Adams (Parsippany, NJ) Micropets, 12.6 cm long, 1.68-mm i.d.] in the middle over a flame and drawing it out. After cooling, the tube is scored with a diamond pencil and broken at the constricted region. This leaves one end of the tube with a narrow constriction and one end fully open. A slurry of glass beads [for example, VirTis Co. (Gardiner, NY) #16-220, 220- μm diameter] in water is drawn into the

¹⁶ D. R. Kellogg, C. M. Field, and B. M. Alberts, *J. Cell Biol.* **109**, 2977 (1989).

open end of the tube using suction on the drawn-out end. The exact volume of beads is not critical, at least in the 5- to 25- μ l range, since beads have not been observed to affect background binding by a column. The slurry of beads, with more water added as needed, is then allowed to settle into the narrow end. Excess liquid is allowed to drain out of the tube. A slurry of the affinity matrix is then drawn into the open end of the tube up to the bed of glass beads, the tube is set upright in a conical 1.5-ml centrifuge tube, and the matrix is allowed to settle and drain by gravity. These columns generally have a column bed that is about 1 cm high; they can be loaded and washed by applying buffers with a 100- μ l Hamilton syringe and collecting the eluted fractions by periodically moving the columns to fresh collection tubes. The flow rate in microcolumns depends on the precise geometry of the drawn-out tube and is therefore somewhat variable, but should be in the range of 200–400 μ l/hr.

Standard Scale Columns. Columns (0.2–3 ml) can be poured in plastic sterile syringes cut to an appropriate length. The bottom of the syringe is plugged with a polyethylene filter disk [for example, Ace Glass (Vineland, NJ) 5848 filter disk support] or with siliconized glass wool (for example, PhaseSep HGC166) and closed with a stopcock (for example, Bio-Rad 732-8107 three-way stopcocks). The matrix is suspended in about 5 vol of column buffer, and then placed into the column with a 9-in. Pasteur pipette; the tip of the pipette is inserted all the way to the bottom of the syringe to avoid trapping bubbles in the barrel of the syringe. As the matrix settles and the clear buffer drains from the bottom of the column, more slurry is added so that the packing column bed is always submerged in the slurry. It is important that the packing be continuous so that the column bed is not disrupted by discontinuous layers that will cause nonuniform flow characteristics. In order to test a column, a small amount of a solution containing a visible dye can be applied to the column. The dye should migrate through the column as a coherent disk.²

Once the bed is settled, it should be topped with a small amount of column buffer, the stopcock closed, and the column cap applied. The matrix bed should never be exposed to air, but rather should always have a small layer of buffer covering it to avoid improper flow characteristics. The column can be capped either with an appropriately sized silicone stopper or with the tip of the plunger of the syringe after removing the plastic handle. A needle is inserted into the stopper with the sharpened end pointed out of the column and the base of the needle is twisted back and forth until it breaks. This leaves the broken end of the needle inside the column and the tip outside. Tubing from a peristaltic pump can be attached to the tip of the needle and used to deliver solutions at uniform,

reproducible flow rates. The optimal flow rate depends on the cross-sectional area rather than the total bed volume; we typically use a rate of about 2 to 3 ml/hr for 1-ml syringes with 0.5- to 1-ml bed volumes.

Preparative Scale Columns. Preparative scale (10–50 ml) work can be done in standard chromatography columns (for example, disposable Bio-Rad Econo-Columns work quite well) run with a peristaltic pump. To assure good flow rates, it is best to avoid a column bed higher than 10 cm. We have used flow rates ranging from one to five bed volumes per hour, again depending on the geometry of the column; columns can generally be loaded with extract at higher flow rates than are appropriate for elution.

Control Columns. Control columns should be prepared in parallel to mimic as closely as possible the experimental matrix. However, we have found little difference between columns containing either no protein or an unrelated protein such as bovine serum albumin (BSA). Ideally, a protein with a similar isoelectric point should be used, although the issue of what constitutes an appropriate control for an affinity column is problematical. “Simple” ionic interactions cannot be discounted as irrelevant because most of the specific interactions that we have detected have an ionic component (see Table I). Would denatured protein be a reliable control, or might relevant domains for interactions reform on the matrix? Ultimately, any interactions detected by affinity chromatography should be validated as biologically relevant by genetic analysis or by other means such as the observation of colocalization of the two proteins *in vivo*, so the issue of an ideal control matrix is perhaps a secondary consideration.

Storing Matrices. If a stable protein is coupled to the matrix, the protein is not degraded by proteases in the extract, and the elution conditions used are mild, then affinity matrices can be equilibrated with buffer containing 50% glycerol, stored at -20° , and reused. High concentrations of glycerol are used to prevent the buffer from freezing, which both denatures proteins and destroys agarose matrices. For example, the T4 gene 32 protein can be coupled to an agarose matrix, then stored and reused for purifying the T4 DNA polymerase many times. Protein ligands on affinity columns are usually not harmed by high ionic strength elution buffers, but treatment with chaotropic agents such as high urea concentrations or SDS probably ruins these matrices permanently. Matrices that are to be reused should therefore not be treated in this way.

Preparing Extracts for Chromatography

Choosing a Method of Detecting Proteins

Detecting small amounts of protein normally involves polyacrylamide gel electrophoresis in the presence of sodium dodecyl sulfate (SDS), fol-

TABLE I
INTERACTIONS DETECTED BY PROTEIN AFFINITY CHROMATOGRAPHY

Ligand	Interacting protein	K_d (M)		Condition for elution	Ref.
		Apparent ^a	Actual		
Phage λ N protein	<i>E. coli</i> NusA	5×10^{-7}	3×10^{-7}	200–300 mM NaCl	9
<i>E. coli</i> NusA	<i>E. coli</i> RNA polymerase core component	10^{-7}	5×10^{-8}	200–300 mM NaCl	10
<i>E. coli</i> RNA polymerase	<i>E. coli</i> σ^{70}	ND	2×10^{-10}	500 mM NaCl	S. McCracken (unpublished data)
	<i>E. coli</i> NusA	ND	5×10^{-8}	500 mM NaCl	S. McCracken (unpublished data)
	<i>E. coli</i> NusG	ND	ND	500 mM NaCl	S. McCracken (unpublished data)
<i>E. coli</i> NusG	<i>E. coli</i> ρ factor	10^{-5}	ND	500 mM NaCl	J. Li (unpublished data)
<i>E. coli</i> ribosomal protein S10	<i>E. coli</i> NusB	3×10^{-6}	ND	1% SDS	S. Mason (unpublished data)
<i>E. coli</i> NusB	<i>E. coli</i> ribosomal protein S10	ND	ND	1% SDS	J. Li (unpublished data)
Mammalian RNA polymerase II	RAP38 (also known as SII or TFIIS)	10^{-7}	ND	200–300 mM NaCl	11
	RAP30/74 (also known as TFIIE or TFIIF)	2×10^{-8}	2×10^{-8}	200–300 mM NaCl	11
Yeast (<i>S. cerevisiae</i>) RNA polymerase II	yRAP37 (also known as P37)	10^{-7}	ND	200–300 mM NaCl	M. Sopta (submitted)
Activation domain of herpes simplex virus trans-activator VP16	Human or yeast TATA box factor TFIID	3×10^{-7}	ND	300–500 mM NaCl	15
Yeast (<i>S. cerevisiae</i>) TFIID	Human TFIIA	10^{-5}	ND	200–500 mM NaCl	J. Greenblatt and B. Honda (unpublished data)

^a The apparent K_d is estimated as one-twentieth of the minimum concentration of ligand required to retain the interacting protein efficiently. ND, not determined.

lowed either by staining gels with silver or by using fluorography to detect radioactively labeled proteins. In the latter case $^{35}\text{SO}_4^{2-}$ or [^{35}S]methionine is usually used to label cells because of the low cost and high specific activity relative to ^{14}C . Radioactive labeling has the advantage of allowing quantitation and detection of fractions containing binding proteins by

scintillation counting before polyacrylamide gel electrophoresis. However, radioactive labeling is relatively expensive and is not more sensitive than silver staining. In addition, the radioactive extracts must be handled more carefully, and fluorography is generally slower than silver staining. However, using radioactive extracts allows a distinction to be made between polypeptides that bind from the extract and proteolytic fragments of the ligand that might be released from the columns during a run, which would otherwise appear to be specific binding proteins. Depending on the ligand protein and the source of the extract, this can be a major consideration that favors the use of radioactive proteins.

Choosing Source of Starting Material

In some cases, a total cell extract is not the optimal starting point for affinity chromatography. For instance, if the ligand protein is known to act in the nucleus, a nuclear extract can provide a 10-fold enrichment of proteins that might interact, allowing much easier detection of the interacting species. In addition, for proteins present in cells in very low abundance, it might be necessary to use another chromatography matrix to fractionate an extract before screening various pools for interacting proteins. For example, a protein involved in DNA replication or transcription might interact with proteins that themselves bind to DNA; in this case, it could be beneficial to enrich for such proteins by DNA-cellulose chromatography to allow very minor interacting species to be detected. Such preliminary chromatographic steps can also assist in removing possible interfering substances from an extract.

Preparation of Whole-Cell Extracts

Whole-cell extracts are usually used as the source of interacting proteins. All procedures are performed at 0–4°.

Yeast cells can be broken by shaking or vortexing with 500- μ m glass beads. We typically break the cells by suspending them in 1 vol of lysis buffer [1 ml/g of cells; 20 mM Tris-HCl (pH 7.5), 1 mM Na₂EDTA, 1 mM 2-mercaptoethanol, and 50 mM NaCl], adding 1 vol of glass beads and vortexing for about 10 min.

Fresh or frozen bacterial cells are most conveniently disrupted by sonication, but they can also be broken by grinding in a mortar with levigated alumina [Sigma (St. Louis, MO) type 305; 2 g/g of bacterial]. The bacteria are suspended (before sonication or after grinding) in 1–2 vol of lysis buffer (same as described above for yeast cells) or in 10 mM Tris-HCl, pH 7.9, 14 mM magnesium acetate, 1 mM dithiothreitol (DTT), and 60–300 mM KCl.

Mammalian cells grown in tissue culture are first washed with phosphate-buffered saline (PBS) and then either used fresh or kept frozen at -70° . We usually homogenize the cells, but have also had good success with sonication.¹¹ The cells are homogenized in 2 ml/g of 10 mM HEPES (pH 7.9), 1.5 mM MgCl_2 , 10 mM KCl, and 0.5 mM dithiothreitol using 10 strokes of a Teflon-coated, motor-driven homogenizer. The nuclei in the extract are then disrupted by adding 1.5 ml/g of 50 mM HEPES (pH 7.9), 75% (v/v) glycerol, 1.26 M NaCl, 0.6 mM Na_2EDTA , 1.5 mM MgCl_2 , 0.5 mM DDT, and homogenizing with 10 more strokes.

All extracts are cleared by centrifugation at high speed (1–3 hr at greater than 100,000 *g* average centrifugal force). It is crucial to remove particulate matter, which will otherwise cause high levels of nonspecific binding that can obscure specific interactions. Glycerol should be present in the fraction loaded onto the columns since this aids in maintaining the stability of many proteins and also reduces background nonspecific binding. [If glycerol is not already present in the extraction buffer, it should be added to at least 10% (w/v) after the high-speed centrifugation step.] We often lyse the cells in chromatography buffer and load the high-speed centrifugation supernatant without further treatment. However, if the extract is not in an appropriate buffer, it can be dialyzed, typically for 4–15 hr, against the affinity column buffer. Particulate matter that forms during dialysis must be removed by centrifugation just prior to loading onto the columns (at least 30 min at 10,000 *g*; 1 hr at 100,000 *g* is preferred).

Both the lysis buffer and the column buffers should contain protease inhibitors to minimize fragmentation of proteins in the extract and to protect the affinity ligand. While the recipe for an effective protease inhibitor cocktail will vary with the application, a good starting point is 0.2 mM phenylmethylsulfonyl fluoride (PMSF) (from a 100 mM stock in 2-propanol), 0.5 mM benzamidine-hydrochloride (from a 500 mM stock in H_2O), 0.5 $\mu\text{g/ml}$ leupeptin (from a 0.5 mg/ml stock in H_2O), and 0.7 $\mu\text{g/ml}$ pepstatin (from a 0.7 mg/ml stock in methanol) in a buffer containing 1 mM Na_2EDTA .

In some cases, a crude lysate will contain small molecules that will interfere with chromatography. This can often be remedied by first precipitating proteins by saturating the extract with ammonium sulfate, collecting the precipitate by centrifugation, and dissolving the precipitated protein in affinity column buffer followed by dialysis. Extracts prepared this way should still be centrifuged at high speed (at least 1 hr at greater than 100,000 *g*) to remove particulate matter that forms during the procedure.

Removing polymers such as DNA and RNA that could interfere with affinity chromatography is more problematical. Passing an extract containing 0.2 M NaCl through a DEAE-cellulose column will remove nucleic

acids; this procedure is recommended provided that the proteins of interest do not bind to either nucleic acids or DEAE-cellulose under these conditions. Our experiences with enzymatic removal using nucleases have not been entirely positive; even relatively pure preparations of DNase I tend to show traces of protease contamination. Attempts to remove DNA from extracts with organic polymers such as polyethylene glycol or polyethyleneimine have often had undesirable side effects—such as the inability to resolubilize particular proteins under low ionic strength conditions, or a drastic increase in background binding, both of which are assumed to be due to the continued presence of the polymers in the extract even after extensive dialysis. If nucleic acids cause a problem, several approaches for removing them should be attempted, since no one method gives satisfactory results in every case.

Loading and Eluting Interacting Proteins

Loading Columns

Once suitable conditions for preparing an extract have been identified, the extract is loaded to the affinity matrix and the control matrix. For microscale columns run by gravity, the flow rate is determined by the geometry of the constricted end and the height of the buffer head. For a 20- μ l column bed, it should be about 100 μ l/15–30 min. For other column sizes, the natural maximal flow rate should be estimated by measuring the rate of flow of buffer while the open column is draining by gravity, then using a lower rate for subsequent operations. Usually an agarose matrix will support a flow rate of at least three column volumes per hour without adverse effects, although the geometry of the column will affect this parameter.

Our standard affinity chromatography loading buffer contains 10–20 mM Tris-HCl or HEPES (pH 7.5–7.9), 10–20% (w/v) glycerol, 1 mM Na₂EDTA, 1–2 mM 2-mercaptoethanol or DDT, and 50–100 mM NaCl. When the buffer contains 100 mM NaCl, there are very few proteins in a whole-cell extract that will bind to a control column and be eluted with salt.

Eluting Columns. After washing with at least 5, and preferably 10, column volumes of the low-salt loading buffer, bound proteins are eluted with increasing concentrations of salt. Since many interactions have at least a small ionic component, the elution of an interacting protein from a column can usually be accomplished with salt. We have observed many cases in which an interacting protein binds to the ligand in 100 mM NaCl and elutes with buffer containing 200–300 mM NaCl (see Table I). Increas-

ing the ionic strength in steps is simple and limits the number of fractions that must be checked for binding proteins. However, information about the strength of interactions can be obtained more reliably by eluting with a gradient of increasing salt. It is also easier to distinguish specific from nonspecific binding using a gradient elution.

In pilot experiments, after a sufficiently harsh salt wash (usually 2 *M* NaCl), the matrix should be stripped by eluting at room temperature with 6 *M* urea containing 0.2% (w/v) SDS to detect any hydrophilic or extremely strong interactions. This of course sacrifices the column.

Strategic Considerations

Advantage can be taken of microcolumns to screen for the effects of a variety of conditions on the binding of proteins from extracts. An important variable to consider is the concentration of the immobilized protein ligand. There is often a critical optimum concentration that must be identified: concentrations of ligand that are too low do not retain enough protein from the extract, while concentrations that are too high can, with highly charged proteins, create ion exchangers that bind hundreds of proteins nonspecifically. An example of a ligand concentration curve using RNA polymerase II columns and mammalian whole-cell extract is shown in Fig. 1A.

The method of extract preparation, the chromatography buffer, and the volume of extract loaded onto the column are also important. It is worth keeping in mind that rare proteins constituting only 10^{-6} – 10^{-5} of the total cell protein will only be detected by silver staining as gel bands containing 10–100 ng protein if the column is loaded with extract derived from at least 100 mg of cells (wet weight). Fortunately, we have generally found that, for rare proteins such as those involved in transcription, the signal-to-noise ratio improves as the extract load increases. This occurs because the background binding capacity of the matrix is saturated with a small amount of extract protein, while the relatively large amount of column ligand cannot be saturated by the small amounts of specific binding proteins in the extract. An example of this phenomenon, again with RNA polymerase II columns and mammalian whole-cell extract, is shown in Fig. 1B. It should be noted, however, that loading excessive amounts of protein can obscure weak interactions if proteins with stronger binding properties are present at high concentrations. The stronger interactions will effectively displace weaker ones if the binding sites overlap. This effect has been observed with actin filament columns.²

The binding capacity of a column for a particular protein can be determined by saturating the matrix with that protein, assuming that it is avail-

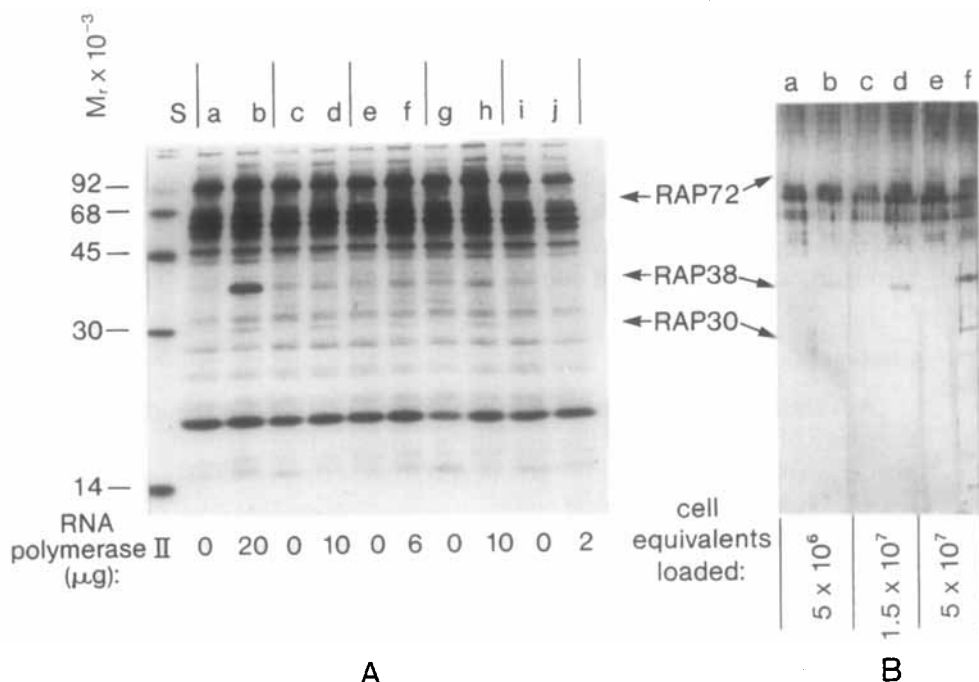


FIG. 1. Effects of ligand concentration and extract load on RNA polymerase II affinity chromatography. (A) Effect of ligand concentration. ^{35}S -Labeled whole-cell extract from murine erythroleukemia cells was loaded onto 20- μl microcolumns prepared with 0 μg (lanes a, c, e, g, i), 20 μg (lanes b, d, f), 10 μg (lane h), or 2 μg (lane j) of calf thymus RNA polymerase II. In lanes d and f the Affi-Gel 10 was inactivated in buffer for 30 or 60 min, respectively, before the addition of RNA polymerase II, resulting in the coupling of about 10 and 6 μg , respectively, to the matrix. Column eluates (0.2 M NaCl) were analyzed by SDS-PAGE and fluorography. S, Protein molecular weight standards. (B) Effect of extract load. Various amounts of murine erythroleukemia whole-cell extract were loaded onto control columns (lanes a, c, e) and RNA polymerase II columns (lanes b, d, f). Eluates (0.2 M NaCl) were analyzed by SDS-PAGE and silver staining. (Taken from Ref. 3.)

able in large quantity. Alternatively, after a column is eluted, the proteins that have flowed through the column are reapplied and the column is reloaded. If more of a given protein is retained in the second pass, then the amount bound in the first pass was probably the maximum that could have bound. If no more binds the second time, no conclusion can be drawn since the column may have been altered by the elution protocol; in this case the experiment can be repeated with two identical columns loaded in series and then eluted separately. If polyacrylamide gel electrophoresis (PAGE) reveals that components are completely missing from the flow-

through fraction that were present in the load fraction, it can be concluded that the column was not saturated for those components. However, the presence of a given protein in the flow-through fraction does not mean the column was saturated, since some of the protein in question might have been denatured or otherwise unavailable for binding. If the binding rate is the problem, the column can be loaded two to four times more slowly.

The dissociation constants of the protein-protein interactions can be estimated from the results of microcolumn affinity chromatography experiments in which the ligand concentration is varied (Fig. 1A, for example). These columns are loaded with 3–10 column volume samples and washed with 10 column volumes of loading buffer before eluting the specifically bound proteins with salt or chaotropic agents. Proteins that remain bound after this extended washing must have dissociation constants at least 10-fold lower than the concentration of ligand coupled to the column. Therefore, the K_d of the interaction should be about one-twentieth of the lowest concentration of ligand that fully retains the binding protein, assuming that all of the ligand is available for binding.

Our experiences and those of others (J. Hosoda, personal communication, 1985) with T4 gene 32 protein columns indicate that about 10% of the 32 protein molecules are available for interacting with other proteins.⁶ Estimates of dissociation constants derived from affinity chromatography experiments on the N–NusA,⁹ NusA–RNA polymerase,¹⁰ and RAP30/74–RNA polymerase II¹¹ interactions have been within factors of two of the dissociation constants measured in other ways (see Table I), suggesting that at least 50% of the N, NusA, and RNA polymerase II are available for binding. We do not know how typical these results are.

Analysis of Binding Fractions

Characteristics of Interacting Proteins

Proteins that interact with an affinity matrix can be characterized by PAGE or by assaying fractions for biochemical activities, assuming the conditions for elution did not alter the activity of the protein. Most proteins can be eluted with high concentrations of salt without affecting their activity, but the use of chaotropic agents such as urea or SDS will normally inactivate enzymes.

If extracts were radioactively labeled, the presence of proteins can be detected by assaying a sample of each fraction in a scintillation counter. Fractions containing protein are pooled—concentrated if necessary by precipitating with acetone or trichloroacetic acid, or by centrifuging against a dialysis membrane [for example, Amicon (Danvers, MA) Centri-

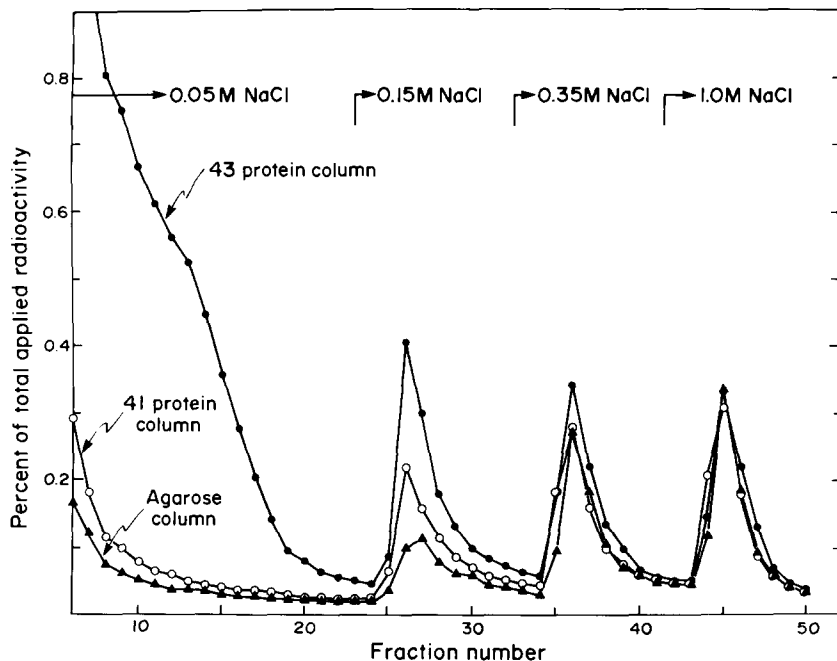


FIG. 2. The elution profile from a T4 bacteriophage DNA polymerase affinity column demonstrates specific interactions in the low-salt wash fractions. Affinity columns containing 1.2 mg/ml T4 bacteriophage gene 43 protein (DNA polymerase) or 0.28 mg/ml gene 41 protein were prepared and used to chromatograph extracts prepared from [35 S]methionine-labeled, T4-infected *E. coli* cells. Lysis and column buffers contained 20 mM Tris-Cl (pH 8.1), 1 mM Na₂EDTA, 1 mM 2-mercaptoethanol, 10% (w/v) glycerol, 10 mM MgCl₂, 0.5% (w/v) Triton X-100, and NaCl as noted. In this case, proteins bound specifically to the gene 43 protein column eluted during the initial 10 column volume wash with the lysis/loading buffer containing 50 mM NaCl. Examination of the eluted fractions by two-dimensional gel electrophoresis confirmed that the specifically bound proteins were known DNA polymerase accessory proteins.⁷

prep]—and then separated by PAGE in the presence of SDS. Proteins are detected by silver staining or fluorography. Weak but specific interactions are often revealed in the initial low-salt wash, so this region of the chromatogram must not be overlooked (see, for example, Fig. 2).

An alternative approach is to assay fractions for enzymatic activities. For instance, if the covalently bound ligand protein has an enzymatic activity, fractions can be assayed for the ability to alter this activity in the test tube. DNA polymerase accessory proteins bind specifically to a T4 DNA polymerase affinity column; these proteins enhance the rate, pro-

cessivity, and accuracy of the polymerase and can be assayed by their effects on any of these properties.⁷

Still another approach is to assay the column flow-through for the loss of an enzymatic activity. For example, a HeLa cell nuclear transcription extract loses a protein that is essential for initiation by RNA polymerase II when it flows through an RNA polymerase II column.¹⁷ The activity that binds to the column can be eluted and assayed.

The proteins that elute from an affinity column can next be chromatographed on standard matrices to obtain further purification. The same results also allow the identification of conditions that can be used to enrich for these species prior to future attempts to study the protein-protein interaction by affinity chromatography. If sufficient amounts of material are available, binding fractions can be used to generate antisera or to perform peptide sequencing. Either of these approaches can lead to the analysis, identification, or cloning of the genes encoding proteins detected by affinity chromatography.

Direct vs Indirect Interactions

If cell extracts were used for affinity chromatography, one must eventually determine whether an interacting protein is binding directly to the ligand or indirectly to the ligand via some other molecule. This can be accomplished, once the interacting protein has been purified, by rechromatographing the pure interacting protein on control and ligand columns.^{10,15}

When a protein machine is held together solely by weak interactions, an affinity column made with any particular component of that machine may detect only interacting proteins that bind directly and sufficiently strongly to the ligand. For protein assemblies that are of low abundance in an extract, this occurs because the concentration of the primary interacting protein always remains too low on the column to bind secondary interacting proteins. However, such secondary interactions are detectable if the interacting protein has multiple subunits held together by strong forces.⁴

Possible Need for Additional Ligand

If a protein undergoes a change in its conformation when it binds to some ligand inside the cell and only then interacts with another protein, the method described above will fail to detect this interaction unless the needed ligand is present in the extract at a sufficient concentration. It is clear that this situation occurs in biological systems: for example, spectrin,

¹⁷ Z. F. Burton, L. G. Ortolan, and J. Greenblatt, *EMBO. J.* **5**, 2923 (1986).

a cytoskeletal protein, binds to actin much more strongly if the spectrin has previously bound to band 4.1 protein.¹⁸ Likewise, *Escherichia coli* RecA protein affinity columns have been found to bind LexA protein only in the presence of DNA.¹⁹ We have encountered the same situation when analyzing the primosome of the T4 bacteriophage DNA replication apparatus.²⁰ Thus second generation protein affinity columns should be considered in which a preformed protein-protein or protein-nucleic acid complex is the immobilized affinity ligand, rather than a single protein alone.

Conclusion

Protein affinity chromatography has many potential uses for studying complex biochemical systems. Starting with a single element of a complex protein machine, it is possible in principle to reassemble the entire machine (see Fig. 3). The technique also has some limitations. For example, even though two proteins interact, they will not necessarily have the opportunity to do so on an affinity matrix. The T4 bacteriophage DNA polymerase binds tightly to a gene 32 protein affinity column, consistent with the specific interaction between these two proteins in solution (Fig. 4). However, although a T4 DNA polymerase column binds several polymerase accessory proteins, it does not bind gene 32 protein.⁷ We assume that this lack of reciprocity reflects a nonrandom orientation of the polymerase molecules on the affinity matrix, with the 32 protein-binding domain in an inaccessible position. Perhaps coupling at a variety of pH values would reveal a condition that provides a more random attachment.

Protein affinity chromatography can detect interactions ranging in strength from K_d 10^{-5} to 10^{-10} M (see Table I). An interacting protein with a $K_d > 10^{-5}$ M will not remain bound to the column when the column is washed by the large volume of buffer that is necessary to lower the background binding, even if the immobilized protein concentration is 10 mg/ml. Alternatively, an interacting protein with a $K_d < 10^{-10}$ M might not be able to bind to the affinity matrix if it failed to dissociate rapidly enough from the endogenous ligand present in the extract. Such tight

¹⁸ V. Ohanian, L. C. Wolfe, K. M. John, J. C. Pindu, S. E. Lux, and W. B. Gratzer, *Biochemistry* **23**, 4416 (1984).

¹⁹ N. Freitag and K. McEntee, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 8363 (1989).

²⁰ T.-A. Cha and B. M. Alberts, in "Cancer Cells 6; Eukaryotic DNA Replication." Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1988.

^{20a} J. Nodwell and J. Greenblatt, submitted.

²¹ C. F. Morris, H. Hama-Inaba, D. Mace, N. K. Sinha, and B. Alberts, *J. Biol. Chem.* **254**, 6787 (1979).

²² J. Rush and W. H. Konigsberg, *Prep. Biochem.* **19**, 329 (1989).

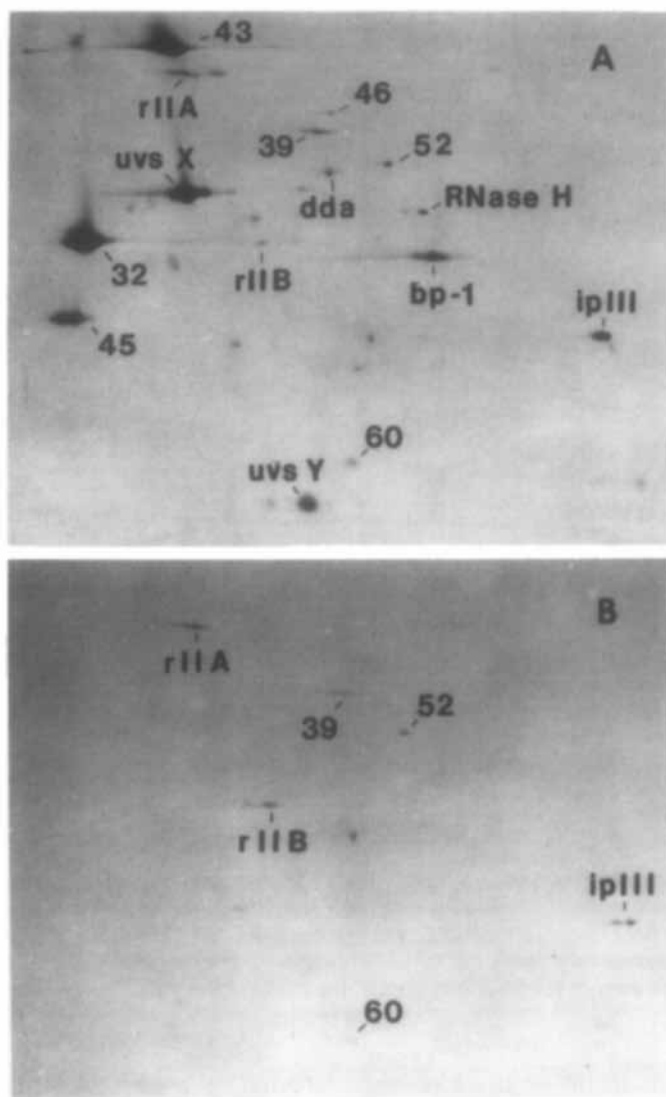


FIG. 4. Analysis of proteins bound to a T4 bacteriophage gene 32 protein affinity column by two-dimensional gel electrophoresis. Proteins from a radioactively labeled extract prepared from bacteriophage T4-infected *E. coli* cells were chromatographed on columns containing 8 mg/ml T4 gene 32 protein (A) or 12 mg/ml bovine serum albumin column (B). Samples from each of the eluted fractions were pooled, concentrated by precipitation, and fractionated by nonequilibrium two-dimensional gel electrophoresis. T4 bacteriophage proteins were detected by fluorography. Consistent with its role in all areas of T4 bacteriophage DNA metabolism, the gene 32 protein binds a large number of proteins known to be involved in DNA replication, genetic recombination, and DNA repair.⁶

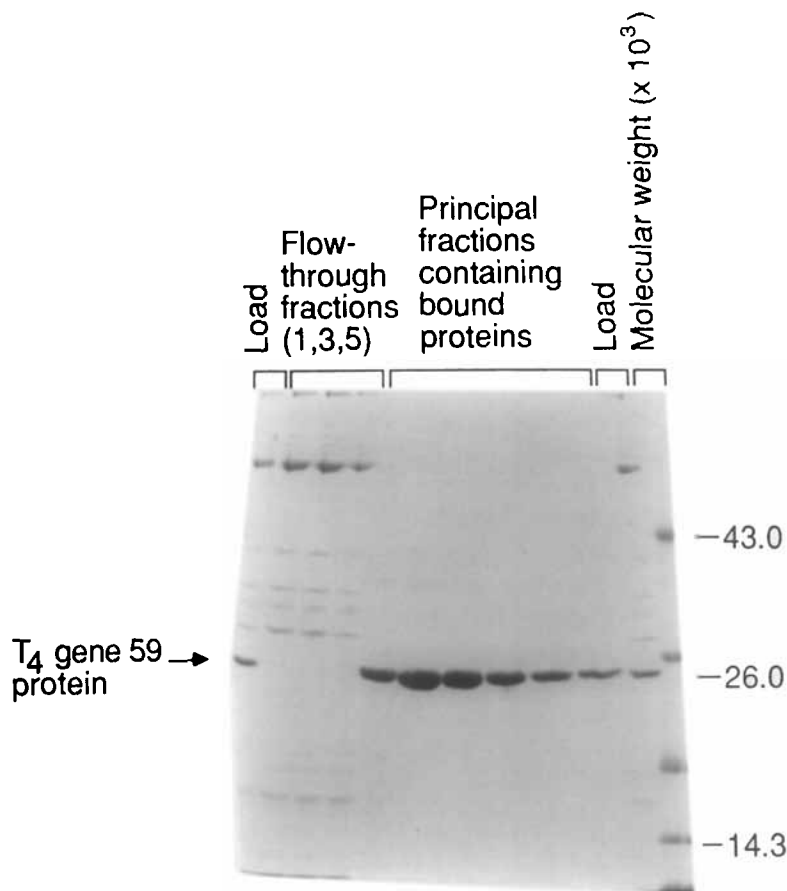


FIG. 5. SDS-polyacrylamide gel analysis of the purification of T4 gene 59 protein on an affinity column containing bound T4 gene 32 protein. Approximately 1 mg of T4 gene 59 protein, estimated to be 40% pure after DNA cellulose chromatography, was fractionated further by taking advantage of its affinity for the T4 gene 32 protein. The 59 protein was in 5 ml of a buffer consisting of 20 mM Tris-HCl (pH 8.1), 1 mM 2-mercaptoethanol, 1 mM Na_2EDTA , 3 mM MgCl_2 , 10% (w/v) glycerol, and 400 mM NaCl. It was loaded at 1 ml/hr onto a 1-ml column of 32 protein-Affi-Gel 10 containing 3.8 mg of bound T4 gene 32 protein. After the column was rinsed with 4 ml of the buffer containing 400 mM NaCl, the bound proteins were eluted with the same buffer containing 2.4 M NaCl and 0.1% Triton X-100. One-milliliter fractions were collected throughout the run. The gel shows that only T4 gene 59 protein is able to bind quantitatively to the column at 400 mM NaCl and the eluted 59 protein is greater than 99% pure (fractions with bound proteins are overloaded on the gel, in order to provide a sensitive estimate of purity). The breadth of the 2.4 M NaCl eluted peak is unusual, and it attests to the strong interaction between T4 gene 59 protein and T4 gene 32 protein. This may also explain why the yield of 59 protein from this column is only about 50%.

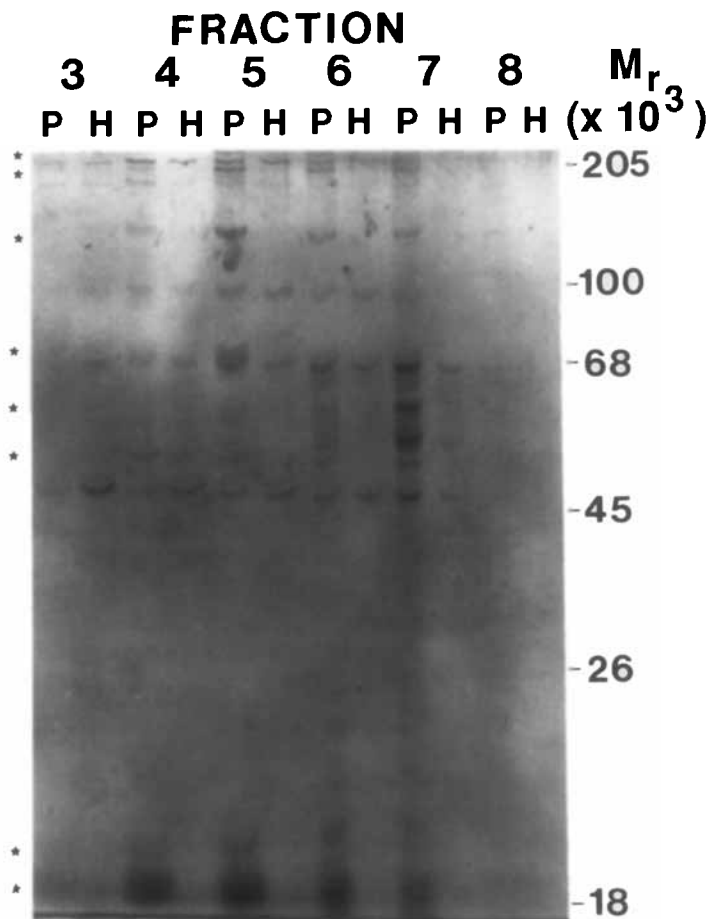


FIG. 6. Affinity chromatography with yeast DNA polymerase I. DNA Polymerase I was purified from a yeast strain overexpressing the *POL1* gene and approximately 400 mg was covalently attached to an Affi-Gel 10 agarose matrix. A control column containing a similar amount of human serum albumin was also constructed. These affinity matrices were used to chromatograph an extract prepared from whole yeast cells. Fourteen grams of a diploid strain lacking the major proteases PEP4 and PRB1 were harvested by centrifugation and stored frozen at -70° , thawed, and mixed with 10 ml of lysis buffer [20 mM Tris-HCl (pH 7.5), 1 mM Na_2EDTA , 1 mM 2-mercaptoethanol, 2 mM MgCl_2 , 50 mM KCl, 0.5 mM PMSF, 0.5 $\mu\text{g/ml}$ leupeptin, and 0.7 $\mu\text{g/ml}$ pepstatin]. The cells were disrupted by vortexing for a total of 9 min with 11 ml of 500- μm glass beads. The lysate was cleared first by centrifuging for 10 min at 27,000 g , then for 2 hr at 160,000 g . The cleared lysate contained about 220 mg of protein in a volume of 12 ml. It was adjusted to 10% (w/v) glycerol and loaded directly to the affinity columns. The columns were eluted with a gradient of increasing KCl in lysis buffer (50–500 mM). Proteins were separated by polyacrylamide gel electrophoresis in the presence of SDS and stained with silver. Proteins binding to the polymerase (P) but not the human serum albumin (H) column are candidates for DNA polymerase accessory proteins (*).

TABLE II
T4 GENE 32 PROTEIN AFFINITY CHROMATOGRAPHY AS
FINAL STEP IN T4 DNA POLYMERASE PURIFICATION^a

Parameter	Before ³² P affinity column	After ³² P affinity column
Endonuclease activity ^b	+	—
Amount of T4 gene 43 protein	20 mg	16.3 mg

^a T4 gene 43 protein (the T4 DNA polymerase), purified by standard procedures,^{21,22} including single-stranded DNA cellulose chromatography, often contains levels of endo-deoxyribonuclease that interfere with DNA enzymology. This nuclease activity can be easily removed by 32 protein affinity column chromatography. For example, 20 mg of 43 protein, estimated to be 98% pure (by SDS-polyacrylamide gel analysis) after standard purification steps^{21,22} and still containing endonuclease, was further purified on a 20-ml column of 32 protein-Affi-gel 10 containing 68 mg of bound 32 protein (this is a molar ratio of 11:1 of bound 32 protein:43 protein). Flow rates throughout are one column volume per hour. The 43 protein was loaded onto the column in 33 ml of a buffer containing 20 mM Tris-HCl, pH 8.1, 1 mM Na₂EDTA, 2 mM 2-mercaptoethanol, 5 mM MgCl₂, 50 mM NaCl, and 10% (w/v) glycerol. Elution was with a 150-ml linear gradient from 50 to 400 mM NaCl in the same buffer; the 43 protein eluted (without endonuclease) in 30 ml with a peak concentration at 170 mM NaCl. The recovery was 82%.

^b Conversion of RFI plasmid to RFII, as assayed by agarose gel electrophoresis.

tein affinity chromatography is very useful, both as an initial method for detecting the potential components of protein machines (Fig. 5) and for the large-scale purification of such components (Fig. 6 and Table II). For a biochemist, it represents an invaluable tool for exploring the complex protein machines that act in so many aspects of cellular metabolism.

[4] Metal Requirements for Nucleic Acid Binding Proteins

By DENISE L. MERKLE and JEREMY M. BERG

Introduction

In recent years, a number of families of nucleic acid binding and gene regulatory proteins have been found to require bound metal ions for full activity.¹ The largest and most well-characterized family is the so-called zinc finger protein family typified by *Xenopus* transcription factor IIIA (TFIIIA). This protein was initially purified as a protein that forms a complex with 5S RNA in *Xenopus* oocytes² and was subsequently shown to be required for the accurate initiation of transcription of 5S RNA genes.³ TFIIIA was found to bind specifically to the internal control sequence found with such genes. In 1983, it was reported that the TFIIIA–5S RNA complex contained several equivalents of zinc as isolated.⁴ Furthermore, removal of zinc by treatment with chelating agents resulted in loss of specific DNA-binding activity; this inactivation could be reversed by addition of zinc.⁴ In 1984, the sequence of a TFIIIA cDNA clone was reported.⁵ Subsequent analysis^{6,7} revealed the presence of nine tandem sequences within the deduced primary structure that approximately matched the consensus (Tyr,Phe)-X-Cys-X_{2,4}-Cys-X₃-Phe-X₅-Leu-X₂-His-X_{3,4}-His-X₂₋₆, where X represents relatively variable amino acids. It was proposed that each of these sequences was capable of binding a zinc ion via the cysteine and histidine residues. Further investigation by a number of methods has revealed that the proposal is correct; these sequences do bind zinc ions via the cysteinate and histidine side chains to form structural domains that are essential for DNA binding.⁸

Since the characterization of TFIIIA, a large number of other proteins that have been implicated in nucleic acid binding or gene regulatory processes have been discovered to either require metal ions such as zinc(II)

¹ J. M. Berg, *J. Biol. Chem.* **265**, 6513 (1990).

² B. Picard and M. Wignez, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 241 (1979).

³ D. R. Engelke, S.-Y. Ng, B. S. Shastry, and R. G. Roeder, *Cell (Cambridge, Mass.)* **19**, 717 (1980).

⁴ J. S. Hanas, D. J. Hazuda, D. J. Bogenhagen, F. H.-Y. Wu, and C.-W. Wu, *J. Biol. Chem.* **258**, 14120 (1983).

⁵ A. M. Ginsberg, B. O. King, and R. G. Roeder, *Cell (Cambridge, Mass.)* **39**, 479 (1984).

⁶ J. Miller, A. D. McLachlan, and A. Klug, *EMBO J.* **4**, 1609 (1985).

⁷ R. S. Brown, C. Sander, and P. Argos, *FEBS Lett.* **286**, 271 (1985).

⁸ J. M. Berg, *Annu. Rev. Biophys. Biophys. Chem.* **19**, 405 (1990).

for full activity or to contain sequences that appear capable of forming metal-binding domains.¹ Some of these sequences are highly similar to the TFIIIA-type consensus whereas others are quite different but share the property of having sets of cysteine and/or histidine residues within short stretches of the primary structure that have the potential of forming metal-binding domains. It is important to realize that this large collection of proteins contains many distinct classes in which the three-dimensional structures of the metal-binding domains and the functions of the proteins as well as the role of the bound metal ion(s) are different.

It is the goal of this account to provide a description of some of the methods that can be used to determine whether a protein suspected of having metal-binding domains does, indeed, bind metal ions and if the presence of the bound metal ions affects the activity of the protein. The methods can be divided into three groups: (1) direct methods, (2) reconstitutive methods, and (3) functional methods. Two methods from each class will be discussed and the advantages and potential pitfalls will be noted. Examples of systems where the methods have been useful will be provided.

Direct Methods

The most direct method is the determination of the total amount of a given metal present in a sample of a purified protein. Atomic absorption spectroscopy provides one convenient and frequently utilized method. A solution of the protein is aspirated into a hot flame. A substantial proportion of the metal ions in the sample are reduced to the neutral atomic state. Atoms absorb light at characteristic frequencies with very small linewidths. This allows the determination of one element in the presence of many other elements. A light source that emits at the appropriate frequency is required for each element. The concentration of the element in the sample is determined by comparison with a standard curve. For zinc, concentrations of less than $1\ \mu M$ can be easily determined using 1-ml sample sizes. The number of equivalents of metal can be evaluated if the protein concentration of the sample is known.

The main advantage of atomic absorption spectroscopy is its directness. The total amount of metal ion in the sample is evaluated without regard to its chemical form. Furthermore, atomic absorption spectroscopy and related methods (e.g., plasma emission spectroscopy) are reasonably sensitive and quite precise. There are, however, several disadvantages. First, the method requires a significant amount of purified protein and is destructive. For many DNA-binding proteins, the many micrograms of purified protein required are not available to burn. Second, the method

measures the total amount of zinc in the sample, be it tightly bound in a metal-binding domain, adventitiously bound to the surface of the protein, or simply a contaminant of the buffer used. It is important always to check a sample of the buffer used in the final purification step (e.g., adjacent fractions from a chromatography column run) to be sure that it does not have the same metal ion content as does the protein sample in question. Finally, it is important to note that the result of a direct metal content determination may be quite dependent on the method of purification. Both false positive and false negative results are possible. Many proteins will bind ions such as zinc even if they do not play a role in the primary function of the protein. Thus, it has traditionally been the practice to dialyze protein samples against solutions containing low concentrations of chelating agents such as ethylenediaminetetraacetic acid (EDTA) prior to analysis. This ensures that the metal ions found are bound in at least a kinetically stable manner. False negative results are also possible if the protein sample is treated too harshly prior to analysis. For many of the DNA-binding proteins that contain metal-binding domains, the metal ions are bound tightly thermodynamically yet are relatively easily exchanged. Thus, dialysis against metal-free buffer or solutions of chelating agents could result of loss of metal ions that are required for function.

An example that illustrates many of these features is the case of TFIIA. In the initial report that this protein contained bound zinc,⁴ samples of the TFIIA–5S RNA complex that had been dialyzed against 5 mM EDTA were found by atomic absorption to contain 2–3 Eq of zinc per protein molecule. When the protein was freed from the RNA by nuclease treatment, dialysis against the same solution resulted in the loss of all zinc from the protein. Purification of the TFIIA–5S RNA complex by another method that was designed to avoid potential chelating agents such as EDTA and dithiothreitol resulted in preparations of the complex that contained 7–11 zinc ions per protein by atomic absorption.⁶ This number of zinc ions was very suggestive in light of the sequence analysis of the protein. The intensity of the X-ray fluorescence produced during extended X-ray absorption fine structure (EXAFS) studies of the TFIIA–5S RNA complex provided independent evidence concerning the zinc stoichiometry.⁹ The intensity observed was consistent with nine zinc ions per complex to within 10%. Since the EXAFS data could be well fit assuming only one type of zinc site with two cysteines and two histidines coordinated to the metal ion, these results strongly suggest that essentially all of the nine sequence repeats do, indeed, bind zinc ions. Quite different results would have been obtained if only two or three of the sequences bound zinc and

⁹ G. P. Diakun, L. Fairall, and A. Klug, *Nature (London)* **324**, 698 (1986).

the remaining zinc present in the preparation was bound to different sorts of sites.

A second direct approach is the method of measurement of metal release. This is applicable to systems in which it is possible to chemically modify the protein in a manner that significantly weakens the protein-metal ion interaction. The metal released from the protein can then be detected in solution with the use of an appropriate metal-sensitive dye. The most generally used method¹⁰ can be applied to proteins with metal-cysteinate coordination, a feature common to many metal-dependent nucleic acid-binding proteins. The modification agent is *p*-hydroxy-mercuribenzoate (PMPS), which reacts with cysteine to form a mercury-thiolate complex. This adduct absorbs at 250 nm so that the modification can be monitored spectrophotometrically. The dye used is 4-(2-pyridylazo)resorcinol (PAR), which forms a 2 : 1 complex with zinc that absorbs at 500 nm. The mercury modification can be reversed by treatment with excess thiol reagents. A disadvantage of this method is that it is dependent on having appropriately reactive ligands available for chemical modification. For example, for a zinc-binding site such as that in carbonic anhydrase, which has only histidine ligands, PMPS titrations would not be effective.

This method has been applied to several zinc-containing nucleic acid-binding proteins, including the bacteriophage gene 32 protein.¹¹ This protein binds single-stranded nucleic acids with high cooperativity. This protein was titrated with PMPS in the presence of PAR. Monitoring the absorbance at 250 nm revealed that 3.2 Eq of the adduct was formed with no additional change in absorbance on the addition of excess PMPS. Concomitant with this modification, an absorbance at 500 nm developed corresponding to the release of 1.12 zinc ions per protein. These observations are consistent with the binding of one zinc ion within the sequence **Cys-Ser-Ser-Thr-His-Gly-Asp-Tyr-Ser-Cys-Pro-Val-Cys**, which had been identified as a potential metal-binding site from a systematic computer-based sequence search.¹² Reversal of the modification with excess dithiothreitol produced the gene 32 apoprotein. Comparative studies of the native and apoproteins suggested that the presence of the bound zinc ion significantly affected the cooperativity of nucleic acid binding but only modestly reduced the binding affinity to a single-site oligonucleotide.¹¹ In addition, the metal ion appears to be important for protein stability as determined by susceptibility to proteolysis.¹¹

¹⁰ J. B. Hunt, S. H. Neece, H. K. Schachman, and A. Ginsberg, *J. Biol. Chem.* **259**, 14793 (1984).

¹¹ D. P. Giedroc, K. M. Keating, K. R. Williams, W. H. Konigsberg, and J. E. Coleman, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 8452 (1986).

¹² J. M. Berg, *Science* **232**, 485 (1986).

Reconstitutive Methods

An alternative approach to the demonstration of metal-binding ability of a protein or peptide involves the ability of metal to bind to the metal-free form. The ion used in the reconstitution can be either the natural metal (either in stable or radioactive form) or a different metal with desired spectroscopic or chemical properties. These methods can be used both to demonstrate the presence of metal-binding activity and to prepare radioactively or spectroscopically labeled materials for other studies.

The first reconstitutive method we shall discuss involves cobalt substitution. Zinc(II) has a filled *d* shell and, therefore, is silent with regard to many spectroscopic probes. In contrast, cobalt(II) has many desirable spectroscopic features and it is structurally and chemically quite similar to zinc(II). Thus, it substitutes into many natural zinc sites to produce proteins that are structurally, and often functionally, similar to the native forms. However, the dissociation constants for the protein–cobalt(II) complexes are often several orders of magnitude larger than those for the zinc(II) complexes, so that care must be taken to ensure that adequate concentrations are used to allow incorporation. This represents a major disadvantage for the cobalt substitution method. Among the advantages is the fact that the position, shape, and intensity of the *d*-to-*d* transitions in the visible region provide a significant amount of information about the nature of the cobalt(II)-binding site.

This method has been applied to single zinc finger domains from a variety of proteins. The first such peptide to be studied corresponded to the second domain of TFIIIA.¹³ This 30-amino acid peptide had been prepared by gene synthesis, expression of a fusion protein, and cleavage. Treatment of this peptide with solutions of cobalt(II) resulted in the production of an absorption spectrum with components in the ultraviolet and visible regions. A set of bands centered near 650 nm is due to *d*-to-*d* transitions. The relatively large extinction coefficients for these bands ($>500 M^{-1} \text{ cm}^{-1}$) indicated that the complex formed was distorted tetrahedral rather than five- or six-coordinated. The spectrum reached nearly its full intensity when 1 Eq of cobalt(II) had been added, indicative of a relatively stable 1:1 complex. Furthermore, the spectrum could be bleached by the addition of 1 Eq of zinc(II), providing evidence for the interaction between the peptide and the ion naturally found in TFIIA. Subsequent studies used these spectrophotometric methods to determine dissociation constants for the peptide–cobalt(II) and peptide–zinc(II) complexes.¹⁴ The values determined indicated that the specificity for zinc(II)

¹³ A. D. Frankel, J. M. Berg, and C. O. Pabo, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 4841 (1987).

¹⁴ J. M. Berg and D. L. Merkle, *J. Am. Chem. Soc.* **111**, 3759 (1989).

over other metal ions in reconstituting the intact protein was reflected in the ion affinities of a single zinc finger peptide.

The second reconstitutive approach is the method of zinc blotting.¹⁵⁻¹⁷ In this method, proteins are separated from one another by sodium dodecyl sulfate-polyacrylamide gel electrophoresis, reduced by soaking in an appropriate buffer, transferred to nitrocellulose filters, and then probed with radioactive zinc (⁶⁵Zn). After suitable washing, the presence of bound zinc can be detected by autoradiography. The conditions to be used at each step have been optimized with use of a series of known zinc-binding proteins together with a set of proteins known not to have a high affinity for zinc.^{16,17} This method has the advantage that small, not necessarily homogeneous, samples can be easily analyzed. Detection of samples smaller than 100 pmol of protein has been demonstrated. Its disadvantages include difficulty in quantitating the number of zinc ions bound per protein as well as some potential for false positive or negative results due to binding at nonfunctionally important sites or the inability to restore the metal-binding activity during the renaturation steps.

This method has been applied to several important systems. These include poly(ADP-ribose)polymerase (NAD⁺ ADP-ribosyltransferase),¹⁷ a protein involved in damaged DNA recognition and repair, and retroviruses, in which case it was demonstrated that the nucleocapsid protein of avian myoblastosis virus had significant zinc-binding activity,¹⁶ providing some of the earliest evidence that the Cys-X₂-Cys-X₄-His-X₄-Cys sequences in these proteins can, indeed, form metal-binding domains as had been proposed.¹²

Functional Methods

If a protein has a function that can be assayed in the laboratory, then studies of loss or gain of activity as a function of metal ion or chelator treatment can be extremely useful in providing evidence for the presence of bound metal ions and in determining their importance for biological function. In addition, if a specific protein sequence has been identified as a potential site for a metal ion interaction by any means, then studies of the effects of disrupting this site can provide support for or against such a hypothesis. Obviously, a wide variety of functional methods are possible, depending on the characteristics of a particular system. The two functional

¹⁵ L. A. Schiff, M. L. Nibert, M. S. Co., E. G. Brown, and B. N. Fields, *Mol. Cell. Biol.* **8**, 273 (1988).

¹⁶ L. A. Schiff, M. L. Nibert, and B. N. Fields, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 4195 (1988).

¹⁷ A. Mazen, G. Gradwohl, and G. de Murcia, *Anal. Biochem.* **172**, 39 (1988).

methods we shall discuss involve the use of chelator inactivation and metal ion reactivation on specific protein–DNA interactions and the use of site-directed metagenesis of proposed metal-binding residues in providing evidence for metal ion binding and identifying metal-binding domain functions.

As noted in the introduction, some of the first evidence that zinc played an important role in the function of TFI_{II}A came from the observation that EDTA treatment resulted in the loss of specific DNA-binding activity.⁴ This approach can easily be applied to other proteins suspected of requiring metal ions for DNA-binding activity. The specific DNA-binding activity can be assayed by footprinting methods, such as the DNase I method used in the TFI_{II}A study, or by gel mobility shift methods. Additional, and stronger, evidence can be obtained if the specific DNA-binding activity can be restored by incubation of the chelator-inactivated preparation with metal salts. The chelators that can be used include EDTA and 1,10-phenanthroline with concentrations from 0.5 to 50 mM required to produce an effect. The chelated metal and free chelating agent can be removed from the protein by dialysis. A reducing agent such as dithiothreitol should be included in the dialysis buffers, as metal-free cysteine-rich proteins are often quite sensitive to air oxidation, which can cause loss of metal-restorable activity. Treatment of these samples with metal ion solutions can potentially produce reconstituted preparations that can be assayed for activity. A variety of metal ions can be tested, although zinc appears to be the most likely ion to be active. Various metal ion concentrations ranging from the micromolar to the millimolar range should be tested. Often an optimum is found as excess metal ion can inhibit some protein–DNA interactions.

This method has been applied to many different systems. For example, two partially purified factors that bind to the human histone H4 promotor were tested in this way using gel mobility shift methods to assay specific DNA binding.¹⁸ Both H4TF-1 and H4TF-2 were found to be inactivated by 0.5 mM 1,10-phenanthroline and reactivated by zinc(II). One of the factors also appeared to be reactivated by iron(II). These relatively simple experiments provided the strong suggestion that these transcription factors contain zinc-binding domains of some sort that are directly or indirectly involved in their specific interactions with DNA. Analogous results were obtained with a 150-amino acid fragment corresponding to the DNA-binding domain of the glucocorticoid receptor that had been expressed in *Escherichia coli*.¹⁹ Treatment of samples of this fragment that had been

¹⁸ L. Dailey, S. B. Roberts, and N. Heintz, *Mol. Cell. Biol.* **7**, 4582 (1987).

¹⁹ L. P. Freedman, B. F. Luisi, Z. R. Korzun, R. Basavappa, P. B. Sigler, and K. R. Yamamoto, *Nature (London)* **334**, 543 (1988).

made metal free with zinc(II) or cadmium(II) resulted in materials that specifically bound to glucocorticoid-responsive elements as assayed by gel mobility shift as well as footprinting methods. Maximal activity occurred at 250–500 μM zinc(II), but at 5 μM cadmium(II) levels.

The second functional method to be discussed involves the use of site-directed metagenesis in probing potential metal-binding domains. One of the reasons for the rapid development of our knowledge of the roles of metal-binding domains in nucleic acid-binding proteins is the fact that they are often recognizable at the deduced protein sequence level. Thus, members of the TFIIIA-like zinc finger protein class are easily identified by characteristic patterns of cysteine, histidine, and other residues whereas other classes of proteins have been proposed to contain metal-binding domains, prior to the availability of experimental data, based on other patterns of potential metal-binding residues. One approach to testing such hypotheses is to change one or more of the residues proposed to interact with the metal ion(s) and to examine the effects in a functional assay. Such an approach has the advantage that such sequence changes can be easily made using current site-directed mutagenesis protocols and that the method can be applied with any available assay. A disadvantage is that mutation of any residue, particularly one that is known to be conserved in a family of proteins, can be expected to have functional effects regardless of the mechanistic basis for these effects. Thus, changing a cysteine may completely disrupt the activity of a protein even if the cysteine is not involved in metal binding.

This method has been applied to several retroviral nucleocapsid proteins^{20–22} that contain potential metal-binding sequences that were discussed briefly above. Mutation of any of the three cysteine residues to serine in an infectious Moloney murine leukemia clone resulted in the production of virus particles that were not infectious.^{20,21} Similar effects were seen when other conserved or conservatively substituted residues within the proposed metal-binding domain were mutated. Further characterization of the virions produced indicated that they lacked detectable amounts of viral RNA although some of the mutant virions contained a significant amount of cellular RNA species. The effects of the same sequence changes on the metal-binding properties of peptides corresponding to this region have been examined and in many cases the magnitude of the effects on metal-binding properties were found to correlate with the

²⁰ R. J. Gorelick, L. E. Henderson, J. P. Hanser, and A. Rein, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8420 (1988).

²¹ C. Méric and S. P. Goff, *J. Virol.* **63**, 1558 (1989).

²² J. E. Jentoft, L. M. Smith, F. Xiangdong, M. Johnson, and J. Leis, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 7094 (1988).

strength of the effect on RNA packaging.²³ These mutagenesis results provide support that metal binding by the nucleocapsid protein occurs and is functionally important and revealed that the metal-binding domain might be involved with a specific interaction with part of the viral RNA required for encapsidation.

Conclusions

The methods discussed represent some of the most widely applicable techniques available for the identification and characterization of the roles of metal ions in nucleic acid-binding proteins. Given the large number of nucleic acid binding and gene regulatory proteins that have been found in recent years to contain metal ions or to require them for activity, initial tests for the presence of metal or for metal requirements should become part of the routine characterization of any newly discovered or purified factor. This continues to be a rapidly developing field and many new systems and new methods can be expected to be discovered in the upcoming years.

Acknowledgments

Work in our laboratory in this area has been supported by grants from The National Institutes of Health (GM-38230) and The National Science Foundation (DMB-8850069).

²³ L. M. Green and J. M. Berg, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 6403 (1990).

[5] Large-Scale Preparation of DNA Fragments for Physical Studies of Protein Binding

By KARLHEINZ TOVAR and WOLFGANG HILLEN

Introduction

The study of sequence-specific protein–DNA interactions profits greatly from the availability of pure proteins and DNA fragments containing the respective target sequences. Because the recognized sequence is only on the order of a few tens of base pairs or less, the DNA substrate is easily obtained in sufficient amounts by chemical synthesis. In many cases, however, cooperative binding distinguishes between the presence

of one or more binding sites on the substrate DNA.^{1,2} These cases require longer DNA fragments, which are difficult to obtain synthetically in sufficient amount and purity. The same holds true for studies of conformational changes of the DNA substrates on protein binding, which are gaining importance.³⁻⁵ At present most of these studies are limited to methods that do not require large amounts of material. It has been our goal to optimize standard procedures of DNA fragment preparation to a state where large (tens of milligrams) amounts of homogeneous, small [approximately 100 base pairs (bp) DNA fragments from the class A through D *tet* regulatory regions^{6,7} became available for physical studies, e.g., neutron scattering,⁸ kinetic determinations using fluorescence spectroscopy,⁹ electrooptics, and melting¹⁰⁻¹² of protein-DNA complexes, and circular dichroism (CD) studies of Tet repressor-*tet* operator interactions.¹³ These methods may be used for efficient large-scale preparations of any desired fragment of DNA. They are based, in part, on previously described procedures.^{14,15}

Outline of Procedure

The general strategy involves known procedures, which have been optimized for handling large amounts of material in a short period of time. In the first step the desired DNA fragment is cloned, possibly in multiple insertions, in a suitable plasmid vector, which should be amplifiable to a high copy number and exhibit genetic stability even without selection. These factors are more important than an easily selectable resistance marker for the cloning protocol. Particular care must be given to the selection of the proper restriction sites flanking the desired fragment be-

¹ A. Hochschild and M. Ptashne, *Cell (Cambridge, Mass.)* **44**, 681 (1986).

² J. D. Gralla, *Cell (Cambridge, Mass.)* **57**, 193 (1989).

³ R. Schleif, *Science* **240**, 127 (1988).

⁴ L. M. De Vargas, S. Kim, and A. Landy, *Science* **244**, 1457 (1989).

⁵ O. Raibaud, *Mol. Microbiol.* **3**, 455 (1989).

⁶ B. Mendez, C. Tachibana, and S. B. Levy, *Plasmid* **3**, 99 (1980).

⁷ K. Tovar, A. Ernst, and W. Hillen, *Mol. Gen. Genet.* **215**, 76 (1988).

⁸ H. Lederer, K. Tovar, G. Baer, R. P. May, W. Hillen, and H. Heumann, *EMBO J.* **8**, 1257 (1989).

⁹ C. Kleinschmidt, K. Tovar, W. Hillen, and D. Pörschke, *Biochemistry* **27**, 1094 (1988).

¹⁰ W. Hillen and B. Unger, *Nature (London)* **297**, 700 (1982).

¹¹ D. Pörschke, K. Tovar, and J. Antosiewicz, *Biochemistry* **27**, 4674 (1988).

¹² M. Wagenhöfer, D. Hansen, and W. Hillen, *Anal. Biochem.* **175**, 422 (1988).

¹³ K. Tovar and W. Hillen, to be published.

¹⁴ S. C. Hardies, R. K. Patient, R. D. Klein, F. Ho, W. S. Reznikoff, and R. D. Wells, *J. Biol. Chem.* **254**, 5527 (1979).

¹⁵ W. Hillen, R. D. Klein, and R. D. Wells, *Biochemistry* **20**, 3748 (1981).

cause these sites must be digested on a large scale with a potentially costly amount of enzyme.

The cultivation of the transformed *Escherichia coli* strain is usually done in batches of 10–300 liters, depending on the facilities available. The amplification of plasmid DNA with chloramphenicol is advantageous, as it does not increase the final yield of DNA much, but rather simplifies the preparation procedure because the ratio of plasmid DNA to chromosomal DNA and to total protein is more favorable. Based on these considerations, amplification should not be omitted.

The protocol used for Brij deoxycholate (DOC) lysis of cells is quite crucial because the contamination with chromosomal DNA can be kept to a minimum at this step. Degradation of RNA and protein can be done in the supernatant of the lysed cells. Extensive degradation of proteins using proteinase K (obtained from Merck, Darmstadt, Germany) is very important for the expediency and efficiency of the following phenol extraction, because it seems to reduce the interface material and to increase the portion of DNA remaining in the aqueous phase.

Restriction of the plasmid DNA should be done with an inexpensive enzyme such as *EcoRI*, because it can be prepared in large amounts and excellent quality from overproducing *E. coli* strains.¹⁶ However, *BamHI* or other readily available enzymes may also be suitable. The results of a time course for *EcoRI* restriction on an analytical scale obtained for the four plasmids described below were reproduced in the preparative digestion. Because the preparative digestion is usually done for 24 hr or longer, it is advisable to split the total amount of plasmid DNA into several independent batches.

After digestion and removal of the restriction enzyme the bulk of the vector DNA is separated from the small insertion fragment by a fractionated polyethylene glycol (PEG) precipitation.^{15,17} In the past, this procedure has been somewhat troublesome because it seemed only poorly reproducible. However, when done exactly as described here we have had repeated success with it. An enrichment of the desired DNA is essential because the expediency of the following HPLC purification depends a great deal on the amount of DNA to be separated.

The final purification of the desired DNA fragment is achieved by high-performance liquid chromatography (HPLC) on Nucleogen (obtained from Diagen, Düsseldorf, Germany). The products obtained this way have been sufficiently pure for physicochemical and enzymatic procedures. The following methods were used to purify *tet* regulatory fragments from four classes (A through D) of tetracycline resistance genes.⁶ They may be applied, however, to any fragment of DNA.

¹⁶ J. Bottermann and M. Zabeau, *Gene* **37**, 229 (1985).

¹⁷ J. T. Lis and R. Schleif, *Nucleic Acids Res.* **2**, 383 (1975).

Methods and Results

Cloning of Desired DNA Fragments

Plasmid pWH802¹⁸ is used as a vector for all four plasmids. It is a 3739-bp DNA derived from pVH51.¹⁹ Its nucleotide sequence is known.²⁰ The *ori* fragment is derived from mini-Col¹⁹ and its selectable marker is a colicin immunity.¹⁴ For colicin E1 preparation²¹ *E. coli* JF290 is grown on a rotary shaker to an OD₆₅₀ of 1 at 37° in 1 liter of L broth (LB). Then 1 mg mitomycin is added and cultivation is continued at 37° for 6 hr. The cells are sedimented by centrifugation in a JA10 rotor at 6000 rpm for 30 min at 4°. To the supernatant 390 g of solid ammonium sulfate is added and the mixture incubated at 4° for 1 hr. The precipitate is spun down in a JA10 rotor at 10,000 rpm at 4°, dissolved in 10 ml of 0.1 M potassium phosphate, pH 7.0, dialyzed against 2 liters of 0.1 M potassium phosphate, pH 7.0, for 12 hr at 4°, lyophilized in aliquots of 500 µl, and stored at 4°. To determine the colicin activity of this preparation 200 µl of competent *E. coli* RR1 cells¹⁴ are grown in 4 ml of LB for 90 min at 37° on a shaker and 200 µl of this culture is incubated without shaking at 37° for 30 min with varying amounts of a batch of colicin dissolved in 500 µl of LB. Then the entire mixture is spread on LB plates containing 10% (w/v) deoxycholate. The amount of colicin preventing growth in this assay is used for selection in all following transformations.²¹ In our hands the colicin could be stored for months without loss of activity. The effort required to perform this somewhat elaborate selection procedure is outweighed by the excellent stability and high copy number of the plasmid, which has been found even with multiple insertions of the extremely strong phage T7 promoter A1.²²

The *tet* regulatory DNA fragments are obtained by elution from polyacrylamide gels of a 76-bp *Fnu*4HI/*Tha*I fragment from pWH320²³ (class A), a 70-bp *Hinc*II/*Taq*I fragment from pWH106²⁴ (class B), a 94-bp *Ban*I/*Hae*III fragment from pBR322²⁵ (class C), and a 66-bp *Sau*3A/*Mn*I fragment from pWH803¹⁸ (class D). The protruding ends are filled in using DNA

¹⁸ B. Unger, G. Klock, and W. Hillen, *Nucleic Acids Res.* **12**, 7693 (1984).

¹⁹ V. Hershfield, H. W. Boyer, L. Chow, and D. R. Helinski, *J. Bacteriol.* **126**, 447 (1976).

²⁰ P. T. Chan, H. Ohmori, J.-I. Tomizawa, and J. Lebowitz, *J. Biol. Chem.* **260**, 8925 (1985).

²¹ L. E. Maquat and W. S. Reznikoff, *J. Mol. Biol.* **125**, 467 (1978).

²² H. Heumann, H. Lederer, W. Kammerer, P. Palm, W. Metzger, and G. Baer, *Biochim. Biophys. Acta* **909**, 126 (1987).

²³ G. Klock and W. Hillen, *J. Mol. Biol.* **189**, 633 (1986).

²⁴ W. Hillen, G. Klock, I. Kaffenberger, L. V. Wray, Jr., and W. S. Reznikoff, *J. Biol. Chem.* **257**, 6605 (1982).

²⁵ J. G. Sutcliffe, *Cold Spring Harbor Symp. Quant. Biol.* **43**, 77 (1978).



FIG. 1. Nucleotide sequences of the constructed restriction fragments containing the regulatory regions of the *tet* classes A to D. The classes and the respective plasmid designations are given on the left side and the length of the fragments (in bp) on the right. The operators are indicated by bars and stars at the center of their palindromic symmetry. The arrows at the top show the direction of *tet* gene transcription.

polymerase Klenow fragment²⁶ and an *EcoRI* linker (5'-GGAATTCC-3') is ligated to the ends. The ligation mixtures are digested with *EcoRI* and the products eluted from a polyacrylamide gel and ligated in 5- through 25-fold molar excess with *EcoRI*-cleaved, dephosphorylated pWH802. The mixtures are transformed to *E. coli* RR1 and recombinants are screened for multiple insertions by their size. This results in plasmids pWH912, pWH923, pWH931, and pWH941. pWH912 carries a double insertion, pWH923 carries a triple insertion, and pWH931 as well as pWH941 contain single inserts of the class A to D *tet* fragments, respectively. Multiple insertions always have parallel orientations. The nucleotide sequences of the DNA fragments are shown in Fig. 1. We did not make further attempts to clone multiple insertions, because the yields turned out to be sufficient from these constructions (see below).

Large-Scale Plasmid Preparation

For plasmid isolation *E. coli* RR1 cells, transformed with the appropriate plasmid, are grown in 10- to 300-liter fermentors at 37°. M9 minimal medium (6 g/liter disodium hydrogen phosphate, 3 g/liter potassium dihydrogen phosphate, 0.5 g/liter sodium chloride, 1 g/liter ammonium chloride, adjusted to pH 7.4) supplemented with 0.4% glucose, 0.4% casein (acid hydrolyzed), 1 mM magnesium sulfate, 20 mg/liter thiamin dichloride is inoculated with 0.01 vol of a culture grown overnight in LB. During

²⁶ R. M. Wartell and W. S. Reznikoff, *Gene* **9**, 307 (1980).

cultivation the pH is observed and, if necessary, adjusted with ammonia. Cell growth is followed by measuring the optical density at 650 nm as a function of time. In the early log phase (OD_{650} 1.4) plasmid amplification is initiated by adding chloramphenicol to a final concentration of 150 mg/liter and cultivation is continued for 14 hr. Cells are harvested by centrifugation and the wet cell paste is stored at -20° . This procedure yields about 4 g of cells/liter of culture. Cells (200 g) are suspended in 1 liter of 25% sucrose, 50 mM Tris-HCl (pH 8.0), 1 mM EDTA. For lysis 16 ml of this cell suspension is mixed with 5 ml of 10 mg/ml lysozyme in 0.25 M EDTA, 50 mM Tris-HCl, adjusted to pH 8.0 in polycarbonate tubes for a Beckman (Palo Alto, CA) Ti55.2 rotor and kept on ice for 15 min. After adding 2 ml of an aqueous 2:1 (v/v) mixture of 10% Brij and 10% deoxycholate the tubes are gently inverted three times and kept on ice for 10 min. The suspension is centrifuged at 44,000 rpm in a Ti55.2 rotor at 4° . The supernatant is collected, RNase A from a stock solution (20 mg/ml in 1.5 M sodium acetate, boiled 10 min to inactivate DNases) is added to a final concentration of 50 μ g/ml, and the mixture is kept on ice for 1 hr. After adjusting this solution to 1% sodium dodecyl sulfate (SDS) by adding the one-twentieth volume from a 20% stock solution the proteins are degraded with proteinase K at a final concentration of 50 μ g/ml (added from a stock solution containing 20 mg/ml proteinase K in water). This mixture clears during incubation at 37° for 12 hr. The DNA is then precipitated with PEG by mixing the solution with 0.5 vol of 30% PEG 6000, 1.5 M sodium chloride, keeping it on ice for 30 min and spinning the precipitate down by centrifugation (JA10 rotor, 10,000 rpm, 4°). The DNA is dissolved in 500 ml of 10 mM Tris-HCl, pH 8.0, 0.1 mM EDTA/400 g wet cells, extracted three times with 1 vol of phenol, four times with 1 vol of ether, and again precipitated with PEG as described above. After dissolving this DNA in 500 ml of 10 mM Tris-HCl (pH 8.0), 0.1 mM EDTA the DNA concentration is determined by spectroscopy (1 OD_{260} = 50 μ g DNA). The yields vary between 1.5 to 3.8 mg DNA/g of cells.

Large-Scale Restriction of Plasmid DNA

DNA is digested with *Eco*RI without any further purification in 100 mM Tris-HCl (pH 7.5), 50 mM sodium chloride, 10 mM magnesium chloride, 0.1 mg/ml bovine serum albumin at 37° for 24 hr. DNA concentrations vary between 1.5 and 2.5 mg/ml. The restriction enzyme stability and the required amount of enzyme for complete digestion are determined in analytical time courses. For that purpose 3 mg plasmid DNA is incubated as described above with 1000 to 5000 U *Eco*RI and tested for complete digestion after 2, 4, 6, 10, 24, and 48 hr. In preparative digestions half of

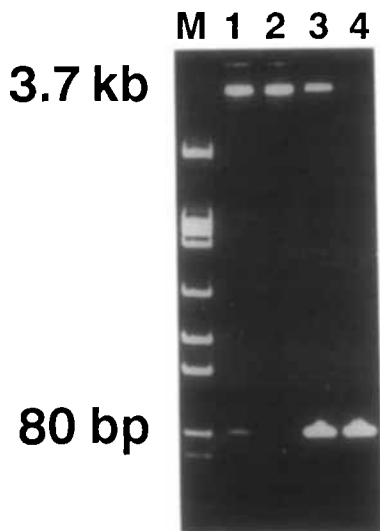


FIG. 2. Gel electrophoresis of the DNA after fractionated PEG precipitation. The analysis was done on a 5% polyacrylamide gel. Lane 1 contains 1 μ g of restricted DNA, lane 2 contains 1 μ g DNA of the PEG precipitate, lane 3 contains 1 μ g DNA of supernatant, and lane 4 contains 1 μ g DNA after final purification by HPLC. Lane M contains a molecular weight standard.

the necessary amount of enzyme is added to start the digestion and the other half after 5 hr. This yields a complete digestion after 24 hr of incubation at 37° with 400 to 1000 U *EcoRI*/mg plasmid DNA, depending on the number of cleavage sites in the respective plasmids. The reaction mixture is then extracted two times with 1 vol of phenol equilibrated with 10 mM Tris-HCl (pH 8.0), 0.1 mM EDTA, four times with 1 vol of ether, and residual ether is removed *in vacuo*.

Fractionated Precipitation of Restricted Plasmid DNA

In order to enrich the small restriction fragments with the *tet* regulatory regions the digested plasmid is selectively precipitated with PEG. This is done by mixing 1 vol 25% (w/v) PEG, 2.5 M sodium chloride with 4 vol of the restricted DNA, which are both preincubated at 37°. During an incubation time of 24 to 48 hr at 37° the precipitated vector DNA sediments to the bottom of the vessel. After this time the solution is decanted carefully without perturbing the vector DNA. The DNA in the supernatant contains nearly all of the small DNA fragment and is precipitated during a 12-hr incubation on ice, after adding 60 g/liter solid PEG. The precipitate

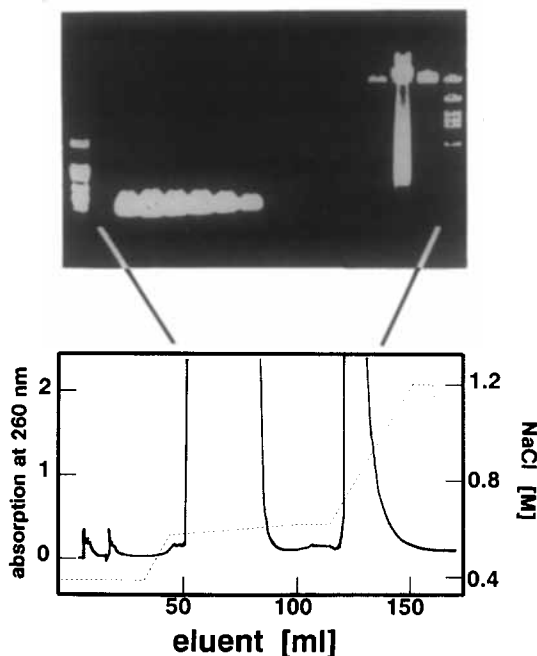


FIG. 3. Separation profile of the DNA on Nucleogen 500-10 IWC. The salt gradient is illustrated by the dashed line and the absorption of the eluent at 260 nm of a preparative run separating 26 mg of the DNA batch analyzed in lane 3 of Fig. 2 is indicated by the solid line. Five microliters of each fraction between the vertical lines was analyzed on the agarose gel shown above the elution profile.

is spun down at 10,000 rpm in a JA10 rotor for 30 min at 4°. A typical result of this procedure is shown in Fig. 2. The enrichment of the small restriction fragments by this procedure varies between 9-fold for the triple insertion plasmids and 80-fold for the single insertion plasmids.

HPLC Purification of Insert DNA

Final purification of the fragments from remaining vector DNA is achieved by HPLC on Nucleogen 500-10 IWC (125 × 10 mm, obtained from Diagen).^{27,28} HPLC is performed on a low-pressure mixing liquid setup (LKB, Bromma, Sweden) equipped with a variable-wavelength UV detector set at 260 nm. All buffers are made from high-purity water (Millipore, Neu Isenburg, Germany) filtered through a 0.2- μ m filter (Schleicher

²⁷ R. Hecker, M. Colpan, and D. Riesner, *J. Chromatogr.* **326**, 251 (1985).

²⁸ M. Colpan and D. Riesner, *J. Chromatogr.* **296**, 339 (1984).

TABLE I
CULTIVATION AND PURIFICATION STEPS

Parameter	<i>tet</i> resistance class			
	A	B	C	D
Transformed plasmid	pWH912	pWH923	pWH931	pWH941
Number and size of inserts (bp)	2 × 86	3 × 80	1 × 84	1 × 78
Total cultivation scale (liters)	200	100	200	200
Wet cell mass (g)	808	420	774	863
Total DNA after lysis (g)	2.2	1.0	3.0	1.3
DNA after PEG precipitation (mg)	110	135	88	44
Total yield of insert DNA (mg)	86	46	54	22
Relative yield of insert DNA (mg/liter of culture)	0.43	0.46	0.27	0.11

and Schuell, Dassel, Germany) and degassed *in vacuo*. Before use the column is washed with 6 vol of water, 4 vol of low-salt buffer (400 mM sodium chloride, 6 M urea, 30 mM potassium phosphate, pH 6.0), 2 vol of high-salt buffer (1.2 M sodium chloride, 5.5 M urea, 30 mM potassium phosphate, pH 6.0), and then equilibrated with low-salt buffer. After use the column is washed with 4 vol of water and stored after equilibration with methanol.

DNA is dissolved in low-salt buffer at concentrations varying between 2 and 4.5 mg/ml and insoluble particles are removed by centrifugation (JA20 rotor, 14,000 rpm, 30 min at 15°). The samples are injected using a loop injector (Rheodyne, Berkeley, CA) with a sample loop of 5-ml volume. In order to determine the optical conditions for the preparative run separating a maximum of 26 mg of DNA, analytical runs may be performed with 15 µg of the DNA samples. In these experiments the optimal slopes of the gradient for separating the insert from the vector DNA is determined. All chromatographic runs are done at a constant flow rate of 3 ml/min and linear gradients with varying slopes. A typical preparative separation profile is shown in Fig. 3. Including column equilibration and sample injection a complete chromatography is done in 80 min and can be very reproducibly repeated many times.

Collected fractions are analyzed by agarose gel electrophoresis and fractions containing only one species of DNA fragment are combined. The DNA is precipitated by adding 120 g/liter solid PEG and collected after a 12-hr incubation on ice by centrifugation (JA14 rotor, 10,000 rpm, 30 min at 4°).

Conclusion

A summary of the results obtained with these methods is presented in Table I. Between 22 and 86 mg of pure DNA fragments varying in size between 78 and 86 bp has been prepared to homogeneity. It may be noted that the triple insertion of the class B *tet* fragment does not lead to the anticipated higher yield compared to the double insert of the class A *tet* fragment, while the plasmids with the two single insertions show lower yields.

[6] Multidimensional Nuclear Magnetic Resonance Spectroscopy of DNA-Binding Proteins

By GRACE PÁRRAGA and RACHEL E. KLEVIT

Introduction

Multidimensional high-resolution nuclear magnetic resonance spectroscopy (2D NMR, 3D NMR, 4D NMR) is a relatively new method for studying macromolecular structure in aqueous and nonaqueous solutions. While the structural information obtained from an NMR analysis of a protein in solution is complementary to that from X-ray diffraction studies of protein crystals, kinetic, thermodynamic, dynamic, and conformational information can also be obtained via NMR spectroscopy.

X-Ray diffraction and high-resolution NMR spectroscopy have different sample and experimental requirements as well as different advantages and disadvantages. The first steps in an X-ray diffraction study involve obtaining crystals of the protein, preparing heavy atom derivatives, and solving the phase problem; these steps yield little structural information until the electron density map is calculated and analyzed. In contrast, a significant amount of information regarding, especially, the secondary structure of a protein is apparent from inspection of a two-dimensional nuclear Overhauser (NOESY) spectrum. For example, it was readily elucidated from the NOESY spectra of transcription factor IIIA (TFIIIA)-like zinc finger peptides ADR1a¹ and ARD1b² that the domains contained α -helical secondary structure. In contrast, analysis of the NOESY spectra

¹ G. Párraga, S. J. Horvath, A. Eisen, W. E. Taylor, L. Hood, E. T. Young, and R. E. Klevit, *Science* **241**, 1489 (1988).

² R. E. Klevit, J. R. Herriott, and S. J. Horvath, *Proteins: Struct. Funct. Genet.* **7**, 215 (1990).

for the human immunodeficiency virus (HIV) metal-binding domain,³ thought to be structurally related to TFIIIA-like zinc fingers, indicated no α -helical secondary structure present. Thus, a qualitative analysis of the NOESY data revealed that the TFIIIA-like zinc finger domains and those in HIV (and by analogy similar sequences found in other viral proteins) were not structurally similar, even before a complete quantitative analysis and structure calculation was performed. Another advantage of NMR spectroscopy in structure-function studies is that once the NMR spectrum of one form of a protein (e.g., wild type) has been assigned, the assignment and structural analysis of altered forms (e.g., mutants, phosphorylated forms) can be quite straightforward.

A variety of DNA-binding proteins and domains have been studied by 2D NMR methods, and application of now well-established methods has led to ¹H assignments for *lac* repressor,^{4,5} *cro* repressor,⁶ zinc finger domains Xfin,⁷ ADR1b⁷ and human enhancer binding protein,⁸ homeodomain *Antennapedia*,⁹ Arc,^{10,11} LexA,¹² glucocorticoid receptor¹³ and estrogen receptor,¹⁴ and incomplete assignments for the λ repressor,¹⁵⁻¹⁸ *trp* repres-

³ M. F. Summers, T. L. South, B. Kim, and D. R. Hare, *Biochemistry* **29**, 329 (1990).

⁴ E. R. P. Zuiderweg, R. Kaptein, and K. Wuethrich, *Proc. Natl. Acad. Sci. U.S.A.* **80**, 5837 (1983).

⁵ R. Kaptein, E. R. P. Zuiderweg, R. M. Scheek, R. Boelens, and W. F. van Gunsteren, *J. Mol. Biol.* **182**, 179 (1985).

⁶ P. L. Weber, D. E. Wemmer, and B. R. Reid, *Biochemistry* **24**, 4553 (1985).

⁷ M. S. Lee, G. P. Gippert, K. V. Soman, D. A. Case, and P. E. Wright, *Science* **245**, 645 (1989).

⁸ J. G. Omichinski, G. M. Clore, E. Appella, K. Sakaguchi, and A. M. Gronenborn, *Biochemistry* **29**, 9324 (1990).

⁹ G. Otting, Y.-Q. Qian, M. Mueller, M. Affolter, W. Gehring, and K. Wuethrich, *EMBO J.* **7**, 4305 (1988).

¹⁰ J. N. Breg, R. Boelens, A. V. E. George, and R. Kaptein, *Biochemistry* **28**, 9803 (1989).

¹¹ M. G. Zagorski, J. U. Bowie, A. K. Vershon, R. T. Sauer, and D. J. Patel, *Biochemistry* **28**, 9813 (1989).

¹² R. M. J. N. Lamerichs, A. Padilla, R. Boelens, R. G. Kaptein, G. Otteleben, H. Ruterjans, J. Granger-Schnarr, P. Oertel, and M. Sharr, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 6863 (1989).

¹³ T. Hard, E. Kellenbach, R. Boelens, B. A. Maler, K. Dahlman, L. P. Freedman, J. Carlstedt-Duke, K. R. Yamamoto, J.-A. Gustafsson, and R. Kaptein, *Science* **249**, 157 (1990).

¹⁴ J. W. R. Schwabe, D. Neuhaus, and D. Rhodes, *Nature (London)* **348**, 458 (1990).

¹⁵ M. A. Weiss, R. T. Sauer, D. J. Patel, and M. Karplus, *Biochemistry* **23**, 5090 (1984).

¹⁶ M. A. Weiss, A. Redfield, and R. H. Griffey, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 1325 (1986).

¹⁷ M. A. Weiss, M. Karplus, and R. T. Sauer, *Biochemistry* **26**, 890 (1987).

¹⁸ M. A. Weiss, C. O. Pabo, M. Karplus, and R. T. Sauer, *Biochemistry* **25**, 897 (1987).

sor dimer,^{19,20} 434 repressor,²¹ and both the *lac* headpiece operator^{22,23} and *Antennapedia*-DNA²⁴ complexes.

A number of excellent monographs^{25,26} as well as Volumes 176 and 177 in this series describe the 2D NMR approach for protein structure in detail. Advances in higher dimension NMR experiments (3D NMR²⁷⁻²⁹ and 4D NMR³⁰) have been reviewed³¹ and detailed elsewhere. This chapter will include a general description of the methodology and will outline the important considerations and limitations that are relevant for protein chemists and molecular biologists contemplating structural studies of DNA-binding proteins and protein-DNA complexes using multidimensional high-resolution NMR.

Some Background on Nuclear Magnetic Resonance

NMR spectroscopy is a powerful tool for the protein chemist and molecular biologist due to the significant amount of structural information offered even from simple one-dimensional nuclear magnetic resonance (1D NMR) experiments. A 1D ¹H NMR spectrum is a plot of intensity (or number of protons) versus resonance frequency or chemical shift (proton environment). In addition to the primary chemical shifts observed for protons in small organic molecules (dependent mostly on chemical bonding patterns), protein NMR spectra contain secondary chemical shifts that are a direct consequence of the *folded* three-dimensional structure of the

¹⁹ C. H. Arrowsmith, L. Treat-Clemons, I. Szilagyi, R. Pachter, and O. Jardetsky, *Proc. 9th Colloque Ampere, Magn. Reson. Polymers* **7**, 10 (1989).

²⁰ C. H. Arrowsmith, J. Carey, L. Treat-Clemons, and O. Jardetsky, *Biochemistry* **28**, 3875 (1989).

²¹ D. Neri, T. Szyperzski, G. Otting, H. Senn, and K. Wuethrich, *Biochemistry* **28**, 7510 (1989).

²² R. Boelens, R. M. Sheek, J. H. van Boom, and R. J. Kaptein, *Mol. Biol.* **193**, 213 (1987).

²³ R. J. J. N. Lamerichs, R. Boelens, G. A. van der Marel, J. H. van Boom, R. Kaptein, R. Buck, B. Fera, and H. Ruterjans, *Biochemistry* **28**, 2985 (1989).

²⁴ G. Otting, Y.-Q. Qian, M. Billeter, M. Mueller, M. Affolter, W. J. Gehring, and K. Wuethrich, *EMBO J.* **9**, 3085 (1990).

²⁵ K. Wuethrich, "NMR of Proteins and Nucleic Acids." Wiley, New York, 1986.

²⁶ K. Wuethrich, *Science* **243**, 45 (1989).

²⁷ H. Oschkinat, C. Griesinger, P. J. Kraulis, O. W. Sorenson, R. Ernst, A. M. Gronenborn, and G. M. Clore, *Nature (London)* **332**, 374 (1988).

²⁸ G. W. Vuister, R. Boelens, and R. Kaptein, *J. Magn. Reson.* **80**, 176 (1988).

²⁹ M. Ikura, L. E. Kay, and A. Bax, *Biochemistry* **29**, 4659 (1990).

³⁰ L. E. Kay, G. M. Clore, A. Bax, and A. M. Gronenborn, *Science* **249**, 411 (1990).

³¹ A. M. Gronenborn and G. M. Clore, *Anal. Chem.* **62**, 2 (1990).

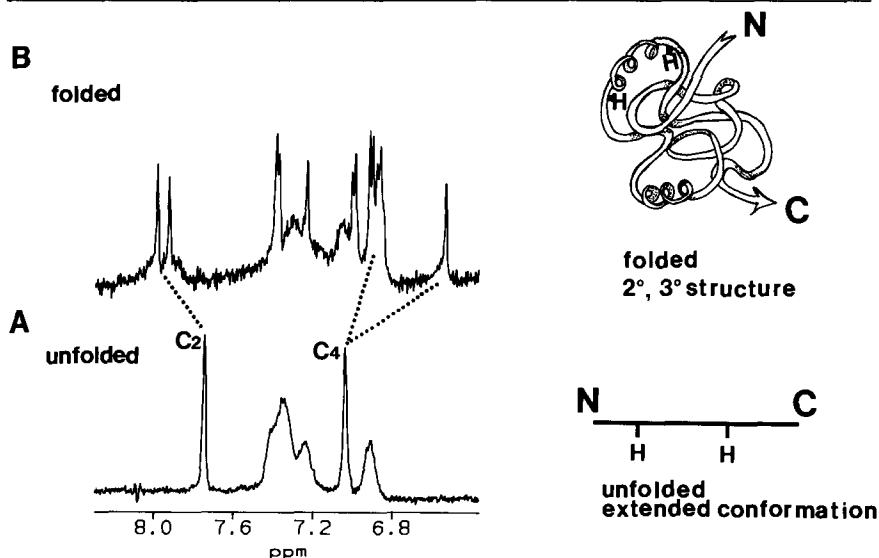


FIG. 1. Comparison of the aromatic regions of spectra of (A) unfolded protein and (B) folded protein. C-2 and C-4 imidazole proton resonances are labeled. N, N terminus; C, C terminus; H, histidine.

protein. A schematic representation of this is shown in Fig. 1. For proteins that are denatured or extended in conformation (Fig. 1A), amino acid side chains are exposed to roughly the same average solution environment so that specific protons of the same amino acid type (for example, the C-2 protons of two different His residues shown in Fig. 1A) resonate at the same chemical shift positions. When the protein assumes a stable folded structure (Fig. 1B), individual residues experience altered chemical environments, resulting in conformation-dependent chemical shift dispersion. As shown in Fig. 1B, C-2 and C-4 protons of the two different His residues experience different environments in the folded structure and now occupy new, unique chemical shift positions. Thus the NMR spectrum of a protein is a direct consequence of its folded secondary and tertiary structure, making NMR spectroscopy an extremely useful tool in structural analyses of protein conformation and conformational changes. The power of 1D NMR spectroscopy in revealing proton chemical shift changes and by inference protein conformational changes has been exploited in studies of another DNA-binding motif, the GCN4-DNA-binding domain.³² On the basis of 1D NMR spectroscopy and circular dichroism (CD) spectroscopy,

³² M. A. Weiss, T. Ellenberger, C. R. Wobbe, J. P. Lee, S. C. Harrison, and K. Struhl, *Nature (London)* **347**, 575 (1990).

Weiss *et al.* have suggested that once the GCN4 basic region has bound to DNA it adopts a folded α -helical structure. Specific chemical shift changes of aromatic protons from unfolded or random coil positions ($-DNA$) to folded chemical shift positions ($+DNA$) in combination with increased α -helical content (CD spectroscopy) were offered as the basis for these suggestions.

The indication that NMR spectroscopy could be useful in solution structure determination³³ was made some time before the method became practicable. The main limitation arose from the necessity to assign each NMR resonance to its specific proton in the protein. However, even for a small protein, most of the peaks in the 1D NMR spectrum contain a contribution from more than one proton, thus making the proton resonance identification difficult if not impossible by 1D NMR. The incorporation of a second dimension in NMR spectra (2D NMR^{34,35}) and recently a third (3D NMR²⁷⁻²⁹) and fourth dimension (4D NMR³⁰) results in the distribution of resonances into two, three, or four dimensions. The extra dimensionality of these experiments serves two important functions. First, nuclear resonances can be distributed into additional frequency dimensions, substantially improving peak resolution. Second, the breakthrough development in 2D NMR was the ability to observe and identify the bonding and spatial relationships between protons. This greatly simplifies the process of resonance identification with respect both to amino acid residue type and to location in the sequence. Thus, it is both the improved resolution and the ability to observe the relationships between resonances that makes the identification of individual proton resonances possible using multidimensional NMR techniques. The structural information that had remained cryptic in protein 1D NMR spectra became accessible, opening the door to the determination of three-dimensional solution structures from NMR data.

The NMR literature is filled with an array of experiments with exotic names such as COSY, NOESY, ROESY, TOCSY/HOHAHA, and INADEQUATE. In this chapter, the spectroscopic details of these experiments will not be discussed. Rather, the type of *information* that is available from specific categories of experiments and the type of *protein sample* required for these experiments will be addressed.

NMR performed on biological molecules can be categorized according to (1) the type and number of nuclei that are observed or (2) the type of

³³ J. Jeener, Ampere International Summer School, Basko Polje, Yugoslavia, 1971, unpublished lecture.

³⁴ W. P. Aue, E. Bartholdi, and R. R. Ernst, *J. Chem. Phys.* **64**, 2229 (1976).

³⁵ J. Jeener, B. H. Meier, P. Bachmann, and R. R. Ernst, *J. Chem. Phys.* **71**, 4546 (1979).

information that is obtained. The first category consists of homonuclear and heteronuclear experiments. In homonuclear experiments, all the information obtained concerns a single kind of nucleus, most commonly ^1H for proteins. Since the ^1H nucleus is the most abundant naturally and is also the most sensitive in NMR spectroscopy, homonuclear ^1H experiments are by far the most widely used. In heteronuclear experiments, more than one type of nucleus is either observed or perturbed. For proteins, these other nuclei are usually ^{15}N and ^{13}C . However, because these isotopes are not abundant naturally, isotopic enrichment of the protein is usually required (discussed later in this section). Another way to categorize NMR experiments is according to the type of structural information that is obtained. Generally, NMR experiments can provide information about *bonding* relationships among nuclei (through bond) or pertaining to *spatial* relationships among nuclei (through space). The through bond-type experiments are also known as coherence transfer, J coupling, or scalar coupling experiments and include COSY (correlated spectroscopy), RELAY (relayed coherence transfer spectroscopy), TOCSY (total coherence spectroscopy), and HOHAHA (homonuclear Hartman–Hahn spectroscopy), HETCOR (heteronuclear correlated spectroscopy), HMBC (heteronuclear multiple bond correlation), and HMQC (heteronuclear multiple quantum correlation) experiments. In such experiments, the spectral information arises from scalar couplings between nuclei and therefore resonances of nuclei that are connected through a finite number of bonds are identified. In general (but not true in some three-dimensional triple-resonance heteronuclear experiments²⁹), information is transferred among nuclei in each amino acid residue but not across the peptide bond. These experiments are therefore used to identify resonances in a spectrum that belong to a single “spin system” which, in the case of proteins, means a single amino acid residue. While this information is sufficient to categorize resonances according to their amino acid residue type (i.e., Gly versus Asp, etc.), it is not sufficient to identify the specific residue in the protein sequence from which that resonance is derived (i.e., Gly-52 versus Gly-68). For sequence-specific assignment of resonances, experiments are used in which the information is transferred through space, via dipolar coupling, regardless of the bonding relationships among the nuclei. The most commonly used experiment of this type is the NOESY (nuclear Overhauser effect spectroscopy), in which nuclei that are within about 5 Å of each other are identified. It is this distance or spatial information from which the details of the three-dimensional structure of the molecule will ultimately be obtained. The goal then is to identify as many peaks as possible in the NOESY spectrum with respect to the pairs of nuclei from which they originate. The greater the number of unambiguous peak assign-

ments that can be made, the more precise and well defined the resulting experimentally determined three-dimensional structure will be.

An elegant new approach has been described for obtaining sequence-specific assignments using only through-bond information.³⁰ The method requires the production of a doubly labeled protein sample in which both ^{15}N and ^{13}C are uniformly incorporated. Advantage is taken of the strong scalar couplings that exist between the heteronuclei and identified spin systems (i.e., amino acid residues) are linked via the heteronuclear scalar coupling directly through the peptide bond. While this approach obviates the need to use through-space NOESY information to *assign* the spectrum, the NOESY information will still be the foundation of the structural determination itself.

General Sample Considerations

Prior to initiating NMR analysis of a DNA-binding protein or domain there are several important points to consider. Some of these considerations derive from limitations of NMR spectroscopy itself (inherently low sensitivity and spectral resolution). Both resolution and sensitivity are dependent on magnetic field strength; with the advent of higher field magnets, some of the current limitations may change. Whenever possible, the link between the sample considerations and the techniques themselves will be discussed so that the relevance of these limitations can be assessed with the development of newer NMR instrumentation.

Molecular Weight

A primary consideration in NMR studies of a macromolecule is molecular weight. If the goal is a detailed solution structure of the protein of interest, NMR analysis is presently limited to molecules of approximate molecular weight <30,000. The actual limit will depend on several properties of the systems of interest. For example, proteins that are mostly α helical tend to have poorly resolved spectra, so spectral overlap becomes a limiting factor at lower molecular weights than for proteins with β -sheet structure. Proteins or domains that form symmetric dimers have half as many resonances as monomeric proteins with the same effective molecular weight, but they will also have the linewidths that correspond to the dimer mass. NMR studies can be performed on larger proteins ($M_r < 100,000$) but determination of a complete three-dimensional solution structure is not currently feasible. In such cases, incorporation of key isotopic labels may allow for selective observation of specific regions of the protein.

The molecular weight limitation stems from several aspects of the

NMR experiment itself. As the molecular weight of a molecule increases, its NMR peaks broaden. The net effect of line broadening is that resolution of individual resonances is decreased in both the one- and multidimensional NMR experiment. As well, broad lines often lead to decreased intensity in the cross-peaks of multidimensional experiments, leading to an effective decrease in obtainable signal to noise. In addition to the dependence of linewidth on molecular weight, with larger molecules there is greater spectral redundancy in the spectrum. For example, in most proteins, all methyl protons will resonate between about 2 and -1 ppm. As the number of methyl groups in the protein increases, the chance that more than one will resonate at any given frequency increases, leading to ambiguities even in the two-dimensional spectrum. This spectral overlap makes resonance identification difficult or impossible in 1D, 2D, and even 3D NMR spectra.

To illustrate the above points, the 1D ^1H NMR spectra of three different proteins are shown in Fig. 2. For the small 30-residue (molecular weight 3562) zinc finger peptide ADR1b, lines are fairly sharp and the resolution of one peak to the next throughout the spectrum is excellent. For comparison, the 1D NMR spectra of *Bacillus subtilis* His-containing protein (HPr; M_r 9100) and *Drosophila* calmodulin (M_r 16,700) are shown in Fig. 2B and 2C. The spectra are plotted so that the intensity of a single proton resonance is the same for all three spectra (see Fig. 2). The heights of other peaks become progressively greater and the widths of the peaks increase with molecular size, resulting in spectral overlap problems. The progressive broadening of peaks and loss of resolution are especially clear in the crowded methyl region (1–2 ppm) of the spectra.

An approach aimed at circumvention of the molecular weight limitation is to study well-defined structural and/or functional domains of a protein or protein–DNA complex by NMR. Because most domains of multidomain proteins consist of 200 residues or less, this is a practicable approach. There are a variety of choices for generating such domains and these will be addressed later in this chapter. Below are additional approaches that address the resolution problem that are useful both for small- to medium-sized proteins or domains and for the study of protein–DNA complexes.

Isotopic Labeling

Resonance overlap caused by either broad lines and/or resonance redundancies can be ameliorated and spectral resolution improved by isotopic labeling methods. There are a number of different approaches available, depending on the specific problem at hand. Isotopic labeling of proteins or domains is a viable course of action when working with bacte-

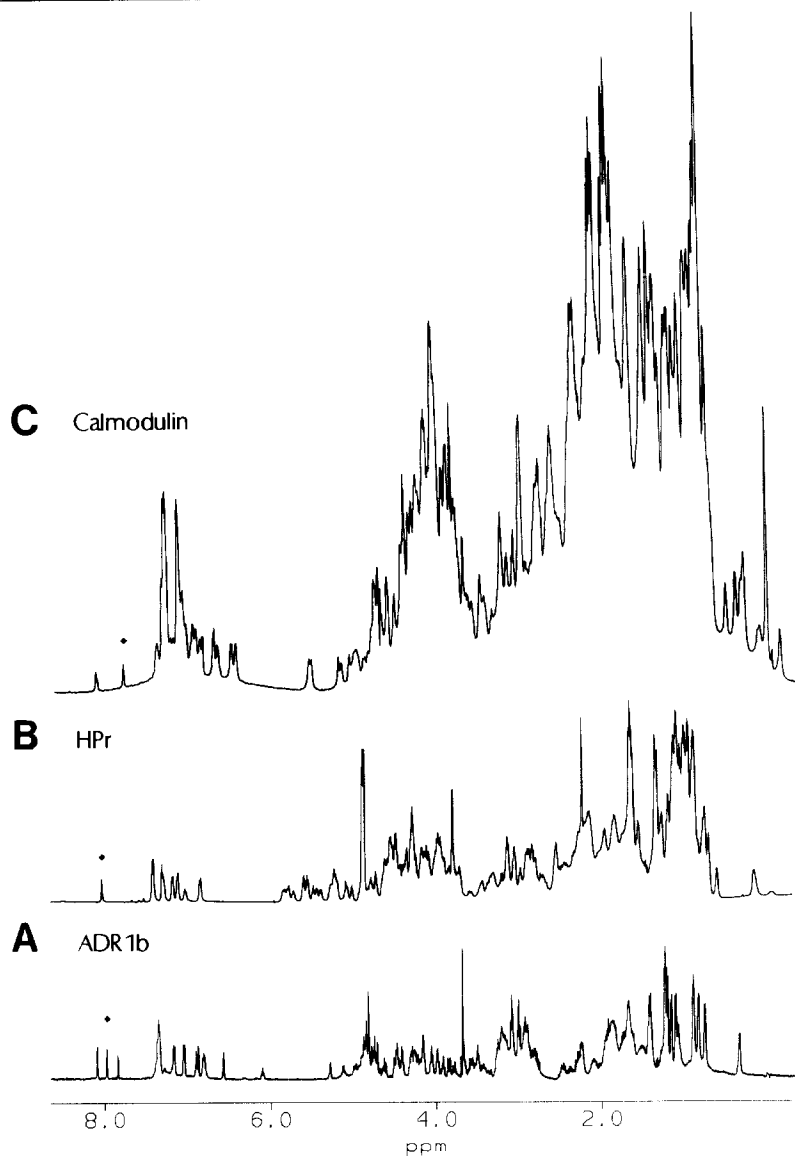


FIG. 2. One-dimensional NMR ^1H spectra of (A) yeast ADR1b (30 residues), (B) *B. subtilis* His-containing protein (HPr; 89 residues), and (C) *Drosophila* calmodulin (147 residues). ◆, Peak of single proton intensity.

rial expression systems as sources of protein. Some excellent reviews and articles describe in detail both selective and nonselective isotopic labeling techniques used in combination with multidimensional NMR to obtain spectral assignments for proteins up to 17 kDa³⁶⁻³⁸ and the details will not be covered here.

There are two different isotopic labeling schemes that involve the incorporation of a nonobserved nucleus, deuterium, into the molecule. Because deuterons resonate at frequencies different from protons, they are "invisible" in an ¹H NMR spectrum. The first deuteration method involves uniform and random deuteration of the protein to about 30% enrichment. In this approach, all the resonances will still be observed in the spectrum, but due to the decrease in dipolar interactions between neighboring ¹H nuclei (now diluted by the incorporated deuterons), the linewidths are significantly reduced, thereby decreasing some of the overlap problems. A second approach is to label a selected group of amino acid residues in the protein with deuterated analogs. The result of this approach is spectral simplification: the spectrum of a deuterated analog is the original proton spectrum minus those peaks that now originate from deuterons. This technique was utilized in an attempt to determine sequential assignments for the *trp* repressor dimer of *M_r* 25K.¹⁹ The scheme is actually better described as "selective protonation" because the bacteria are grown on a deuterated medium (an algal hydrolysate) in which the amino acids to be observed are supplemented in their natural, protonated forms. Thus, one can produce a protein in which only a limited subset of residues are observable. With five such selectively protonated analogs of the *trp* repressor, about 70% of the amide backbone assignments were determined. This approach has the added bonus that since most of the hydrogen atoms in the engineered molecules are deuterons, the resonances from the remaining protons have narrower linewidths, due to the decrease in dipolar coupling that results from the dilution of protons in the sample.

Another powerful approach to dealing with the resonance overlap problem involves isotopic labeling with ¹⁵N and/or ¹³C. As with the deuteration methods, both selective and random labelings are useful, depending on the particular problem to be addressed. Selective labeling is useful in the resonance assignment stage of NMR analysis. For example, by specifically incorporating ¹⁵N-labeled Leu residues into a protein and using heteronuclear correlation experiments, the ¹H resonances derived from

³⁶ D. M. LeMaster, this series, Vol. 177, p. 23.

³⁷ D. C. Muchmore, L. P. McIntosh, C. B. Russell, D. E. Anderson, and F. W. Dahlquist, this series, Vol. 177, p. 44.

³⁸ D. W. Hibler, L. Harpold, M. Dell'Acqua, T. Pourmotabbed, J. A. Gerlt, J. A. Wide, and P. H. Bolton, this series, Vol. 177, p. 74.

Leu residues can be immediately identified. This approach may require the generation of several different selectively labeled samples and can therefore be rather expensive in both time and materials. However, a judicious choice of a subset of amino acid types can usually be made, for example, residues that are well dispersed throughout the sequence or potential DNA-binding residues. In a different kind of selective labeling approach, ^{13}C nuclei can be incorporated into specific positions of specific residues (controlled by the position of ^{13}C label in the glucose added to the growth medium). This scheme has been used to determine stereospecific assignments for the methyl groups in Val residues.²¹ The specific incorporation of stable isotopes into side chains may also be a useful approach for observing specific intermolecular contacts, especially those observed in complexes formed between labeled DNA-binding proteins and their cognate DNA.

Uniform labeling with ^{15}N and/or ^{13}C allows one to perform a variety of 2D, 3D,^{29,39} and 4D NMR³⁰ experiments that are capable of overcoming a significant amount of the overlap problem present in homonuclear 2D ^1H NMR spectra. These experiments allow for more unambiguous assignments of the information contained in the through-space (NOESY) spectra that are the basis for the three-dimensional structure determination.

Amount of Material Required

The question most often asked of an NMR spectroscopist engaged in the determination of protein structures concerns how much protein will be required for a complete NMR analysis. A *minimum* of one sample (natural isotopic abundance) with a concentration of at least 1–5 mM is required. In order to be expeditious with samples that are unstable or prone to aggregation and/or oxidation, it is preferable to have at least two such samples so that H_2O and D_2O experiments can be performed with a minimum of sample manipulation. As well, in our own experience we have noted that fresh protein samples tend to give higher quality spectra than older samples, so it is extremely useful to have a good supply of protein on hand in order to obtain the best spectra possible. It should be noted that NMR is a nondestructive technique, so samples can be reclaimed for other experiments. The required sample volume will vary from 350 to 700 μl , depending on the NMR instrument used. For a low-molecular-weight domain like the zinc finger from ADR1b (M_r 3562), a single 5 mM sample in 500 μl requires about 9 mg of material, whereas for a higher molecular weight protein like calmodulin (M_r 16,700) about 40 mg of purified protein

³⁹ E. R. P. Zuiderweg and S. W. Fesik, *Biochemistry* **28**, 2387 (1989).

is required for the same 5 mM sample. Of course, if heteronuclear experiments are to be performed, similar samples of isotopically labeled material will also be required.

Why is so much protein required for NMR spectroscopy when much less material is required for other types of spectroscopies? The inherent insensitivity of NMR spectroscopy derives from the small energy difference between the two states between which transitions are detected, resulting in a small net population difference between the two states. As the applied external magnetic field is increased (i.e., stronger magnets) the energy difference between states is increased and so is the population difference between states, resulting in a stronger signal and, hence, increased sensitivity. Such an increase has been realized with the availability of 600-MHz spectrometers. High-quality multidimensional NMR spectra of proteins at submillimolar concentrations have been obtained on these instruments.

Source of Proteins

Since NMR studies require large quantities of biological materials, the source and its cost and yields are important considerations. Several approaches are available for generating samples and their advantages and disadvantages are discussed below.

Protein Purification from Tissue Sources. Prior to advances in peptide synthesis and recombinant DNA technologies made in the last decade, the only sources for large amounts of protein for structural studies were tissues and organisms. In general, this source of biological material is less favored because of its inherent lack of flexibility toward important approaches such as isotopic labeling and mutagenesis. However, tissue sources remain an option when (1) the protein domain is too large to be synthesized chemically by solid-phase peptide synthesis, (2) a cDNA clone and/or expression system for the protein are not available, (3) overexpression of the protein in yeast or bacteria is lethal to the cells, (4) the bacterially expressed protein does not contain functionally and/or structurally critical posttranslational modifications found in the eukaryotically expressed protein.

Bacterial Expression of Cloned Genes. To date, most of the DNA-binding proteins and domains studied by NMR have been bacterial and phage repressor proteins (Lac,^{4,5} Cro,⁶ λ ,¹⁵⁻¹⁸ Trp,¹⁹ Arc,^{10,11} LexA¹²). Usually, these proteins have been purified from bacteria after transcriptional induction of the gene encoding the protein (overexpression using strong inducible promoters). Recombinant DNA technology makes these overexpression systems available for nonbacterial gene expression as well.

DNA-binding domains from the eukaryotic proteins *Antennapedia*,⁴⁰ the glucocorticoid receptor,¹³ and the estrogen receptor¹⁴ have been generated and studied by NMR.

There are many advantages to both the NMR spectroscopist and the molecular biologist in using an expression system as a source for protein. The most obvious advantage is the fact that with most expression systems there is a huge increase in protein synthesis. For DNA-binding proteins and transcription factors that are usually found in minute amounts in the cell, protein concentrations may be elevated to as high as 2–10% of the total cell protein. Some examples of DNA-binding proteins that have been purified in large amounts from bacterial expression systems are λ repressor,⁴¹ *lac* repressor,⁴² *cro* repressor,⁴³ *trp* repressor,^{44,45} and *Antennapedia* homeodomain,⁴⁰ Arc,⁴⁶ and LexA.⁴⁷

Another advantage to working with a bacterial expression system is that mutant and deletion forms of the protein can also be expressed and purified. As stated earlier, one of the greatest advantages of NMR in structural studies is the relative ease with which mutant forms of proteins can be studied once the spectrum of the wild-type protein has been assigned. Thus, an efficient bacterial expression system and multidimensional NMR are an extremely powerful combination of methods when applied to structure–function studies of wild-type and mutant molecules.

The use of bacterial expression systems also makes labeling of proteins with selective and nonselective isotopic labels quite straightforward. As described earlier, these approaches can increase the size of protein that can be studied by NMR. In general, then, the ability to label domains and proteins with different nuclei and the ease with which mutant proteins can be generated are probably the most advantageous or important reasons for using bacteria to express proteins of interest.

Chemically Synthesized Peptides and Domains. Recent advances in automated solid-phase peptide chemistry have opened a new era in peptide

⁴⁰ M. Mueller, M. Affolter, W. Leupin, G. Otting, K. Wuethrich, and W. J. Gehring, *EMBO J.* 4299 (1988).

⁴¹ R. T. Sauer, K. Hehir, R. S. Stearman, M. A. Weiss, A. Jeitler-Nelsson, E. G. Suchanek, and C. O. Pabo, *Biochemistry* 25, 5992 (1986).

⁴² R. M. Sheek, E. R. P. Zuiderweg, K. J. M. Klappe, J. H. van Boom, R. Kaptein, H. Ruterjans, and K. Beyreuther, *Biochemistry* 22, 228 (1983).

⁴³ P. L. Weber, D. E. Wemmer, and B. R. Reid, *Biochemistry* 24, 4553 (1985).

⁴⁴ A. Joachimiak, R. L. Kelley, R. P. Gunsalus, P. Sigler, and C. Yanofsky, *Proc. Natl. Acad. Sci. U.S.A.* 80, 668 (1983).

⁴⁵ J. L. Paluh and C. Yanofsky, *Nucleic Acids Res.* 14, 7851 (1986).

⁴⁶ A. K. Vershon, J. Bowie, T. Karplus, and R. T. Sauer, *Proteins: Struct. Funct. Genet.* 1, 302 (1986).

⁴⁷ M. Schnarr, J. Pouyet, M. Granger-Schnarr, and M. Daune, *Biochemistry* 24, 2812 (1985).

research. Coincidental to the advances made in peptide synthetic techniques was the observation (via deletion and genetic analyses) that very short protein fragments or domains contained specific and discrete protein activities (phosphorylation sites, DNA-binding and activation domains). Thus, peptide synthesis has become an important source of substrates for structure–function studies of small domains and proteins.

In general, very short peptides (<30 residues) do not form stable secondary or tertiary structures in solution, with the exception of peptides that contain disulfide bridges. Thus, while it was generally acknowledged that some small domains like the helix–turn–helix domain of the *lac* repressor headpiece (60 residues) retained structure, it was thought that reconstitution of even smaller domains from synthetic peptides was not possible *in vitro*. There are now several examples of very small domains that possess stable structure in solution. One of the first examples of a short, nondisulfide-bridged peptide with stable structure was the 30-residue zinc finger domain. In their studies with a fragment from the transcription factor TFI_{II}A (TFI_{II}A-2), Frankel *et al.*⁴⁸ showed that the fragment could be folded *in vitro* on the addition of zinc. The zinc-bound peptide was also both proteolytically and thermally stable, supporting the notion that metal binding was essential for folding into a stable domain. More recently, an even smaller domain, the 17-residue viral zinc-binding domain, was also shown to possess a stable folded structure in the presence of zinc.³ Thus, it would appear that in the absence of disulfide bonds, high-affinity metal binding is an efficient way to stabilize the structure of small polypeptides.

Molecules the size of either of the two aforementioned zinc-binding domains are quite amenable to either manual or automated chemical synthesis using solid-phase technique.^{7,8,49,50} Currently, peptides of up to about 30 residues can be synthesized routinely in sufficient yield for structural studies. With special care, peptides of about 60 to 90 residues can be synthesized⁵¹ but coupling must be *extremely* efficient to yield sufficient material for structural studies.

One advantage to peptide synthesis is the relative ease of purification. A one-step purification involving reversed-phase high-performance liquid chromatography (HPLC) is often sufficient for purifying synthetic pep-

⁴⁸ A. D. Frankel, J. Berg, and C. O. Pabo, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 4841 (1987).

⁴⁹ D. Roise, S. J. Horvath, J. Tomich, H. Richards, and G. Schatz, *EMBO J.* **5**, 1327 (1986).

⁵⁰ M. F. Bruist, S. J. Horvath, L. Hood, T. A. Steitz, and M. I. Simon, *Science* **235**, 777 (1987).

⁵¹ A. Wlodawer, M. Miller, M. Jaskolski, B. K. Sathyanarayana, E. Baldwin, I. T. Weaver, L. M. Silk, L. Clauson, J. Schneider, and S. B. H. Kent, *Science* **245**, 616 (1989).

tides. The caveat remains, however, that purification is straightforward only when the synthesis has proceeded cleanly and satisfactorily. HPLC purification becomes much more difficult when the HPLC profile of the synthesis product consists of many peaks rather than one single major product. This can occur with sequences containing duplicated Arg or Lys residues, commonly found in DNA-binding motifs. As well, Cys and His residues can be problematic. It is also advisable to avoid amino-terminal Met residues as these often oxidize irreversibly to Met sulfone.⁵²

Many institutions have core peptide synthesis facilities with equipment for synthesis, sequencing, amino acid compositional analysis, and peptide purification. Alternatively, there are several companies that will produce purified peptides in sufficient quantities for structural analysis by NMR. The cost varies by several thousand dollars for about 30 mg of a purified 30-residue peptide.

Some disadvantages of chemical synthesis are as follow: (1) yield is finite and depends on how the synthesis proceeded, (2) mutant varieties of the peptide must be synthesized *de novo*, (3) isotopic labeling (²H, ¹⁵H, ¹³C) during chemical synthesis can be an expensive enterprise,³⁶ (4) synthesis of peptides longer than 30 residues is difficult on most automated equipment and requires double and triple coupling of some residues, (5) reconstitution and solubility may be problematic (see Solubility and Reconstitution, below), (6) purification of mutant varieties or resynthesized batches may require different purification protocols. Given these limitations, though, we have found that chemically synthesized peptides can be useful substrates for NMR spectroscopic studies of small, stable DNA-binding domains.

Purity and Composition

The inherent insensitivity of NMR spectroscopy is almost an advantage when dealing with sample purity. Although it is necessary that samples be as pure as possible, as much as a 5% impurity (proteinaceous) in an NMR sample will not usually interfere with the 1D or 2D NMR spectrum. The most common source of impurities in samples is small organic molecules from HPLC columns, lyophilizers, and contaminated laboratory water. We have found it useful to perform all purification of NMR samples in three-times distilled and filtered H₂O [Millipore (Bedford, MA) Milli Q system]. It is also helpful to avoid lyophilizing voluminous amounts of different proteins and buffers simultaneously on the same lyophilizer.

⁵² R. A. Houghten and C. H. Li, this series, Vol. 91, p. 549.

Another source of contamination is bacterial growth in samples causing proteolysis and degradation of the sample. The normal care taken with any protein sample during purification should be sufficient to avoid bacterial contamination and sample degradation. One must also be aware that paramagnetic ions are usually not compatible with NMR spectroscopy (Co^{2+} , Mn^{2+} , etc.) and their use should be avoided in the course of the sample purification and preparation. A low (0.1–0.5 mM) concentration of ethylenediaminetetraacetic acid (EDTA) can be added to the final NMR sample to chelate any contaminating paramagnetic ions to avoid the spectral broadening they might otherwise cause. Alternatively, distilled and purified water can be run over a Chelex column to rid the laboratory water of paramagnetics and other minerals and metals.

The final step in sample preparation is the removal of all protonated species (buffers, reducing reagents, and chelators) from the sample by exhaustive dialysis and/or gel filtration. This step is critical for optimal signal-to-noise and spectral quality. Even small amounts of protonated buffers will be observed in the NMR spectrum as sharp, intense lines and these contribute to the “dynamic range” problem in NMR. Dynamic range refers to the ability of the detection system of the instrument to measure and digitize signals from low concentration solutes (i.e., the protein) in the presence of other species present at higher concentration. In addition, an artifact known as t_1 streaking—strong vertical lines of noise in a multidimensional NMR spectrum—arises from the sharp signals of buffers. Deuterated buffers, acids, and bases are available and should be used whenever possible when preparing a sample either in D_2O or H_2O for an NMR experiment.

Accurate knowledge of the protein sequence, metal content, pH, and molecular weight are also important as sample heterogeneity or irreproducibility can complicate analysis of the resulting spectra. For samples that have been prepared via bacterial expression vectors, the expressed protein should be subjected to nondenaturing polyacrylamide gel electrophoresis (PAGE) and denaturing sodium dodecyl sulfate (SDS)–PAGE under reducing conditions. The protein sequence or at least an amino acid compositional analysis should be obtained to ensure that the composition and sequence is correct. For peptides that have been chemically synthesized, both the sequence and composition should be determined prior to structural studies. It is not advisable to attempt to sequence peptides directly using NMR, although in principle this can be done using the sequential assignment procedure.^{53,54} Moreover, since deletion peptides or peptides

⁵³ K. Wuethrich, G. Wider, G. Wagner, and W. Braun, *J. Mol. Biol.* **155**, 3111 (1982).

⁵⁴ M. Billeter, W. Braun, and K. Wuethrich, *J. Mol. Biol.* **155**, 321 (1982).

that are incompletely deblocked (containing tosyl or other blocking groups on side chains) can sometimes be generated during synthesis and purification, the real molecular weight (M_r) of the peptide should be measured by time-of-flight, ion spray, or fast atom bombardment (Fab) mass spectrometry. Mass spectrometry should also clarify any sequence inconsistencies that have surfaced after sequence and amino acid compositional analysis.

Solubility and Reconstitution

Because highly concentrated samples are required for NMR analysis, solubility is a primary concern. Unfortunately, there is very little information in the literature on how such concentrated samples are dissolved or reconstituted into folded domains. Of course, each protein, domain, or peptide will have its own optimum conditions for solubility and stability, but there are some general considerations that must be kept in mind if high-quality spectra are to be obtained. In general, one would like to optimize solubility in order to gain sensitivity, but this must be achieved under conditions (pH, ionic strength) that do not promote aggregation. As well, as discussed below, there are certain pH limitations dictated by the NMR experiment itself (see pH and Stability, below).

Generally a lyophilized protein sample that has been treated by either gel filtration (Sephadex G-15 or G-25) or dialysis to rid the sample of protonated buffers and reagents is dissolved in 400 μ l of D₂O and deuterated buffers. Initial characterization of the protein by NMR is undertaken under pH and buffer conditions shown optimal for the protein solubility and activity. When such information is not available, a program aimed at optimizing the desired properties must be undertaken. As an example, we offer our own experience with the ADR1 zinc finger peptides. For reconstitution of the peptides with ZnCl₂, we found (by trial and error) that Tris-HCl buffer at concentrations no greater than 50 mM with at least 10 mM acetate was ideal for peptide and ZnCl₂ solubility. We also found that heating this mixture to 70–80° for 5 min and slow cooling increased final solubility at room temperature. For domains or proteins that have been purified in the native state (for example, baculovirus expression), reconstitution of the folded state is not necessary. However, solubility/folding problems encountered on purification from bacterial inclusion bodies may be overcome by purification in guanidine hydrochloride or guanidine isothiocyanate.⁵⁵

⁵⁵ K. Nagai, Y. Nakaseko, K. Nasmyth, and D. Rhodes, *Nature (London)* **332**, 284 (1988).

Aggregation State

It is important to know the aggregation state of the protein under the conditions to be used in the NMR experiments. This can be accomplished by analytical ultracentrifugation. Furthermore, evidence for oligomerization or aggregation can often be obtained from the 1D NMR spectrum of the protein. Proton linewidths are dependent on the effective molecular weight of the protein, as was shown in Fig. 2A–C. Some evidence for concentration-dependent aggregation in the submillimolar–millimolar range can be obtained by measuring linewidths as a function of protein concentration. For the zinc finger peptide ADR1a we monitored aromatic, C α , and methyl proton peaks with respect to linewidths at 0.5 and 5 mM in the presence and absence of zinc in order to ensure that no concentration-dependent aggregation or dimerization was taking place. This was especially important for the initial studies on zinc finger domains, where the possibility existed for aggregation due to oxidation of Cys thiols or for metal coordination by more than one peptide.

pH and Stability

In order to determine structures by NMR, the backbone amide proton resonances must be observed in the NMR experiment. These amide protons are labile and often exchange rapidly with solvent, necessitating the collection of spectra in H₂O rather than the deuterated solvent. Even in H₂O, observation of labile protons is feasible only when the exchange with solvent is slow on the NMR time scale (exchange rate $< 1 \times 10^3 \text{ min}^{-1}$). The rate of exchange is strongly pH dependent, as amide-solvent exchange is acid/base catalyzed.⁵⁶ Thus, it is desirable to work with samples that are at an acidic pH, where the solvent-exchange rate is slower. This explains the large number of protein NMR studies that are performed at pH values between 3 and 4.5.

Many proteins, however, are not structurally or functionally native at low pH values. As well, other properties of interest such as metal ion binding may also be pH dependent. The necessity to work at low pH is also problematic for proteins with *pI* values in the acidic range, as proteins tend to precipitate at pH values near their *pI*. Thus, NMR analysis must be a compromise between optimal conditions for the function and structure of the protein and optimal conditions for NMR analysis. In practice, we have found that nearly all amide proton resonances can be observed for *some* proteins at pH values as high as ~ 7.0 . When “native” structural characteristics are of interest, it is essential to assess the dependence of

⁵⁶ K. Wüthrich and G. Wagner, *J. Mol. Biol.* **155**, 321 (1982).

the structure and/or function under the buffer and pH conditions where the NMR spectra are acquired. This can be done by investigating NMR chemical shift perturbations as pH or buffer conditions are altered to those conditions optimal for NMR experiments. Other than changes in the resonances from side chains that are ionizable over the pH range of interest, significant chemical shift changes should not be observed in the spectrum. This can be taken as evidence that the three-dimensional structure has not changed under the experimental conditions. To that end we monitored the 1D NMR spectra of the zinc finger peptides ADR1a and ADR1b as a function of pH in order to assess their stability and structure.⁵⁷ pH stability was an especially important concern for these molecules because Zn^{2+} binding by His imidazole nitrogens was proposed to be a driving force for domain folding and the normal pK_a of a His imidazole N is in the range pH 6–7. The pH dependence of the NMR spectra showed that a mixture of folded and unfolded species existed in the NMR tube below pH 5.5, indicating that the Zn^{2+} -stabilized tertiary structure begins to unfold below pH 5.5. Therefore, 2D NMR spectra for the determination of the zinc finger structure were acquired at pH 5.5 as a compromise between structural stability and amide exchange rates.

Protein–DNA Complexes

Ultimately, the application of multidimensional NMR techniques should allow determination of the three-dimensional structures of more DNA-binding proteins and protein–DNA complexes. To date, studies involving protein–DNA complexes have been reported in some detail for the *lac* repressor–operator system^{22,23} and the *Antennapedia*–DNA complex.²⁴ Both studies have resulted in the determination of the approximate positioning or orientation of these domains onto the DNA. Both analyses involved homonuclear and heteronuclear 2D NMR experiments on 600-MHz spectrometers, using ^{15}N and ^{13}C isotopically labeled proteins that were overexpressed using bacterial expression vectors. The oligonucleotides utilized in both studies were chemically synthesized.

The addition of DNA resonances to already crowded protein NMR spectra further complicates 2D and even 3D and 4D NMR spectra. As described earlier, it is this spectral overlap that severely limits the interpretation/identification of observed NOEs and chemical shift perturbations. For studies of DNA–protein complexes, all the considerations described in this chapter will still be extant, with some additional ones as well. In order to restrict the molecular weight of the complex, the shortest func-

⁵⁷ G. Párraga, S. J. Horvath, E. T. Young, L. Hood, and R. E. Klevit, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 137 (1990).

tional sequence of DNA is desirable. However, in order to avoid end-fraying effects on the DNA, an oligonucleotide slightly longer than the identified binding sequence is advisable. As well, binding affinities tend to decrease with decreasing DNA length for many DNA-binding proteins, presumably due to the loss of nonspecific protein–DNA backbone interactions. Since a stable complex is required, DNA length must be optimized versus complex stability. Synthetic oligonucleotides can be generated in high purity and yield on commercially available synthesizers. As well, protein–DNA complexes have been found to be of lower solubility,^{58,59} than the individual components, presumably due to electrostatic neutralization. This problem will have to be addressed individually for each system.

It is anticipated that these considerations will be overcome for other protein–DNA complexes. The ever-growing list of multidimensional NMR techniques available to the spectroscopist, along with the many different biological molecules that can be generated by the molecular biologist and synthetic chemist, bode well for the elucidation of the structural details of these important molecular interactions.

⁵⁸ J. D. Baleja, Ph.D. Thesis, University of Alberta, Edmonton, Alberta, Canada.

⁵⁹ B. R. Reid, personal communication.

[7] Crystallization of Protein–DNA Complexes

By ANDRZEJ JOACHIMIAK and PAUL B. SIGLER

Introduction

X-ray crystallography is the most powerful tool with which to visualize the time- and space-averaged details of specific protein–nucleic acid interactions at the atomic level. Crystallographic analysis of protein–DNA complexes has been made possible by advances in the chemical synthesis of simulated DNA targets and the overexpression of the relevant proteins and protein domains from clones of their genes or cDNAs. A critical prerequisite of this method is the necessity of preparing well-ordered specific crystalline complexes that provide X-ray diffraction data of sufficient accuracy and resolution to assure a stereochemically useful model of the molecular interface. There is evidence that the difference between specific and nonspecific protein–DNA complexes can be very small,¹ therefore in order to understand the discriminatory features of a molecular

¹ B. Luisi, W.-X. Zu, Z. Otwinowski, L. Freedman, K. Yamamoto, and P. B. Sigler, *Nature (London)* in press (1991).

interface one needs accurate, high-resolution X-ray diffraction data. There is no hard and fast rule that relates crystal size, crystalline order, and accuracy of data acquisition to biochemical information; however, diffraction data to 2.5–2.8 Å of three-sigma accuracy will usually lead to a molecular model of the protein–nucleic acid complex that displays the chemistry of the interface in a useful way. Properly refined models, with data to 2.0-Å resolution or higher (especially if augmented with the repeated occurrence of the same chemical entity in the crystallographic asymmetric unit), should provide an unequivocal model that includes fixed solvent molecules and ions. In this chapter we identify considerations and recommend procedures that may enhance one's ability to produce X-ray quality crystals of specific protein–DNA complexes.

General Considerations

Physiological Relevance

Many crystalline specimens used in structural biology face the criticism that the nonphysiological conditions often required to produce X-ray quality crystals diminish if not negate the biological relevance of the study. This criticism is especially sharp where the stereochemistry of a specific protein–nucleic acid interface as seen in the crystal structure of the complex conflicts seriously with the expectations of most workers in the field. Although this concern is important to keep in mind in the design of the crystallization experiments, there are precious few cases in protein crystallography where a specific complex formed from sensibly chosen components has been rendered useless or misleading by the conditions of crystallization. This is especially true in the case of protein–DNA complexes where no challenge to the validity of the chemical relevance of a crystalline complex² has been substantiated by independent studies.

Special attention must be paid to designing and evaluating the DNA target used in cocrystallization. Obviously DNA targets must be chosen that include not only the recognition sequence (or an idealized version thereof) but the appropriate span of DNA with correctly identified boundaries. Careful attention to this issue paid handsome dividends to Phillips and co-workers in their study of the *met* repressor–operator complex that revealed tandem and overlapping binding involving an unexpected

² D. Staacke, B. Walter, B. Kisters-Woike, B. v. Wilcken-Bergmann, and B. Müller-Hill, *EMBO J.* **9**, 1963 (1990).

interface of the repressor.³ To avoid mistakes the crystallographer and collaborators must experimentally establish the size and sequence of the optimal DNA segment with care (e.g., see Refs. 4 and 5). This includes the correct symmetry if any is suggested.¹ Reliance on others could be misleading, as evidenced by the flawed gel shift analysis of the *trp* repressor–operator complex by Staacke et al.² Last, consideration must be given to the possibility of crystallographic disorder resulting from the packing of helices in alternate directions. DNA duplexes that are nearly symmetrical and have chemically similar terminal base pairs can sometimes pack in two almost equivalent modes, leading to a confusing mixture of molecular images.

Plasticity and Crystalline Order

DNA-binding proteins and their DNA targets are usually flexible to facilitate their biological functions. This plasticity does not necessarily dim the prospects for producing well-ordered crystals, because a specific complex may often be more rigid than either of its components. Whereas the “flexible reading heads” of the uncomplexed *trp* repressor have a different orientation in various independent crystallographic representations, they are well fixed in a single specific orientation in all four independent representations of the *trp* repressor–operator complex.⁶ DNA targets, especially the rather long ones, are structurally better ordered in cocrystals of regulatory complexes than in crystals of their uncomplexed counterparts.^{6,7} Indeed, well-ordered crystals of duplexes longer than 12 bp are rare, whereas complexes of 14 up to 32 bp can be clearly visualized in cocrystalline complexes. The protein–DNA interface is more akin to the interior of the protein than to the solvent-exposed surface of the individual macromolecules.⁶ Therefore, in addition to the fact that the macromolecular backbones are more restrained in the complex, the flexibility of the surface side chains of the protein is also significantly reduced when it forms an interface with its DNA partner.⁶

³ S. E. V. Phillips, *Curr. Opinion Struct. Biol.* **1**, 89 (1991).

⁴ B. K. Hurlburt and C. Yanofsky, *J. Biol. Chem.* **265**, 7853 (1990).

⁵ T. Haran, A. Joachimiak, and P. B. Sigler, *Nature (London)* Submitted (1991).

⁶ Z. Otwinowski, R. W. Schewitz, R.-G. Zhang, C. L. Lawson, A. Joachimiak, R. Marmorstein, B. F. Luisi, and P. B. Sigler, *Nature (London)* **335**, 321 (1988).

⁷ R. Marmorstein, Ph.D. thesis, University of Chicago, Illinois, 1989.

Multiple Equilibria

Protein-DNA cocrystallizations are complex multicomponent systems whose behavior is described by more than just the dissociation constant of the protein-DNA complex. For example, there is likelihood of forming "hairpins" from single strands of a palindromic self-complementary duplex. Protein subunits that are in a monomer/dimer equilibrium may dimerize only when they bind to the DNA target, giving rise to the possibility that only one subunit of the protein will be bound to a symmetrical duplex if the latter is in excess. Recognition of such pitfalls is intrinsic to the strategy of growing crystals. Fortunately, because of the nearly millimolar concentration of macromolecules used in cocrystallization, a judicious choice of relative concentrations can help to drive the equilibrium toward homogeneity. There, concentrations, that are typically thousands times greater than the dissociation constant of the complex, can help overpower the potentially disruptive effects of precipitating agents (such as high salt and alcohols) used in cocrystallization.

Special Considerations

General Comments

The basic techniques used for the cocrystallization of DNA-binding proteins with their DNA targets are essentially the same as those used for any macromolecule or macromolecular complex. Thoroughly described vapor and liquid-diffusion techniques⁸ that require only 1–20 μ l of the macromolecular solution are used to conserve valuable materials while screening a wide panorama of crystallization conditions. The usual concern for purity and functional homogeneity in macromolecular crystallizations applies equally to the crystallization of protein and DNA in specific complexes. Special considerations that apply to protein-DNA cocrystals are discussed below.

Stoichiometry

The desired protein : DNA stoichiometry should be established in advance by solution experiments. Empirically it has been found that an excess of DNA is often, but not always, desirable. Most bacterial repressor-operator complexes, as well as the CAP-CAP site, have been crystallized best in systems where the concentration of DNA duplex was in excess

⁸ See, for example chapters by A. McPherson [5] and G. N. Phillips [8] in "Methods in Enzymology," Vol. 114 (1985).

of the dimeric protein (1.2 : 1–1.6 : 1).^{9–12} Several reasons are offered for this observation, all of which are based on the idea that the concentration of desired duplexes is actually less than expected. One cause is defective oligonucleotides, such as those damaged by acid depurination. Second is the fact that short DNA helices are more readily destabilized, especially symmetrical ones composed of self-complementary oligonucleotides that can form competing hydrogen-bonded hairpins. A third reason is the failure to supply equivalent amounts of each strand when forming asymmetric duplexes. This error is likely to be encountered unless care is taken to establish the extinction coefficient of each strand. Extinction coefficients can be easily calculated from the known absorbance of each base at a particular wavelength and the base sequence of the strand. The absorbance of an oligonucleotide solution must either be measured under conditions where there is no hypochromic effect (high temperature, complete hydrolysis) or corrected for the hypochromic effect. In the case of the cocrystallization of the glucocorticoid receptor DNA-binding domain with a modified version of its symmetrical DNA target, it was necessary to provide the protein and DNA in the exact stoichiometry of two domains per target since each subunit bound independently to each half-site and without positive cooperativity (see Ref. 1). An excess of DNA would lead to the formation of complexes in which only one protein subunit would be bound to one of the two half-sites of the symmetrical target.

The Optimal Protein

Segmentation. It is generally advisable to work with as compact a protein as possible consistent, of course, with the preservation of the function under study. This reflects an intuitive view that restricting the degrees of conformational freedom predisposes to well-ordered lattices. The wisdom of this approach is underscored by the success of those studying antibody structure and antibody–antigen interactions by crystallography. The Fab domain—a compact globular structure containing the antigen-binding site—has produced many well-ordered crystal structures whereas the entire immunoglobulin, composed of flexibly linked domains, does not crystallize readily in well-ordered lattices. To date, the stereochemical details of antigen–antibody interactions have been derived exclusively from complexes formed by the compact Fab fragment with its

⁹ R. S. Jordan, T. V. Whitecomb, J. M. Berg, and C. O. Pabo, *Science* **230**, 1383 (1985).

¹⁰ A. Joachimiak, R. Q. Marmorstein, R. W. Schevitz, W. Mandecki, J. L. Fox, and P. B. Sigler, *J. Biol. Chem.* **262**, 4917 (1987).

¹¹ H. C. Pace, P. Lu, and M. Lewis, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 1870 (1990).

¹² S. C. Schultz, G. C. Shields, and T. A. Steitz, *J. Mol. Biol.* **213**, 159 (1991).

antigen and not from intact immunoglobulins. The utility of segmenting a multidomain protein either proteolytically or genetically as an aid to crystallization has been so well established that not to do so would be delinquent. One of the first of the bacterial DNA-binding regulatory proteins to be studied crystallographically was the N-terminal DNA-binding domain of the much larger and more elaborate λ repressor protein. It is also this domain that has given well-ordered crystals of the λ repressor-operator complex.⁹ The complex of a template DNA and the Klenow fragment of *Escherichia coli* DNA polymerase I is another example of the utility of employing a proteolytically trimmed fragment that preserves important functions in crystallographic studies.¹³

Most eukaryotic DNA-binding genetic regulatory proteins are large polypeptide chains composed of several flexibly linked domains, each responsible for distinct functions. It can be a formidable task to cocrystallize such a multidomain, multifunctional, flexible protein with its cognate DNA sequence. Fortunately in most cases the DNA-binding domain constitutes an independently stable substructure that is solely responsible for the specific affinity of the intact protein for a DNA-binding site. Such domains can be separated from the rest of the protein either proteolytically or genetically. Typical examples are the DNA complexes formed by homeodomains of transcriptional regulators of development,^{15,16} the zinc finger-containing DNA-binding domains of the steroid receptors¹ and Zif 268.¹⁷ Overexpressing bacterial clones that produce such fragments avoid the microheterogeneity of proteolytic digestion or posttranslational modifications and can be used to prepare pure protein in very large quantities.¹⁸

Thermophiles. Often a homolog of the protein of interest can be obtained from thermophilic organisms and may be more suitable for cocrystallization. Although there is neither systematic evidence nor rigorous logic behind this prejudice, there is a general feeling that such proteins (and presumably their DNA complexes) form better crystals. *Bacillus stearothermophilus*¹⁹ and the strict thermophiles, *Thermus thermophilus*

¹³ P. S. Freemont, J. M. Friedman, L. S. Beese, M. R. Sanderson, and T. A. Steitz, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8924 (1988).

¹⁴ J. Anderson, M. Ptashne, and S. C. Harrison, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 1307 (1984).

¹⁵ C. Wolberger, C. O. Pabo, A. K. Vershon, and A. D. Johnson, *J. Mol. Biol.* **217**, 11 (1991).

¹⁶ B. Liu, C. Kissinger, C. O. Pabo, E. Martin-Blanco, and T. B. Kornberg, *Biochem. Biophys. Res. Commun.* **171**, 257 (1990).

¹⁷ N. Pavelatich and C. Pabo, personal communication (1991).

¹⁸ F. W. Studier and B. A. Moffatt, *J. Mol. Biol.* **189**, 113 (1986).

¹⁹ A. Yonath, F. Frolow, M. Shoham, J. Mussig, I. Makowski, C. Glotz, W. Jahn, S. Weinstein, and H. G. Wittmann, *J. Cryst. Growth* **90**, 231 (1988).

and *Thermus aquaticus*, have become common sources of bacterial proteins and can be expressed from overproducing *E. coli* strains.

Mutational Changes Involving Sulfur Atoms. Protein produced in bacteria (or, in principle, in yeast) offers the opportunity to exploit mutations that can aid crystallization and/or the structure determination. For example, complications caused by oxidation of cysteines can be circumvented by replacing or deleting the sensitive residues, provided the change does not affect function. Conversely, mutational changes of single Ser residues to Cys can be very helpful in obtaining heavy atom derivatives.²⁰ Replacement of Cys or Met with their selenium counterparts using Cys/Met auxotrophs is a gentle and elegant way to introduce a special scattering atom into a protein that, in conjunction with tunable synchrotron radiation, can provide phase information by the method of multiwavelength anomalous diffraction (MAD).²¹ Such phases are, in principle, sufficient to determine the structure of a protein–DNA complex using only the selenium-substituted protein.

The Optimal DNA Fragment

The Sequence and Length of the DNA Target. The DNA fragment used for cocrystallization is selected on the basis of a careful analysis of genetic and biochemical data. The goal is to identify the DNA sequence elements that determine the specificity of protein binding and the segment of the DNA that is in contact with the protein. Whereas genetic experiments are valuable in determining the sequence determinants of specificity, biochemical and physicochemical studies are required to define the boundaries of the contact site. Methylation protection and footprinting, especially those conducted with small, “groove-specific” probes such as hydroxyl radicals, are helpful, but ethylation interference and primer extension analysis of gel retardation are clearly the most powerful methods to define the segment of DNA in contact with the protein. Excellent accounts of these latter techniques are given by Gartenberg and Crothers.²²

The span of the DNA target is sometimes mistakenly equated with the span of the specificity determinants. This reflects a misunderstanding of how DNA binds to proteins. Most of the stabilizing contacts made at the specific interface between a protein and a nucleic acid involve the phosphate backbone. These phosphate interactions often involve nucleotides that extend beyond the bases that define the identity of the target. If one

²⁰ K. Nagai, C. Oubridge, T. H. Jenssen, J. Li, and P. R. Evans, *Nature (London)* **348**, 515 (1990).

²¹ W. A. Hendrickson, J. R. Horton, and D. M. LeMaster, *EMBO J.* **9**, 1665 (1990).

²² M. R. Gartenberg and D. M. Crothers, *Nature (London)* **333**, 824 (1988).

is to fully understand the chemistry of the specific complex one must cocrystallize a DNA fragment that spans the entire target, not just the segment bearing the identity elements.

The length of the DNA fragment in relation to the length of the contact site has significant effect on the nature of crystal. Harrison and co-workers initially cocrystallized relatively short 14-bp DNA fragments with 434 repressor¹⁴ and 434 *cro* protein.²³ They found that the protein makes appropriate contacts with its operator, but in doing so it interacts with a neighboring duplex, causing a potential distortion at the junction between segments of DNA and complicating the interpretation of the protein-DNA interface.²³ In case of cocrystallization of the *engrailed* homeodomain with its DNA target, the DNA duplex was longer than the dimensions of the protein and therefore two proteins were seen to be bound to the DNA duplex. One protein molecule bound to the predicted site and the second bridged two adjacent DNA segments.²⁴

Often DNA fragments are designed to produce a truly symmetrical target duplex. For such cases the "top" strand and "bottom" strand of the duplex are the same, therefore only one sequence defines the duplex. In cases where pseudopalindromic symmetry exists, it is not possible to use just one sequence. In such systems two different strands of the DNA target duplex must be designed to avoid the self-complementarity of individual strands. This problem is invariably encountered when the target has an odd number of base pairs and would be symmetrical except for a pseudo dyad that "skewers" a central base pair.

For experimental simplicity the sequence of the DNA fragment is often designed to contain idealized recognition elements, special flanking sequences, and symmetrical spacers that would be expected to aid cocrystallization. Experience indicates the need to check the binding behavior of such synthetic DNA elements before proceeding with cocrystallizations.

Optimizing the Crystal Packing. The design of DNA fragments for cocrystallization must consider the following issues. In all protein-DNA cocrystals except one (λ *cro* repressor-operator complex²⁵) the DNA segments pack "end to end" to form a more or less distorted, extended helix. In order that the sequence repeats in register with the DNA helical repeat of 10.5 bp/turn, Anderson *et al.* suggested the $7n$ rule: that is, a sequence repeat every 7, 14, or 21 bp.¹⁴ End-to-end packing would thereby produce an exact repeat every two (21 bp) or four (42 bp) helical turns.

²³ C. Wolberger, Y. Dong, M. Ptashne, and S. C. Harrison, *Nature (London)* **335**, 789 (1988).

²⁴ C. R. Kissinger, B. Liu, E. Martin-Blanco, T. B. Kornberg, and C. O. Pabo, *Cell* **63**, 579 (1990).

²⁵ R. G. Brennan, S. L. Roderick, Y. Takeda, and B. W. Matthews, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 8165 (1990).

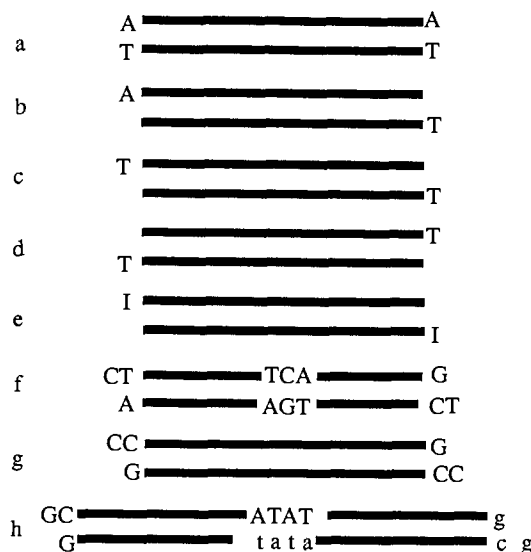


FIG. 1. Sequence termini of the DNA targets cocrystallized with the proteins listed in the Table I. Bars represent the span of a specificity determinant and the flanking regions. Top strand reads from 5' end on the left to 3' end on the right and the bottom strand reads in the opposite direction. Large space in h indicates a break in the strand.

For example, a 3_1 screw of a 14-bp sequence repeat will occur in exactly four helical repeats. This will produce a unit cell length along the 3_1 screw axis of 42 bp or about 143 Å. Although this is an attractive way to use an idealized DNA helix as a basis for crystallographic screw symmetry, this packing scheme has been observed only once. Clearly deviations from ideal B-DNA helices and end-to-end packing that is out of helical register subvert the application of this rule.

Figure 1 shows the DNA termini that have been used successfully in cocrystals and thus serve as a guide in designing duplexes for cocrystallization: (1) blunt-ended double-helical units, either forming an extended helix register ($7n$) or without any end-to-end stacking, (2) complementary overhanging nucleosides such as a 5' A on the top strand and a corresponding 5' T on the bottom strand, (3) single overhanging nucleosides that are the same on the top and bottom strands, such as 5' T, 3' T, 5' C 5' I, etc.; although these obviously cannot form Watson-Crick base pairs, these identical nucleotides can base pair⁶ or can stack to form a hooklike structure.²⁶ Identical overhanging bases provide the potential for symmetrical interactions such as those that relate the operator segments in the *trp*

²⁶ Y. C. Kim and J. M. Rosenberg, personal communication (1991).

repressor-operator cocrystals⁶; (4) overhanging bases can interact with a base pair at the end of the neighboring helix to form a "base triple" (Fig. 2). In this way, adjacent helices can be connected end to end by two steps of a triple helix. An example is shown in Fig. 3, which also shows how different base triplets were used to form nearly isomorphous packing interactions.¹ A particularly successful example of DNA design for crystallization was described by Luisi *et al.*¹ These authors observed a triple helix mode of end-to-end packing in a complex between the DNA-binding domain of the glucocorticoid receptor and an artificially symmetrical DNA target. In order to grow isomorphous crystals containing the more natural, asymmetric DNA target, Luisi *et al.*¹ tailored the termini to form two different type of triple helices that were isostructural but ensured a unique polarity to the packing and hence to the orientation of the asymmetric complex in the lattice (Figs. 1, 2, and 3).

It seems quite clear that there is no general trend that specifies the optimal length and terminal structure of the oligonucleotides to be used in the cocrystallization of proteins and nucleic acids. In most cases the optimal DNA fragment must be determined experimentally by systematic variation of the internal sequence, length, and termini of the DNA duplex, guided by genetic and biochemical evidence.^{9,27}

In order to scan a series of DNA targets more effectively (applying the guidelines described above) some investigators break the DNA segment into "modules" that are linked within the complex by "sticky ends." This combinatorial approach is a particularly effective strategy when dealing with long DNA targets. Fig. 1h shows the scheme by which Schultz *et al.*¹² successfully combined modules to produce X-ray quality crystals of the CAP-CAP site complex.

Halogenated Bases. The DNA component of a cocrystallized complex offers a simple and direct way to introduce atomic substitutions for solving the crystallographic phase problem. With increasing frequency 5-halogenated (brominated or iodinated) pyrimidines have been introduced into the DNA target. Typically 5-bromo-deoxyuridine and 5-iodo-deoxyuridine are introduced as an isomorphous substitutions for thymidine and 5-bromo-deoxycytosine and 5-iodo-deoxycytosine as nearly isomorphous substitutions for deoxycytosine. The bromine atom is a rather "light" heavy atom for phase determination by multiple isomorphous replacement (MIR) but it is an excellent anomalous scattering atom for the application of MAD phasing. Conversely the iodine atom, which possesses 53 electrons (44 more than the 5-methyl of T) is a suitable heavy atom for MIR phasing. Although the iodine anomalous signal is relatively strong for

²⁷ A. Joachimiak, *J. Cryst. Growth* **90**, 201 (1988).

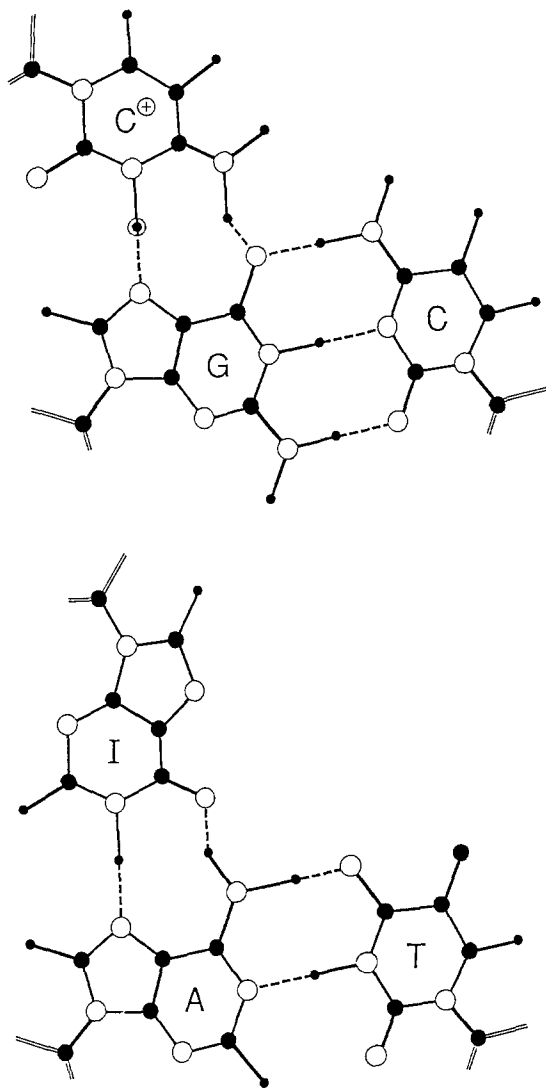


FIG. 2. Hoogsten-like interactions between a GC base pair and a protonated C base, and an inosine (I) interaction with an AT base pair, as observed in the cocrystals of the DNA-binding domain of the glucocorticoid receptor in a complex with its target DNA.¹

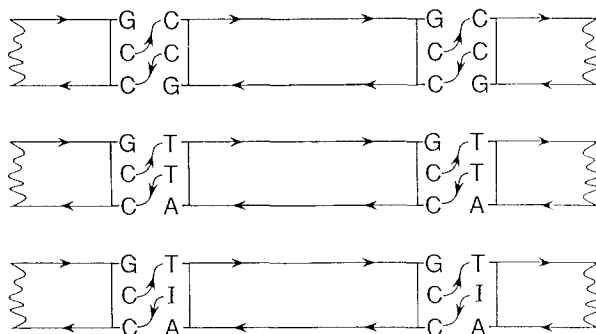


FIG. 3. *Top*: A schematic diagram of the end-to-end packing of the DNA helices observed in the crystal structure of the DNA-binding domain of the glucocorticoid receptor with DNA target showing symmetric C-G : C triple helix formation. *Middle and Bottom*: The proposed packing of the DNA duplexes using an asymmetric DNA target. Base triplets are formed by the interactions I-A : T T-A : T and C-G : C.

CuK α radiation (1.54 Å) from laboratory sources, it is not a good atom for application of MAD phasing with synchrotron radiation.

Preparation of DNA for Cocrystallization

Chromatographic Purification. DNA fragments of various length are usually synthesized with a commercially available DNA synthesizer on a 1 to 10 μ mol scale. The polynucleotide is released from the column and the alkaline-labile base-protecting groups are removed in one step by treatment with a saturated ammonia solution at 55° for 8 hr. The oligonucleotides (including halogenated bases) seem to survive this treatment intact. It is best to remove the by-products of ammonia treatment from the lyophilized ammonium salt with ethanol precipitation of the oligonucleotide (2.5 vol EtOH : 1.0 vol of a concentrated solution of oligonucleotides) from 0.3 M sodium acetate (pH 5.2).

DNA fragments can be purified in several alternative ways. The method we have found to be the most reliable and general, as well as least damaging, is preparative reversed-phase high-performance liquid chromatography (RP-HPLC) (C-4 or C-8 Vydac) at 50°. The procedure involves two chromatographic steps; the first is performed on the 5'-dimethoxytritylated (DMT) oligonucleotide. The oligonucleotide is then detritylated and subjected a second time to RP-HPLC on the same column matrix. The method depends on the retardation of the hydrophobic DMT group and hydrophobic ion pairing in the presence of triethylammonium as counterion (0.1 M,

pH 6.2). Both the triethylammonium acetate and acetonitrile used as eluant are volatile; however, the triethylammonium acetate used as the counterion/buffer can be damaging on lyophilization because a residue of less volatile acetic acid can form that will depurinate the oligonucleotide. This problem is best avoided by first exchanging the salt with triethylammonium bicarbonate. The elevated temperature serves to destabilize unwanted secondary structure. An alternative to the second column step that is sure to eliminate secondary structure is ion-exchange chromatography in 10 mM sodium hydroxide (pH 11.8) on a quaternary amine column [e.g., Pharmacia (Piscataway, NJ) Mono Q fast protein liquid chromatography (FPLC)]. Unfortunately, DNA fragments longer than 20 bp do not resolve well by this ion-exchange step and halogenated bases are at some risk.

Between the two RP-HPLC purification steps the dimethoxytrityl group, used to protect the 5' terminus, is removed by acid treatment. This procedure, if carried too far, can depurinate DNA; therefore, this reaction is potentially harmful and should be strictly controlled. Greater than 90% detritylation is achieved by treatment with a 4:1 (v/v) mixture of glacial acetic acid and water at room temperature for 20 min. The reaction is stopped by neutralization on dry ice with the addition of equimolar amounts of cold triethylamine. The sample is dialyzed against 25 mM triethylammonium bicarbonate, then against water, and lyophilized. It is advisable to treat the detritylated oligonucleotide with saturated ammonia (55°, 12–18 hr). The ammonia treatment will split the chain at the sites of fortuitous depurination, thus facilitating their removal on the second chromatographic step. Properly done, this ammonia treatment should also remove all residual base-protecting groups. It is important not to overreact with ammonia when bromine- and particularly iodine-substituted oligonucleotides are involved. The halogenated oligonucleotides are also quite sensitive to photochemical reactions, therefore exposure to UV light (shorter than 350 nm) should be avoided.

Annealing Duplexes. Self-complementary or nearly self-complementary oligonucleotides can base pair to form either palindromic duplexes or hairpins. The formation of duplexes is a second order reaction favored at high concentrations of oligonucleotide. The formation of hairpins is an intramolecular zero-order reaction that occurs equally well irrespective of concentration. One can drive the base pairing toward the duplex and minimize hairpinning of single strands by annealing at as high an oligonucleotide concentration as is practicable.

Self-complementary oligonucleotides are annealed as follows. The lyophilized sodium salt of DNA is dissolved in neutral 10 mM cacodylate and 50 mM sodium chloride at a concentration of 2–4 mM in duplex and heated for 5 min to a temperature much above the melting point. For the 18-bp self-complementary strands of the *trp* operator self-complementary

nonadodecamer this temperature is 85°. The sample is then allowed to cool slowly to 4° in a water bath in the cold room at a rate of 1° every 4–6 min and is stored at 4°.

Surveying Crystallization Conditions

General Considerations. Screening conditions for the crystallization of protein–DNA complexes are carried out in much the same way as they are for other macromolecules and macromolecular assemblies. In addition to stoichiometry, there are special problems associated with the cocrystallization of specific protein–DNA complexes. Many DNA-binding proteins require high salt concentration (100–300 mM NaCl) to prevent their precipitation while in storage. This can necessitate the introduction of unwanted amounts of salt into the crystallization solution. For this purpose it is wise to use stock solutions of very high protein concentrations so as to minimize the volumes that are introduced into the drops. In some instances, high salt increases the solubility of the protein–DNA complex. One may take advantage of this property to grow cocrystals. One can lower the ionic strength in a vapor diffusion crystallization drop by using volatile salts such as ammonium acetate or ammonium bicarbonate.¹ Liquid diffusion offers the advantage that nonvolatile components can be readily removed from—or added to—the macromolecular solution and the system can be better maintained in a reducing environment. Unfortunately, liquid diffusion techniques are more cumbersome than vapor diffusion systems.

Crystallization conditions can be surveyed by vapor diffusion using droplets hanging from plastic coverslips over reservoirs sealed in Linbro boxes (Flow Labs, McLean, Va). This system allows one to use small droplets and thereby save expensive material. Initial concentrations of the complex in the droplet are generally chosen to be between 0.1 and 0.5 mM. Buffer, monovalent and divalent salts, and precipitating agents are usually at half the reservoir concentration. Droplets are equilibrated against one or more milliliters of the reservoir. In one instance, a crystalline complex was obtained by limited evaporation of the drop over an empty well.²⁸

Precipitating Agents. Precipitating agents used for crystallization of protein–nucleic acid complexes fall into three categories: (1) high salt (typically ammonium sulfate and sodium chloride), (2) organic solvents [various alcohols such as ethanol, 2-methyl-2,4-pentanediol (MPD)], and (3) polyethylene glycols (PEG) of different molecular weights. It was once thought that high salt systems would not be useful for complex

²⁸ R. G. Brennan, Y. Takeda, J. Kim, W. F. Abderson, and B. Matthews, *J. Mol. Biol.* **188**, 115 (1986).

crystallization because a sufficiently high concentration of salt would ultimately dissociate even specific protein–nucleic acid complexes. This has proven not to be the case. Moderately well-ordered crystals of the 434 *cro* repressor–operator complex were obtained from concentrated solutions of ammonium sulfate,¹⁴ the λ *cro* repressor–operator DNA complex was crystallized from sodium chloride,²⁸ and the *lac* repressor–operator DNA was crystallized from sodium acetate.¹¹ However, Table I shows that, so far, high-salt systems have not produced cocrystals that diffract to better than 3-Å resolution (in the best direction).

Organic solvents and PEG have been more effective for growing well-ordered crystals of protein–DNA complexes. The precipitating agents used include MPD, ethanol, and PEGs of average M_r 400, 3000, 4000, 6000, and 8000. PEG has the unique feature of being excluded from the crystal interstices, therefore this precipitating agent never intrudes on the macromolecular interface or crystal packing contacts.

As mentioned previously, protein–nucleic acid complexes are often much less soluble at low ionic strength. This fact has been exploited in the growth of crystals of the complex containing the DNA-binding domain of the glucocorticoid receptor and its DNA target by using a volatile salt such as ammonium bicarbonate, which can diffuse out of the droplet.

The hydrogen ion concentration seems to have a profound effect on the solubility, as well as the stability of the complex and the packing interactions that stabilize the crystal. The special effect of pH on the crystallization of protein–nucleic acid complexes results from the fact that both adenine and cytosine are partially titrated within two units of neutrality. Thus, stabilizing interactions in the complex of the DNA-binding domain of the glucocorticoid receptor–DNA target that involved a protonated “overhanging” cytidine (Figs. 1 and 3) are disrupted at a pH above 6.1. The profound effect of pH on solubility and lattice stability can be exploited to grow X-ray quality crystals. Crystals of MAT α 2–DNA complex, the *engrailed* homeodomain–DNA complex, and the glucocorticoid receptor DNA-binding domain–target DNA complex were obtained by forming a pH gradient between the drop and reservoir.^{15,16}

Multivalent Ions. Multivalent cations play a dominant role in crystallizing specific protein–DNA complexes, presumably by forming stabilizing interactions with the phosphate oxygens of the DNA. The impact of such ions is reminiscent of the dominant influence of these ions on the crystallization of uncomplexed DNA. For example, in the *trp* repressor–operator system millimolar incremental changes in the concentration of Ca^{2+} , Mg^{2+} , Sr^{2+} , Ba^{2+} , $\text{Co}(\text{NH}_3)^{3+}$, Mn^{2+} , Cu^{2+} , and Fe^{2+} had dramatic effects on the solubility, crystal form, and degree of crystalline order.^{10,27} Similarly, Schultz *et al.*,¹² in their study of the CAP–CAP site complex, showed

TABLE I
CRYSTALS OF PROTEIN-DNA COMPLEXES

Protein-DNA complex		pH	Effector molecules	Precipitating agent	Diffraction limit (Å)	Ref.
Protein	DNA					
<i>EcoRI</i>	TCGCGAATTCGCG GCGCTTAAGCGCT	7.4	EDTA	PEG 400	2.7	26
λ <i>cro</i> repressor	TATCACCGCGGTGATA ATAGTGGCGCCACTAT	6.9		NaCl	3.4	25
434 <i>cro</i> repressor	ACAATATATATTGT TGTATATATAACA	8.0	Co(NH ₃) ₆ Cl ₃ MgCl ₂	(NH ₄) ₂ SO ₄ PEG 400	3-5 2.5	23 9
λ repressor	TATATACCGCCAGTGTAT TATAGTGGGTCACCATAA	7.0		PEG 3000	2.5-3.0	29
434 repressor	TATACAAAGAAAGTTTGACT TATGTCTTTCAAACATGAA	7.0	MgCl ₂ , Spermine	PEG 3350 or PEG 8000	3.0	12
CAP	GCGAATGTGATAcacatcg GCTTACACiatagtaagcg	5-6	cAMP, CaCl ₂ , Spermine	MPD	1.9	6
<i>trp</i> repressor	TGTACTAGTTAACTAGTAC CATGATCAATTGATCATGT	7.2	CaCl ₂ , L-Trp			
	CGTACTAGTTAACTAGTACG GCATGATCAATTGATCATGC	5.0 7.2	MgCl ₂ L-Trp CaCl ₂ L-Trp	MPD MPD	3.2 3.0	10 7
<i>arg</i> repressor	AAGTGAATAATTATTCACTTT TTTCACCTATTATAAGTGAA	7.2	CaCl ₂ L-Arg	PEG 3400	2.9	Joachimiak (unpublished)
DNase I	GCGATCgcgacgc CGCTAGCGctagcgc	7	EDTA	PEG 6000	2.0	30
<i>lac</i> repressor	TTGTGAGCGGTCACAA AACACTCGGAGTGT	6.5	2-Nitrophenyl- β -D-fucoside	CH ₃ COONa	6.5	11
<i>engrailed</i> homeodomain	ATTAGGTAATTACATGGCAA AATCCATTAAATGACCGTTT	6.7		pH gradient 8-9 to 6.7	2.0	16
<i>MATa2</i>	ACATGTAATTCATTACACGC GTACATTAAAGTAAATGTGCGT	7.5	CaCl ₂	PEG 400; pH gradient 7.5 to 5	2.9	15
Glucocorticoid receptor	CCAGAACATCGATGTTCTG GTCTTGTAGCTACAAAGACC	6.0	MgSO ₄ , Spermine ZnCl ₂	MPD or Ethanol MPD	2.5 3.0	1 1
Zif268	ICAGAACATCATGTTCTGA GTCTTGTAGCACAAAGACTC TGCgggtgcg CGCCACGCT	6.0 6.5	NaCl	PEG 400	2.1	17

that Sr^{2+} , Ba^{2+} , spermine, and spermidine had significant effects on the type and quality of the crystals.

Calcium and magnesium ions are used most often to grow good quality crystals of the protein–nucleic complexes; cobalt hexamine chloride and polyamines are also useful. Schultz *et al.*¹² made the interesting observation that the presence of Ca^{2+} , spermine, spermidine, *n*-octyl- β -D-glucopyranoside, and MPD in the *stabilizing* medium can improve the limit of diffraction and reduce the amount of diffuse scattering and mosaicity. These observations suggest that it is necessary to search for optimal stabilization conditions as well as growth conditions if one is to achieve the highest resolution. The value of modifying the stabilizing supernatant can be extended one step further. Appropriate use of alcohols, diols, and the like can enable one to lower the temperature to as far as -40° . In favorable cases this causes a significant improvement in the degree of crystalline order and/or radiation resistance. This, in turn, can significantly enhance the resolution of the study.

Physiologically relevant cofactors are important for cocrystallization of specific complexes, presumably because they stabilize the correct stereochemistry. The corepressor ligand, L-tryptophan, was essential to grow the high-quality crystals of the *trp* repressor–operator complex¹⁰ and *lac* repressor–operator DNA cocrystals have been grown in the presence of the anti-inducer 2-nitrophenyl- β -D-fucoside.¹¹ Metal ion cofactors have played a useful role in obtaining crystals of nuclease–DNA complexes. For example, depleting the system of magnesium ion cofactor enabled Rosenberg and co-workers to grow X-ray quality crystals of the *EcoRI*–target DNA complex.²⁶ Removing EDTA protection and supplying magnesium ions to the crystal causes *in situ* hydrolysis of the substrate. Similar observations have been made for cocrystals of the DNase I–DNA complex.³⁰

The Future

We are at a very early point on the learning curve in our ability to grow X-ray quality crystals of specific protein–DNA complexes. Special problems have been recognized and strategies have been developed to deal with them. The case of the nucleosome has not been treated here because of its special architectural and biochemical features. As developments in overexpression continue to enhance our ability to obtain adequate

²⁹ A. K. Aggarwal, D. W. Rogers, M. Drott, M. Ptashne, and S. C. Harrison, *Science* **242**, 899 (1988).

³⁰ D. Suck, A. Lahm, and C. Oefner, *Nature (London)* **332**, 464 (1988).

amounts of valuable proteins and with continued improvement in production and purification of oligonucleotides, the prospects for crystallizing increasingly complicated assemblies improve. With yearly advances in the application of synchrotron radiation and the development of new phasing techniques and data acquisition devices, the availability of crystalline specimens will remain the limiting experimental factor in developing an understanding of the stereochemistry of high-affinity interfaces between proteins and their DNA targets.

[8] Gel Retardation

By JANNETTE CAREY

I. Introduction

Understanding gene regulatory mechanisms and other biological processes based on protein–nucleic acid interactions requires biochemical analysis of the binding reactions. While X-ray crystallography and nuclear magnetic resonance (NMR) are increasingly useful in revealing the *structural* details of such interactions, they cannot reveal the nature and magnitude of the *forces* that contribute to specific complex formation. The affinity, stoichiometry, and cooperativity of binding must also be analyzed to develop a complete picture of how the typically extraordinary specificity of these interactions is achieved. A fundamental difficulty in making such measurements for protein–nucleic acid interactions is the very high affinities that generally characterize these reactions. Such high affinities—often with subnanomolar dissociation constants—preclude direct observation of the binding equilibrium by spectroscopic methods, since few spectroscopic signals are detectable at concentrations comparable to the dissociation constants. This constraint has spurred development of numerous indirect and/or nonequilibrium methods, the most popular of which are gel retardation, filter binding, and footprinting.

Gel retardation is a powerful approach for both qualitative and quantitative analysis of protein–nucleic acid interactions. This method is based on the observation that the electrophoretic mobility of a nucleic acid through a polyacrylamide gel can be altered when a protein is bound to it. The current popularity of the technique dates from the 1981 reports of Garner and Revzin¹ and Fried and Crothers,² although the origins of the method can be traced to earlier literature.^{3,4}

The gel retardation method has several advantages compared to other common assays for protein–nucleic acid interactions. First, a specific binding protein can be detected by its effect on mobility even when other binding proteins are present, as in a crude cell extract. Second, the specific nucleic acid target site of a given binding protein can be identified from a mixed population of fragments. Although both these applications can be

¹ M. M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).

² M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

³ B. K. Chelm and E. P. Geiduschek, *Nucleic Acids Res.* **7**, 1851 (1979).

⁴ H. W. Schaap, M. Green, and C. G. Kurland, *Mol. Gen. Genet.* **109**, 193 (1970).

carried out with other methods, gel retardation is quick and simple, and the direct visualization of complexes is often informative. Third, because free protein is separated from complexes, only *active* protein will be present in the complex bands.⁵ Fourth, complexes can often be resolved even if they differ only in their stoichiometries or in the physical arrangement of their components. These last two are the most important advantages of the gel method, and are unique to it.

Development of a gel retardation assay for a new binding protein can be approached systematically.⁵ The virtues of systematic study include greater understanding, not only of the specific protein–nucleic acid system, but of the gel method itself. This chapter provides guidelines for such an approach, beginning with basic directions that work for a majority of proteins, and proceeding through troubleshooting steps to optimize the method for a new protein. These guidelines apply equally well to proteins that bind RNA or DNA, although in this chapter DNA binding is assumed for convenience. Presumably, these guidelines coupled with chemical intuition would work equally well for nonprotein DNA-binding molecules and/or non-nucleic acid target molecules, although these applications have apparently not yet been demonstrated. Also presented here are methods for quantitative analysis of gel assay results, including determination of binding constants, stoichiometries, cooperativity, and specificity.

A phenomenological theory of gel retardation has been developed, which gives quite good agreement between theory and experiment for several simple systems.⁶ In future, this approach may yield new practical insights as well as providing a rigorous foundation for analyzing gel retardation data.

II. Methods

A. Pilot Conditions

The choice of starting conditions for a gel retardation assay of a previously uncharacterized nucleic acid-binding protein is necessarily arbitrary. However, an exhaustive search for optimal conditions for the *trp* repressor of *Escherichia coli*⁵ led to certain useful generalizations. The table summarizes starting conditions for the experimentally controllable variables that should work for many, but not all, protein–DNA systems.

140 × 140 × 1.5 mm gel (~35 ml)
Wells 0.5 to 1.0 cm wide with 3-mm spacing

⁵ J. Carey, *Proc. Natl. Acad. Sci.* **85**, 975 (1988).

⁶ J. R. Cann, *J. Biol. Chem.* **264**, 17032 (1989).

10% acrylamide (29:1 acrylamide:bis)
0.5× Tris-borate-EDTA buffer (TBE; Ref. 6), pH 8.3
Run in cold room (4°)
5% glycerol loading dye
100 V, 2 hr for 100-bp DNA

B. Detailed Protocol

1. *Preparation of Gel.* The optimal acrylamide concentration depends on the length of the DNA fragment used in the experiment. As the length of the DNA fragment increases, the acrylamide concentration should be reduced. A 10% gel works well for DNAs of up to 200 to 300 bp in length. The gel is poured as usual (see, e.g., Ref. 7), allowed to polymerize, mounted in the gel box, and equilibrated at 4° (preferably overnight). For reproducibility in quantitative work, the times between pouring, polymerization, and running should be kept constant. The gel box ideally should permit free circulation of coolant (air or water) on both surfaces of the gel plates to allow efficient heat dissipation. It should also allow for access to the wells so that the gel can be loaded while the current is turned on. Before loading, the gel is preelectrophoresed until the current drops to a constant value.

2. *Reaction Mixes.* If pure protein is available, a good first experiment is to titrate a small, constant amount of labeled DNA with increasing amounts of the protein. The concentration of DNA should be as low as is practical. High specific radioactivity DNAs labeled with ³²P by nick translation can be used at concentrations of 10 pM or less, as only a hundred or so counts per minute per band is adequate for detection.

A reaction mix is assembled containing the DNA plus necessary buffers, cofactors, salts, or other components if these requirements are known. If they are unknown, a reasonable choice of buffer to start with is 0.5× TBE with no added salt. The reaction mix need not have the same composition as the gel buffer. The reaction mix should also contain sucrose or glycerol at up to 5% final concentration, and loading dyes as desired, to facilitate loading. The mix is then aliquotted into reaction tubes. The protein is diluted over a wide range, say from the highest possible concentration down to a concentration equal to that of the DNA, and aliquots of protein are added to the reaction tubes. For a first experiment, 10-fold serial dilutions should be tested over this entire protein concentration range. One control sample should contain only the protein at the highest concentration tested, another only DNA at the chosen fixed concentration.

⁷ T. Maniatis, E. F. Fritsch, and J. Sambrook, "Molecular Cloning: A Laboratory Manual," p. 174. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1982.

The samples are mixed and allowed to equilibrate. For unknown reaction rates, 15 min is a compromise length of time, and attainment of equilibrium can be verified by comparison with a much longer incubation time (1 to 2 hr). Sample volumes can range from 5 to 75 μ l; the smaller surface-to-volume ratio of larger samples reduces reequilibration when sample and running buffers are different.⁸

3. *Gel Loading.* Immediately before loading, the wells of the prerun gel are cleaned out by flushing with reservoir buffer. The voltage is turned up to ~ 300 V (or more; see Ref. 9), and the samples are rapidly loaded. (*Caution:* Care must be exercised to avoid electrocution!) A glass capillary and mouth pipette, or a sequencing pipette tip and adjustable microliter pipettor, are convenient for loading. As soon as the last sample dyes have just entered the gel, the voltage is reduced to avoid heating, and the gel is run long enough for the free DNA to migrate about half the length of the gel. Longer running times may lead to excessive dissociation of complexes.

4. *Detection.* The gel is dismantled, and the lane containing only protein is cut off if it is to be stained with Coomassie Blue or silver. The rest of the gel is transferred to a sheet of filter paper for drying. A typical 1.5-mm gel requires about 1 hr under vacuum at 80° to dry completely. The dried gel is exposed to autoradiographic film. Overnight exposure is sufficient to detect several hundred ³²P counts per minute in a single gel band when an intensifying screen is used.

C. Troubleshooting

The conditions outlined above generally resolve complexes from free DNAs for many binding proteins. However, if the expected mobility shift is not observed, the protocol may require modification (as outlined in Section V). Troubleshooting can be carried out in a logical way by considering what occurs during the gel retardation process.

Before electrophoresis, the samples should contain an equilibrium mixture of free DNA, free protein, and complexes. When this mixture is loaded onto the running gel, all species begin to migrate under the influence of the electric field, each according to its own characteristic mobility, which is determined by its size, shape, and charge. Separation of bound and free components thus begins at the moment of loading, unless by chance two components have very similar electrophoretic mobilities.

Under most circumstances, free DNA has the highest mobility. As free DNA begins to enter the gel, the preexisting equilibrium may be perturbed by mass action in the direction of complex dissociation. On the other hand,

⁸ D. Senear and M. Brenowitz, *J. Biol. Chem.* (in press).

⁹ M. G. Fried, *Electrophoresis* **10**, 366 (1989).

the interface between the gel and the sample causes a discontinuity in the rate of electrophoresis, from the very fast rate in the sample to a much slower rate in the gel. Thus, focusing of the sample occurs at this interface, as is well illustrated by many reports in which samples initially occupying a volume of $1 \times 5 \times 10$ to 20 mm give rise after electrophoresis to free DNA bands of only $1 \times 5 \times 1$ to 5 mm. This focusing must concentrate the sample at the boundary, and could enhance complex formation by mass action, thereby opposing the effect of DNA depletion. Perturbations in either direction will depend on the relative rates of association and dissociation, and the clearance time for free DNA to fully enter the gel.

Once all the free DNA in the sample has entered the gel as a focused band, there can be no further increase in the amount of DNA in that band, provided it is the highest mobility species present. Even if *all* the complexes subsequently liberated their DNA by dissociation during electrophoresis, the DNA thus freed could not catch up with the initial free DNA band. Therefore, the free DNA band accurately reports on the equilibrium fraction of free DNA, except for perturbations during the clearance time. The clearance time for all free DNA to enter the gel is therefore the "dead time" of the method. Fortunately, this time is under experimental control via the variables of DNA length, acrylamide concentration, and loading voltage. In addition, clearance time is equal for all samples on a gel that is loaded with the current turned on.

The species with the next fastest gel mobility is ordinarily the protein-DNA complex. The remarkable fact that this complex migrates as a discrete band in gels was the founding observation of gel retardation. Bands corresponding to complexes are unexpected because the time required to run a gel can exceed the typical complex half-life by orders of magnitude. To explain their surprising persistence, the idea of "caging" was invoked.² The term caging has turned out to be a misunderstood, although not incorrect, way to describe what happens in the gel, because it is sometimes incorrectly interpreted as the cause of the apparently slow dissociation of complexes in the gels. In fact, all evidence suggests that the gel has *no* effect on dissociation rates. Rather, caging probably works by causing an increase in the effective concentration of the interacting components by slowing their electrophoretic escape after dissociation, leading to enhancement of the bimolecular (and thus concentration-dependent) reassociation reaction. It seems likely that the low water activity in polyacrylamide gels contributes to this effect as well. Thus, although complexes may dissociate many times during the gel run, reassociation is strongly favored, leading to persistence of complex bands even when the gel run time greatly exceeds the half-time for dissociation.

Mobility of the complex is related to mobility of the free protein.⁵ The

stained protein-only lane may thus provide a clue as to how to proceed in modifying the protocol if no shift is observed. If the protein forms a band in the gel, then it is negatively charged at the pH of the TBE buffer. If its mobility is too similar to that of the DNA, little or no retardation of the DNA will occur on binding.⁵ In this case it is advantageous to run gels at lower pH values. Buffers should be chosen that have high buffering capacity at each pH value. Some buffers, such as dilute phosphate, are poor choices as they lose their buffering capacity during electrophoresis and must be recirculated. On the other hand, if the protein band is at the well rather than in the gel, the protein may either be very basic or highly aggregated. The following paragraphs give more details for troubleshooting (following the guide in Section V).

The pilot experiment described above may fail if the protein is too dilute or only partially active. Proteins can be concentrated by precipitation with ammonium sulfate, or by dialysis against solid Sephadex (Pharmacia, Piscataway, NJ) or Aquacide (Calbiochem, Los Angeles, CA), or by ultrafiltration (e.g., Amicon, Danvers, MA). Determination of protein activity is essential for accurate measurement of binding constants, and is discussed in Section III,D.

If there is no detectable free *or* bound DNA in the sample lanes, nuclease activity may be present as a contaminant of the protein preparation. On the other hand, this result could also be consistent with complex formation: if complexes are present initially, but dissociate slowly during electrophoresis, the liberated DNA can become so diffuse as to be undetectable as a band or even as a smear. In some cases the DNA may disappear at intermediate protein concentrations and reappear at higher concentrations, as high stoichiometry complexes form. Disappearance of the band corresponding to free DNA can be taken as evidence of complex formation,^{1,5,10} and can be quantitated as described in Section III, if the possibility of nuclease activity is rigorously eliminated first. To accomplish this, a second DNA fragment lacking the binding site can be titrated simultaneously with the specific fragment (i.e., in the same reactions) if the two DNAs are of different lengths.⁵

Dissociation of complexes during electrophoresis is not, in and of itself, a problem for many purposes because the disappearance of free DNA can be quantitated instead of the appearance of complexes. However, dissociation may be minimized by increasing the concentration of acrylamide and/or bis in the gel, and reducing running times as much as possible. Dissociation of complexes during the dead time (while the DNA is entering the gel) is minimized by loading the gel while it is running at high

¹⁰ A. Revzin, *BioTechniques* 7, 346 (1989).

voltage^{5,9} and/or by loading larger samples¹¹; on the other hand, smaller samples may provide more focused bands.

Smearred, streaked, or uneven running of DNA can be due to several kinds of problems with the gel itself. If polymerization was too rapid it may have been uneven, or the gel may have been run too hot. The acrylamide concentration may be too low, or a stacking gel may be required to provide a second interface where the complexes can become further focused. The samples may have entered the gel too slowly, or a component of the loading dye may have interfered with migration (e.g., polyethylene glycol). All these potential sources of failure should be explored until the operator is confident of the technique.

If none of the suggested modifications succeeds, it may be necessary to explore a wider range of gel and/or solution conditions. If cofactors are required for binding, these may have to be incorporated into the gel when it is cast and included in the running buffer. Some compounds may become covalently incorporated into the gel by this method, forming, in effect, an affinity matrix for the protein; alternatively, cofactors may be introduced by electrophoresis or diffusion. The submarine horizontal gel system of Hendrickson and Schleif¹² allows for more efficient diffusion. Charged cofactors may become redistributed during electrophoresis. This can be minimized by recirculating the gel running buffer, or, in the case of titratable cofactors, by judicious choice of pH. In the case of *trp* repressor, the corepressor L-tryptophan is required in the gel to observe operator binding at 10% acrylamide, but at 20% acrylamide the corepressor is required only in the reaction mixes and not in the gel. This result implies that even small ligands may be caged to some extent by the polymeric gel matrix.

III. Quantitation

A. Affinity

Once adequate resolution is obtained of the free DNA from the complex band, and/or from any smear resulting upon its dissociation, it is possible to use the gel method to determine the quantitative parameters describing the binding reaction. The first point to establish is the protein concentration required to bind half the DNA. An experiment similar to the one described above can be used, with the protein concentration varied over the binding range such that at least two protein concentrations are tested in each log unit. The midpoint concentration can be estimated by eye from the

¹¹ M. Brenowitz, E. Jamison, A. Majumdar, and S. Adhya, *Biochemistry* **29**, 3374 (1990).

¹² W. Hendrickson and R. F. Schleif, *J. Mol. Biol.* **178**, 611 (1984).

autoradiogram, and gives an indication of the approximate magnitude of the dissociation constant, K_d . Visual estimation of K_d is generally accurate to within a factor of 10 or so, although precision is considerably better than this. This estimate of K_d is valid only if the DNA concentration is negligible compared to the protein concentration at the midpoint (at least 10-, and preferably 100-fold, lower).

To obtain a more accurate value for K_d , quantitation of the gel results can be done by film densitometry, direct scanning of gel radioactivity, or scintillation counting of excised gel bands. Smaller increments in protein concentration will improve the precision of the K_d value determined by these methods. The lane separation given in the protocol is essential to reliable quantitation by any of these methods. If the autoradiogram is to be scanned, it is important that the optical density of the bands be within the linear response range of the film. This is usually achieved through preflashing of the film to overcome the response threshold.¹³

The gel can also be sliced and counted in a scintillation spectrometer. For ^{32}P the quenching due to the acrylamide and filter paper is not generally a problem, but for lower energy isotopes it is necessary to slice the wet gel (no filter paper) and to dissolve the polyacrylamide by oxidation. This is accomplished using 17% HClO_4 and 21% H_2O_2 in tightly closed plastic scintillation vials at 60° overnight.^{5,13} Chemiluminescence from peroxide reactions can be a problem, and can be minimized by altering the gel cross-linker so that oxidative dissolution is unnecessary.¹⁴ Chemiluminescence is detectable as an instability of the cpm data with repeated measurement. An alternative treatment, therefore, is to allow chemical reactions to run their course, as determined by reproducibility of cpm values. Vials should be stored at least 24 hr in the dark, preferably in the scintillation counter, to reduce the background from photoluminescence as well. Direct scanning of the gels is much simpler because of its speed, linear response, and freedom from luminescence interference, although sensitivity and resolution may be less good than with film.

The fraction of DNA that is bound and free is determined after normalization, either using an internal standard⁵ as described in Section II,C or by summing the total number of counts in each lane.⁸ If not all the DNA is bound at very high protein concentrations, a correction should be made for this inactive population of DNA molecules. Partial activity is not generally found with restriction fragments, but can sometimes be a problem with synthetic DNA. Fractional activity is also a common, although poorly understood, problem with nucleic-acid binding proteins, and cor-

¹³ R. A. Laskey and A. D. Mills, *Eur. J. Biochem.* **56**, 335 (1975).

¹⁴ B. W. Fox, "Biochemical and Biophysical Methods," Vol. 5, Elsevier, New York, 1976.

rection must be applied for it. The correction factor is determined under quite different binding conditions, as described in Section III,D.

The data are plotted as fraction of free DNA vs protein concentration to determine K_d , which is approximately equal to the protein concentration at which half the free DNA has become bound, provided the DNA concentration is vanishingly small compared to the protein concentration at the midpoint. It may seem more logical to measure the appearance of complexes than the disappearance of free DNA, and in practice both should probably be plotted. The two resulting measurements of K_d will differ if the equilibrium is perturbed by the gel run. For good quantitation, it is often necessary to span six or more orders of magnitude in protein concentration in order to have well-determined baselines before and after the binding zone. The best way to plot such data is probably the so-called Bjerrum plot of fraction free DNA vs log of protein concentration. The principal advantage of this plot is that the log scale compresses the large range of protein concentrations into a convenient size. *All* binding data appear sigmoid on such plots, and the exact shape of the curve depends on the binding parameters, as described in Section III,D.

A very simple test can be applied to determine if the binding reaction is complicated by multiple binding sites or cooperativity. The experimental data are compared to a theoretical binding curve calculated from the binding expression for a model with independent and equivalent sites.⁵ When good data do not fit this simple model, the information in the complex bands can be used to determine the correct binding isotherm,^{8,11} as discussed in Section III,E.

Some discussion is in order about the validity of binding constants measured by gel retardation. As with any other method in which free and bound components are separated, the possibility exists of perturbing the equilibrium by mass action, as discussed above. Although this effect is minimized by analyzing the disappearance of free DNA rather than the appearance of complexes,^{1,5-7} it may not be eliminated completely. For example, significant dissociation may occur during the dead time while free DNA is entering the gel. Although the dead time can be reduced by raising the loading voltage, it may still be long relative to the dissociation rate of the complexes or to the rate of reequilibration between sample buffer and loading buffer. Thus, it is essential to have a way to evaluate the assay before interpreting the absolute values of binding constants.

The best practical solution to this problem is to measure the binding constants by a second method that does not perturb the equilibrium. Ackers and co-workers¹⁵ have shown that DNase I footprinting can be

¹⁵ M. Brenowitz, D. F. Senear, M. A. Shea, and G. K. Ackers, this series, Vol. 130, p. 132.

used as a quantitative assay that meets this criterion for certain proteins. For uncharacterized proteins, it must be established independently that the footprinting conditions do not perturb the DNA binding equilibrium. This was shown to be the case for *trp* repressor,⁵ and the DNase I assay in turn was used to validate the gel assay results. However, even when such tests are met, caution is advised in interpreting very large differences in binding constants determined by the gel method.

B. Kinetics

Rate measurements are useful for their role in studies of mechanism, and as a means of confirming a binding constant, as $K_d = k_{\text{off}}/k_{\text{on}}$. Although measurements of association and dissociation rates by gel retardation are straightforward, they are subject to the same perturbations as equilibrium measurements. Useful rate information can be obtained from gel retardation,^{9,10} but it should not be the sole criterion used to validate a binding constant made by the same method.

Association rates are determined by mixing the components and loading samples at time intervals. Because the free DNA begins to be separated from complexes as soon as the sample is loaded onto the running gel, the loading process itself acts to quench the association reaction. For dissociation rates, complexes are preformed and allowed to reach equilibrium. The association reaction is quenched by either dilution or by addition of a quenching agent so that the dissociation reaction can be observed. Samples are loaded at time intervals after quenching. Typical quenching agents include unlabeled target DNA or nonspecific DNA, which are silent in autoradiograms of the gels. It is important to note that the quenching agent is competing with the labeled DNA for dissociated protein, which is usually in vast excess over the labeled DNA. Thus, the competitor must be in molar excess over the *protein* in order to observe competition. To verify that the quenching agent acts only as an inert sink for excess protein, k_{off} should be shown to be independent of competitor concentration.

C. Specificity

Use of a nontarget DNA as an internal control as described in Section II,C also allows simultaneous, independent observation of nonspecific binding. The nonspecific binding constant is evaluated identically, and it must be corrected for the statistical nature of nonspecific binding to DNA.¹⁶ The ratio of affinity constants for specific and nonspecific binding is a measure of the specificity: the ability of a given protein to discriminate

¹⁶ J. D. McGhee and P. H. von Hippel, *J. Mol. Biol.* **86**, 469 (1974).

between the true target site and nontarget sites. This ratio is expected to vary with solution conditions because the two kinds of complexes typically have differing contributions to their free energy of binding.¹⁷ The gel retardation method is not the best way to determine specificity over the required wide range of conditions, however, because the gels themselves vary over these conditions. For example, as the salt concentration is raised, the gels run hotter, promoting dissociation as well as smearing of the bands. If it can be established that the gel method provides a valid measure of the equilibrium constant, then it may be adequate to vary conditions in the reaction mixes only, rather than in both the gel and the reactions. However, this approach is not recommended. It is preferable to use a true equilibrium method or footprinting under nonperturbing conditions when possible for these experiments.

D. Stoichiometry

The ability of the gel method to resolve complexes that differ only by their stoichiometries is a unique advantage of the technique, and can be exploited in a double-label experiment to determine the molar ratios of protein to DNA in each complex. Obviously, a requirement for this experiment is that discrete complex bands be detected in the gel. An experiment similar to that described in Section II,B above can be used with both labeled protein and labeled DNA.

The choice of isotopes for such an experiment is made according to the usual criterion for a double-label experiment¹⁴: the two isotopes should be quantifiable separately when present together, preferably with minimal corrections. All pairwise combinations of ³²P, ³⁵S, ¹⁴C, and ³H should work except for ³⁵S and ¹⁴C, which have energy spectra that are too similar. However, quenching may shift the energy spectrum of an isotope¹⁴; therefore, counting efficiencies and spillover for the chosen isotope pair should be determined prior to the experiment under the exact conditions that will be used for scintillation counting. It is convenient if only one of the two isotopes is detectable by autoradiography, to facilitate preparing a template for fractionation of the gels,⁵ but this is not essential.

³²P and ³H are used as an example here for illustration, and they are a good choice: spillover corrections are not excessive, and counting is reasonably efficient. ³²P-Labeled DNAs are easily prepared in high yield and high specific radioactivity. The short half-life of this isotope confers a unique advantage: the complex correction factors sometimes required in double-label scintillation counting can be confirmed after a suitable

¹⁷ M. T. Record, Jr., C. F. Anderson, and T. M. Lohman, *Q. Rev. Biophys.* **11**, 103 (1978).

decay period and will give an internally consistent result if they were accurate.⁵ ^3H -Labeled precursors for protein labeling *in vivo* or *in vitro* by biosynthesis or chemical modification are readily available at high specific radioactivities. Proteins labeled with ^3H should be stable for long periods when stored at 4° in dilute solution.

It is essential to have sufficient amounts of protein and DNA present in each complex band for accurate and precise quantitation. Because ^3H is somewhat inefficiently counted and sensitive to quenching, it is necessary to have well over 1000 dpm of ^3H /band. This amount of radioactivity is difficult to achieve in a single band under the conditions given in Section II,B, even with theoretical maximum specific radioactive labeling of the protein. The factor that limits the amount of protein in a complex band is the amount of DNA present. Thus, to improve detection of ^3H -labeled protein, the amount of DNA in the reaction mixes should be increased. Preliminary calculations can be carried out to determine the amount of DNA required to give a detectable amount of protein in the complex bands, using the estimated binding constant and quenching factors and the known specific radioactivity of the protein, and assuming a minimum stoichiometry of 1 : 1. The exact DNA concentration must be known to determine its specific radioactivity, and can be measured sensitively by any of several fluorescent methods.

After the gel is run, it is not dried but exposed to autoradiographic film while still wet. The pattern of ^{32}P -labeled DNA distribution is used as a template for excising bands from the wet gel.⁵ The specific radioactivities of the protein and DNA are determined by excising empty regions of the gel and adding known chemical amounts of DNA or protein to each slice, and control samples are prepared to determine quenching and spillover correction factors.¹⁴ Gel slices are oxidized as in Section II,A, and counted in scintillation fluid. The results are converted from counts per million to moles via the specific radioactivity and counting efficiency of each component, after correction for background and spillover.

In addition to excising individual complex bands, it is useful to count each entire lane of the gel to account for all the applied radioactivity and determine its distribution. This is especially important for the protein-only lane: the position of free protein must be known before a valid stoichiometry can be determined for the protein-DNA complexes. If the free protein migrates too closely to the complexes, the gel conditions (pH, acrylamide concentration, run time) can be altered to achieve better separation. If the protein is in large excess over the DNA, then the distribution of free protein will be approximately the same in the protein-DNA lanes as in the protein-only lane. In this case, the radioactivity in the protein-only slices can be subtracted from that in the equivalent complex slices.

A stoichiometry determination by gel retardation can also be used to establish if the protein is fully active for DNA binding. This application takes advantage of the fact that only *active* protein will be present in the complex bands; this is a unique attribute of the gel method compared to other DNA-binding assays. When the DNA concentration is at least 10- to 20-fold above the K_d , the input amount of protein required to bind half the DNA should be approximately equal to half the DNA concentration times the stoichiometry in the complex band. If more protein is required to achieve half saturation, this is an indication of partially active protein.

E. Cooperativity

Preliminary indications of cooperativity can be obtained simply by comparing the observed binding curve with the theoretical binding curve calculated from the expression for the binding constant, using a model of independent and equivalent sites. The presence of cooperativity is most easily assessed from the breadth of the transition on a semilog plot: for noncooperative interactions, an increase of 1.81 log units in protein concentration is required to increase the bound fraction from 10 to 90%. Positive cooperativity reduces this span, while negative cooperativity increases it. The degree of cooperativity is determined by fitting the data to theoretical curves calculated for various degrees of cooperativity. However, the presence of multiple classes of independent binding sites with different affinities cannot easily be distinguished from negative cooperativity unless the affinities differ by about 100-fold, and it may be necessary to physically separate the binding regions to decide between these two models.

A more sophisticated treatment of the data takes advantage of the unique ability of the gel to resolve individually liganded states. Binding constants for each ligation step can be extracted by fitting the data to an appropriate binding expression.⁸ All complex bands as well as the free DNA band must be quantitated. High-quality data and discrete, well-resolved bands for each ligation state are required for this approach. Several reports have appeared illustrating its success in quantitative analysis of cooperativity, at least in simple cases.^{8,11}

IV. Identification of New Binding Proteins

The gel method is ideal for purifying new proteins that bind to a defined target DNA. Two approaches can be taken: the gel method can be used to assay classical biochemical purification steps, or the protein can be purified directly from the complex in the gel. The most important element for success in the latter approach is an appropriate concentration of DNA.

As discussed in Section III,D above, the factor that limits the amount of binding protein in a complex is the amount of DNA present. Thus, raising the DNA concentration will increase the amount of bound protein, as long as the protein itself is not limiting.

In crude cell extracts, detection of binding activity toward a specific target DNA is complicated by the presence of other binding proteins. It is possible to detect a specific binding protein against this background of nonspecific binding activities because the specific and nonspecific proteins are competing for binding to the target DNA, but the specific complex has higher affinity under most conditions. The best way to exploit this difference in practice is to first do a trial experiment, like the one described in Section II,B, using two labeled fragments of different sizes simultaneously, one specific and one nonspecific, to determine the concentration of extract required to shift the DNAs out of their free positions. Then do a second experiment in which the reaction mix contains the labeled DNAs plus the chosen concentration of extract, and aliquots receive increasing amounts of an unlabeled competitor DNA. At some level of input competitor DNA, the nonspecific fragment will dissociate while the specific fragment remains bound. It may be necessary to examine a vast range of competitor concentrations, first in very coarse and then in very fine increments, to find the right concentration. Mixed sequence DNA is preferable to homopolymers as a competitor.

The binding protein can be partially purified from the complex band by denaturing gel electrophoresis. The complex band is excised from the wet retardation gel using the ^{32}P -labeled DNA pattern as a template. The gel slice is soaked briefly in a minimal volume of loading solution appropriate for the denaturing gel system that will be used; the gel slice is then stuffed into a well of the denaturing gel in contact with the bottom of the well.⁵ An adjacent lane of the retardation gel should contain crude extract only. A slice should be excised from this protein-only lane at a position equivalent to that of the complex, and prepared for the denaturing gel in parallel with the complex band. A sample of the free labeled DNA should also be run alongside, preferably also in an excised gel slice. Localization of the protein can be by autoradiography for radioactively labeled proteins, silver staining in cases where enough protein is present, or by Western blotting if antibodies are available.⁵ If protein cannot be detected in an ordinary Laemmli-type¹⁸ gel, other gel systems can be tried (e.g., see Ref. 19 for proteins too small to be resolved by Laemmli gels).

¹⁸ U. K. Laemmli, *Nature (London)* **227**, 680 (1970).

¹⁹ A. L. Shapiro, E. Vinuela, and J. B. Maizel, *Biochem. Biophys. Res. Commun.* **28**, 815 (1967).

An innovation²⁰ has been to cross-link the protein to bromodeoxyuridine-containing DNA using UV irradiation before running the retardation gel, to enhance detection of the protein in the denaturing gel. The cross-linking step also works *in situ* in the retardation gel, whether agarose²¹ or acrylamide (D. W. Ballard, personal communication, 1991).

V. Troubleshooting Guide

Problem	Possible causes
All DNA in free position at all protein concentrations	Inactive protein Too little protein Very acidic protein Gel pH too high
All DNA in bound position at all protein concentrations	Too much protein
No DNA in any position at all protein concentrations	Nuclease activity Dissociation of complexes Samples entered gel too slowly Acrylamide concentration too low Gel run too long
Some DNA remains unbound at all protein concentrations	Too little protein Partially inactive DNA
Complexes do not enter gel	Very large protein, aggregation, or cooperativity Very basic protein Gel pH too low Acrylamide concentration too high
No protein detectable in complex band	Too little DNA Very small protein
DNA smeared or unevenly streaked	Gel run too hot Uneven gel polymerization Dissociation of complexes Acrylamide concentration too low Samples entered gel too slowly Gel run too long

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²⁰ C. Wu, S. Wilson, B. Walker, I. Dawid, T. Paisley, V. Zimarino, and H. Ueda, *Science* **238**, 1247 (1987).

²¹ D. W. Ballard, E. Böhnlein, J. W. Lowenthal, Y. Wano, B. R. Franza, and W. C. Greene, *Science* **241**, 1652 (1988).

[9] DNA Bending in Protein–DNA Complexes

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Introduction

The realization that bound proteins may greatly distort nucleic acids is not new to molecular biology. The nucleosome, the earliest protein–DNA complex to be structurally characterized,¹ remains the most dramatic example of protein-induced DNA deformation. In more recent years non-crystallographic techniques have revealed that induced DNA bending is a common but not universal characteristic of regulatory proteins. The functional purpose of DNA flexure is in a simplistic sense obvious: the bend stabilizes a particular tertiary structure of a larger complex between DNA and one or more proteins. However, this facile answer sidesteps the issue of why DNA should be required to curve in such complexes when they might equally well be organized with proteins arrayed linearly along the DNA, especially in view of the significant energetic cost of DNA bending. It seems plausible that a compact shape facilitates the structural stabilization and regulatory signals contributed by interactions at protein–protein interfaces in multiprotein–DNA complexes. As experimental understanding of intrinsic and induced DNA bending increases, we can expect more sophisticated explanations for the phenomenon to emerge. This chapter is directed at experimental approaches to the problem that make use of standard molecular biological techniques, enabling widespread application to the numerous protein–DNA interaction systems that are now under study.

We divide the primary methods of interest into two categories: *comparative electrophoresis* and *sequence perturbation*. In principle both are capable of specifying the locus and direction of bends induced in DNA by proteins. The first method relies on the empirical fact that the electrophoretic mobility of DNA molecules depends on their shape, as well as on their contour length.² Even when a protein is bound, DNA shape still affects the mobility of the complex in nondenaturing gel electrophoresis,² as long as the DNA molecule is severalfold larger than the segment that interacts directly with the protein. The basic strategy of comparative

¹ T. J. Richmond, T. J. Finch, B. Rushton, D. Rhodes, and A. Klug, *Nature (London)* **311**, 532 (1984).

² H.-M. Wu and D. M. Crothers, *Nature (London)* **308**, 509 (1984).

electrophoresis is to alter DNA shape by changing the position of a bend in the molecule,² or by changing the helical phasing between two bends,³ and to determine the relative electrophoretic mobility of the resulting constructs. Interpretation of experiments of this kind has a strongly empirical basis, requiring careful evaluation of alternative explanations as the technique is "bootstrapped" to new applications.

By sequence perturbation we mean systematic study of bending or binding strength of a DNA-protein complex as a function of the base sequence in a region where a DNA bend is suspected; this approach applies only outside the strongly conserved sequence that forms specific hydrogen bonds to the protein. The basis for interpretation of these experiments is provided by the rules developed for nucleosomes and *Escherichia coli* CAP protein, according to which A/T-rich sequences are preferred when DNA bends toward its minor groove, and G/C-rich segments are favored when the bend is toward the major groove.⁴⁻⁷ The degree of bending is most sensitive to sequence at the locus where one of the grooves faces the protein,⁶ corresponding to sites where roll between the base pairs causes the helix to bend. Hence, from the position of maximum sequence sensitivity, and the nature of the sequences that provide maximal bending, it is possible in principle with this relatively labor-intensive method to specify the bend locus and direction with high accuracy.

Comparative Electrophoresis

The three major properties that characterize a bend are (1) its locus, as described by the position of its center, (2) its direction, defined, for example, relative to the roll and tilt axes of a specific helix position such as the center of the bend, and (3) its magnitude, usually measured relative to a standard bend. Comparative electrophoresis is able to provide answers to all these questions, although different constructs are required for each of the three cases. However, the experiments share a common gel electrophoresis technology.

Methodology: Electrophoresis of Protein-DNA Complexes

To detect conformation-dependent changes in mobility, DNA molecules must migrate through the gel matrix by reptation, or snakelike motion, rather than by a sieving mechanism. The empirical fact is that bent

³ S. S. Zinkel and D. M. Crothers, *Nature (London)* **328**, 178 (1987).

⁴ H. R. Drew and A. A. Travers, *J. Mol. Biol.* **186**, 773 (1985).

⁵ S. C. Satchwell, H. R. Drew, and A. A. Travers, *J. Mol. Biol.* **191**, 659 (1986).

⁶ M. R. Gartenberg and D. M. Crothers, *Nature (London)* **333**, 824 (1988).

⁷ T. E. Shrader and D. M. Crothers, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 7418 (1989).

molecules migrate more slowly than linear ones in high-percentage acrylamide gels⁸; as the gel percentage is reduced, or if agarose gels are used, the anomaly in the mobility of the bent form lessens or disappears. This fact has been interpreted,² perhaps simplistically, as reflecting the predicted dependence of mobility on the mean square end-to-end distance for molecules of fixed contour length.^{9,10} It is likely that motion through the pores of the gel acts to straighten the molecule out, since the entire chain must follow the path chosen by its leading segment; the energy needed to straighten bent molecules may be an important factor contributing to their relatively slow motion.¹¹ On the other hand, the gel retains some local elasticity, allowing the molecule to move toward the shape that is favored in free solution, thereby reducing its end-to-end distance. This tendency to return to solution shape may account for the qualitative success of theories that predict a correlation between mean square end-to-end distance and electrophoretic mobility.^{9,10} However, the problem is a difficult one, and theory is not yet a quantitatively reliable guide to the electrophoretic properties of DNA molecules of complex shape.

The protein-DNA complexes we have examined are quite stable and insensitive to moderate changes in electrophoresis conditions. Complexes are preformed in binding buffers appropriate for each protein, and commonly electrophoresed in nondenaturing polyacrylamide gels, of acrylamide percentage between 5 and 10%; the optimum acrylamide-to-bis-acrylamide ratio has been found empirically¹² to be about 75 : 1. We prefer this ratio, used with the highest acrylamide percentage that gives adequate mobility to the complex, since we have found that these conditions yield optimally sharp bands for the complexes at a fixed migration distance.¹² This observation provides support for the view that high acrylamide concentration provides a "cage" around the complex that stabilizes it against dissociation during electrophoresis, although it is also possible that the phenomenon results from other factors, such as an influence of high gel percentage on the thermodynamic activity of water. (If bound water molecules are released from DNA or protein in forming the complex, then a lowered concentration of water in the gel will result in an increased binding affinity.)

The gels are generally 16 cm in length with a 1.5 mm by 14 cm cross-

⁸ J. C. Martini, S. D. Levene, D. M. Crothers, and P. T. Englund, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 7664 (1982).

⁹ L. S. Lerman and H. L. Frisch, *Biopolymers* **21**, 995 (1982).

¹⁰ O. J. Lumpkin and B. H. Zimm, *Biopolymers* **21**, 2315 (1982).

¹¹ S. Levene and B. H. Zimm, *Science* (in press).

¹² H.-N. Lui-Johnson, M. R. Gartenberg, and D. M. Crothers, *Cell (Cambridge, Mass.)* **47**, 995 (1986).

section. Loading dyes containing glycerol or sucrose may be added to binding reactions immediately before electrophoresis. To obtain sharp protein-DNA complex bands, the final concentration of glycerol should be kept to a minimum ($\leq 10\%$, v/v). We generally electrophorese the complexes at 420 V in $0.5 \times$ TBE buffer [45 mM Tris, 45 mM boric acid, 1 mM ethylenediaminetetraacetic acid (EDTA), pH 8.3], maintaining a temperature of 20–25° in the gel with a constant temperature gel apparatus (Hoefer, San Francisco, CA) and a circulating refrigerated water bath (Neslab, Portsmouth, NH). Electrophoresis at high voltage and low ionic strength also contributes to band sharpening, since those factors act to minimize complex dissociation during electrophoresis. Modifications to the electrophoresis buffer are tolerated and often necessary. EDTA may be eliminated if divalent cations such as Zn(II) are essential for protein binding. Buffers with alternate pH may be used, for example if the protein is small and has an acidic isoelectric point, causing it to confer little or no retardation on the mobility of the DNA to which it is attached.¹³ DNAs are generally detected by ³²P autoradiography, β scanning, or ethidium fluorescence.

Mapping Bend Locus by Circular Permutation of Sequence

A DNA bend reduces the end-to-end distance of a DNA fragment and therefore its electrophoretic mobility if it is located near the center of the molecule; a bend near a molecular end influences neither appreciably (see Fig. 1). This simple determinant of gross molecular shape is the basis of the circular permutation assay used to map the locus of DNA bending.² A tandem dimer or multimer of a binding site fragment is cloned and cleaved with restriction enzymes that cut only once within the sequence, resulting in a set of circularly permuted DNA molecules that differ in the position of the binding site relative to the molecular ends. Figure 2 shows the variation in gel mobility for two series of circularly permuted *lac* promoters, one wild type (lanes 1–5) and one mutated (lanes 6–10), bound by *E. coli* CAP protein.¹² The slowest moving species (lanes 1 and 6) position the protein-binding site near the molecular center.

CAP-induced DNA bending and not the influence of bound protein is thought to be responsible for the observed position-dependent variation in gel mobility; complexes with *lac* repressor, another globular protein, migrate equivalently regardless of binding site position.² However, DNA-binding proteins with extended and negatively charged domains, as pro-

¹³ J. Carey, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 975 (1988).

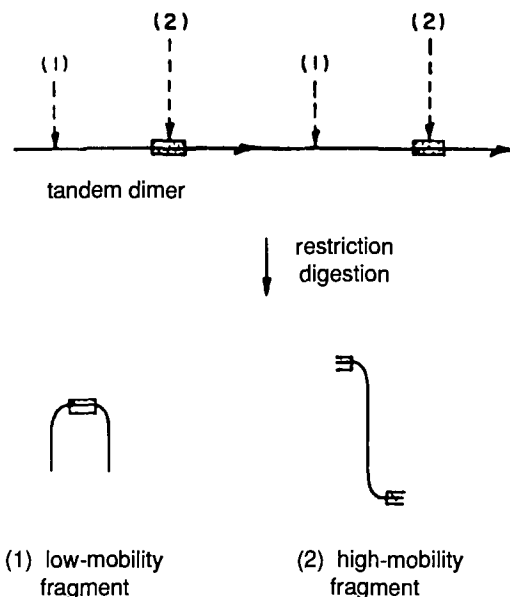


FIG. 1. Logic of the circular permutation experiment for determination of bend position.² A cloned tandem dimer or higher order repeat is cleaved with a series of restriction enzymes that cut once in the sequence of interest. Fragment (1), having a bend in the center, is highly bent and therefore of low gel mobility. Fragment (2) is nearly linear since the bend is near the end, and hence nearly normal in electrophoretic mobility.

posed for GCN4, may modulate electrophoretic mobility of protein–DNA complexes in a position-dependent manner without inducing DNA bends.¹⁴

For identification of the bend center one can use a plot such as that shown in Fig. 3, which is based on the data in Fig. 2. The mobilities of the complexes with circularly permuted DNAs are plotted against position of the DNA molecular end in the numbering system of the parent fragment. Distance moved down the gel reaches a maximum (corresponding to the minimum in the plot of Fig. 3) when the binding site, and hence the bend, is near the molecular end. Extrapolation of the lines on both sides of the minimum to their intersection enables identification of the bend center, often to within a few base pairs. If the DNA molecule has a more complex shape than provided by a single symmetric bend, the curve in Fig. 3 can be asymmetric, and the minimum need not correspond to the center of the local bend. For example, the minimum in Fig. 3 is displaced about half a

¹⁴ M. R. Gartenberg, C. Ampe, T. R. Steitz, and D. M. Crothers, *Proc. Natl. Acad. Sci. U.S.A.* in press (1990).

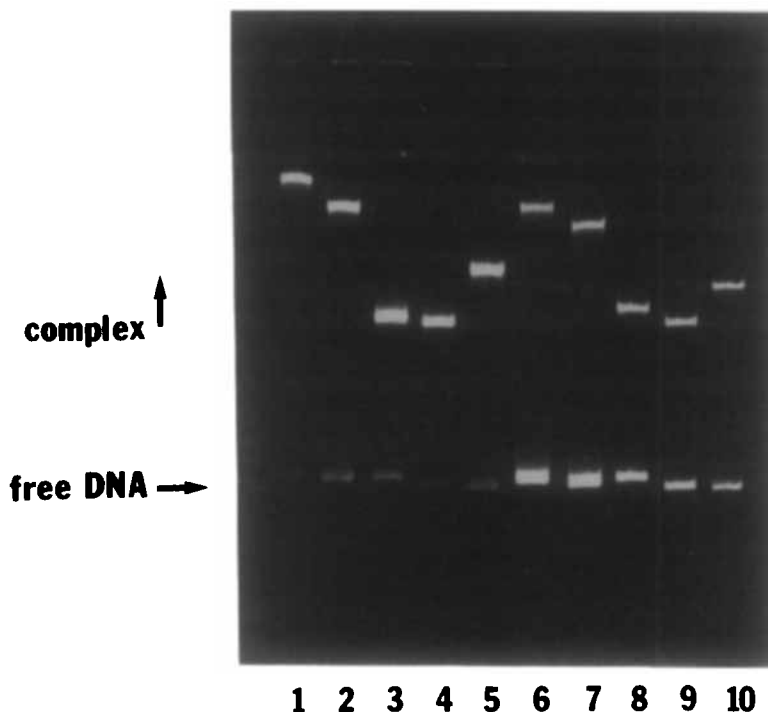


FIG. 2. Illustration of the circular permutation mobility assay.¹² Circularly permuted 203-bp *lac* promoter fragments complexed with CAP protein are shown in lanes 1–5, and a similar series is shown in lanes 6–10 for a mutated sequence (sy203). Electrophoresis is on a 10% polyacrylamide gel, 75 : 1 (w/w) acrylamide : bisacrylamide in TBE buffer.

helical turn downstream in the *lac* promoter from the center of the CAP binding site, an effect thought to be due to additional in-phase curvature in the promoter.¹²

The mobility of CAP protein complexed with the mutant DNA sequence in Figs. 2 and 3 is larger than that of the wild-type complex when the bend is near the center of the molecule, implying a smaller bend with the mutant sequence. In contrast, when the bend is near the end of the molecule, the mutant sequence provides a complex with lower gel mobility, an effect that was interpreted as a reflection of a bend in excess of 90°.¹²

Assigning DNA Bend Direction by Phase-Sensitive Detection

Static DNA bends are characterized by their direction relative to distal features, such as other DNA bends or bound proteins, as well as local helical parameters, including the major and minor grooves within the

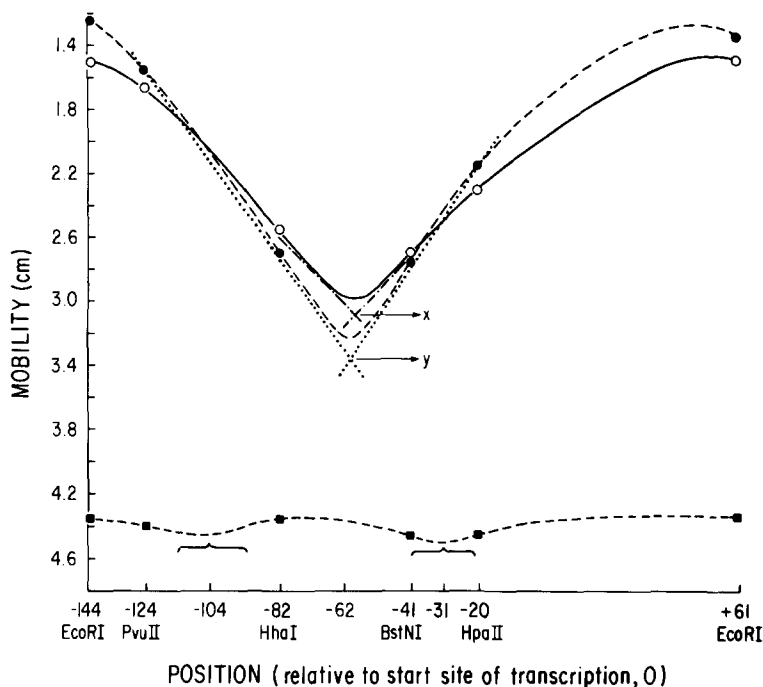


FIG. 3. Mapping the bending locus.¹² Filled circles show the mobility of the CAP-wt203 complexes (Fig. 2); open circles indicate the CAP-sy203 complexes. Filled squares show the mobility of the naked DNAs, which run identically. The center of sequence symmetry is between -61 and -62 . The bend center in the sy203 fragment, labeled x , is at about -58 , and the estimated center of the wt203 bend (y) is at about -60 . The regions of the naked DNA where a bend is suspected are indicated by braces.

binding site. Most protein-induced DNA bends examined are directed toward the protein; however, the structure of the DNase I-DNA cocrystal demonstrates that protein-induced bends need not be oriented in this fashion.¹⁵ Structural variability extends further: at the center of some DNA-binding sites, such as those for $\gamma\delta$ resolvase¹⁶ and the nucleosome,¹ the major groove faces the protein, while at others, such as those for CAP¹⁷ and typical dimeric prokaryotic repressors,¹⁸⁻²⁰ the minor groove faces the protein.

¹⁵ D. Suck, A. Lahm, and C. Oefner, *Nature (London)* **332**, 464 (1988).

¹⁶ G. F. Hatfull, S. Z. Noble, and N. D. F. Grindley, *Cell (Cambridge, Mass.)* **49**, 103 (1987).

¹⁷ J. Waricker, B. P. Engelman, and T. A. Steitz, *Proteins* **2**, 283 (1988).

¹⁸ C. Wohlberger, Y. Dong, M. Ptashne, and S. C. Harrison, *Nature (London)* **335**, 789 (1988).

¹⁹ S. R. Jordan and C. O. Pabo, *Science* **242**, 893 (1988).

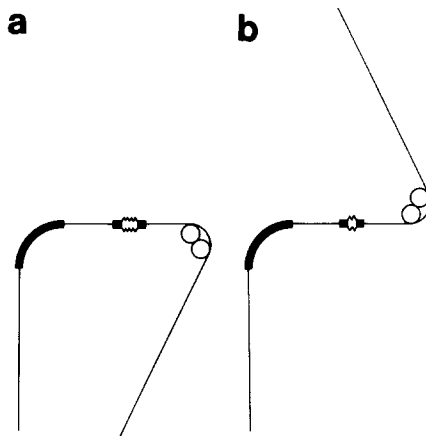


FIG. 4. Logic of the phase-sensitive experiment for determination of bend direction.³ Shown are (a) cis and (b) trans isomers of DNA constructs containing both A tract (heavy line) and CAP-induced bends. Phasing between the two bends is controlled by the length of the variable linker region in the center of the molecule.

The relative directions of two DNA bends located in the same fragment may be determined from the electrophoretic mobilities of isomers that differ by the length of the intervening DNA; as the orientation of the bends varies, the mobility of the protein-DNA complexes should increase from a minimum when the end-to-end distance is short to a maximum when the end-to-end distance is long.^{3,21} Figure 4 provides a schematic representation of the cis and trans isomeric structures on which this assay depends. In this example, a fragment containing a CAP binding site is fused to a fragment containing phased adenine (A) tracts. The spacing between the two loci is varied over a full turn of the double helix by the insertion of linkers of variable length. Adenine tracts provide an excellent standard because the direction²² and magnitude²³ of the A tract bend have been determined. Exogenous bends are not required if only the relative orientation of multiple endogenous bends is of interest.^{24,25}

²⁰ A. K. Aggarwal, D. W. Rodgers, M. Drott, M. Ptashne, and S. C. Harrison, *Science* **242**, 899 (1988).

²¹ J. J. Salvo and N. D. F. Grindley, *Nucleic Acids Res.* **15**, 9771 (1987).

²² H.-S. Koo and D. M. Crothers, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 1763 (1988).

²³ H.-S. Koo, J. Drak, J. A. Rice, and D. M. Crothers, *Biochemistry* **29**, 4227 (1990).

²⁴ J. J. Salvo and N. D. F. Grindley, *EMBO J.* **7**, 3609 (1988).

²⁵ U. K. Snyder, J. F. Thompson, and A. Landy, *Nature (London)* **341**, 255 (1989).

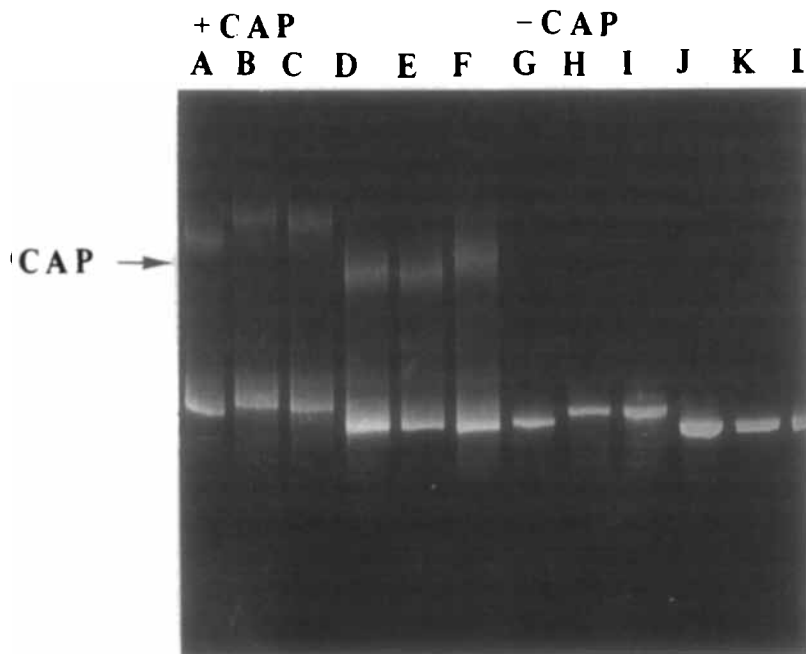


FIG. 5. Illustration of mobility shifts for the binding of CAP to the variable-linker constructs.³ Lanes A-F, CAP-DNA complexes in which the distance between the centers of the CAP and A tract bends varies from 104 to 94 bp. Electrophoresis was in a 5% polyacrylamide gel, 39:1 acrylamide : bisacrylamide, in TBE buffer.

As shown in Figs. 5 and 6, the gel mobilities of CAP-bound spacer constructs vary sinusoidally with a period of approximately 10 bp in their dependence on linker length. The CAP-bound construct with an 18-bp insert (lane B, Fig. 5) migrates slowest, and is thereby identified as the *cis* isomer, in which the molecule is nearly planar and the two bends are in essentially the same direction (see Fig. 4). Determination of bend direction in the local coordinate frame of the nucleotides at the bend center requires that one know the number of helical turns between bend centers.

This experiment was originally designed to remove a residual ambiguity concerning the direction of curvature of A tracts, namely whether the bend is directed toward the major or the minor groove at the A tract center.³ Given that DNA bends toward the *minor* groove at the center of the CAP binding site, only integral or half-integral numbers of helical turns need be considered between the positions of the CAP and A tract bends in the *cis* isomer. The centers are separated by 101.5 bp; with a helical repeat of 10.5 to 10.7 bp, the only acceptable answer is 9.5 helical turns.

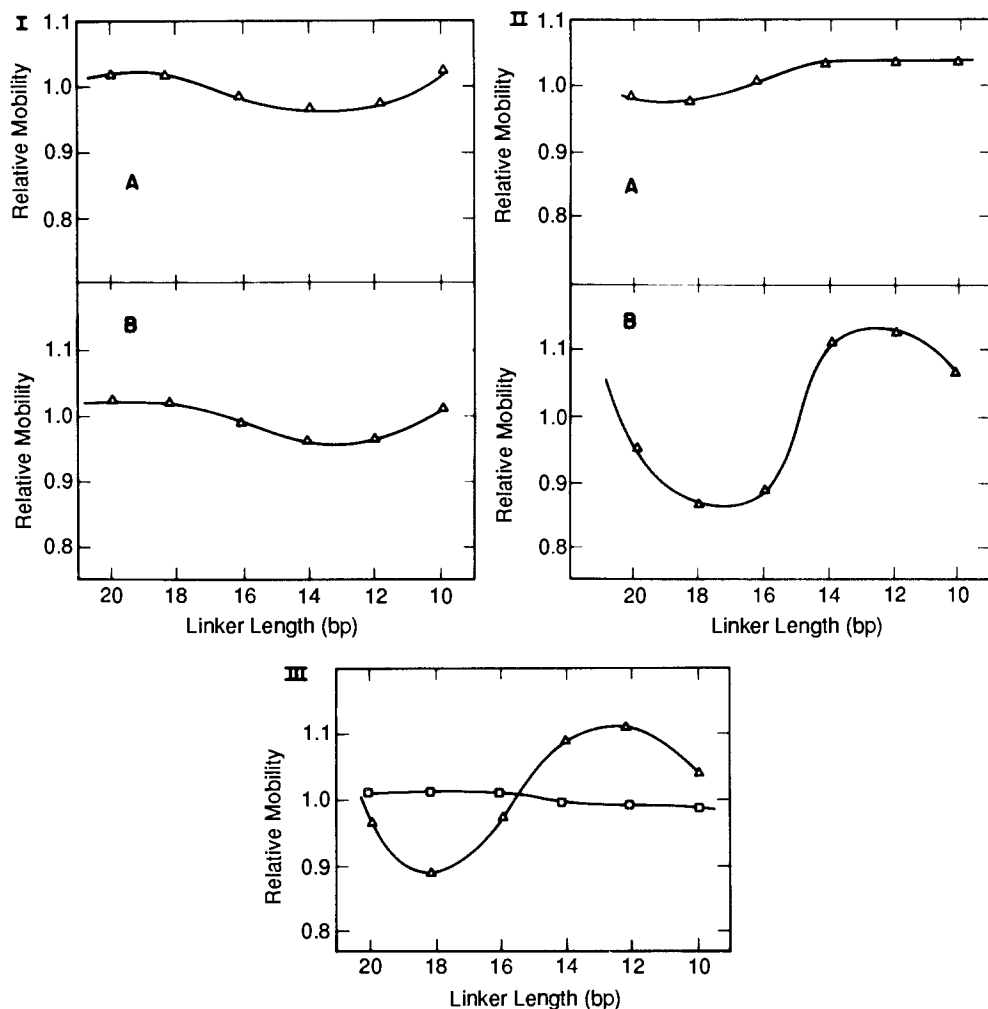


FIG. 6. Quantitative analysis of the phase dependence of the electrophoretic mobility.³ (I) and (II) show relative mobility, normalized to the average for the set of values, for the *lac* repressor (I) and CAP protein (II) DNA complexes. In (I) and (II) the top graph (A) is for the free DNA, and the lower graph (B) for the protein-DNA complex. Note that for *lac* repressor, unlike the CAP-DNA complex, the fractional variation in gel mobility is very similar for free DNA and the complex. This is the behavior expected when a DNA fragment has some intrinsic curvature, which is unaltered by protein binding. (III) shows the ratio of the normalized mobilities for the protein-DNA complex, divided by the corresponding normalized free DNA mobility. Squares show values for CAP-DNA, and triangles are for repressor-DNA complexes. Note the strong periodic variation of relative mobility for the CAP-DNA complex normalized values. This is the behavior expected when a protein-induced bend is superimposed on intrinsic fragment curvature. From the mobility minimum, the construct with linker length 18 is the *cis* isomer. The lack of significant variation of normalized mobility in (III) confirms the lack of DNA bending by *lac* repressor.

Hence, with a half-integral number of turns, the coordinate frame for the A tract bend is half a turn displaced from that for the CAP bend, and the overall center of the A tract bend is directed toward the *major* groove at its center. (Individual A tracts bend toward the minor groove at their centers, but the center of the collective bend produced by four A tracts is toward the major groove in the central 5-bp segment that lies between A tracts.) As a result of this and related experiments, the A tract bend can now be used as a comparative standard in experiments designed to determine the direction of bending induced by proteins whose structure is less well understood than CAP. Almost no variation in mobility is observed for the same constructs when bound by *lac* repressor,³ as expected for a nonbending protein.

The circular permutation and phase-sensitive detection assays are complementary in that the first maps bend loci irrespective of bend direction while the second assigns bend direction but provides insufficient information to map bend loci precisely.

Determination of Bend Magnitude by Comparison to Calibrated Standards

The magnitude of the protein-induced bends in a DNA fragment can be estimated by comparison of the electrophoretic mobility of the fragment with those of molecules containing well-characterized bends.²⁶ Specifically, a calibration curve is generated from the mobilities of DNA constructs in which the protein-binding site has been replaced by DNA sequences containing from three to nine adenine tracts, repeated with a period of 10.5 bp (Fig. 7). These "in-phase" bends add to produce molecules with increasing curvature, known to correspond to a deflection of the helix axis of about $18^\circ/\text{CA}_6\text{C}$ repeat.²³ To ensure that the differences in mobility are caused predominantly by the shape of the DNA molecule, the total lengths of the DNA fragments are held nearly constant and a protein-binding site is included at the end of the calibration fragments (Fig. 7).

In Fig. 8, a typical calibration curve shows a plot of relative mobility versus the square of the number of resident A tracts. For molecules with less than six A tracts, the data are adequately fit by a second order polynomial. The mobility of the fragment with the centrally bound CAP is also shown. The interpolated bend magnitude is roughly 5.6 A tract equivalents or 101° , a value that is nearly invariant to changes in acrylamide concentration.²⁶

²⁶ S. S. Zinkel and D. M. Crothers, *Biopolymers* **29**, 29 (1990).

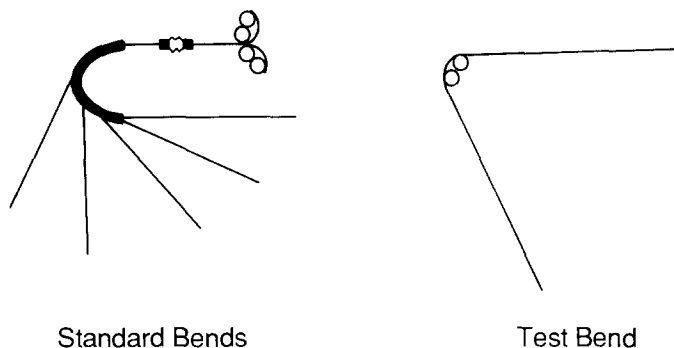


FIG. 7. Logic of the experiment to determine the magnitude of the CAP-induced bend by comparison with A tract bends.²⁶ The standard bends consist of a set of DNA molecules containing runs of from three to nine A tracts phased at 10.5-bp intervals, located in the center of the molecule; the A tract bend in these sequences is known to be $18^\circ/\text{A tract}$.²³ A CAP-binding site at the end of the standard fragments corrects for gel retardation due to bound CAP protein, which is shown both cis and trans to the A tract bends; experiments showed that the relative orientation has little effect when the CAP bend is at the end of the molecule.²⁶ The magnitude of the CAP-induced bend is taken to be equal to that provided by A tracts in the standard-bend DNA molecule, the mobility of which (with CAP bound) matches the mobility of the test bend molecule with CAP bound.

An alternative approach²⁷ compares the ratio of the mobilities of molecules that have a bend of unknown magnitude in the center or at the end to the same ratio for molecules with standard bends.

Sequence Perturbation

Experimental evidence continues to accumulate showing that DNA flexibility and bending anisotropy play a role in protein recognition. DNA sequences isolated from chicken erythrocyte nucleosomal core particles are probably selected for favorable DNA binding and bending, and show a corresponding sequence signature in their nonrandom distribution of A · T- and G · C-rich sequences.^{4,5} The observations include an increased probability for G · C-rich regions to occupy positions where the major groove is compressed as the DNA curves around the protein, whereas A · T-rich regions are favored in regions subjected to minor groove compression. These sequences preferentially occupy similar orientations in small, highly strained DNA circles,⁴ suggesting that DNA flexibility may be a driving force in nucleosome placement. Similar rules have been deduced from the properties of mutants of the DNA-binding site for

²⁷ J. F. Thompson and A. Landy, *Nucleic Acids Res.* **16**, 9687 (1988).

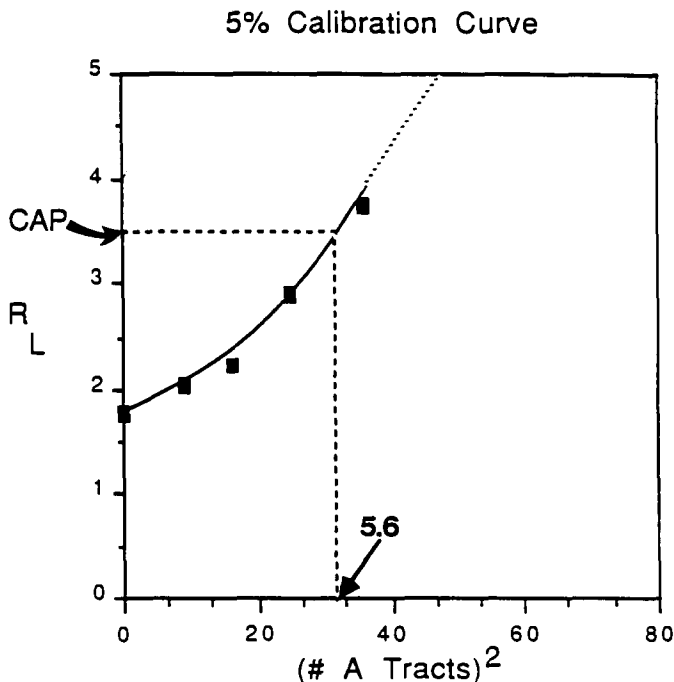


FIG. 8. Calibration curve for measurement of the magnitude of the CAP-induced bend.²⁶ The R_L value is the apparent number of base pairs in the CAP–DNA complex, determined by comparison with a ladder of normal DNA fragments, divided by the actual number of base pairs in the test fragment. Deviation of R_L from 1 when the number of A tracts is 0 reflects the gel retardation due to CAP protein bound at the end of the corresponding standard bend fragment. The plot of R_L versus the square of the number of A tracts was fitted to a quadratic polynomial for the purpose of interpolation to find the number of A tracts (5.6) that corresponds to the R_L value (3.49) measured for CAP bound to the center of the test bend fragment. Electrophoresis was on a 5% polyacrylamide gel (acrylamide : bisacrylamide, 39 : 1).

E. coli CAP protein⁶; this system and the sequence preferences of nucleosomes are discussed more extensively below.

Another example is provided by the phage 434 repressor system, for which the protein-binding constant is affected by mutations in the central 4 bp of the binding site, shown by diffraction studies not to contact the protein.²⁸ These effects correlate with the stiffness of the central 4-bp segment, suggesting that the ability of this segment to bend and twist affects the binding constant by providing optimal protein–DNA contacts

²⁸ G. B. Koudelka, S. C. Harrison, and M. Ptashne, *Nature (London)* **326**, 886 (1987).

in more distal regions of the binding site.²⁹ However, effects due to bending may be of secondary importance relative to twisting in this case, since the mutated segment may have a critical effect on the helical phasing between the two DNA segments whose interactions dominate specific binding. We have reduced the importance of this factor in the case of CAP protein by focusing on bends that lie distal to the consensus recognition sequence, as measured from the central axis of the binding site. In the case of nucleosomes, it was possible to design sequences to test directly the importance of helical twist in determining relative binding constants.³⁰

Our reported studies on CAP and nucleosomes were designed primarily for the purpose of elucidating the sequence dependence of DNA bending. In retrospect, it is clear that the results further refine our understanding of the nature and extent of the bent regions in the protein-DNA complexes. For example, the sites one helical turn and 1.5 helical turns from the dyad axis of the CAP-DNA complex showed maximal sensitivity in the dependence of bending on sequence.⁶ However, no sequence-dependent modulation of bending could be detected at the locus two full turns removed (except when a permanent DNA bend resulting from an A tract was inserted). These results indicate that DNA is bent primarily by alternately compressing its major and minor grooves as it wraps around the protein, and that the bent region covers 1.5 helical turns on each side of the bending locus. In the case of nucleosomes, we were able to show that sequences that favor bending have little preference for any special site on the surface of the histone core, implying relatively uniform bending.³⁰ We expect that future studies of other complexes by these methods will be similarly informative about the position and direction of induced bends.

Escherichia coli CAP Protein

The CAP binding site consists of a consensus binding domain that interacts with the helix-turn-helix recognition motif of the protein and a distal binding domain that flanks the consensus and presumably interacts nonspecifically with a ramp of positive electrostatic potential that is present on the protein surface.^{6,12,17,31} To examine the influence of DNA sequence on induced bending and binding, we generated an extensive library of CAP binding site mutants. We deliberately restricted the size of randomized regions in each mutant, so that the sequences examined can be regarded as local perturbations on the wild-type sequence. The extent

²⁹ M. E. Hogan and R. H. Austin, *Nature (London)* **329**, 263 (1987).

³⁰ T. E. Shrader and D. M. Crothers, *J. Mol. Biol.* in press.

³¹ T. A. Steitz, *Q. Rev. Biophys.* **23**, 205 (1990).

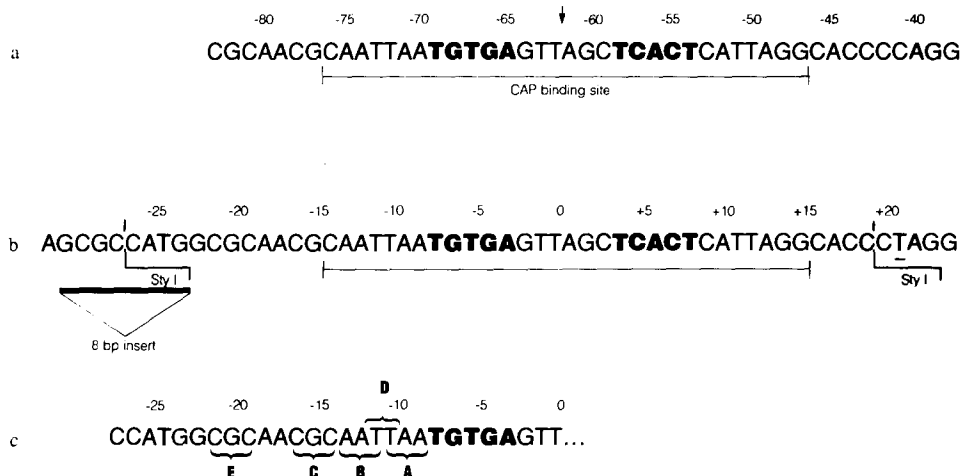


FIG. 9. Constructs for sequence-perturbation studies of CAP-induced bending.⁶ (a) The sequence and features of the wild-type CAP-binding site in the *lac* promoter. The underlined region denotes the thermodynamically defined binding domain,¹² boldface type indicates nucleotides in the consensus sequence region, and an arrow identifies the pseudodyad axis of symmetry. Positions are numbered relative to the transcription start site. (b) The sequence changes introduced to create *StyI* restriction sites; dinucleotide steps are numbered relative to the pseudodyad axis. A 211-bp *lac* promoter fragment containing this sequence was cloned into the *EcoRI* site of pGEM-2 (Promega). (c) The location of sites fully randomized by oligonucleotide replacement. Each site was mutagenized independently of the others.

of bending and stability of complexes of the mutant binding sites were compared to those of the wild type by nondenaturing gel electrophoresis.

Methods. Engineering of unique *StyI* restriction sites flanking the CAP binding site in the *lac* promoter permitted derivatization of the natural sequence by replacement with synthetic oligonucleotides (Fig. 9a). Sequence changes were confined to the distal binding domains to avoid disruption of critical recognition contacts. The oligonucleotides contained regions of randomized sequence two or three consecutive base pairs in length, produced by degenerate synthesis. Each of the five sites A–E (Fig. 9b) were mutagenized independently of one another. Individual binding site clones were resolved from the pool of degenerate constructs by bacterial transformation and identified by sequencing of supercoiled miniprep DNA of individual colonies.³²

Mutant binding sites, centrally located within 211-bp *lac* promoter fragments, were analyzed both for extent of induced DNA bending, taken

³² Promega Biotech Technical Manual (1986).

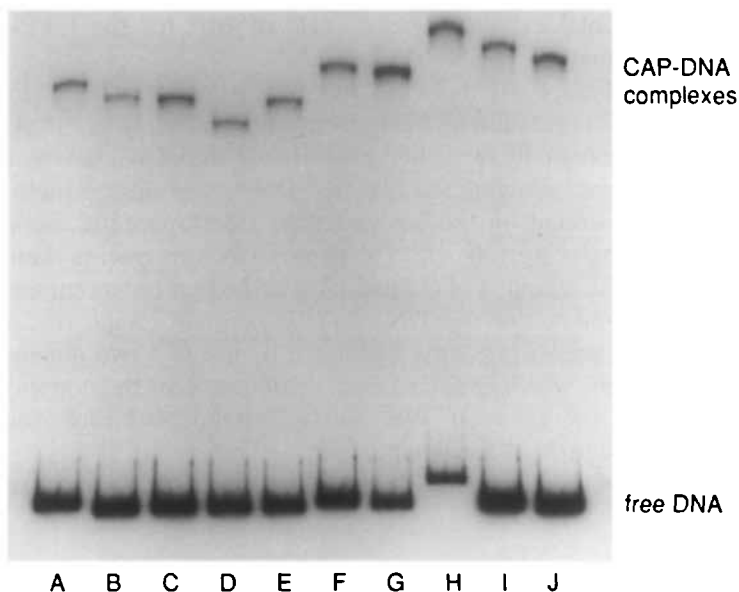


FIG. 10. Electrophoresis gel showing the variability of the mobility of CAP protein bound to DNA-binding site mutants.⁶ Electrophoresis was on a 10% polyacrylamide gel, 75:1 acrylamide:bisacrylamide. The sample in lane H has an A tract bend centered at position -22, in phase with the CAP-induced bend, lane E contains an out-of-phase A tract bend centered at position -27, and lane F contains the wild-type sequence. The other lanes illustrate the electrophoretic properties of some of the mutants A-D in Fig. 9.

to be a simple function of the electrophoretic mobility anomaly, and relative CAP binding affinity. In order to calibrate the dependence of mobility on extent of bending, constructs were made in which A tracts were placed in phase (A tract center two helical turns from the central dyad axis) and out of phase (A tract 2.5 turns from the dyad). These constructs had mobilities about 20% smaller and larger, respectively, than the wild-type CAP complex. Figure 10 illustrates this analysis: lanes E, G, and H contain the wild-type binding sequences except that in E an out-of-phase A tract has been added in the region flanking the binding site, and in lane H the added A tract is in phase with the CAP bend; the fragment in lane G has no added A tract. (The lengths of the fragments in E, G, and H were held constant within 1 bp.) Since an A tract bends the helix by about 18° , we could, by extrapolating the nearly linear relationship between the three calibration points, assign a change in the CAP-induced bend angle to each of the mutants. The variation ranged from increased bending of about 10° to decreased bending by about 30° , the latter representing a substantial

fraction of the total estimated bend angle of 101° for the CAP-DNA complex in solution.

Relative binding constants (K_{rel}) can be measured with high accuracy using a competition method and gel electrophoresis analysis.^{33,34} A comparison sequence (usually the wild type) is incorporated in a shorter (<211 bp) DNA fragment, allowing the two free DNAs and their protein complexes to be separated on the basis of their electrophoretic mobilities. Standard nondenaturing gels with ^{32}P -labeled DNA fragments then yield lanes with variable amounts of radioactivity in the four bands corresponding to wild-type and mutant free DNA and their protein complexes. Analysis of such experiments is greatly facilitated by use of a two-dimensional β scanning device, which enables direct quantitation of the counts (β) in each band from the gel scan. We used a Betscope 603 Blot Analyzer (Betagen, Waltham, MA) for this purpose.

Calculation of K_{rel} from the count values β for each band uses the equation

$$K_{\text{rel}} = K_{\text{mut}}/K_{\text{wt}} \\ = \beta_{\text{mut}}(\text{complex}) \beta_{\text{wt}}(\text{free DNA}) / \beta_{\text{mut}}(\text{free DNA}) \beta_{\text{wt}}(\text{complex})$$

where mut and wt refer to mutant and wild type, respectively. From the K_{rel} values, the difference in standard free energy ($\delta\Delta G^\circ$) of binding between mutant and wild-type DNAs can be calculated from the thermodynamic equation $\delta\Delta G^\circ = -RT \ln K_{\text{rel}}$.

Results. The data resulting from measurements of mobility changes and relative binding constants for a series of mutants provide a quantitative view of the influence of DNA sequence on these parameters. Figure 10 shows a polyacrylamide gel analysis of the mobility of a number of DNA binding site mutants complexed with DNA fragments mutated at the CAP binding site. We have described elsewhere how the results can be analyzed mathematically to rank particular dinucleotides for their contribution to bending and binding.^{6,35}

A simpler, nonmathematical way to present the data is presented in Fig. 11, which shows the sequence of the dinucleotide changes at each position that yield the largest perturbations from the mobility of the wild-type complex. From the extremes, which are evident at positions one and 1.5 turns from the dyad axis, we deduce that bending is most sensitive to mutations at those positions, where the minor and major grooves turn successively to face the protein. Note that the mobility is highest (least

³³ M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

³⁴ M. M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).

³⁵ M. R. Gartenberg, Ph.D. Thesis, Yale University, New Haven, Connecticut (1990).

bending) when G · C-rich dinucleotides are found at positions 10 and 11 nucleotides from the dyad, whereas mobility is lowest (greatest bending) when A · T-rich dinucleotides are found there. At sites 15 and 16, the rules are inverted, corresponding to the opposite direction of bending the DNA, toward the major groove in this case. (Although the diagram in Fig. 10 features only the dinucleotides at the extremes, we found that the properties of other A · T-rich and G · C-rich dinucleotides cluster near the values shown at the extremes.) Hence the data define the bending site and direction to base pair accuracy, and show that it does not extend to the locus two turns from the dyad axis.

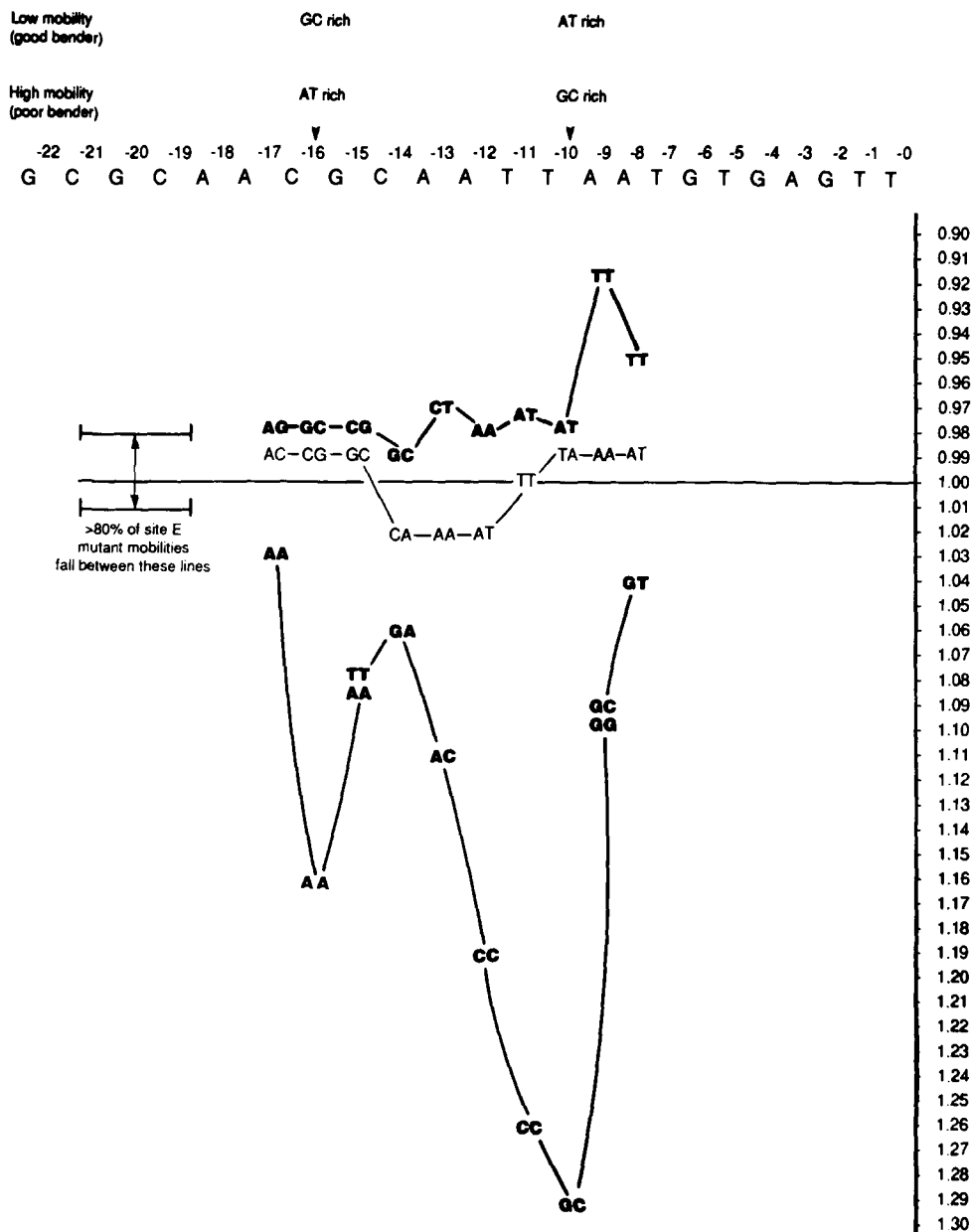
Measurement of the relative binding constant for 56 DNA-binding sites mutated at sites A and D in Fig. 9 shows that there is substantial although imperfect correlation between binding strength and magnitude of bending.^{35a} In general, A · T-rich dinucleotides at positions 10 and 11 from the dyad yield stronger binding, by as much as a factor of 50, than do G · C-rich sequences.

Nucleosomes

Nucleosomal core particles contain about 146 bp, or 14 turns of the DNA duplex, in close association with a protein octamer containing two each of the core histones H2A, H2B, H3, and H4. The DNA is wrapped around the outside of the octamer in about 1.75 superhelical turns, and bends toward the protein primarily by alternately compressing the major and minor grooves. As a result, the structure contains about 14×2 or 28 bending loci, which are phased at the helical repeat of DNA in the nucleosome. By use of synthetic DNA molecules one can investigate the relative preference when bulk DNA is replaced by special sequences having G · C- and A · T-rich segments that alternate every ~ 5 bp; we have called such sequences "anisotropically flexible" DNA because they are designed to conform to the rules for preferred bending alternately toward the major and minor grooves.⁴⁻⁷

The flexible sequences can be placed in various positions relative to the dyad axis of the core particle to determine whether the energetic demands on anisotropic bending are more pronounced in some regions than others. In addition, the sequence repeat can be varied to investigate the helical repeat of the nucleosome; one expects optimal incorporation of DNA molecules in which the anisotropically flexible regions are phased in exact compliance with the preferred helical repeat of DNA on the nucleosome. We describe here experiments illustrating the general

^{35a} —, Dalma-Weiszhausz, M. R. Gartenberg, and D. M. Crothers, to be published.



method, which involves competitive reconstitution of different DNA sequences into nucleosomal core particles, using a labeled DNA fragment to compete against a standard unlabeled sample. The extent of incorporation is determined from the percentage of a labeled DNA fragment that is converted to nucleosomal complex, which is separated from free DNA by electrophoresis on a polyacrylamide gel.

Methods

The artificial nucleosome positioning sequences used in this study are constructed by cloning multiple copies of synthesized oligonucleotides into a modified pGEM2 vector (Promega Biotech, Madison, WI). A unique, nonsymmetric *Ava*I restriction site is introduced into the original vector; *Ava*I cleavage of the site produces four-nucleotide, 5'-overhanging ends³⁶ that are complementary to each other but not self-complementary. Synthetic duplex oligonucleotides must have corresponding unique ends, each designed to hybridize with only one end of the cleaved vector, thus ensuring that all copies of the monomer assemble in a head-to-tail orientation rather than in a random manner.

In addition to forcing direct repeats on ligation, the oligonucleotides are constructed with a complete *Ava*I recognition site at only one end; incomplete ends contain the proper four-nucleotide overhang but not the terminal recognition base pair. Direct repeat of these sequences produces a single *Ava*I site at one end of the multimer. As a result of recreation of a single endonuclease site, different types of oligonucleotides may be combined in a defined order by successive cloning steps.

Oligonucleotides were synthesized on an Applied Biosystems (Foster City, CA) DNA synthesizer and purified by gel electrophoresis on 12% polyacrylamide gels in TBE buffer (45 mM Tris, 45 mM boric acid, 1 mM EDTA, pH 8.3) containing 50% (w/v) urea. This procedure removes most

³⁶ J. L. Hartley and T. J. Gregori, *Gene* **13**, 347 (1981).

FIG. 11. Sequence dependence of the mobility of CAP-DNA complexes. The dinucleotide estimated to confer the largest (bottom of the graph) and smallest (top of the graph) mobility on the CAP-DNA complex. The high-mobility sequences show extremes at positions -10, -11, and again at -16, sites where the DNA minor and major grooves, respectively, face the protein. This observation implies maximum sensitivity of bending to sequence at the positions where the DNA bends alternately by roll into the minor and major grooves. Note that at positions -10, -11, the worst benders (greatest mobility) are the G/C-rich dinucleotides GC and CC, whereas the best benders are AT and AT. At the -16 locus AA is the worst bender and GC is the best. The wild-type sequence is shown near the middle line in the figure.

of the longer and shorter oligonucleotides that represent ~10% of a crude DNA synthesis. Approximately 10 OD₂₆₀ units of oligonucleotide is loaded per gel lane. The correct oligonucleotide band is identified by its UV shadow, excised from the gel, and isolated by soaking the crushed gel slice 8 hr at 65°. Acrylamide gel fragments are removed by two rounds of spinning the sample in an Eppendorf microfuge to pellet insoluble material. No further purification is required to anneal the single-stranded oligonucleotides and ligate the resulting duplexes into multimers.

Ten micrograms of the purified single strands is phosphorylated using 2–5 U of T4 polynucleotide kinase in T4 ligase buffer (25 mM Tris, pH 7.8, 10 mM MgCl₂, 4 mM 2-mercaptoethanol, 0.4 mM ATP). The strands are then mixed and annealed by slow cooling from 90 to 4°. Approximately 10 µg of the double-stranded oligonucleotide is polymerized by ligation for 15 min at room temperature (18–30°) and the ligation mixture is separated on a 6% polyacrylamide TBE gel. The correct multimer bands are visualized by staining in a 1 µg/ml ethidium bromide solution (exposure to ethidium bromide is kept as short as possible, ~5 min) and purified as above.

Multimers are cloned into the dephosphorylated asymmetric *Ava*I site of the pGEM2-*Ava* vector. One-half of the DNA isolated from each multimer band is used in the cloning step. Clones are screened first by restriction mapping and finally by dideoxy sequencing of DNA samples purified by the alkaline lysis method.³² From the DNA isolated from 3 ml of an overnight growth of DH1 *E. coli* cells, 10% is required to restriction map and 40–50% is required for dideoxy sequencing. The sequences of the clones revealed that errors in the body of the multimer are extremely rare, while errors at the junction between the oligonucleotide multimer and the phosphatased vector occurred in 5–10% of the clones.

For the clones to be used for nucleosome reconstitution, DNA from 0.5 liter of cells grown 12–24 hr is isolated by an expanded alkaline lysis procedure (technical manual, Promega Biotech). The average yield from this procedure was about 1 mg plasmid DNA/liter of cells.

To purify a fragment for reconstitution, several different pairs of restriction enzymes are used. These enzymes remove the oligonucleotide-containing section surrounded by enough flanking vector DNA to create a fragment of the overall correct size. The *Eco*RI and *Hind*III sites are separated by 62 bp in pGEM2/*Ava*. Excision of a fragment constructed of five repeats of a 20-bp oligonucleotide with these enzymes would produce a 162-bp fragment, suitable for mononucleosome reconstitution, which could be labeled at both ends by the Klenow fragment of DNA polymerase I. To isolate fragments for footprinting studies, the pairs of enzymes *Eco*RI + *Pvu*II or *Sac*I + *Hind*III are used. These pairs of enzymes

generate fragments that can be singly end labeled by Klenow fragment of DNA polymerase I and an appropriate choice of ^{32}P -labeled deoxynucleotide triphosphates. Digestions are made on $\sim 50\ \mu\text{g}$ of plasmid DNA. A 5% polyacrylamide TBE gel is used to separate the insert from the vector. Inserts are visualized by their UV shadows and purified as above.

Nucleosome reconstitution has recently been extensively reviewed.^{7,30,37} The competitive reconstitution procedure that we and others have used is similar to previous salt exchange procedures with the addition of bulk competitor DNA to lower the total protein-to-DNA ratio.³⁸ The general approach (Fig. 12) is to mix chromatin (lacking its linker histones) or core nucleosomes with a labeled DNA fragment at a high salt concentration, where the histone octamers are free to exchange. After a suitable incubation time the exchange is stopped by lowering the ionic strength to a level where the histones are locked in place. Our recipe involves mixing $\sim 5\ \mu\text{g}$ of stripped chromatin³⁹ with $20\ \mu\text{g}$ of bulk DNA and $0.1\ \mu\text{g}$ of the labeled DNA fragment whose reconstitution is being tested. These components are incubated 30–45 min in $70\ \mu\text{l}$ of buffer containing $1\ \text{M}$ NaCl, $100\ \mu\text{g/ml}$ albumin, and 0.1% Nonidet P-40. The ionic strength is lowered by three $210\text{-}\mu\text{l}$ additions of low salt buffer [$1\ \text{mM}$ Tris (pH 8.0), $0.1\ \text{mM}$ EDTA] to bring the final NaCl concentration to $0.1\ \text{M}$.

The fraction of a DNA fragment that has been reconstituted into nucleosomes is determined by measuring the amount of radioactivity in the free and complexed band of a native polyacrylamide gel. This approach provides more information concerning possible degeneracy in the position of the histone octamer on the DNA fragment than agarose gels. We use a 5% gel [75 : 1 (w/w), acrylamide : *N,N'*-methylenebisacrylamide] to separate free from complexed DNA. In this system, as in others we have examined, a high acrylamide-to-bisacrylamide ratio produces the sharpest bands. The fraction of DNA reconstituted can be measured by using an autoradiogram of the dried gel to locate and excise the desired bands, which are then counted in a scintillation counter. Alternatively, the appropriate regions of the dried gels are counted directly on a Betascope 603 blot analyzer.

Control experiments from several laboratories suggest that the fraction of nucleosomes that are reconstituted reflects the equilibrium binding ratios at $1\ \text{M}$ NaCl. First, neither lengthening the time of incubation in high salt buffer, nor the rate at which the ionic strength is lowered, changes the ratio of labeled fragment that is reconstituted into nucleosomes.⁷ Sec-

³⁷ S. D. Jayasena and M. J. Behe, *J. Mol. Biol.* **208**, 320 (1986).

³⁸ N. Ramsay, *J. Mol. Biol.* **189**, 179 (1986).

³⁹ L. C. Lutter, *J. Mol. Biol.* **124**, 391 (1978).

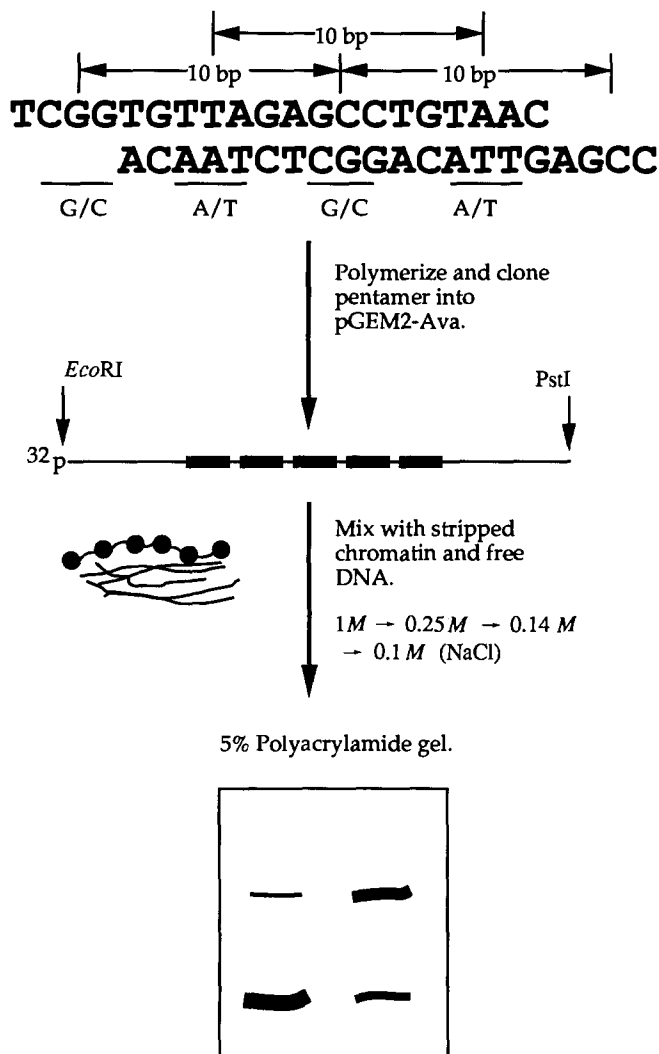


FIG. 12. Design and testing of anisotropically flexible DNA sequences for favorable incorporation into nucleosomes.⁷ The anisotropically flexible sequence TG is shown at the top, with emphasis on its alternation of A/T and G/C-rich regions every 5 bp. Head-to-tail clones of this sequence are excised as restriction fragments of ~170 bp, and subjected to a competitive salt gradient procedure for chromatin reconstitution, followed by analysis of the extent of nucleosome formation on a polyacrylamide gel.

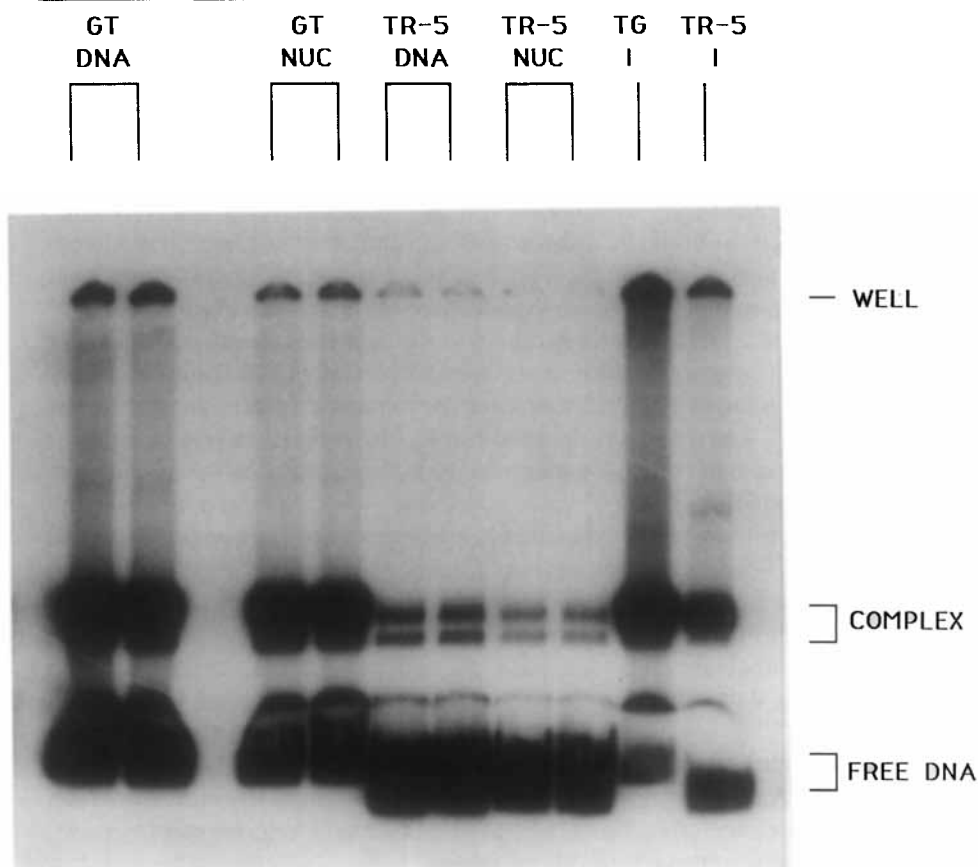


FIG. 13. Nucleosome reconstitution reflects equilibrium histone DNA binding. Fragments containing five copies of the GT (5'-TCGGGTTTAGAGCCTGTAAC-3') or the TR oligonucleotide (5'-TCGGAAGACTTGCAACTGT-3') were reconstituted starting from free DNA on nucleosomal complexes. The fraction of a labeled fragment in the final nucleosomal complex is independent of its addition as free DNA (GT-DNA and TR-DNA) or as nearly completely complexed with histones (GT-NUC and TR-5-nuc). The nucleosomes used to add labeled fragments as preexisting complexes are shown in the two rightmost lanes. Additionally, it is clear that a very small fraction (3–5%) of the labeled DNA fragment exists in aggregated forms that do not enter the gel.

ond, the same final labeled nucleosome-to-labeled free DNA ratio is reached regardless of whether the labeled fragment is introduced to the high salt incubation as free DNA or in nucleosomal complexes (Fig. 13). Finally, when a trace amount of labeled free nucleosomal DNA was incubated in high salt with otherwise identical unlabeled DNA and unlabeled nucleosomes at several protein to DNA ratios, the final fraction of labeled

nucleosomes, after the ionic strength was lowered, reflected the percentage of free DNA in the original mixture.³⁷

Results

To calibrate the size of the nucleosome positioning signal of our synthetic sequences, we determined the amount of anisotropically flexible DNA needed to mimic the histone binding affinities of some natural nucleosome positioning sequences. The fragments studied (Fig. 14) contain increasing amounts of artificial sequences roughly centered in fragments of nearly constant overall length. The molecule derived from two repeats of the 20-bp TG oligonucleotide in a 176-bp fragment binds histones as tightly as do the natural 5S RNA-encoding nucleosome-positioning DNA sequences.^{40,41} Interestingly, the required 40-bp synthetic region is approximately the size of the region identified as essential for positioning in some natural fragments.^{39,42}

In a second series of experiments we determined the optimal repeat of flexible A/T- and G/C-rich regions for histone binding.^{30,43} We reasoned that this optimum should coincide with the most favored helical repeat of DNA on the nucleosome, whose value remains a point of controversy. Based on nuclease cleavage patterns and periodicities extracted from sequencing data, nucleosomal DNA has long been argued to be overwound.^{4,5,44} However, more recent experiments involving the reconstitution on nucleosomes on small DNA circles argue for an unaltered value of the DNA twist.^{45,46} Our system of repeated oligonucleotides, for which precisely defined changes of the phasing of flexible regions can be made, offered a new approach to this problem.

Figure 15 shows the series of oligonucleotides of differing lengths that were synthesized to study this problem. Nucleosomes were reconstituted onto DNA fragments containing five tandem repeats of each sequence; Fig. 15 shows the reconstitution energies, relative to a standard flexible sequence. The free energy minimum near a sequence repeat of 10.1 bp clearly supports the idea of DNA overwinding on the histone surface,

⁴⁰ R. T. Simpson and D. W. Stafford, *Proc. Natl. Acad. Sci. U.S.A.* **80**, 51 (1983).

⁴¹ D. Rhodes, *EMBO J.* **4**, 3473 (1985).

⁴² P. C. Fitzgerald and R. T. Simpson, *J. Biol. Chem.* **260**, 15318 (1985).

⁴³ T. E. Shrader, Ph.D. Thesis, Yale University, New Haven, Connecticut (1990).

⁴⁴ J. T. Finch, L. C. Lutter, D. Rhodes, R. S. Brown, B. Rushton, M. Levitt, and A. Klug, *Nature (London)* **269**, 29 (1977).

⁴⁵ I. Goulet, Y. Zivanovic, and A. Prunell, *J. Mol. Biol.* **200**, 253 (1988).

⁴⁶ Y. Zivanovic, I. Goulet, —, Revet, M. Le Bret, and A. Prunell, *J. Mol. Biol.* **200**, 267 (1988).

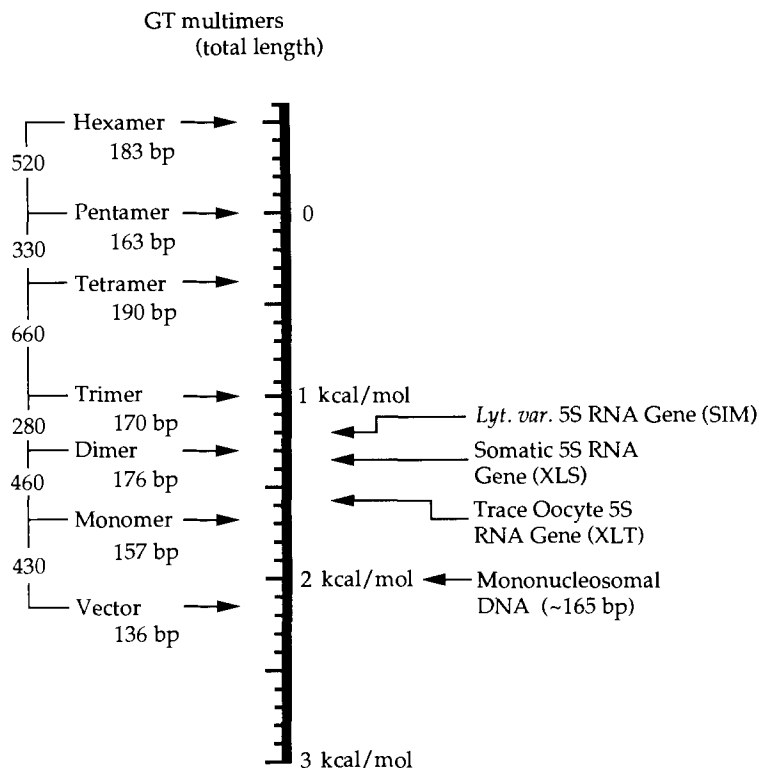
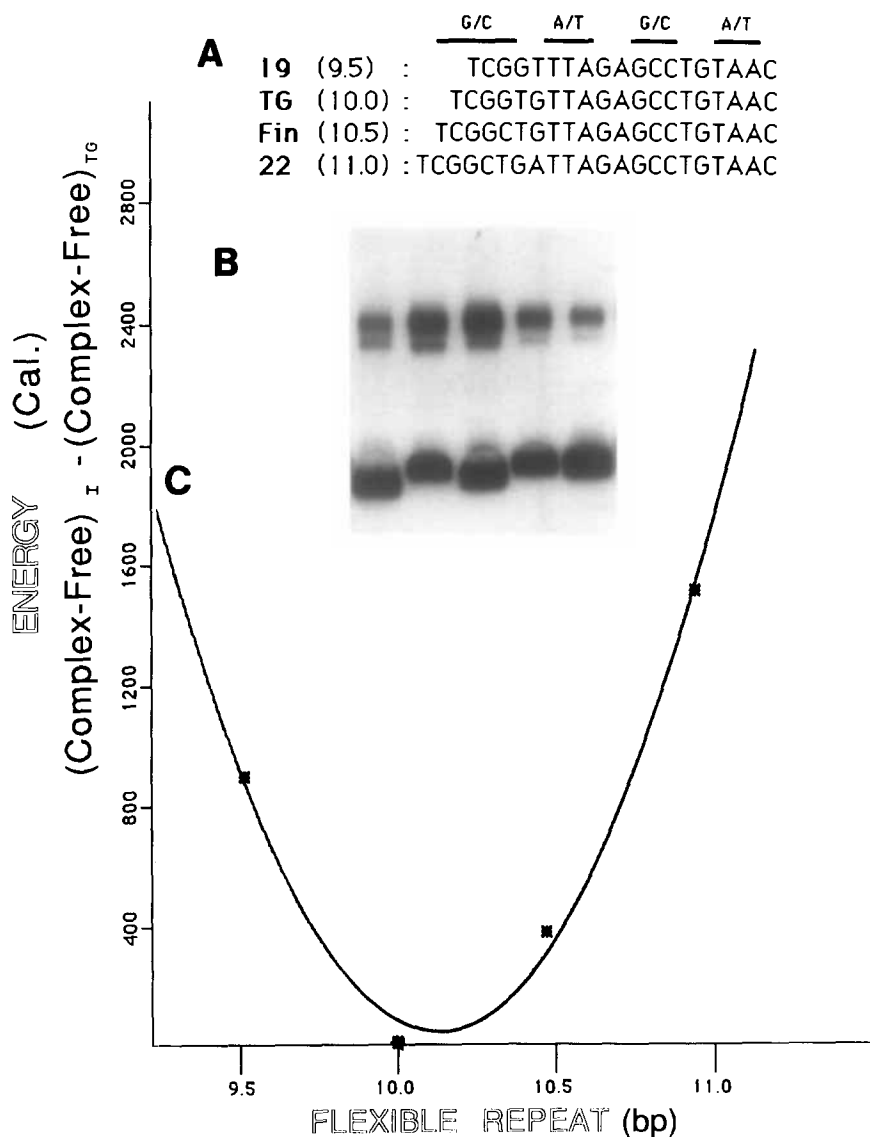


FIG. 14. Nucleosome binding as a function of the length of anisotropically flexible DNA.⁷ The free energies of reconstitution of three natural nucleosome-positioning sequences compared with designed sequences containing various lengths of flexible DNA. Sequences that reconstitute most favorably are at the top of the diagram. The binding of the best natural sequences can be mimicked with approximately 40 bp of repetitive DNA. Additionally, the binding free energy increment for each additional flexible region is quite constant. The total length of each multimer (vector DNA plus the indicated number of 20-bp GT oligonucleotides) is given below the multimer name. The difference in binding free energy between fragments is given at the extreme left. These free energies are all relative to the GT pentamer, which binds histones with essentially the same affinity as the TG pentamer.

since the optimal repeat is significantly shorter than the 10.5- to 10.6-bp helical repeat of B-DNA in solution.^{47,48} This difference between optimal phasing for nucleosome formation and solution helical repeat is too large to be explained solely on the basis of the helical path of the DNA on the nucleosome surface. Consequently we believe that there is a true change

⁴⁷ D. Rhodes and A. Klug, *Nature (London)* **292**, 378 (1981).

⁴⁸ L. J. Peck and J. C. Wang, *Nature (London)* **292**, 375 (1981).



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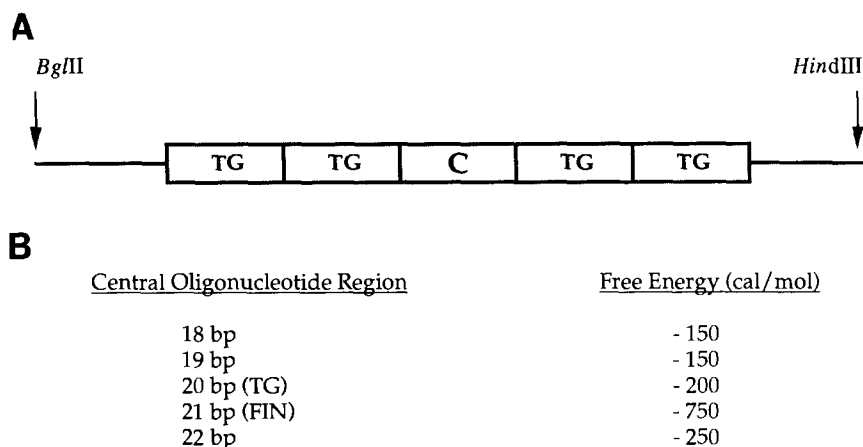


FIG. 16. Helical repeat near the nucleosome dyad.³⁰ (A) Schematic sequence of fragments used to determine the helical repeat near the nucleosome dyad. Each *Bgl*III plus *Hind*III fragment contains DNA sequences derived from five oligonucleotides. The two TG oligonucleotides and vector DNA at either end of each fragment make up constant regions, common to all sequences. The central region is variable and is derived from one of several oligonucleotides. (B) Central oligonucleotides for the molecules tested. Also given are reconstitution free energies relative to a fragment containing five 20-bp TG oligonucleotides. For the nucleotide sequences of the oligonucleotides, see Fig. 11. Several of these sequences form nucleosomes more favorably than the reference sequence, including 20(TG), which contains the same oligonucleotide-derived region as the reference sequence. This is presumably due to the use of the pJC vector, rather than pGEMA2/*ava*, in the construction of these fragments.

FIG. 15. Optimal phasing of flexible regions.³⁰ (A) Oligonucleotides used to determine the optimal phasing of flexible (A/T or G/C rich) regions for nucleosome reconstitution. Only the top strand of each molecule is shown. All oligonucleotides were double-stranded with 4-bp asymmetric *Ava*T overhangs. (B) Polyacrylamide gel of competitive reconstitution of fragments constructed from five copies of the above oligonucleotides centered in a fragment of ~160- to 170-bp total length. From left to right, the fragments contain five copies of the oligonucleotides 19, TG, GT, FIN, and 22. The sequence of GT is identical to that of TG with inversion of the first TpG dinucleotide. These sequences (TG and GT) have essentially the same free energies of nucleosome reconstitution (≈ 0). (C) Free energy of nucleosome reconstitution plotted as a function of flexible repeat. Flexible repeat is defined as one-half the length of the oligonucleotide monomer that makes up the central 95- to 110-bp of each fragment. The curve represents a least-squares fit of a parabola to the data. Free energies are relative to the free energy for nucleosome reconstitution of a fragment containing five copies of the TG oligonucleotide. The minimum of the curve, at about 10.1 bp of the sequence repeat, should coincide with the preferred helical repeat of bending sites on DNA in nucleosomes.

in average DNA twist as it is packaged into nucleosomes in the correct direction to lessen the "linking number paradox."^{49,50}

The degree of overwinding of DNA on the nucleosome surface determined from our experiments is similar to the values for this parameter measured in other studies. However, the appropriateness of a single average value for twist over the ~145 bp of nucleosomal DNA is unclear. Like the uneven bending of DNA seen in the nucleosomal crystal structure,¹ the overall DNA twist may be the average of disparate values from different regions of the structure. To explore the optimal repeat near the nucleosome dyad, the series of molecules depicted in Fig. 16 was constructed. These molecules all contain common vector sequences and two 40-bp "arms" derived from two copies of the TG oligonucleotide (10.0-bp sequence repeat), flanking a central region of variable length. The fragment containing the 21-bp central region reconstitutes most efficiently. The extra base pair, averaged over the 10 turns of DNA in the oligonucleotide-derived region, adjusts the overall helical repeat from 10.0 to 10.1 bp, leading to optimal alignment for the 40-bp arms. The fragments containing 19-, 20-, or 22-bp central regions reconstitute less well. This suggests that the local value for the helical repeat near the nucleosome dyad is not more than 10.5 bp/turn, close to the average of 10.1 bp.

⁴⁹ J. H. White and W. R. Bauer, *J. Mol. Biol.* **189**, 329 (1986).

⁵⁰ A. A. Travers and A. Klug, *Philos. Trans. R. Soc. London B* **317**, 537 (1987).

[10] Footprinting Protein–DNA Complexes *in Vivo*

By SELINA SASSE-DWIGHT and JAY D. GRALLA

This chapter describes the use of primer extension procedures to probe nucleoprotein complexes *in vivo* in *Escherichia coli* and *in vitro*. Included are an overview of the procedure (when it is appropriate and what materials are required), a description of the choice and preparation of materials, step-by-step protocols for primer extension probing with dimethyl sulfate and potassium permanganate, and a troubleshooting guide.

Overview of Primer Extension Probing

The technique of primer extension footprinting analysis¹ has been used for footprinting various regulatory regions *in vitro* on linear and su-

¹ J. D. Gralla, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 3078 (1985).

percoiled DNA¹⁻³ and for examining protein-DNA interactions *in vivo* on both plasmid²⁻⁸ and genomic DNA⁹ in different bacteria.¹⁰ It has been used to study viral transcription complexes in infected mammalian cells.¹¹⁻¹³ It has also been applied more generally to the study of specific modifications induced by various anticancer drugs.¹⁴

The protocol, which is illustrated in Fig. 1, involves three major steps. First, the DNA is covalently modified with an attacking reagent. Next, the modified DNA is isolated and, if necessary, broken at the sites of modification. Third, the sites of modification are detected by primer extension procedures. In the primer extension reaction a specific ³²P end-labeled synthetic oligonucleotide is hybridized to the modified template DNA and extended with a DNA polymerase. The resulting extension products, which all have the same ³²P end-labeled 5' end and a 3' end that varies according to the position of the template modification at which the primer extension is inhibited, are analyzed on a DNA sequencing gel.

If the original modification took place on protein-bound DNA, the attacking reagent may have had altered access to the DNA. In the simplest case this results in a "protected" band on the gel autoradiograph compared to a lane in which the protein was not present. The interpretation is just as in conventional footprinting techniques that involve using end-labeled template DNA rather than an end-labeled primer.^{15,16}

The use of an end-labeled primer rather than end-labeled template DNA is advantageous in that it allows regulatory regions to be footprinted on circular templates either *in vivo* or *in vitro*; it avoids the isolation and end labeling of specific modified DNA fragments. The same end-labeled primer can be used to probe a given regulatory region either *in vivo* or *in vitro*. Another advantage is that various regions of the same DNA sample can be analyzed simply by splitting the sample and probing with different

² J. A. Borowiec, L. Zhang, S. Sasse-Dwight, and J. D. Gralla, *J. Mol. Biol.* **196**, 101 (1987).

³ S. Sasse-Dwight and J. D. Gralla, *J. Biol. Chem.* **264**, 8074 (1989).

⁴ J. A. Borowiec and J. D. Gralla, *Biochemistry* **25**, 5051 (1986).

⁵ Y. Flashner and J. D. Gralla, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8968 (1988).

⁶ Y. Flashner and J. D. Gralla, *Cell (Cambridge, Mass.)* **54**, 713 (1988).

⁷ S. Sasse-Dwight and J. D. Gralla, *J. Mol. Biol.* **202**, 107 (1988).

⁸ S. Sasse-Dwight and J. D. Gralla, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8934 (1988).

⁹ S. Sasse-Dwight and J. D. Gralla, *Cell* **62**, 945 (1990).

¹⁰ E. Morett and M. Buck, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 9401 (1988).

¹¹ R. L. Buchanan and J. D. Gralla, *Mol. Cell. Biol.* **7**, 1554 (1987).

¹² L. Zhang and J. D. Gralla, *Nucleic Acids Res.* **18**, 1797 (1990).

¹³ L. Zhang and J. D. Gralla, *Genes Dev.* **3**, 1814 (1989).

¹⁴ J. D. Gralla, S. Sasse-Dwight, and L. Poljak, *Cancer Res.* **47**, 5092 (1987).

¹⁵ D. Galas and A. Schmitz, *Nucleic Acids Res.* **5**, 3157 (1978).

¹⁶ G. M. Church and W. Gilbert, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 1991 (1984).

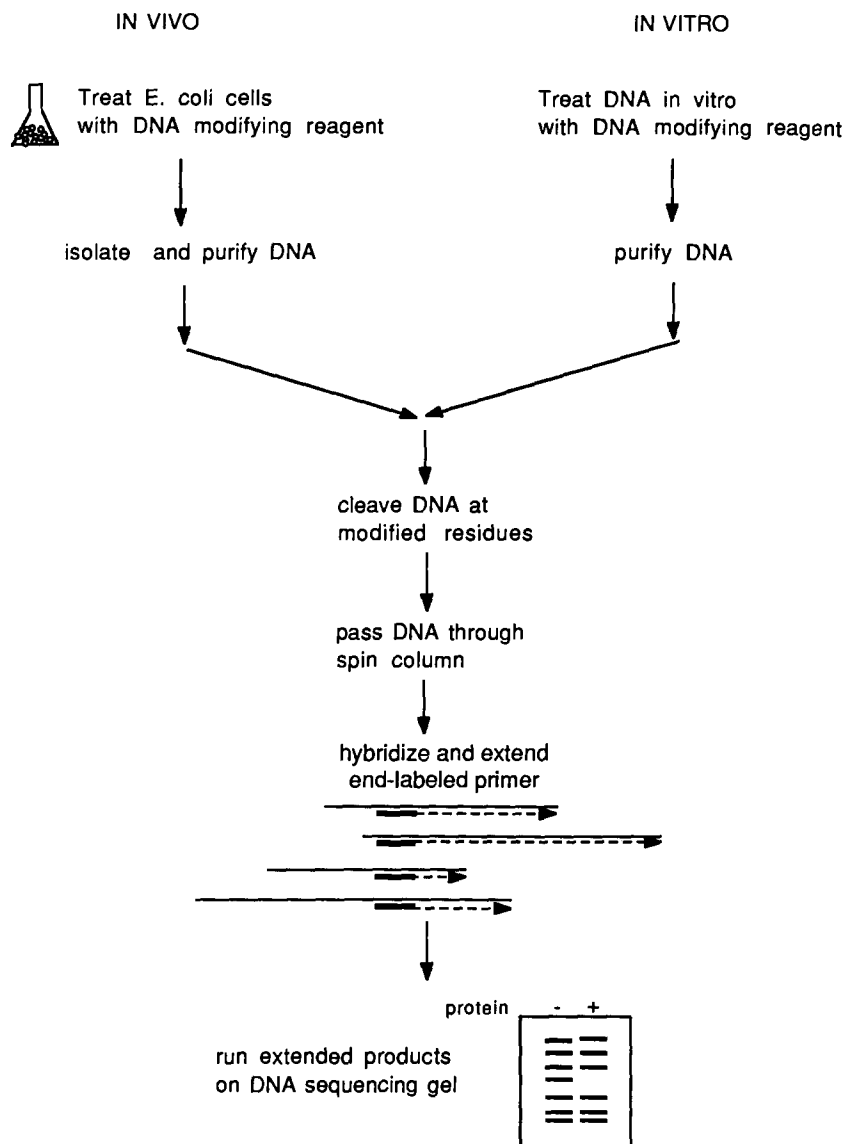


FIG. 1. Outline of the primer extension footprinting analysis technique *in vivo* and *in vitro*.

primers designed to read through different regions. Both strands can be probed on the same DNA template simply by designing two primers, each of which hybridizes to a different strand. In addition, reagents whose DNA modifications inhibit extension can be used without the necessity for DNA cleavage chemistry, theoretically increasing the diversity of reagents that can be employed for DNA footprinting. Last, primer extension footprinting is a relatively fast technique, as it eliminates the need for end-labeling each sample of modified template DNA.

Choice of Reagents

Since primer extension footprinting relies on extension blockage, the DNA modification reagents chosen must produce lesions that either directly impede extension or that can be specifically altered to serve as blocking sites. In addition, if the regulatory region is to be footprinted *in vivo*, the reagent must be able to penetrate the cell. Two chemicals have been used *in vivo*, dimethyl sulfate (DMS)^{1,2,4} and potassium permanganate (KMnO₄).^{3,8} The first has been useful for identifying residues protected and enhanced by protein binding while the second is most useful in identifying distorted DNA or DNA melted in transcription complexes. The procedure for using each of these reagents *in vitro* or *in vivo* on either plasmid or chromosomal DNA in bacterial cells will be described in the following sections. Although not described here, these techniques have also been applied in preliminary experiments in mammalian cells where the range of attacking reagents is broader due to more facile entry as compared with bacteria.¹¹⁻¹³ The technique has been used successfully with the enzymes DNase I and micrococcal nuclease.

Dimethyl sulfate (DMS), which has been employed for sequencing¹⁷ and other footprinting protocols,^{15,16} and potassium permanganate (KMnO₄) are two chemicals that have been used in primer extension probing. Both reagents are capable of penetrating the bacterial cell and have therefore been useful in studying protein-DNA interactions *in vivo*. In addition, both reagents meet the important requirement for primer extension analysis of modifying the DNA in such a way that the modifications can be made to serve as blocking sites for extension by a DNA-copying enzyme. Specifically, DMS modifies, among other sites, the N-7 position of guanine residues.¹⁷ When treated appropriately with piperidine, the DNA strand becomes susceptible to cleavage at these modified guanine residues¹⁷ and this break serves as a block to the DNA-copying enzyme during primer extension. Potassium permanganate, on the other hand,

¹⁷ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

modifies primarily T and to a lesser extent C residues.¹⁸ This modification results in oxidation of the 5,6-double bonds to form thymine glycols.^{18,19} The Klenow fragment of DNA polymerase I has been shown to stop extension after inserting a base across from a thymine glycol-modified residue.^{19,20} Alternatively, treatment of the modified DNA with alkali results in the conversion of the thymine glycol residue to urea, which the polymerase is unable to copy.¹⁹

In addition to these two chemicals, several enzymes have been used in primer extension footprinting. These include DNase I¹¹⁻¹³ and micrococcal nuclease,²¹ which have different modes of cleavage and thus give complementary information. They are not suitable for use in bacteria since they do not penetrate the cells, but have been used in primer extension protocols *in vitro* and in permeabilized mammalian cells.^{11-13,21} Their use will not be described in detail here, since it is a matter of using standard digestions followed by the primer extension probing procedures that will be described below.

One advantage to using a variety of reagents is that each attacks the DNA differently and their joint use can give a very detailed picture of the DNA bound by the protein. For example, DNase I cleavage occurs by attack through the DNA minor groove, DMS attacks guanines in the DNA major groove, micrococcal nuclease attacks exposed positions on the DNA backbone, and permanganate attacks melted DNA preferentially (see references 4, 12, and 21). In principle, many chemicals and enzymes can be adapted for primer extension probing. The main limitation is that they must either break the DNA, modify it so as to allow breakage, or form lesions that block the progress of a copying enzyme. As an example, the following section describes the different information that can be obtained by using the two reagents DMS and KMnO_4 .

Dimethyl Sulfate Footprinting

With the chemical dimethyl sulfate, we have used primer extension probing to reveal residues protected and enhanced by protein binding both *in vitro*^{2,5,6} and *in vivo*.^{2,4,7,8} For instance, DMS was used *in vitro* to determine the occupancy of the *lac* O_1 and O_3 operators on both linear and supercoiled DNA.² These results demonstrated that a single *lac* repressor molecule can bind simultaneously to these two operators only on supercoiled DNA. DMS was also used as a probe for *lac* repressor binding

¹⁸ H. Hayatsu and T. Ukita, *Biochem. Biophys. Res. Commun.* **29**, 556 (1967).

¹⁹ H. Ide, Y. W. Kow, and S. S. Wallace, *Nucleic Acids Res.* **13**, 8035 (1985).

²⁰ J. M. Clark and G. P. Beardsley, *Biochemistry* **26**, 5398 (1987).

²¹ L. Zhang and J. D. Gralla, *Nucleic Acids Res.* **17**, 5017 (1989).

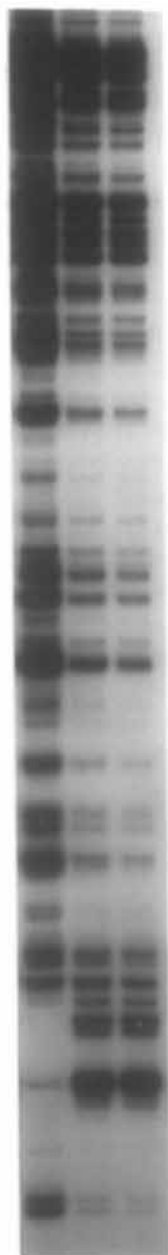
on the same plasmid DNA *in vivo* in experiments that demonstrated that this protein also binds cooperatively to the *lac* O_1 and O_3 operators *in vivo* and that this cooperativity requires DNA supercoiling.⁷ These results point to some advantages associated with the primer extension footprinting protocols in probing supercoiled DNA and DNA in the living cell. In addition, in these and related experiments it was also important that the relative occupancy of the three *lac* operators on the same template be known reliably. The primer extension procedure allowed the very same sample to be split and probed with different primers. Such experiments demonstrated that DNA looping is an important control mechanism in regulating *lac* expression.

As described above, the primer extension technique is designed for facile probing of supercoiled DNA. However, the initial attack on DNA should not lead to a loss of supercoiling. For instance, DMS modifies purine residues without causing strand cleavage. These modified residues serve as sites for DNA cleavage by piperidine subsequent to DNA purification. Strictly speaking, if a DNA-breaking enzyme is used as a probe, the template should be broken at less than one site per template so that each attack occurs on a supercoiled template. This can be checked by agarose gel electrophoresis to show the incomplete loss of form I DNA. In practice, if supercoiling is required only for the formation but not the maintenance of the complex, this consideration is unimportant.

Thus, DMS serves as an effective probe for examining protein-DNA interactions *in vivo* by primer extension analysis. As has been observed previously for DMS, it is valuable in detecting specific contacts that the protein makes with individual guanine residues. In principle, other breakage protocols could be used to probe adenine and other modifications.¹⁷

Probing with Potassium Permanganate

In contrast to DMS, the reagent potassium permanganate preferentially attacks single-stranded DNA.¹⁸ It serves as a probe for detecting sharply distorted DNA² and for DNA melted in an open complex with RNA polymerase.^{3,8} This second property is illustrated in Fig. 2, which shows the pattern of protection obtained at the *lacUV5* promoter when potassium permanganate is used in a primer extension analysis. Lane 1 (Fig. 2) illustrates the control pattern where no proteins were present during modification. When RNA polymerase is allowed to bind to the *lac* promoter DNA prior to modification, the bands from -10 to +4 become very hyperreactive (lanes 2 and 3, Fig. 2). It was demonstrated that this hyperreactivity represents those bands that are melted in the open complex formed between RNA polymerase and *lac* promoter DNA.³ Thus, perman-

1 2 3

ganate serves as an efficient probe of open complex formation *in vitro* using primer extension footprinting analysis.

Figure 3 illustrates that permanganate can also detect open promoter complexes *in vivo*. When the same plasmid DNA used for the above *in vitro* experiment was probed under activating conditions for the *lac* promoter *in vivo*, a pattern essentially identical to that obtained *in vitro* was observed (lane 3). In this instance, rifampicin, which traps RNA polymerase in an open promoter complex *in vitro*²²⁻²⁴ and *in vivo*,⁴ was added to the cells just prior to permanganate addition in order to collect open complexes. This signal can be compared with that obtained under repression conditions in the absence of rifampicin (lane 1). These experiments and others led to the demonstration that permanganate can be used to determine the ratio of the rate of open complex formation to the rate of transcription initiation *in vivo*.³

More recently, it has also been observed that permanganate can be used as a footprinting reagent *in vitro* using end-labeled DNA rather than an end-labeled primer.²⁵ In these experiments it is necessary to cleave the DNA strand with piperidine at the sites of permanganate modification. This step is not required during primer extension analysis, but the results using the two techniques should differ only in unusual circumstances, where atypical permanganate lesions block extension but are not piperidine sensitive.

Probing on Chromosomal DNA

Thus far, these examples of primer extension analysis have been confined to footprints along multicopy plasmid DNA *in vitro* or *in vivo*. Because *E. coli* genomic DNA contains far more sequence complexity than

²² A. Sippel and G. Hartmann, *Biochim. Biophys. Acta* **157**, 218 (1968).

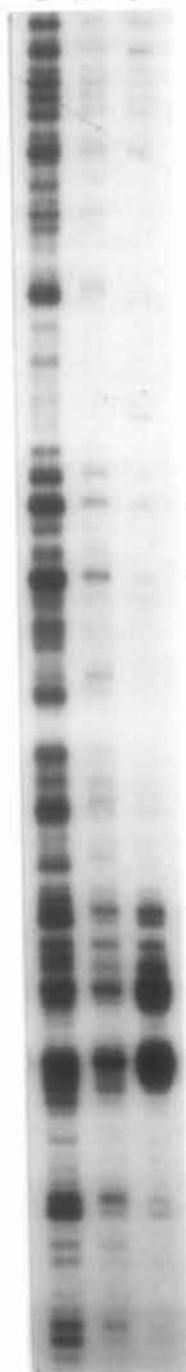
²³ D. C. Hinkle, W. F. Mangel, and M. J. Chamberlin, *J. Mol. Biol.* **70**, 209 (1972).

²⁴ C. Bordier and J. Dubochet, *Eur. J. Biochem.* **44**, 617 (1974).

²⁵ K. R. Fox and G. W. Grigg, *Nucleic Acids Res.* **16**, 2063 (1988).

FIG. 2. *In vitro* potassium permanganate pattern obtained at the *lacUV5* promoter along supercoiled plasmid DNA in the presence and absence of prebound RNA polymerase. Lane 1, *in vitro* pattern obtained in the absence of any proteins; lane 2, *in vitro* pattern obtained in the presence of RNA polymerase, which was allowed to bind the promoter DNA for 5 min before KMnO_4 was added; lane 3, *in vitro* pattern obtained with permanganate when the cAMP receptor protein (CRP) was allowed to bind in the presence of cAMP for 5 min followed by RNA polymerase binding for 5 min before treatment with permanganate. R refers to a reference band. (For a more detailed description of experimental procedures, see Fig. 1A of Ref. 3, from which this figure was reprinted with permission.)

1 2 3



←R

←R

←-10

←+4

plasmid DNA and is present in only one copy per cell, it presents several problems for primer extension footprinting analysis. These include a far greater potential for nonspecific hybridization and a much weaker overall signal intensity. In order to surmount these problems, we have developed a modification of the polymerase chain reaction (PCR)²⁶ in which only one primer, the end-labeled primer, is used for extension. Because Taq polymerase is used, this permits both the hybridization and extension temperatures to be higher, lowering the probability of nonspecific hybridization. In addition, the multiple rounds of hybridization and extension result in a much stronger signal. These multiple rounds, although they do not exponentially amplify the signal as in a standard PCR reaction, lead to a theoretical doubling of the signal with every round. This allows a reasonable footprinting signal to be obtained on *E. coli* chromosomal DNA *in vivo* using primer extension analysis.

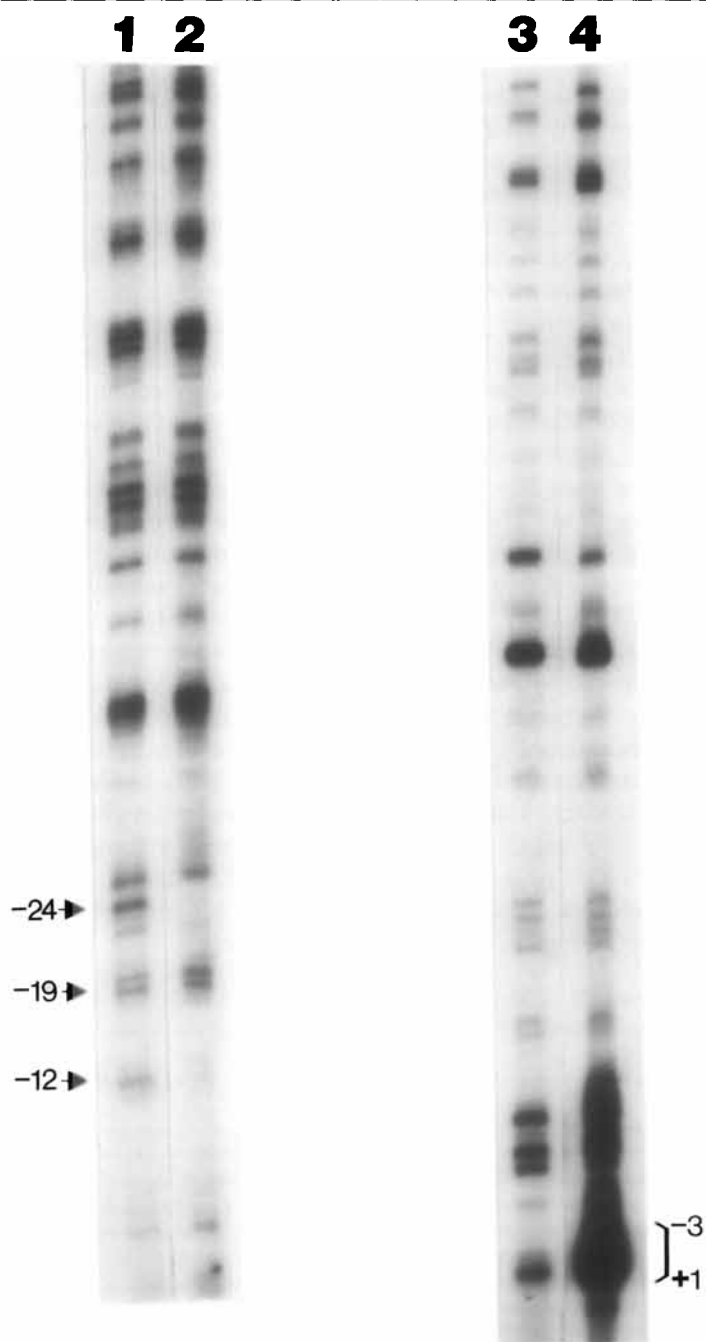
Results of such experiments using either DMS or KMnO₄ as the footprinting reagent are shown in Fig. 4. In these experiments, the *E. coli* glnAp2 promoter was probed under activating conditions. The DMS experiments demonstrate that protection of the promoter region (bands marked -12 and -24) by the RNA polymerase known to bind at this sequence *in vitro*^{27,28} can be seen in the lane in which a wild-type strain was probed (lane 2) compared with a lane in which a strain deficient in the sigma factor (σ^{54}) of the polymerase was probed (lane 1). Similarly, lanes 4 and 3 show the permanganate patterns obtained at this promoter in the presence and absence of the required RNA polymerase. Strongly hyperreactive bands at the DNA transcriptional start site are observed only when the σ^{54} RNA polymerase is present in the cell. Thus, primer extension analysis of genomic DNA using a modification of the PCR can be used to detect specific protein-DNA contacts and open promoter complexes *in vivo* using DMS and potassium permanganate, respectively.

²⁶ K. B. Mullis and F. A. Faloona, this series, Vol. 155, p. 335.

²⁷ E. Garcia, S. Bancroft, S. G. Rhee, and S. Kustu, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 1662 (1977).

²⁸ T. P. Hunt and B. Magasanik, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 8453 (1985).

FIG. 3. *In vivo* potassium permanganate pattern obtained at the *lacUV5* promoter along plasmid DNA under different metabolic conditions. Lane 1, permanganate pattern obtained under nonactivating conditions *in vivo*; lane 2, permanganate pattern obtained when KMnO₄ was added to the shaking cells just subsequent to induction of the *lac* operon with cAMP and IPTG; lane 3, permanganate pattern obtained when rifampicin was added to trap open complexes for 5 min following induction with cAMP and IPTG, isopropyl thiogalactoside. R refers to a reference band. (For a more detailed description of experimental procedure, see Fig. 3A of Ref. 3, from which this figure was reprinted with permission.)



Preparation and Choice of Materials: General Considerations

DNA Preparation

The modified template DNA should be free of proteins and other contaminants that may induce artifactual stops during the primer extension. This requires that the DNA be carefully extracted with phenol in order to remove protein, preferably to the point where there is little or no contaminating interface. In all cases, either *in vitro* or *in vivo*, ultrapure or redistilled, neutralized phenol should be used in order to avoid introducing contaminants or DNA damage that might interfere with the extension reaction. In particular, the primer extension procedure is theoretically more sensitive to unwanted DNA damage since such modifications could block extension, thereby producing bands that, if not cleavable by piperidine, would not be observed on end-labeled DNA. In procedures involving piperidine (Aldrich, Milwaukee, WI), the reagent should be of the highest purity and retested periodically for induction of stop-inducing modifications in DNA.

In order to ensure removal of various ions and chemicals such as the reagent used to cleave the DNA at the modification sites, we strongly recommend passing the DNA through a spin column (Sephadex G-50-80), equilibrated in sterile, doubly distilled water, directly before the extension reaction. This removes contaminants such as magnesium, which can strongly inhibit extension at high concentrations. For this reason, if it is necessary to concentrate the final DNA stocks following column purification, precipitations should be carried out with 0.3 M sodium acetate rather than with magnesium salts.

Normally, 0.5–1.0 μg of whole-plasmid DNA is used per primer exten-

FIG. 4. DMS (lanes 1 and 2) and KMnO_4 (lanes 3 and 4) patterns obtained using a modification of the PCR when the chromosomal *glnAp2* promoter was footprinted under activating conditions *in vivo*. Lane 1, *in vivo* DMS pattern obtained in a strain that is deficient in the σ factor, which is required for binding at the *glnAp2* promoter, σ^{54} ; lane 2, *in vivo* DMS pattern obtained for the wild-type strain in which σ^{54} is present. The -24 and -24 regions, representing the two major grooves to which the σ^{54} RNA polymerase binds, are marked by arrows, as is the -19 band, which serves as a reference band as it does not change in relative intensity when the protein is present. Lane 3, *in vivo* permanganate pattern at the chromosomal *glnAp2* promoter obtained in a strain that is deficient for the σ^{54} factor, which is required for binding at the *glnAp2* promoter, σ^{54} ; lane 4, *in vivo* KMnO_4 pattern obtained in the wild-type strain in which σ^{54} is present. The positions of the melted residues are marked to the right of the figure. The growing cells were split into two samples just prior to modification so that the permanganate-treated cells (lanes 3 and 4) are identical to those treated with DMS (lanes 1 and 2, respectively).

sion analysis reaction. This gives a signal equivalent to approximately 2.5 ml of midlog *E. coli* cells when multicopy plasmid DNA is footprinted *in vivo*. In the case of chromosomal DNA, approximately 3 ml of midlog-phase cells will yield the desired amount of DNA for one sample of primer extension analysis using a modification of the polymerase chain reaction (PCR, see below).

Primer Preparation

The primer extension footprinting technique calls for the use of a 5'-³²P end-labeled primer. Although labeled dNTPs could theoretically also be used, we have found their use to be associated with high background signals. The primer is chosen based on location, end sequences, and its theoretical hybridization temperature. In general, oligonucleotides that are 17–20 bases in length and have a 3' end that hybridizes to a position at least 15–20 bp away from the sequence to be analyzed have been found to be most reliable. In addition, it is also best to design a primer such that its 5' and 3' ends are composed of either G or C residues. If the primer extension analysis is to be carried out on plasmid DNA, the hybridization temperature should be chosen to be approximately 50–55°, as determined by the formula

$$T_m = 69.3 + 0.41(G + C)\% - 650/L$$

where L is the length of the oligonucleotide in nucleotides and $(G + C)\%$ represents the percentage G/C content.²⁹ In the case of chromosomal DNA, where a modification of the PCR is employed, we have found good results for primers with a T_m of around 57°.

Crude preparations of oligonucleotide should be purified on a polyacrylamide gel. Precautions should be taken to remove acrylamide by passing the primer through a 0.2- μ m Acrodisc (Gelman Sciences, Ann Arbor, MI). During purification of the primer, precipitation with ammonium ions should be avoided, as they strongly inhibit the subsequent kinasing reaction. Instead, primers should be precipitated with sodium acetate.

For the kinasing reaction, 20–30 pmol of oligonucleotide is end labeled with [γ -³²P]ATP.²⁹ The final reaction (50- μ l volume) is then passed through Sephadex G-50–80 spin columns equilibrated in TE buffer pH 8.0 (10 mM Tris-HCl, 1 mM EDTA) in order to remove unincorporated label. This generally results in an end-labeled primer that is anywhere from 0.7 to 5×10^6 cpm/ μ l. As a rule, 0.3 – 0.5×10^6 cpm of ³²P end-labeled primer is added per primer extension reaction. As the primer decays, however,

²⁹ T. Maniatis, E. F. Fritsch, and J. Sambrook, eds., "Molecular Cloning: A Laboratory Manual." Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1983.

the molar amount of primer required to give the desired amount of radioactivity increases. Since excessive molar amounts of unlabeled primer reduce hybridization specificity and compete with labeled primer, the labeled primers are generally discarded after they decay to a level of 0.3×10^6 cpm/ μ l or less. In general, a high specific activity for the end-labeled primer is advantageous.

Extension Enzyme

In the procedure discussed below we use the Klenow form of DNA polymerase I for extension. In order to minimize artifactual sequence-induced extension stops the reaction is done at 50–52°. We also describe a modified protocol in which Taq DNA polymerase is used. In principle, any DNA-copying enzyme can be used.

Detailed Protocols for Chemical Probing and Primer Extension

Note. A convenient way to prepare the Sephadex G-50–80 spin columns that are required for these experiments is to stopper the bottom of a 1-ml syringe with a polyethylene disk cut by a cork borer to be the diameter of the 1-ml syringe. Next, pour the Sephadex G-50–80, preequilibrated in the desired solution, into the syringe and let the Sephadex solution settle to 1 ml. Then pack the column to the 0.7-ml mark by spinning for 2 min in a clinical centrifuge at setting 4. Finally, load the sample carefully on top and spin again for 2.5 min at the same setting.

In Vitro Treatment with DMS or KMnO₄

DMS as the Modifying Reagent

1. Bring DNA samples to be treated with DMS (0.1–10 μ g) to a final volume of 100 μ l, where one-third of the volume is composed of $3 \times$ transcription salts ($3 \times$: 9 mM MgCl₂, 0.3M KCl, 0.6 mM DTT, 90 mM Tris-HCl, pH 8.0, 0.3 mM EDTA) or other salts required for the regulatory protein being examined. Prewarm the DNA for 2 min at 37° before adding the desired protein, also diluted to give $1 \times$ transcription salts or other desired salts.

2. While the protein-DNA mixture is incubating at 37°, dilute the DMS to 150 mM (from 10.6 M stock) with water and vortex well. Use care in handling DMS stock solutions, since the chemical is toxic.

3. Add 6.7 μ l of 150 mM DMS to the protein-DNA sample and allow the modification to proceed for 5 min at 37°.

4. Then quench the reaction with 200 μ l cold stop buffer (3 M ammo-

nium acetate, 1 M 2-mercaptoethanol, 250 $\mu\text{g}/\text{ml}$ tRNA, 20 mM EDTA) and add 600 μl cold 95% (v/v) ethanol to precipitate the DNA for 10 min at -70° . Spin the samples for 10 min in a microfuge at 4° . The pellets are washed with 70% (v/v) ethanol and dried.

5. Following this, resuspend the samples in 100 μl 1 M redistilled piperidine, place black electrical tape over the microfuge tube top, close the lid, and place the tubes in a clamped holder for 30 min at 90° .

6. Place the piperidine-treated modified DNA on ice and then spin the sample through a 1-cm³ Sephadex G-50-80 column preequilibrated in distilled water as described at the beginning of this section. Approximately 70 μl of cleaved DNA in water should be recovered and adjusted to give 0.5 $\mu\text{g}/35 \mu\text{l}$.

Potassium Permanganate Treatment

0.37 M KMnO_4 [formula weight (FW) of $\text{KMnO}_4 = 158.04 \text{ g/mol}$, solubility limit = 60 g/liter or approximately 0.37 M]: Weigh out 12 g permanganate (since KMnO_4 is a strong oxidant, gloves and safety glasses should be worn when working with this reagent). Bring to just over 200 ml with distilled water and heat to boiling. Allow to boil for 3–5 min until the volume reaches 200 ml. Allow solution to cool and store in brown jar. This solution should be good for 1–2 months

1. Bring 0.5–2.0 μg DNA to 17.5 μl such that one-third of the volume is $3 \times$ salts (if a protein is to be added, bring it to only one-half this volume, 8.75 μl).

2. Prewarm the DNA samples 2 min at 37° . Add protein, diluted to give $1 \times$ transcription salts, and incubate for the desired time.

3. Add 2.5 μl 80 mM KMnO_4 (freshly diluted from 0.37 M stock) for exactly 2 min. Then quench the permanganate reaction with 2.0 μl 2-mercaptoethanol (14.7 M) and place on ice.

4. Add 6 μl 0.2 M EDTA and 27 μl water; extract samples by adding an equal volume of phenol and vortexing. Then place samples on ice for 2 min, heat for 3 min at 90° , and centrifuge. Remove DNA aqueous layer.

5. Spin the resulting aqueous sample through a Sephadex G-50-80 spin column preequilibrated in water as described at the beginning of this section. Dilute the resulting DNA, if necessary, such that its concentration is approximately 0.5 $\mu\text{g}/35 \mu\text{l}$.

In Vivo Treatment with DMS and KMnO_4

Note. Certain rich medias can quench permanganate. Therefore, minimal medium is generally used for KMnO_4 footprinting. It is advisable to check the medium itself to determine whether it quenches permanganate

by adding KMnO_4 to the medium at the desired footprinting concentration. If the permanganate immediately turns brown, the medium is quenching this reagent. If the brown color takes a few minutes to develop, the medium can generally be used. When the experiment is done with medium containing cells, the cell pellet will often be brown and the supernatant will remain a purplish color. If the pellet remains white, this is another indication that the KMnO_4 has been quenched by the medium.

1. Cells to be footprinted are grown overnight in appropriate medium and diluted 1 : 100 the next morning.

2. *DMS*: At approximately 0.5–0.7 OD_{660} , 10 ml of cells is treated with DMS for 5 min to give a final DMS concentration of 2–10 mM (the optimal concentration may vary, so it is advisable to initially test the DMS concentration over this small range). The cells should remain shaking at 37° during DMS treatment.

KMnO₄: KMnO_4 is generally added to 10 ml of growing midlog-phase cells for 2 min to a final concentration of 10 mM from a 0.37 M stock. As with DMS, however, this concentration should be adjusted as needed. Since permanganate is useful in detecting open complexes, freshly made rifampicin (50 mg/ml in methanol), which traps the bacterial RNA polymerase in an open complex, can be added to a final concentration of 0.2 mg/ml to the shaking cells for 5 min just prior to permanganate addition when desired. This is necessary if the rate of open complex formation is much slower than the rate of initiation.³

3. Following chemical treatment of the cells with either DMS or KMnO_4 , the samples are removed from the shaking 37° water bath and poured immediately into prechilled tubes. Following this, they are pelleted for 5 min at 5K rpm in a cold SS34 rotor and the supernatant is discarded.

4. DNA isolation:

Plasmid DNA. Plasmid DNA is normally isolated according to the method of Holmes and Quigley³⁰ with the following modifications: (1) the lysate is treated with RNase A (0.2 mg/ml for 30 min at 40°) and proteinase K (0.2 mg/ml for 60 min at 52–57°) before precipitation with 2-propanol; (2) following 2-propanol precipitation the pellets are resuspended in 550 μl of TE buffer, pH 8.0, and extracted once with phenol, three or four times with phenol : chloroform : isoamyl alcohol (25 : 24 : 1, v/v/v), and once with chloroform : isoamyl alcohol (24 : 1, v/v) (the phenol : chloroform : isoamyl alcohol extractions are performed by vortexing the extractions, heating the samples for 10–20 min at 50–55°, cooling the samples, and spinning

³⁰ D. S. Holmes and M. Quigley, *Anal. Biochem.* **114**, 193 (1981).

them for 2 min to separate the interfaces); and (3) following extraction the samples are precipitated with 0.1 vol 3 M sodium acetate and 2.0 vol cold 95% (v/v) ethanol.

DMS: Samples are resuspended in 100 μ l 1.0 M piperidine and cleaved at 90° for 30 min as described by Maxam and Gilbert.¹⁷ Following piperidine cleavage they are spun through a 1-cm³ Sephadex G-50–80 spin column preequilibrated in distilled water. For 10 ml of starting cells the final volume is diluted to 105 μ l with distilled water.

KMnO₄: Samples are resuspended in 150–300 μ l of distilled water (with heating at 50–55° when necessary). Up to 200 μ l is spun on a 1-cm³ Sephadex G-50–80 column preequilibrated in water. The resulting solution is ready for further analysis without dilution.

Chromosomal DNA. *Escherichia coli* genomic DNA is isolated according to the procedure of Owen and Borman³¹ with the following modifications: (1) the samples are allowed to incubate overnight at 37–50° following proteinase K addition; (2) following the first chloroform extraction the samples are extracted two to four times more with chloroform, once with phenol, and once again with chloroform. This procedure is rapid and results in sufficiently pure DNA for primer extension analysis. The resulting pure pellets are treated as described above for the plasmid DNA pellets except that the final volume of DNA in water should be kept closer to 70 μ l for DMS samples and 150 μ l for KMnO₄ samples when 10 ml of cells is footprinted.

Primer Extension Analysis of DNA Treated in Vitro or in Vivo

At this point, all samples, irrespective of the DNA modification reagent or whether they are plasmid or chromosomal DNA, should be at an approximate concentration of 0.5 μ g/35 μ l in distilled water following passage through a spin column (see previous section). It is very important that the DNA be passed through this spin column in order to remove any contaminants that might interfere with the extension reaction. There are two different procedures that can be used for primer extension analysis of the modified DNA samples. The most common, alkaline denaturation, is recommended for *in vitro* or *in vivo* plasmid DNA samples. It involves denaturation of the modified DNA with alkali, neutralization, hybridization, and extension with Klenow fragment of DNA polymerase I. The second extension technique involves a modification of the PCR in which only one primer, the labeled primer, is used. Because this allows multiple rounds of denaturation, hybridization and extension, it has been developed

³¹ R. J. Owen and P. Borman, *Nucleic Acids Res.* **15**, 3631 (1987).

for footprinting the more complex chromosomal DNA samples *in vivo*. In addition, the higher temperature of extension allowable with Taq polymerase mitigates the possibility of nonspecific hybridization.

The following describes the protocol for each of these extension techniques. In general, primer extension analysis should be performed for the first time on *in vitro* modified DNA. This is recommended as, although the overall strategy of primer extension is fast and straightforward, there are several steps that require attention to detail. In addition, we have found that adjustments sometimes need to be made for different primers.

Alkaline Denaturation Primer Extension (for Analysis of in Vitro or in Vivo Modified Plasmid DNA)

1. Begin with 35 μ l of DNA in distilled water (approximately 0.5 μ g). Add 1 μ l 32 P end-labeled primer (diluted in distilled water, if necessary, to give $0.3\text{--}0.5 \times 10^6$ cpm/ μ l).

2. Add 1/9 vol (4 μ l) 0.01 M NaOH to each extension reaction and mix well.

3. Heat samples for 2 min at 80° in order to denature template DNA; transfer directly to ice and allow to cool for at least 5 min.

4. Add 1/9 vol (5 μ l) freshly made 10 $\times$ TMD buffer [0.5 M Tris-HCl, pH 7.2, 0.1 M MgSO₄, 2 mM dithiothreitol (DTT)] to each sample and mix well.

5. Hybridize the end-labeled primer to the modified DNA template by heating the samples for 3 min at 45–50°; return the samples directly to ice. *Note:* the hybridization temperature may need to be altered slightly in accordance with the calculated T_m (see previous section) of the primer. It is best to keep the hybridization temperature at or just under the T_m of the primer in order to minimize nonspecific binding.

6. Add 1/9 vol (5 μ l) of 4 $\times$ 5 mM dNTPs (a mixture containing the 4 dNTPs, each at a concentration of 5 mM). Tap samples well and return to ice.

7. Dilute the Klenow fragment of DNA polymerase I to 0.5–1.0 U/ μ l using Klenow diluent [50% glycerol (w/v), 50 mM KH₂PO₄, 1 mM DTT, 100 μ g/ μ l bovine serum albumin (BSA); make 1 ml and store at –20°]. The enzyme should be placed at the bottom of a cold microcentrifuge tube without creating bubbles and the cold diluent slowly added. The enzyme can be mixed by gently tapping the bottom of the tube.

8. At time t , add 1 μ l diluted Klenow fragment to a given sample and tap gently for 5–10 sec. Place the tube at 50° for exactly 10 min.

9. At time t plus 10 min, add 1/3 vol quench (4 M ammonium acetate, 20 mM EDTA) to the sample and mix thoroughly.

10. Precipitate the sample with 2.5–3.0 vol cold 95% ethanol and incu-

bate at -70° for 20 min or overnight at -20° . Pellet the DNA in a microfuge for 10 min at 4° and wash the DNA pellet gently with 70% ethanol by rolling the microcentrifuge tube. Respin for 2 min, remove the liquid, and dry the pellet. Then proceed with the steps described under Polyacrylamide Gel Electrophoresis (below).

Extension by PCR Modification (for Analysis of Chromosomal DNA Modified in Vivo)

As stated above, this technique is often necessary when analyzing chromosomal DNA, as it magnifies the normally low signal by allowing multiple rounds of extension and permits the extension to occur at a higher temperature, thereby minimizing nonspecific hybridization. It is equivalent to the PCR except that only one primer is added to the reaction mix.

1. Prepare the samples by adding the following solutions in order:

Distilled water	49.5 μ l
10 $\times$ Taq reaction buffer (as provided with enzyme)	10.0 μ l
4 $\times$ 5 mM dNTP mix (each of the 4 dNTPs present at 5 mM)	4.0 μ l
32 P End-labeled primer (diluted to 0.5×10^6 cpm/ μ l with distilled water)	1.0 μ l
DNA in distilled water	35.0 μ l
Taq polymerase (2.5 U/ μ l)	0.5 μ l

2. Overlay samples with 100 μ l mineral oil.

3. Treat samples as follows:

- A: 1.5 min at 94°
- B: 2 min at 57° (or T_m of primer)
- C: 3 min at 72°
- D: 1 min at 94°
- E: Repeat steps B through D 15–20 times
- F: Heat 2 min at 57° (or T_m of primer)
- G: Heat 10 min at 72°
- H: Let samples cool to room temperature

4. Separate the sample from the mineral oil by pipetting it up from the tube bottom and placing in a fresh tube.

5. Extract samples once with chloroform : isoamyl alcohol (24 : 1).

6. Precipitate DNA samples with 1/3 vol quench (4 mM ammonium acetate, 20 mM EDTA) and 2.5 vol cold 95% ethanol for 20 min at -70° or overnight at -20° .

7. Pellet DNA by spinning for 10 min at 4° . Wash the pellet with 70%

ethanol by gently rolling the tube, respin for 2 min, remove the ethanol, and dry the pellet. The samples are now ready for polyacrylamide gel electrophoresis.

Polyacrylamide Gel Electrophoresis

The resulting primer extension pellets are now ready for resuspension in 4–6 μ l of a loading dye composed of alkali/EDTA, formamide, urea, and marker dyes [0.075 g ultrapure urea, 100 μ l deionized formamide, 8 μ l 50 mM NaOH/1 mM EDTA, 8 μ l 0.5% (w/v) of each xylene cyanol and bromphenol blue]. For consistent results, it is best to treat each sample in an identical fashion when resuspending. This is best accomplished by placing the samples with dye at 45–50° and removing each to tap forcefully for 30 sec. Following this, the samples should be vortexed in pairs for 30 sec. This technique ensures that most of the sample resuspends in the dye mix. This can be checked by making sure that most of the counts (over 90%) are now in the solution.

Following a brief spin in order to collect the dye, the samples are ready for loading. Following electrophoresis, the gel is dried and exposed on X-ray film using an intensifying screen, if necessary. Normally, a strong signal can be observed for the autoradiograph after 1 to 2 days.

Although approximately the same number of counts should be loaded per lane, the signal intensities observed on the autoradiograph can differ from one lane to another if different amounts of DNA template are present in the samples. This is not usually a problem *in vitro*, where exact DNA amounts are known, but can be a problem *in vivo*, where the efficiency of DNA isolation varies from sample to sample. This variability can often make comparison of lanes difficult. In order to reduce the difficulty in interpreting such footprinting patterns, it is advisable to pick one or more bands outside of the region where the protein binds and use this as a reference band. In this manner, taking the ratio of the intensity of a band whose signal changes in response to protein binding to the intensity of the reference band within the same lane allows internal normalization of signal intensity for a given lane. This ratio can then be compared with the same ratio for a different lane in order to make a fair comparison of band intensities between the two lanes. If a band is heavily protected by a protein during the modification reaction, this comparison can often be done by eye. However, when protein binding results in a subtle change in pattern or when quantitative information on binding is desired, the autoradiograph can be scanned in order to determine the desired ratio of the changing band to the reference band for each lane. This procedure was used to calculate the occupancies of the *lac* O_1 and O_3 operators *in vivo*

over a range of effective repressor concentrations, making it possible to determine the relative binding affinities of the two operators for *lac* repressor *in vivo*.⁷

Troubleshooting

It is highly recommended that primer extension footprinting analysis first be performed on *in vitro*-modified plasmid DNA. This is best done on linearized DNA treated with a range of concentrations of the footprinting reagent. Once this procedure has been shown to work, it is ensured that the buffers are correct and that the primer design is acceptable. The following is a list of things that can go wrong with the extension analysis either *in vitro* or *in vivo* and suggestions on correcting them.

No Signal Observed

An autoradiograph in which no pattern is observed could indicate that the buffers are incorrect (such as the extension buffer or the dNTPs) or could occur due to problems with the primer. For instance, if the primer sequence is incorrect, hybridization will not occur. It is also important to check the predicted T_m of the primer (see General Considerations) and make sure that the hybridization temperature is just under or equal to, and not significantly above, this number. Another possibility concerns the purity of the DNA. If this DNA contains contaminants that interfere with the extension reaction, it could result in either a loss of signal or a signal that contains artifactual stops and/or smears. No signal can also result from either too little or too much KMnO_4 . It is for this reason that we strongly recommend initial titration on naked DNA.

Smearing

Primer extension analysis should result in a very clean signal, especially when a multicopy plasmid serves as the template DNA. Although the chromosomal footprints also result in a relatively clean footprint, they can sometimes have a light background smear caused by the complexity of the genome, which may be unavoidable. However, a very bad, smeary background pattern that interferes or occludes the real signal may be preventable for either plasmid or chromosomal DNA *in vitro* or *in vivo*.

As stated above, extra bands in the primer extension pattern could result if there are contaminants that interfere with the extension reaction and result in random stopping, for instance high concentrations of magnesium ion. In addition, the primer could hybridize nonspecifically to the DNA. In this instance, it is advisable to see whether there are similar

sequences along the template DNA at which the primer could hybridize. Extension along different sequences would give an indiscernible and often smeary signal. Smeary signals can also be avoided by choosing a primer with the recommended predicted T_m values and hybridizing the primer to the modified DNA at this temperature. Several temperatures can be tested to determine the optimal temperature for hybridization. However, we have found that some primers work better than others even when all the desired conditions have been met, so there are other unknown factors that may contribute to the quality of the primer with respect to primer extension analysis.

A smeary signal can sometimes result if the DNA cleavage chemistry employed to break the DNA strand at the modification sites also damages DNA in such a way that they now interfere with the extension reaction. This is an important consideration to take into account when designing new chemical probes. Although the piperidine cleavage used for DMS footprinting can result in some DNA damage the extent of damage is far less than the desired reaction, such that a smeary pattern should not be observed. A very common cause of extra bands is the use of less than ultrapure piperidine. We normally redistill the piperidine (Aldrich) and store for a few months and then discard. However, it is probably not necessary to redistill the piperidine and control lanes in which DMS or KMnO_4 were not added will reveal if there is a problem.

Nonspecific hybridization and therefore a smeary pattern can also result when the end-labeled primer decays to a value less than 0.3×10^6 cpm/ μl (this value may be higher for chromosomal footprinting). This is due to the fact that the molar amount of primer that must be added to give the desired amount of radioactivity must be increased as the primer decays. It is therefore best to use the primer soon after the kinasing reaction.

Extra bands can also occur due to sequence-induced stopping of the copying enzyme. These can be prominent if the DNA template is supercoiled and will often lessen in intensity if the DNA is linearized before copying. As with DNA sequencing, such artifacts of DNA structure can be reduced by various means, including using different enzymes and modified deoxynucleotides. If such bands persist, they will appear falsely as unprotected nucleotides on footprinting.

Bottom- or Top-Heavy Signal throughout Lane

The pattern obtained within a lane should be relatively even with respect to signal intensity. However, it is sometimes observed that the extension looks like it does not proceed past a certain point, resulting in

a pattern that is strongest near the bottom of the autoradiograph and very faint or lacking at the top of the autoradiograph. In some cases, it is possible that this is caused by contaminants that interfere with the extension reaction. However, the most likely possibility is that the DNA was overdigested during the modification reaction. This can be corrected by reducing the concentration of the modifying reagent and testing it over a given range until the pattern becomes more even.

In contrast, when the signal is most intense near the top of the autoradiograph, this is likely to be due to underdigestion of the template DNA during modification. Thus, the concentration of the DNA modifying reagent should be increased.

[11] Electron Microscopy of Protein–DNA Complexes

By MARK DODSON and HARRISON ECHOLS

Introduction

The electron microscope has proven to be an extremely valuable tool for studying protein–DNA interactions. The instrument provides a powerful means of obtaining both qualitative and quantitative information concerning the nature of specific and nonspecific binding of proteins to DNA. Although commonly used to corroborate conclusions about protein–DNA interactions obtained by other biochemical and biophysical techniques, electron microscopy often can provide critical information about protein–DNA interactions that is not obtained by conventional biochemical and biophysical techniques. Some examples are given in the final section.

There are numerous examples illustrating the use of electron microscopy in analyzing nucleoprotein structures. These include the visualization and mapping of the site-specific binding to DNA of RNA polymerases,^{1–4} transcription factors,^{5–12} site-specific recombinases,^{13–16} and replication

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initiator proteins.¹⁷⁻²⁵ Electron microscopy has shown that many site-specific DNA-binding proteins bind to several target sites on DNA and self-associate by protein-protein interactions to generate highly organized nucleoprotein structures. The DNA in many of these specialized nucleoprotein structures has been observed to assume some type of topological deformation in which the DNA appears to be bent, wrapped, looped, or unwound.²⁶⁻²⁸ Moreover, electron microscopy has been used to thoroughly characterize the role of specialized nucleoprotein structures in the biochemical pathways that mediate site-specific recombination and DNA replication.^{18,20,26-33} Immunoelectron microscopy has been used to eluci-

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²² D. K. Chatteraj, R. J. Mason, and S. Wickner, *Cell (Cambridge, Mass.)* **52**, 551 (1988).

²³ S. Mukherjee, H. Erickson, and D. Bastia, *Cell (Cambridge, Mass.)* **52**, 375 (1988).

²⁴ I. A. Mastrangelo, P. V. C. Hough, J. S. Wall, M. Dodson, F. B. Dean, and J. Hurwitz, *Nature (London)* **338**, 658 (1989).

²⁵ M. Schnos, K. Zahn, F. Blattner, and R. B. Inman, *Virology* **168**, 370 (1989).

²⁶ H. Echols, *BioEssays* **1**, 148 (1984).

²⁷ H. Echols, *Science* **233**, 1050 (1986).

²⁸ H. Echols, *J. Biol. Chem.* **265**, 14697 (1990).

²⁹ S. Eisenberg, J. Griffith, and A. Kornberg, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 3198 (1977).

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³² T. A. Baker, B. E. Funnell, and A. Kornberg, *J. Biol. Chem.* **262**, 6877 (1987).

³³ M. Dodson, F. B. Dean, P. Bullock, H. Echols, and J. Hurwitz, *Science* **238**, 964 (1987).

date the protein composition of nucleoprotein intermediates in DNA replication and site-specific recombination.^{16,20,34-37}

The interactions of nonspecific DNA-binding proteins with DNA have also been very thoroughly characterized. In this area, electron microscopy has been crucial in contributing to our understanding of the function of RecA in homologous recombination.³⁸⁻⁴⁰ The propensity for RecA to form thick filaments when bound to DNA has allowed the visualization of the overlying and underlying strands of knotted and catenated DNA; this technique has permitted critical topological insights into the pathway of strand exchange during site-specific recombination.⁴¹ The interaction of single-stranded DNA-binding proteins with single-stranded DNA⁴²⁻⁴⁵ and topoisomerase with double-stranded DNA⁴⁶ have been thoroughly characterized by electron microscopy. The development of a technique for spreading chromatin⁴⁷ has provided insight into the structural organization of chromatin and nucleosomes,⁴⁸⁻⁵¹ and has permitted the direct visualization of transcription and splicing.^{52,53}

³⁴ S. I. Reed, J. Ferguson, R. W. Davis, and G. R. Stark, *Proc. Natl. Acad. Sci. U.S.A.* **72**, 1605 (1975).

³⁵ M. Wu and N. Davidson, *Nucleic Acids Res.* **5**, 2825 (1978).

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⁴⁰ J. D. Griffith and L. Harris, *Crit. Rev. Biochem.* **23**, 543 (1988).

⁴¹ S. A. Wasserman and N. R. Cozzarelli, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 1079 (1985).

⁴² H. Delius, N. J. Mantell, and B. Alberts, *J. Mol. Biol.* **67**, 341 (1972).

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⁴⁷ O. L. Miller, *Science* **164**, 955 (1969).

⁴⁸ J. D. Griffith and G. Christiansen, *Annu. Rev. Biophys. Bioeng.* **7**, 19 (1978).

⁴⁹ J. L. Cartwright, S. M. Abmayr, G. Fleischman, K. Lowenhaupt, S. C. R. Elgin, M. A. Keen, and G. C. Howard, *Crit. Rev. Biochem.* **13**, 1 (1982).

⁵⁰ I. Igo-Kemenes, W. Hörz, and H.-G. Zachau, *Annu. Rev. Biochem.* **51**, 89 (1982).

⁵¹ H. Zentgraf and C.-T. Bock, and M. Schrenk, in "Electron Microscopy in Molecular Biology: A Practical Approach" (J. Somerville and U. Scheer, eds.), p. 81. IRL Press, Washington D.C., 1987.

⁵² M. F. Trendelenburg and F. Puvion-Dutilleul, in "Electron Microscopy in Molecular Biology: A Practical Approach" (J. Somerville and U. Scheer, eds.), p. 101. IRL Press, Washington D.C., 1987.

⁵³ Y. N. Osheim and A. L. Beyer, this series, Vol. 180, p. 481.

The following sections describe the procedures that we have used to visualize and analyze nucleoprotein structures by electron microscopy. The protocols listed here are by no means the only ones available for preparing and examining such structures by electron microscopy. A more detailed review describing additional and specialized applications may be found in Ref. 54.

The method used for preparing nucleoprotein complexes for electron microscopy employs a protein free spreading technique. This method differs from the original Kleinschmidt method, in which a solution of DNA with a small basic protein (usually cytochrome *c*) is spread as a monolayer on a water surface.⁵⁵⁻⁵⁸ DNA spread in this manner is covered by the positively charged protein and can be adsorbed to a plastic support film mounted on a grid; the protein-covered DNA is then stained with uranyl salts followed by rotary shadowing with Pt-Pd or tungsten. DNA treated by this procedure is easily observed in the electron microscope with single-stranded DNA being easily discernible from double-stranded DNA. However, this technique is generally unsatisfactory for visualizing the interaction of DNA-binding proteins with DNA because these interactions are obscured by the protein used for spreading and thickening the DNA. Consequently, several techniques that permit visualization of DNA-binding proteins bound to DNA were developed.^{4,54,59-65} Two of these techniques will be presented in detail in this section. The techniques are easy to use, fast, reproducible, and require only nanogram quantities of DNA. A protocol for immunoelectron microscopy of nucleoprotein complexes is also presented. In the descriptions, we assume that the reader is familiar with the operation of both the electron microscope and the vacuum shad-

⁵⁴ L. Coggins, in "Electron Microscopy in Molecular Biology: A Practical Approach" (J. Somerville and U. Scheer, eds.), p. 1. IRL Press, Washington D.C., 1987.

⁵⁵ K. A. Kleinschmidt, this series, Vol. 12B, p. 361.

⁵⁶ R. W. Davis, M. Simon, and N. Davidson, this series, Vol. 21, p. 413.

⁵⁷ H. J. Burkhardt and A. Puhler, in "Methods in Microbiology" (P. M. Bennet and J. Grinstead, eds.), Vol. 17, p. 133. Academic Press, New York, 1984.

⁵⁸ L. T. Chow and T. R. Broker, this series, Vol. 180, p. 239.

⁵⁹ J. Dubochet, M. Ducommun, M. Zollinger, and E. Kellenberger, *J. Ultrastruct. Res.* **35**, 147 (1971).

⁶⁰ J. D. Griffith, in "Methods in Cell Biology" (D. M. Prescott, ed.), Vol. 7, p. 129. Academic Press, New York, 1973.

⁶¹ T. Koller, J. M. Sogo, and H. Bujard, *Biopolymers* **13**, 995 (1974).

⁶² P. J. Highton and M. Whitfield, *J. Microsc. (Oxford)* **100**, 299 (1974).

⁶³ U. K. Laemli, *Proc. Natl. Acad. Sci. U.S.A.* **72**, 4288 (1975).

⁶⁴ H. J. Vollenweider, J. M. Sogo, and T. Koller, *Proc. Natl. Acad. Sci. U.S.A.* **72**, 83 (1975).

⁶⁵ P. Lahart and T. Koller, *Eur. J. Cell Biol.* **24**, 309 (1980).

owing device used for preparing and shadowing the electron microscope grids.

The preparation of nucleoprotein complexes for electron microscopy (EM) typically involves five steps. In the first step, the samples are treated with a reagent that cross-links functional groups of proteins. This "fixing" step serves to rigidify the bound protein structures and keep them trapped onto the DNA. However, certain DNA-protein complexes are destroyed by the cross-linking reagent, and so this step should be used with care. In the next step, unbound proteins are separated from the nucleoprotein complexes by gel filtration. This step removes excess protein that might obscure the nucleoprotein complex or confuse the interpretation. Unreacted cross-linking reagents are also removed so that any subsequent enzymatic steps, such as restriction enzyme digests, are not interfered with. In the third step, the nucleoprotein complexes are adsorbed to an electron microscope grid that is either covered with a plastic-supported carbon film or a carbon film alone. Prior to adsorption of the DNA, the grids are pretreated in a manner that renders the carbon surface hydrophilic so that the DNA will adhere. In the fourth step, excess sample is removed, and the grid is washed and stained with uranyl salts to enhance contrast. The washing procedure is necessary to prevent salts in reaction buffers from precipitating during staining or drying and interfering with the visualization of the nucleoprotein. Finally the specimen is rotary shadowed with metal and examined in the microscope.

Coating Grids with Formvar

Formvar is a plastic resin that is soluble in organic compounds such as dichloroethane. In this coating protocol, a thin layer of Formvar is formed on a water surface and then stretched over electron microscope grids. In the following protocol, the Formvar film is stabilized by coating it with a thin layer of carbon. Without the carbon overlay the Formvar would disintegrate in the electron beam of the microscope. The carbon-coated Formvar, transparent to the electron beam, serves as support for the specimens.

Required Materials

- Large Pyrex dish or glass bowl either painted flat black or covered with black electrical tape
- Round-bottom glass tube, 38 × 430 mm (Bellco, Vineland, NJ, glass pipette canister, round bottom)
- Pasteur pipettes (9 in.) and rubber bulb
- Solution of Formvar (1.5% in dichloroethane, Ted Pella, Redding, CA)

Electron microscope grids; Cat. No. 3HGC300, 3HGC400, or 3HGC500 (Ted Pella) are satisfactory

Fine-tipped forceps with a rubber O ring for keeping the jaws clamped together

Either an in-house vacuum line or small vacuum apparatus for picking up grids

Oven at 55°

Portable fluorescent lamp

Glass microscope slide

Dissecting microscope (optional, but highly recommended)

1. Wash out the glass dish with detergent, and then rinse well with high-quality distilled water. There must be no residual detergent after washing or the technique will not work.

2. Fill the bowl to about 1 or 2 in. from the top with high-quality distilled water at room temperature. If the water is too warm the technique will not work.

3. Wash off the round bottom of a 38 × 430 mm glass tube with detergent and rinse off with distilled water. Again, there must be no residual detergent. Dry the tube in the oven and then let it cool to room temperature. Do not touch the bottom so as to prevent contamination from finger grease.

4. Position the fluorescent lamp slightly to one side of the bowl so that it illuminates the surface of the water. The black coating on the bowl prevents glare and also contrasts with the silver-colored Formvar film that is to be spread on the water surface.

5. Fill the drawn-out portion of a 9-in. Pasteur pipette (approximately 200 μ l) with Formvar solution.

6. This step is difficult, but crucial to the success of the technique. Hold the pipette so that the tip is about an inch or two above the surface of the water and to the side of the bowl. Next, quickly move the tip to the opposite side, keeping the tip the same distance above the water surface, and, while the tip is in motion, squirt the Formvar onto the water.

7. Look at the water surface. There should be a thin film on the surface. Look for a region that is silvery gray in color. Regions that are purple, blue, or gold may also appear. These regions are too thick for use and should be avoided. If a silvery gray area at least 0.5 by 0.5 in. cannot be found, then discard the water and begin again from step 1. The silvery gray area may appear somewhat streaked.

8. Lay about a dozen grids, shiny side down, onto a silvery gray patch. Make a tight pattern. Do not lay down more than 16 grids or the film will sink in the next step.

9. Hold a clean 38 × 430 mm glass tube vertically over the grids and *gently* bring the rounded bottom down onto the grids and submerge them about half an inch or less under water.

10. Maintain the glass tube in a vertical position with the grids submerged and *gently* move the tube to the side of the bowl.

11. Twist the remaining Formvar film up onto the sides of the tube.

12. Bring the tube back to the center of the bowl, tilt it steeply, and then remove it from the water. The grids should remain stuck to the bottom of the tube.

13. Examine the grids under the dissecting microscope. If there are drops of water on the grid surfaces, gently aspirate them off the grid with a finely drawn-out Pasteur pipette hooked up to a vacuum line.

14. Dry the grids for 2 min in an oven at 55°.

15. Using fine-tipped forceps, remove the grids from the glass tube, and lay them shiny side up on the end of a clean glass microscope slide. The grids are now ready for carbon coating.

Sometimes the grids consistently sink when attempting to adhere them to the glass tube in step 12. We are not sure what the cause of this is; but if this step does not work, then try picking the grids up with Parafilm. Do this by gently laying a 2-in. square of Parafilm over the grids and then carefully drape the remaining Formvar on top of the Parafilm. Gently remove the Parafilm and grids from the water surface, and then dry the grids overnight at room temperature. Once dry, the grids can be carbon-coated right on the Parafilm.

Carbon Coating Formvar-Covered Grids

Required Materials

Aluminum foil

Disk of Whatman (Clifton, NJ) filter paper

Graphite rods

Fine sand paper

Pencil sharpener

1. Sharpen the end of a graphite rod and insert into the spring-loaded holder of the vacuum shadower.

2. Flatten the end of another graphite rod and place it in the opposite holder so that the flattened surface of the rod contacts the sharpened rod.

3. Adjust the spring on the holder containing the sharpened rod so that the point continuously maintains pressure against the flattened surface of the adjacent carbon rod.

4. Fold a piece of aluminum foil over several times into a rectangle

that is about 1 in. wide $\times$ 5 in. long. Next, fold the rectangle into a $\sqcap$ shape.

5. Cut a 1 $\times$ 1 in. square of filter paper and place it onto the foil so that only half of the paper is under the "roof" of the foil.

6. Place the glass slide with the Formvar-covered grids onto the foil at the end opposite to the "roof."

7. Place the foil with the paper and grids on it directly under the carbon rods. The rods should be about 3 to 4 in. above the grids.

8. Evacuate the chamber to about 10^{-4} torr. Slowly apply current to the carbon electrodes until they just begin to spark, wait a few seconds, then stop the current flow.

9. Look at the filter paper. The portion underneath the foil roof should still be white and the area not protected by the roof should be a light brown. If it is not brown, repeat step 8 until the paper becomes a light brown, which indicates that the grids have been sufficiently coated with carbon.

10. Remove the grids from the vacuum shadower and store them in a glass Petri dish with a cover. Formvar-covered grids coated with carbon can be stored indefinitely.

Carbon Coating Grids without Use of Formvar Support

In this method the Formvar support film is dispensed with. Instead a thin layer of carbon film is directly applied onto the grids. The advantage of this method is that the contrast of specimens mounted on grids coated with carbon alone is better than that with grids coated with both Formvar and carbon. The disadvantage is that the carbon tends to easily tear without the Formvar support, which necessitates the use of finer meshed grids. This in turn reduces the visible area of view on the grid.

Required Materials

Büchner funnel with a length of plastic hose and a screw clamp near the end of the tubing. Attach the funnel to a heavy ringstand so that it does not wobble or vibrate when filled with water

Stainless steel mesh that has been bent into a level platform that stands about an inch high from the bottom of the Büchner funnel

Either 400- or 500-mesh tabbed electron microscope grids (3HGC400 or 3HGC500, Ted Pella)

Mica sheets (about 1 $\times$ 3 in., Ted Pella)

Fresh razor blade

Squares (1 in.) of Whatman filter paper

Clean glass microscope slides

1. Cleave several sheets of mica with a fresh razor blade. Lay the cleaved sides face up onto glass microscope slides. Use gloves or forceps to handle the sheets so as not to contaminate them with finger grease.
2. Place a square of filter paper between the mica and glass slide so that a portion of the paper sticks out from underneath the mica. The paper will be used to access the extent of carbon deposition on the mica.
3. Place a few of the mica sheets into the evaporator at varying distances from the carbon electrode. This arrangement will yield a gradient of differing carbon thicknesses at the carbon is evaporated onto the mica sheets.
4. Evacuate the chamber to at least 10^{-4} torr. Slowly apply current to a freshly sharpened carbon electrode just until sparks form and then stop the current about 5 sec later. Repeat this procedure until the paper becomes a light brown.
5. Clamp the end of the hose that is attached to the Büchner funnel and fill it with distilled water. Squeeze the tubing several times to remove air bubbles. (If there is any residual air in the tubing or funnel air bubbles may inadvertently travel up the tubing and displace the grids or carbon film.)
6. Using clean forceps, place about a dozen grids shiny side up onto the immersed mesh platform. When submerging the grids be sure to dislodge any air bubbles clinging to the grids or forceps, otherwise the grids may float or stick to the forceps.
7. Take a carbon-coated sheet of mica and gently breathe on it. This thin layer of moisture will assist in the separation of the carbon film from the mica. With the carbon side facing up, begin to submerge the mica at an angle of about 45° . As the carbon begins to separate from the mica and float, position it over the submerged grids.
8. Drain the water from the funnel. The carbon film should gently settle down onto the grids.
9. Allow the mesh and grids to drain for a few minutes. Place the mesh and grids into a desiccator with some Drierite and then evacuate and seal the desiccator. Allow the grids to dry overnight. Check the quality and contrast of a sample from each batch in the electron microscope.

Adsorbing Nucleoprotein Complexes to EM Grids

Two methods for applying nucleoprotein complexes to electron microscope grids are presented here. In the first,⁴ the carbon-coated grids are made hydrophilic by a glow discharge, then a solution of polylysine is applied, and finally the nucleoprotein complexes are adsorbed to the

treated grid. In the second method,^{16,18,65,66} an organic dye is used to provide the grid surface with a positive charge, and the nucleoprotein complexes are then adsorbed to the dye-treated grid. The second method is useful if the vacuum shadower is not equipped with a glow discharge circuit.

Adsorbing Nucleoproteins to EM Grids Using Polylysine Technique

Required Materials

Formvar-covered grids coated with carbon

Lead buckshot that has been slightly flattened with a hammer

A small metal block

Pasteur pipette with a finely drawn-out tip about 0.5 mm in diameter; make by drawing out a 9-in. Pasteur pipette over a flame, score with a diamond-tipped scribe, and then break the tip off; hook the pipette up to a vacuum line and safety trap (clamp the modified Pasteur pipette to a ring stand and attach it to the vacuum via the safety trap)

Incandescent lamp

Dissecting microscope (optional, but highly recommended)

Fine-tipped forceps with a rubber O ring for holding the jaws shut

Parafilm

Two small glass Petri dishes with covers

0.1 and 0.01 M solutions of ammonium acetate

Polylysine, 1 $\mu\text{g/ml}$; prepare fresh from a 1 mg/ml stock solution of polylysine [poly(L-lysine), M_r 2000, Sigma (St. Louis, MO) #PO879] in distilled water; store 1-ml aliquots of 1 mg/ml stock polylysine solutions at -20°

Uranyl acetate solution ($\sim 5\%$); make by mixing 0.5 g uranyl acetate with 10 ml distilled water, vortex 3 min, remove undissolved salt by centrifugation for 5 min in a tabletop centrifuge at full speed, withdraw 8 ml of supernatant, and filter through a $0.45\text{-}\mu\text{m}$ filter; store refrigerated in the dark

DNA dilution buffer⁴: Tris, HEPES, phosphate, or triethanolamine buffers at pH values ranging from 7.5 to 8.0 at concentrations ranging from 10 to 50 mM plus Mg^{2+} at 3 to 10 mM are usually suitable for diluting and adsorbing DNA to grids. KCl or NaCl at $\leq 150\text{ mM}$, formamide at $\leq 85\%$ (v/v), glycerol $\leq 20\%$ (v/v), or glyoxal $\leq 0.5\text{ M}$ may be present in the dilution buffer. If desired, EDTA and dithiothreitol may also be included at low concentrations as well ($\leq 1\text{ mM}$)

⁶⁶ J. Sogo, A. Stasiak, W. DeBernardin, R. Losa, and T. Koller, in "Electron Microscopy in Molecular Biology: A Practical Approach" (J. Somerville and U. Scheer, eds.), p. 61. IRL Press, Washington D.C., 1987.

Glow-Discharge Treatment of Grids. 1. Place grids onto a small metal block and cover the handles with the flattened buckshot. Covering the handles keeps them from becoming charged, which prevents fluid from running onto the jaws of the forceps in subsequent manipulations.

2. Place the block with the grids onto the center platform of the vacuum shadower, and insert the glow discharge electrode into its socket so that it is positioned about 4 to 5 in. above the grids. Be certain that the electrode does not come into contact with any other parts inside the chamber.

3. Evacuate the chamber to about 75-mtorr air pressure, then activate the glow discharge circuit. A pinkish-blue glow should be visible above the grids. After glow discharging for about 10 to 15 sec, switch off the glow discharge circuit.

4. Admit air to the chamber and remove the glow-discharged grids, which are now ready to be coated with polylysine. The grids will begin to lose their charge after several minutes, so it is important to coat them with polylysine as soon as possible.

Application of Polylysine to Glow Discharge-Treated Grids. 1. Pick up a glow-discharged grid by its handle with forceps, clamp the jaws shut with the O ring, and lay the forceps onto the benchtop so that the glow-discharged side is facing up.

2. Place a 7- μ l drop of freshly prepared 1 μ g/ml polylysine onto the grid.

3. About 30 sec later aspirate the polylysine drop off the grid by touching the edge of the grid to the drawn-out Pasteur pipette tip hooked up to a vacuum. Use a dissecting microscope and side illumination with the incandescent lamp to do this. If the grid has been properly glow discharged, it should take several seconds for the grid to dry as the fluid is aspirated off its surface, and interference colors should be visible at the trailing edge of the fluid.

Adsorption of Nucleoprotein Complexes to Polylysine-Treated Grids.

1. Dilute an aliquot of the nucleoprotein complexes so that the concentration of DNA is approximately 1 to 2 μ g/ml.

2. Place a 5- μ l droplet of the diluted nucleoprotein solution onto a polylysine-treated grid.

3. About 1 min later invert the grid and drag it several times over the surface of a solution of 0.1 M ammonium acetate contained in a small glass Petri dish. Do not submerge the back surface of the grid.

4. Touch the inverted grid to a 10- μ l droplet of uranyl acetate (previously placed on the surface of a small square of Parafilm). The drop will stick to the grid as it is picked up.

5. Drag the grid two or three times over the surface of a solution of 0.01 M ammonium acetate contained in a small glass Petri dish. Again, do not submerge the back surface of the grid.

6. Aspirate the grid to dryness using the drawn-out Pasteur pipette hooked up to a vacuum line or blot excess fluid off the grid by touching the edge to a piece of filter paper. The grid is now ready to be rotary shadowed as described below.

Adsorbing Nucleoproteins to EM Grids Using the Alcian Blue Dye Technique

Required Materials

Fresh 0.2% solution of Alcian Blue 8GX⁶⁷ in 3% acetic acid

Formvar-covered grids coated with carbon

Fine-tipped forceps with a rubber O ring for holding the jaws shut
Parafilm

Uranyl acetate solution (same as above)

DNA dilution buffer: Same as for polylysine technique except that DNA will not stick if formamide is present⁵⁴

Application of Alcian Blue to Grids and Adsorption of Nucleoprotein Complexes. 1. Dilute 0.2% Alcian Blue 8GX to 0.002% in doubly distilled water.

2. Pick up a carbon-coated grid by its handle with forceps and clamp the jaws shut with the O ring.

3. Place a 10- μ l droplet of the diluted Alcian Blue solution onto the grid for 5 min.

4. Wash the grid by gently dragging it over the surface of double-distilled water in a Petri dish several times.

5. Remove any remaining water off the grid either by aspiration with a finely drawn Pasteur pipette or by touching a piece of filter paper to the side of the grid.

6. Adsorb DNA to grids as described above for grids that have been treated with polylysine.

Rotary Shadowing Nucleoprotein Complexes

We generally use a tungsten filament (0.02-in. diameter) for rotary shadowing. However, evaporation of a Pt-Pd wire wrapped around a tungsten filament works just as well.

1. Loosen both of the screws that clamp the tungsten filament onto the copper electrodes.

2. Loosen one of the copper electrodes from its retaining block by loosening the screw that holds the electrode in place.

⁶⁷ M. A. Hayat, in "Positive Staining for Electron Microscopy," p. 187. Van Nostrand-Reinhold, New York, 1975.

3. Slide the electrode out of its holder. Insert a 4.2-cm long tungsten filament into it and tighten the screw so that the tungsten filament is firmly clamped into the electrode.
4. Insert the electrode back into its holder.
5. Adjust the height of the electrodes so that the tungsten filament is at an 8° angle with respect to the center of the rotating platform, which will be 9.5 cm distant from the tungsten filament.
6. Insert the other end of the filament into the other electrode and tighten the screw so that it is held firmly in place.
7. With one hand squeeze the two vertical posts holding the electrode retaining blocks about a millimeter toward one another, and with the other hand tighten the screw that holds the electrode in place. This places tension on the filament and keeps it from bending as it heats up and expands when current is applied.
8. Place the grids to be shadowed onto the center of the rotating platform and move the platform so that the distance from its center to the tungsten filament is 9.5 cm.
9. Evacuate chamber to at least 10^{-5} torr.
10. Turn on the rotating platform and, if it is adjustable, set it to about 30 rpm.
11. Apply current to the filament and in one motion adjust the current to about 32 to 34 A. The filament should burn white hot for at least 1.5 to 2 min and then burn out. As the filament burns the current will decrease. Do not adjust current once it is set. If the filament either burns out too soon or burns too long, the shadowing will be too light. The optimal current setting required to achieve the correct filament burn time must be determined empirically for each vacuum shadowing apparatus.
12. Shut off the current and admit air to the chamber. The grids are now ready for examination in the electron microscope.

Preparing Nucleoprotein Complexes for Electron Microscopy

Required Materials

- Glutaraldehyde, 8% aqueous, EM grade (Polysciences, Miles, IL)
- Polyethylene microcentrifuge tubes (400 μ l; Cat. #19002, Denville Scientific, Inc., Denville, NJ) for drip columns or 1-ml plastic syringes for spin columns
- Spectra mesh, either polyethylene or nylon, 70- to 100- μ m mesh size (Fisher Scientific, Pittsburgh, PA)
- Sephacrose 4B or CL-4B
- Hypodermic needles, 25- or 22-gauge

Ring stand and thermometer clamp
Razor blad or scalpel
Hemostat
Parafilm
Safety glasses

Fixing Nucleoprotein Complexes with Glutaraldehyde

We generally use glutaraldehyde to fix proteins to DNA. Formaldehyde fixation followed by glutaraldehyde fixation can also be used.⁷ The fixatives cross-link lysine residues with one another, and serve to cross-link and rigidify large multiprotein structures. Glutaraldehyde fixation does not cross-link protein to DNA. Sometimes aldehyde fixation fails to work, and may in fact alter a protein so that it dissociates from DNA. In this event it may be necessary to simply dilute the sample just prior to adsorbing it to the grid. The protein concentration of the diluted sample must be low enough so that it does not interfere with visual interpretation of the micrographs. Alternatively, other cross-linking reagents can be tried.

1. Open a fresh ampoule of 8% glutaraldehyde, dilute an aliquot into distilled water to yield a concentration of 1.5% solution, and store on ice.

2. DNA and proteins are mixed in reaction buffer so that the concentration of DNA is about 10 to 20 $\mu\text{g/ml}$ in a volume of 15 to 25 μl . Try to avoid buffers such as Tris, which may react with the glutaraldehyde in the next step. If bovine serum albumin is to be included in the reaction (which usually is not necessary), be sure that its final concentration is less than 3 $\mu\text{g/ml}$ when the sample is to be applied to a grid. After incubating at the desired temperature and time, add the freshly diluted glutaraldehyde to a final concentration of 0.1%, and continue to incubate an additional 10 to 15 min at the same temperature used for the reaction.

3. Remove excess unbound proteins and glutaraldehyde by gel filtration over a Sepharose 4B microcolumn.

Preparation of Sepharose 4B Microcolumns

Excess unbound protein can be removed either by passing the samples through drip columns¹⁹ or by passing through spin columns.^{20,37} Drip columns are slow, but if taken care of they can be used several times. Spin columns offer the advantage of speed, but on occasion excess protein may leak through.

Drip Column Gel Filtration

1. Grip a 25-gauge needle with a hemostat and cut off half of the plastic Luer (the hemostat prevents accidental injury to the fingers; wear safety glasses when cutting the Luer to prevent accidental eye injury).

2. Cut the tip off of a 400- μ l microcentrifuge tube so that a hole about 2 mm in diameter is obtained.

3. Cut a piece of the flexible plastic mesh about 4×4 mm and plug the hole. A straightened paper clip is useful for this. Don't pack the mesh too tightly or the column will drain too slowly.

4. Fill the severed needle with distilled water and slip it onto the bottom of the microcentrifuge tube so that the Luer fits snugly.

5. Fill the tube with water and clamp it to a ringstand with a thermometer clamp. If the needle has been properly seated and the mesh is not too tightly packed into the hole, water should drain freely from the needle, and there should be no leakage at the Luer/tube joint.

6. Fill the microcolumn with a slurry of Sepharose 4B or CL-4B. A bed height of 3 cm is sufficient for most work. Do not become alarmed if the column runs dry, as there will still be enough water or buffer retained by capillary action to keep the bed wet for a couple of hours.

7. Make a tower for holding a reservoir of buffer by cutting the tip of another 400- μ l microcentrifuge tube and connecting it to the top of the microcolumn.

8. The nucleoprotein complexes will be eluted in the void volume of the microcolumn. Determine the void volume as follows. Wrap an index card with a piece of Parafilm, which will be used to collect fractions from the microcolumn. Let the column drain and remove the reservoir tower. Layer 10 μ l of a solution of blue dextran onto the surface of the drained bed, allow it to run in, and then chase it with 10 μ l of reaction buffer. Fill the microcolumn with buffer, attach a reservoir tower, and then fill the tower with buffer. The first five fractions containing the blue dextran will be the fractions to pool when the nucleoprotein complexes are run over the microcolumn. Pass two column volumes of 4 M NaCl followed by two column volumes of distilled water to remove residual dye from the column. Radiolabeled DNA can also be used instead of blue dextran to determine the void volume.

9. Store the column until use by filling with distilled water, capping it, and then placing the needle tip into a 1.5-ml microcentrifuge tube filled with distilled water. Columns can be reused several times so long as they are washed with high salt and buffer or water after each use.

10. Equilibrate a calibrated Sepharose 4B microcolumn with two column volumes of buffer.
11. Drain the column and layer the solution containing the fixed nucleoprotein complexes onto the gel bed.
12. After the solution enters the bed, elute with buffer and pool the five peak fractions. After gel filtration the fixed nucleoproteins are usually stable for several hours at room temperature. At this stage the complexes are ready for electron microscopy, restriction enzyme digestion, or labeling with antibody. Gel filtration over a 2-cm high bed dilutes samples about two to threefold.

Spin Column Gel Filtration

1. Plug a 1-ml syringe with a swatch of Spectra Mesh.
2. Fill with a 1-ml slurry of Sepharose CL-4B in DNA dilution buffer.
3. Centrifuge at 2000 rpm in a tabletop centrifuge for 2 min, using a swinging bucket rotor.
4. Place a fresh microcentrifuge tube under the column, and load the fixed samples (50 μ l) onto the column.
5. Recover the sample by centrifugation at 2000 rpm for 2 min.

Immunoelectron Microscopy of Nucleoprotein Complexes

Some nucleoprotein structures may consist of several proteins. So long as the appropriate antisera are available, the protein composition of a multiprotein complex can be qualitatively determined. The method is similar to other techniques, such as Western blotting, that probe with antibody and a visible secondary label. The major difficulty is optimizing the proper concentrations of both the antibody and the particular batch of protein A-colloidal gold (see below). The technique is extremely powerful once conditions have been optimized.

Required Materials

Purified IgG antibody

Protein A-colloidal gold, either 10- or 15-nm diameter gold particles, 5–10 μ g/ml (E-Y Laboratories, San Mateo, CA)

Buffer A: 3-cm bed height Sepharose 4B microcolumn equilibrated with 10 mM Tris-HCl, pH 8.4, 150 mM NaCl, 10 mM MgCl₂

Buffer B: 3-cm bed height Sepharose 4B microcolumn equilibrated with 40 mM HEPES-KOH, pH 7.6, 100 mM KCl, 10 mM magnesium acetate

Buffer C: 4-cm bed height Sepharose 4B microcolumn equilibrated with 40 mM HEPES-KOH, pH 7.6, 100 mM KCl, 10 mM magnesium acetate. (Note: Spin columns may be substituted; however, the final column should be plugged with glass wool, which specifically traps excess colloidal gold.³⁷⁾)

1. Apply glutaraldehyde-fixed nucleoprotein complexes to a 3-cm bed height Sepharose 4B microcolumn, elute with buffer A, and pool the five peak fractions.

2. Add 1 μ l of purified antibody (0.5 to 2.5 μ g) that has been diluted in equilibration buffer A. The dilution of IgG added to the nucleoprotein complexes to be probed is such that the complexes are not aggregated by the bound antibody. The correct dilution must be determined empirically for each purified IgG. Usually a 2- to 10-fold dilution of 5 to 25 mg/ml DEAE-Sephacel-purified IgG is adequate.

3. Incubate for 30 min at 37°.

4. Apply antibody-bound nucleoprotein complexes to a 3-cm bed height Sepharose 4B microcolumn, elute with buffer B, and pool the five peak fractions. This step removes unbound excess antibody.

5. Add 1 μ l of protein A-colloidal gold conjugate particles and incubate for about an hour at room temperature.

6. Apply antibody-gold-bound nucleoprotein complexes to a 4-cm bed height Sepharose 4B microcolumn, elute with buffer C, and pool the five peak fractions. This step removes unbound excess colloidal gold. The samples are now ready for adsorption to grids.

Examination of Nucleoproteins in Electron Microscope and Analysis

We usually use an accelerating voltage of 80 kV. Lower acceleration voltages will enhance contrast but will reduce resolution. Magnifications ranging from $\times 20,000$ to $\times 45,000$ are the most useful for observing nucleoprotein complexes. Using the wobbler to focus does not always work well with older microscopes. We usually focus by eye at a magnification of $\times 35,000$. If a molecule is to be photographed, it is best to take a through focus series of photographs, and then *immediately* develop the plates. If the photographs do not turn out well, it is easy to go back and rephotograph the molecule of interest. If possible, try to take photographs that depict several molecules in the same field of view. Presenting several similar molecules in a field of view is incontrovertible proof of the frequent occurrence of the structure. Always include a size bar on the micrographs as a means of giving the scale of magnification.

In order to measure the length of a segment of DNA an internal standard of known length must be included in the sample. The internal standard can be nicked circular, relaxed covalently closed, or linear DNA. The length of the internal standard should be as close as possible to that of the unknown but sufficiently different so that it can be unmistakably discerned.

Data that map the locations of protein-DNA binding sites and boundaries along a linear DNA segment are presented as a histogram. A homogeneous population will approximate a normal curve when plotted as a histogram in intervals of about half of a standard deviation. A heterogeneous population will have a large standard deviation, and the histogram may be skewed or bimodal. Measuring errors, such as inadvertently measuring molecules from micrographs of different magnification, will also exhibit these discrepancies. The data are displayed with the distances in base pairs or fractions of unit length from the end(s) of the DNA to either the edge(s) or the center of the protein moiety on the abscissa. The number of measurements in an interval is plotted on the ordinate.

If the numerical value of a quantity such as the distance from the end of a DNA segment to the edge of a nucleoprotein is to be stated, present the value as a mean plus or minus the sample standard deviation and include the number of measurements in the sample. Generally sample sizes ranging from 30 to 100 are satisfactory.

The frequency of a particular class of nucleoprotein can be determined by directly examining the specimen grids in the microscope so long as the structure of interest is easily discernable by eye or can be tagged with an identifiable marker such as colloidal gold. With a little practice in moving the stage with one hand and scoring molecules with a multikey hand tally with the other hand it is possible to scan several thousand molecules per hour. To scan a grid rapidly, reduce the magnification to a point where it is still possible to identify specific structures and distinguish separate DNA molecules. Score only nucleoprotein complexes that are completely in a field of view. Once all the interpretable molecules in a field of view have been scored move the stage just enough so that the previous field of view is not visible and score the next set of molecules. Follow a pattern so that the same area of the grid is scored only once. Score molecules in sets of 100. After accumulating at least 3 sets of 100 counts, a mean score and standard deviation for the frequency of each class of molecule scored can be calculated. For structures or classes not discernable by eye, it is necessary to take photographs and measure the size and frequencies from enlarged prints or projected images.

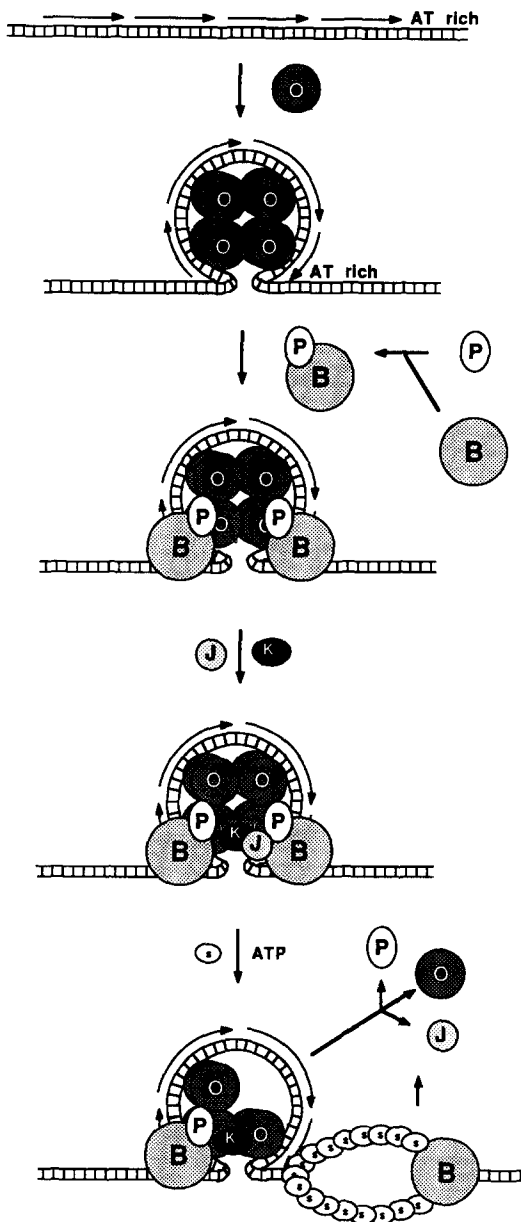


FIG. 1. Model for the prepriming pathway for localized initiation of DNA replication by phage λ . A sequential series of protein assembly and disassembly reactions at *ori* λ culminate in localized unwinding of the origin DNA, providing for the specific initiation of DNA replication. Electron micrographs depicting each of these stages are shown in Figs. 2–4. See text for details.

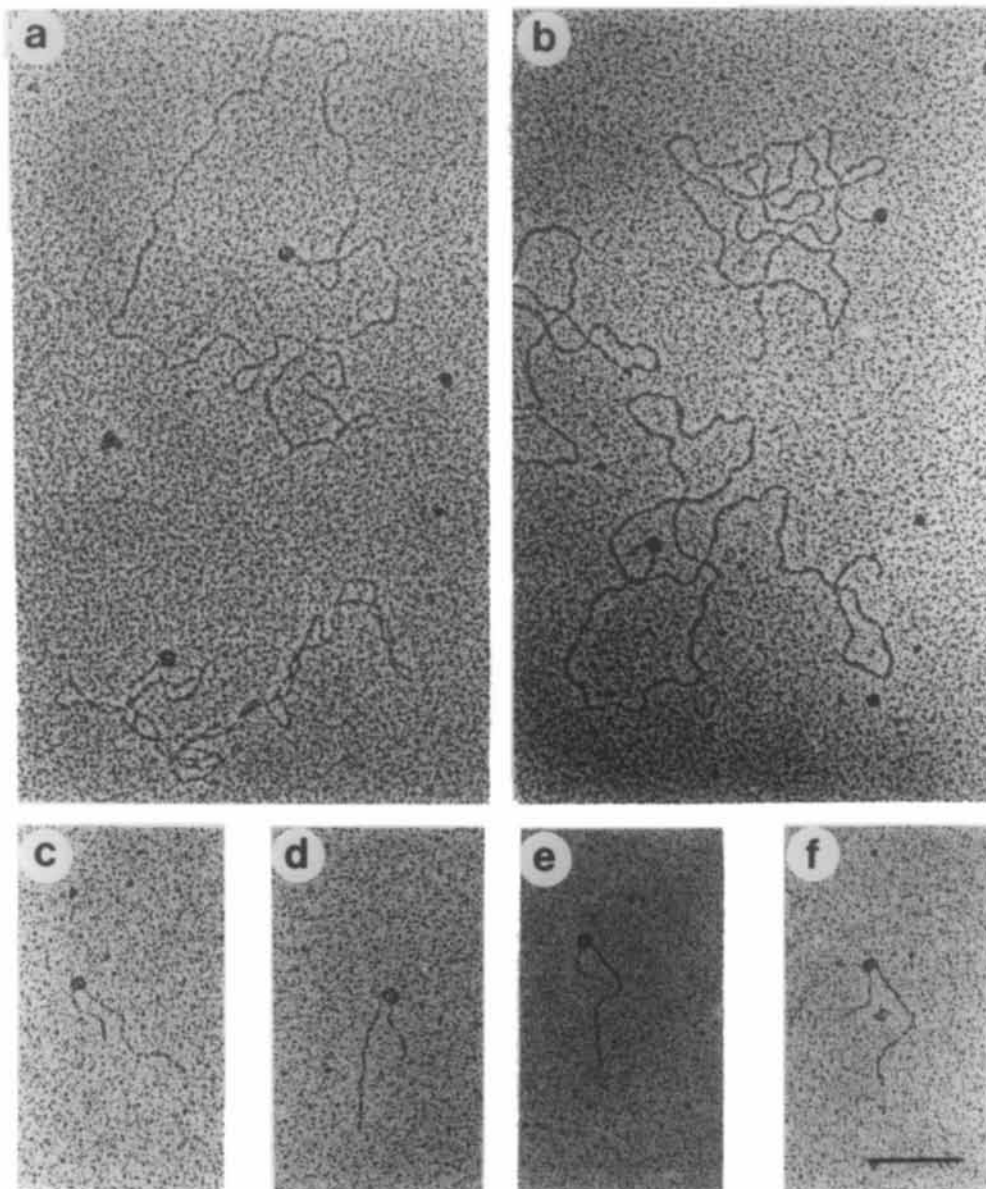
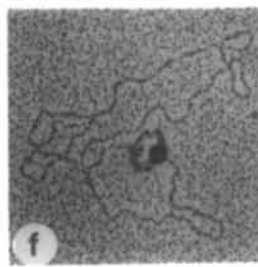
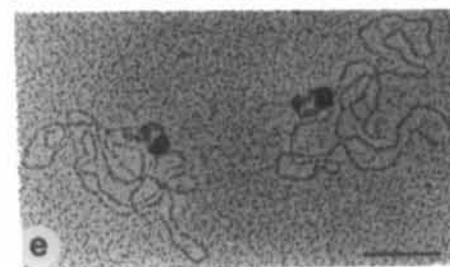
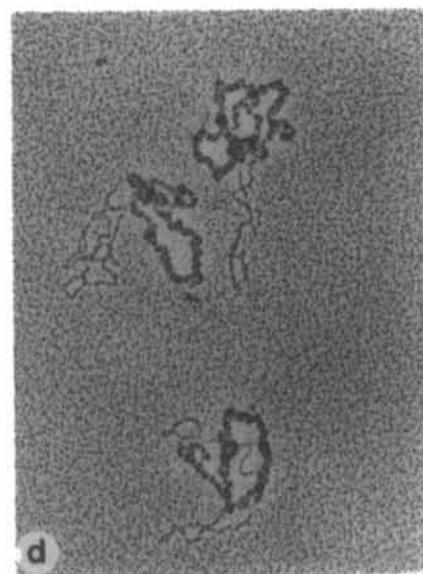
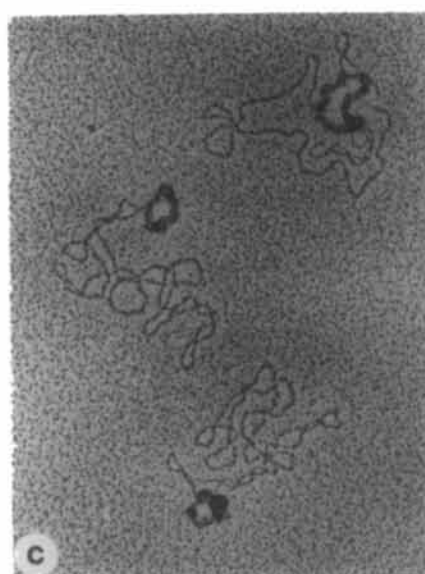
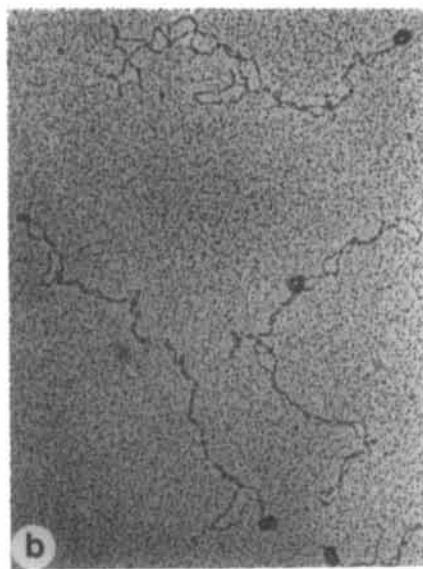
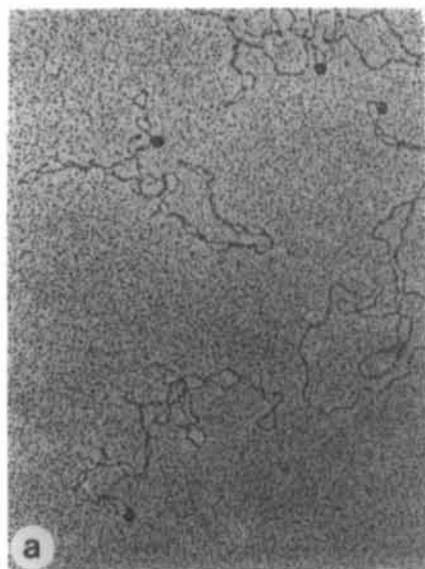


FIG. 2. Electron micrographs of phage λ O protein bound to *ori* λ DNA. (a) O protein bound to supercoiled and nicked circular plasmid DNA. (b) O protein bound to plasmid DNA linearized by restriction enzyme that cleaves adjacent to *ori* λ . (c-f) O protein bound to a restriction fragment containing *ori* λ . Note that the DNA enters and exits the O-some from the same side, which indicates that the DNA is either folded or wound about the protein core. The DNA-winding concept is also supported by measurements of the length of DNA associated with the nucleoprotein structure (bar: 0.1 μ m). (Reprinted from Ref. 19.)



Some Examples of Exceptional Power of Electron Microscopy

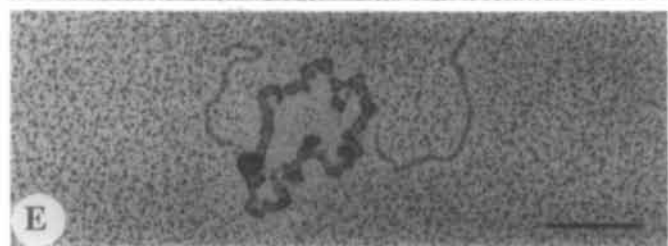
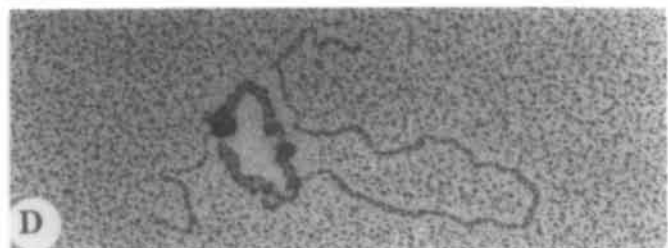
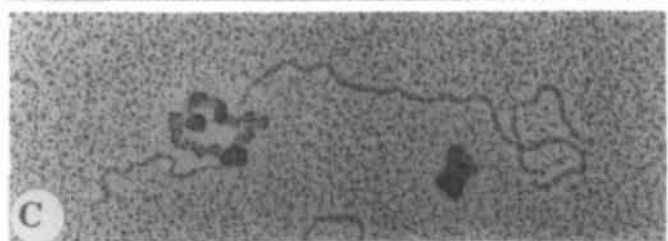
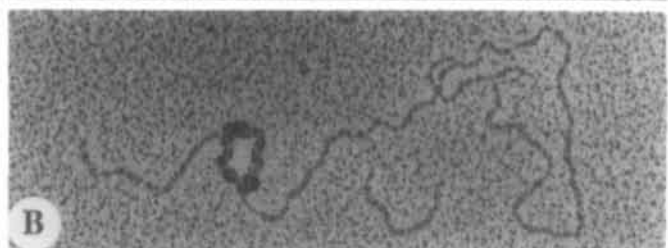
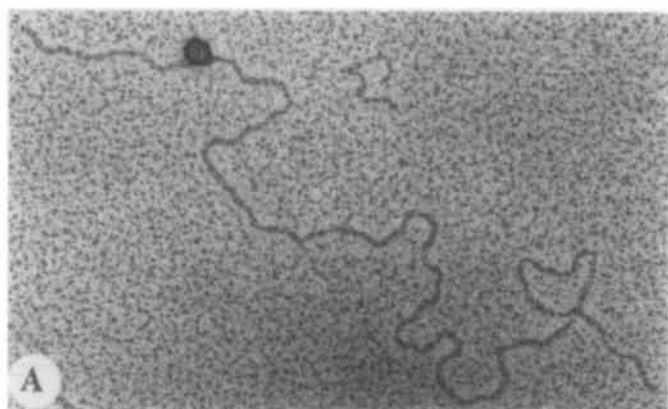
The electron microscope is an enormously powerful tool for the analysis of the structure and dynamics of DNA–protein complexes. One special power of the technique is its ability to determine the path of the DNA in nucleoprotein complexes, information not available from other techniques of DNA–protein analysis (except for X-ray crystallography). Thus, electron microscopy has led to the realization that DNA-wound nucleoprotein structures are critical intermediates in the initiation of DNA replication and site-specific recombination. Electron microscopy has also been used to demonstrate self-association of transcriptional regulators bound to distant sites, and the association of a distant regulator with RNA polymerase (DNA-looping interactions). By the use of immunoelectron microscopy, the sequential assembly and disassembly of nucleoprotein complexes has been followed. To illustrate the special value of electron microscopy, we will present two examples from our work: the sequential pathway that initiates DNA replication by phage λ ; and the DNA-looping interaction involved in transcriptional activation of the *glnA* promoter from the NtrC enhancer site.

Initiation of DNA Replication by Bacteriophage λ

We used two complementary electron microscopic approaches to show that a series of protein–DNA and protein–protein assembly/disassembly reactions precede the initiation of DNA replication at the bacteriophage λ origin of replication, *ori* λ . In the first approach, multiprotein structures were visualized by electron microscopy of rotary-shadowed nucleoprotein complexes. Using this technique, we visualized the successive assembly of three distinct nucleoprotein complexes that precede origin-specific initiation of DNA replication. In the second approach, antibody labeling of individual proteins with colloidal gold revealed the protein complement of these structures. The inferred pathway is shown in Fig. 1.

The assembly of the first nucleoprotein structure that catalyzes the cascade of assembly and disassembly reactions results from the binding

FIG. 3. Electron micrographs of nucleoprotein structures at *ori* λ . (a) O protein alone bound to *ori* λ plasmid DNA. (b) O, P, and DnaB proteins bound to *ori* λ plasmid DNA. (c) Unwound nucleoprotein structures generated by the interactions of DnaJ, DnaK, and single-stranded DNA-binding proteins and ATP with the O–P–DnaB–*ori* λ complex. (d) Highly unwound nucleoprotein complexes formed by the six-protein reaction of (c) in the presence of DNA gyrase. (e–g) *ori* λ DNA and the six-protein reaction of (c). Note the difference in size of the nucleoprotein structures at the two forks. The smaller structure is presumably DnaB because of its location at the unwinding fork (bar: 0.1 μ m). (Reprinted from Ref. 31.)



of a phage-encoded protein, λ O, to the λ origin of replication, *ori* λ .⁶⁸ O protein binds to an inverted sequence that is repeated four times in *ori* λ .⁶⁹ Because each binding sequence is inverted, O is likely to bind as a dimer.⁶⁹ Thus a minimum of eight O monomers binds to linear *ori* λ DNA. However, electron microscopy showed that the functional unit is not a linear array, but a DNA-wound specialized nucleoprotein structure (snup) called the O-some¹⁹ (Fig. 2).

In order for λ to initiate replication, the DnaB helicase must first be captured from the *Escherichia coli* host.⁶⁹ This is done by the λ -encoded P protein, which binds tightly to the DnaB helicase.⁷⁰ The second structure that is assembled in the pathway to initiating DNA replication arises from the interaction of the P-DnaB complex with the O-some. This second stage nucleoprotein intermediate is larger and more asymmetric than the O-some, and the structure includes all of the repeated O-binding sites plus an adjacent (A + T)-rich region^{19,71} (Fig. 3b).

λ also requires several heat-shock proteins of the host to initiate replication.⁶⁹ In the presence of single-stranded DNA-binding protein and ATP, the heat-shock proteins DnaJ and DnaK interact with the second stage nucleoprotein to generate the key intermediate in initiating replication^{31,71} (Fig. 3c-g). This structure consists of locally unwound DNA that is stabilized by single-stranded DNA-binding protein. Nearly complete unwinding occurs in the presence of a topoisomerase. The unwound entity is capable of associating with DnaG primase and DNA polymerase holoenzyme in

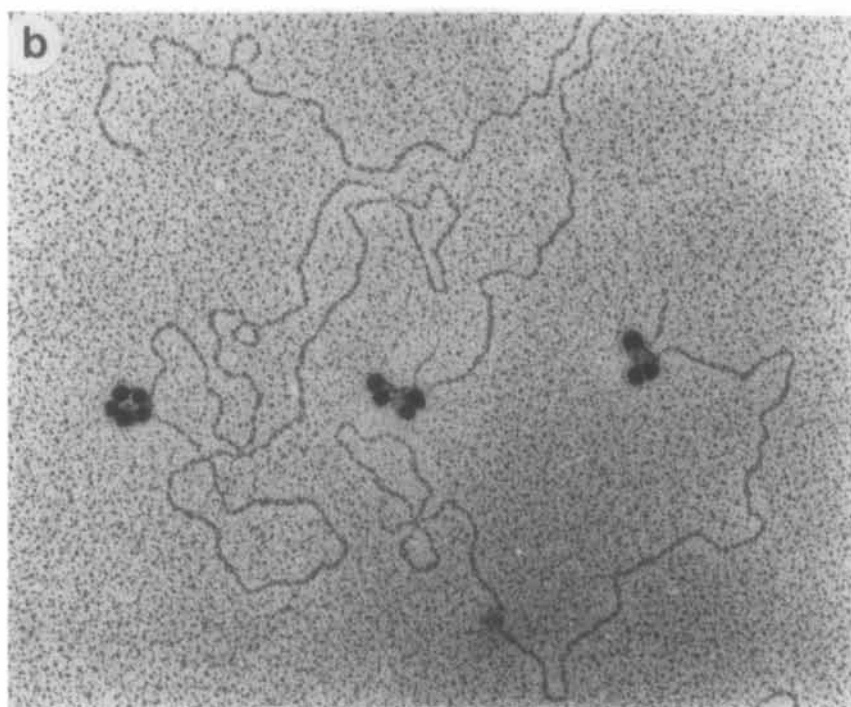
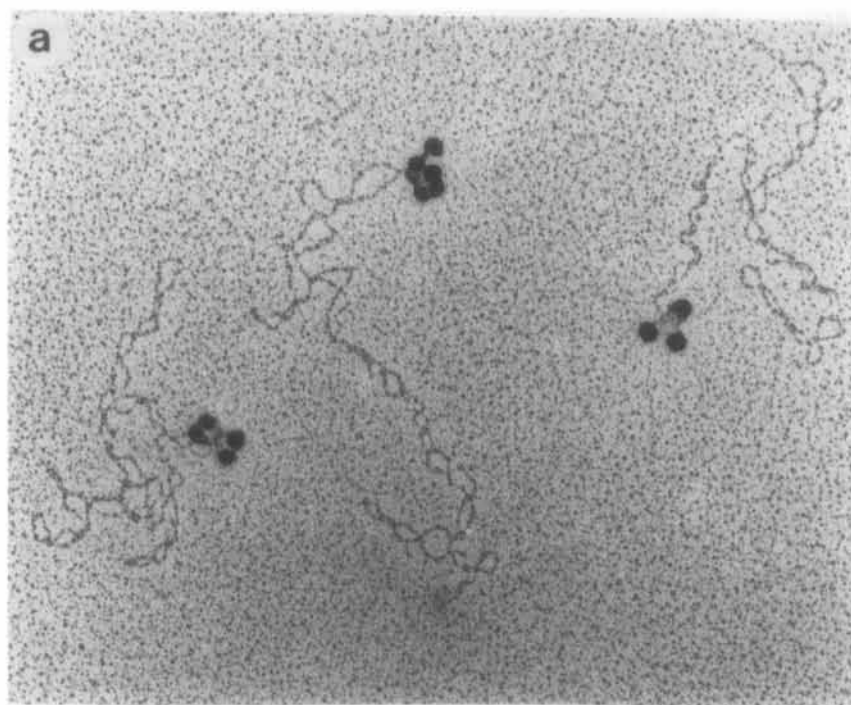
⁶⁸ T. Tsurimoto and K. Matsubara, *Nucleic Acids Res.* **9**, 1789 (1981).

⁶⁹ M. Furth and S. Wickner, in "Lambda II" (R. Hendrix, R. Weisberg, F. Stahl, and J. Roberts, eds.), p. 145. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1983.

⁷⁰ S. Wickner, *Cold Spring Harbor Symp. Quant. Biol.* **43**, 303 (1979).

⁷¹ C. Alfano and R. McMacken, *J. Biol. Chem.* **264**, 10709 (1989).

FIG. 4. Electron micrographs of nucleoprotein structures generated on supercoiled *ori* λ DNA followed by cleavage with restriction enzyme. *ori* λ plasmid DNA was incubated with O, P, DnaB, DnaJ, DnaK, and single-stranded DNA-binding proteins and ATP, and then cut with a restriction enzyme that cleaves asymmetrically with respect to the origin. (A) Presumed O-P-DnaB-DnaJ-DnaK complex. (B) Partially unwound nucleoprotein structure formed by the six-protein unwinding reaction. (C-E) Highly unwound structures formed by the six-protein unwinding reaction in the presence of DNA gyrase. Note that unwinding is unidirectional from the left, and that a large nucleoprotein complex, presumably the incompletely disassembled initiation structure, is visible at the left fork. The second DnaB needed for bidirectional unwinding is presumably stalled in the initiation structure (bar = 0.1 μ m). (Reprinted from Ref. 31.)



initiating DNA replication.^{72,73} Unwinding and replication are unidirectional^{31,72} (Fig. 4), which contrasts with the predominantly bidirectional mode of replication *in vivo*.⁷⁴ However, bidirectional unwinding and replication occur *in vitro* if a third heat-shock protein, GrpE, is included with the repertoire of components that yield the unidirectionally unwound structure.⁷⁵

Simple rotary shadowing could not reveal the actual protein composition of these mutiprotein replication intermediates. Nor did it reveal the function of the heat-shock proteins in the pathway to unwinding and initiation of replication other than to show a requirement for their participation. Immunoelectron microscopy was used to address these problems.³⁶ Nucleoprotein structures were probed with specific antibody and secondarily labeled with colloidal gold-protein A conjugates (Fig. 5). With this powerful analytical tool, the protein composition of the nucleoprotein complexes comprising each stage of the pathway leading to the prepriming complex was discerned, and the function of the heat-shock proteins was determined (Fig. 1). Immunoelectron microscopy clearly showed that the second stage nucleoprotein structure is in fact composed of O, P, and DnaB proteins bound to *ori* λ . This technique also showed that the heat-shock proteins DnaJ and DnaK associate with the *ori* λ -O-P-DnaB structure, resulting in the assembly of a further intermediate stage. In the presence of ATP, the bound heat-shock proteins function in a disassembly reaction to eject P and some O from the initiation structure, allowing DnaB to proceed rightward in its unwinding reaction, with the newly unwound DNA stabilized by single-stranded DNA-binding protein. The bidirectional unwinding reaction presumably requires more extensive disassembly mediated by the participation of GrpE, so that a second DnaB helicase can proceed leftward.

As for origin-specific initiation of DNA replication, the precise regulation of promoter-localized initiation of transcription is mediated by snups. A DNA-looping model has been proposed as a mechanism for both positively and negatively regulating transcription (reviewed in Ref. 28). In this

⁷² K. Mensa-Wilmot, R. Seaby, C. Alfano, M. S. Wold, B. Gomes, and R. McMacken, *J. Biol. Chem.* **264**, 2853 (1989).

⁷³ M. Zylitz, D. Ang, K. Liberek, and C. Georgopoulos, *EMBO J.* **8**, 1601 (1989).

⁷⁴ M. Schnos and R. Inman, *J. Mol. Biol.* **51**, 61 (1970).

⁷⁵ C. Vasilikiotis, D. Ang, C. Georgopoulos, and H. Echols, unpublished work (1991).

FIG. 5. Tagging of nucleoprotein structures with IgG antibody and *Staphylococcus* protein A-gold conjugate. (a) Gold-tagged O-somes on supercoiled *ori* λ plasmid DNA. (b) Gold-tagged O-somes on *ori* λ plasmid DNA linearized with restriction enzyme. (Reprinted from Ref. 36.)

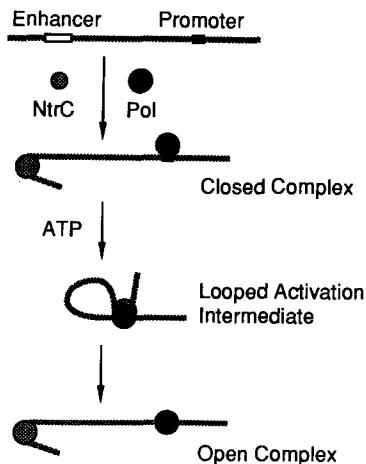


FIG. 6. Model for the action of NtrC from the enhancer site. NtrC protein binds to the enhancer and associates with promoter-bound RNA polymerase. The association facilitates the transition of RNA polymerase from an inactive closed complex into a productive open complex. Transcription of the *glnA* gene by polymerase ensues, and NtrC remains enhancer bound to initiate another round of activation. (Reprinted from Ref. 12.)

model, regulatory DNA sequences distant to the promoter exert their influence by first specifically binding to a regulatory protein. The DNA-bound regulatory protein then associates via direct protein–protein contacts with other regulatory proteins and/or with promoter-bound RNA polymerase to generate a snup. The snup then either represses or stimulates the initiation of transcription.

Transcriptional Activation of glnA Promoter from NtrC Enhancer Site

We used electron microscopy to directly visualize the central biochemical prediction of the DNA-looping model for the positive regulation of the prokaryotic *glnA* operon from its enhancer (Fig. 6).¹² The enhancer element of the *glnA* operon has two sites that tightly bind to the NtrC regulatory protein.⁷⁶ As with other enhancers, the *glnA* enhancer functions independently of distance and orientation to the promoter.⁷⁶ The NtrC regulatory protein facilitates the transition of promoter bound RNA polymerase from an inactive closed complex into an active open complex.⁷⁷ By electron microscopy, NtrC was seen bound at the enhancer, and RNA polymerase was seen bound at the promoter under conditions for open

⁷⁶ L. J. Reitzer and B. Magasanik, *Cell (Cambridge, Mass.)* **45**, 785 (1986).

⁷⁷ D. L. Popham, D. Szeto, J. Keener, and S. Kustu, *Science* **243**, 629 (1989).

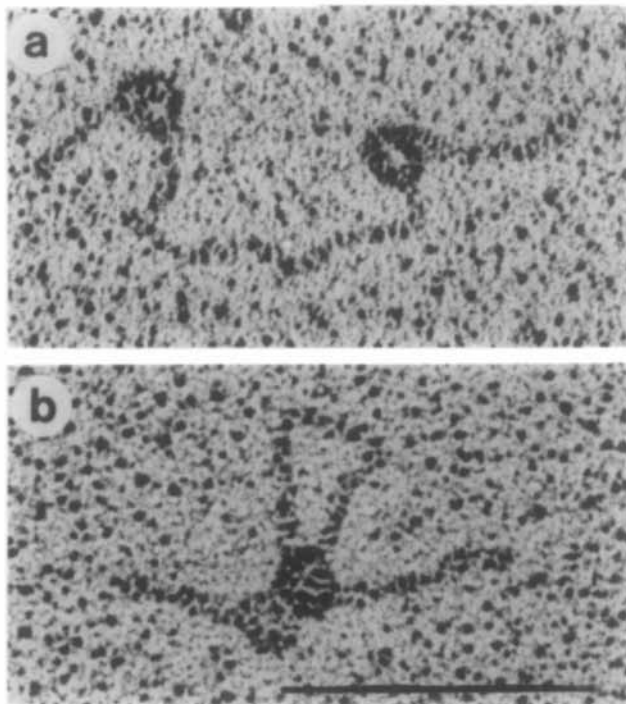


FIG. 7. Electron micrographs of NtrC bound to the enhancer and RNA polymerase bound to the promoter. (a) The product of the activation reaction in which NtrC is bound to the enhancer and RNA polymerase is stably bound to the promoter. (b) Loop formed by the association of enhancer-bound NtrC and promoter-bound RNA polymerase—the presumed activation intermediate. Note the higher electron density of the RNA polymerase, which presumably is shadowed more efficiently with tungsten than is NtrC (bar: 0.1 μm). (Reprinted from Ref. 12.)

complex formation (Fig. 7a). About 15% of these doubly bound DNA molecules exhibited a distinct looped structure in which the length of the loop was the engineered distance between the enhancer and promoter (Fig. 7b). Since these looped structures disappear when RNA polymerase is allowed to transcribe the DNA, it is likely that the structures are activation intermediates. The direct visualization of the heterologous interaction over distance between the DNA-bound NtrC regulatory protein and RNA polymerase shows that this enhancer system operates through a snu-type mechanism. Other workers have used electron microscopy to visualize DNA-looping between homologous transcriptional regulatory proteins.^{7–10}

Acknowledgments

We thank Chris Vasilikiotis and Wen Su for figures, Richard Eisner for editorial help, and Reid Johnson and Karen Heichman for expert advice. Research at Berkeley was supported by a grant from the National Institutes of Health (GM17078). M.D. was supported by Damon Runyon-Walter Winchell Cancer Fund Fellowship DRG-961.

[12] Analysis of DNA-Protein Interactions by Affinity Coelectrophoresis

By WENDELL A. LIM, ROBERT T. SAUER, and ARTHUR D. LANDER

Introduction

Gel retardation is a useful and convenient approach for analyzing protein-nucleic acid binding. Methods in current use have limitations, however, particularly when applied to the analysis of weak or rapidly dissociating interactions. We describe here a novel gel retardation method, referred to as affinity coelectrophoresis (ACE), which is generally applicable for studying protein-nucleic acid binding, and is particularly well suited for analyzing those interactions that are difficult to study using traditional gel retardation approaches. ACE is performed by electrophoresing a labeled nucleic acid fragment through an agarose gel with precast lanes containing protein at different concentrations. Unlike traditional gel retardation, binding is measured under equilibrium conditions, as the gel is running. The extent of binding can be measured from the extent of retardation in each lane.

In this chapter, we describe the theory and practice of ACE and its use in analyzing the binding of the N-terminal domain of the λ repressor protein to the O_L1 operator site. The advantages and disadvantage of ACE, as compared with other gel retardation methods, are also discussed.

Methods

ACE is performed using horizontal agarose gels that are cast with two types of wells: lanes, which are rectangular wells into which proteins mixed with low gelling temperature agarose are introduced and allowed to gel, and a slot, into which a nucleic acid sample is introduced just prior to electrophoresis. An apparatus for casting such gels may be assembled as shown in Fig. 1.

Briefly, a 75 × 100 mm Plexiglas casting tray ("t" in Fig. 1A) is

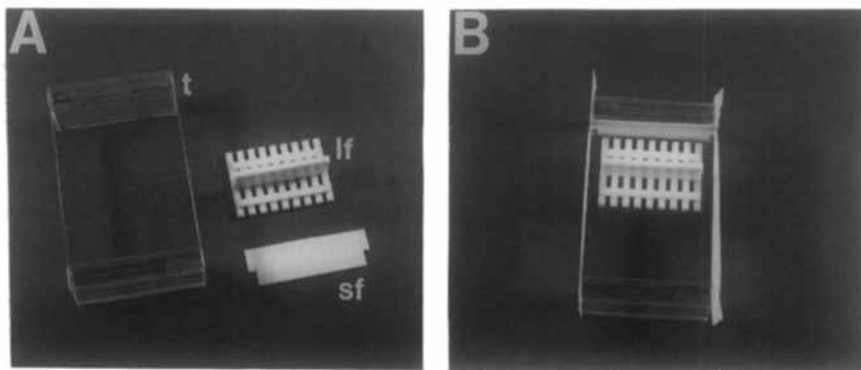


FIG. 1. Casting apparatus. (A) Gels are cast using a tray (t), a lane-forming comb (lf), and a slot-forming comb (sf), assembled as shown in (B). For details, see text.

fitted with a piece of GelBond film (FMC Bioproducts, Rockland, ME) and the sides are taped. Two types of combs are then placed into the tray. The lane-forming comb ("lf") is machined from a single piece of Teflon and consists of nine parallel bars, each $45 \times 4 \times 4$ mm, held together by cross-bracing with a spacing of 3 mm between each bar. This comb also contains a handle to permit its easy removal from formed gels. It is centered in the casting tray with the long sides of the rectangle parallel to the long sides of the tray.

The slot-forming comb is cut from a 25×75 mm rectangle of 1-mm thick teflon. Small 4.5×10 mm rectangles are removed from two corners to produce a comb with one 66-mm edge, as shown ("sf" in Fig. 1A). This comb is stood on its short edge, perpendicular to and at a distance of approximately 4 mm from the lane-forming comb (Fig. 1B). The slot-forming comb is stabilized in the upright position by pressing the tape used to seal the casting tray against the overhanging tabs of the comb.

One percent (w/v) low gelling temperature agarose (Bethesda Research Laboratories, Bethesda, MD) is prepared in electrophoresis buffer and heated until dissolved. In the experiments described below, a buffer of 50 mM potassium HEPES, pH 7.5, 1.5 mM CaCl_2 , 0.1 mM EDTA, and 50, 100, or 200 mM KCl (as indicated) was used. Molten agarose (15–20 ml) is poured into the casting tray and allowed to cool. After the agarose has fully gelled, both combs are removed. Removing the lane-forming comb requires care in order to avoid tearing the lanes or slot. The best approach is to slide the comb a few millimeters forward and backward several times to break the seal between the Teflon and the agarose, then to pivot it about one end until it can be lifted easily out of the lanes. This process is made easier if detergent [0.1% Nonidet

P-40 (NP-40)] or nonspecific protein (0.1 mg/ml bovine serum albumin) is added to the molten agarose before pouring. Alternatively, flooding the top of the gel with electrophoresis buffer will help lubricate the lane-forming comb during its removal.

Next, proteins are prepared for embedding in the lanes. Protein samples are diluted in electrophoresis buffer to twice their desired concentrations (if detergent or nonspecific protein is used in pouring the gel, it should also be added here). Approximately 0.32 ml of each protein sample is mixed with an equal volume of 2% (w/v) low gelling temperature agarose prepared in the same buffer, previously melted and held at 37°. Immediately after mixing, each sample is pipetted into a lane, filling until level with the rest of the gel (to ensure that the mixture does not gel prematurely, protein samples should be warmed to room temperature, or if desired, to 37°, prior to addition of molten agarose).

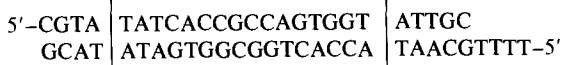
For measurement of protein–nucleic acid affinities, it is convenient to introduce the same protein sample, at nine different concentrations, into each of the lanes. In the experiments described here, the protein used was the N-terminal domain (residues 1–102) of the bacteriophage λ repressor protein (for review, see Sauer *et al.*¹), purified as described by Lim and Sauer.² Protein concentrations were determined using a calculated molar extinction coefficient at 276 nm of $7250\text{ M}^{-1}\text{ cm}^{-1}$. Protein samples were introduced into the gel lanes at final concentrations of 50, 100, 150, 200, 300, 400, 600, and 1200 nM.

After the samples in the lanes have solidified (typically 1–2 min), the gel is removed from the casting tray and submerged under buffer (minus detergent and protein) in the electrophoresis chamber. Gels are oriented with their long axis parallel to the direction of electrophoresis, with the slot end of the gel facing the cathode. Any appropriate-sized submarine gel box will do, provided that sufficient cooling and buffer recirculation can be provided. The experiments described here were performed using a Hoefer SuperSub (Hoefer Scientific Instruments, San Francisco, CA), which has built-in cooling and recirculation, and which can hold up to four ACE gels at once.

Radioactively labeled nucleic acid is then diluted into electrophoresis buffer containing a tracking dye (0.5% bromphenol blue) and 6% sucrose, and introduced into the slot by underlaying. Depending on the amount of agarose used in casting the gel, the slot will hold between 0.1 and 0.2 ml. In the experiments described here, a synthetic 29-bp double-stranded oligodeoxyribonucleotide was used. The sequence of the oligonucleotide was

¹ R. T. Sauer, S. R. Jordan, and C. O. Pabo, *Adv. Protein Chem.* **40**, 1 (1990).

² W. A. Lim and R. T. Sauer, *J. Mol. Biol.* **219**, 359 (1991).



The boxed region corresponds to the O_L1 operator site of bacteriophage λ . The duplex DNA was radiolabeled by filling in the four-nucleotide overhang using [α - 32 P]ATP and the enzyme Sequenase (U.S. Biochemical Corporation, Cleveland, OH).

After introduction of the nucleic acid sample, electrophoresis is carried out at 3–4 V/cm (90–120 V in the Hoefer SuperSub) for approximately 1.5–2 hr, or sufficient time for the nucleic acid to migrate most of the way through the lanes (once the mobility of a nucleic acid sample is known relative to bromphenol blue, electrophoresis times may be judged from the migration of the dye front). Resistive heating during electrophoresis will depend on buffer composition, but in buffers at or near physiological ionic strengths, internal cooling within the gel box or buffer chambers is necessary to avoid overheating. In the experiments described below, a flow of cold tap water through the coolant ports of the electrophoresis box was sufficient to maintain gel temperatures at approximately 22° throughout the runs.

After electrophoresis, gels are air dried and autoradiographed. Air drying may be accomplished using forced warm air (2–3 hr in a Hoefer Easy-Breeze), or a warm vacuum oven containing desiccant.

Results

Figure 2 illustrates the use of ACE to measure the affinity of a protein for a nucleic acid. The protein sample is cast in the nine lanes at concentrations ranging from high (far left) to low (far right), and labeled nucleic acid introduced into the slot (Fig. 2A). During electrophoresis, nucleic acid migrates through each of the lanes, slowing down in lanes containing high concentrations of binding protein. The amount by which nucleic acid migration is retarded is not the same in each lane but varies with the concentration of protein (Fig. 2B).

Figure 2C shows an example of this behavior. An amino-terminal fragment of λ repressor was included in each of the lanes at concentrations from 50 nM to 1.2 μ M, and a radiolabeled oligonucleotide containing the O_L1 operator site was introduced into the slot and electrophoresed. The buffer, in this case, contained 50 mM KCl. As shown, the mobility of the oligonucleotide was shifted progressively with increasing protein concentration, up to about 300 nM, above which no further retardation was observed.

The pattern of progressive electrophoretic mobility retardation displayed in Fig. 2 can be understood as resulting from the progressive

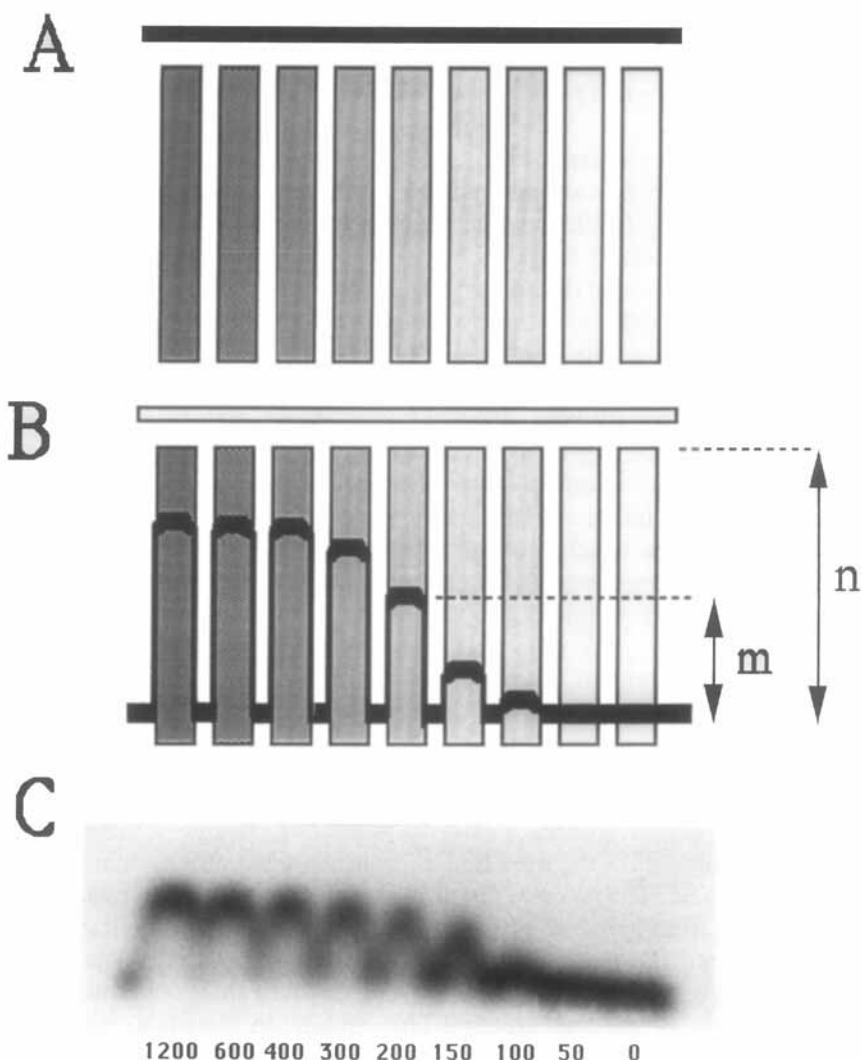


FIG. 2. Mobility retardation patterns obtained with ACE. (A) Diagram of ACE gel prior to electrophoresis. Proteins have been cast into each of the lanes at decreasing concentrations from left to right, and labeled DNA has been introduced into the slot. (B) Diagram of the same gel following electrophoresis. The DNA front has moved toward the anode (at bottom), and has been progressively retarded with increasing protein concentration. m and n , the measurements used in determining the value of R , are shown. Note that it is preferable to measure n from the top of the lanes, not from the slot. (C) Autoradiogram of an ACE gel in which the protein was an amino-terminal fragment of the λ repressor and the DNA was a ^{32}P -labeled oligonucleotide containing the O_L1 operator site (see text). The protein concentrations in the lane are given in nanomolar units. In this experiment the electrophoresis buffer contained 50 mM KCl.

saturation of nucleic acid with protein. At high protein concentrations, the observed electrophoretic mobility presumably corresponds to that of the nucleic acid-protein complex. At intermediate protein concentrations, each nucleic acid molecule spends only a fraction of its time bound to protein. Provided that the kinetics of association and dissociation are reasonably rapid relative to electrophoresis times, a single intermediate mobility should be observed. If there are effectively only two electrophoretically different states of the nucleic acid (i.e., a single bound state and single free state), then the fraction of time that the nucleic acid spends in the bound state should be the same as the fraction by which its observed mobility is shifted from that of free nucleic acid to that of the bound complex.

This observation may be stated more precisely after defining a unitless number, R , the retardation coefficient, with which to quantify mobility shifts. In any lane of an ACE gel, R is found by measuring the difference between nucleic acid migration in that lane and the migration of free nucleic acid (i.e., its migration in a lane containing no protein), and normalizing that value to the migration of free nucleic acid (in Fig. 2B, $R = m/n$). If R_∞ is taken to represent the maximum value of R (i.e., when the protein concentration is saturating), then the above argument suggests that, in any lane, $R/R_\infty = \Theta$, where Θ represents the fractional saturation of nucleic acid with protein (i.e., bound DNA/total DNA) in that lane.

Because the dependence of R on protein concentration should parallel the dependence of Θ on protein concentration, data from ACE can be analyzed using several common graphic methods to model the binding reaction and measure the equilibrium binding constant. The simplest model for binding is a first order reaction:



where P is protein and D is the DNA fragment. Since $\Theta = R/R_\infty$, the expression for binding can be written as

$$R = R_\infty/[1 + (K_d/[P])] \quad (1)$$

In general, it is more convenient to plot values for the total rather than the free protein concentration in each lane. In this case, Eq. (1) becomes

$$R = R_\infty/\{1 + [K_d/([P_{tot}] - [PD])]\} \quad (2)$$

which reduces to

$$R = R_\infty/[1 + (K_d/[P_{tot}])] \quad (3)$$

whenever the concentration of nucleic acid is low ($<0.1 K_d$). In practice,

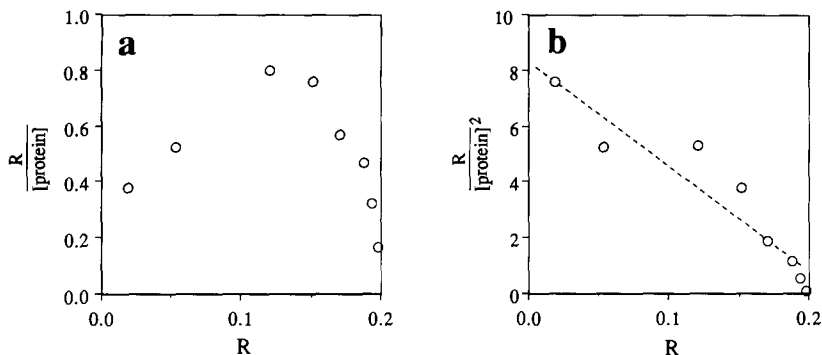


FIG. 3. Scatchard analysis. The data in Fig. 2C have been graphed on plots of R versus $R/[P_{\text{tot}}]$ (a) and R versus $R/[P_{\text{tot}}]^2$ (b). Only in (b) are the data points reasonably well fit by a straight line, suggesting that binding is effectively second order.

this condition is easy to meet when using radiolabeled nucleic acids, since picomolar amounts are usually sufficient to produce rapid autoradiographic results. In the experiment shown in Fig. 2C, the concentration of radiolabeled oligonucleotide was approximately 100 pM.

If the mechanism of binding is higher order, with more than one protein molecule binding in a strictly cooperative manner to each nucleic acid molecule, i.e.,



then Eq. (3) can be generalized to

$$R = R_{\infty}/[1 + (K_d/[P_{\text{tot}}]^n)] \quad (3')$$

where n equals the number of protein molecules binding to each molecule of nucleic acid.

Experimental data may be fit to Eq. (3') most accurately using nonlinear regression analysis, or transformed to the form of a Scatchard equation:

$$R/[P_{\text{tot}}]^n = -R/K_d + R_{\infty}/K_d$$

which permits the value of K_d to be calculated from the inverse slope of a plot of R versus $R/[P_{\text{tot}}]^n$.

Figure 3 presents the results of graphing the data from Fig. 2C on a plot of R versus $R/[P_{\text{tot}}]^n$ for $n = 1$ to model first order binding (Fig. 3A) and $n = 2$ to model second order binding (Fig. 3B). As shown, the data points in Fig. 3B are better fit by a straight line than those in Fig. 3A. This behavior is in agreement with the fact that the amino-terminal fragment of λ repressor, which is monomeric at the concentrations tested ($K_{\text{dimer}} =$

TABLE I
BEST FIT VALUES OF K_d AND R_x FOR THE
BINDING OF THE N-TERMINAL DOMAIN OF λ
REPRESSOR(1-102) TO O_L1 OPERATOR DNA^a

[KCl] (mM)	K_d (M^2)	R_x
50	1.9×10^{-14}	0.21
100	5.0×10^{-14}	0.17
200	9.8×10^{-14}	0.07

^a As shown in Fig. 4, assuming second order binding [Eq. (3'), $n = 2$]. Under the conditions of these experiments, half-maximal binding is observed at a total protein concentration equal to $\sqrt{K_d}$.

$3 \times 10^{-4} M$; Weiss *et al.*³), binds strongly to the O_L1 oligonucleotide only as a dimer (for review, see Sauer *et al.*¹).

Direct fits of the same data to Eq. (3') were also made by nonlinear least-squares analysis, and the derived curves for $n = 1$ and $n = 2$ are shown in Fig. 4A. Data were also analyzed from experiments in which the electrophoresis buffer contained either 100 or 200 mM KCl (Fig. 4B and C). It is clear that, in every case, the data are better fit by the second order than the first order model. The values of K_d derived from the best fit second order equations are given in Table I. Errors were calculated by assuming a 0.5-mm measurement error in the estimation of mobility shifts, and an insignificant error in the preparation of protein dilutions. Also presented in Table I are the derived values of R_x , a measure of the maximum degree of mobility retardation under each of the buffer conditions tested.

From Table I it can be seen that the affinity of the N-terminal domain for O_L1 DNA decreases (K_d increases) with increasing salt concentration (the binding of intact λ repressor to operator shows a similar salt sensitivity⁴). At 50 mM KCl, the calculated value of K_d is $1.9 \times 10^{-14} M^2$. This value is very close to the value of $1.2 \times 10^{-14} M^2$ that was measured under similar buffer conditions by DNase I footprinting analysis.⁵ It is also apparent that the value of R_x decreases with increasing salt concentration; the explanation for this phenomenon is not yet clear.

³ M. A. Weiss, C. O. Pabo, M. Karplus, and R. T. Sauer, *Biochemistry* **26**, 897 (1987).

⁴ H. C. M. Nelson and R. T. Sauer, *Cell (Cambridge, Mass.)* **42**, 549 (1985).

⁵ J. F. Reidhaar-Olson, D. A. Parsell, and R. T. Sauer, *Biochemistry* **29**, 7563 (1990).

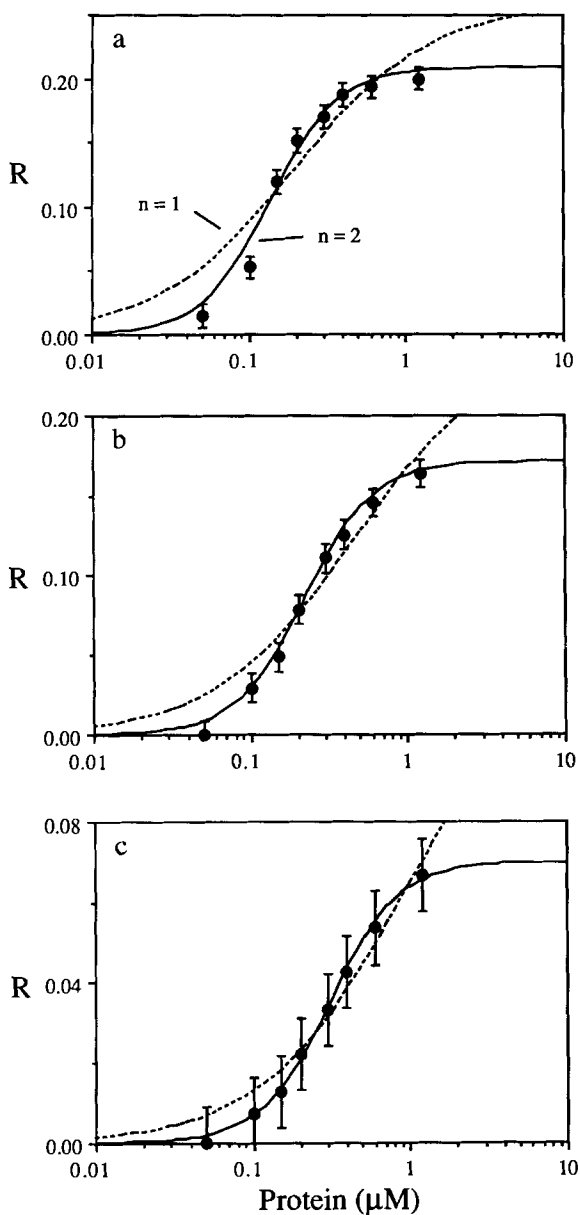


FIG. 4. Effect of KCl concentration on binding. R has been plotted as a function of protein concentration for the data shown in Fig. 2C (a), and for two other experiments in which the protein and DNA samples were identical, but the KCl concentration was increased from 50 mM (a) to 100 mM (b) and 200 mM (c). Nonlinear least-squares analysis was used to fit the data to first order ($n = 1$) or second order ($n = 2$) binding models, as described by Eq. (3'). Nonlinear analysis was performed using the NONLIN module of the statistical package SYSTAT (Systat, Inc., Evanston, IL).

Discussion

ACE is a simple electrophoretic method for analyzing protein–nucleic acid binding. In principle, ACE is a form of affinity electrophoresis,⁶ and is generally applicable to the study of macromolecular binding. For example, ACE has been used to study the binding of heparin, an anionic oligosaccharide, to a variety of proteins.⁷

Differences between ACE and Other Gel Retardation Methods

For analyzing protein–DNA interactions, ACE differs from gel retardation methods that are in current use (referred to below as “traditional gel retardation”) in several important ways.

Measurements are made under equilibrium conditions. Traditional gel retardation assays require separation of free DNA from DNA–protein complexes. When the dissociation of such complexes is too rapid, traditional gel retardation cannot be used (see [8], this volume). In contrast, the mobility shifts that occur in ACE develop without disturbing equilibrium; rapid dissociation therefore does not create difficulties. In practice, this means that ACE is well suited to measuring moderate to weak affinities. The data reported above illustrate this point. The amino-terminal fragment of the λ repressor exhibits a relatively weak affinity for DNA and its binding to operator is difficult to quantify by traditional gel retardation (W. A. Lim and R. T. Sauer, unpublished observations, 1991). In theory, there is no lower limit to the affinities that can be measured by ACE, although the ability to prepare lanes containing proteins at sufficiently high concentrations may be limited by the availability of material, or by protein solubility. Also, at very high protein concentrations (e.g., >10 mg/ml), the effects of protein on the electric field strength within the gel may become nonnegligible. So far, in the λ repressor system, values of K_d as low as $6 \times 10^{-9} M^2$ (indicating that half-maximal saturation occurs at $80 \mu M$) have been measured by ACE (W. A. Lim, R. T. Sauer, and A. D. Lander, unpublished observations, 1991).

In ACE, data are derived from measurements of electrophoretic mobility, rather than film densitometry or radioactive counting. In traditional gel retardation assays, quantitative measurements of binding require determining the fraction of radioactivity in electrophoretic bands representing free or bound nucleic acid. If film densitometry is used for this purpose, errors can result from nonlinearity or film saturation. In the event that the

⁶ V. Horejsi, *Anal. Biochem.* **112**, 1 (1981).

⁷ M. K. Lee and A. D. Lander, *Proc. Natl. Acad. Sci. U.S.A.* **88**, 2768 (1991).

nucleic acid fragment used in a traditional gel retardation assay contains a subpopulation of inactive molecules (apparently a common occurrence with synthetic oligonucleotides), corrections to the data must be made (cf. [8], this volume). In contrast, data from ACE are obtained from the electrophoretic pattern alone. Measurements can be made manually, and measurement errors estimated from the widths of the bands. Labeled, nonbinding molecules have no influence on the pattern of mobility shifts, but rather migrate independently as an unshifted front (W. A. Lim, R. T. Sauer, and A. D. Lander, unpublished observations, 1991).

ACE can measure binding over a wide range of conditions. Traditional gel retardation assays require the use of relatively sieving gels (e.g., 10% acrylamide) and high electric field strengths (at least during the time that samples enter the gel). In theory, the porosity of the gel matrix and the field strength can both affect binding. ACE is carried out in 1% agarose, a relatively nonsieving medium, and at low field strengths (e.g., 3–4 V/cm; experiments may also be carried out to test the effect of voltage on binding). Provided that an apparatus with adequate cooling is used, ACE may also be carried out in buffers of widely varying ionic strength (cf. Fig. 4).

ACE can fractionate oligonucleotides by affinity. If a mixture of oligonucleotides is electrophoresed through lanes containing a nucleic acid-binding protein, each oligonucleotide will produce a pattern of shifts reflecting its individual affinity. By recovering, amplifying, and identifying (sequencing) DNA molecules located at different positions within the lanes of an ACE gel, the relative affinities of multiple DNA sequences for DNA-binding proteins can be assessed simultaneously. By extension, this method can be used to identify, from a mixture of completely random oligonucleotides, those sequences that bind best to any DNA-binding protein (it has been reported that this type of approach can also be used with traditional gel retardation⁸). The need to be able to determine directly the DNA sequences to which proteins bind is likely to increase, as more DNA-binding proteins are identified simply on the basis of homology to other known DNA-binding proteins.

Limitations on the Uses of ACE

Although ACE should be generally applicable to the study of protein–nucleic acid interactions, there are several limitations.

The amount of protein used in ACE is likely to be greater than that

⁸ T. K. Blackwell and H. Weintraub, *Science* **250**, 1104 (1990).

used in traditional gel retardation assays. In measuring affinity either by ACE or by traditional gel retardation, the same series of protein concentrations will be required, but significantly larger volumes will be needed for ACE, in order to fill the lanes. In the experiments described above, lanes of 45×4 mm were used and were filled to a height of approximately 3 mm (0.54 ml final volume). Protein samples may be conserved by reducing gel height (thereby reducing the amount of labeled DNA that can be loaded, and increasing the time required for autoradiography), or by using a lane-forming comb that produces shorter lanes and electrophoresing for a shorter time. The minimum lane length and electrophoresis time required for an ACE experiment depends on how substantial the mobility shifts are. In general, the greatest mobility shifts will occur when the oligonucleotides used are small and the proteins are large and/or basic. The data in Table I also suggest that mobility shifts are increased by decreasing the ionic strength. In the experiments described above, the protein was rather small ($M_r \sim 11,000$) and slightly acidic (Coomassie blue-stained ACE gels revealed a small degree of protein migration toward the anode), suggesting that many DNA-binding proteins could be expected to produce significantly greater mobility shifts. Thus for many ACE experiments, shorter lanes may be adequate. It should also be pointed out that, although maximal mobility shifts can be achieved by carrying out electrophoresis until all DNA has passed completely through each lane, Eq. (3') is valid only when electrophoresis has been stopped while the free oligonucleotide front remains within the region of gel containing the proteins (a method is described below to correct for errors that arise when this condition is not met).

Protein-nucleic acid complexes that dissociate very slowly may fail to produce useful patterns in ACE gels. The assertion that ACE measures binding at equilibrium depends on the assumption that equilibrium is reasonably rapid compared with the time of electrophoresis. The effects of binding kinetics on affinity-electrophoretic separations in general have been modeled previously⁹; the results imply that, in ACE gels, slow kinetics will be manifested as spreading of the electrophoretic bands observed at intermediate R values. In general, significant spreading should occur when the value of $1/k_{\text{off}}$ is at least of the same order as the electrophoresis time. In practice, with electrophoresis times in the hours, only very high-affinity interactions would be expected to display this effect.

Exposure to molten low gelling temperature agarose might lead to denaturation of some proteins. Very heat-labile proteins, particularly at

⁹ V. Matousek and V. Horejsi, *J. Chromatogr.* **245**, 271 (1982).

high concentrations, may be intolerant of spending even a short time at 37° (the temperature of the agarose that is mixed with protein samples prior to addition to the lanes of an ACE gel).

Graphical Analysis of ACE Data

As described in the previous section, by using Eqs. (3) or (3') or linear transformations of them, data from ACE experiments may be manipulated graphically so as to model binding and derive values for K_d and R_∞ . The validity of these equations requires that several conditions be met, only some of which may be under the control of the investigator. These conditions are enumerated below. Where appropriate, corrections or alternate formulas are presented that may be used when these equations are not valid.

The nucleic acid concentration should be low compared with the value of K_d . As already described, the substitution of total for free protein concentration in the derivation of Eqs. (3) and (3') is valid only when the concentration of nucleic acid is significantly lower than the value of K_d . The effect of a higher DNA concentrations on the relationship between R and $[P_{tot}]$ may be illustrated by transforming Eq. (2) into Scatchard form:

$$\frac{R}{[P_{tot}]} = \frac{-R}{K_d + [D_{tot}](1 - R/R_\infty)} + \frac{R_\infty}{K_d + [D_{tot}](1 - R/R_\infty)}$$

When $[D_{tot}]$ is significant compared with K_d , a plot of R vs $R/[P_{tot}]$ will be concave downward, with a slope that depends on the values of K_d , $[D_{tot}]$, and R/R_∞ .

Electrophoresis should be terminated before the nucleic acid front has migrated completely through any of the protein-containing zones. The retardation coefficient R is defined as the mobility shift normalized to the migration of the free oligonucleotide (m/n in Fig. 2B). Normalization assures that R should not depend on the length of time of electrophoresis. However, if electrophoresis is conducted long enough so that nucleic acid molecules pass completely through protein containing zones, the nucleic acid molecules will speed up again to their free mobility, and calculated values of R will fall with further running. Because nucleic acid fronts will exit the protein-containing regions at different times (the most retarded will exit last), values of R will be changed by different amounts for each lane. It therefore becomes necessary in the case of such "overruns" to correct R values to what they would have been at the time just before the emergence of the nucleic acid fronts. It should be pointed out that overruns

are measured not by whether the free nucleic acid front has traveled beyond the boundaries of the cast *lane*, but by whether it has traveled past the position of the *protein*, which was initially cast within the lane, but which may have moved somewhat, either toward the anode or toward the cathode, during electrophoresis. Accordingly, to determine the exact amount by which to correct R , it is necessary to know how much the proteins in the lanes move during electrophoresis. In many cases, this can be determined easily by protein staining. Dried ACE gels may be stained by immersing in Coomassie Brilliant Blue [2.5 g/l in 45% (v/v) methanol, 5% (v/v) acetic acid] for 5 min, followed by destaining in 25% methanol. Protein migration can then be measured as the amount by which the stained rectangles are shifted from the lanes themselves (with anodal migration taken as positive and cathodal as negative). Corrected values of R may then be determined as follows:

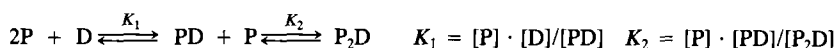
$$\begin{aligned} \text{If } R < 1 - (L + p)/n, \quad R_{\text{corr}} &= R[(n - p)/(L + m)] \\ \text{If } R \geq 1 - (L + p)/n, \quad R_{\text{corr}} &= R \end{aligned}$$

where L stands for the lane length, p for the measured amount of protein migration, R_{corr} for the corrected values of R , and m , n , and R are as defined previously. Thus, although it is simpler to obtain usable values of R when overruns are avoided, it is relatively easy to correct for the effects of overruns. Indeed, since overruns give the largest possible mobility shifts for any given lane length, it may be desirable when trying to conserve on protein to cast short lanes and deliberately overrun the nucleic acid front.

Nucleic acid and protein should combine to form only a single type of complex. Equating R/R_∞ with Θ presupposes that there are only two electrophoretic species in equilibrium, a bound species, which has one mobility and a free species, which has a greater mobility. This assumption can break down when more than one protein molecule can bind to a single nucleic acid molecule, or when more than one nucleic acid molecule can bind to a single protein. In such cases, each type of complex will possess its own electrophoretic mobility, and observed mobility shifts will be a weighted average of these mobilities. More precisely, if there are n such bound species, and θ_i represents the fractional saturation of the i th species, and r_i represents the mobility shift associated with the i th species (i.e., the mobility of the i th species subtracted from and normalized to the mobility of free nucleic acid), then it should be the case that

$$R = \sum_{i=1}^n \theta_i r_i$$

By substituting into appropriate equilibrium equations, one can determine the appropriate relationship between R and $[P_{\text{tot}}]$. It is important to note, however, that the relationship obtained will not necessarily match binding curves obtained by other techniques. For example, if a molecule of nucleic acid is capable of binding either one or two molecules of protein, i.e.,



then the relationship of Θ to $[P_{\text{tot}}]$ is described by

$$\Theta = \frac{1 + \frac{K_2}{[P_{\text{tot}}]}}{1 + \frac{K_2}{[P_{\text{tot}}]} + \frac{K_1 K_2}{[P_{\text{tot}}]^2}} \quad (4)$$

where Θ is defined as bound DNA over total DNA. In contrast, in ACE, the relationship of R to $[P_{\text{tot}}]$ will be

$$R = r_1 \left(\frac{\frac{r_2}{r_1} + \frac{K_2}{[P_{\text{tot}}]}}{1 + \frac{K_2}{[P_{\text{tot}}]} + \frac{K_1 K_2}{[P_{\text{tot}}]^2}} \right) \quad (5)$$

The isotherms defined by Eqs. (5) and (4) will be equivalent only when $r_2/r_1 = 1$. In practice, the value of r_2/r_1 is likely to vary between 1 and 2, depending on the size of the DNA fragment and the properties of the protein being studied. For example, with a very large protein, binding of just a single protein molecule to the DNA is likely to result in nearly maximal retardation, and little further change should occur on binding of a second protein molecule; as a result, r_2/r_1 would be very close to 1. Conversely, with a very small protein, the mobility shift induced by the binding of a single molecule to DNA is likely to be such that a second molecule of protein shifts the complex almost as much as the first; as a result, r_2/r_1 should be around 2 [it is noteworthy that, when $r_2/r_1 = 2$, Eq. (5) generates an isotherm equivalent to what is obtained if Θ is expressed in terms of bound *protein* over total DNA, rather than bound *DNA* over total DNA].

Acknowledgments

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[13] Laser Cross-Linking of Protein-Nucleic Acid Complexes

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PETER H. VON HIPPEL

Introduction

Ultraviolet (UV) light-induced photochemical cross-linking of proteins to nucleic acids has been demonstrated for a variety of nucleic acid-binding proteins.¹⁻¹⁶ Irradiation with UV light brings about the formation of a "zero-length" covalent bond between the protein and the nucleic acid to which it is bound, thereby "freezing" the interaction between the two biopolymers. Many model systems have been studied in attempting to understand the chemistry of this photochemical process.¹⁷⁻²⁵ These studies have shown that the mechanistic pathways involved are often complex.

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- ¹¹ A. Havron and J. Sperling, *Biochemistry* **16**, 67 (1977).
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- ¹⁷ K. C. Smith, *Biochem. Biophys. Res. Commun.* **34**, 354 (1969).
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- ²⁰ A. E. Reeve and T. R. Hopkins, *Photochem. Photobiol.* **31**, 297 (1980).
- ²¹ A. E. Reeve and T. R. Hopkins, *Photochem. Photobiol.* **31**, 413 (1980).
- ²² M. D. Shetlar, K. Hom, J. Carbone, D. Moy, E. Steady, and M. Watanabe, *Photochem. Photobiol.* **39**, 135 (1984).
- ²³ M. D. Shetlar, J. Carbone, E. Steady, and K. Hom, *Photochem. Photobiol.* **39**, 141 (1984).
- ²⁴ K. C. Smith and D. H. C. Meun, *Biochemistry* **7**, 1033 (1968).
- ²⁵ M. D. Shetlar, J. Christensen, and K. Hom, *Photochem. Photobiol.* **39**, 125 (1984).

that reaction yields vary widely, and that competing processes may intervene. However, these studies have also shown that any amino acid residue can, in principle, be induced to form a UV-induced covalent cross-link with any nucleotide residue of DNA or RNA. Thus, despite the fact that reaction mechanisms are not fully understood, the introduction of covalent bonds between proteins and nucleic acids by UV irradiation represents a useful and reproducible technique that has been applied successfully *in vivo* and *in vitro*.^{1-16,26-32}

Our primary motivation for developing a UV laser-induced version of this cross-linking methodology was to find a way to study dynamic aspects of rapid (millisecond) protein–nucleic acid interactions under conditions that freeze the “instantaneous” (on the time scale of the reaction) binding equilibria involved. Providing that the equilibria could be frozen, we also hoped to use such procedures to characterize thermodynamically and kinetically the sequential interactions that occur within protein–nucleic acid complexes of biological interest.

Our initial studies were carried out using purified components of the bacteriophage T4 DNA leading strand replication system.^{33,34} This system is capable of incorporating several hundred nucleotide residues per second into new DNA.³⁵ Thus each template-dependent one-step reaction cycle, which involves a number of sequential reactions to insert a single nucleotide residue at the 3' end of the nascent DNA chain, must go to completion in milliseconds. Clearly, such processes are not amenable to analysis by conventional UV cross-linking methodology, which involves the use of fairly low-power, broad-band germicidal lamps and requires many minutes of UV irradiation to produce reasonable quantities of product.

Consequently we decided to develop a cross-linking procedure based on laser irradiation, which could, in principle, introduce many photons

²⁶ D. S. Gilmour and J. T. Lis, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 4275 (1984).

²⁷ D. S. Gilmour and J. T. Lis, *Mol. Cell. Biol.* **5**, 2009 (1985).

²⁸ D. S. Gilmour and J. T. Lis, *Mol. Cell. Biol.* **7**, 3341 (1987).

²⁹ J. P. Schouten, *J. Biol. Chem.* **260**, 9916 (1985).

³⁰ D. Angelov, V. Y. Stefanovsky, S. I. Dimitrov, V. R. Russanova, E. Keskinova, and I. G. Pashev, *Nucleic Acids Res.* **16**, 4525 (1988).

³¹ G. Dreyfuss, Y. D. Choi, and S. A. Adam, *Mol. Cell. Biol.* **4**, 1104 (1984).

³² S. Piñol-Roma, S. A. Adam, Y. D. Choi, and G. Dreyfuss, this series, Vol. 180, p. 410.

³³ C. F. Morris, H. Hama-Inaba, D. Mace, N. K. Sinha, and B. M. Alberts, *J. Biol. Chem.* **254**, 6787 (1979).

³⁴ N. G. Nossal and B. M. Peterlin, *J. Biol. Chem.* **254**, 6032 (1979).

³⁵ B. Alberts, C. F. Morris, D. Mace, S. Sinha, M. Bittner, and L. Moran, in “DNA Synthesis and Its Regulation” (M. Goulian, P. Hanawalt, and C. F. Fox, eds.), p. 241. Benjamin, Menlo Park, California, 1975.

into the system in very short periods of time.^{36,37} The laser cross-linking techniques presented in this chapter are fast and efficient, and can be used to link a wide variety of binding proteins (specific or nonspecific) to either single- or double-stranded nucleic acids. This technology has proven useful in exploring both static and dynamic protein–nucleic acid complexes. It can yield information about protein–nucleic acid binding constants and binding site sizes, and can be used to define actual points of contact between protein and nucleic acid species within complexes. Under some conditions this method can also be used to examine protein–protein interactions, protein–ligand interactions, and some aspects of nucleic acid structure. Procedures for designing experiments to yield such information are described below.

Experimental Procedures

Lasers and Irradiation Techniques

In principle a laser is merely another source of ultraviolet light, albeit more powerful than standard UV lamps. Its main advantage for our purposes is that it delivers this power very rapidly in monochromatic form. Although we have used a number of different lasers in our work, we find that a neodymium–yttrium–aluminum–garnet (Nd:YAG) laser (model DCR-3G, purchased from Spectra-Physics, Inc., Mountain View, CA) is the most practical. The DCR-3G laser emits light in a Gaussian mode, with the most intense light located at the center of the beam. We have also used a Spectra-Physics DCR-2 Nd:YAG laser successfully, but it should be noted that the DCR-2 emits light in a collimated output with a “hole” or “dot” directly in the middle of the beam; i.e., the beam is shaped like a doughnut. When using the DCR-2, the DCR-3D, or any laser with a “dot” profile, it is important to recognize that the most intense light is *not* located at the center of the beam and that samples must be positioned accordingly.

A schematic diagram of the laser setup we usually use is shown in Fig. 1. The power delivered is dependent on the number of pulses per unit time, and each Spectra-Physics Nd:YAG laser is optimized to operate at 1, 10, 20, or 30 pulses per second (Hz). The Nd:YAG lasers used for the studies presented here had repetition rates of 10 Hz. The Nd:YAG laser produces pulsed radiation at 1064 nm, with a pulse energy of 900 mJ (110-MW peak, DCR-3G). This high peak power permits efficient conver-

³⁶ C. A. Harrison, D. H. Turner, and D. C. Hinkle, *Nucleic Acids Res.* **10**, 2399 (1982).

³⁷ J. W. Hockensmith, W. L. Kubasek, W. R. Vorachek, and P. H. von Hippel, *J. Biol. Chem.* **261**, 3512 (1986).

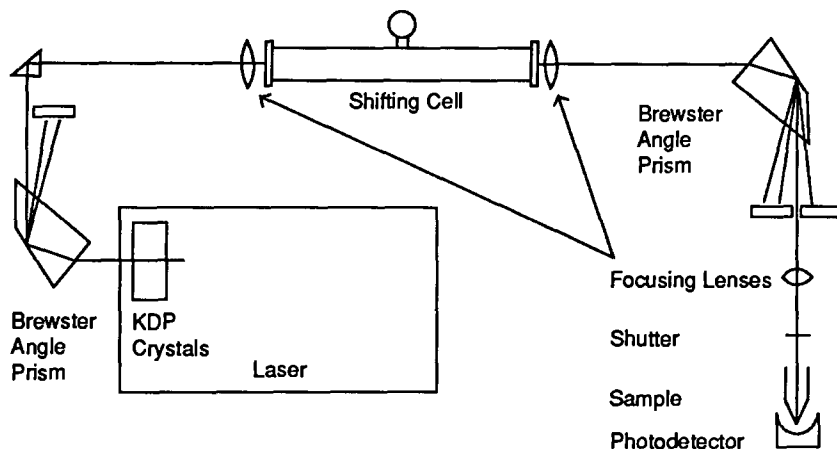


FIG. 1. Schematic diagram of the laser irradiation apparatus. The H_2 -shifting cell is shown for completeness. It is not needed and can be omitted from the setup when protein-nucleic acid cross-linking is performed at an irradiation wavelength of 266 nm. See text for details.

sion of the 1064-nm radiation to 532-, 355-, and 266-nm photons, using a combination of nonlinear optical (potassium dihydrogen phosphate or potassium dideuterophosphate) "doubling" crystals that are generally provided as part of the apparatus. The average power output generated at these wavelengths is typically 360, 160, and 80 mJ/pulse, respectively. Pulses in the UV are typically about 5 nsec in duration.

Laser frequencies other than the "primary" ones listed above are generated by focusing one of the primary beams into a wavelength-shifting cell, which is filled with hydrogen gas at 10 atm.³⁸ The wavelengths of the emergent beams are shifted from that of the incident radiation to frequencies determined by the vibration spectrum of the hydrogen gas. This stimulated Raman-shifting process produces radiation that is displaced in both the Stokes (lower frequency) and anti-Stokes (higher frequency) directions relative to the incident radiation. These shifted beams carry significantly less power than does the incident irradiation. Wavelengths of 282.3, 252.7, and 228.7 nm are obtained using anti-Stokes shifts of the 532-nm band; wavelengths of 309.1, 273.9, 245.9, 223.1, and 204.2 nm result from anti-Stokes shifts of the 355-nm band; and wavelengths of 239.5 and 217.8 nm are obtained from anti-Stokes shifts of the 266-nm band.

Radiation of the desired frequency is isolated using a dispersing quartz prism (see Fig. 1). We use Brewster angle prisms (also known as modified

³⁸ S. P. A. Fodor, R. P. Rava, R. A. Copeland, and T. G. Spiro, *J. Raman Spectrosc.* **17**, 471 (1986).

Pellin Broca prisms and obtainable from Continental Optical Corporation, Hauppauque, NY) to turn the beam by 90° while maintaining good dispersion. The beam is directed through a focusing lense, through a shutter, and into the sample. The sample itself is held by capillary action in the bottom of an open, horizontal Eppendorf tube. The desired incident radiation is focused just in front of the sample surface, so that the diverging circular beam strikes the full surface area of the solution (~3-mm diameter for a 10- μ l sample).

This configuration was selected to minimize differences in focal position of the lens as a function of wavelength. Operating the laser in the constant pulsing mode produces the best pulse-to-pulse energy stability ($\pm 4\%$ at 266 nm). Since the laser pulses continuously, it was necessary to devise a shutter apparatus that could withstand many pulses of irradiation, but could also operate rapidly enough to select and let through a single pulse from the incident "photon stream." We have found that a homemade shutter composed of a copper plate attached to a tubular solenoid (Guardian Electric Manufacturing Co., Chicago, IL, Cat. #LT4x12-C-12D) that is synchronized with the laser and controlled by a momentary switch is most efficient and cost effective. The copper plate is simply attached to the solenoid with epoxy and can be replaced as necessary.

The relative power at each irradiation wavelength was determined using a volume-absorbing disk calorimeter (Sciencetech, Inc., Boulder, CO, model 380103 or 380105) in conjunction with a power indicator (Sciencetech model 36-2002). At very low power levels we typically use an isoperibol enclosure (Sciencetech model 36-0203) around the calorimeter to protect it from ambient heat exchange due to air currents and incidental thermal input from the environment. In addition, we use a high-speed photodetector (Sciencetech model 301-020) in conjunction with a 520-MHz frequency counter (B&K Precision, Chicago, IL, model 1851) to count the number of pulses introduced into the sample.

While some of this sounds complicated, we emphasize that most laboratories that use lasers for spectroscopy will have much of this equipment. As a consequence molecular biologists interested in using this technique can often make arrangements to irradiate their samples in the laboratories of colleagues, and thus need not purchase all this apparatus. In typical applications only a few minutes of actual irradiation time are required to produce enough cross-linked material to occupy days or even weeks of subsequent analysis.

Proteins and Nucleic Acids

The proteins and nucleic acids used in the "calibrating" studies of this methodology that are described or mentioned here were prepared as follows. Bacteriophage T4 gene 43, 45, and 44/62 proteins were prepared

by a modification of the method of Morris *et al.*³³ T4 gene 32 protein was prepared according to the procedures of Nossal and Bittner *et al.*^{39,40} The proteolytic degradation products of gene 32 protein (G32P*I and G32P*III) were purified according to Lonberg *et al.*⁴¹ *Escherichia coli* ρ protein was prepared according to Finger and Richardson⁴²; *E. coli* RNA polymerase was prepared by the method of Burgess and Jendrisak,⁴³ and *E. coli* RecA protein was a gift of Dr. Stephen Kowalczykowski (Northwestern University Medical School, Chicago, IL). *Escherichia coli* DNA polymerase I was purchased from New England Biolabs (Beverly, MA) bacteriophage T4 polynucleotide kinase from New England Nuclear (Boston, MA) and horse heart myoglobin from Sigma (St. Louis, MO). Oligonucleotides were purchased from either Pharmacia (Piscataway, NJ) or New England Biolabs. The oligo(rC) and oligo(rU) series of oligonucleotides were a gift of Dr. James McSwiggen (USB, Cleveland, OH). Reagents for electrophoresis were purchased from International Biotechnologies (New Haven, CT) and [γ -³²P]ATP from New England Nuclear. Oligonucleotides were 5'-³²P end-labeled according to the method of Maxam and Gilbert.⁴⁴

Analysis of Cross-Linking Experiments

Sample Volumes and Beam Attenuation by Other Adsorbing Materials. In our experiments samples have been irradiated at volumes ranging from 10 to 300 μ l. Within this range the sample volume is not critical. However, the absolute number of "absorbing centers" (i.e., macromolecules with large extinction coefficients at the irradiation wavelength) within the sample is critical, since this directly affects the number of incident photons that are available to bring about cross-linking. This issue is discussed further below (see Fig. 11), but represents an important aspect of setting up an experiment and is here illustrated with an example. If one is irradiating a typical protein-nucleic acid sample of volume 10 μ l, 0.25 μ M nucleic acid and 0.4 μ M protein at 266 nm, the presence of (e.g.) $\sim$ 1 mM ATP will reduce the efficiency of cross-linking by 50% compared with a no-ATP-containing control, because (averaged over the sample volume) the ATP will absorb 50% of the incident photons. Thus dilution of the sample should not present a problem, but holding the macromolecule concentration con-

³⁹ N. G. Nossal, *J. Biol. Chem.* **249**, 5668 (1974).

⁴⁰ M. Bittner, R. L. Burke, and B. M. Alberts, *J. Biol. Chem.* **254**, 9565 (1979).

⁴¹ N. Lonberg, S. C. Kowalczykowski, L. S. Paul, and P. H. von Hippel, *J. Mol. Biol.* **145**, 321 (1981).

⁴² L. R. Finger and J. P. Richardson, *Biochemistry* **20**, 1640 (1981).

⁴³ R. R. Burgess and J. J. Jendrisak, *Biochemistry* **14**, 4634 (1975).

⁴⁴ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

stant while increasing the volume may decrease the number of photons absorbed by other "absorbing centers" (e.g., ATP) present at constant concentration.

Separation and Identification of Cross-Linked Species. After irradiation, a variety of strategies may be used for separating cross-linked macromolecules, including filter binding, phenol extraction, column chromatography, and gel electrophoresis. We generally use relatively small oligonucleotides (single- or double-stranded) when we are trying to quantify cross-linking and establish the identity of a cross-linked protein subunit derived from an irradiated complex containing multiple protein species. This allows us to use the rate of migration of the cross-linked protein in a denaturing polyacrylamide gel to identify the protein species involved. Alternatively, nuclease digestion of larger, internally labeled oligonucleotide adducts prior to gel electrophoresis separation can also be used.

Typically, we work with 10- μ l sample volumes, which are denatured by heating to 100° in 40 mM Tris-HCl, pH 6.8, 4% 2-mercaptoethanol, 1.6% (w/v) sodium dodecyl sulfate (SDS), and 10% (w/v) glycerol. Gel electrophoresis in the presence of SDS is performed essentially as described by Dreyfuss *et al.*,⁴⁵ using a 4% stacking gel and a 10% separating gel. The gels were run at a constant temperature of 37°, using a heater apparatus consisting of an aluminum plate, gasket, Plexiglas spacer, and Plexiglas sheet sandwiched such that water could be circulated between the Plexiglas sheet and the aluminum plate. The water temperature is controlled with a Lauda circulator and the aluminum surface of the heater plate is placed against one glass surface of the gel apparatus. This assembly reduces the background radioactivity in the gel and is critical for the analysis of some cross-linked complexes containing double-stranded DNA (SDS may promote renaturation or aggregation of certain DNAs). The gels are always prepared on a plastic support (GelBond) supplied by the FMC Corporation (Princeton, NJ). The plastic support allows the gels to be frozen and thawed without distintegrating and also preserves the size and shape of the gel during staining procedures, thereby allowing direct comparison to the autoradiographs. After electrophoresis, gels were covered with Saran wrap and autoradiography was performed at -80° using Kodak (Rochester, NY) XAR-5 film and Du Pont Cronex Lightning-Plus enhancing screens (X-Ray Corp., Portland, OR). Following autoradiography, the gels were stained with silver and dried.⁴⁶

Silver Staining and Acid Lability of Cross-Links. Staining is always carried out *after* autoradiography, because the cross-linked adducts are

⁴⁵ G. Dreyfuss, S. A. Adam, and Y. D. Cho, *Mol. Cell. Biol.* **4**, 415 (1984).

⁴⁶ J. H. Morrissey, *Anal. Biochem.* **117**, 307 (1981).

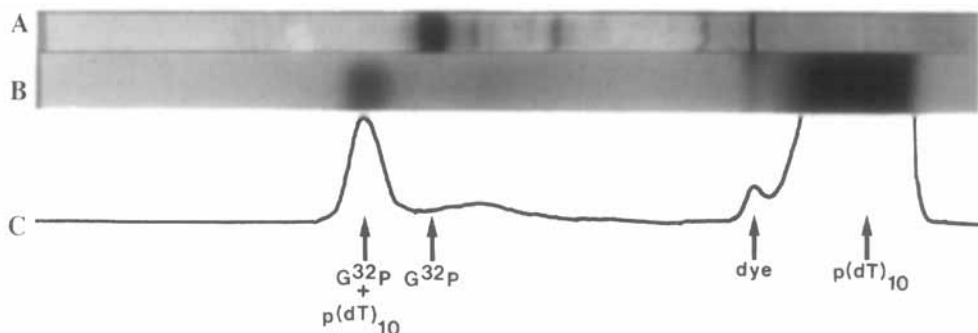


FIG. 2. Silver-stained polyacrylamide gel (A), autoradiogram (B), and densitometric scan of autoradiogram (C) for a typical photocross-linking experiment. Reaction conditions were 25 mM Tris-acetate, pH 7.5, 60 mM potassium acetate, 6 mM magnesium diacetate, 5 mM 2-mercaptoethanol, 3.9 μ M T4 gene 32 protein, and 0.25 μ M [5'-³²P](dT)₁₀ (in nucleotide residues). The sample was irradiated with 1.4×10^{17} photons at 245.9 nm.

not stable under the acidic conditions used to silver stain SDS-polyacrylamide gels. Initially, we found that gels stained prior to autoradiography had reduced band intensities and higher backgrounds, suggesting that some of the cross-linked adduct was being released during the staining procedure. Earlier results of others had led to the suggestion that UV-induced cross-links between protein and nucleic acid might be acid labile.^{2,4} We have shown, under conditions similar to those used for fixation and staining of a gel, that only 25% of the cross-linked adduct remains intact after a 30-min incubation in 10% (1.58 M) acetic acid and 50% (v/v) methanol. We have not investigated the mechanism of the adduct lability but suggest that the lability may be the result of depyrimidination.

Analysis. Band intensities from the autoradiographs and stained gels can be determined in a number of ways. Initially, we carried out densitometric scanning and integration of the area under the peaks using a soft laser scanning densitometer (model SL-504-XL) from Biomed Instruments (Richmond, CA), coupled to an Apple IIe computer. A typical set of gel patterns and densitometer traces obtained in this way for an experiment in which bacteriophage T4 gene 32 protein was cross-linked to (dT)₁₀ is shown in Fig. 2. More recently, we have adopted a video-based scanning procedure, in which the autoradiographic images are digitized using a Dage video camera, an AT&T 6300 computer, and software supplied by the Computer Technology Center at the University of Virginia (Charlottesville, VA). An alternate procedure that we now also use avoids the use of films altogether by scanning the radioactive gels directly in real time using an Ambis radioactive gel scanning system (San Diego, CA).

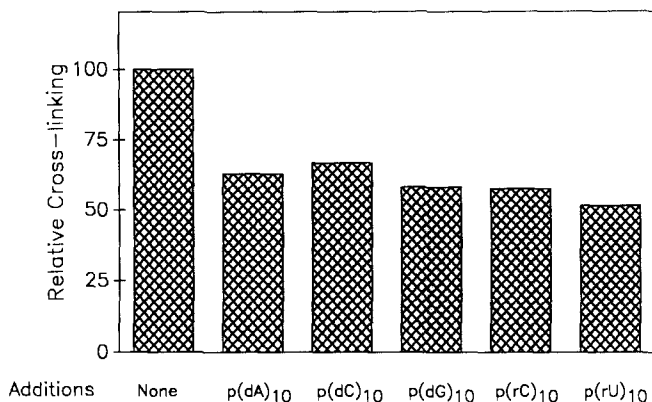


FIG. 3. Oligonucleotide competition experiments. The concentration of T4 gene 32 protein was $0.4 \mu M$, and that of $[5'-^{32}P](dT)_{10}$ was $35 \mu M$ (in nucleotide residues). Prior to irradiation, the gene 32 protein-(dT)₁₀ complex was challenged with $35 \mu M$ (dA)₁₀, (dC)₁₀, (dG)₁₀, (rC)₁₀, or (rU)₁₀. Irradiation was with a single pulse of 5.9×10^{16} photons at 266 nm. Reaction conditions are as described in Fig. 2. Quantitation of data is described in the text. The data are normalized to 100% for the amount of photoadduct formed in the unchallenged reaction.

Calibration and Characterization of UV Laser Cross-Linking Reaction

T4-Coded Gene 32 Protein-Oligonucleotide Model System. We initially calibrated our procedures by monitoring the cross-linking of a variety of single-stranded 10-mer oligonucleotides to T4-coded gene 32 protein.⁴⁷ This set of protein-nucleic acid complexes had been previously characterized by standard solution thermodynamic methods, and the binding of the protein to these oligonucleotides has been shown to be independent of nucleotide composition.⁴⁷ As a consequence, we could interpret the efficiency of cross-linking of these oligonucleotide-gene 32 protein complexes as a function of laser power and wavelength directly in terms of the relative photoreactivities of the various oligonucleotides tested. This conclusion was directly confirmed with the cross-linking system by performing competition experiments in which gene 32 protein was mixed with $5'-^{32}P$ -labeled (dT)₁₀ and then challenged with an equal concentration of an unlabeled competing oligomer. As shown in Fig. 3, all of the competing oligonucleotides reduce the yield of the gene 32 protein-(dT)₁₀ adduct in a proportionate fashion, thereby demonstrating directly that the binding affinity of gene 32 protein is indeed the same for all of the 10-mer oligonucleotides tested.

⁴⁷ S. C. Kowalczykowski, N. Lonberg, J. W. Newport, and P. H. von Hippel, *J. Mol. Biol.* **145**, 75 (1981).

TABLE I
UV CROSS-LINKING OF HOMOPOLYMER OLIGOMERS TO BACTERIOPHAGE
T4-CODED GENE 32 PROTEIN^a

Wavelength (nm)	(dT) ₁₀ cross-linked (%)	Cross-linking relative to (dT) ₁₀			
		(dT) ₁₀	(dC) ₁₀	(dG) ₁₀	(dA) ₁₀
204.1	0.13	100	ND ^b	12.9	ND
217.8	1.8	100	1.2	ND	ND
223.0	0.76	100	ND	ND	ND
228.7	0.91	100	ND	ND	ND
239.5	2.2	100	3.1	ND	0.4
245.9	3.7	100	0.8	ND	ND
252.7	4.1	100	0.5	ND	ND
266.0	2.6	100	4.3	ND	2.7
273.9	3.6	100	1.0	ND	ND
282.3	3.1	100	0.6	ND	ND
309.1	0.06	100	ND	ND	ND

^a Reaction conditions are described in Fig. 2. Oligonucleotide concentrations were held constant at 0.25 μ M nucleotide residues. The gene 32 protein concentration was held constant at 0.4 μ M. Irradiation was performed with a total flux of 1.4×10^{17} photons/sample at each wavelength.

^b ND, No cross-linking detected, which at these levels of sensitivity corresponds to less than 0.005% of the input oligomer cross-linked.

Optimal Cross-Linking Wavelengths and Nucleotide Residues. To establish optimal wavelengths for cross-linking, as well as to define the nucleotide residues through which cross-linking can effectively occur, gene 32 protein was then mixed with (dT)₁₀, (dC)₁₀, (dA)₁₀, and (dG)₁₀, and irradiated at various wavelengths. The results are summarized in Table I (taken from Ref. 37).

As shown in Table I, cross-linking is most effective with the pyrimidine-containing oligonucleotides, (dT)₁₀ and (dC)₁₀. In Table I the (dT)₁₀ results are presented as the percentage of input oligomer that is cross-linked at each wavelength, while the results for the other oligonucleotides are normalized to the yield of (dT)₁₀-gene 32 protein cross-links obtained at each wavelength. Thymidine is, by far, the most efficient cross-linker at all wavelengths, and except at one or two isolated wavelengths effective cross-linking through purine nucleotides is not observed. At wavelengths greater than 245.9 nm, cross-linking efficiency follows the order: dT (1) $\gg$ dC (0.0084) > rU (0.0041) > rC, dA, dG (<0.001). The level of cross-linking, relative to dT, is shown for oligos (dC)₁₀, (dG)₁₀, and (dA)₁₀. The efficiency of cross-linking of gene 32 protein to the ribose-containing

TABLE II
CROSS-LINKING OF VARIOUS NUCLEIC ACID-BINDING PROTEINS
TO OLIGONUCLEOTIDES^a

Protein (μM)	Cross-linking relative to gene 32 protein with (dT) ₁₀		
	(dT) ₁₀	(rU) ₁₀	(rC) ₁₀
Bacteriophage T4 gene 32 protein (0.4)	100	0.41	ND ^b
<i>E. coli</i> ρ protein (0.4)	1.53	0.21	0.82
<i>E. coli</i> RNA polymerase (0.4)			
σ subunit (0.4)	3.97	1.99	0.09
β and β' subunits (0.4)	1.88	0.84	ND
Rabbit muscle lactate dehydrogenase (5.0)	ND		

^a Reaction conditions are described in Fig. 1. Oligonucleotide concentrations were held constant at 0.25 μM nucleotide residues. Irradiation was performed with a total flux of 1.4×10^{17} photons/sample using 245.9-nm light.

^b ND, No cross-linking detected, <0.005% of oligomer cross-linked.

oligonucleotides (rU)₁₀ and (rC)₁₀ was also examined; the results are included in Table II.

It is clear from these findings that proteins can be cross-linked to either deoxyribose- or ribose-containing nucleic acids, but that the efficiency of cross-linking is very dependent on the nature of the nucleotide residue involved. Much of the following discussion applies equally to DNA- and to RNA-binding proteins, although for RNA cross-linking (through rU and rC residues), the specific activity of the ³²P-labeled RNA may need to be higher to yield a detectable signal.

Table II (taken from Ref. 37) shows that other proteins known to bind nucleic acids can also be cross-linked. The relative ratios of cross-linked products obtained for proteins with (dT)₁₀ and (rC)₁₀ vary from protein to protein, and may, in part, be related to differences in the relative binding constants of these proteins for the two oligonucleotides. This is clearly demonstrated in Table II for the *E. coli* ρ protein (an RNA-dependent ATPase), which can be efficiently cross-linked to (rC)₁₀. The binding affinity of this protein for (rC)₁₀ is more than 100 times greater than that of gene 32 protein for this oligomer,⁴⁸ and this tight binding is presumably responsible for the greater than 150-fold increase in cross-linking observed for the ρ -(rC)₁₀ complex, relative to the gene 32 protein-oligomer calibration system.

⁴⁸ J. A. McSwiggen, D. G. Bear, and P. H. von Hippel, *J. Mol. Biol.* **199**, 609 (1988).

It is also clear, however, that relative binding affinities are not the only source of differences in cross-linking efficiency observed for different systems. The major factor in many cases may well be the relative "intimacy of contact" (in terms of the cross-linking mechanism) between the protein and a properly placed thymidine residue (for cross-linking with 266-nm radiation). An example of such an effect of detailed interaction geometry on the efficiency of cross-linking can be taken from our work on the cross-linking of site-specific binding proteins to DNA.^{48a} Using single-pulse 266-nm radiation, we have shown that the saturating efficiency of cross-linking *E. coli* Cro protein to a specifically bound double-stranded oligonucleotide fragment carrying the canonical Cro-binding sequence, is approximately 12%. The efficiency of cross-linking Cro protein to a (nonspecifically bound) (dT)₂₀ · (dA)₂₀ fragment is <1%. In contrast, the observed efficiency of cross-linking *E. coli* cyclic AMP-binding protein (CAP) to its specific target sequence on a double-stranded DNA oligomer is ≪1% (we believe that this sequence contains no dT residues that are properly positioned to interact with the protein in the specific complex). Clearly, *both* binding affinity and the "intimacy" of appropriate (ideally involving dT residues if irradiating at 266 nm) protein–nucleic acid contacts are important in establishing cross-linking efficiency.

Cross-Linking to Define Protein–Nucleic Acid Contact Positions. Also shown in Table II are the relative cross-linking efficiencies for oligonucleotides with the α , β , and β' subunits of the *E. coli* RNA polymerase.^{5–8,36} These RNA polymerase data show that laser cross-linking can be effective in discriminating contacts between nucleic acids and the various subunits of a protein complex. Similar results have now been obtained with more than 20 different nucleic acid-binding proteins (data not shown). In contrast to the nucleic acid-binding proteins, a protein such as lactate dehydrogenase, which has no intrinsic affinity for DNA, shows no adduct (cross-link) formation activity.

Cross-Linking Efficiency. Determination of the *yield* of cross-linked product is important in many types of analyses. As shown in Table I, the yield of gene 32 protein–(dT)₁₀ adduct with 252.7-nm irradiation corresponds to ~4% of the input oligonucleotides (in this experiment oligonucleotide is the limiting species, with protein present in considerable excess). Calculating from component concentrations and the known binding constant for this interaction ($K_{\text{assoc}} \approx 10^6 \text{ M}^{-1}$),⁴⁷ we estimate that ~25% of the input oligomer is complexed with the gene 32 protein in these solutions. Therefore the *normalized* efficiency of cross-linking of (dT)₁₀ to

^{48a} E. M. Evertsz, W. L. Kubasek, D. Kuninger, R. Brennan, and P. H. von Hippel, manuscript in preparation.

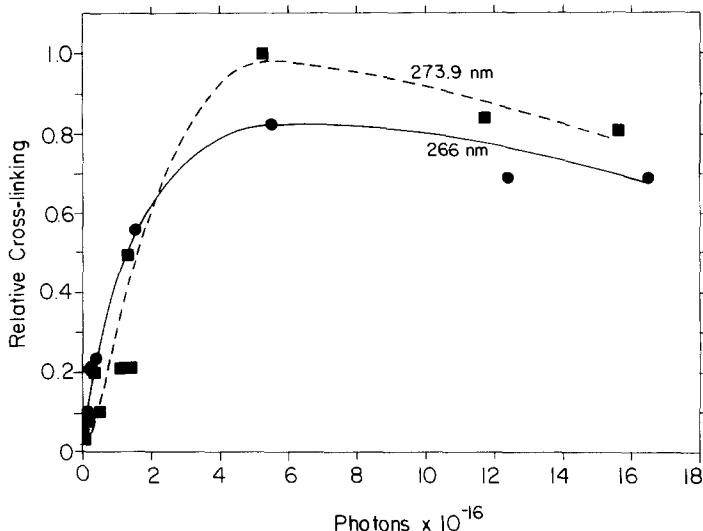


FIG. 4. Dose-response curve for the photocross-linking of T4 gene 32 protein to $(dT)_{10}$. Reaction conditions are as described in Fig. 2. The concentration of T4 gene 32 protein was $0.4 \mu M$ and that of $[5'-^{32}P](dT)_{10}$ was 72 nM (in nucleotides). Irradiation was with photons at either 266 nm ($\bullet$) or 273.9 nm ($\blacksquare$). A relative cross-linking yield of 1.0 corresponds to $\sim 18\%$ of the complex present in this set of experiments. The data are normalized to unity for the amount of photoproduct formed by 5.2×10^{16} photons at 273.9 nm.

gene 32 protein corresponds to an $\sim 16\%$ yield of the complex present during irradiation.

Quantum Yield for Cross-Linking. Making a few additional assumptions, we can also calculate the quantum yield (defined as the number of complexed oligomers that are cross-linked per absorbed photon) at each wavelength. For example, if we assume only a single dT residue per oligomer need be involved to obtain a cross-link (this has been demonstrated in Ref. 37), and that 25% of the oligomers are complexed in the reaction mix, we obtain a quantum yield (per nucleotide residue) of $\sim 1 \times 10^{-4}$ at 266 nm. This is comparable to quantum yields previously estimated with related model systems.²²

In Fig. 4, we demonstrate that the amount of cross-linked product in a given experiment is a function of the total photon flux, as well as of the wavelength of the incident light. In general, at low levels of incident photons, the extent of cross-linking is a linear function of the total incident flux. At higher photon inputs the yield of cross-links reaches a maximum value, and then starts to decline. The data plotted in Fig. 4 were generated by using constant pulses of fixed power and

increasing the total photon flux by increasing the total number of pulses. The same curves can be generated by using a fixed number of pulses and varying the power per pulse.³⁷ The data of Fig. 4 indicate that total photon fluxes exceeding 5×10^{16} photons do not significantly increase the yield of cross-linked product, and that higher levels may decrease the yield of adduct (see below).

Advantages of Single-Pulse Irradiation. The laser wavelengths used to generate the above data (with gene 32 protein and oligonucleotides of various composition) vary widely in their power per pulse; consequently, most samples were irradiated with more than one pulse so that the total number of incident photons could be kept constant in our quantum yield experiments. Having established that the highest quantum yields for cross-linking are associated with wavelengths between 245 and 280 nm, we have chosen to pursue single-pulse experiments using 266-nm light. Use of this wavelength provides ample power for single-pulse experiments, and also simplifies the experimental setup, since both the H₂ shifting cell and the second Pellin-Broca prism (see Fig. 1) are not needed. Our rationale for preferring single-pulse experiments involves consideration both of the quantum yields of the various changes that can be induced by UV irradiation of protein–nucleic acid complexes, and of the rates of association–dissociation reactions between, and conformational transition rates within, the macromolecular species.

Quantum Yields for Competing Photochemical Processes. We have measured a quantum yield of $\sim 10^{-4}$ for protein–(dT)₁₀ cross-link formation (per dT residue) with 266-nm radiation. Reported quantum yields for other UV-induced alterations of proteins and nucleic acids include (1) 4×10^{-2} for pyrimidine dimer formation, (2) 5×10^{-6} for backbone breakage of single-stranded DNA,⁴⁹ (3) 1×10^{-5} for backbone breakage of single-stranded RNA,⁵⁰ (4) 8×10^{-7} for DNA–DNA cross-link formation,⁴⁹ and (5) 7×10^{-6} for RNA–protein cross-link formation.⁵⁰ These data suggest that pyrimidine dimers will be introduced into DNA at approximately 400 times the rate at which protein–DNA cross-links are formed, while the other photochemical reactions mentioned above will be much less probable and not likely to complicate cross-linking studies of this type. However, pyrimidine-dimer formation can certainly

⁴⁹ G. B. Zavilgelsky, G. G. Gurzadyan, and D. N. Nikogosyan, *Photochem. Photobiophys.* **8**, 175 (1984).

⁵⁰ E. N. Dobrov, Z. H. Arbieva, I. S. Khromov, and S. V. Kust, *Photochem. Photobiol.* **43**, 493 (1986).

introduce problems into the interpretation of multipulse cross-linking experiments.^{50a}

It is apparent that pyrimidine dimers will be introduced into DNA with high probability during the first pulse of a multiple-pulse laser cross-linking experiment, or during the initial phase of conventional ultraviolet irradiation. These pyrimidine dimers can cause significant distortion of the DNA to which the protein is bound. Since the rates of conformational transitions in macromolecular complexes are usually in the 100- μ sec to 10-msec time range,⁵¹ the geometries or affinities of protein-nucleic acid interactions would not be expected to be altered during the initial 5-nsec laser pulse. However, the $\sim$ 100-msec interval *between* laser pulses certainly does provide ample time for conformations and contacts to be altered. Hence, the second incoming pulse could "freeze" an interaction that is substantially different from the interaction "seen" by the first pulse. This argues that the single-pulse irradiation mode should be used in experiments where protein-nucleic acid contacts might be altered by pyrimidine-dimer formation.

Estimated Rate of Protein-DNA Cross-Link Formation. Clearly photochemical *activation* of cross-link formation (by a single 5-nsec laser pulse) is fast compared to macromolecular association-dissociation and transconformation reactions. However, this does not prove that the subsequent chemical reaction that converts this photoactivation process into a protein-nucleic acid cross-link is also sufficiently fast to "freeze" labile macromolecular equilibria. Photochemical reactions can be quite slow (in the millisecond to second time range) and translational or rotational diffusion of macromolecular components occurring *after* the initial photochemical activation event could result in the appearance of significant nonspecific cross-linking artifacts.

To examine this question, we conducted the following experiment (details in Ref. 48a). Solutions of proteins that are not expected to interact significantly with DNA (e.g., horse heart myoglobin, alcohol dehydroge-

^{50a} These comparisons of quantum yields with other competing photochemical processes include several that have been extensively studied only by low-intensity UV lamp irradiation procedures. The photon flux of a laser source is enormously greater than that achievable with a UV lamp. As a consequence quantum yields and photoproducts obtained by these methods may be very different. These issues are developed further in Evertsz *et al.*^{48a} and have been reviewed in detail by D. N. Nikogosyan, *Int. J. Radiat. Biol.* **57**, 233 (1990). These potential differences between the products of laser and UV-lamp irradiation should also be considered in comparing the cross-linking results obtained here with those reported elsewhere in this volume [25].

⁵¹ G. Careri, P. Fasella, and E. Gratton, *Crit. Rev. Biochem.* **3**, 141 (1975).

nase) were mixed with 5'-³²P-labeled (dT)₂₀, and subjected to single-pulse irradiation at 266 nm. We then looked for cross-link formation by our standard polyacrylamide gel electrophoresis procedures (see above) as a function of protein concentration. The results showed that no nonspecific cross-linking was observed at protein concentrations below approximately 100 μ M. This is in contrast to the *specific* cross-linking in interactions of high affinity, which we observe at DNA-binding protein concentrations in the 1 μ M concentration range in our "standard" cross-linking experiments.

Thus it appears that binding interactions involving association constants weaker than $\sim 10^3 M^{-1}$ will not be detected by our cross-linking procedures. Furthermore, calculations of average mean free paths on the basis of estimated protein and nucleic acid translational diffusion constants suggest that the chemical part of the UV laser-induced protein-nucleic acid cross-link formation reaction goes to completion in less than 1 μ sec. Thus photoactivated thymidine residues that are *not* in intimate contact with protein at the instant of activation will tend to be "quenched" by reaction with the solvent, and it appears that we can indeed "freeze" macromolecular binding equilibria and transconformation reactions of the sort we seek to study by this technique.

Analysis of Various Types of Protein-Nucleic Acid Interactions by UV Laser Cross-Linking Technique

The above discussion shows that single-pulse ultraviolet laser cross-linking can "freeze" protein-nucleic acid binding equilibria. As a further test of this proposition, we have examined a number of such equilibria that have previously been characterized by standard thermodynamic techniques. In each case we have measured binding isotherms and association constants comparable to those obtained by standard methods. In addition we have taken advantage of the ability of the laser cross-linking procedure to "freeze" a number of dynamic equilibria that are *not* easily studied by other techniques. Such results are briefly summarized in the following paragraphs to provide the reader with a further indication of the range and scope of problems that can be attacked by the UV laser cross-linking technique.

Binding Constant Measurements—Calibration with Gene 32 Protein. Using an association constant of $4 \times 10^6 M^{-1}$ for the binding of T4-coded gene 32 protein to (dT)₁₀ (see Ref. 47), we can calculate the concentration of the protein-DNA complex that should be formed in the presence of varying concentrations of (dT)₁₀. Assuming that the efficiency of cross-linking is constant for all samples, the amount of complex produced should

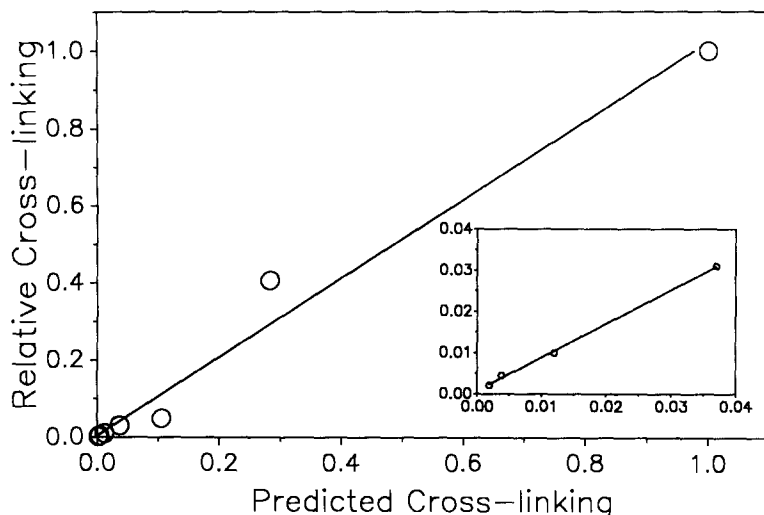


FIG. 5. Correlation between theoretical and actual cross-linking of $[5'\text{-}^{32}\text{P}](\text{dT})_{10}$. T4 gene 32 protein ($0.1 \mu\text{M}$) was incubated with $1 \mu\text{M}$ $[5'\text{-}^{32}\text{P}](\text{dT})_{10}$ under the reaction conditions described in Fig. 2. The complex was titrated with increasing amounts of $(\text{dT})_{10}$ resulting in concentrations of $(\text{dT})_{10}$ ranging from 1 to $100 \mu\text{M}$ (in nucleotides). Samples were irradiated with a single pulse of 4×10^{16} photons at 266 nm. The data were quantitated and normalized to unity for the amount of photoproduct formed at $1 \mu\text{M}$ $(\text{dT})_{10}$. Using a binding constant of 4×10^6 , the fraction of $(\text{dT})_{10}$ bound to the gene 32 protein was "predicted" as a fraction of the total $(\text{dT})_{10}$ present and normalized to unity for the $1 \mu\text{M}$ reaction. *Inset:* Expanded view of first four points.

be directly proportional to the amount of cross-linked adduct formed. Figure 5 demonstrates that the predicted cross-linking is exactly proportional to the actual cross-linking. Hence the complementary experiment, in which the concentration of $(\text{dT})_{10}$ is held constant and the protein is varied, should also yield a binding curve that can be fit to a standard titration isotherm. Figure 6 shows this to be the case.

The general utility of determining binding constants by this laser cross-linking methodology is confirmed by the fact that the same binding constants are obtained over a 25-fold range of $(\text{dT})_{10}$ concentration (10^{-8} to $2.5 \times 10^{-7} \text{ M}$ nucleotide). Clearly, the small reaction volume we use, coupled with the low concentrations of nucleic acid required, means that only very small quantities of material are needed to measure binding constants by this technique. (In addition, of course, these binding constants can be measured in impure solutions.)

The general applicability of such laser cross-linking determinations of protein-binding constants to nucleic acids, and more specifically for pro-

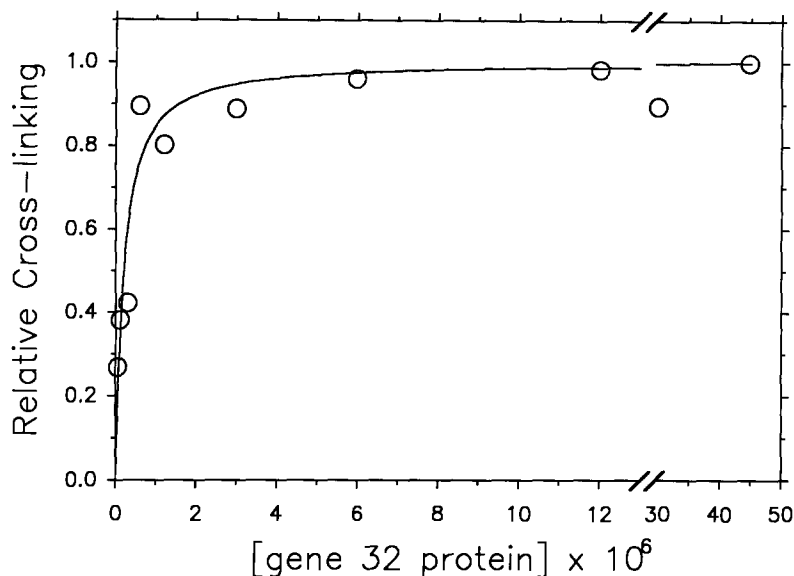


FIG. 6. T4 gene 32 protein titration of $(dT)_{10}$. Reaction conditions were 50 mM Na_2HPO_4 , pH 7.7, 1 mM Na_2EDTA , 1 mM 2-mercaptoethanol, 50 nM $[5'\text{-}^{32}\text{P}](dT)_{10}$ (in nucleotides) and from 60 nM to 45 μM T4 gene 32 protein. Each sample was irradiated with a single pulse of 5.7×10^{16} photons at 266 nm.

teins that tend to translocate rather rapidly on the nucleic acid component (e.g., nucleases and polymerases) is demonstrated by our previously reported binding measurements of T4-coded DNA polymerase (gene 43 protein) to various DNA substrates.⁵²

Salt Titrations. The binding of proteins to nucleic acids is supported by a variety of specific and nonspecific interactions. These interactions include the attraction of oppositely charged ionic groups; such attraction can, of course, be modulated by increasing salt concentrations. Figure 7 shows a salt concentration titration curve for the binding of gene 32 protein to $(dT)_{10}$, determined by the laser cross-linking technique. Again, standard methods confirm the profile that is generated.⁴⁷

Nucleic Acid-Binding Protein Site Size Determinations. In most cases, the binding affinity of protein for nucleic acid should be maximal when the nucleic acid-binding domain of the protein is fully occupied. When the nucleic acid is too short to fill the site, favorable interactions will be

⁵² J. W. Hockensmith, W. L. Kubasek, T. Cross, M. K. Dolejsi, W. R. Vorachek, and P. H. von Hippel, in "DNA Replication and Recombination" (R. McMacken and T. J. Kelly, eds.), Vol. 47, p. 111. Alan R. Liss, New York, 1987.

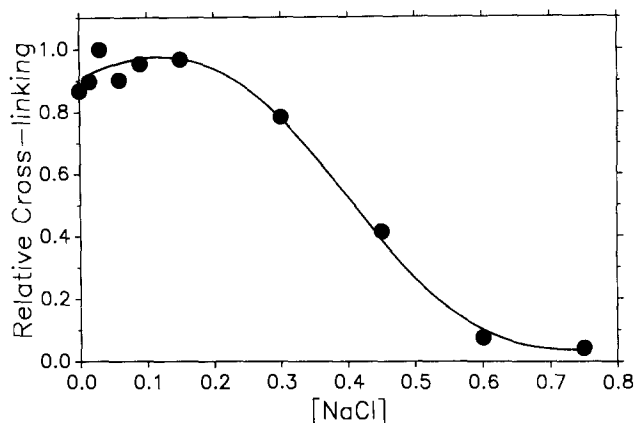


FIG. 7. Salt titration of T4 gene 32 protein-(dT)₁₀ interaction. Reaction conditions and irradiation are as described in Fig. 6, with the gene 32 protein concentration at 0.4 μ M.

diminished; hence binding (and thus cross-linking) should be diminished. Nucleic acid moieties smaller than the actual nucleic acid-binding site may also be able to bind to the site in more than one way; measured binding constants in such cases will represent an average of the available binding modes. Consequently, if a nucleic acid-binding protein is bound to successively longer oligonucleotides, the most favorable binding should occur



FIG. 8. Site size determination for the T4 gene 32*III protein. Reaction conditions are as described in Fig. 6, with gene 32*III concentration at 1.9 μ M and [5'-³²P]oligo(dT) concentration at $\sim$ 0.2 μ M (in nucleotides). Irradiation was at 266 nm with a single pulse of 2.1×10^{16} photons.

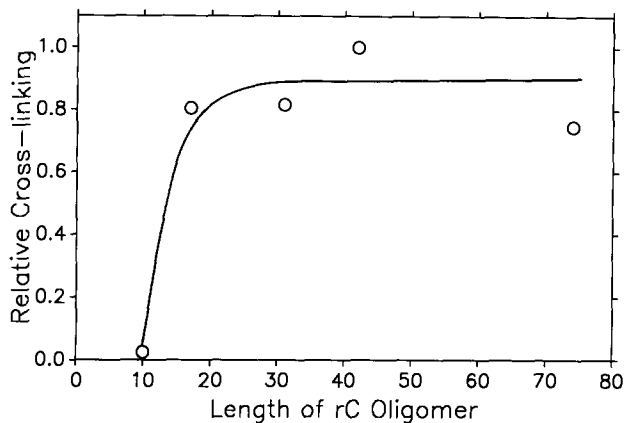


FIG. 9. Site size estimate for *E. coli* ρ protein. Reaction conditions are as described in Fig. 2. ρ protein concentration was $0.4 \mu\text{M}$ and the $[5'\text{-}^{32}\text{P}]\text{oligo(rC)}$ concentration was $\sim 30 \text{ nM}$ (in nucleotides). Irradiation was with 7.6×10^{16} photons at 245.9 nm .

when the site on the protein is filled. A dramatic example illustrating this point is presented in Figs. 8 and 9.

Figure 8 shows the cross-linking of the T4-coded gene 32 protein to oligo(dT) molecules of increasing length. There is a dramatic increase in cross-linking between (dT)₆ and (dT)₇, suggesting that an oligonucleotide of length seven exhibits the most thermodynamically favorable interaction with the nucleic acid-binding site of the protein. A binding site size of seven has been previously reported for the gene 32 protein binding cooperatively to single-stranded DNA.⁵³

***E. coli* ρ Protein RNA-Binding Site Size.** Cross-linking techniques can also be used to demonstrate RNA binding to proteins. Thus, we have cross-linked *E. coli* ρ protein to oligo(rC)_n; the results summarized in Fig. 9 suggest an RNA site size from 10 to 17 nucleotide residues/ ρ protein monomers. This result is in good agreement with other data that suggest that the monomer site size for this protein ranges from 11 to 18 nucleotide residues, defined either as the site size needed to achieve full RNA binding, or the site size needed to achieve maximal ATPase activation.^{48,54}

Bacteriophage T4-Coded Polymerase Accessory Proteins Interactions. While site size determination or "footprinting" by the laser cross-linking methodology provides information that can be obtained by other methods,

⁵³ D. E. Jensen, R. C. Kelly, and P. H. von Hippel, *J. Biol. Chem.* **251**, 7215 (1976).

⁵⁴ D. G. Bear, P. S. Hicks, K. W. Escudero, C. L. Andrews, J. A. McSwiggen, and P. H. von Hippel, *J. Mol. Biol.* **199**, 623 (1988).

TABLE III
T4 ACCESSORY PROTEIN-DNA CONTACTS^a

Nucleic acid ^a	Addition	Relative cross-linking		
		G45P alone	G44/62P alone	G45P + G44/62P
(dT) ₁₆	ATP	4	3/0	2, 2/170
(dT) ₂₀	ATP	14	70/9	13, 6/310
(dT) ₂₀ · (dA) ₁₆	ATP	47	9/1	2, 0/24
(dT) ₁₆	ADP	37	0/0	11, 5/0
(dT) ₂₀	ADP	81	53/12	4, 0/0
(dT) ₂₀ · (dA) ₁₆	ADP	105	274/10	0, 0/0

^a The number of molecules of (dT) oligomer was held constant for each of these experiments. Therefore the (dT)₂₀ · (dA)₁₆ experiment contained only one-fifth as much single-stranded DNA as the (dT)₂₀ experiment.

the cross-linking technique also has potential for providing information that is *not* easily obtained by other approaches.

For example, the bacteriophage T4 DNA replication system involves a DNA-dependent ATPase composed of genes 44, 62, and 45 proteins. When these three proteins are assembled on DNA in the presence of ATP, no footprint can be detected by neocarzinostatin or DNase I cleavage, suggesting that the hydrolysis of ATP greatly weakens the complex.⁵⁵ Laser cross-linking, however, is sufficiently fast to trap the transient complexes formed by the ATPase with the DNA in the presence of ATP (Table III, taken from Ref. 52). Experiments were performed using (1) gene 45 protein and DNA, (2) gene 44/62 proteins and DNA, or (3) a combination of gene 45 protein, gene 44/62 proteins, and DNA. Additionally, either ATP or ADP was added to the reaction. We have found that the gene 62 protein alone does not cross-link to (dT)₁₆ in the presence of ATP, but does cross-link to (dT)₂₀. Addition of the gene 45 protein results in a dramatic increase in cross-linking of the gene 62 protein to the three DNAs tested, but causes an apparent decrease in gene 44 protein cross-linking.

Cross-linking in the presence of ADP yields quite a different set of results. While the gene 62 protein still shows a preference for (dT)₂₀, the addition of gene 45 protein in the presence of ADP abolishes the gene 62 protein cross-linking. In addition, the gene 44 protein cross-links much better to (dT)₂₀ in the presence of ADP than in the presence of ATP. On

⁵⁵ H. E. Selick, J. Barry, T.-A. Cha, M. Munn, M. Nakanishi, M. L. Wong, and B. Alberts, in "DNA Replication and Recombination" (R. McMacken and T. J. Kelly, eds.), p. 183. Alan R. Liss, New York, 1987.

addition of the gene 45 protein, the gene 44 protein cross-linking is totally abolished in the presence of ADP.

These results demonstrate several things. First, the gene 62 protein cross-linking results show that an optimal length of DNA may be required to fill a binding site on the protein, thereby suggesting a site size between 16 and 20 nucleotide residues. Second, these results demonstrate that the laser cross-linking technique can be used to analyze multiple protein complexes and that proteins in such complexes may contact nucleic acids in a significantly different way than do the individual protein components in isolation. Finally, proteins that bind mononucleotides can have their polynucleotide contacts altered significantly by changing the nucleotide that is present.

In addition, as indicated above, these results show that the laser cross-linking technique can be used to "freeze" dynamic or rapidly interconverting protein-nucleic acid complexes. Such studies have been used to demonstrate rapid *changes* in protein-DNA contacts, and show that complexes such as the T4 DNA replication system cycle through altered protein conformations as proteins, substrates, and/or products (nucleotides) are added to or removed from the system.

Protein Photodegradation

Experimentation with the UV laser cross-linking procedure has also provided additional information that should be considered when attempting to cross-link proteins to nucleic acids by either conventional or laser ultraviolet irradiation. Photodegradation of proteins (protein chain breakage) after irradiation has been noted previously, although the reaction wavelengths and reaction mechanism have not been well characterized.¹ We have investigated photodegradation in more detail, since under certain circumstances it can significantly complicate UV laser cross-linking experiments.

Figure 10 presents an action spectrum for the photodestruction of protein. After laser irradiation at each wavelength, the protein was electrophoresed on an SDS-polyacrylamide gel and the protein remaining in the original protein band was quantitated. The photodegradation products are sufficiently random that they spread over the entire gel below the parent protein band, hence disappearing into the background. The decrease in the amount of stainable protein is related to both the wavelength and the input photon level, with maximum loss of stainable protein occurring at high photon flux in the region where peptide bonds absorb, a result that is certainly consistent with protein degradation.

Three sets of data are shown in Fig. 10. The high power data for

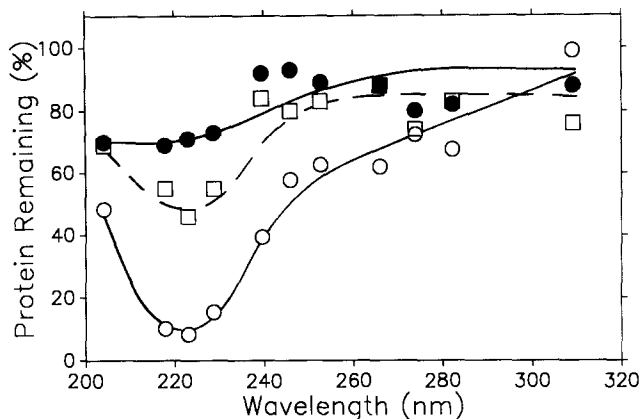


FIG. 10. Action spectrum for photodegradation. Reaction conditions are as described in Fig. 2. (O) T4 gene 32 protein ($0.4 \mu M$) was mixed with $0.25 \mu M$ $[5'-^{32}P](dT)_{10}$ (in nucleotides) and irradiated with 1.4×10^{17} photons; (●) $3.9 \mu M$ T4 gene 32 protein was mixed with $0.22 \mu M$ $[5'-^{32}P](dT)_{10}$ and irradiated with 2.1×10^{16} photons; (□) $84 \mu g/ml$ T4 gene 45 protein was mixed with $0.22 \mu M$ $[5'-^{32}P](dT)_{10}$ and irradiated with 2.1×10^{16} photons. After irradiation samples were treated as described under Analysis of Cross-Linking Experiments.

the gene 32 protein shows considerable photodestruction, and we would suggest that this process accounts for the decreased cross-linking seen at the higher power levels of Fig. 4. Our data are consistent with a previous report that irradiation with $\sim 1.5 \times 10^{17}$ photons at 254 nm inactivates approximately one-third of the molecules in a gene 32 protein solution.¹⁶ Alternatively the lower power level data for the gene 32 protein demonstrate that photodegradation can be minimized while maximizing the cross-linking (compare with Fig. 4). We have included photodegradation data obtained with gene 45 protein to illustrate that this phenomenon can occur with all proteins.

Photoreversal of protein-DNA cross-links has been suggested to occur at low wavelengths of irradiation.² We do not have any evidence for photoreversal of protein-nucleic acid cross-links, but should point out that photodegradation of complexes may mimic photoreversal. Consequently the optimization of laser cross-linking experiments should include experiments to find conditions that maximize cross-linking while minimizing protein photodegradation.

We note that photodegradation can be greatly minimized simply by avoiding the use of irradiating wavelengths below 245 nm. Users of low-pressure Hg fluorescent lamps should be aware that 14% of the emitted ultraviolet light is at wavelengths other than 254 nm, and high-pressure

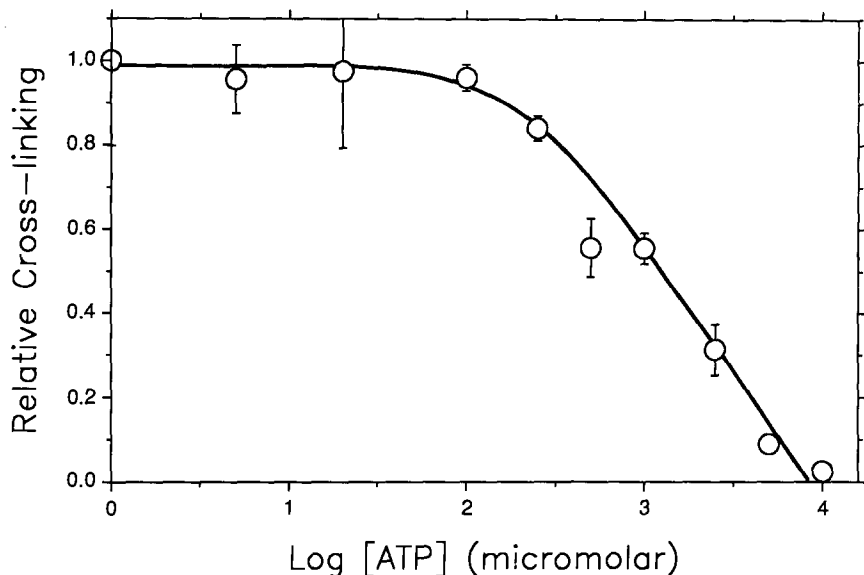


FIG. 11. Attenuation of cross-linking by ATP. Reaction conditions are as described in Fig. 2. Gene 32 protein ($0.4 \mu M$) was incubated with $0.25 \mu M$ $[5'-^{32}P](dT)_{10}$ and increasing concentrations of ATP prior to irradiation by a single pulse of 3.3×10^{16} photons at 266 nm.

Hg-Xe arc lamps have <1% of their ultraviolet output at 254 nm.⁵⁶ When using these light sources precautions should be taken to eliminate photodestructive wavelengths and effects.

Photoattenuation

The overall yield of cross-linked products may also be affected by the absorption of incoming photons by macromolecules other than those directly involved in the formation of protein-nucleic acid cross-links. Figure 11 shows the cross-linking of gene 32 protein to $(dT)_{10}$ in the presence of increasing amounts of ATP. At levels of ATP approaching 1 mM, the total cross-linking starts to diminish.

Viewing the depth of the sample to be irradiated as a series of very thin layers, we find that under normal conditions there is sufficient ultraviolet light to penetrate all the layers with some light remaining. When high levels of an absorbing center such as ATP are added, then most of the photons are removed by these absorbing centers within the first few layers below the surface. Layers located deeper in solution receive less light,

⁵⁶ T. P. Coohill, *Photochem. Photobiol.* **41**, 501 (1985).

and hence cross-linking is diminished in these layers, resulting in an overall decrease in cross-linking for the total solution.

While this attenuation is detrimental, in that it diminishes overall cross-linking, it has a beneficial effect in that it also reduces photodegradation of the protein (data not shown). In effect, the added adsorbing centers have resulted in fewer photons being available per mole of cross-linkable species, and consequently the experimental curve of Fig. 4 has been shifted so that the experiment is now being performed on the initial linear section of that curve. This situation may be "corrected" by merely increasing the input power. We suggest that when calibrating a system with respect to laser cross-linking it is useful to prepare a power curve such as that of Fig. 4.

Summary and Perspectives

This methodological chapter shows that ultraviolet laser cross-linking can be a very valuable tool in exploring protein-nucleic acid, protein-protein, and nucleic acid-nucleic acid interactions. The information obtained is often complementary to, and can be confirmed by, other techniques. However, in some instances this method will yield information that is difficult to obtain by other techniques.

The laser cross-linking technique is very simple and readily adaptable to a variety of systems. Typically laser cross-linking at 266 nm with a single pulse of $2-5 \times 10^{16}$ photons/sample works very well for protein concentrations ranging from 10^{-8} to 5×10^{-5} M, and for nucleic acid (in units of nucleotide residues) concentrations of 10^{-8} to 10^{-4} M. Higher concentrations of proteins and nucleic acids or nucleotide ligands can be used, but the input power level must be adjusted to take attenuation effects into account. Various other irradiation wavelengths can also be used, but should be evaluated for potential absorption by and photodegradation of the macromolecule of interest. Additionally, the technique permits examination of dynamic and static equilibria that have been perturbed in a variety of ways, such as by the addition of nucleotide cofactors and other complexing ligands.

We hope that this discussion of the UV laser cross-linking technique and its background will help others to devise additional variants of this powerful approach that may be better suited to other purposes and other experimental problems.

Acknowledgments

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[14] Kinetic Studies on Promoter–RNA Polymerase Complexes

By MALCOLM BUCKLE, ALEXANDRE FRITSCH, PASCAL ROUX,
JOHANNES GEISELMANN, and HENRI BUC

Introduction

An understanding of the nature of the contacts between a protein and a nucleic acid existing in a given stereospecific complex at equilibrium is generally not enough to account for its function. In many instances knowledge of the corresponding pathway is a prerequisite for the elucidation of the mechanism by which a selected function is exerted. From a methodological point of view this is a much more difficult task.

A classical case is that of the interaction between RNA polymerase and its target site on the DNA, the promoter, where the overall rate of formation of the kinetically competent complex usually determines the frequency of initiation of full-length transcripts or the strength of the promoter.^{1–3} In this respect, it is necessary to use the time dimension to determine how protein–DNA contacts are sequentially established. A simple recording of changes in footprinting patterns on the DNA is not sufficient for a full characterization of the driving forces involved in the overall kinetic process, even when complemented with detailed genetic studies. Even though the protein component plays a predominant role, current technology is unfairly biased toward the DNA in that it is easier to investigate and interpret sequential changes in footprinting patterns than to follow modifications or subtle alterations in the protein occurring within the nucleoprotein complex.

This chapter will focus on two related aspects. First, an overview will be given of technological developments that allow the characterization of

¹ W. R. McClure, *Annu. Rev. Biochem.* **54**, 171 (1985).

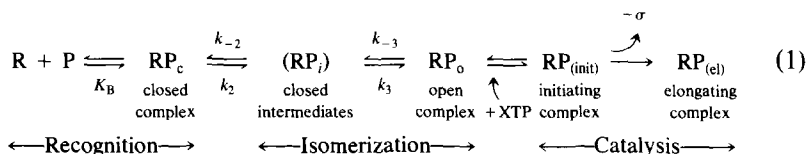
² H. Bujard, M. Brunner, U. Deuschle, W. Kammerer, and R. Knaus, in "RNA Polymerase and the Regulation of Transcription. Proceedings of the 16th Steenbock Symposium" (W. S. Reznikoff, *et al.*, eds.), p. 95. Elsevier, New York, 1987.

³ S. Leirimo and M. T. J. Record, *Nucleic Acids Mol. Biol.* **4**, in press (1990).

protein conformational changes taking place on discrete and separable nucleoprotein complexes representative of sequential changes in an overall reaction. The corresponding technology for the nucleic acid counterpart of the complex is rather well known and will not be reviewed in detail, except to stress the utility of a thorough characterization of the chemical reaction involved in the footprinting technique.

Second, methodologies used to characterize the change in the structure of the nucleic acid component, grouped here under the heading of footprinting techniques, may be adapted to obtain information about species only transiently present during the process of protein-nucleic acid recognition. This implies several technological advances (notably in the design of a rapid mixing device adapted to handle microvolumes and in the analysis of changes of chemical reactivity profiles as the reaction proceeds) that will be discussed in the second section of the chapter.

The interaction between *Escherichia coli* RNA polymerase (R) and the *lacUV5* promoter (P) provides a convenient system that illustrates the variability and versatility of these approaches. The process by which recognition and transcription of DNA into messenger RNA occurs may be schematically represented as, e.g.,^{2,4}:



It should be taken into consideration that Eq. (1) is derived primarily from kinetic studies that determine association constants (K_B) for promoter recognition (see Ref. 4) and forward rate constants (k_2 , k_3 , etc.) for the isomerization processes and from snapshot experiments conducted as the polymerase moves away from the promoter and into active transcription.⁵⁻⁷ This minimal scheme will be challenged and further detailed. Characterization of the various intermediates requires a combination of three approaches: (1) the isolation of species appearing sequentially in the pathway after the open complex is formed, (2) changes in conditions that allow the isolation of certain intermediates prior to the appearance of RP_o , and (3) the recording of transient intermediates during the formation of RP_o .

At equilibrium, in the absence of ribonucleotides, at 37° and in the presence of 100 mM KCl at pH 8.0, the open complex, RP_o , is the predomi-

⁴ H. Buc and W. R. McClure, *Biochemistry* **24**, 2713 (1985).

⁵ A. Spassky, K. Kirkegaard, and H. Buc, *Biochemistry* **24**, 2723 (1985).

⁶ A. Spassky, *J. Mol. Biol.* **188**, 99 (1986).

⁷ J. R. Levin, B. Krummel, and M. J. Chamberlin, *J. Mol. Biol.* **196**, 85 (1987).

nant binary species. As detailed in Ref. 8 for the interaction between RNA polymerase and several variants of the T7A1 and *lac* promoters, various elongating complexes can also be isolated by the addition of a suitable choice of ribonucleotides. Hence a set of various elongating complexes, $RP_{(el)}$, where specified lengths of mRNA are present in the product site of the enzyme, can be obtained. The comparative studies of RP_0 and of the subsequent $RP_{(el)}$ species are referred to below in the section Sequential Approach; when the enzyme is in the abortive mode $RP_{(init)}$, it is continuously synthesizing short transcripts; consequently, in this situation, only an average picture of the recycling process can be obtained.

The same methodology can be applied to the study of intermediates leading to the open complex, RP_0 , if conditions can be found under which the intermediate of interest is either the predominant species or in rapid equilibrium with one of the intermediates on the pathway. Such is the case for a binary complex quenched at 16° at the *lacUV5* promoter (RP_0)_{LT} (LT meaning low temperature), which undergoes a rapid isomerization with a postulated intermediate (RP_i) when the temperature is raised to 37°. ⁴ To a reasonable approximation, therefore, the "quenched" species RP_{LT} can be assumed to provide a picture of the short-lived species (RP_i). Note however that a careful kinetic study is required to justify this approach.

The RP_c binary complex, which, contrary to RP_i , has not yet been trapped, is the major species appearing transiently at 37°. At this temperature the kinetic scheme [Eq. (2)] may be reduced to



Here again preliminary steady state kinetic studies allow prediction of the enzyme concentration $[R]$ (in excess over $[P]$) and of the time scale required to study the decay of the RP_c binary complex into the final open complex under optimal conditions at 37° (Fig. 1; for more details see Ref. 5). The corresponding characteristic time is of the order of 0.1 sec⁻¹.

For most promoters, the complete formation of an open complex will require between 5 sec and 20 min. The classical static techniques will have to be adapted to a time scale ranging from 1 sec to 1 min, depending on the promoter considered.

Sequential Approach: Characterization of Stable Intermediates

Identification of Protein Components in Nucleoprotein Assemblies

The most commonly used technique to show an interaction between a protein and a nucleic acid is the band shift or gel retardation assay. The mobility of a radioactively labeled nucleic acid is altered by protein bind-

⁸ B. Krummel and M. J. Chamberlin, *Biochemistry* **28**, 7829 (1989).

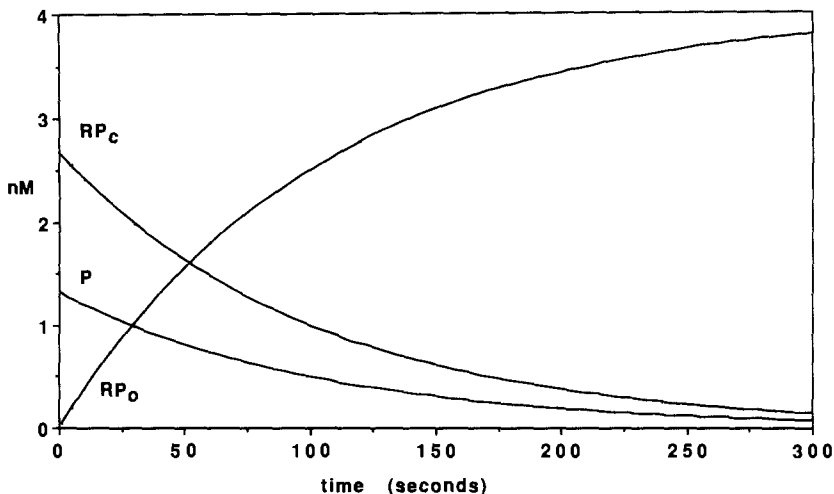


FIG. 1. Theoretical kinetic profile for open complex formation involving RNA polymerase and a weak promoter. The formation of an open complex is assumed to obey the simplified scheme given in Eq. (2). The relative concentrations of RNA polymerase (R), promoter (P), closed complex (RP_c), and open complex (RP_o) are calculated from the expressions $[P_o]_t = \lambda P_T e^{-t/\tau}$, $[RP_c]_t = (1 - \lambda)e^{-t/\tau}$, and $[RP_o]_t = P_T(1 - e^{-t/\tau})$, where P_T represents the total promoter concentration, $\lambda = (1 + K_B[R])^{-1}$ and $\tau^{-1} = (1 - \lambda)k_2$. Values of $K_B = 10^7 M^{-1}$, $k_2 = 0.01 \text{ sec}^{-1}$, $[R] = 200 \text{ nM}$, and $[P] = 4 \text{ nM}$ were used for this calculation.

ing. The main advantage of this technique is that it demonstrates a strong interaction between a given DNA fragment and a protein or protein assembly.^{9,10} A disadvantage is that no details of the interactions involved are revealed. In its usual form identification of a nucleoprotein complex is carried out by visualization of a radioactively labeled DNA fragment following autoradiography. An innovation of this technique has been developed in this laboratory¹¹ that provides more detailed information from such assays. Nonlabeled DNA is used and after separation on native nondenaturing gels the complexes are electrotransferred to nitrocellulose sheets. Immunodecoration with antibodies and revelation with protein A labeled with ^{125}I allows visualization of the protein component of the complexes. The choice of ^{125}I allows quantification either by excision of the labeled bands and counting or by careful densitometric scanning of the autoradiogram. In order to illustrate this technique we will use the example of RNA polymerase in RP_o and $RP_{(el)}$ complexes.

Gel Electrophoresis. Polyacrylamide gels of approximately 1 mm thick-

⁹ M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

¹⁰ M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).

¹¹ A. Kolb, M. Buckle, and H. Buc, *J. Cell. Biochem.* **12D**(Suppl.), 179 (1988).

ness and a length of 10 cm are prepared in TE buffer as described in Ref. 12. Complexes between 203-base pair (bp) linear fragments of *lacUV5* DNA (20 nM) and RNA polymerase (30 nM) are formed at 37° as described in Ref. 12. Aliquots of 2 μ l are taken into 2 μ l of 22% Ficoll containing bromphenol blue and applied to a gel running in TBE (Tris-borate, EDTA) at a constant voltage of 100 V at room temperature. The length of migration and the percentage of acrylamide are empirically determined to allow visualization of retarded complexes as well as the free DNA fragment.

Electrotransfer. On completion of electrophoresis the gel is removed to Whatman (Clifton, NJ) filter paper and placed against a nitrocellulose sheet. Transfer is equally effective if carried out by placing the gel/nitrocellulose sandwich in solution between two electrodes, as classically used for Western blots,¹³ or by arranging the gel/nitrocellulose between two graphite slabs, as in the so-called semidry blot.¹⁴ Transfer is usually carried out by applying a potential of 7 V/cm distance between the electrodes for 1 hr independent of the technique of transfer used. Around pH 8.0, the efficiency of transfer greatly varies from one protein to the next. We found empirically that it can be maximized by varying first the pH of the transfer buffer containing a low amount of sodium dodecyl sulfate (SDS) (<0.01%) and then by increasing the SDS content. Immediately following electrotransfer, the protein component of the complexes reacts with dyes such as Ponceau Red or China ink as would a single protein. Such a coloration may be used to confirm the completion of transfer. Silver staining of the polyacrylamide gel subsequent to electrotransfer can also be used for this purpose.

Immunodecoration. The nitrocellulose sheets are incubated at 37° in a blocking solution such as phosphate-buffered saline containing either 5% nonfat powdered milk or a suitable nonionic detergent (0.05% Tween 20, rather than Triton X-100) and are then exposed to the specific antibody raised against one or more of the protein components of the complex. To date we have used polyclonal antibodies, although a powerful alternative would be the use of monoclonal antibodies whose efficiency of detection should reflect the accessibility of epitopes of the protein in various complexes with DNA.¹⁵ If a quantification of the protein component within different complexes in the same gel is desired then subsequent detection with protein A labeled with ¹²⁵I¹⁶ is favored, since this allows either efficient

¹² D. C. Straney and D. M. Crothers, *Cell (Cambridge, Mass.)* **43**, 449 (1985).

¹³ H. Towbin, T. Staehelin, and J. Gordon, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 4350 (1979).

¹⁴ J. Kyshe-Anderson, *J. Biochem. Biophys. Methods* **10**, 203 (1984).

¹⁵ X. M. Li and J. S. Krakow, *J. Biol. Chem.* **262**, 8383 (1987).

¹⁶ S. W. Kessler, *J. Immunol.* **115**, 1617 (1975).

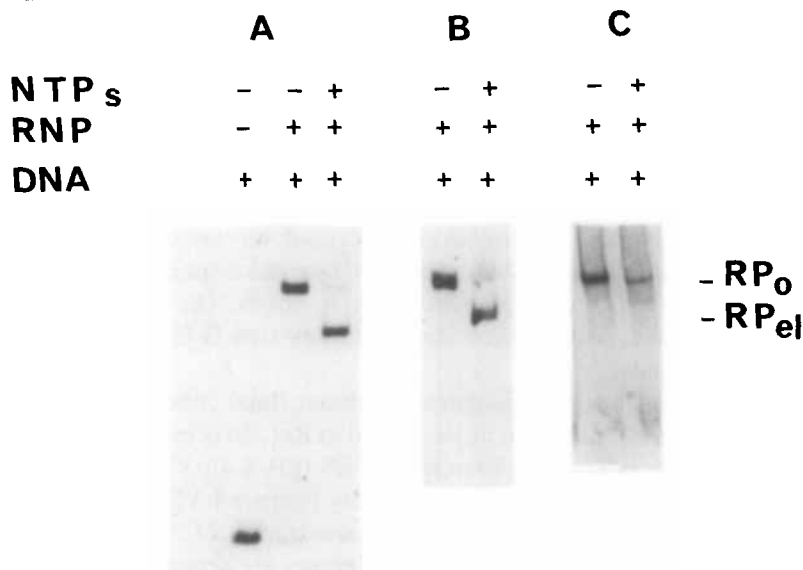


FIG. 2. Gel retardation assay of open and elongating complexes formed between RNA polymerase and the *lacUV5* promoter. Complexes between RNA polymerase (RNP) and the *lacUV5* promoter (DNA) in the presence or absence of nucleotide triphosphate substrates (NTPs) are formed as described in the text, and applied to nondenaturing 4% polyacrylamide gels. In (A), where ^{32}P -labeled DNA was used, the gel was autoradiographed; when nonlabeled DNA was present (B and C), the contents of the gel were electrotransferred to nitrocellulose sheets and incubated with either anti- α subunit antibodies (B) or anti- σ subunit antibodies (C) followed by ^{125}I -labeled protein A and autoradiography as discussed in the text. RP_0 , open complex, no substrate; RP_{el} , elongating complex with substrate such that polymerase has synthesized a 10-mer ribonucleotide.

densitometric scanning of the resulting autoradiogram or counting of the excised bands. On the other hand, a second anti-Fc antibody, covalently labeled with biotin, may be incubated with the nitrocellulose sheet followed by a streptavidin-alkaline phosphatase complex, which in turn binds to the biotinylated anti-antibody.¹⁷ A simple and rapid assay using 5-bromo-4-chloro-3-indolyl phosphate (PCIB) and nitroblue tetrazolium (NBT) gives a red-colored band on the nitrocellulose sheet.

Figure 2 shows how the presence or absence of several subunits of RNA polymerase can be seen at various stages of the reaction, following the reaction with protein A labeled with ^{125}I (cf. similar methods in Ref. 18).

¹⁷ M. Wilchek and E. Bayer, *Anal. Biochem.* **171**, 1 (1988).

¹⁸ D. C. Straney, S. B. Straney, and D. M. Crothers, *J. Mol. Biol.* **206**, 41 (1989).

Recording Protein Conformational Changes in Series of Consecutive Binary Complexes

At a higher level of complexity one would like to investigate important changes occurring in the protein itself. The initial rate of proteolytic digestion has been widely used for such a purpose (cf. Ref. 19 for an example). We will use here as an example the time course of digestion of the σ subunit (M_r 70,100) of *Escherichia coli* RNA polymerase with the protease V8 from *Staphylococcus aureus*. This protease has a specificity for D-G^{19a} or E-X (where X is a polar residue) peptide bonds. On the σ subunit of RNA polymerase, V8 can theoretically attack two D-G sites as well as several E-X bonds.

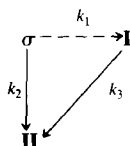
Proteolytic Attack. The isolated σ subunit (final concentration in the reaction 0.5 mg/ml) prepared as described in Ref. 20 is incubated at 37° in 50 μ l of a buffer containing 20 mM HEPES (pH 8.4), 3% (v/v) glycerol, 100 mM KCl, and 10 mM MgCl₂. Freshly prepared V8 protease in the same buffer is then added to a final concentration of 3 μ g/ml [ratio of V8/ σ , 1/160 (w/w)]. At time intervals, aliquots are removed to a quench solution containing 0.1 mM phenylmethylsulfonyl fluoride (PMSF) in SDS (0.5%, w/v). The time course of digestion is chosen as a function of the rate of disappearance, in this case, of the σ subunit. A typical profile of attack occurs over a 2-min period at 37°; however, in order to trap certain intermediates (RP_i) the temperature of the reaction must be decreased to around 16°. In comparing the digestion profile of these aptly named "frozen" complexes, the time scale of attack must necessarily be extended and clearly the comparisons with free subunit or in this case free RNA polymerase holoenzyme must be conducted at the same (16°) temperature. For example, the digestion of σ subunit in the holoenzyme at 37° took place over a period of 10 min under the conditions used for the more rapid digestion of isolated σ subunit. The digestion products are then separated by SDS-polyacrylamide gel electrophoresis (PAGE) and either stained with Coomassie Blue or electrotransferred to nitrocellulose and immunodecorated with anti- σ antibodies and ¹²⁵I-labeled protein A, (Fig. 3a). Quantitative densitometric scanning of these gels was carried out using a Bio-Rad (Richmond, CA) densitometer equipped with a Shimadzu (Columbia, MD) integrator. Normalization is then performed on a molar basis. It is not always possible to detect all the reaction products using SDS-PAGE. In this particular instance the three bands shown on the gel represent the larger fragments resulting from digestion.

¹⁹ C. J. Lazdunski, J. Pages, and D. Louvard, *J. Mol. Biol.* **97**, 309 (1975).

^{19a} D, Aspartic acid; E, glutamic acid; G, glycine.

²⁰ M. Gribskov and R. R. Burgess, *Gene* **26**, 109 (1983).

Quantitative Analysis of Proteolytic Products. It was first noted that the sum of the molar concentrations of the three major bands remains constant. Second, the time course of appearance and disappearance of the three major bands (Fig. 3b) indicates that the decay of the initial species follows a single exponential process, whereas two exponential phases are required to fit the two other curves. Within experimental error the characteristic time constants are equal, thus suggesting a scheme of the following type.



Hence for the σ subunit the rate of proteolysis by V8 can be characterized by three parameters, two of which (k_1 and k_2) represent conditions of initial attack, and the remaining one (k_3) a rate of interconversion between two products. It was verified that the same major products and the same scheme for their interconversion apply when σ was engaged in the RP_i , the RP_o , or various $RP_{(el)}$ species at *lacUV5*. The changes in the rate constants k_1 , k_2 , and k_3 are used to measure the conformational rearrangements in the σ subunit as elongation proceeds until σ is released.

Changes in Conformation of Cognate DNA-Binding Site as Probed by Chemical and Nucleolytic Reagents

The method of choice for the study of conformational changes in the nucleic acid component of nucleoprotein or nucleic acid-drug complexes, is to monitor the differential accessibility of base or sugar residues to chemical and nucleolytic probes. Ideally several conditions should be fulfilled.

1. The complex should adopt a *single* conformation so that the pattern of attack may be interpreted in simple terms (one pattern—one species). In default of this, one should search for conditions that favor the limiting species.
2. The reagent should attack the DNA and leave the protein intact. The differential rate of modification ($\pm$ protein) should reflect the DNA deformation in the *intact* binary complex. In particular, the reagent should neither displace nor perturb the complex that is probed.
3. In order to facilitate a structural interpretation one must be able to decide whether the reagent probes the major conformational state of the DNA or whether it reacts with another conformation in rapid equilibrium

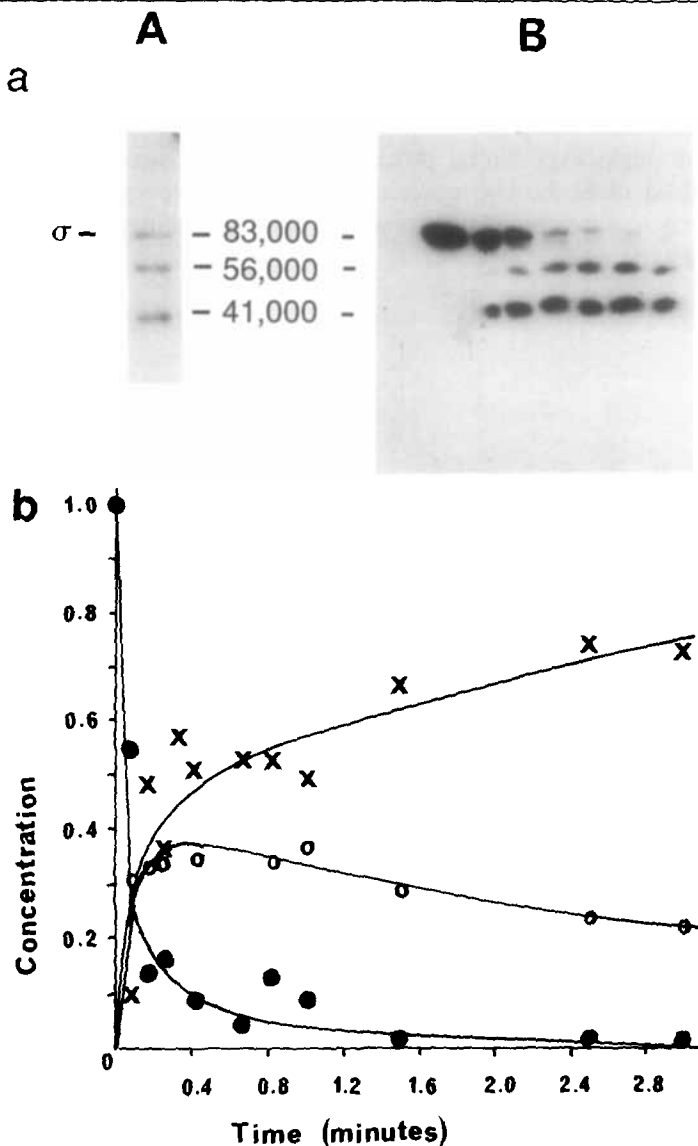


FIG. 3. V8 proteolytic digestion of the σ subunit of RNA polymerase. (a) Isolated σ subunit (A) or σ in the RNA polymerase holoenzyme (B) was digested with the protease V8 from *Staphylococcus aureus* over a limited time period as described in the text. The products were separated on SDS-PAGE and electrotransferred to nitrocellulose. Revelation was carried out using immunodecoration by anti- σ antibodies and ^{125}I -labeled protein A. (A) isolated σ alone following 30 sec of digestion with V8 protease at a protein-to-protease ratio of 160/1 (w/w) prior to quenching with 0.1 mM PMSF in 0.5% SDS. (B) Time course

with this species. An analysis of the rate of attack at each position allows one to distinguish between these possibilities.

In reality, none of these conditions is perfectly satisfied. Particular efforts to fulfill requirements (1) and (2) have been made and are described in other chapters of this volume (cf. also Refs. 21 and 22), but it should be stressed that even when binary complexes isolated from retardation gels are mildly attacked by a given probe, one is never certain that important rearrangements taking place during electrophoresis are not perturbing the initial complex. (A comparison of footprints before and after gel electrophoresis when a single retarded band is present should always be performed as a control.²³)

For quantitative analysis, conditions of single-hit kinetics should be respected (this means that for a 200-bp DNA fragment more than 90% of the DNA should remain intact); as a consequence the intensity of a given band will be independent of the position of the band with respect to the label (located at one end of a DNA strand). In many cases, however, a slight departure from single-hit kinetics can be accommodated by applying secondary corrections. A more crucial problem that cannot be easily circumvented in this way occurs in the case of a protein-DNA complex (cf. criterion (2) above). A first modification of the DNA can affect the reactivity at another position in the DNA. Single-hit conditions ensure that only *one* modification is present on each DNA molecule. The most obvious illustration of this situation is the attack of a complex formed on

²¹ L. C. Lutter, *J. Mol. Biol.* **124**, 391 (1978).

²² M. Brenowitz, D. F. Senear, M. A. Shea, and G. K. Ackers, this series, Vol. 130, p. 132.

²³ M. D. Kuwabara and D. S. Sigman, *Biochemistry* **26**, 7234 (1987).

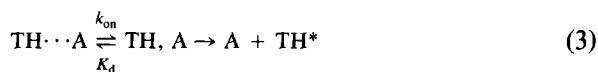
of digestion of σ in the RNA polymerase holoenzyme. Conditions were as described in the text, each lane representing increments of 1 min of attack prior to quenching. (b) Kinetic profiles for the attack of V8 protease on isolated σ subunit. Conditions were carried out as described above. The relative proportions of each band were calculated from densitometric scans of the type of gels shown in (a). ●, Native σ subunit; ○, 56-kDa peptide; and ×, 41-kDa peptide. Curves were fitted according to the scheme shown in the text by these equations:

$$\begin{aligned}\sigma(t) &= e^{-kt} \\ \text{I}(t) &= \frac{k_2}{k_3 - k} (e^{-kt} - e^{-k_3t}) \\ \text{II}(t) &= 1 + \frac{k_1 - k_3}{k_3 - k} e^{-kt} + \frac{k_2}{k_3 - k} e^{-k_3t}\end{aligned}$$

where $k = k_1 + k_2$.

a negatively supercoiled template by a nicking reagent.²⁴ Furthermore, single-hit considerations should apply to the attack on both the DNA and the protein when the reagent (such as diethyl pyrocarbonate or osmium tetroxide) can affect the two partners.²⁵ The initial rate of attack of DNA at any of the positions considered should be significantly larger than the rate of attack at a protein residue that, once modified, could affect the free energy of association and/or the conformation of the complex.

The third condition (3) is rarely taken into consideration. The simplest possible case, imino-proton exchange from a base pair in a free duplex,²⁶ serves to make the point: for an AT base pair, the reaction can be summarized as



TH is a protonated thymine hydrogen bonded to a cognate adenine on the opposite strand. In the initial situation only the open state (TH, A) can react, the exchange on the thymine leading to TH* being catalyzed by the base components of the buffer.

A correct interpretation of the exchange time can only be made by following its dependence with respect to the catalyst concentration and comparing with the exchange time of a freely exposed base. At low catalyst concentration, many openings are required before exchange occurs and the exchange time is equal to that observed for the monomer measured under the same conditions of catalysis, divided by the base pair dissociation constant K_d . If the catalyst concentration is high enough, the exchange time is equal to the base pair lifetime, k_{on}^{-1} , since exchange occurs for each opening event.²⁶ In an analogous fashion one can carry out chemical modification of a base in a DNA molecule and measure the rate of attack as a function of the reagent concentration (C). We will examine two situations characterized by their respective rate profiles.

Hyperbolic Profile. When an overall profile is generated that shows a hyperbolic dependence of the rate on c then the same formalism developed for proton exchange will be used. At high reagent concentrations, an initial rate of attack that is independent of the concentration and nature of the reagent will be interpreted as the rate of access to a distorted, reactive intermediate [k_{on} in Eq. (3)]. At low reagent concentrations the initial rate of attack (now proportional to the reagent concentration) reflects the deformability of the complex. If, as in the case of imino-proton exchange, a model representation of the reactive species can be designed, a two-

²⁴ D. M. J. Lilley and B. Kemper, *Cell (Cambridge, Mass.)* **36**, 413 (1984).

²⁵ M. Buckle and H. Buc, *Biochemistry* **28**, 4388 (1989).

²⁶ J. L. Leroy, D. Broseta, and M. Guéron, *J. Mol. Biol.* **184**, 165 (1985).

step reaction scheme will be used to yield the corresponding equilibrium constant, K_d .

Linear Profile. Rapid reaction rates that are strictly first order with respect to the reagent at *all* reagent concentrations are classically interpreted as reflecting a property of the ground state. We define the ground state as being the major populated species that is the target for the reagent used at the initial equilibrium conditions. Such has been the case for a long time for DNase I attack on a double-stranded DNA fragment. This could turn out to be wrong, and to represent rather a minor population of conformations in rapid equilibrium with the ground state. As in the preceding case model systems representative of the putative reactive intermediate will be useful to clarify the problem. The practical problems that are then encountered in dealing with the comparison of reactivity profiles, obtained at various reagent concentrations and on several discrete binary complexes, arise from the quantitative scanning of gels of electrophoresis and the comparison of the corresponding optical density profiles (see below).

An example in which many of these considerations are examined in some detail is given in Ref. 25. In this study the reactivity of bases in the locally melted region of an open complex was examined using the reagents dimethyl sulfate (DMS) and diethyl pyrocarbonate (DEP). The reactivity of DMS with the N-7 of guanine residues in DNA occurs readily since no important distortion of the DNA is required to render the methylation site accessible. The kinetics of attack are always proportional to the concentration of the reagent. Conversely, the rate of DMS attack at position N-1 of cytosines that are not base paired presents a slower, hyperbolic profile. A similar situation is encountered for the reactivity of DEP with adenines in single-stranded DNA or in open complexes.²⁵ In both cases the reactive cytosine or adenine residues have to be in conformations that depart extensively from classical B-DNA structures. For bases in an open complex, the situation is, in part, analogous to that described in Eq. (3); the rate of modification measured at high concentrations of reagent reflects the k_{on} controlled formation of the distorted reactive intermediate. This deformation or isomerization step was shown to be a function of the position of a given base within the putative single-stranded region of the open complex.²⁵

These considerations impose certain criteria governing the choice of a given reagent as a probe of DNA structures. Table I^{27-40a} lists a selection of DNA-modifying or DNA-cleaving reagents that are currently in use.

²⁷ D. Suck and C. Oefner, *Nature (London)* **321**, 620 (1986).

²⁸ H. R. Drew and A. A. Travers, *J. Mol. Biol.* **186**, 773 (1985).

²⁹ D. S. Sigman, D. R. Graham, V. D'Aurora, and A. M. Stern, *J. Biol. Chem.* **254**, 12269 (1979).

³⁰ D. S. Sigman, A. Spassky, S. Rimsky, and H. Buc, *Biopolymers* **24**, 183 (1985).

TABLE I
MODIFYING AND CLEAVING REAGENTS AS PROBES

Reagent	Base	Site on base	Structure attacked	Ref.
DNase I	Nonspecific	—	Minor groove	27, 28, 29
OP_2Cu^+	Nonspecific	—	Minor groove	6, 30
Fe-EDTA	Nonspecific	—	Minor groove	31, 32
Dimethyl sulfate (DMS)	G	N-7, N-3, N-1	B-DNA	33
	C	N-3	Unpaired cytosines	25, 34
Diethyl pyrocarbonate (DEP)	A	N-7 (major groove)	Z-DNA Cruciform loops Hogsteen base pairing Open complexes	35 36 37 25
KMnO_4	T	C-5-C-6 (double bond)	Cruciforms Open complexes	37a 38
OsO_4	T	C-5-C-6 (double bond)	Supercoiled B-DNA B-Z junctions Open complexes	39 40 40a

The reagents are broadly classified into two general, nonmutually exclusive groups in a compilation that is by no means exhaustive. The division is based on the consideration that Fe-EDTA, $(\text{OP})_2\text{Cu}^+$, DNase I, and finally DMS (for guanine residues) all represent reagents that, while not necessarily base nonspecific, probe a DNA structure in a fashion that reflects as closely as possible either the ground state of the DNA structure or a structure in rapid equilibrium with the ground state. One expects, therefore, a linear dependence on the concentration of the reagent, within the limits of single-hit conditions.

The reactivity profiles generated by the second group require an interpretation of the type afforded by Eq. (3). The plateau of reactivity observed for cytosines or adenines in an open complex may be interpreted as reflecting the rate of access to the reactive conformation of the base.

³¹ T. D. Tullius, *Trends Biochem. Sci.* **12**, 297 (1987).

³² T. D. Tullis, *Nature (London)* **332**, 663 (1988).

³³ W. Gilbert, A. Maxam, and A. Mirzabekov, in "Alfred Benzon Symposium XIII" (N. O. Kjeldgaard and O. Maaloe, eds.), p. 139. Munksgaard, Copenhagen, 1976.

³⁴ A. Spassky and D. Sigman, *Biochemistry* **24**, 8050 (1985).

³⁵ W. Herr, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 8009 (1985).

³⁶ J. C. Furlong and D. M. J. Lilley, *Nucleic Acids Res.* **14**, 3995 (1986).

³⁷ D. Mendel and P. B. Dervan, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 910 (1987).

^{37a} D. M. J. Lilley and E. Palecek, *EMBO J.* **3**, 1187 (1984).

³⁸ S. Sasse-Dwight and J. D. Gralla, *J. Biochem. (Tokyo)* **264**, 8074 (1989).

³⁹ E. Lukasova, F. Jelen, and E. Palacek, *Gen. Physiol. Biophys.* **1**, 53 (1984).

⁴⁰ G. Galazka, E. Palecek, R. D. Wells, and J. Klysik, *J. Biol. Chem.* **261**, 7093 (1986).

^{40a} M. Budde, personal communication.

Chemical Reactivity of Single-Stranded DNA. To illustrate the methodology involved in the kinetic approach, we will describe a series of experiments on putative single-stranded DNA regions with two reagents, diethyl pyrocarbonate (DEP) and osmium tetroxide (OsO_4). The single-stranded regions examined were either 57-mer oligomers or the same oligomers hybridized in complexes with complementary strands of DNA containing stretches of mismatches that produced fragments having "open" unpaired regions of varying sizes.

Attack by Osmium Tetroxide. DNA fragments (10 $\mu\text{g}/\text{ml}$) (5'-end-labeled by polynucleotide kinase with $[\gamma\text{-}^{32}\text{P}]\text{ATP}$) were incubated in a 20 mM HEPES buffer, pH 8.4, at 37° in the presence of 3% glycerol and 0.1 mg/ml sonicated calf thymus DNA. An aliquot of a stock solution (20 mM) of OsO_4 was added to a final concentration of 1 mM. At zero time the reaction was begun by the addition of pyridine to a final concentration of 0.1 M. Aliquots of the reaction were withdrawn at defined time intervals and quenched by precipitation of the DNA in ethanol. Samples were vacuum dried, resuspended in 1 M piperidine, heated at 90° for 30 min, lyophilized, and after denaturation in formamide at 90° for 3 min separated on 12% (w/v) polyacrylamide sequencing gels. Visualization of the bands was effected by classical autoradiography, quantification was carried out by scanning densitometry. A gel is shown in Fig. 4.

Attack by Diethyl Pyrocarbonate. DNA fragments as above were incubated in the HEPES buffer as described for the OsO_4 reaction. A stock solution of DEP, freshly prepared in ethanol, was diluted into water and added to the reaction mixture at zero time to give a final DEP concentration of 20 mM. Aliquots of the reaction mixture were removed at defined time intervals and quenched by the addition of 10 mM imidazole solution. The DNA was ethanol precipitated, vacuum dried, resuspended in 1 M piperidine heated at 90° for 30 min, and lyophilized. Polyacrylamide gel electrophoresis, autoradiography, and densitometry were carried out as described above.

The results confirmed that both DEP and OsO_4 react either with all bases in a single-stranded DNA, or exclusively with only those bases that lie within the mismatched region of the hybrids. The kinetic analysis demonstrated that single-hit conditions may be met only during a relatively short interval of time, but that in both cases the reaction was essentially second order. The reaction of adenines in an open complex with DEP has already been shown to be a first order process, limited by an isomerization that exposes the base.²⁵ A similar study of OsO_4 on thymines in RP_0 is currently under investigation. Furthermore, it has been shown⁴¹ that the

⁴¹ E. Palacek, A. Krejcová, M. Votjšková, V. Podgorodnichenko, T. Ilynia, and A. Poverenny, *Gen. Physiol. Biophys.* **8**, 491 (1989).

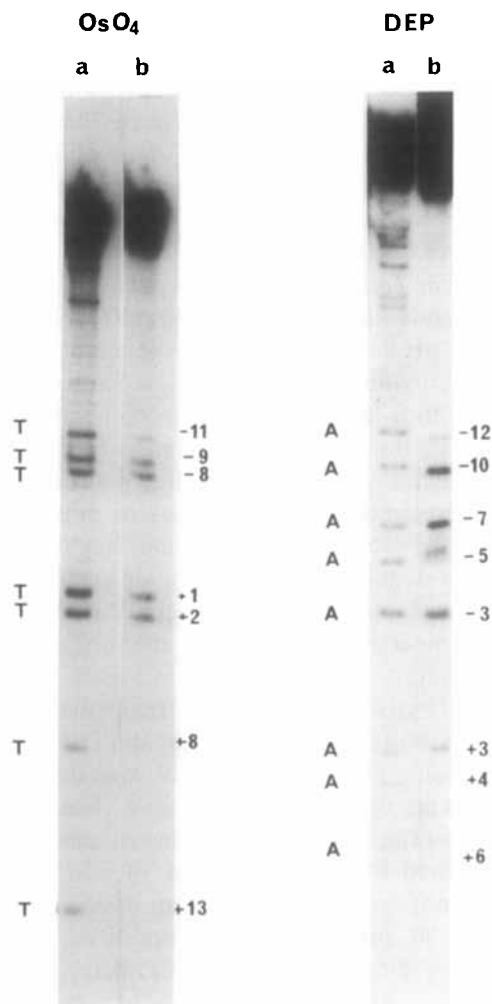


FIG. 4. Chemical cleavage of DNA by diethyl pyrocarbonate and osmium tetroxide. Single-stranded DNA (a) or duplex DNA (b) containing an internal 13-bp stretch of unpaired cytosines was modified by diethyl pyrocarbonate or osmium tetroxide as described in the text. Cleavage of the modified DNA at the sugar phosphate backbone was carried out by piperidine treatment at 90° for 30 min and the end-labeled fragments were separated on a 12% sequencing polyacrylamide gel. The 57-mer sequence was derived from a region of the *lac* promoter and the numeration on the gel refers to the position of a base with respect to the transcription start site at +1.

nature of the base, such as piperidine, is important in determining the reactivity of OsO_4 toward a given nucleotide.

Characterization of Transient Intermediates

When a given amount of RNA polymerase is added to a mixture containing a DNA fragment and the substrates necessary for abortive cycling of the enzyme at the corresponding promoter, a latency time, τ , is observed during the time course of product accumulation (Fig. 1). The rate profile for the formation of the kinetically competent binary complex RP_o can be traced from the results of the enzymatic assay using a simple derivation with respect to time (Fig. 1). In simple cases, based on that depicted in Eq. (3), it is expected to follow the relation

$$[\text{RP}_o]_t = [\text{P}]_T(1 - e^{-t/\tau}) \quad (4)$$

where $[\text{P}]_T$ is the concentration of the promoter containing DNA fragment. Analysis of the dependence of τ on RNA polymerase concentration in the enzymatic assay leads to an estimation of the constants K_B and k_2 for the two-step model given in Eq. (2), and thus allows the prediction, at each concentration of enzyme, of the concerted decay of the RP_c intermediate, as well as that of the free promoter (curves RP_c and P in Fig. 1; the two species being assumed to be in rapid equilibrium with respect to the time scale in Fig. 1).

Independent determinations of the deformations occurring in the DNA fragment and the enzyme during the same time interval will challenge and hence further define this model. Indeed, within this context, relatively rapid chemical attack on the DNA, at time intervals suitably spaced with respect to τ , can be used for this type of study. In practice, to date, this approach has been used only in a qualitative manner, using fairly weak promoters⁵ where τ is generally longer than 3 min. We describe here improvements allowing a more rigorous and quantitative test of current models.

Analysis of Establishment of Chemical Reactivity Profiles

A typical footprinting experiment would yield a series of radioactively end-labeled DNA fragments generated by chemical attack for a short period (Δt) at various times after the mixing of a promoter and the enzyme RNA polymerase. Analysis of the cleavage pattern requires polyacrylamide gel electrophoresis and densitometric scanning. This approach is currently used to test and specify the model given in Eq. (2) and Fig. 1. For each concentration of RNA polymerase a single characteristic time τ will define the decay and appearance of the various species:

$$\begin{aligned} [P_o]_t &= \lambda P_T e^{-t/\tau} \\ [RP_c]_t &= (1 - \lambda) P_T e^{-t/\tau} \\ [RP_o]_t &= P_T (1 - e^{-t/\tau}) \end{aligned}$$

where $\lambda = 1/(1 + K_B[R])$.

Besides checking the consistency of the two-step model, this kinetic footprinting experiment will generate a reactivity profile characteristic of the free promoter I_0 , of the final complex I_2 , and of the intermediate I_1 at each position of the sequence.

If the model is obeyed, then the intensity of a given band I_x should either stay constant or vary between the initial and the final state according to an exponential law:

$$I_x(t) - I_x(0) = [I_x(\infty) - I_x(0)]e^{-t/\tau} \quad (5)$$

Errors can be estimated by a reconstitution experiment, as explained in Ref. 5. DNA attacked in the absence of enzyme or in the final complex are mixed in known proportions and applied to separate lanes of the gel. That the values of τ are consistent may be checked by examining all the sensitive bands corresponding to the kinetic system. Extrapolation to an infinite time yields $I_x(\infty)$, which corresponds to τ_2 , the reactivity pattern of the open complex. Extrapolation to zero time yields $I_x(0)$, which is

$$I_x(0) = P_T[\tau_0\lambda + \tau_1(1 - \lambda)] \quad (6)$$

Experiments performed at various RNA polymerase concentrations affect λ according to Eq. (4). I_0 , I_1 , and K_B can then be estimated from the change of $I_x(0)$ with respect to RNA polymerase concentration. Conversely, if the two-step model does not apply, then the minimal number of exponential components of the pattern will be derived using, for example, the Padé-Laplace approximants.⁴² Eventually, the observations will yield the pattern of the reactivity profile of several consecutive intermediates. The following two sections describe technical details of signal acquisition and data analysis.

Rapid Mixing: A Microvolume Continuous/Stopped Flow Apparatus

In order to follow reactions that have a τ of around 10 sec, and to satisfy the requirements outlined above, a rapid mixing technique was required. A classical stopped-flow apparatus generally allows only one product sample per experiment to be assayed and requires a large amount of starting material in order to analyze several consecutive samples. For our purposes we are obliged to work with relatively small volumes (<100

⁴² E. Yeramian and P. Claverie, *Nature (London)* **326**, 169 (1987).

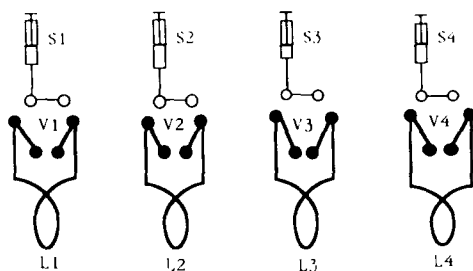
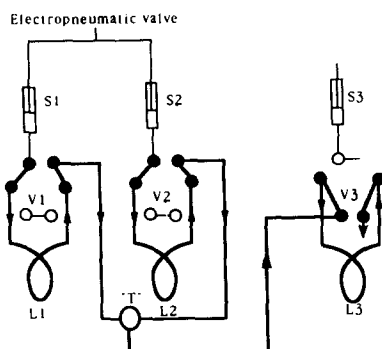
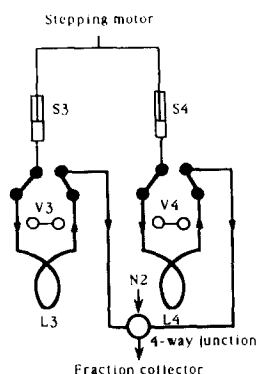
a**b****c**

FIG. 5. Rapid mixing stopped-flow apparatus. Schematic representation of the operating principle. A detailed explanation is given in the text. S1 and S2 are electropneumatically driven water-filled syringes, S3 and S4 are water-filled syringes driven by a step motor. L1, L2, L3, and L4 are reaction loops. All valves labeled V are HPLC-type valves having negligible dead volumes.

μl) that should be used, however, to generate several data points on a time curve. In order to satisfy these requirements we designed the apparatus described in Fig. 5.

The system may be used either to follow a time course of, for example, a classical enzyme substrate reaction, or to study the effects of a reagent on a complex formed at $t = 0$ over a relatively short time course, as

typified by a chemical attack on a complex between RNA polymerase and a promoter containing DNA fragment. The operating principle of the apparatus relies on the use of high-performance liquid chromatography (HPLC) valves having a negligible dead volume. For a simple rapid mixing and time course quench, the overall operation may be summarized as described below.

The loops L1, L2, and L4 (Fig. 5) are initially loaded with, for example, enzyme, radioactive substrate, and quenching solution, respectively. At time $t = 0$, the contents of loops L1 and L2 are propelled via a mixing T junction into a third loop, L3, in which the enzyme reaction now takes place. The contents of loop L3 are then ejected as discrete fractions as a function of time into a second T junction, where they are mixed with an equal volume of quenching solution. The fractions are then carried by a continuous flow of an inert gas such as nitrogen into a waiting fraction collector. Prior to the reaction cycle all loops and connecting tubes are purged with air to prevent dilution due to lamellar flow. All reaction and mixing loops are immersed in a cryogenic fluid maintained at any given temperature. The volume of the loops L1 and L2 can be varied from 20 to 200 μl . To facilitate smooth flow through the tubing, bores of 0.3- to 0.8-mm diameter are used.

The force that propulses the contents of loops L1 and L2 through the T junction into loop L3 is provided via two water-filled syringes whose pistons are simultaneously driven by an electropneumatic valve. As soon as loop L3 is filled by the mixed reaction a second electropneumatic valve is activated. This operation places the loops L3, containing the reaction, and L4, containing the quench, under the control of a stepping motor. The controlled descent of this motor progressively injects small volumes of water into loops L3 and L4, which consequently expel the required amounts of reaction and quench into the exit four-way junction from where the stream of inert gas carries the mixture to the fraction collector. The inert gas flux also prevents premature mixing of the reaction with the quench solution. The stepping motor that drives the fraction collector is programmed to displace successive tubes in synchrony with the ejection of each sample.

The whole operation is commanded by an IBM PS2 microcomputer. The operating program allows selection of different volumes of the reaction to be removed to the quench and collector at various time points. The rapid mixing does not inactivate any of the enzymes examined to date nor does it induce shearing in linear DNA fragments.

In the second mode, RNA polymerase is placed in L1, DNA fragment in L2, and at $t = 0$ the two are propelled into L3. Loop L4 contains a specified concentration of the modifying chemical reagent, which at given

time intervals is mixed with the current nucleoprotein complex and ejected into the fraction collector. The awaiting collecting tubes can either contain a predetermined volume of a suitable quench or an aliquot of quench may be manually or mechanically added at the required times to the tubes in the fraction collector.

The advantages offered by this type of machine include short (millisecond) mixing times, negligible or nonexistent dead volumes (a logical consequence of using HPLC valves and of having a stream of inert gas that ejects all the samples into awaiting collecting tubes), no losses due to lamellar flow-induced mixing as solutions flow through narrow-bore tubes (avoiding here by having air present in all the mixing and connecting loops prior to reaction), and a reproducibility of time and volumes.

Quantitative Determination of Radioactivity Profile in Given Lane of Polyacrylamide Gel

The establishment of a reactivity profile assigned to the initial, transient, or final species (see above) implies quantitative determination of the bands resolved on a polyacrylamide gel. There are three principal techniques that are in current use. The first of these involves direct counting of excised radioactive bands and remains the most accurate and direct approach. It is necessary, however, that the sample loaded onto the gel is uniformly labeled (i.e., compare the pattern obtained from an end-labeled linear DNA fragment as opposed to that from a digest of a nonhomogeneously reductively methylated ^{14}C -labeled protein) and that the bands are sufficiently resolved so as to permit discrete excision of each band. Both conditions are in practice rarely fulfilled. Consequently, the remaining two techniques, involving scanning of either an autoradiogram of the gel or the gel itself, must be used. The problems involved in densitometric scanning of autoradiograms have been treated elsewhere and the introduction of new devices is constantly improving the methodology.⁴³ We shall restrict ourselves here to four considerations.

1. It should be emphasized that it is the nature of the scanning device that dictates the type and quality of the electrophoretic gel, and not the converse. There are two types of apparatus that are in current use for densitometric scanning. For the sake of argument we will classify these into two categories, which we call line densitometers and band densitometers.

Line densitometers: This type of machine scans a narrow window (0.05 to 0.3 mm) down the length of a lane on a gel or autoradiogram. In practice the bands are aligned such that the slit passes through the center of each

⁴³ G. M. Smith and D. G. Thomas, *CABIOS* 6, 93 (1990).

band. The scanner then displays or traces a series of peaks that correspond in surface area to the optical density profile across that portion of each band that is sampled. In general there is accompanying software that integrates the peaks thus obtained. The major specific reservation with this system is that the sampling of each band must provide a true representation of the distribution of material in the band. In order to be compatible with the relatively simple algorithms used in the integration process, the optical profile of a band should be as close to a Gaussian as possible and all the bands are assumed to possess identical profiles independent of their length of displacement through the gel.

Band densitometers: In this instance the whole gel is photographed and digitalized such that an associated computer can reconstruct an image in which the pixel density is proportional to the recorded optical density of the photograph. The experimenter then chooses, with the aid of appropriate software, the region that corresponds to a discrete peak that is then easily quantified for comparative purposes. This system thus eliminates the problem of representative sampling that flows the line densitometers, since the whole of a band is actually sampled.

2. Independent of the type of scanning apparatus used, the peak signal obtained must be clearly differentiated from background noise in a uniform fashion that does not bias the integration procedure subsequently used to quantitate the area contained within the peak. This phenomenon is independent of the saturating capacity of the film.

3. Photographic films currently popular for autoradiography of sequencing type gels have a dynamic range of approximately 10^2 ; simply stated this means that the measured optical density should be linear over a range of 0.01 to 1.0 OD units (this is an approximate range and will vary among film types and autoradiographic conditions). Below the lower limit errors arise due to an increasingly subjective analysis of the signal-to-noise ratio on behalf of the operator. Above an OD of approximately 1.0 the response of the film is nonlinear and continued ignorance of this phenomenon induces large quantitative errors. Realistic scanning thus necessitates autoradiography of relatively short exposure times compared to the half-life of the isotope used and under constant conditions such that a series of exposure times allows the comparison of different bands on a gel (having ascertained that the response of a film to the radioactivity is constant with time).

4. The problems associated with densitometric analysis exemplified above, and the obviously powerful incentive to resolve them, have resulted in two relatively recent innovations in this field. The first approach retains the idea of scanning densitometry and addresses itself to the problem of the dynamic range of the photographic film. The answer adopted was to

use a special film that is an integral part of the scanner.⁴⁴ The film itself has a dynamic range of 10^5 ; furthermore, the bands are scanned in their entirety, thus eliminating sampling errors arising from anomalous migration. The film has the added advantage of being reusable. The second technique returns to a direct determination of the radioactivity in a gel. The polyacrylamide gel is placed just below a network of thin wires, the intersections of which record changes in electrical potential due to the presence of ionizing radiation. This grid is gradually displaced over a gel so as to map the distribution of radioactivity. Again, a computer equipped with the relevant algorithms reconstitutes an image where the pixel density reflects the counts acquired per unit area. The counts per unit area are used to carry out the comparative analysis between different regions (i.e., bands) on the gel. This system eliminates all the disadvantages of optical systems outlined above. However, use of this approach is restricted to the quantification of uniformly labeled samples. Furthermore, to date, the resolution between bands is no greater than 4 mm, i.e., insufficient for the separation of bands characteristic of a DNA sequencing gel.

Conclusion

We have attempted to provide an overview of how kinetic considerations may be merged with existing and emerging technologies to study conformational changes in nucleoprotein complexes. Reaction conditions can be adjusted so as to favor a given species at equilibrium. A variety of techniques can then be used for the characterization of this reaction intermediate. Both the protein and the nucleic acid component may be studied in considerable detail.

A kinetic study provides additional information concerning both the actual number of components in a given system as well as the rates of interconversion between the species. This is the only approach by which transient intermediates may be characterized. The most widely used techniques to date involve "footprinting." The time resolution is determined by the duration of signal acquisition (on the order of seconds, for most footprinting techniques).

A major advantage of footprinting techniques is that they allow the recording of changes occurring on all the bases of a nucleic acid target (~50 bp in the case of an RNA polymerase promoter interaction) and therefore provide a sample of data allowing quantitative testing of simple kinetic models. The most delicate part is probably the interpretation of changes in chemical reactivity patterns in terms of the deformation of the

⁴⁴ R. F. Johnston, S. C. Pickett, and D. L. Barker, *Electrophoresis* **11**, 355 (1990).

nucleic acid template. Future technological advances will certainly include the development of equally powerful approaches to furnish details about the protein component responsible for these deformations.

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[15] Thermodynamic Methods for Model-Independent Determination of Equilibrium Binding Isotherms for Protein–DNA Interactions: Spectroscopic Approaches to Monitor Binding

By TIMOTHY M. LOHMAN and WŁODZIMIERZ BUJALOWSKI

Introduction

A detailed understanding of macromolecular interactions, such as those involving proteins and nucleic acids, requires information about the energetics (thermodynamics) and kinetics of the interacting molecules as well as their structures. In particular, thermodynamic studies provide the information required to determine and quantify the forces that stabilize these interactions and any functionally important structural changes that occur within the complex. A quantitative study of the equilibrium binding of a protein to a nucleic acid is usually designed to determine the following information: (1) the stoichiometry of the interaction, (2) the existence and degree of cooperativity (positive or negative), (3) model-independent macroscopic binding constants and possibly model-dependent intrinsic binding constants, and (4) the dependence of the equilibrium binding parameters on solution variables (e.g., temperature, pH, and salt concentration), which can yield information about the molecular forces involved in the binding interaction.

In order to achieve these goals, equilibrium binding isotherms must be measured over a range of binding densities. An equilibrium isotherm for the binding of a ligand to a macromolecule is simply the relationship

between the extent of ligand binding (ligand bound per macromolecule) and the free ligand concentration. True binding isotherms are model independent and reflect only this relationship. From such an isotherm one can obtain information about the macroscopic affinity of the ligand for the macromolecule and the stoichiometry of binding. However, for systems that involve the interaction of multiple ligands with a macromolecule, one can also analyze the binding isotherms to obtain physically meaningful *intrinsic* interaction constants that are related to the free energies of interaction. This is accomplished by comparing the experimental isotherms to theoretical predictions based on statistical thermodynamic models that incorporate the known molecular features of the interaction (cooperativity, overlap, etc.).

A number of techniques have been used to examine protein–nucleic acid binding equilibria. Sequence-specific binding has been examined quantitatively by filter binding,¹ footprinting,^{2,3} and gel retardation techniques,^{4,5} whereas nonspecific binding has been examined quantitatively using transport and chromatographic techniques.^{6–11} Some of these techniques are thermodynamically rigorous, while others involve assumptions that must be verified for each system. Techniques such as equilibrium dialysis are thermodynamically rigorous and yield absolute binding isotherms without assumptions; however, this approach is time consuming and may not be applicable to studies of interactions between macromolecules.

More convenient methods become available if a change in a spectroscopic property of either the ligand or the macromolecule occurs on formation of a complex and these have been used to examine the binding of a variety of proteins, and other ligands, to both DNA and RNA^{12–29}; how-

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¹³ D. E. Draper and P. H. von Hippel, *J. Mol. Biol.* **122**, 321 (1978).

¹⁴ S. C. Kowalczykowski, N. Lonberg, J. W. Newport, and P. H. von Hippel, *J. Mol. Biol.* **145**, 75 (1981).

ever, these are indirect methods. If an "apparent" binding isotherm is obtained by indirect methods that involve assumptions, then the apparent isotherm and the interaction parameters that one obtains will be no more accurate than the assumptions. This may cause particular problems if an isotherm will be used to differentiate between alternative models for an interaction, since if one of the models does not reproduce the isotherm, this may be due either to the failure of the model or the failure of the assumptions on which the calculated isotherm is based. As a result, it is important to obtain a correct, model-independent binding isotherm, independent of assumptions.

In this chapter, we discuss the use of spectroscopic approaches to study binding phenomena and their rigorous thermodynamic analysis to obtain true, model-independent binding isotherms. In these approaches, one monitors a spectroscopic signal [e.g., absorbance, fluorescence, circular dichroism, nuclear magnetic resonance (NMR) linewidth, or chemical shift] from either the macromolecule or the ligand that changes on formation of the ligand-macromolecule complex. In order to use this signal change to calculate a binding isotherm, it is commonly *assumed* that a linear relationship exists between the signal change and the fractional saturation of the ligand or macromolecule. However, if a linear relationship does not hold, then the calculated apparent isotherm will be incorrect. In the general case, in which multiple ligands (e.g., proteins) bind to a macromolecule (e.g., DNA), the fractional signal change and the extent of binding may not (and often do not) have a simple linear relationship. This may occur if the ligand can bind in several modes, each possessing a different spectroscopic signal (e.g., if cooperativity exists). The extent of any deviation from the ideal linear behavior is usually unknown *a priori*,

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²⁹ T. M. Lohman and W. Bujalowski, *Biochemistry* **27**, 2260 (1988).

hence the degree of error introduced into the apparent isotherm and the resulting binding parameters is also unknown.

If an indirect spectroscopic approach is used, it is necessary to determine the relationship between the spectroscopic signal change and the degree of saturation of the ligand or the macromolecule. We will review thermodynamic approaches to analyzing such data that allow the determination of true binding isotherms in the absence of assumptions. The use of these methods allows one to use a spectroscopic signal to obtain a true binding isotherm, even when a direct proportionality does not exist between the fractional signal change and the degree of binding. Since the methods reviewed in this chapter are thermodynamically rigorous, they do not require calibration by other techniques. In some cases one can even determine spectroscopic parameters for a ligand interaction at individual binding sites.

The only constraint on the methods of analysis reviewed here is that they are not valid if the ligand or macromolecule undergoes a change in aggregation or self-assembly state within the experimental concentration range examined. Therefore, as in any case, knowledge of the self-assembly properties of the protein or nucleic acid under study is essential before a rigorous analysis of a binding phenomenon can be undertaken. However, these approaches are valid for binding systems that exhibit cooperativity of any type.

Previous discussions of the use of these rigorous approaches have appeared, both for the case in which the spectroscopic signal is from the macromolecule^{24,25,29-32} as well as when the signal is from the ligand.^{21,33-35} The methods of analysis differ for these two cases, as is discussed below. These methods of analysis are not constrained to the study of interactions that can be monitored by spectroscopic signals. Rather, they are general and can be used to analyze titrations that monitor any property that reflects binding (e.g., viscosity changes^{31,36}).

Theory

In our discussions of binding, the term *macromolecule* refers to the species that binds multiple ligands. We first discuss the case in which the observed signal is from the macromolecule, since the results are fairly

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³⁶ B. M. J. Revet, M. Schmir, and J. Vinograd, *Nature (London) New Biol.* **229**, 10 (1971).

intuitive. We then discuss the case in which the observed signal is from the ligand, which requires a slightly different analysis. In each case, we derive the general binding density function (BDF) and show that the BDF is dependent on only the binding density distribution of ligand on the macromolecule, thus demonstrating that the BDF analysis is thermodynamically rigorous and without assumptions. The BDF equations are derived without specifying the actual signal that is monitored in order to emphasize their general character; however, specific cases are treated as examples.

In studies of protein–DNA and all ligand–macromolecule equilibria, two types of titrations can be performed to examine the binding equilibrium. In one case, the macromolecule is titrated with ligand and we refer to this as a “normal” titration, since the binding density, ν , (average moles of ligand bound per mole of macromolecule) increases during the progress of the titration. In the second case, the ligand is titrated with the macromolecule, which we refer to as a “reverse” titration, since the binding density decreases during the progress of the titration. Generally, the type of titration that is performed will depend on whether the signal being monitored is from the macromolecule (normal) or the ligand (reverse). Although this is not a strict requirement, the use of a normal titration when monitoring a signal from the ligand generally results in a titration curve that is less precise. The following sections discuss the rigorous methods that can be used to convert a titration curve, i.e., a change in signal as a function of titrant into a true, model-independent binding isotherm, which can then be analyzed to obtain binding and thermodynamic parameters for that interaction.

Thermodynamic Basis for Binding Density Function Method of Analysis

The methods of analysis reviewed here should be applied in cases in which an indirect signal is used to monitor binding and multiple equilibria may exist, i.e., multiple ligands can bind per macromolecule. The thermodynamic basis for these methods is that the distribution of ligands bound in different states i , symbolized by $\Sigma \nu_i$ (average number of ligands bound per macromolecule), is strictly determined at equilibrium by the free ligand concentration, L_F (more precisely the chemical potential of the free ligand).³⁷ Therefore, if the free ligand concentration is the same for two (or more) solutions at different total macromolecule concentrations (M_T), then the average ligand binding density distributions, $\Sigma \nu_i$, will also be the same

³⁷ T. Hill, “Cooperativity Theory in Physical Biochemistry.” Springer-Verlag, New York, 1985.

in each solution. As a result, constant values of L_F and hence $\Sigma \nu_i$ will exist for a number of different combinations of *total* ligand and *total* macromolecule concentrations (L_T, M_T), that satisfy the mass conservation equation [Eq. (1)]:

$$L_T = (\Sigma \nu_i)M_T + L_F \quad (1)$$

Therefore, if the set of concentration pairs (L_T, M_T) can be found for which $\Sigma \nu_i$ and L_F are constant, then $\Sigma \nu_i$ and L_F can be determined from the slope and intercept, respectively, of a plot of L_T vs M_T , based on Eq. (1). The methods to determine the set of concentration pairs (L_T, M_T) that correspond to a constant L_F and hence $\Sigma \nu_i$ are described in the following sections and depend on whether the signal is from the ligand or the macromolecule.

Signal from Macromolecule

In this case, the signal reflects the average degree of saturation of the macromolecule and a normal titration (addition of ligand to a constant macromolecule concentration) is generally performed. Any physicochemical intensive property of the macromolecule (e.g., fluorescence, absorbance, circular dichroism, and viscosity) can be used to monitor binding if this property is affected by the state of ligation of the macromolecule.

Macromolecule Binding Density Function (MBDF) Expressions for Multiple Ligand Binding. We first consider the general relationship between the concentration of each macromolecular species with i ligands bound, M_i , and the experimentally observed signal from the macromolecule, S_{obs} . For a total macromolecule concentration, M_T , the equilibrium distribution of the macromolecule among its different ligation states, M_i , is determined by the free ligand concentration, L_F . Therefore, at each L_F , S_{obs} is the algebraic sum of the concentrations of the macromolecule in each state, M_i , each weighted by the value of the physicochemical property for that state, S_i . For example, if only one ligand can bind per macromolecule, then a two-state system as described in Eq. (2) applies,

$$S_{\text{obs}} = S_F M_F + S_B M_B \quad (2)$$

where the subscripts F and B refer to the free and bound states, respectively.

In general, a macromolecule will have the ability to bind r ligands, hence the general, multistate equation for the observed signal, S_{obs} , of a sample containing ligand at total concentration, L_T , and macromolecule at total concentration, M_T , is given by Eq. (3),

$$S_{\text{obs}} = S_F M_F + \sum_{i=1}^r S_i M_i \quad (3)$$

where S_F is the molar signal of the free macromolecule and S_i is the molar signal of the macromolecular complex with i bound ligands, M_i ($i = 1$ to r). Equation (3) is valid in the *absence* of macromolecular aggregation.

The mass conservation equation, which relates M_F and M_i to M_T , is given in Eq. (4):

$$M_T = M_F + \sum_{i=1}^r M_i \quad (4)$$

From the definition of the binding density, ν_i (i moles of ligand bound per mole of macromolecule), given in Eq. (5),

$$\nu_i = iM_i/M_T \quad (5)$$

one obtains the expression for M_i given in Eq. (6):

$$M_i = (\nu_i/i)M_T \quad (6)$$

On introducing Eqs. (4) and (6) into Eq. (3), one obtains Eq. (7), which can be rearranged to a general form as in Eq. (8):

$$S_{\text{obs}} = S_F M_T + \sum M_T (S_i - S_F) (\nu_i/i) \quad (7)$$

$$\Delta S_{\text{obs}} \equiv (S_{\text{obs}} - S_F M_T) / S_F M_T = \sum (\Delta S_i/i) (\nu_i) \quad (8)$$

In Eq. (8), $\Delta S_{\text{obs}} \equiv (S_{\text{obs}} - S_F M_T) / S_F M_T$ is the experimentally determined fractional molar signal change observed in the presence of total ligand and macromolecule concentrations, L_T and M_T , and $\Delta S_i/i$ is the average molecular signal change *per bound ligand* in the complex containing i ligands. Note that the quantity, $S_F M_T$, is simply the initial signal from the unliganded macromolecule, i.e., the signal at the start of the titration, before addition of ligand.

Equation (8) indicates that ΔS_{obs} , the macromolecular binding density function (MBDF), is a function *only* of the binding density distribution, $\sum \nu_i$, since $\Delta S_i/i$ is an intrinsic molecular property of the system. Therefore, the average degree of saturation of the macromolecule, $\sum \nu_i$, must be the same for any pair of total concentrations, L_T and M_T , for which ΔS_{obs} is constant. Since $\sum \nu_i$ is a unique function of the free ligand concentration, then the value of L_F at the same degree of saturation (constant MBDF) must also be constant. As a result, the MBDF is the appropriate function to be used to determine model-independent values of L_F and $\sum \nu_i$. The details for how this is accomplished are described below. The above derivation is rigorous and independent of any binding model and as such can be applied to any binding system, with or without cooperative interactions or overlapping binding sites (see below).

TABLE I
MACROMOLECULAR BINDING DENSITY FUNCTIONS FOR DIFFERENT SPECTROSCOPIC SIGNALS

Method	Macromolecular binding density function (MBDF)	Comments
Absorbance	$\Delta A_{\text{obs}} = \Sigma(\Delta \epsilon_i/i)\nu_i$ where $\Delta A_{\text{obs}} = (A_{\text{obs}} - \epsilon_F M_T)/\epsilon_F M_T$ $\Delta \epsilon_i = (\epsilon_i - \epsilon_F)/\epsilon_F$	$\epsilon_F M_T$ is initial absorbance of macromolecule solution in absence of ligand; ϵ_F and ϵ_i are extinction coefficients for free macromolecule and each state of bound macromolecule
Fluorescence	$Q_{\text{obs}} = \Sigma(Q_i/i)\nu_i$ where $Q_{\text{obs}} = (F_{\text{obs}} - F_F M_T)/F_F M_T$ $Q_i = (F_i - F_F)/F_F$	$F_F M_T$ is initial fluorescence of macromolecule solution in absence of ligand

In Table I, we give the expressions for the macromolecular binding density functions (MBDF) when the experimentally observed signal is due to ligand-dependent changes in the fluorescence or absorbance of the macromolecule; however, Eq. (8) is general and not restricted to spectroscopic changes.

Data Analysis When Signal Is from Macromolecule and "Normal" Titrations Are Performed. We will demonstrate the use of this method of analysis with data from studies of the binding of the oligonucleotide, dT(pT)₁₅, to the *Escherichia coli* single-strand binding (SSB) protein, which is a stable homotetramer under the conditions of the experiment.^{24,25} The reader should consult the original papers for the experimental details of the fluorescence measurements, inner filter corrections, etc., as we will focus on the analysis of the experimental titrations once they have been obtained. In this case, the SSB tetramer is the macromolecule, since it binds multiple molecules of the ligand, [dT(pT)₁₅].^{24,25} The intrinsic SSB tryptophan fluorescence is substantially quenched on binding the oligonucleotide, hence this signal was used to monitor the interaction. Therefore, in this case, the MBDF is the observed degree of fluorescence quenching, Q_{obs} , and the appropriate relationship [from Eq. (8)] is given in Eq. (9) (see Table I),

$$Q_{\text{obs}} = \Sigma(Q_i/i)(\nu_i) \quad (9)$$

where Q_i/i is the average degree of quenching per bound oligonucleotide, with i oligonucleotides bound per SSB tetramer.

Figure 1 shows the results of two titrations with dT(pT)₁₅ (ligand), at different SSB tetramer (macromolecule) concentrations, in which the values of Q_{obs} (MBDF) have been plotted as a function of total dT(pT)₁₅

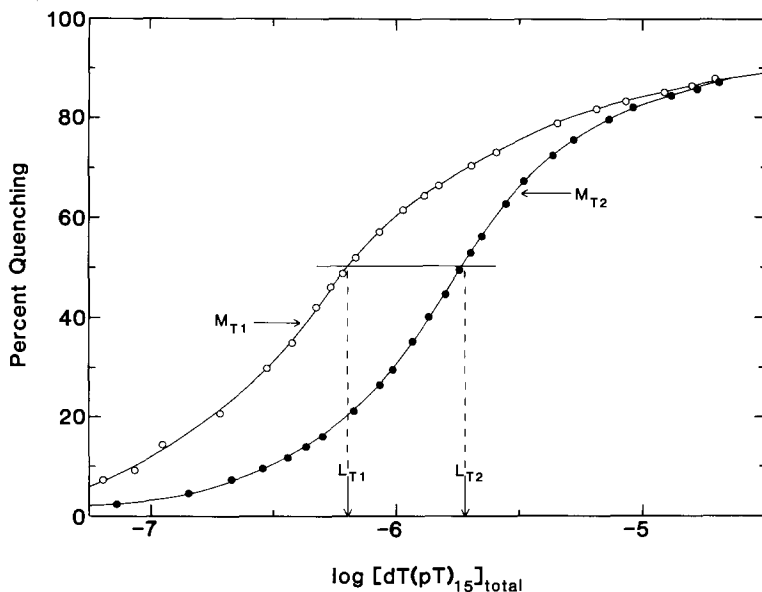


FIG. 1. Titrations of the SSB tetramer with $\text{dT}(\text{pT})_{15}$, as monitored by the quenching of the intrinsic tryptophan fluorescence of the SSB protein in 20 mM KH_2PO_4 , 0.2 M KCl (pH 7.4, 8°) at two different SSB tetramer concentrations: (○) 2.74×10^{-7} M and (●) 9.45×10^{-7} M. (Data from Bujalowski and Lohman.²⁴)

concentration (L_T). Also depicted in Fig. 1 is the process by which *one set* of values of $\Sigma \nu_i$ and L_F can be obtained from these data. A horizontal line intersecting both curves is drawn, defining *one* constant value of the MBDF (constant Q_{obs}). The point of intersection of this horizontal line with each titration curve defines the two sets of values of L_{T_x} , M_{T_x} ($x = 1$ and 2) for which L_F and $\Sigma \nu_i$ are constant, as discussed above. The example in Fig. 1 shows only two titrations, hence only two sets of values of L_{T_x} , M_{T_x} are obtained. For the analysis of n titrations, n sets of concentration pairs, L_{T_x} , M_{T_x} , are obtained (see below).

For the two titrations at macromolecular (SSB tetramer) concentrations, M_{T_1} and M_{T_2} ($M_{T_2} > M_{T_1}$), two mass conservation equations can be written in the form of Eq. (1) for each pair of concentrations, L_{T_x} , M_{T_x} . These two equations can be solved for $\Sigma \nu_i$ and L_F , with the results given in Eqs. (10a) and (10b):

$$\Sigma \nu_i = (L_{T_2} - L_{T_1}) / (M_{T_1} - M_{T_2}) \quad (10a)$$

$$L_F = L_T - \Sigma \nu_i (M_T) = (M_{T_1} L_{T_2} - M_{T_2} L_{T_1}) / (M_{T_1} - M_{T_2}) \quad (10b)$$

This process is then repeated for a series of m horizontal lines that can be

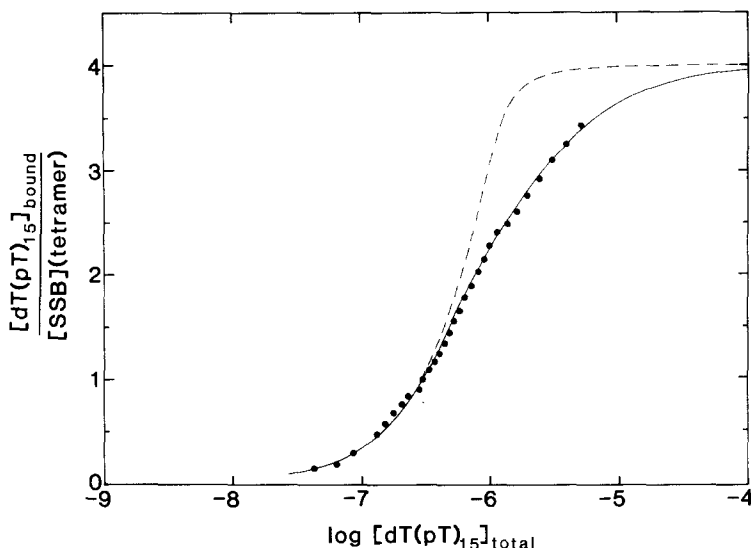


FIG. 2. The average number of $\text{dT}(\text{pT})_{15}$ molecules bound per SSB tetramer as a function of the total concentration of $\text{dT}(\text{pT})_{15}$, in 20 mM KH_2PO_4 (pH 7.4, 8°), 0.2 M KCl. This plot was constructed from an analysis of the two titration curves shown in Fig. 1, using Eqs. (10a) and (10b). The solid line is a theoretical curve based on the "square" model,²⁴ using the best-fit values of $K_{16} = 3.0 \times 10^7 \text{ M}^{-1}$ and $\sigma_{16} = 0.17$. The tetramer concentration is $2.74 \times 10^{-7} \text{ M}$. The dashed curve represents the theoretical isotherm for $K_{16} = 3.0 \times 10^7 \text{ M}^{-1}$ and $\sigma_{16} = 1$. (Data from Bujalowski and Lohman.²⁴)

drawn to intersect both titration curves, each line yielding one set of values of L_F and $\Sigma \nu_i$. The analysis of the *series* of m horizontal lines will therefore yield a set of m values of L_F and $\Sigma \nu_i$, which can then be used to construct a *model-independent* binding isotherm. These data can also be used to determine the relationship between the signal change (Q_{obs} , in this case) and the average degree of ligand binding.

Figure 2 shows the model-independent binding isotherm ($\Sigma \nu_i$ vs L_T) constructed from an analysis of the two titration curves in Fig. 1, whereas Fig. 3 shows the relationship between the degree of fluorescence quenching (Q_{obs}) and the average degree of binding ($\Sigma \nu_i$). As can be seen from the deviation of the data from the dashed line in Fig. 3, Q_{obs} is *not* a linear function of the degree of binding, rather the average quenching per bound $\text{dT}(\text{pT})_{15}$ decreases as the degree of binding increases.²⁴ Therefore, an incorrect apparent isotherm would have been calculated if a linear relationship had been assumed for this system. The solid line that describes the experimental isotherm in Fig. 2 is a theoretical isotherm based on a thermodynamic model that incorporates negative cooperativity among

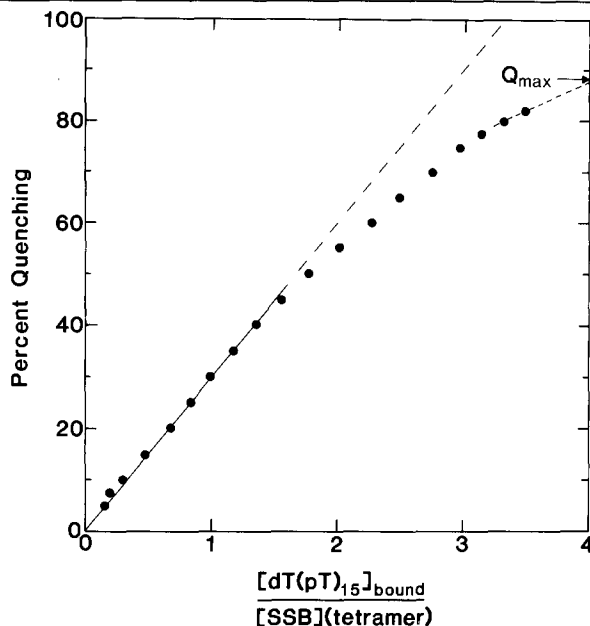


FIG. 3. The dependence of the quenching of the SSB protein fluorescence on the average number of $\text{dT}(\text{pT})_{15}$ molecules bound per SSB tetramer in 20 mM KH_2PO_4 , 0.2 M KCl (pH 7.4, 8°). The number of bound $\text{dT}(\text{pT})_{15}$ molecules at each value of the fluorescence quenching was determined from the two titrations shown in Fig. 1, using Eqs. (10a) and (10b) in the text. The dashed line represents a linear extrapolation from the initial part of the curve. (Data from Bujalowski and Lohman.²⁴)

the four binding sites for $\text{dT}(\text{pT})_{15}$ on the SSB tetramer, with intrinsic equilibrium constant $K_{16} = 3.0 (\pm 0.5) \times 10^7 \text{ M}^{-1}$ and negative cooperativity parameter $\sigma_{16} = 0.17 (\pm 0.02)$.²⁴ The model-independent analysis of these binding data indicates that the first two molecules of $\text{dT}(\text{pT})_{15}$ bind to two protomers of the SSB tetramer with approximately equal affinity, whereas the third and fourth molecules bind with lower affinity due to the negative cooperativity. A detailed discussion of this phenomenon and its implications for SSB binding to single-stranded (ss) DNA has been reported.^{24,25}

In principle, as shown in Figs. 1 and 2, only two titration curves are needed to obtain values of $\Sigma \nu_i$ and L_F that cover the full range of binding densities, if one is able to cover the entire range of the signal change with each titration curve [this is not the case when a reverse titration is performed (see the section Signal from Ligand, below)]. This was possible for the data in Fig. 1 since the signal change (tryptophan fluorescence

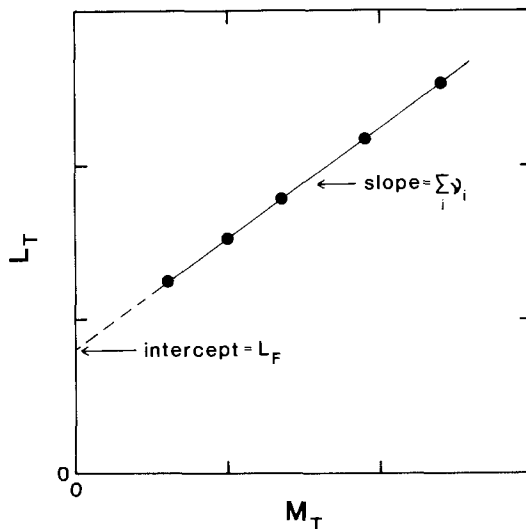


FIG. 4. An example of the graphical determination of the free ligand concentration, L_F , and the average binding density, $\sum \nu_i$, from a theoretical set of values (L_T, M_T) ($x = 1$ to 5) that would be determined from a binding density analysis of five titration curves.

quenching) for the SSB tetramer-dT(pT)₁₅ interaction is large and the affinity is fairly high under the conditions used. In this case, the data can be analyzed using Eqs. (10a) and (10b). However, more accurate values of $\sum \nu_i$ and L_F can be obtained if more than two titrations are performed and the data are analyzed graphically. For a case in which n titrations are performed, then n sets of values of $(L_{T_x}, M_{T_x}, x = 1 \text{ to } n)$ will be obtained, one for each titration curve that is intersected by each horizontal line (constant MBDF). A plot of L_T vs M_T can then be constructed (see Fig. 4), which will result in a straight line, according to Eq. (1), from which the values of L_F and $\sum \nu_i$ can be obtained from the intercept and slope, respectively (see also Fig. 12). If a plot of L_T vs M_T determined in this manner is not linear, this may indicate one or more inconsistent sets of titration data or aggregation phenomena associated with the ligand or the macromolecule and these data should be viewed with caution.

We also note that this method can be used to analyze titrations for which one is not able to saturate the macromolecule with ligand (e.g., in cases of weak affinity or solubility limitations). This is due to the fact that the analysis does not require knowledge of the signal change at saturation. Of course, in such cases, data will be obtained only over a limited binding density range.

Signal from Ligand

In this case, some intensive property, S , of the ligand changes on binding to the macromolecule, hence the signal monitors the apparent degree of saturation of the ligand and a "reverse" titration (addition of macromolecule to a constant ligand concentration) is generally performed. A general discussion of the uses of this approach has been presented by Bujalowski and Lohman.²¹

Ligand Binding Density Function (LBDF) Expressions for Multiple Ligand Binding. Consider the equilibrium binding of a ligand [total concentration, L_T (moles per liter)] to a macromolecule [total concentration, M_T (moles per liter)] such that there can be r states of bound ligand ($i = 1$ to r), with each state possessing a different molar signal, S_i . The observed signal, S_{obs} , from the ligand solution in the presence of macromolecule has contributions from the free ligand and the ligand bound to the macromolecule in any of its r possible bound states and can be expressed by Eq. (11),

$$S_{\text{obs}} = S_F L_F + \sum_{i=1}^r S_i L_i \quad (11)$$

where S_F and L_F are the *molar* signal and concentration of free ligand, respectively, and S_i and L_i are the molar signal and concentration of the ligand bound in state i , respectively. Equation (11) is valid when the molar signal of each species is independent of concentration (i.e., in the absence of ligand aggregation). The concentrations of free and bound ligand are related to the total ligand concentration by conservation of mass [Eq. (12)],

$$L_T = L_F + \sum_{i=1}^r L_i \quad (12)$$

and furthermore

$$L_i = \nu_i M_T \quad (13)$$

where ν_i is the degree of ligand binding (moles of ligand bound per mole of macromolecule) for the i th state. Substituting Eqs. (12) and (13) into Eq. (11) yields Eq. (14):

$$S_{\text{obs}} - S_F L_T = M_T \sum (S_i - S_F) \nu_i \quad (14)$$

Dividing both sides by $S_F L_T$, the initial signal from the ligand before titration with macromolecule, and next multiplying by L_T/M_T yields Eq. (15),

$$[(S_{\text{obs}} - S_F L_T)/S_F L_T](L_T/M_T) = \sum [(S_i - S_F)/S_F] \nu_i \quad (15)$$

which can be rewritten as Eq. (16):

$$\Delta S_{\text{obs}}(L_T/M_T) = \sum_{i=1}^r (\Delta S)_i \nu_i \quad (16)$$

TABLE II
LIGAND BINDING DENSITY FUNCTIONS FOR DIFFERENT SPECTROSCOPIC SIGNALS

Method	Ligand binding density function (LBDF)	Comments
Absorbance	$\Delta A_{\text{obs}}(L_T/M_T) = \Sigma(\Delta \varepsilon_i)\nu_i$ where $\Delta A_{\text{obs}} = (A_{\text{obs}} - \varepsilon_F L_T)/\varepsilon_F L_T$ $\Delta \varepsilon_i = (\varepsilon_i - \varepsilon_F)/\varepsilon_F$	$\varepsilon_F L_T$ is initial absorbance of ligand solution in absence of the macromolecule; ε_F and ε_i are extinction coefficients for free ligand and each state of bound ligand
Fluorescence	$Q_{\text{obs}}(L_T M_T) = \Sigma(Q_i)\nu_i$ where $Q_{\text{obs}} = (F_{\text{obs}} - F_F L_T)/F_F L_T$ $Q_i = (F_i - F_F)/F_F$	$F_F L_T$ is initial fluorescence of ligand solution in absence of macromolecule

$\Delta S_{\text{obs}} \equiv (S_{\text{obs}} - S_F L_T)/S_F L_T$ is the experimentally observed fractional change in the signal from the ligand (normalized to the initial signal from the ligand, $S_F L_T$) in the presence of *total* ligand and *total* macromolecule concentrations, L_T and M_T , and $(\Delta S)_i = (S_i - S_F)/S_F$ is the intrinsic fractional signal change from the ligand when it is bound in state i . Note that the quantity, $S_F L_T$, is simply the initial signal from the free ligand in the absence of macromolecule, i.e., the signal at the start of the titration, before addition of macromolecule. If fluorescence of the ligand is being monitored, then ΔS_{obs} corresponds to the quenching of the ligand fluorescence. In Table II, we give the expressions for the ligand binding density functions (LBDF) for the cases in which the signal is due to fluorescence or absorbance changes; however, Eq. (16) is general and not restricted to only spectroscopic changes.

Equation (16) indicates that the quantity $\Delta S_{\text{obs}}(L_T/M_T)$ is equal to $\Sigma(\Delta S)_i \nu_i$, the sum of the binding densities for all i states of ligand binding, weighted by the intrinsic signal change for each bound state. Therefore, the quantity, $\Delta S_{\text{obs}}(L_T/M_T)$, is the ligand binding density function (LBDF). The weighting factor, $(\Delta S)_i$, is a molecular quantity that is constant for a particular binding state i , under a given set of experimental conditions. Therefore the quantity, $\Sigma(\Delta S)_i \nu_i$, and hence the LBDF, is constant for a given binding density distribution, $\Sigma \nu_i$. At equilibrium, the values of L_F and $\Sigma \nu_i$ (and each separate value of ν_i) are constant for a given value of $\Delta S_{\text{obs}}(L_T/M_T)$, independent of the macromolecule concentration, M_T . Hence, one can obtain model-independent estimates of $\Sigma \nu_i$ and L_F from plots of $\Delta S_{\text{obs}}(L_T/M_T)$ vs M_T for two or more titrations performed at different total ligand concentrations, under identical solution conditions. This is accomplished by obtaining the set of concentration pairs (L_{T_x}, M_{T_x}) , from each titration, for which the binding density function is constant and

solving for $\Sigma \nu_i$ and L_F , graphically as shown in Figs. 4 and 12, using the relationship given in Eq. (1). The procedure is therefore analogous to the case in which a signal from the macromolecule is monitored during the titration. The main difference is that a different formulation of the binding density function is required.

It is useful to compare Eqs. (8) and (16), which are the corresponding binding density functions for the macromolecule (MBDF) and ligand (LBDF), respectively. When the signal from the *macromolecule* is monitored, the MBDF is simply equal to ΔS_{obs} ; however, when the signal from the *ligand* is monitored, the LBDF is equal to $\Delta S_{\text{obs}}(L_T/M_T)$. Therefore, the form of the BDF that is used in the analysis of binding data depends on whether the signal is from the ligand or the macromolecule. However, in both cases, the BDF is a function only of the molecular signal, which is constant for a given distribution of bound ligand, and the average binding density on the macromolecule.

Theoretical Behavior of LBDF When Data Are Collected by "Reverse" Titrations. It is informative to consider the general properties of the LBDF when experiments are performed as reverse titrations, since these differ considerably from the behavior of the MBDF when experiments are performed as normal titrations.²¹ Figures 5 and 7 show the behavior of the LBDF, obtained from reverse titrations, and these should be compared to the behavior of the MBDF, obtained from normal titrations, shown in Fig. 1. Figure 5 represents a series of theoretical LBDF curves, when reverse titrations are performed, whereas Fig. 7 shows actual data for the binding of the *E. coli* SSB tetramer (ligand) to poly(U) (macromolecule), in which binding was monitored by the quenching of the intrinsic fluorescence of the SSB protein. In Fig. 5, the noncooperative, overlap model of McGhee and von Hippel³⁸ for large ligand binding to a linear nucleic acid [see Eq. (20) with $\omega = 1$] was used to generate the theoretical plots. The curves in Fig. 5 were generated for a single binding mode with the observed signal from the ligand with $(\Delta S)_i = 0.5$. However, the qualitative features of the LBDF plot are independent of this model and hold even for ligand binding to independent and identical sites on proteins (i.e., no overlap of potential binding sites) or if multiple binding modes or cooperative binding exists. Equation (16) indicates that the ordinate, $[\Delta S_{\text{obs}}(L_T/M_T)]$, of the plot in Fig. 5, is equal to the average binding density, ν , multiplied by $(\Delta S)_i$, hence the ordinate is a relative binding density scale. For a large ligand binding to a nucleic acid with site size, n , the maximum value of ν at saturation of the nucleic acid, ν_{max} , equals $1/n$ (0.1 in this case), hence the

³⁸ J. D. McGhee and P. H. von Hippel, *J. Mol. Biol.* **86**, 469 (1974); J. D. McGhee and P. H. von Hippel, *J. Mol. Biol.* **103**, 679 (1976).

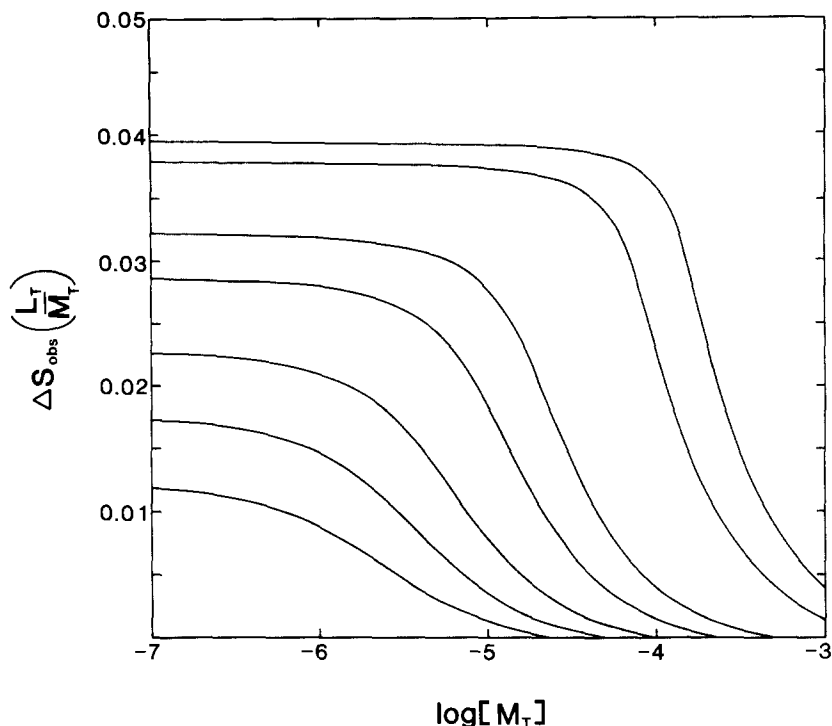


FIG. 5. Plots of the ligand binding density function, $\Delta S_{\text{obs}}(L_T/M_T)$, vs the logarithm of the total nucleic acid concentration, M_T , generated for the theoretical case of a "reverse" titration of a large ligand that binds noncooperatively to a homogeneous linear nucleic acid lattice. The noncooperative binding isotherm of McGhee and von Hippel³⁸ was used to generate these curves using the following parameters: ligand site size, $n = 10$ nucleotides, $K = 10^6 M^{-1}$, and $\Delta S_{\text{max}} = 0.5$. The total ligand concentrations, L_T , are, from left to right: 5×10^{-8} , 10^{-7} , 2×10^{-7} , 5×10^{-7} , 10^{-6} , 5×10^{-6} , and $10^{-5} M$. (Taken from Bujalowski and Lohman.²¹)

maximum possible value of the ordinate is 0.05, since we have specified $(\Delta S)_i = 0.5$ in the calculations for Fig. 5.

As is evident from a comparison of Fig. 1 with Figs. 5 and 7, one major difference between the behavior of the BDF when the data are collected by reverse titrations is that the curves for lower total ligand concentrations, L_T , do not cover the same range of binding densities as the curves for higher values of L_T . As L_T decreases, the largest value of ν attainable at that concentration of ligand also decreases, hence $\Delta S_{\text{obs}}(L_T/M_T)$, which equals $\nu(\Delta S)_i$, also decreases. This is a direct result of the use of reverse titrations and is a consequence of the fact that the binding density is solely determined by the free ligand concentration (for a given K), hence in the

limit of low M_T , the value of ν is limited by the value of L_T , since L_F can never exceed L_T . As a result, a single binding density function curve obtained at one value of L_T usually cannot span the same range of binding densities as other curves obtained at different values of L_T . Only when $L_T \gg 1/K$ will the binding density function span the full range of binding densities. Hence, in order to obtain values of $\Sigma \nu_i$ and L_F that cover the full range of binding densities, a series of overlapping binding density function curves must be obtained. The behavior shown in Figs. 5 and 7 occurs even for ligands that bind with high cooperativity. We emphasize that this distinctive behavior results only when data are collected by a series of reverse titrations.

As discussed above and shown in Fig. 1, when normal titrations are performed and a signal from the macromolecule is monitored, all of the curves for different concentrations of macromolecule can span the full range of binding densities. Normal titrations of macromolecule with ligand can also be performed when the signal change is from the ligand and this may permit the full range of binding densities to be covered in each titration. However, in this case, each normal titration must be directly compared with a control titration in which buffer is titrated with ligand in order to calculate the binding density function, hence the relative accuracy of the data is generally reduced.

Data Analysis When Signal Change Is from Ligand and "Reverse" Titrations Are Performed. As an example of the use of the binding density function analysis when a signal from the ligand is monitored, we discuss a study of the interaction of *E. coli* SSB tetramer (ligand) with the single-stranded homopolynucleotide, poly(U) (macromolecule), in which the quenching of the intrinsic SSB tryptophan fluorescence was used to monitor binding.^{21,22} Under the conditions of these experiments, the SSB tetramer binds to single-stranded nucleic acids at equilibrium in its (SSB)₆₅, binding mode, in which approximately 65 nucleotides of the nucleic acid wraps around the SSB tetramer, interacting with all 4 of the SSB proto-mers.³⁹⁻⁴³ This method has also been used to analyze the binding of a series of oligopeptides containing lysine and tryptophan to single-stranded polynucleotides, with binding monitored by the fluorescence quenching of the peptide.²⁶

A series of reverse titrations of *E. coli* SSB tetramer with poly(U) at

³⁹ T. M. Lohman and L. B. Overman, *J. Biol. Chem.* **260**, 3594 (1985).

⁴⁰ W. Bujalowski and T. M. Lohman, *Biochemistry* **25**, 7799 (1986).

⁴¹ W. Bujalowski, L. B. Overman, and T. M. Lohman, *J. Biol. Chem.* **263**, 4629 (1988).

⁴² S. Chrysogelos and J. Griffith, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 5803 (1982).

⁴³ J. D. Griffith, L. D. Harris, and J. Register, *Cold Spring Harbor Symp. Quant. Biol.* **49**, 553 (1984).

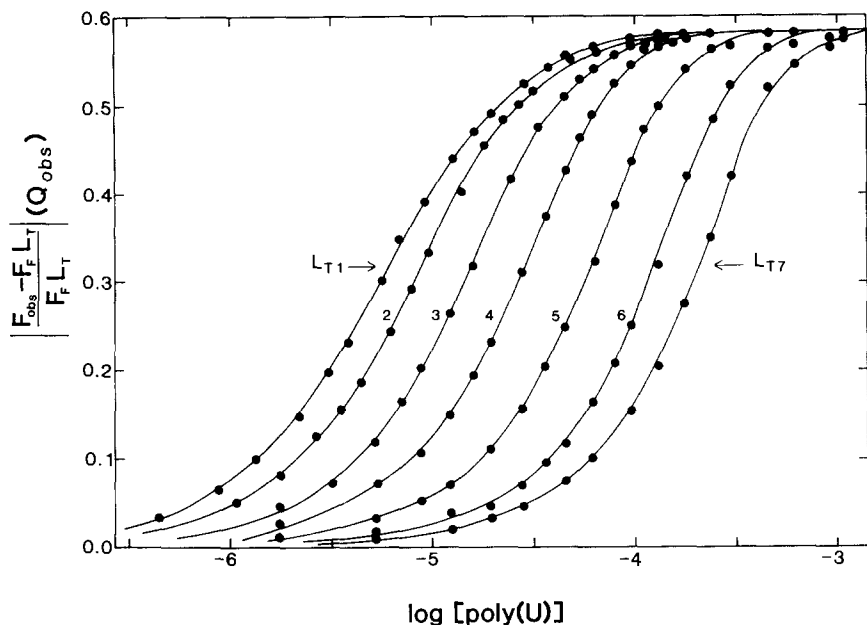


FIG. 6. "Reverse" titrations (addition of nucleic acid to protein) of *E. coli* SSB protein with poly(U) at different SSB concentrations in 10 mM Tris (pH 8.1), 0.2 M NaCl, 25.0°. The extent of SSB fluorescence quenching is plotted vs the logarithm of the poly(U) nucleotide concentration. The SSB tetramer concentrations for each curve are (1) 5.69×10^{-8} M; (2) 1.22×10^{-7} M; (3) 2.44×10^{-7} M; (4) 5.28×10^{-7} M; (5) 1.30×10^{-6} M; (6) 2.84×10^{-6} M; (7) 5.20×10^{-6} M. (Data from Bujalowski and Lohman.²¹)

different SSB tetramer concentrations is shown in Fig. 6. These data were used to construct the LBDF [$Q_{\text{obs}}(L_T/M_T)$] and the plot of $Q_{\text{obs}}(L_T/M_T)$ vs $\log[M_T]$ is shown in Fig. 7. As mentioned above, even though all of the reverse titrations shown in Fig. 5 span the same full range of SSB fluorescence quenching, not all of them span the full range of binding densities as seen from the LBDF plot in Fig. 7. As a result, multiple titrations at different values of L_T are required to obtain data over the full binding density range. At this point, the set of LBDF curves in Fig. 7 is analyzed in a similar manner to the analysis of the MBDF curves discussed above (see Fig. 1). A series of horizontal lines that intersect the LBDF curves are drawn, each line defining a constant value of the LBDF, $Q_{\text{obs}}(L_T/M_T)$. Each horizontal line will yield a value of L_F and its associated value of $\Sigma \nu_i$. The following describes the analysis for one such horizontal line. The points of intersection of one horizontal line with each binding density function curve determines the set of concentration pairs (M_{T_i} ,

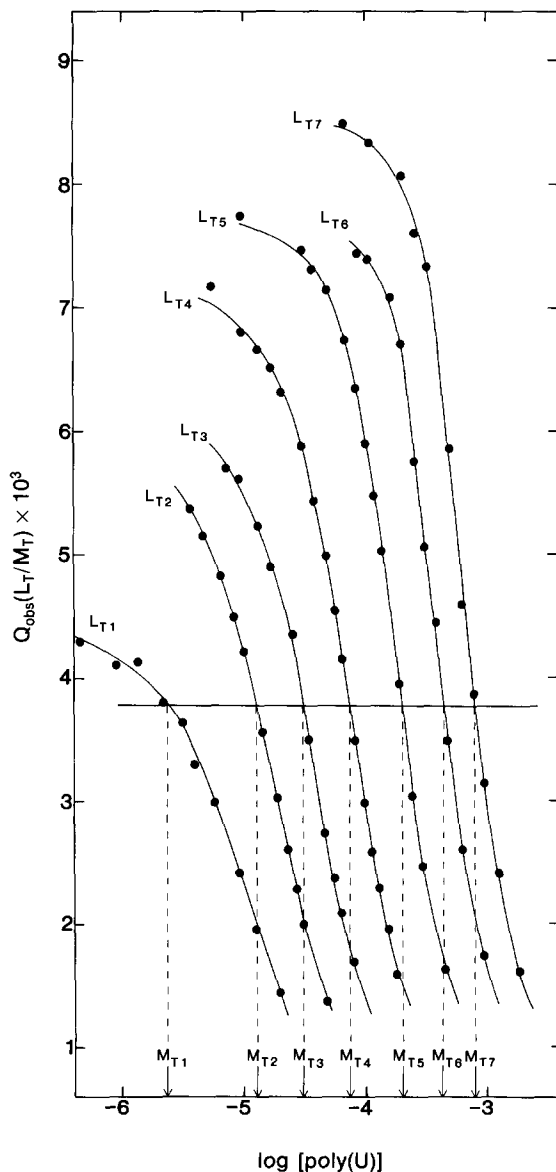


FIG. 7. The dependence of the ligand binding density function, $Q_{\text{obs}}(L_T/M_T)$ on the logarithm of the poly(U) concentration at different SSB protein concentrations in 10 mM Tris (pH 8.1) plus 0.2 M NaCl, 25.0°. The points of intersection with the separate curves of any horizontal line drawn at a constant value of the binding density function [e.g., $Q_{\text{obs}}(L_T/M_T) = 3.8 \times 10^{-3}$ in the figure], determine the set of total protein and macromolecule concentrations, L_{T_i} , M_{T_i} , for which the average binding density, $\Sigma \nu_i$ (and free protein

L_{T_x} ($x = 1$ to 7) for which L_F and $\Sigma\nu_i$ are constant, as shown in Fig. 7 for one constant value of the LBDF. Based on Eq. (1), the average binding density, $\Sigma\nu_i$, and L_F , can then be determined, graphically, from the slope and intercept of a plot of L_T vs M_T . By repeating this procedure for a series of horizontal lines that span the range of values of $Q_{\text{obs}}(L_T/M_T)$, values of $\Sigma\nu_i$ as a function of L_F can be obtained and a model-independent binding isotherm can be constructed. A series of plots of L_T vs M_T , each generated from a different value of $Q_{\text{obs}}(L_T/M_T)$ from Fig. 7, is shown in Fig. 8.

Theoretically it is enough to analyze only two different protein concentrations to obtain two equations of the form shown in Eq. (1) (for L_{T_1} , M_{T_1} and L_{T_2} , M_{T_2}) from which the values of $\Sigma\nu_i$ and L_F can be determined using Eqs. (10a) and (10b). However, as mentioned above, since not all of the LBDF curves fully overlap, only a subset of them will intersect each horizontal line. Therefore, not all of the curves can be used to determine each value of $\Sigma\nu_i$ and L_F , especially at the higher binding densities. In addition, significantly more accurate determinations of L_F and $\Sigma\nu_i$ are obtained when titrations at several ligand concentrations (L_T) are performed. In the construction of a series of LBDF plots as shown in Fig. 7, one should cover as wide a range of ligand concentrations as possible; however, care should be taken to avoid large changes in total ligand concentration between two successive titrations, since this may bias the determination of L_F and $\Sigma\nu_i$. In our experience, six to eight titrations using successive total ligand concentrations that differ by a factor of two will generate an accurate set of data.

The model-independent binding isotherm constructed from the full analysis of the data in Fig. 7 for the binding of the *E. coli* SSB tetramer to poly(U) in 0.2 M NaCl, is plotted as a Scatchard plot⁴⁴ in Fig. 9. One can obtain intrinsic binding parameters by comparing this binding isotherm to theoretical isotherms generated by various models. The solid line in Fig. 9 is the best-fit theoretical isotherm, based on a model in which *E. coli* SSB tetramers bind to single-stranded nucleic acids with "limited" cooperativity to form a mixture of tetramers and octamers (dimers of tetramers)²² [see Eq. (21)]. The best-fit theoretical isotherm for an "unlimited" cooperativity model³⁸ [see Eq. (20)] (dashed curve in Fig. 9), which as-

⁴⁴ G. Scatchard, *Ann. N.Y. Acad. Sci.* **51**, 660 (1949).

concentration, L_F) is constant. The SSB tetramer concentrations for each curve are as follow: (L_{T_1}) 5.69×10^{-8} M; (L_{T_2}) 1.22×10^{-7} M; (L_{T_3}) 2.44×10^{-7} M; (L_{T_4}) 5.28×10^{-7} M; (L_{T_5}) 1.30×10^{-6} M; (L_{T_6}) 2.84×10^{-6} M; (L_{T_7}) 5.20×10^{-6} M. (Data from Bujalowski and Lohman.²¹)

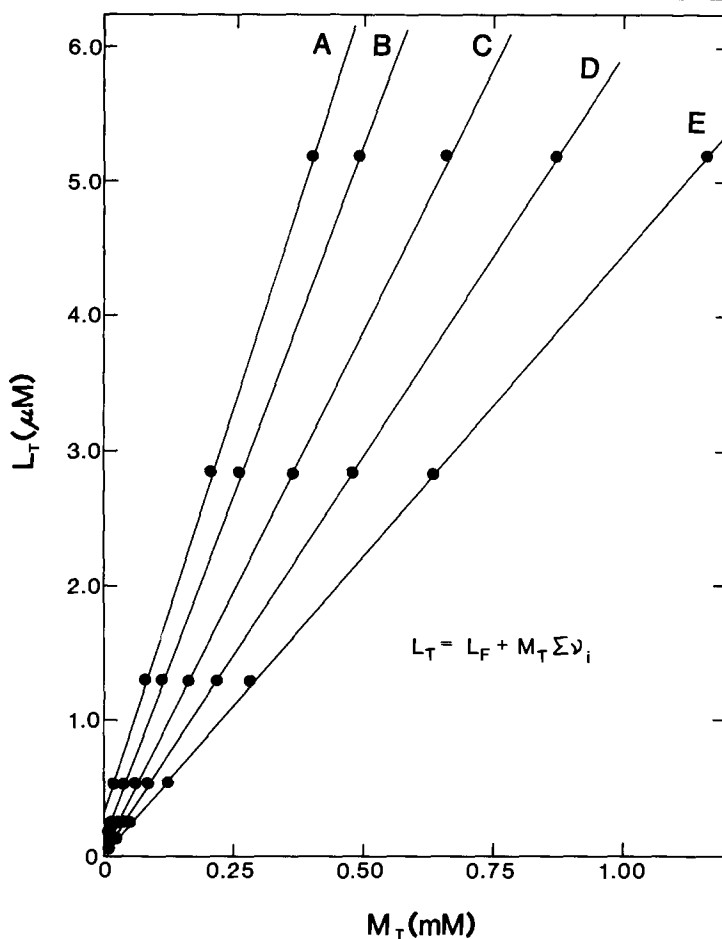


FIG. 8. Examples of the determination of the free SSB tetramer concentration (L_F) and the average binding density, $\sum \nu_i$ (SSB tetramers bound per nucleotide) from a plot of the total SSB tetramer concentration (L_T) vs the total poly(U) concentration (M_T) (nucleotides). Each straight line yields one set of values of L_F and $\sum \nu_i$ from the intercept and slope, respectively [see Eq. (1) in text]. The values of L_T and M_T for the five plots (A–E) shown were obtained from analysis of the ligand binding density function (LBDF) plots shown in Fig. 7. The values of the LBDF [$Q_{\text{obs}}(L_T/M_T)$] for each plot are (A) 6.4×10^{-3} ; (B) 5.6×10^{-3} ; (C) 4.4×10^{-3} ; (D) 3.4×10^{-3} ; (E) 2.6×10^{-3} . (Data from Bujalowski and Lohman.²¹)

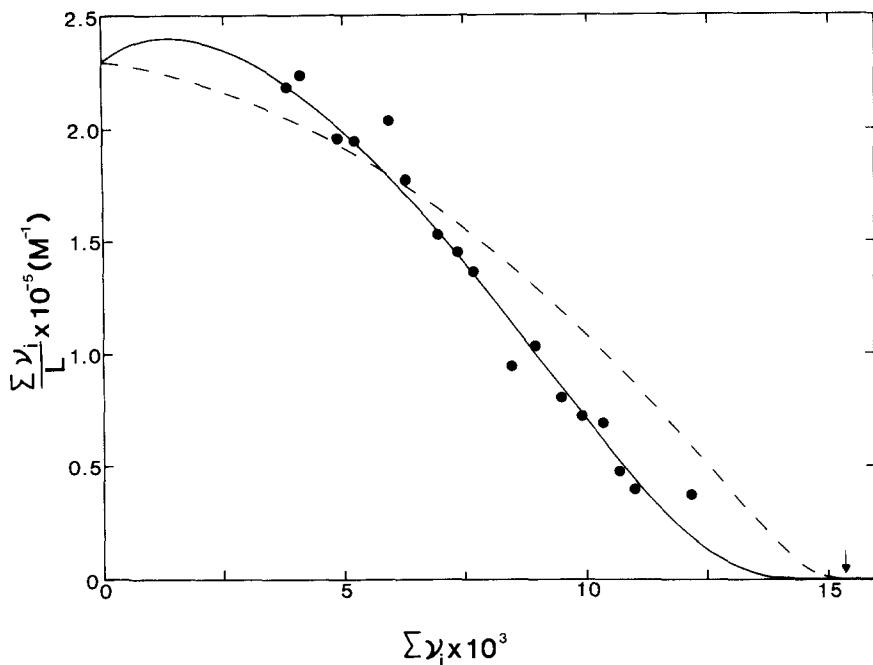


FIG. 9. Scatchard plot for *E. coli* SSB tetramer binding to poly(U) in its (SSB)₆₅ binding mode in 10 mM Tris (pH 8.1), 25°. The plot was constructed from data that were determined using the general method of analysis of all of the binding density function curves in Fig. 7. The solid line represents the best-fit isotherm for a model that describes the limited cooperative binding of SSB to single-stranded nucleic acids in its (SSB)₆₅ binding model (Bujalowski and Lohman²²), with binding parameters $K = 1.15 \times 10^5 \text{ M}^{-1}$; $\omega_{\text{lim}} = 420$, and $n = 65$ nucleotides per SSB tetramer. The arrow indicates the point of full saturation, $\nu = 1/n$. (Data from Bujalowski and Lohman.²²)

sumes that SSB tetramers can form a distribution of clusters, limited only by the length of the nucleic acid, does not describe the isotherm nearly as well.²²

Correlation between Signal Change, ΔS_{obs} , and Average Binding Density. Measurements of the average binding density, $\Sigma \nu_i$, as a function of the free ligand concentration enable one to determine the relationship between the average signal change and the fraction of bound ligand. This is not necessary in order to obtain a binding isotherm as discussed above; however, if ΔS_{obs} is found to be directly proportional to the fraction of bound ligand (L_B/L_T), then binding isotherms can be constructed with much greater ease from a titration at a single ligand concentration (see below). Figure 10 shows the relationship between Q_{obs} and L_B/L_T [=

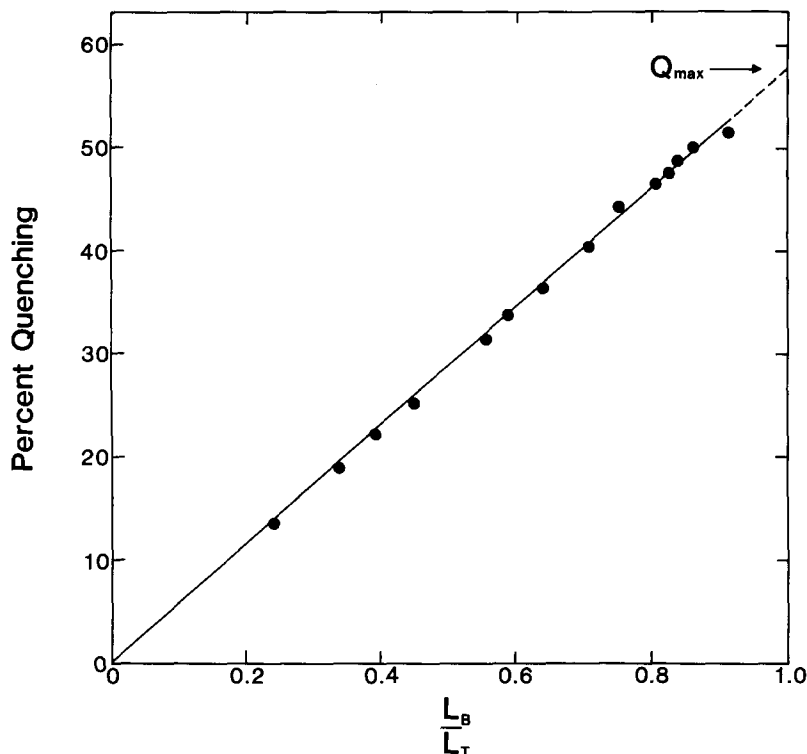


FIG. 10. The relation between the fluorescence quenching, Q_{obs} , and the fraction of protein bound L_B/L_T for SSB tetramer binding to poly(U) in 10 mM Tris (pH 8.1), 0.2 M NaCl, 25°; [SSB] = 2.44×10^{-7} M (tetramer) (curve L_{T_3} in Figs. 6 and 7). The concentration of bound protein, L_B , was calculated from $L_B = M_T(\Sigma \nu_i)$, where the average binding density, $\Sigma \nu_i$, was determined using the LBDF binding analysis method described in the text. The quenching of SSB fluorescence is observed to be directly proportional to the fraction of bound SSB, L_B/L_T . Therefore, for this case, $Q_{obs}/Q_{max} = L_B/L_T$. A linear extrapolation of the data to $L_B/L_T = 1$ (fully bound protein) yields a value of 57.7% for Q_{max} , under these solution conditions. (Data from Bujalowski and Lohman.²¹)

$(\Sigma \nu_i)M_T/L_T]$, for the *E. coli* SSB tetramer–poly(U) data shown in Figs. 6–9. Under these conditions, Q_{obs} is directly proportional to the fraction of bound SSB tetramer over the range of L_B/L_T that was investigated. However, it is important to check the relationship between quenching and L_B/L_T over a wide range of binding densities, since the signal change can be dependent on binding density. In the case of *E. coli* SSB tetramer–single-stranded nucleic acid interactions in NaCl concentrations ≥ 0.2 M (pH 8.1), this direct proportionality holds for binding densities up to at least 0.012 ($\sim 78\%$ saturation of the nucleic acid²¹). Therefore, under these

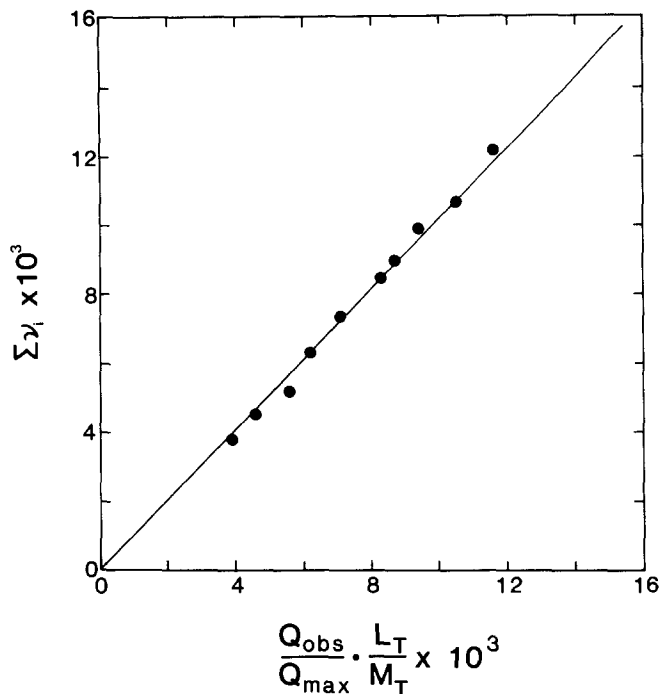


FIG. 11. The average binding density determined from a single SSB-poly(U) titration curve, $\nu_{app} = (Q_{obs}/Q_{max})(L_T/M_T)$ correlates well with the average binding density, $\Sigma\nu_i$, determined from the LBDF method of analysis. The slope of the line is 1.02. Data from curve 3 in Figs. 6 and 7 and a value of $Q_{max} = 57.7\%$ were used for the determination of ν_{app} . (Data from Bujalowski and Lohman.²¹)

conditions, the fractional fluorescence quenching of the SSB tetramer is equal to the fraction of bound SSB, i.e., $Q_{obs}/Q_{max} = L_B/L_T$. If it can be demonstrated that Q_{obs} is directly proportional to L_B/L_T , one can determine the maximum extent of protein fluorescence quenching, Q_{max} , from a linear extrapolation of a plot of Q_{obs} vs L_B/L_T to $L_B/L_T = 1$ as shown in Fig. 10.

For a protein-nucleic acid interaction, it is possible that the relationship, $Q_{obs}/Q_{max} = L_B/L_T$ is valid only over a limited range of binding densities; hence one could miscalculate the extent of binding by assuming that a direct proportionality exists for all binding densities. A useful check of this relationship can be made over a wide range of binding densities by plotting $(Q_{obs}/Q_{max})(L_T/M_T)$ vs the true value of $\Sigma\nu_i$ obtained from the LBDF analysis. $(Q_{obs}/Q_{max})(L_T/M_T)$ is an *apparent* binding density, based on the assumption that $Q_{obs}/Q_{max} = L_B/L_T$. This type of plot is shown in Fig. 11 for the *E. coli* SSB-poly(U) data presented in Fig. 6. In this case

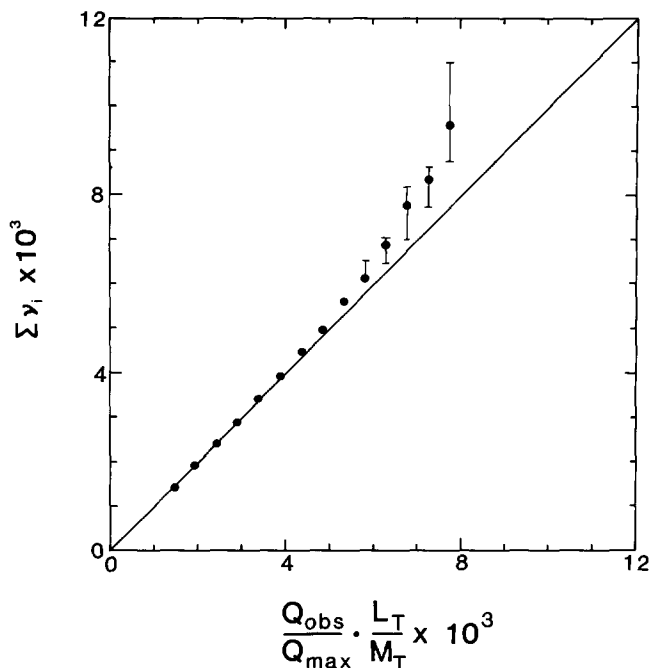


FIG. 12. Demonstration that $Q_{\text{obs}}/Q_{\text{max}}$ is *not* equal to L_B/L_T for the SSB tetramer binding to poly(dT) in 2.0 M NaBr (10 mM Tris, pH 8.1, 25°). Data from five reverse titrations were analyzed using the LBDF method of analysis.²¹ The apparent average binding density $\nu_{\text{app}} = (Q_{\text{obs}}/Q_{\text{max}})(L_T/M_T)$, calculated assuming $L_B/L_T = Q_{\text{obs}}/Q_{\text{max}}$, with $Q_{\text{max}} = 0.825$, deviates from the true average binding density, $\Sigma \nu_i$, determined by the binding density function analysis. The line shows the expected relationship if $Q_{\text{obs}}/Q_{\text{max}} = L_B/L_T$. (Data from Overman *et al.*²³)

(0.2 M NaCl, pH 8.1, 25°), there is a direct correspondence between these two values over the range from 25 to 80% coverage of the nucleic acid, hence it can safely be concluded that $Q_{\text{obs}}/Q_{\text{max}} = L_B/L_T$ in this range.

On the other hand, it would be unwise to conclude that $Q_{\text{obs}}/Q_{\text{max}} = L_B/L_T$ for all solution conditions, based on an analysis under only one set of solution conditions (Fig. 11). In fact, we have found that Q_{obs} is *not* directly proportional to L_B/L_T for SSB–single-stranded nucleic acid binding in NaBr²³ as shown in Fig. 12. We have also observed a nonlinear relationship between Q_{obs} and L_B/L_T for the SSB–poly(C) interaction in 15 mM NaCl (T. Lohman, unpublished, 1991). The nonlinearity in the latter case reflects the binding density-dependent shift in SSB binding modes that has been previously observed,^{41,43} since the (SSB)₃₅ mode exhibits a lower extent of quenching than either the (SSB)₅₆ or the (SSB)₆₅ binding modes.^{29,39,40}

Generation of Binding Isotherms from Single Titration when $\Delta S_{\text{obs}}/\Delta S_{\text{max}} = L_{\text{B}}/L_{\text{T}}$. The LBDF analysis allows one to rigorously determine a model-independent binding isotherm and the relationship between the ΔS_{obs} and the fraction of bound ligand, $L_{\text{B}}/L_{\text{T}}$. The LBDF method is time consuming since six to eight titrations are required to construct a single binding isotherm with good precision over a wide range of binding densities, although this is necessary if the relationship between ΔS_{obs} and $L_{\text{B}}/L_{\text{T}}$ is not known *a priori*. However, if it is determined from the LBDF analysis that a linear relationship exists between ΔS_{obs} and $L_{\text{B}}/L_{\text{T}}$ over a wide range of binding densities, then one can use this relationship to determine the average binding density and free ligand concentration from a single titration curve.²¹ For this simple case, Eq. (16) reduces to Eq. (17),

$$\Delta S_{\text{obs}}/\Delta S_{\text{max}} = L_{\text{B}}/L_{\text{T}} \quad (17)$$

and it follows that

$$L_{\text{F}} = (1 - \Delta S_{\text{obs}}/\Delta S_{\text{max}})L_{\text{T}} \quad (18)$$

and

$$\nu = (\Delta S_{\text{obs}}/\Delta S_{\text{max}})(L_{\text{T}}/M_{\text{T}}) \quad (19)$$

Thus, a single titration can be used to obtain ν as a function of L_{F} . However, we stress that one *should not assume* that the fractional signal change is equal to the fraction of bound ligand, since this can lead to significant errors if it is not true. On the other hand, if a direct proportionality does not exist between the signal change and the fraction of bound ligand over a wide range of binding densities, the true binding isotherm can still be constructed without any assumptions using the LBDF analysis.

Experimental Aspects

Since the methods of analysis reviewed here are applicable to a variety of spectroscopic (and nonspectroscopic) approaches, we will not review the details of data collection for each possible approach. However, there are some general experimental details that we can address concerning the collection of a set of titration curves to be used in a BDF analysis. We discuss the case of an LBDF analysis obtained by reverse titrations as a specific example. If the titrations require the addition of large volumes of titrant, then one must be careful to account for the dilution of the total ligand concentration, L_{T} , which will occur throughout the titration; i.e., the value of L_{T} will vary along the LBDF curve. Serious systematic errors may result if this dilution is not considered in the calculation. Alternatively, one can preadd ligand to the titrant (macromolecule) solu-

tion, at concentration L_T , in order to maintain the same total ligand concentration, L_T , throughout each titration. In this manner, no change in L_T occurs and the graphical analysis (see Fig. 7) is straightforward. The same procedure of "doping" the ligand titrant with macromolecule can be used to avoid changes in the total macromolecule concentration, M_T , throughout a normal titration (see Fig. 1).

Overlapping Binding Site Models for Nonspecific Binding of Protein to a Linear Nucleic Acid

The discussions presented above have focused on the generation of true, model-independent binding isotherms (i.e., the relationship between the average binding density and the free protein concentration). However, once a true binding isotherm has been obtained by a rigorous method, the next step is the analysis of this isotherm to obtain either model-independent macroscopic binding parameters or to use a statistical thermodynamic model to obtain intrinsic binding parameters. For site-specific binding of proteins to regulatory sites, this is a relatively straightforward process, although complications such as cooperativity among multiple binding sites within an operon may also need to be considered.^{3,45} However, for nonspecific binding of proteins to a linear nucleic acid lattice, the analysis is more complex, since overlap of potential protein-binding sites must be considered.^{38,46-49} Closed form expressions for the isotherms for noncooperative and unlimited nearest neighbor cooperativity have been derived for an infinite lattice^{38,48} and for a finite lattice.⁵⁰ For the noncooperative case the binding is characterized by two parameters, the intrinsic equilibrium binding constant, K , and the site size, n , which is the number of nucleotides or base pairs occluded by the bound ligand. The unlimited cooperative case requires an additional parameter, ω , the nearest neighbor cooperativity parameter, which characterizes the interactions between two nearest neighbor proteins on the nucleic acid lattice.^{38,47} In the original formulation of the unlimited cooperativity model, separate equations were obtained for the noncooperative vs cooperative cases³⁸; however, a single equation for the binding isotherm, which is applicable for both cooperative and noncooperative binding has been reported.⁵¹ The generalized isotherm

⁴⁵ M. Brenowitz, D. F. Senear, M. A. Shea, and G. K. Ackers, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 8462 (1986).

⁴⁶ D. M. Crothers, *Biopolymers* **6**, 575 (1968).

⁴⁷ J. A. Schellman, *Isr. J. Chem.* **12**, 219 (1974).

⁴⁸ A. S. Zasedatelev, G. V. Gurskii, and M. V. Volkenshtein, *Mol. Biol.* **5**, 245 (1971).

⁴⁹ T. Tsuchiya and A. Szabo, *Biopolymers* **21**, 979 (1982).

⁵⁰ I. R. Epstein, *Biophys. Chem.* **8**, 327 (1978).

⁵¹ W. Bujalowski, T. M. Lohman, and C. F. Anderson, *Biopolymers* **28**, 1637 (1989).

for the unlimited cooperativity model (applicable for all $\omega > 0$) is given in Eq. (20), in the Scatchard form:

$$\nu/L_F = K(1 - n\nu)\{[2\omega(1 - n\nu)]/[(2\omega - 1)(1 - n\nu) + \nu + R]\}^{n-1} \times \{[1 - (n + 1)\nu + R]/[2(1 - n\nu)]\}^2 \quad (20)$$

where

$$R = [((1 - (n + 1)\nu)^2 + 4\omega\nu(1 - n\nu))]^{1/2}$$

The analysis of experimental binding isotherms using this overlap binding model with nearest neighbor unlimited cooperativity has been reviewed.⁵² An alternative overlap model considers a limited nearest neighbor cooperativity between bound proteins (i.e., the formation of clusters of dimers). The closed form expression for this limited cooperativity model²² is given in Eq. (21):

$$\nu/L_F = K[q^2 - 2\nu q + (1 - \omega_{\text{lim}})\nu^2]^n/q^{(2n-1)} \quad (21)$$

where $q = [1 - (n - 1)\nu] + \{[1 - (n - 1)\nu]^2 - \nu(1 - \omega_{\text{lim}})[2 - (2n - 1)\nu]\}^{1/2}$. This limited cooperativity model has been used to analyze the equilibrium cooperative binding of the *E. coli* SSB tetramer to long, single-stranded DNA in its (SSB)₆₅ binding mode to obtain the intrinsic equilibrium constant, K , and the limited cooperativity parameter, ω_{lim} , for the formation of dimers of SSB tetramers on the DNA²² (see Fig. 9).

We emphasize that in all cases the *essential* starting point before embarking on such model-dependent analyses is a model-independent equilibrium binding isotherm. Only if this is available will the application of a specific statistical thermodynamic model, based on the known molecular aspects of the protein–nucleic acid system under study, provide meaningful interaction parameters.

Pitfalls in Analysis of Binding Data Obtained by Monitoring Spectroscopic Signals

Problems When the Assumption that $\Delta S_{\text{obs}}/\Delta S_{\text{max}} = L_B/L_T$ Is Not Valid: One Ligand Can Bind to a Nucleic Acid Lattice in Two Modes, Each Mode Possessing a Different Signal

We now discuss a theoretical case in order to illustrate the problems that one can encounter if a linear relationship between the fractional signal change and the fraction of bound ligand is incorrectly assumed to hold. We consider the nonspecific, noncooperative binding of a single protein to

⁵² S. C. Kowalczykowski, L. S. Paul, N. Lonberg, J. W. Newport, J. A. McSwiggen, and P. H. von Hippel, *Biochemistry* **25**, 1226 (1986).

a linear nucleic acid (modeled as an infinite one-dimensional homogeneous lattice). However, the protein is able to bind in two different modes, each characterized by different site sizes (nucleotides covered per ligand), $n = 5$ and $k = 10$. Although the case treated here is hypothetical, there are several examples of proteins that behave in this manner under some conditions, e.g., the *E. coli* SSB protein^{39,41,53} and the fd gene 5 protein.⁵⁴

The theoretical treatment of this equilibrium binding problem requires a three-state lattice model. The treatment of the three-state lattice problem has been discussed,⁵¹ using the sequence generating function approach.^{55,56} The general secular equation for the three-state lattice model, in the absence of cooperativity, but considering overlap of potential binding sites, is given in Eq. (22):

$$x^{(n+k+1)} - x^{(n+k)} - K_2 L x^{(n+1)} - K_1 L x^{(k+1)} = 0 \quad (22)$$

where K_1 and K_2 are the intrinsic binding constants for the ligand in the first ($n = 5$) and second ($k = 10$) modes, respectively, and L is the free ligand concentration. The binding densities (ligand bound per nucleotide) in the two modes, ν_1 and ν_2 , are then given in Eqs. (23) and (24):

$$\begin{aligned} \nu_1 &= \partial \ln x_1 / \partial \ln(K_1 L) \\ &= K_1 L / [(n+k+1)x_1^{(n+k+1)} - (n+k)x_1^{(n+k)} - (n+1)K_2 L x_1^{(n+1)} \\ &\quad - (k+1)x_1^{(k+1)} K_1 L] \end{aligned} \quad (23)$$

$$\begin{aligned} \nu_2 &= \partial \ln x_1 / \partial \ln(K_2 L) \\ &= K_2 L / [(n+k+1)x_1^{(n+k+1)} - (n+k)x_1^{(n+k)} - (n+1)K_2 L x_1^{(n+1)} \\ &\quad - (k+1)x_1^{(k+1)} K_1 L] \end{aligned} \quad (24)$$

where x_1 is the largest root of the secular equation.⁵¹ Given values for n , k , K_1 , K_2 , and L , then ν_1 and ν_2 can be obtained as a function of the free ligand concentration, L .

We consider the hypothetical case in which the fluorescence of the ligand is partially quenched on binding to the nucleic acid and this signal is used to monitor binding. In this case, a reverse titration would be performed. Then, in the presence of the nucleic acid, Eq. (25) relates the observed quenching, Q_{obs} , to the average binding densities in the two modes.

$$Q_{\text{obs}}(L_T/M_T) = Q_1 \nu_1 + Q_2 \nu_2 \quad (25)$$

where Q_1 and Q_2 are the molecular fluorescence quenchings of the ligand

⁵³ T. M. Lohman, W. Bujalowski, and L. B. Overman, *Trends Biochem.* **13**, 250 (1988).

⁵⁴ J. W. Kansy, B. A. Clack, and D. M. Gray, *J. Biomol. Struct. Dyn.* **3**, 1079 (1986).

⁵⁵ S. Lifson, *J. Chem. Phys.* **40**, 3705 (1964).

⁵⁶ D. L. Bradley and S. Lifson, in "Molecular Associations in Biology" (B. Pullman, ed.), p. 261. Academic Press, New York, 1964.

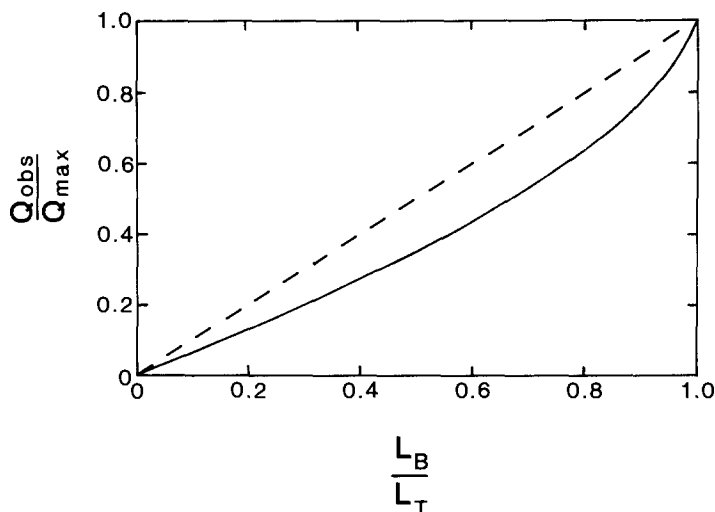


FIG. 13. The relationship between the fractional fluorescence quenching, $Q_{\text{obs}}/Q_{\text{max}}$, and the fraction of bound ligand, L_B/L_T , for the hypothetical case of a protein binding to a linear nuclear acid in two binding modes, possessing different site sizes (n and k) and degrees of fluorescence quenching (Q_1 and Q_2). For this case, the following parameters were used: mode 1, $n = 5$ nucleotides, $K_1 = 10^6 M^{-1}$, $Q_1 = 0.2$; mode 2, $k = 10$ nucleotides, $K_2 = 3 \times 10^6 M^{-1}$, $Q_2 = 0.9$. The dashed line represents a direct proportionality between $Q_{\text{obs}}/Q_{\text{max}}$ and L_B/L_T .

when bound in the two modes. Furthermore, the total ligand concentration, L_T , is given in Eq. (26):

$$L_T = L_F + (\nu_1 + \nu_2)M_T \quad (26)$$

For the case in which $n = 5$ nucleotides, $K_1 = 1 \times 10^6 M^{-1}$, $Q_1 = 0.2$, $k = 10$ nucleotides, $K_2 = 3 \times 10^6 M^{-1}$, and $Q_2 = 0.9$, Fig. 13 indicates that the fractional change in ligand fluorescence quenching, $Q_{\text{obs}}/Q_{\text{max}}$, is *not* a linear function of the fraction of bound ligand, L_B/L_T . In Fig. 14 we have plotted the actual total binding density, $\Sigma \nu_i$ ($= \nu_1 + \nu_2$) vs the "apparent" binding density that is calculated if it is assumed that $Q_{\text{obs}}/Q_{\text{max}} = L_B/L_T$, i.e., " ν_{app} " $= (Q_{\text{obs}}/Q_{\text{max}})(L_T/M_T)$. At low values of the actual binding density ($\Sigma \nu_i$), the apparent binding density is lower than the actual binding density; however, as the actual binding density increases, the apparent binding density decreases further due to a shift in the distribution of bound ligand toward the mode possessing the lower site size ($n = 5$), which also has a lower fluorescence quenching. It is clear for this case that the apparent binding isotherm, calculated by assuming a direct proportionality between $Q_{\text{obs}}/Q_{\text{max}}$ and L_B/L_T , would be incorrect and

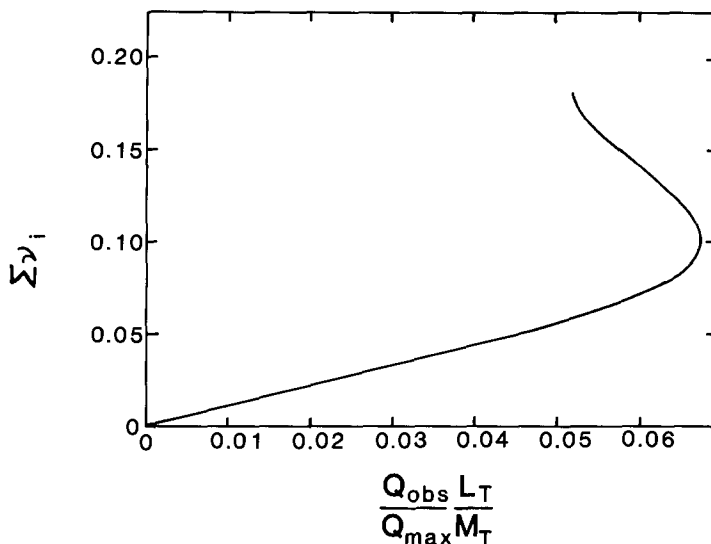


FIG. 14. The relationship of the actual average binding density, $\Sigma \nu_i$ (protein bound per nucleotide) to the apparent binding density, $\nu_{app} = (Q_{obs}/Q_{max})(L_T/M_T)$, calculated by assuming that $Q_{obs}/Q_{max} = L_B/L_T$. These theoretical data are based on a hypothetical model in which a protein can bind to a linear nucleic acid lattice in two binding modes, possessing different site sizes ($n = 5$ and $k = 10$ nucleotides), different extents of fluorescence quenching ($Q_1 = 0.2$, $Q_2 = 0.9$), and different binding constants ($K_1 = 10^6 M^{-1}$, $K_2 = 3 \times 10^6 M^{-1}$).

quite different than the true isotherm. Therefore, the analysis of this type of system would require the use of the binding density function analysis.

Potential Problems with the Use of Double-Reciprocal Plots to Determine Binding Parameters and Spectral Characteristics of Bound Ligand

A common procedure used to analyze titrations, especially those monitored by spectroscopic techniques, is to obtain the signal for fully bound ligand by performing experiments as a function of excess macromolecule concentration and plotting the signal vs $1/M_T$ and extrapolating to $1/M_T = 0$.⁵⁷ Correctly applied, the extrapolation is carried out in the range of low binding density and, as a result, this method will yield the maximal signal change *only* in the limit of low binding density. However, the signal change for the bound ligand at low binding density may not be the same as that for the bound ligand at intermediate or high binding densities, especially for a ligand that binds in a cooperative manner (i.e., there may

⁵⁷ H. H. Benesi and J. H. Hildebrand, *J. Am. Chem. Soc.* **71**, 2703 (1949).

be different signals associated with different binding modes that dominate at different binding densities). Therefore, the use of this approach to construct apparent binding isotherms always has these associated limitations and assumptions.

Detection of Binding Modes without Associated Signal Change

It is possible, for a complex macromolecular binding system, that not all of the ligand–macromolecule binding modes have an associated change in the physicochemical signal being monitored. In this case, the question arises as to whether these binding modes will be detected on monitoring that physicochemical signal. In fact, a true binding isotherm that incorporates all existing binding modes can still be obtained as long as at least one of the binding modes has an associated signal change. This results from the fact that the different binding modes are all thermodynamically coupled through the free ligand concentration, L_F , as shown in Eqs. (27) and (28):

$$L_F = L_T - (\Sigma \nu_i) M_T \quad (27)$$

$$M_T = M_F + \Sigma M_i \quad (28)$$

Therefore, even if there is no signal change on formation of a particular binding mode, the ligands or macromolecules bound in this “invisible” mode will still be detected since they affect L_F through mass conservation [Eq. (27)], hence the free ligand concentration, L_F , is linked to all binding modes, independent of whether they are accompanied by a signal change.

Analysis of Systems in Which Different Binding Modes Yield Signal Changes of Opposite Sign

An unlikely, although possible, situation is one in which different binding modes might have signal changes of opposite sign (i.e., fluorescence quenching vs enhancement). In such a case, a maximum or minimum in the observed signal would occur as a function of binding density and in the binding density function itself (i.e., $\partial S / \partial \nu = 0$). In such a case, the thermodynamic methods of analysis reviewed here can still be used to analyze a series of BDF curves; however, the analysis must be restricted to regions of each curve that have the same sign (e.g., $\partial S / \partial \nu > 0$). The analysis cannot be performed in the regions where $\partial S / \partial \nu = 0$, since no net signal change accompanies binding, although information can be obtained on either side of these regions.

Summary

The measurement of equilibrium binding constants for ligand–macromolecule interactions by monitoring a change in some spectral property of the ligand or the macromolecule is a common method used to study these interactions. This is due to the high sensitivity of the spectroscopic methods and general ease in applying these experimental procedures. In addition, binding can be monitored continuously, thus facilitating kinetic measurements. The main problem with these methods results from the fact that the spectroscopic signal is an indirect measure of binding, since the relationship between the change in the spectroscopic signal and the extent of binding is unknown, *a priori*. A common recourse is to assume a strict proportionality between the signal change and the fractional saturation of the ligand or macromolecule; however, it is often the case that such a direct proportionality does not hold. In this chapter we have reviewed the use of methods to analyze ligand–macromolecule equilibrium titrations that are monitored by indirect spectroscopic techniques. These methods of analysis yield thermodynamically rigorous, model-independent binding isotherms, hence assumptions concerning the relationship between the signal change and the extent of binding are not required. In fact, these methods can also be used to determine quantitatively the relationship between the signal change and the average degree of binding. In addition, the approaches discussed here are general and not limited to spectroscopic signals and therefore can be used with any intensive physicochemical property that reflects binding.

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[16] Analysis of Equilibrium and Kinetic Measurements to Determine Thermodynamic Origins of Stability and Specificity and Mechanism of Formation of Site-Specific Complexes between Proteins and Helical DNA

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Introduction

One of the central goals of molecular biology is to elucidate the basis of the control of gene expression at the level of molecular interactions. This requires understanding the origins of specificity and stability of protein–nucleic acid complexes. Specificity and stability are determined not only by the sequences and functional groups on the macromolecular species but also by the physical environment (e.g., temperature, pH, and salt concentration). Structural and equilibrium binding studies have been used to probe the molecular and thermodynamic origins of stability and specificity. In some systems, specificity is kinetically determined, and mechanistic studies are required to understand its basis. In this chapter, thermodynamic and mechanistic conclusions derived from equilibrium and kinetic studies on *Escherichia coli lac* repressor and RNA polymerase ($E\sigma^{70}$) are used as examples of these analyses.

Harrison and Aggarwal¹ have reviewed the structural conclusions from X-ray crystallographic investigations of complexes between DNA oligomers and site-specific binding proteins (434 repressor, 434 Cro, λcI repressor, *E. coli trp* repressor), where the binding subdomain on the protein contains the helix–turn–helix (HTH) motif. Sequence homology and NMR data (for the *lac* repressor headpiece binding to an operator sequence) suggest that the HTH motif is also used by *lac* repressor^{2–5} and *E. coli*

¹ S. C. Harrison and A. K. Aggarwal, *Annu. Rev. Biochem.* **59**, 933 (1990).

² R. Kaptein, E. R. P. Zuiderweg, R. M. Scheek, R. Boelens, and W. F. van Gunsteren, *J. Mol. Biol.* **182**, 179 (1985).

³ R. Boelens, R. M. Scheek, J. H. van Boom, and R. Kaptein, *J. Mol. Biol.* **193**, 213 (1987).

⁴ R. Boelens, R. M. Scheek, R. M. J. N. Lamerichs, J. de Vlieg, J. H. van Boom, and R. Kaptein, in "DNA Ligand Interactions" (W. Guschlbauer and W. Saenger, eds.), p. 191. Plenum, New York, 1987.

⁵ R. M. J. N. Lamerichs, R. Boelens, G. A. van der Marel, J. H. van Boom, R. Kaptein, F. Buck, B. Fera, and H. Rüterjans, *Biochemistry* **28**, 2985 (1989).

RNA polymerase^{6,7} in site-specific binding. At the molecular level, the specificity of protein–DNA interactions may involve both (1) direct readout of the DNA sequence, defined as the contribution to specific recognition from interactions between functional groups on the protein and functional groups on the base pairs that are accessible in the grooves of the DNA helix; and (2) indirect readout, defined as the contribution to specific recognition from the effects of nucleotide sequence on the conformation and physical properties of the specific DNA site.¹

Direct readout is generally assumed to be the primary determinant of specificity. In the structures that have been solved to high resolution, direct readout is accomplished by hydrogen bonds and nonpolar contacts between amino acid side chains and functional groups on the base pairs that are accessible in the major groove. Factors that may contribute to indirect readout include sequence-dependent variations in the conformation of the sugar phosphate backbone, the twist angle of the base pairs, or the width of grooves, and/or differences in physical properties of DNA, including lateral and torsional stiffness and degree of static curvature.

Harrison and Aggarwal¹ point out that a recurring theme in these structures is the complementarity that exists between the proteins and both the sugar phosphate backbone and the base pairs of the DNA site. Record and collaborators^{8–10} provide an analogous discussion from a thermodynamic perspective. To obtain a tightly bound complex, complementarity must exist at both the steric level and at the level of specific functional groups. Such complementarity is a hallmark of all noncovalent complexes, including those of enzymes with substrates and especially with transition states.¹¹ [The transition state stabilization theory of enzyme catalysis states that the surface of the enzyme catalytic site and of the substrate are sufficiently complementary to form a weakly bound enzyme–substrate complex, and that the chemical and physical changes in the substrate that convert it to the transition state (activated complex) on the path to product are driven in large part by the much greater degree of complementarity between the transition state and the enzyme catalytic site.¹¹] In order to understand the origins of specificity and stability of

⁶ M. Gribskov and R. R. Burgess, *Nucleic Acids Res.* **14**, 6745 (1986).

⁷ D. A. Siegel, J. C. Hu, W. A. Walter, and C. A. Gross, *J. Mol. Biol.* **206**, 591 (1989).

⁸ M. T. Record, Jr., and M. C. Mossing, in "RNA Polymerase and Regulation of Transcription" (W. Reznikoff *et al.*, eds.), p. 61. Elsevier Science Publishing, New York, 1987.

⁹ M. T. Record, Jr., and R. S. Spolar, in "Nonspecific DNA-Protein Interactions" (A. Revzin, ed.), p. 33. CRC Press, Boca Raton, Florida, 1990.

¹⁰ M. T. Record, Jr., in "Unusual DNA Structures" (R. D. Wells and S. C. Harvey, eds.), p. 237. Springer-Verlag, New York, 1988.

¹¹ See for example the recent review by J. Kraut, *Science* **242**, 533 (1988).

noncovalent macromolecular associations, it will be important to quantify the extent of complementarity of a pair of recognition surfaces, the extent of local and global changes in conformation that accompany the attainment of this degree of complementarity, and the thermodynamic consequences thereof. Even in the case of enzyme–substrate complexes, this information is not yet available.

To date, no general molecular rules for specificity of protein–DNA complexes beyond the concept of mutual complementarity have emerged. Harrison and Aggarwal¹ review the complexity of site-specific recognition and conclude that “the variety of protein–base pair contacts is more evident than their uniformity.” There is no evidence for a unique or even a degenerate “recognition code” involving amino acid side chains and base pairs. A single base pair may simultaneously interact with several different amino acids. For example, in the complex formed between 434 O_R1 and R(1–69) (the DNA-binding domain of 434 repressor), the AT at position 1 is contacted by Gln-17, Asn-36, and Gln-28.^{1,12} A single amino acid may interact with different bases, may simultaneously interact with two or more base pairs, and/or may interact with different functional groups on a given type of base at different locations in the complex. For example, in the R(1–69)/O_R1 complex, different Gln residues contact A, G, C, and T.^{1,12} Gln-28 makes two hydrogen bonds with A at position 1 and also makes a nonpolar contact with T at position –1. Gln-29 forms two hydrogen bonds with G at position 2 and also makes nonpolar contacts with T at positions 3 and 4. Gln-33 forms a hydrogen bond to T at position 4 and also makes a nonpolar contact with C at position 5. These structural observations imply that site-specific protein–DNA interactions will in at least some cases be strongly context dependent.^{1,8,13}

In many discussions of specificity and stability of protein–DNA interactions based on structural data and/or genetic studies, it has been assumed (implicitly or explicitly) that the stability of protein–DNA complexes results from the same interactions that determine the specificity of recognition, and that the contributions of these interactions are “additive” (i.e., independent of context). Parallel measurements of equilibrium and kinetic parameters are required to gain insight into the thermodynamics (“energetics”) and mechanisms of the process of forming these site-specific complexes, and to relate specificity and stability to the patterns of direct and indirect readout. To date no system has been extensively

¹² A. K. Aggarwal, D. W. Rodgers, M. Drottar, M. Ptashne, and S. C. Harrison, *Science* **242**, 899 (1988).

¹³ M. C. Mossing and M. T. Record, Jr., *J. Mol. Biol.* **186**, 295 (1985).

investigated at *both* the structural and thermodynamic level, although a number of such studies are currently in progress.

Interpretations of structural data and of thermodynamic data obtained under a single set of solution conditions are complicated by the fact that functional groups on both proteins and DNA are capable of interacting not only with each other, but also with water and solutes. The latter interactions (including hydrogen bonds with water and Coulombic interactions of phosphates on the DNA polyelectrolyte with cations) may be intrinsically more or less favorable than the macromolecular interactions that replace them in the complex. The nature and consequences of small molecule-macromolecule interactions in solution cannot generally be ascertained from crystal structures of the complexed and uncomplexed species. The sites occupied by water, ions, and other solutes in the solid-state structure may not be relevant for the effects of these species in solution, and certainly provide little information about the strength of small molecule-macromolecule interactions. Such information is necessary in order to understand the thermodynamics of the process of pairing the mutually complementary macromolecular recognition surfaces, which must involve the rearrangement or loss of preexisting interactions with solutes and water. Interactions of proteins with nucleic acids are exchange reactions, and the stability and specificity of protein-DNA interactions are strong functions of the relevant environmental variables (temperature, pH, salt concentration, etc.) that affect these exchange processes. The exchange character of protein-DNA interactions must be considered in interpreting the dependence of equilibrium "constants" and rate "constants" on temperature and solution variables. In particular, characteristic effects of salt concentration and temperature are observed that differ at both a qualitative and a quantitative level from the effect of these variables on covalent reactions involving low-molecular-weight ionic solutes.

A well-known classical conceptual framework exists to interpret the effects of changes in salt concentration and temperature on reactions of low-molecular-weight ionic solutes. However, these reactions are not suitable model systems for the interactions of proteins with nucleic acids, or for other noncovalent interactions of biopolymers. For example, the composite variable "ionic strength" obtained from Debye-Hückel theory has been applied with some success to analyze effects of salt concentration on the thermodynamics and kinetics of reactions of small, charged solutes. However, "ionic strength" has been shown both theoretically and experimentally to be an inappropriate variable for analysis of ion effects on the interactions of oligocations and proteins with nucleic acid polyanions. As a second example, van't Hoff or calorimetric analysis of covalent reactions of small molecules typically yields values of ΔH° and ΔS° for the corre-

sponding process that are relatively temperature independent. On the other hand, large heat capacity effects are observed in most noncovalent processes involving proteins. Consequently van't Hoff plots are highly nonlinear and ΔH° and ΔS° are strong functions of temperature.

To date only a few appropriate model systems for noncovalent interactions of proteins and nucleic acid have been characterized in detail. These include the transfer of small, nonpolar solutes from the pure state (liquid, solid, or vapor) to water and the binding of small oligocationic ligands to nucleic acid polyanions. Results with these model systems demonstrate that quantitative measurements of the effects of temperature and salt concentration on protein–DNA interactions provide detailed insight into the types of noncovalent exchange interactions involved and the magnitude of their contributions to the changes in standard-state thermodynamic functions for both nonspecific and site-specific binding. This review chapter provides a conceptual overview of the thermodynamic, mechanistic, and, in some cases, molecular information that can be deduced from the dependence of equilibrium and kinetic quantities (observed equilibrium “constants” and rate “constants”) on temperature and salt concentration. For the thermodynamic discussion, we have chosen the interaction of *lac* repressor with its operator as our primary example. The kinetic studies reviewed here involve *lac* repressor and RNA polymerase. Although the examples that we discuss come from only a limited number of systems, this conceptual approach is generally applicable to all other site-specific and nonspecific protein–DNA and ligand–DNA interactions.

Equilibria and Thermodynamics of Site-Specific Protein–DNA Interactions

Thermodynamic Background

Specificity of Protein–DNA Interactions at the Thermodynamic Level. At a thermodynamic level, questions regarding the origins of stability and of specificity of site-specific protein–DNA interactions are interrelated. The stabilities of site-specific (**RS**) and nonspecific (**RD**) complexes are defined by the standard free energy changes (ΔG°) for the appropriate processes of complex formation:

1. Protein + specific DNA site $\rightarrow$ specific complex:



2. Protein + nonspecific DNA site $\rightarrow$ nonspecific complex:



We use the symbol **R** for protein, since both *lac* repressor and RNA polymerase are typically designated **R**. We use **S** for the specific DNA site, which differs from the standard abbreviation for operator (**O**) or promoter (**P**) DNA sites.] At a thermodynamic level, specificity is defined as the standard free energy change $\Delta G_{\mathbf{RD} \rightarrow \mathbf{RS}}^{\circ}$ for the process of transferring a protein from a nonspecific site (**D**) to a specific site (**S**):

$$\begin{aligned} \Delta G_{\mathbf{RD} \rightarrow \mathbf{RS}}^{\circ} &= \Delta G_{\mathbf{RS}}^{\circ} - \Delta G_{\mathbf{RD}}^{\circ} \\ &= -RT \ln(K_{\mathbf{RS}}/K_{\mathbf{RD}}) \\ &= (\bar{G}_{\mathbf{RS}}^{\circ} - \bar{G}_{\mathbf{RD}}^{\circ}) - (\bar{G}_{\mathbf{S}}^{\circ} - \bar{G}_{\mathbf{D}}^{\circ}) \end{aligned} \quad (3)$$

[In a biological sense, regulation (the consequence of specificity) may be under either thermodynamic or kinetic control.⁸ In the latter case, the thermodynamic definition of specificity ($K_{\mathbf{RS}}/K_{\mathbf{RD}}$) is replaced by the appropriate ratio of rate constants.] For a DNA molecule with $n_{\mathbf{S}}$ specific sites and $n_{\mathbf{D}}$ nonspecific sites, the distribution of protein at low binding density is determined by the site-weighted specificity ratio $n_{\mathbf{S}}K_{\mathbf{RS}}/n_{\mathbf{D}}K_{\mathbf{RD}}$. The origins of stability of both site-specific and nonspecific protein–DNA complexes must be understood in order to deduce the specificity of the interactions as a function of environmental variables.

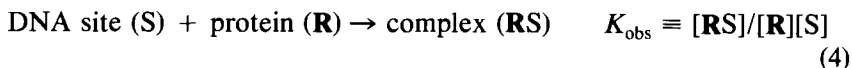
To understand the stability of protein–DNA complexes one must dissect at thermodynamic and molecular levels the contributions of various noncovalent interactions to the complicated process of bringing two macromolecular sites together and creating two complementary recognition surfaces. The process of juxtaposing these recognition surfaces involves disruption of preexisting interactions between individual functional groups on each biopolymer and water or solutes (e.g., ions), and replacement of these interactions by contacts between complementary functional groups on each macromolecule. Thus both nonspecific and specific binding are exchange processes involving both water and solutes. The stability of noncovalent complexes in aqueous solution is totally different from what would be predicted *in vacuo*, and is highly dependent on the details of the solution environment.

Binding Constants (K_{obs}) for Site-Specific Interactions. Thermodynamics provides information regarding processes, often gleaned from measurements of the corresponding equilibrium constants, and their dependence on physical variables. In the discussion that follows the process is the noncovalent association of protein with DNA. Typically, for the association of a protein and a DNA molecule containing one (or more) specific site(s), the extent of formation of the specific complexes at equilibrium is assayed as a function of total biopolymer concentrations in a titration of

protein with DNA (or vice versa) under reversible binding conditions. (References in the following sections are representative but not inclusive; the bibliographies in these references should also be consulted.)

Quantitative equilibrium studies of site-specific binding have been performed with many prokaryotic proteins involved in regulation of gene expression, including the *E. coli* proteins *lac* repressor,¹³⁻¹⁸ *trp* repressor,¹⁹ *gal* repressor,²⁰ CAP,²¹⁻²⁵ RNA polymerases ($E\sigma^{70}$)^{26,27} and ($E\sigma^{32}$),^{28,29} *EcoRI* endonuclease^{14,30,31} and the bacteriophage proteins Mnt³² and Arc³³ from P22 and *cI* repressor³⁴⁻³⁶ and Cro^{37,38} from phage λ . Commonly used assays include filter binding,^{18,39} footprinting,^{40,41} and the gel retardation assay.^{21,22} The equilibrium extent of binding for a specified set of solution conditions is used to define a binding constant K_{obs} for the process:

- ¹⁴ J.-H. Ha, R. S. Spolar, and M. T. Record, Jr., *J. Mol. Biol.* **209**, 801 (1989).
- ¹⁵ M. D. Barkley, P. A. Lewis, and G. E. Sullivan, *Biochemistry* **20**, 3842 (1981).
- ¹⁶ R. B. Winter and P. H. von Hippel, *Biochemistry* **20**, 6948 (1981).
- ¹⁷ P. A. Whitson, J. S. Olson, and K. S. Matthews, *Biochemistry* **25**, 3852 (1986).
- ¹⁸ A. D. Riggs, H. Suzuki, and S. Bourgeois, *J. Mol. Biol.* **48**, 67 (1970).
- ¹⁹ J. Carey, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 975 (1988).
- ²⁰ M. Brenowitz, E. Jamison, A. Majumdar, and S. Adhya, *Biochemistry* **29**, 3374 (1990).
- ²¹ M. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).
- ²² M. M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).
- ²³ R. H. Ebright, Y. W. Ebright, and A. Gunasekera, *Nucleic Acids Res.* **17**, 10295 (1989).
- ²⁴ R. H. Ebright, A. Kolb, H. Buc, T. A. Kunkel, J. S. Krakow, and J. Beckwith, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6083 (1987).
- ²⁵ T. Heyduk and J. C. Lee, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 1744 (1990).
- ²⁶ H. S. Strauss, R. R. Burgess, and M. T. Record, Jr., *Biochemistry* **19**, 3504 (1980).
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- ²⁸ M. A. Fisher, Ph.D. Thesis, University of Wisconsin-Madison (1990).
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- ³⁰ D. R. Lesser, M. R. Kurpiewski, and L. Jen-Jacobson, *Science* **250**, 776 (1990).
- ³¹ B. J. Terry, W. E. Jack, R. A. Rubin, and P. Modrich, *J. Biol. Chem.* **258**, 9820 (1983).
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- ³³ A. K. Vershon, S.-M. Liao, W. R. McClure, and R. T. Sauer, *J. Mol. Biol.* **195**, 323 (1987).
- ³⁴ H. C. M. Nelson and R. T. Sauer, *Cell (Cambridge, Mass.)* **42**, 549 (1985).
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- ³⁷ J. G. Kim, Y. Takeda, B. W. Matthews, and W. F. Anderson, *J. Mol. Biol.* **196**, 149 (1987).
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- ⁴⁰ R. T. Ogata and W. Gilbert, *J. Mol. Biol.* **132**, 709 (1979).
- ⁴¹ M. Brenowitz, D. F. Senear, M. A. Shea, and G. K. Ackers, this series, Vol. 130, p. 132.



This definition of K_{obs} requires knowledge of the equilibrium concentrations of both the complex and the free reactants. In practice, these quantities are often difficult to determine independently, and K_{obs} is more typically (but less appropriately) defined by neglecting all possible coupled or competitive equilibria that may be simultaneously occurring involving protein, DNA, and/or complex, and writing:

$$K_{\text{obs}} \equiv \frac{[\text{RS}]}{(\mathbf{R}_T - [\text{RS}])(\mathbf{S}_T - [\text{RS}])} \quad (5)$$

where $\mathbf{R}_T$ and $\mathbf{S}_T$ are total concentrations (on the molar scale) of protein and of specific DNA sites, respectively. If coupled or competitive equilibria (e.g., nonspecific binding, protein aggregation) are occurring and/or if significant concentrations of intermediate complexes are present at equilibrium, then $[\mathbf{R}] \neq \mathbf{R}_T - [\text{RS}]$ and/or $[\text{S}] \neq \mathbf{S}_T - [\text{RS}]$, and the definitions of K_{obs} in Eqs. (4) and (5) are not equivalent. In these cases K_{obs} as defined by Eq. (5) may depend on the total concentration of protein and/or DNA.⁴² Moreover, any dependence of K_{obs} on temperature, pH, salt concentration, or other variables (as discussed below) may contain contributions from the effects of these variables on the coupled process as well as on the process of protein–DNA association. While a significant amount is known about potential coupled equilibria for $\text{E}\sigma^{70}$ RNA polymerase,²⁷ the extent to which these aggregation and nonpromoter binding events contribute to the equilibria and kinetics of $\text{E}\sigma^{70}$ –promoter interaction is not known, although *in vitro* experiments can be designed to minimize these contributions (cf. Kinetics and Mechanisms of Site-Specific Protein–DNA Interactions, below).

In most quantitative protein–DNA binding studies using the nitrocellulose filter assay or footprinting assays, the equilibrium concentration of complex has been determined as a function of $\mathbf{R}_T$ and $\mathbf{S}_T$ and the value of K_{obs} then calculated by Eq. (5). For RNA polymerase $\text{E}\sigma^{70}$, where multiple models of nonpromoter binding complicate the direct investigation of equilibrium binding at promoters, values of K_{obs} have been determined by competitive equilibrium^{26,27} and kinetic⁴³ assays. Alternatively, K_{obs} has been evaluated from the ratio of the second-order association rate constant to the first-order dissociation rate constant, where the rate laws are ex-

⁴² O. G. Berg, R. B. Winter, and P. H. von Hippel, *Biochemistry* **20**, 6929 (1981).

⁴³ T. R. Kadesh, S. Rosenberg, and M. J. Chamberlin, *J. Mol. Biol.* **155**, 1 (1982).

pressed in terms of the concentrations in Eq. (5)⁴⁴⁻⁴⁸ (cf. Kinetics and Mechanisms of Site-Specific Protein–DNA Interactions, below). In some applications of the gel shift assay, the free DNA concentration has been determined instead of the concentration of the protein–DNA complex.^{19,22} In other applications of the gel shift assay, concentrations of both free DNA and complex have been measured and used to define K_{obs} .^{21,24,32,33} Because of these differences in the definition of K_{obs} and the possible dependence of K_{obs} on protein and/or DNA concentration, values of K_{obs} should be determined by several different methods and compared over a range of macromolecular concentrations at each specified set of environmental variables (temperature, pressure, pH, salt concentration, etc.).

Interpretation of Standard Free Energy Changes $\Delta G_{\text{obs}}^\circ$. The standard free energy change ($\Delta G_{\text{obs}}^\circ$) is evaluated from K_{obs} defined according to Eq. (4) [or, if necessary, Eq. (5)]:

$$\Delta G_{\text{obs}}^\circ = -RT \ln K_{\text{obs}} \quad (6)$$

and provides the thermodynamic measure of stability of the protein–DNA complex relative to the reactants under the conditions of the experiment. When K_{obs} is defined according to Eq. (4), then $\Delta G_{\text{obs}}^\circ$ is the free energy change for the process of converting reactants to products in a hypothetical mixture where the concentrations of reactant and product macromolecules are 1 M but the environment is that of a dilute solution, and where the temperature, pH, and concentrations of electrolyte ions and buffer components are those specified in determining K_{obs} .

$\Delta G_{\text{obs}}^\circ$ contains information regarding the difference in molar free energies of the pure species and the differences in their interaction with water. Consider for simplicity the hypothetical case where no competitive multiple equilibria involving **R**, **S**, and/or **RS** are present, and where a thermodynamic equilibrium constant K_{eq} (expressed in terms of equilibrium activities on the molar scale and therefore a function of temperature and pressure only) and the corresponding standard free energy change ΔG° can be unambiguously determined for the process **R** + **S** $\rightarrow$ **RS**. For this case:

$$\Delta G^\circ = \bar{G}_{\text{RS}}^\circ - (\bar{G}_{\text{R}}^\circ + \bar{G}_{\text{S}}^\circ) = -RT \ln K_{\text{eq}} \quad (7)$$

where $\bar{G}_i^\circ$ is the dilute solution standard state chemical potential (partial

⁴⁴ J.-H. Roe, R. R. Burgess, and M. T. Record, Jr., *J. Mol. Biol.* **176**, 495 (1984).

⁴⁵ J.-H. Roe, R. R. Burgess, and M. T. Record, Jr., *J. Mol. Biol.* **184**, 441 (1985).

⁴⁶ J.-H. Roe and M. T. Record, Jr., *Biochemistry* **24**, 4721 (1985).

⁴⁷ C. J. Dayton, D. E. Prosen, K. L. Parker, and C. L. Cech, *J. Biol. Chem.* **259**, 1616 (1984).

⁴⁸ G. Duval-Valentin and R. Ehrlich, *Nucleic Acids Res.* **15**, 575 (1987).

molar Gibbs free energy) of macromolecular species i ($i = \mathbf{R}, \mathbf{S}, \mathbf{RS}$). Each $\bar{G}_i^\circ$ may be decomposed into contributions from the molar free energy of the pure liquid state of that species ($\bar{G}_i^\bullet$, obtained by extrapolation) and the contribution to the free energy of the solution per mole of added macromolecular solute resulting from its interaction with solvent ($RT \ln \gamma_i^{\text{ds}}$):

$$\bar{G}_i^\circ = \bar{G}_i^\bullet + RT \ln \gamma_i^{\text{ds}} + RT \ln \bar{V}_1^\bullet \quad (8)$$

where γ_i^{ds} is the composition-independent activity coefficient describing the interaction with solvent in the (ideal) dilute solution (relative to the ideal solution) and $\bar{V}_1^\bullet$ is the molar volume of the solvent.¹⁴ The $RT \ln \bar{V}_1^\bullet$ term appears because a molar concentration scale is employed in the definitions of K_{eq} [Eq. (7)]. If a vapor phase reference state is chosen for $\bar{G}_i^\bullet$, then the interaction coefficient γ_i^{ds} is replaced by the Henry's law constant for that species.

Interpretation of $\Delta G_{\text{obs}}^\circ$ defined by Eq. (6) differs from ΔG° of Eq. (7) because of the participation of electrolyte ions and buffer components in the interactions of proteins with nucleic acids in aqueous solution, as well as the presence of coupled macromolecular equilibria, if Eq. (5) is used to define K_{obs} . The stability of a noncovalent complex is determined by the difference in the quality and quantity of noncovalent interactions involving both the macromolecular recognition surfaces and solvent components (water, ions, etc.) in the complex ($\mathbf{RS}$) and in the uncomplexed macromolecular species ($\mathbf{R}, \mathbf{S}$). The use of macromolecular concentrations in the definition of K_{obs} that do not take into account differences in the extent of association of ions, other small solutes, and solvent with the complex and with the reactants, as well as the neglect of activity coefficients describing nonideality arising from solute-solute interactions, results in a dependence of K_{obs} (and standard thermodynamic quantities derived from it) on solution variables (such as pH and ion concentrations) as well as on temperature and pressure. The classic biochemical study of this type is Alberty's⁴⁹ work on the effect of environmental variables (pH, pMg) on the thermodynamics of ATP hydrolysis. This dependence on solution conditions is indicated by the subscript "obs."

In summary, since stability is a free energy difference between complexed and uncomplexed species and depends to a major extent on solution conditions, it must be considered a relative rather than an absolute quantity. Stabilities of noncovalent complexes in solution cannot be deduced directly from structural data or molecular modeling exercises on the com-

⁴⁹ R. A. Alberty, *J. Biol. Chem.* **244**, 3290 (1969).

plex. Specificity [cf. Eq. (3)] is also a relative quantity that, in general, is highly sensitive to solution conditions.

Analyses of the effects of temperature and pressure on K_{obs} yield the standard enthalpy change ($\Delta H_{\text{obs}}^\circ$) and the standard volume change ($\Delta V_{\text{obs}}^\circ$), respectively. Determination of the dependence of K_{obs} on pH and salt concentration allows the evaluation of the stoichiometries of participation of protons ($\Delta \nu_{\text{H}^+}$) and electrolyte ions, respectively, in the association reaction. This thermodynamic information in turn provides insight into the molecular origins of stability of the site-specific protein–DNA complex. With the association reaction of *lac* repressor with the *lac* operator as our primary example, we discuss the analysis and interpretation of experimental data concerning the dependence of K_{obs} on temperature and on solution variables.

Van't Hoff Analysis of Effects of Temperature on K_{obs}

Information regarding the nature of the thermodynamic driving force for noncovalent macromolecular binding processes may be obtained from calorimetric determinations of $\Delta H_{\text{obs}}^\circ$ and $\Delta C_{\text{p,obs}}^\circ$ or from van't Hoff analysis of the temperature dependence of K_{obs} . Both approaches allow $\Delta G_{\text{obs}}^\circ$ for the association process to be dissected into enthalpic and entropic contributions at the temperature of interest. Although enthalpies of interaction can in principle be measured directly and more accurately by isothermal mixing calorimetry than by the van't Hoff method, the calorimetric method has not yet been successfully applied to characterize site-specific protein–DNA interactions, presumably because its requirement for high (micromolar) concentrations of reactants may introduce competing interactions (e.g., aggregation of the protein and/or formation of nonspecific protein–DNA complexes).

In the van't Hoff method, applicable at any reactant concentration where product formation is accurately measurable, $\Delta H_{\text{obs}}^\circ$ is calculated from the slope of a plot of $\ln K_{\text{obs}}$ versus $1/T$:

$$\left(\frac{\partial \ln K_{\text{obs}}}{\partial 1/T} \right)_P = - \frac{\Delta H_{\text{obs}}^\circ}{R} \quad (9)$$

If $\ln K_{\text{obs}}$ is found to vary linearly with reciprocal absolute temperature, the van't Hoff $\Delta H_{\text{obs}}^\circ$ is constant over the temperature range studied. This is the behavior exhibited by most covalent reactions over the relatively narrow range of absolute temperatures where protein–DNA interactions are usually examined. If $\Delta H_{\text{obs}}^\circ$ is independent of temperature (i.e., if $\Delta C_{\text{p,obs}}^\circ = 0$), then the standard entropy change $\Delta S_{\text{obs}}^\circ$ will also be indepen-

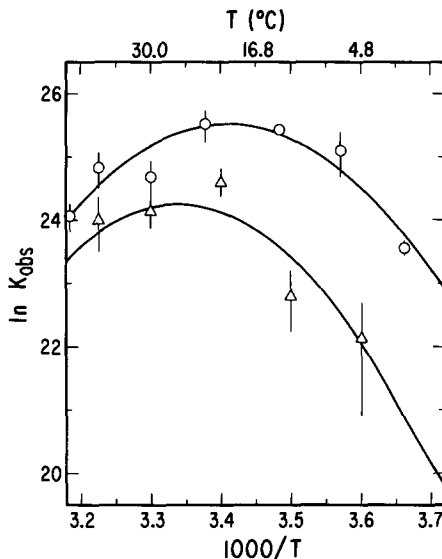


FIG. 1. Effects of temperature on site-specific binding of *lac* repressor to the O^{sym} operator ($\circ$) and *EcoRI* to its recognition sequence (Δ) in the form of van't Hoff plots. The curves are fits to the general relationship $\ln K_{\text{obs}} = (\Delta C_{\text{P,obs}}^{\circ}/R)[(T_{\text{H}}/T) - 1 - \ln(T_{\text{S}}/T)]$, useful when $\Delta C_{\text{P,obs}}^{\circ}$ is large in magnitude and independent of temperature. [From J.-H. Ha, R. S. Spolar, and M. T. Record, Jr., *J. Mol. Biol.* **209**, 801 (1989).]

dent of temperature and the only effect of temperature on $\Delta G_{\text{obs}}^{\circ}$ will be the explicit linear dependence:

$$\Delta G_{\text{obs}}^{\circ} = \Delta H_{\text{obs}}^{\circ} - T\Delta S_{\text{obs}}^{\circ} \quad (10)$$

Contrary to this simple behavior, all site-specific protein–DNA interactions investigated as a function of temperature to date exhibit highly nonlinear van't Hoff behavior.^{14,20} Van't Hoff plots of equilibrium constants for site-specific binding of *lac* repressor and *EcoRI* are shown in Fig. 1. In all site-specific protein–DNA interactions examined to date, K_{obs} exhibits a relative maximum in the experimentally accessible temperature range. For the interaction of *lac* repressor with the symmetric *lac* operator (O^{sym}), the maximum in K_{obs} occurs at 20°, and for specific binding of *EcoRI* the maximum is at 26°.¹⁴ For the association of RNA polymerase $E\sigma^{70}$ with the λP_{R} promoter to form an open complex, the maximum is near 42°. ^{14,45} The maximum in K_{obs} defines a characteristic temperature T_{H} at which $\Delta H_{\text{obs}}^{\circ} = 0$. At lower temperature ($T < T_{\text{H}}$), $\Delta H_{\text{obs}}^{\circ}$ is positive; at higher temperatures ($T > T_{\text{H}}$), $\Delta H_{\text{obs}}^{\circ}$ is negative.

The strong temperature dependence of $\Delta H_{\text{obs}}^{\circ}$ allows one to apply the

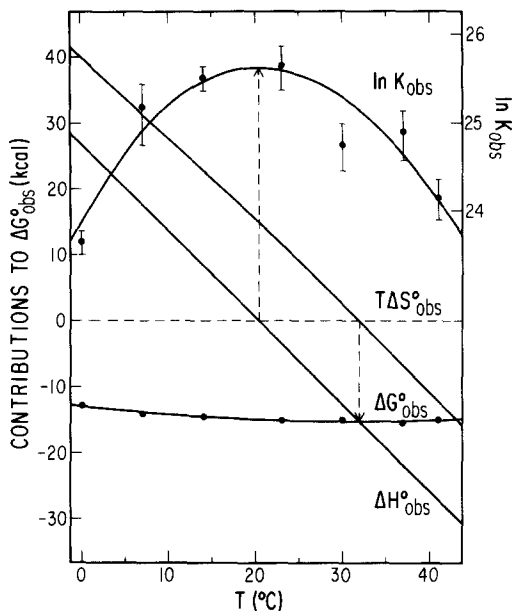


FIG. 2. The thermodynamics of the interaction of *lac* repressor with an isolated symmetric operator (O^{sym}) site. Values of $\ln K_{\text{obs}}$ and $\Delta G_{\text{obs}}^{\circ}$ are plotted as a function of temperature. Enthalpic ($\Delta H_{\text{obs}}^{\circ}$) and entropic ($T\Delta S_{\text{obs}}^{\circ}$) contributions to $\Delta G_{\text{obs}}^{\circ}$, as well as theoretical fits to $\ln K_{\text{obs}}$ and $\Delta G_{\text{obs}}^{\circ}$, were obtained assuming a constant $\Delta C_{\text{p,obs}}^{\circ}$ of $-1.3 \text{ kcal mol}^{-1} \text{ K}^{-1}$ over the temperature range investigated. [From J.-H. Ha, R. S. Spolar, and M. T. Record, Jr., *J. Mol. Biol.* **209**, 801 (1989).]

van't Hoff analysis to estimate the standard heat capacity change $\Delta C_{\text{p,obs}}^{\circ}$ for the process of complex formation, since

$$\Delta C_{\text{p,obs}}^{\circ} = \left(\frac{\partial \Delta H_{\text{obs}}^{\circ}}{\partial T} \right)_{\text{p}} \quad (11)$$

For all site-specific protein–DNA complexes investigated, the standard heat capacity change for the association process is a large negative quantity.^{14,20} In fact $\Delta C_{\text{p,obs}}^{\circ}$ is much larger in magnitude than the entropy change ($\Delta S_{\text{obs}}^{\circ}$) throughout the experimentally accessible temperature range. As a result, enthalpy–entropy compensation is observed, leading to characteristic temperature dependences of the standard thermodynamic functions (see Fig. 2) with the following features: (1) Both $\Delta H_{\text{obs}}^{\circ}$ and $T\Delta S_{\text{obs}}^{\circ}$ decrease rapidly with increasing temperature with nearly identical slopes, leaving $\Delta G_{\text{obs}}^{\circ}$ relatively unchanged. Consequently there is a shift in the nature of the thermodynamic driving force with temperature. The site-specific

binding of proteins to DNA is entropically driven at low temperature, and shifts toward enthalpically driven with increasing temperature; (2) at the characteristic temperature T_H where $\Delta H_{\text{obs}}^\circ = 0$, K_{obs} exhibits a relative maximum. At the characteristic temperature T_S where $\Delta S_{\text{obs}}^\circ = 0$, $\Delta G_{\text{obs}}^\circ$ exhibits a relative minimum.

Sturtevant⁵⁰ has examined possible origins of $\Delta C_{\text{p,obs}}^\circ$ for a variety of processes involving proteins, including denaturation, oligomerization, and binding of small ligands. In general, $\Delta C_{\text{p,obs}}^\circ$ may include contributions from the hydrophobic effect (i.e., a change in the amount of water-accessible nonpolar surface area), from creation or neutralization of charged groups, from changes in internal vibrational modes, and/or from the existence of temperature-dependent changes in equilibria between two or more conformational states or states of aggregation of the protein.⁵⁰ For proteins, Sturtevant concluded that the contribution of the hydrophobic effect is approximately four times larger than that caused by changes in vibrational modes, and that these two factors are in general the dominant contributors to the $\Delta C_{\text{p,obs}}^\circ$ of noncovalent interactions involving proteins. Using Sturtevant's semiempirical relationships, Ha *et al.*¹⁴ obtained similar estimates of the relative contributions to $\Delta C_{\text{p,obs}}^\circ$ from the hydrophobic effect and from stiffening of vibrational modes of protein-DNA interactions and proposed that the large negative $\Delta C_{\text{p,obs}}^\circ$ results primarily from the hydrophobic effect. Quantitative interpretation of the heat capacity change provides estimates both of the reduction in water-accessible nonpolar surface area on the protein and DNA that occur in formation of the specific complex, and of the contribution of the hydrophobic effect to the standard free energy change $\Delta G_{\text{obs}}^\circ$.¹⁴

To calibrate the heat capacity effect in terms of the reduction in water-accessible nonpolar surface, Spolar *et al.*⁵¹ and Livingstone *et al.*⁵² examined heat capacity changes for processes in which the change in water-accessible nonpolar surface area can be calculated from structural data. Heat capacity changes for the process of hydrocarbon transfer from the dilute aqueous solution standard state to the pure liquid phase and the process of protein folding in aqueous solution are large and negative. In both cases $\Delta C_{\text{p,obs}}^\circ$ is proportional to the reduction in water-accessible nonpolar surface area (ΔA_{np}); moreover the proportionality constant is the same within error for both hydrocarbon transfer and protein folding^{51,52}:

$$\Delta C_{\text{p,obs}}^\circ / \Delta A_{\text{np}} \cong -0.3 \text{ cal K}^{-1} \text{ \AA}^{-2} \quad (12)$$

⁵⁰ J. M. Sturtevant, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 2236 (1977).

⁵¹ R. S. Spolar, J.-H. Ha, and M. T. Record, Jr., *Proc. Natl. Acad. Sci. U.S.A.* **86**, 8382 (1989).

⁵² J. R. Livingstone, R. S. Spolar, and M. T. Record, Jr., *Biochemistry* **30**, 4237 (1991).

Consequently the hydrophobic effect must be the dominant factor contributing to the $\Delta C_{p,obs}^\circ$ of protein denaturation.⁵⁰⁻⁵² Spolar *et al.*⁵¹ proposed that this relationship can be applied to any noncovalent process involving a globular protein that exhibits a large $\Delta C_{p,obs}^\circ$. For example, Ha *et al.*¹⁴ used Eq. (12) to estimate the reduction in nonpolar water-accessible surface area that occurs in the association of protein with DNA. For all protein-DNA interactions examined, the value of ΔA_{np} calculated from the experimental $\Delta C_{p,obs}^\circ$ is significantly larger than the estimated water-accessible nonpolar surface area of the complementary surfaces of the protein and the DNA. Consequently Ha *et al.*¹⁴ proposed that conformational changes that remove additional nonpolar surface of the protein and/or DNA from water must occur as part of site-specific binding. Since the heat capacity effect accompanying DNA denaturation ($\Delta C_{p,obs}^\circ \approx 0.04$ to $0.06 \text{ cal g}^{-1} \text{ K}^{-1}$) is substantially smaller on a weight basis than that observed in protein denaturation ($\Delta C_{p,obs}^\circ \approx 0.09$ to $0.15 \text{ cal g}^{-1} \text{ K}^{-1}$),^{50,53} it is reasonable to expect that changes in conformation of the protein are the primary origin of the effect.

For a process with a large $\Delta C_{p,obs}^\circ$, the contribution of the hydrophobic effect (ΔG_{hyd}°) to ΔG_{obs}° can be estimated using model compound transfer data. Baldwin⁵⁴ utilized thermodynamic data for the transfer of liquid hydrocarbons to water to estimate the contribution of the hydrophobic effect to the driving force for protein folding. Ha *et al.*¹⁴ applied this approach to protein-DNA interactions. In the physiological temperature range, they demonstrated that

$$\Delta G_{hyd}^\circ \approx (8 \pm 1) \times 10^1 \Delta C_{p,obs}^\circ \quad (13)$$

which may be used to estimate ΔG_{hyd}° for a protein-DNA interaction from measurements of $\Delta C_{p,obs}^\circ$.

For the interaction of *lac* repressor with the symmetric *lac* operator DNA site, $\Delta C_{p,obs}^\circ = -1.3(\pm 0.3) \text{ kcal mol}^{-1} \text{ K}^{-1}$, leading to an estimate of ΔG_{hyd}° of approximately $-10^2 \text{ kcal mol}^{-1}$. ΔG_{hyd}° is much larger in magnitude than ΔG_{obs}° , which is $\sim -15 \text{ kcal mol}^{-1}$ at the solution conditions examined. Therefore Ha *et al.*¹⁴ proposed that the hydrophobic effect drives the thermodynamically unfavorable processes involved in forming the site-specific protein-DNA complex. These include the translational and rotational restrictions on the reactants in complexation^{55,56} and various costs associated with any intrinsic lack of steric and/or functional group

⁵³ P. L. Privalov, O. B. Ptitsyn, and T. M. Birshstein, *Biopolymers* **8**, 559 (1969).

⁵⁴ R. L. Baldwin, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 8069 (1986).

⁵⁵ M. I. Page and W. P. Jenks, *Proc. Natl. Acad. Sci. U.S.A.* **68**, 1678 (1971).

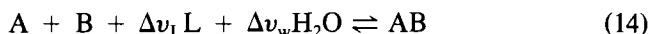
⁵⁶ C. Chothia and J. Janin, *Nature (London)* **256**, 705 (1975).

complementarity between the interacting surfaces, which may require deformations of the protein and/or DNA surfaces or net loss of favorable contacts on removal from water in the process of complexation.⁸⁻¹⁰

Effect of Solutes (Ligands) on K_{obs}

An equilibrium binding constant defined in terms of the equilibrium activities of all individual chemical species participating in the reaction [cf. K_{eq} in Eq. (7)] is a function of temperature and pressure only. However, the observed equilibrium constant K_{obs} [Eq. (4) or (5)] is formulated in terms of concentrations of reactant and/or product macromolecules, without regard to their extent of interaction with solutes or solvent. This experimentally convenient definition of K_{obs} neglects the possible participation of site-bound solutes (e.g., H^+ , anions) in the binding process (and/or in coupled macromolecular equilibria) and neglects the effects of extreme macromolecular thermodynamic nonideality (e.g., the interaction of cations with nucleic acid polyanions). Therefore K_{obs} is generally a strong function of the corresponding compositional variables. In this and the following section, we examine effects on K_{obs} of the concentrations of uncharged solutes, protons, and electrolyte ions.

General Analysis of Effects of Solute (Ligand) Concentration on K_{obs} . When an uncharged solute acts as a ligand (L) and binds to different extents to complexed (AB) and free states of uncharged biopolymers (A, B), changes in concentration of the ligand will affect the equilibrium extent of association of A and B. The effect of such a ligand may be represented by the mass action of equation for the process:



If K_{obs} is defined in terms of the equilibrium concentrations of A, B, and AB, then it is readily shown that^{57,58}

$$\left(\frac{\partial \ln K_{obs}}{\partial \ln [L]} \right)_{T,P} = \Delta v_L - \frac{[L]}{[H_2O]} \Delta v_w \quad (15)$$

where v_L and v_w are the average number of ligands and water molecules bound per biopolymer. [For both ligand ($x = L$) and water ($x = w$), $\Delta v_x \equiv v_{x,AB} - v_{x,A} - v_{x,B}$.] At low concentrations of the ligand ($<0.1 M$), the term $\Delta v_w [L]/[H_2O]$ is usually negligible.

If the N sites for a ligand on a biopolymer (A, B, or AB) are independent and identical:

⁵⁷ J. Wyman, Jr., *Adv. Protein Chem.* **19**, 223 (1964).

⁵⁸ C. Tanford, *J. Mol. Biol.* **39**, 539 (1969).

$$v_L = \frac{NK_L[L]}{1 + K_L[L]} \quad (16)$$

where K_L is the intrinsic binding constant for association of L with an individual site. In this simplest possible situation, the concentration of ligand L will affect K_{obs} [cf. Eq. (15)] if K_L and/or N changes on complexation, since either of these will lead to a nonzero Δv_L . In general Δv_L is a function of ligand concentration and hence a plot of $\ln K_{\text{obs}}$ vs $\ln[L]$ will be nonlinear over an extended range of $[L]$, with Δv_L obtained from the tangent to the curve. From Δv_L , information regarding changes in the intrinsic association constant (K_L) and/or in the number of ligand-binding sites (N) that accompany the process $A + B \rightarrow AB$ can be obtained. If binding of L is weak, then Δv_L is proportional to $[L]$ and a plot of $\ln K_{\text{obs}}$ vs $[L]$ may be the more suitable means of investigating changes in K_L and/or N .

Protons as Ligands: Differential Protonation Accompanying Binding.

The dependence of K_{obs} on pH in the presence of excess electrolyte has been studied for a few site-specific protein–DNA interactions. Although in this case both the ligand and the macromolecules are charged, effects of pH on K_{obs} in excess salt may be analyzed using Eq. (15). Because $[H^+] \ll [H_2O]$, the net proton stoichiometry Δv_{H^+} is directly obtained from the pH dependence of K_{obs} :

$$\left(\frac{\partial \ln K_{\text{obs}}}{\partial \ln[H^+]} \right)_{T,P} = - \left(\frac{\partial \log K_{\text{obs}}}{\partial \text{pH}} \right)_{T,P} = \Delta v_{H^+} \quad (17)$$

deHaseth *et al.*⁵⁹ and Lohman *et al.*⁶⁰ discuss two extreme scenarios to interpret Δv_{H^+} : (1) the requirement for protonation (possibly applicable to nonspecific binding of *lac* repressor) and (2) the shifted titration curve [applicable to (Lys)₅–DNA interactions]. If r independent and identical sites on the protein must be protonated as a prerequisite for binding (case 1), then:

$$\Delta v_{H^+} = \frac{r}{1 + K_H[H^+]} \quad (18)$$

where K_H is the intrinsic binding constant for a proton to an individual site. If protonation is required for complex formation, then at high pH where $K_H[H^+] \ll 1$, $\log K_{\text{obs}}$ is predicted to decrease linearly with increasing pH, with a slope of $-r$ (i.e., $\Delta v_{H^+} = r$). At sufficiently low pH, K_{obs} is insensitive to pH, because all ionizable groups are fully protonated

⁵⁹ P. L. deHaseth, T. M. Lohman, and M. T. Record, Jr., *Biochemistry* **16**, 4783 (1977).

⁶⁰ T. M. Lohman, P. L. deHaseth, and M. T. Record, Jr., *Biochemistry* **19**, 3522 (1980).

($\Delta v_{H^+} = 0$). At intermediate pH, Δv_{H^+} varies with pH, equaling $r/2$ at the pK_a of the protonating group.

As an example of the pH dependence of protein–DNA interactions, $\log K_{obs}$ for nonspecific binding of *lac* repressor to DNA at 0.13 M NaCl is found to be a linear function of pH over the pH range 7.7–8.4.⁵⁹ From the slope of this line [cf. Eq. (17)], Δv_{H^+} is determined to be 2.1 ± 0.2 . The specific binding constant for formation of a repressor–operator complex is much less pH dependent (corresponding to $\Delta v_{H^+} < 1$ over the pH range 7.1–8.0, and $\Delta v_{H^+} \approx 0$ in the pH range 8.0–8.4).¹⁵ These results are consistent with a requirement for protonation at two sites ($pK < 7$) in nonspecific interactions, but a smaller requirement (if any) for protonation in the formation of the specific repressor–operator interaction.

Effects of Concentration and Nature of Electrolyte on K_{obs} of Protein–DNA Interactions

General Analysis of Effects of [MX] on K_{obs} . With exception, equilibrium constants K_{obs} for noncooperative site-specific and nonspecific association of proteins with double helical DNA are found to decrease *strongly* with increasing salt concentration [MX]. The dependence of K_{obs} on [MX] is significantly reduced by the presence of even low concentrations of competitive divalent (e.g., Mg^{2+}) or trivalent (e.g., spermidine) cations, and may of course be difficult to detect if the extent of binding is close to saturation. For both site-specific and nonspecific binding of proteins to double-helical DNA, plots of $\ln K_{obs}$ vs $\ln[MX]$ are typically linear over the experimentally accessible range, in the absence of competitive equilibria involving the protein, DNA, or the electrolyte. Many studies of site-specific binding of *lac* repressor and RNA polymerase have been carried out with mixtures of univalent and divalent cations (M^+ , Mg^{2+}). At a fixed $[Mg^{2+}]$, K_{obs} is found to be relatively independent of $[M^+]$ at low $[M^+]$. With increasing $[M^+]$, the effect of $[M^+]$ on K_{obs} increases greatly. Only at very high $[M^+]$ does $\ln K_{obs}$ decrease linearly with increasing $\ln[M^+]$. The nonlinearity of $\ln K_{obs}$ as a function of $\ln[MX]$ results from the competitive association of Mg^{2+} with DNA, the extent of which is itself a strong function of [MX].^{9,26,27,60–62}

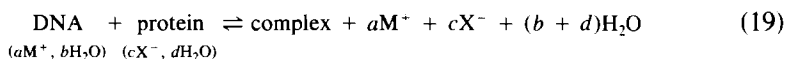
In the absence of Mg^{2+} or other competitors, the slopes (SK_{obs}) of plots of $\ln K_{obs}$ vs $\ln[MX]$ (where $SK_{obs} \equiv (\partial \ln K_{obs} / \partial \ln[MX])_{T,p}$), are typically large in magnitude, negative, and independent of salt concentration. For the various site-specific and nonspecific interactions of *lac* repressor and RNA polymerase with DNA, $-5 \geq SK_{obs} \geq -20$ (cf. reviews in Refs. 9, 10,

⁶¹ M. T. Record, Jr., P. L. deHaseth, and T. M. Lohman, *Biochemistry* **16**, 4791 (1977).

⁶² M. T. Record, Jr., C. F. Anderson, and T. M. Lohman, *Q. Rev. Biophys.* **11**, 102 (1978).

and 27). Consequently the salt acts as if it participated stoichiometrically as a product of the process of complex formation, with a large and relatively constant stoichiometric coefficient [cf. Eq. (15)]. Analysis of this behavior demonstrates that it results primarily from the Coulombic interactions of cations with the nucleic acid polyanion, the extent of which depends primarily on the axial charge density of the nucleic acid,⁶³⁻⁶⁵ and is therefore reduced in processes that reduce the axial charge density.^{9,59,62,64-67} This polyelectrolyte behavior, which is fundamentally different from site binding at both molecular and thermodynamic levels, nevertheless may be summarized as the release of a stoichiometric number of cations from the DNA when its axial charge density is reduced by binding of an oligocation (L^{z+}) or a protein with a positively charged DNA-binding domain. The large stoichiometry of cation release is usually the primary determinant of SK_{obs} . This contribution to SK_{obs} is denoted the "polyelectrolyte effect".^{9,10,59,66}

More generally, extension of Eq. (15) to interpret effects of salt concentration on K_{obs} of protein-DNA interactions also requires consideration of anion-protein interactions and differential hydration effects. In forming the protein-DNA complex, site-bound or preferentially accumulated anions and water as well as cations may be displaced from the vicinity of the interacting surfaces. Neglecting differential protonation, a general stoichiometric description of protein-DNA interactions in an aqueous univalent (MX) salt solution is^{9,62}



The apparent stoichiometric coefficients a , b , c , and d are related to differences in the preferential interaction coefficients of the electrolyte ions and of H_2O with DNA, protein, and the protein-DNA complex. The dependence of K_{obs} on salt concentration is given by^{9,62}

$$SK_{\text{obs}} \equiv \left(\frac{\partial \ln K_{\text{obs}}}{\partial \ln [MX]} \right)_{T,P} \approx - \left[a + c - (b + d) \frac{2[MX]}{[H_2O]} \right] \quad (20)$$

The use of salt concentration rather than mean ionic activity as the independent variable has been justified.^{62,66} Note that Eq. (20) differs in a

⁶³ G. S. Manning, *J. Chem. Phys.* **51**, 924 (1969).

⁶⁴ C. F. Anderson and M. T. Record, Jr., *Annu. Rev. Phys. Chem.* **33**, 191 (1982).

⁶⁵ M. T. Record, Jr., M. Olmsted, and C. F. Anderson, in "Theoretical Biochemistry and Molecular Biophysics" (D. L. Beveridge and R. Lavery, eds.), p. 285. Adenine Press, New York, 1990.

⁶⁶ M. T. Record, Jr., T. M. Lohman, and P. L. deHaseth, *J. Mol. Biol.* **107**, 145 (1976).

⁶⁷ G. S. Manning, *Q. Rev. Biophys.* **11**, 179 (1978).

nontrivial manner from Eq. (15) in that the *sum* of the stoichiometries of cations and anions is determined from the experimentally measurable value of SK_{obs} .

Protein–DNA interactions are typically investigated at low to moderate salt concentration ($[MX] = 0.05\text{--}0.20\text{ M}$), where the magnitude of $2[MX]/[H_2O]$ is small and $-SK_{\text{obs}}$ is approximately equal to the sum of the stoichiometric coefficients of the electrolyte ions ($a + c$), where typically $a \gg c$ for interactions of proteins with double-stranded DNA. At higher salt concentration, where the contribution of water release to SK_{obs} is no longer negligible, SK_{obs} is expected to decrease in magnitude (i.e., become less negative) as a result of the linkage of water and salt activities required by the Gibbs–Duhem equation. (An increase in salt concentration reduces the water activity and makes the displacement of water to the bulk solution more favorable than it was at lower salt concentration.) Careful measurement and analysis of the behavior of SK_{obs} at high salt concentration in principle provides a useful route to determining the net stoichiometry of water participation ($b + d$) in the binding reaction. These experiments are difficult to perform because the binding constant K_{obs} is typically low, and are difficult to interpret because of uncertainties regarding effects of high salt concentration on the cation stoichiometry (a) and the anion stoichiometry (c).⁶⁸

The extent of binding of some proteins to single-stranded nucleic acids has been shown to increase with increasing salt concentration. These apparent exceptions to the general behavior discussed in this section must result from one or more of the additional roles of salt concentration, including stabilizing specific conformations of the nucleic acid, favoring release of water, and/or in driving cooperative binding. For example, the extent of interaction of single-stranded oligonucleotides with site II of the *E. coli* ribosomal protein S1 increases with increasing salt concentration above 0.1 M NaCl ,⁶⁹ and the extent of interaction of single-stranded oligonucleotides with the second site of *E. coli* single strand-binding (SSB) tetramers increases with increasing salt concentration below 0.1 M NaCl .⁷⁰ In the latter case, the origin of the effect is a decrease in anticooperativity of binding to the second site that occurs with increasing salt concentration. The intrinsic binding constant for interaction of the oligonucleotide with this site decreases with increasing salt concentration, but the salt concentration dependence of K_{obs} is dominated by the anticooperativity effect.

⁶⁸ J.-H. Ha, Ph.D. Thesis, University of Wisconsin-Madison (1990).

⁶⁹ D. E. Draper and P. H. von Hippel, *J. Mol. Biol.* **122**, 339 (1978).

⁷⁰ W. Bujalowski and T. M. Lohman, *J. Mol. Biol.* **207**, 269 (1989).

Cation-DNA Interactions and Effect of Electrolyte Concentration on Binding of Oligocationic Ligands to DNA (Polyelectrolyte Effect). Because DNA is a polyanion of high axial charge density, the extent of interaction of proteins with DNA is typically more sensitive to salt concentration than to the concentration of any other solution component, including the protein concentration! With two negatively charged phosphate groups per 3.4 Å, the axial charge density of B-DNA is so high as to cause the local accumulation of a high concentration of cations and the exclusion of anions from the vicinity of DNA.^{63-65,71} At low salt concentration, these ion concentration gradients are predicted by all polyelectrolyte theories to be thermodynamically equivalent to the neutralization of approximately 88% of the structural charge on B-DNA.⁶⁴⁻⁶⁶ As a consequence, B-DNA behaves like the weak electrolyte $(M_{0.88}^+ \text{DNAP}^-)_n$, where M^+ is a univalent cation and DNAP^- is a phosphate charge of DNA.^{9,64-66} As judged by cation nuclear magnetic resonance (NMR) experiments,⁷² the molecular extent of neutralization of DNA phosphates by associated cations is relatively independent of the bulk univalent salt concentration ($[\text{MX}]$). Theoretical analyses predict that the thermodynamic degree of neutralization decreases with increasing salt concentration,^{65,73} and with a decrease in the axial structural charge density of DNA.^{62,64-66}

Binding of an oligocation (L^{z+}) to DNA in a univalent salt solution is a cation-exchange process in which the DNA structural charge density is reduced and cations (M^+) are released from the DNA to the bulk solution. Experimental determinations of the stoichiometry of univalent cation release [cf. Eqs. (19) and (20)] on binding of a variety of oligocations L^{z+} to double-helical DNA and RNA [including Mg^{2+} , polyamines, $(\text{Lys})_N$, etc.] indicate that $a = (0.88 \pm 0.05)z$, independent to a first approximation of the bulk concentration of MX.^{60,66,74-76} The anion stoichiometry in these cases is apparently negligible ($c \approx 0$). For the interaction of oligolysines with single-stranded polynucleotides, $a = (0.71 \pm 0.03)z$, as anticipated from the lower axial charge density of these polyanions, and the anion

⁷¹ C. F. Anderson and M. T. Record, Jr., *Annu. Rev. Biophys. Biophys. Chem.* **19**, 423 (1990).

⁷² S. Padmanabhan, M. Paulsen, C. F. Anderson, and M. T. Record, Jr., in "Monovalent Cations in Biological Systems" (C. Pasternak, ed.), p. 321. CRC Press, Boca Raton, Florida, 1990.

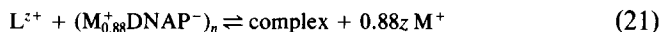
⁷³ P. Mills, C. F. Anderson, and M. T. Record, Jr., *J. Phys. Chem.* **90**, 6541 (1986).

⁷⁴ W. H. Braunlin, T. J. Strick, and M. T. Record, Jr., *Biopolymers* **21**, 1301 (1982).

⁷⁵ W. H. Braunlin, C. F. Anderson, and M. T. Record, Jr., *Biochemistry* **26**, 7724 (1987).

⁷⁶ G. E. Plum and V. A. Bloomfield, *Biopolymers* **29**, 13 (1990).

stoichiometry is negligibly small ($c = 0$).⁷⁷ Association of an oligocation L^{z+} with helical B-DNA may be represented by the cation-exchange equation:



Consequently ligand–DNA association is favored by low salt concentration, and K_{obs} increases strongly with decreasing salt concentration.

At low salt (MX) concentration, in the absence of specific anion effects, the stoichiometry of cation release (0.88z) on binding of L^{z+} to the helical B-DNA polyanion is obtained from the derivative [cf. Eq. (20)]:

$$SK_{\text{obs}} \equiv \left(\frac{\partial \ln K_{\text{obs}}}{\partial \ln [MX]} \right)_{T,P} = -0.88z \quad (22)$$

For association of an oligocation (L^{z+}) with the B-DNA polyanion in excess univalent salt MX,

$$\ln K_{\text{obs}} = \ln K^{\circ} - 0.88z \ln [MX] \quad (23)$$

and

$$\Delta G_{\text{obs}}^{\circ} = \Delta G_0^{\circ} + 0.88zRT \ln [MX] \quad (24)$$

where $\ln K^{\circ}$ and the corresponding standard free energy change ΔG_0° are extrapolated values at the 1 M MX pseudostandard state. Since the extrapolated K° for binding of oligocations [e.g., (Lys)_N, polyamines] to nucleic acid polyanions is within an order of magnitude with unity, we conclude that the intrinsic preference of DNA for these oligocations (relative to M^{+}) is small and that these binding interactions are driven at low salt concentrations by M^{+} release.^{9,60,62,66,67,77} The resulting favorable free energy of dilution (an entropic effect) from counterion displacement into the bulk solution is called the polyelectrolyte effect.^{9,10,59,66} Therefore, the contribution of the polyelectrolyte effect to the stability of an L^{z+} –B-DNA complex in an univalent salt solution ($\Delta G_{\text{PE}}^{\circ}$) may be estimated from the relationship:^{9,10,66}

$$\Delta G_{\text{PE}}^{\circ} = 0.88zRT \ln [MX] \quad (25)$$

In general, the interactions of oligocations [including polyamines with 2–4 positive charges, Mg^{2+} , $Co(NH_3)_6^{3+}$, and oligolysines with 3–10 positive charges] with nucleic acids behave as pure cation-exchange processes, without evidence of significant anion effects at low salt concentration.^{60,66,74–77}

⁷⁷ D. P. Mascotti and T. M. Lohman, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 3142 (1990).

If the binding of L^{z+} to DNA is investigated in a mixed salt buffer containing both univalent and di- or trivalent cations [e.g., binding of (Lys)₅ in an NaCl/MgCl₂ mixture], interpretation of the apparent stoichiometric coefficient a of univalent cation release is complicated by competition between the univalent and di- or trivalent cations.^{60,61,74} These cations compete more effectively than monovalent cations for local accumulation near the surface of the DNA. Even low concentrations of a multivalent cation [e.g., Mg²⁺, spermidine (3+)] will reduce both K_{obs} and $|SK_{\text{obs}}|$ for the binding of L^{z+} to DNA, especially at low salt ([MX]) concentration. Effects of Mg²⁺ concentration on K_{obs} and $|SK_{\text{obs}}|$ for the interaction of pentyllysine with B-DNA decrease with increasing salt concentration.⁶⁰ This behavior can be accounted for with a simple competitive binding-polyelectrolyte model.^{61,62}

Anion-Protein Interactions Involving Site-Binding or Preferential Interactions, Fundamentally Different from Interactions of Cations with Nucleic Acid Polyanions. Both the large number and high axial density of negative charges on the DNA polyanion are necessary to give rise to its characteristic polyelectrolyte behavior.^{65,78} On the other hand, ligands like L^{z+} and the domains of proteins that interact with DNA are typically relatively short oligocations with a much smaller number of charges (and possibly a lower axial charge density) than that of polyanionic DNA. The thermodynamic extent of counterion association with a charged group on an oligocation or oligoanion is predicted to be far smaller than that characteristic of a polyion of the same axial charge density.⁷⁸ Experimental investigation of the interactions of oligolysines with single-stranded polynucleotides⁷⁷ demonstrates that Coulombic interactions of anions with oligocations (L^{z+} ; $z \leq 10$) make only a minor contribution to the salt concentration dependence of K_{obs} for L^{z+} -DNA interactions [$c \approx 0$ in Eq. (20)].

For proteins, other relevant modes of interaction of anions with the DNA-binding domain may occur and lead to an effect on the anion stoichiometry c . Anions may bind to sites on the protein, generally with relatively small binding constants ($\leq 10^2 M^{-1}$).⁶² Evidence for a more diffuse type of anion-protein interaction also exists in which a variety of anions are excluded (relative to water) from the vicinity of protein surfaces.^{79,80} Both of these interactions become negligible at low salt concentrations. Although both weak binding and exclusion of anions may occur, effects of

⁷⁸ M. Olmsted, C. F. Anderson, and M. T. Record, Jr., *Proc. Natl. Acad. Sci. U.S.A.* **86**, 7766 (1989).

⁷⁹ T. Arakawa and S. N. Timasheff, *Biochemistry* **21**, 6545 (1982).

⁸⁰ T. Arakawa and S. N. Timasheff, *J. Biol. Chem.* **259**, 4979 (1985).

the nature of the anion on protein–DNA interactions are most simply modeled as resulting from competitive displacement of the anion from weak sites on the protein.^{68,81–83} This competitive mode of action of anions will decrease K_{obs} and increase $|SK_{\text{obs}}|$ of the protein–DNA interaction relative to a situation in which no anions are displaced and SK_{obs} is determined only by the stoichiometry of cation release.⁸³ All the effects of anions are predicted to become more pronounced at higher anion concentration, and for anions that are at the strong-interacting end of the Hofmeister series.^{83–85}

Effects of Salt Concentration on lac Repressor–Operator Interaction. The large effects of salt concentration on K_{obs} for interactions of *lac* repressor with restriction fragments containing either the wild-type primary *lac* operator (O^+) or a mutant *lac* operator sequence (O^{c666} , O^{c640}) are shown in Fig. 3. For O^+ and O^{c666} in NaCl, $-SK_{\text{obs}} = 5 \pm 1$, whereas for O^{c640} , $-SK_{\text{obs}} = 10 \pm 1$.¹³ For the interaction of *lac* repressor with nonoperator DNA, $-SK_{\text{obs}} = 11 \pm 1$.⁸⁶ Because these studies were conducted at relatively low salt concentration (0.1–0.2 M), differential hydration effects can be neglected so that $-SK_{\text{obs}}$ is equal to the sum of the stoichiometries of cation and anion release [$-SK_{\text{obs}} = a + c$; Eq. (20)].

Before proceeding to interpret these values of SK_{obs} , it is important to emphasize their practical consequences. First, since the sum of the stoichiometric coefficients of the cation and anion ($a + c = 5$ –10) is far larger than that of *lac* repressor (with a stoichiometry of unity), errors in salt concentration (although typically much smaller than errors in protein concentration) are magnified and may contribute as much or more as uncertainties in protein concentration to the uncertainty in K_{obs} . Second, it is possible to obtain virtually any desired value of K_{obs} (if competing processes do not interfere) by appropriate choice of salt concentration. Therefore, values of K_{obs} (and of standard thermodynamic functions derived from K_{obs}) are only meaningful if all ion concentrations are accurately specified. Comparisons of values of K_{obs} for different DNA sequences or different systems generally must be made at the same concentrations of all ionic species. Extrapolations to a different salt concentration will be

⁸¹ S. Leirmo, C. Harrison, D. S. Cayley, R. R. Burgess, and M. T. Record, Jr., *Biochemistry* **26**, 2095 (1987).

⁸² L. B. Overman, W. Bujalowski, and T. M. Lohman, *Biochemistry* **27**, 456 (1988).

⁸³ S. Leirmo and M. T. Record, Jr., in "Nucleic Acids and Molecular Biology" (D. M. J. Lilley and F. Eckstein, eds.), Vol. 4, p. 123. Springer-Verlag, Berlin, 1990.

⁸⁴ P. H. von Hippel and T. Schleich, *Acc. Chem. Res.* **2**, 257 (1969).

⁸⁵ K. D. Collins and M. W. Washabaugh, *Q. Rev. Biophys.* **18**, 323 (1985).

⁸⁶ T. M. Lohman, C. G. Wensley, J. Cina, R. R. Burgess, and M. T. Record, Jr., *Biochemistry* **19**, 3516 (1980).

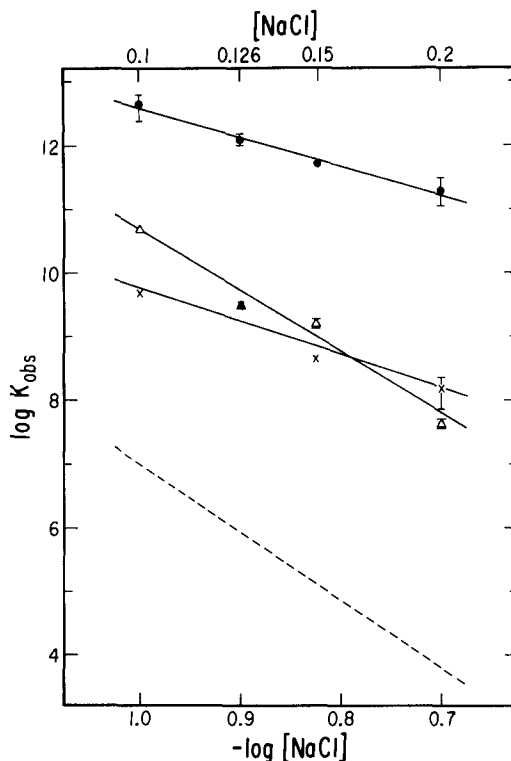


FIG. 3. Effects of NaCl concentration on the binding of *lac* repressor to specific and nonspecific DNA sequences. The logarithm of the specific binding constant $K_{\text{obs}}^{\text{RS}}$ at (23°) is plotted vs the negative logarithm of the NaCl concentration for (●) O⁺, (△) O^c640, and (X) O^c666. [Data from M. C. Mossing, and M. T. Record, Jr., *J. Mol. Biol.* **186**, 295 (1985).] Dashed line depicts the dependence of the logarithm of the nonspecific binding constant $K_{\text{obs}}^{\text{RD}}$ on the negative logarithm of the NaCl concentration: $\log K_{\text{obs}}^{\text{RD}} = -(10.7 \pm 1.0)\log[\text{Na}^+] - (3.7 \pm 1.0)$. [From T. M. Lohman, C. G. Wensley, J. Cina, R. R. Burgess, and M. T. Record, Jr., *Biochemistry* **19**, 3516 (1980).]

nonlinear (even on a log–log plot) if mixed salts are present (e.g., MgCl₂ and NaCl), and generally will be inaccurate. “Ionic strength” is demonstrably *not* a useful composite ionic variable for protein–DNA interactions, either in theory or in practice, and should not be used in these analyses.^{9,46,61,62,87,88}

Analysis and interpretation of cation effects on K_{obs} : In order to decom-

⁸⁷ M. T. Record, Jr., S. J. Mazur, P. Melancon, J.-H. Roe., S. L. Shaner, and L. Unger. *Annu. Rev. Biochem.* **50**, 997 (1981).

⁸⁸ T. M. Lohman, *Crit. Rev. Biochem.* **19**, 191 (1985).

pose values of SK_{obs} in univalent salt (MX) solutions into contributions from the individual cation and anion stoichiometric coefficients a and c , it is useful to compare these values of SK_{obs} with those determined in the pure salt of a divalent cation and the same univalent anion X^- (e.g., MgX_2). A much lower range of MgX_2 concentrations yields the same experimentally accessible range of values of K_{obs} as investigated in MX. At the lower concentration of X^- in MgX_2 , the anion stoichiometry c should be reduced to an extent consistent with a mass action model. Replacement of M^+ with Mg^{2+} is expected to reduce the cation stoichiometry to approximately one-half its value in M^+ (unless Mg^{2+} plays an additional specific role in complex formation).⁵⁹ For nonspecific binding of *lac* repressor in $MgCl_2$, $-SK_{\text{obs}} = 6 \pm 1$.⁵⁹ This reduction in SK_{obs} is consistent with the interpretation $-SK_{\text{obs}} = a$ and $c = 0$ for nonspecific binding of *lac* repressor. Therefore it appears that this process is well described by a pure cation-exchange model in Cl^- salts. For site-specific binding of *lac* repressor to the *lac* operon (containing two pseudooperators and the primary *lac* operator), Barkley *et al.*¹⁵ observed that SK_{obs} was approximately halved when investigated in $MgCl_2$ as compared to $NaCl$, again consistent with an interpretation involving primarily cation release.

From a survey of structures of prokaryotic protein-DNA complexes solved to high resolution, Harrison and Aggarwal¹ conclude that "hydrogen bonds from positively charged side chains to DNA phosphates appear with only modest frequency. Coulombic interactions from less strongly anchored arginines and lysines do appear to be important, however, since in each structure a number of such residues lie near the DNA backbone." This observation suggests that neutralization of DNA phosphates by these proteins is generally not accomplished by classical salt bridges (ion pairs). However, the often large stoichiometry of cation displacement from DNA that accompanies binding of most proteins clearly indicates that the charge density of DNA is reduced on complexation. Presumably this is the consequence of the above-mentioned Coulombic interactions between DNA phosphates and positively charged groups on the protein, but the molecular details of these interactions in solution are unknown. Close juxtapositions of DNA phosphates and arginine and lysine residues also occur in the crystal structure of the eukaryotic *engrailed* homeo domain-DNA complex of Pabo and collaborators.⁸⁹

If it is assumed that SK_{obs} for binding of *lac* repressor to these operator sequences is primarily an expression of the polyelectrolyte effect, then the thermodynamic contribution of cation release ($\Delta G_{\text{PE}}^\circ$) to the observed

⁸⁹ C. R. Kissinger, B. Liu, E. Martin-Bianco, T. B. Kornberg, and C. O. Pabo, *Cell (Cambridge, Mass.)* **63**, 579 (1990).

net standard free energy of association ($\Delta G_{\text{obs}}^{\circ}$) can be estimated from Eq. (25). At $[\text{NaCl}] = 0.2 \text{ M}$, the polyelectrolyte effect contributes approximately -5 kcal to $\Delta G_{\text{obs}}^{\circ}$, which is approximately -15 kcal at this salt concentration.¹³ The contribution of $\Delta G_{\text{PE}}^{\circ}$ to $\Delta G_{\text{obs}}^{\circ}$ varies widely for different specific and nonspecific protein–DNA interactions. For formation of the specific open complex between *E. coli* RNA polymerase $E\sigma^{70}$ and the λP_R promoter, the estimated $\Delta G_{\text{PE}}^{\circ}$ is actually more favorable than $\Delta G_{\text{obs}}^{\circ}$ at 0.2 M NaCl .^{9,10}

Anion effects on K_{obs} and SK_{obs} : For both nonspecific and specific binding of *lac* repressor to the wild-type *lac* operon, replacement of Cl^- by acetate (Ac^-) leads to an approximately 30-fold increase in K_{obs} at constant salt concentration without a significant ($<10\text{--}15\%$) change in SK_{obs} .^{15,59} Detailed investigations of anion effects on K_{obs} and SK_{obs} for the interaction of *lac* repressor with the *lac* operon on $\lambda\text{plac DNA}$ ¹⁵ and with an isolated synthetic (strong-binding) *lac* operator⁶⁸ have been performed. For the interaction with $\lambda\text{plac DNA}$ at a fixed Na^+ concentration of 0.13 M , K_{obs} decreases by a factor of approximately 10^4 in the order $\text{Ac}^- \geq \text{F}^- > \text{Cl}^- > \text{Br}^- > \text{NO}_3^- > \text{SCN}^- > \text{I}^-$.¹⁵ Comparisons of K_{obs} for the interaction of *lac* repressor with linear plasmid DNA carrying the symmetric strong-binding *lac* operator (O^{sym}) in K^+ salts [Cl^- , Ac^- , and glutamate (Glu^-)] demonstrate that K_{obs} decreases in the order $\text{Glu}^- > \text{Ac}^- > \text{Cl}^-$, and that the magnitude of the effect on K_{obs} due to the choice of anion increases with increasing salt concentration above 0.1 M .⁶⁸ Even though the nature of the anion has a detectable effect on SK_{obs} , the stoichiometry of cation release is in all cases the primary determinant of SK_{obs} .

No unique interpretation of these large anion effects is possible, primarily because the reference point of a noninteracting anion is not known. If it is assumed that all the effects are due to competitive anion binding, then perhaps Glu^- is a noninteracting anion [$c = 0$ in Eq. (20)]. Alternatively, it is possible that Cl^- does not interact with the DNA-binding site of *lac* repressor, as inferred from comparison of SK_{obs} in NaCl and MgCl_2 . In this scenario, the effects of replacing Cl^- by Glu^- must be attributed to preferential hydration in Glu^- (i.e., preferential exclusion of Glu^-)⁸⁰ and/or to anomalous nonideality of glutamate salts.⁹⁰ Interpretation of the data as purely a preferential hydration effect leads to the deduction that as many as 200 mol of water may be displaced per mol of complex formed.⁶⁸

In summary, effects of the nature of the anion on this (and other) protein–DNA interactions are *very* large, but their origin remains difficult

⁹⁰ D. Mascotti and T. M. Lohman, personal communication (1990).

to diagnose. These effects must be understood in order to refine the estimate of the contribution of the polyelectrolyte effect to stability and specificity, and to begin to ascertain the stability and specificity of protein–DNA complexes in the *in vivo* environment [a concentrated polyelectrolyte solution where the total concentration of electrolyte cations (primarily K^+ in *E. coli*) greatly exceeds that of electrolyte anions (primarily glutamate in *E. coli*) as a result of the high concentration of nucleic acid (RNA, DNA) polyanions].^{91,92}

Kinetics and Mechanisms of Site-Specific Protein–DNA Interactions

Introduction

Relating Kinetics to Mechanism. Kinetic studies are required to determine the “mechanism,” i.e., the path or sequence of elementary kinetic steps, by which a protein binds to a specific site on DNA to form a complex. From kinetic studies, information is obtained regarding the minimum number of significant steps in the mechanism, the number and (in some cases) thermodynamic or kinetic characteristics of kinetically significant intermediates, and the nature of the activation barriers between intermediates. For any reaction, these are difficult questions to address. Site-specific protein–DNA interactions are certainly no exception.

Rate Laws of Site-Specific Protein–DNA Interactions. Association typically exhibits second order (or pseudofirst order) kinetics: The kinetics of association of several proteins (including *lac* repressor and RNA polymerase of $E\sigma^{70}$) with specific DNA sites are observed to follow a second order rate law, with an experimentally determined second order rate constant k_{assoc} ($M^{-1} \text{ sec}^{-1}$). (Our notation for rate constants is internally consistent in this section, although not always consistent with the notation in the cited work.) For the overall process $R + S \rightarrow RS$, the kinetics of association are found to conform to the rate equation:

$$\frac{d[RS]}{dt} = k_{\text{assoc}}(R_T - [RS])(S_T - [RS]) \quad (26)$$

Note that it is by no means obvious *a priori* that this rate law must be applicable, except for the situation in which association occurs in a single elementary (i.e., diffusion–collision limited) bimolecular reaction step, and in the absence of any competing interactions that might significantly reduce the concentrations of the free protein and the free DNA site.

⁹¹ B. Richey, S. Cayley, M. Mossing, C. Kolka, C. F. Anderson, T. C. Farrar, and M. T. Record, Jr., *J. Biol. Chem.* **262**, 7157 (1987).

⁹² D. S. Cayley, B. A. Lewis, H. J. Guttman, and M. T. Record, Jr., *J. Mol. Biol.* in press (1991).

The mechanism of association of a protein with a specific DNA site embedded in a molecule containing nonspecific DNA sites will involve some or all of the following general classes of steps:

1. Changes in the state of aggregation of the protein, conformational changes in the protein or DNA, and/or nonspecific binding of the protein to DNA, prior to the elementary bimolecular association step
2. An elementary bimolecular step in which an initial complex at the specific DNA site is formed at the diffusion–collision rate or at a rate that is slower than the estimated maximum diffusion–collision rate because of orientation (entropic) effects
- 2'. An elementary bimolecular step in which an initial complex is formed at a distant nonspecific site at the (orientation-corrected) diffusion–collision rate, followed by a mechanistically distinct diffusional process in the domain of the DNA molecule to locate the specific site
3. Local and/or global conformational changes that occur in the initial complex at the specific site subsequent to the elementary bimolecular step, and result in formation of the functional specific complex

In the following sections, mechanisms involving these classes of steps are introduced, and interpretations of the experimentally determined second order association rate constant k_{assoc} in terms of these potential mechanisms are discussed. In general, the approach to the analysis of second order kinetic data in terms of mechanisms is the following: The rate constant k_{assoc} [Eq. (26)] is measured as a function of temperature and solution conditions (especially salt concentration, pH, and, where appropriate, solvent viscosity; DNA length is also an important variable if the process is facilitated as in class 2' above). If the magnitude of k_{assoc} is appropriate for a diffusion–collision reaction and k_{assoc} is only weakly dependent on temperature and salt concentration, then the interaction is probably diffusion limited. If k_{assoc} is only weakly dependent on temperature and salt concentration but its magnitude is significantly less than the diffusion limit, then a diffusion–collision mechanism with severe orientation restrictions may be appropriate. If k_{assoc} is smaller than the diffusion limit and strongly dependent on temperature and/or salt concentration (especially if k_{assoc} decreases with increasing temperature) then intermediates before and/or after the elementary diffusion–collision step must be considered. On the other hand, if k_{assoc} is significantly greater than the diffusion–collision limit, various facilitating mechanisms involving nonspecific DNA sites on the same molecule must be considered. The importance of a facilitating mechanism may be diagnosed from measurements of the dependence of k_{assoc} of DNA length and on salt concentration.

Dissociation typically first order (or pseudofirst order): Kinetics of

dissociation of site-specific protein–DNA complexes follow the expected first order rate law:

$$-\frac{d[\text{RS}]}{dt} = k_{\text{dissoc}}[\text{RS}] \quad (27)$$

where k_{dissoc} is the observed (generally composite) first order rate constant. The mechanism of dissociation must involve passage through the same steps (in the reverse direction) as in the association mechanism, according to the principle of microscopic reversibility. For example, if association is facilitated by the presence of contiguous nonspecific sites, these sites and facilitating mechanisms will also play a role in dissociation, and the equilibrium constant and thermodynamic quantities derived therefrom will be independent of this path-dependent effect. Observed deviations from this situation can usually be attributed to complications arising from differences in the way that the kinetics of association and dissociation are investigated.

In investigations of the kinetics of dissociation of site-specific complexes, a polyanionic competitor (e.g., heparin) is often added to sequester the free protein. In this case, it is important to examine whether k_{dissoc} is a true first order dissociation rate constant, or whether k_{dissoc} contains contributions from both dissociation (first order) and from competitor-induced displacement (typically pseudofirst order if competitor is in excess). The intrinsic contribution from dissociation may be separated from competitor-induced displacement by extrapolation to zero competitor concentration.⁹³

Since cations may be considered fundamental and omnipresent competitors with proteins for the vicinity of the DNA polyanion, all elementary protein–DNA dissociation rate constants are actually pseudofirst order, because cations are reactants in the elementary step of dissociation of the protein from the DNA, and the cation concentration is in vast excess.^{46,88,94} Hence these elementary rate constants generally vary as some large positive power of the cation concentration. This effect, which can serve as a diagnostic probe of mechanism, need not be (and indeed cannot be) extrapolated away, as long as measurements of association and dissociation kinetics and of equilibrium extents of binding are compared at identical concentrations of electrolyte ions.

Characterization of Intermediates. Two different and complementary approaches have been used to probe mechanisms of site-specific binding, as illustrated by studies on *E. coli* RNA polymerase. One approach has

⁹³ C. L. Cech and W. R. McClure, *Biochemistry* **19**, 2440 (1980).

⁹⁴ T. M. Lohman, P. L. deHaseth, and M. T. Record, Jr., *Biophys. Chem.* **8**, 281 (1978).

been to isolate and characterize intermediates using low-temperature or other "quenching" methods to block subsequent steps. This approach assumes that intermediates that can be isolated as a function of temperature (or other variables) bear a direct relationship to the transient intermediates that exist on the kinetic path for conversion of reactants to products at higher temperature. With RNA polymerase, structural methods, including electron microscopy⁹⁵ and various footprinting and chemical modification/chemical protection assays,⁹⁶ have been applied. Polymerase-promoter complexes investigated include those of σ^{70} -RNA polymerase ($E\sigma^{70}$) with the T7 A3,⁹⁷ *tetR*,⁴⁸ and *lacUV5* promoters.^{97,98} Similar studies have been carried out on intermediates in the interaction of σ^{32} -RNA polymerase ($E\sigma^{32}$) and the *groE* promoter.²⁹ The complementary approach is to look for the characteristic dependencies of the individual elementary rate constants or overall composite rate constants on such variables as DNA length, sequence (of either protein or DNA), salt concentration, pH, and temperature. The analysis of effects of these variables in terms of mechanisms of site-specific interactions is discussed in the remainder of this chapter.

Use of Temperature and Salt Concentration to Investigate Mechanisms of Interactions of Proteins and DNA

Our focus in the following sections is on the interpretation of experimental rate constants of association and dissociation in terms of the mechanism of formation of the site-specific complex. In general, temperature, salt concentration, and pH are the important diagnostic probes of mechanism.

Investigations of the kinetics of site-specific protein-DNA interactions constitute a relatively young field, beginning with the elegant studies of Riggs, Bourgeois, and Cohn⁹⁹ on the *lac* repressor-operator interaction and of Hinkle and Chamberlin¹⁰⁰ on RNA polymerase $E\sigma^{70}$ -promoter interactions. Analysis of kinetic data in terms of mechanisms of protein-DNA interactions is also a new area, in which apparent precedents provided by mechanistic studies on the covalent reactions between small ions or molecules are often not of direct relevance. This is particularly the case in analyzing the effects of temperature and salt concentration on the observed association and dissociation rate constants.

⁹⁵ R. C. Williams and M. J. Chamberlin, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 3740 (1977).

⁹⁶ T. D. Tullius, *Annu. Rev. Biophys. Biophys. Chem.* **18**, 213 (1989).

⁹⁷ R. T. Kovacic, *J. Biol. Chem.* **262**, 13654 (1987).

⁹⁸ A. Spassky, K. Kirkegaard, and H. Buc, *Biochemistry* **24**, 2723 (1985).

⁹⁹ A. D. Riggs, S. Bourgeois, and M. Cohn, *J. Mol. Biol.* **53**, 401 (1970).

¹⁰⁰ D. Hinkle and M. Chamberlin, *J. Mol. Biol.* **70**, 187 (1972).

For both elementary and composite rate constants characterizing chemical reactions between low-molecular-weight solutes, the logarithm of the observed rate constant is typically a linear function of the reciprocal absolute temperature. The Arrhenius activation energy E is obtained from the slope. Interpretation of the magnitude and temperature dependence of an elementary rate constant for a covalent chemical reaction using the approximations of the Eyring quasithermodynamic model yields the activation parameters $\Delta H^{\ddagger}$ (which is approximately equal to E) and (with further assumptions) $\Delta G^{\ddagger}$ and $\Delta S^{\ddagger}$ for formation of the transition state from the reactants. In the case of an elementary single-step reaction, E and $\Delta H^{\ddagger}$ must be positive. Association processes in water where the rate-determining step is diffusion-collision should exhibit small activation energies of approximately 5 kcal, arising primarily from the temperature dependence of the diffusion coefficient ($D \propto T/\eta$, where η is the temperature-dependent solvent viscosity).

Rate constants (k) for second order association reactions between two low-molecular-weight ions (A^{Z_A} , B^{Z_B}) exhibit a modest dependence on the ionic strength (I) of the supporting electrolyte. For reactants of like charge, k increases with increasing ionic strength as a result of the more effective screening of the like charges of the reactants by the ionic atmospheres formed by the added salt. In the Brønsted-Bjerrum-Debye-Hückel theory, valid at low salt concentration ($<0.01\text{ M}$), $\log k$ is predicted to vary linearly with $I^{1/2}$: $\log k/k_0 \approx 1.02 Z_A Z_B I^{1/2}$, where k_0 is the rate constant at $I = 0$.¹⁰¹

Differences in the conceptual framework and methods of analysis of thermodynamic data on covalent reactions of small ionic solutes and non-covalent interactions of large and highly charged solutes were discussed in Equilibria and Thermodynamics of site-specific Protein-DNA Interactions, above. In the analysis of effects of temperature and salt concentration on the kinetics of noncovalent interactions of proteins and nucleic acids, analogous conceptual and methodological differences occur. Arrhenius plots of the logarithms of association and dissociation rate constants versus reciprocal absolute temperature must be nonlinear if the van't Hoff plot of the logarithm of the corresponding equilibrium constant versus reciprocal absolute temperature is nonlinear. Interpretation of the resulting heat capacities of activation ($\Delta C_p^{\ddagger}$) in terms of the hydrophobic effect (i.e., in terms of changes in the amount of water-accessible nonpolar surface in the various intermediates and/or transition states that determine the corresponding rate constant) provides molecular information about

¹⁰¹ E. A. Moelwyn-Hughes, "Chemical Statics and Kinetics of Solutions." Academic Press, New York, 1971.

mechanism.⁴⁵ Likewise, effects of the concentration, valence, and species of electrolyte on the magnitude of the rate constant and on the log-log derivatives Sk_{assoc} and Sk_{dissoc} may be interpreted using polyelectrolyte theory (but *not* Brønsted-Bjerrum-Debye-Hückel theory) in terms of the participation of these ions in the various steps and transition states of the mechanisms.^{46,88,94} These mechanistic analyses and molecular interpretations are discussed further in subsequent parts of this section.

Another fundamental issue, which remains to be resolved, regards the nature of a noncovalent transition state or activated complex, and more specifically the factors that determine its frequencies of decomposition into products and reactants. In the Eyring model, the frequency of decomposition of the transition state is the quantum mechanical frequency of vibration ν along the reaction coordinate ($\nu = k_{\text{B}}T/h$, where k_{B} is Boltzmann's constant and h is Planck's constant). Almost certainly this frequency factor is completely inappropriate for noncovalent transition states. In the absence of a theory to predict frequency factors for noncovalent transition states, one cannot obtain meaningful values of activation entropies ($\Delta S^{\ddagger}$) or activation free energies $\Delta G^{\ddagger}$ for noncovalent reaction steps from rate constants and their temperature dependences. If it is assumed that the frequency factors are invariant to changes in the recognition surfaces or in solution conditions, then relative values of activation entropies ($\Delta\Delta S^{\ddagger}$) and activation free energies ($\Delta\Delta G^{\ddagger}$) can be estimated.

A key similarity between attempts to relate kinetic data to mechanisms for noncovalent and covalent reactions is that no mechanism can be proven to be complete. Instead, one must proceed by demonstrating a minimal number and order of contributing steps, and by excluding possible alternative mechanisms on the basis of inconsistency with the kinetic data. A second similarity between analyses of mechanisms of noncovalent and covalent reaction is the reliance on mathematical approximations, such as the steady state assumption or rapid equilibrium approximations, in the analysis of a proposed mechanism. Even in the current era of powerful nonlinear fitting routines, such analytical approximations have their place, because of the insight provided by analytical expressions for the overall rate constants in terms of contributions from elementary rate constants. Strictly speaking, the assumptions involved in these approximations must be tested for all macromolecular sequences (i.e., site variants of the protein and/or DNA molecules) and environmental conditions (temperature, salt concentration, etc.) employed to be sure of their applicability. The nature and relative importance of individual kinetic steps are likely to be unique for each system investigated, and moreover may be different for different sequence variants of the recognition regions of the DNA and protein, and/or for different choices of environmental variables, since, in general, each

step of the mechanism of formation of a specific complex displays a unique dependence on these variables.

Elementary Bimolecular Step in Protein–DNA Association

Magnitude of Diffusion–Collision Rate Constant. The elementary bimolecular diffusion–collision rate constant k_{dc}^o for the reaction of two uncharged spheres, A and B, is given by the Smoluchowski equation:

$$k_{dc}^o = 4\pi(N/10^3)(D_A + D_B)(r_A + r_B) M^{-1} \text{ sec}^{-1} \quad (28)$$

where the factor 4π is the spherical solid angle (indicating that all directions of approach of the spheres lead to reaction), D_A and D_B are diffusion coefficients (in $\text{cm}^2 \text{ sec}^{-1}$), r_A and r_B are hydrodynamic radii (in cm), and N is Avogadro's number.

Proteins and nucleic acids are of course not adequately modeled as uncharged, uniformly reactive spheres. DNA is a highly charged locally cylindrical polyanion. Proteins may or may not be spherical, and more importantly are polyampholytes with an overall charge that is not uniformly distributed and is a function of pH. Only a small fraction of the molecular surface of either protein or DNA is "reactive." Long-range Coulombic interactions may increase or decrease the probability of collision. Introduction of these effects leads to an improved estimate of the diffusion–collision rate constant for noncovalent interactions of biopolymers (**R** and **S**)^{88,102,103}:

$$k_{dc} = 4\pi\kappa f(N/10^3)(D_R + D_S)(r_R + r_S)M^{-1} \text{ sec}^{-1} \quad (29)$$

where κ is the probability that the collision has the correct mutual orientation to lead to interaction, and f is a dimensionless factor that accounts for nonspherical geometry and long-range Coulombic effects ($f < 1$ for reactants of the same charge; $f > 1$ for reactants of opposite charge).¹⁰² For the interaction of *lac* repressor (modeled as a sphere with a radius of 40 Å and an interaction surface that is approximately 20% of the total surface) with λ DNA (of negligible diffusion coefficient relative to *lac* repressor) containing the *lac* operator site (modeled as a cylinder with a radius of 10 Å and an interaction surface that is approximately 25% of the total surface of the operator site), von Hippel and Berg¹⁰³ estimate that $\kappa \approx 0.05$ and that the orientation-corrected diffusion-limited rate constant is on an order of magnitude of $10^8 M^{-1} \text{ sec}^{-1}$ (neglecting geometrical and electrostatic corrections, which are expected to be relatively small). Since

¹⁰² O. G. Berg and P. H. von Hippel, *Annu. Rev. Biophys. Biophys. Chem.* **14**, 131 (1985).

¹⁰³ P. H. von Hippel and O. G. Berg, *J. Biol. Chem.* **264**, 675 (1989).

the diffusion coefficients of spheres decrease slowly with increasing molecular weight ($D \propto M^{-1/3}$), this "order of magnitude" estimate should be generally applicable for site-specific interactions of proteins with macromolecular DNA; a somewhat higher estimate of the diffusion-limited rate constant will be obtained for the interactions of proteins with small DNA fragments, which diffuse more rapidly than the protein.

Rate constants for elementary bimolecular steps in mechanisms of noncovalent interactions may in principle be significantly less than those calculated as the diffusion-collision limit if the severity of orientation effects (an entropic activation barrier) has been underestimated in the use of Eq. (29). Enthalpic activation barriers in excess of that expected from the temperature dependence of the diffusion coefficient are inconsistent with an elementary diffusion-limited process. Observation of a large enthalpic barrier for a second order rate constant k_{assoc} indicates that it is not an elementary rate constant, and that additional steps, either preceding or following the bimolecular step, contribute to the second order rate constant (see below). For multistep mechanisms of association, if complex formation is investigated at sufficiently high dilution of reactants, if the reactants maintain the correct conformation and state of association upon dilution, and if any unimolecular steps subsequent to the bimolecular step are sufficiently fast in comparison to the rate of dissociation of the initial bimolecular complex, then (and only then) the overall rate of association should be determined by the orientation-corrected diffusion-collision rate of formation of the initial complex.

Effects of Temperature and Salt Concentration on Diffusion-Collision-Limited Reactions. The diffusion-collision rate constant k_{dc} [eq. (29)] is expected to be only weakly dependent on temperature, since the only enthalpic activation barrier to the process is that intrinsic to diffusion. Diffusion coefficients of solutes are proportional to the absolute temperature and inversely proportional to the temperature-dependent solvent viscosity η ($D \propto T/\eta$). In water, the temperature dependence of T/η leads to an estimated activation energy for a diffusion-collision interaction of approximately 5 kcal, corresponding to a 1.4-fold increase in a diffusion-collision-limited association rate constant between 25 and 37°. In addition to this relatively small dependence of k_{dc} on temperature, a related diagnostic is the inverse proportionality predicted to exist between k_{dc} and solvent viscosity. While changes in η have been used to examine whether reactions of small molecules are diffusion limited, additives such as sucrose or glycerol used to vary the viscosity of water over the range necessary to affect the rate constant to a significant extent also may affect the conformations and stabilities of interacting biopolymers as a result of

weak, surface area-dependent preferential interactions.^{104,105} Hence the effects of these solutes on the kinetics of association must be interpreted with caution.

Effects of long-range Coulombic interactions on the diffusion-limited rate constant for interactions of oligocations L^{z+} with the DNA polyanion have been examined by Lohman *et al.*⁹⁴ in the context of the two-state polyelectrolyte theory of Manning.^{63,67} By analogy with the Brønsted–Bjerrum–Debye–Hückel theory of salt effects on reactions of low-molecular-weight charged species, Lohman *et al.* proposed that the electrolyte screens the interaction between DNA and L^{z+} , and hence reduces the rate of interaction of these oppositely charged reactants. For double-helical DNA, screening of phosphates is equivalent to the association of 0.12 univalent cation/phosphate.⁶⁶ The elementary diffusion–collision reaction is assumed to proceed via a low-entropy, a high-free energy “transition state” in which L^{z+} has penetrated the screening ion atmosphere surrounding the DNA and is localized in the vicinity of its surface, but in which the condensed cations have not yet been displaced. Assuming rapid equilibrium between transition state and reactants (as in covalent transition state theory), Lohman *et al.* predicted that screening effects should result in a modest linear dependence of the logarithm of k_{dc} on the logarithm of the univalent salt concentration⁹⁴:

$$\frac{d \ln k_{dc}}{d \ln [MX]} \equiv Sk_{dc} = -0.12z \quad (30)$$

(This is probably a maximum estimate of the magnitude of Sk_{dc} , since use of a more realistic steady state analysis of the process of passing through the screening-controlled transition state will reduce the magnitude of Sk_{dc} .) As long as the ligand is an oligocation and not a polycation, screening effects are asymmetric and the consequences of screening of the charges on L^{z+} by MX are negligible by comparison.⁷⁸

In the case of proteins like *lac* repressor, the situation is in principle more complicated because the net charge on the protein is negative, although the net charge on the DNA-binding domain is apparently positive. Goeddel *et al.*¹⁰⁶ found that the second order rate constant (k_{assoc}) for the interaction of *lac* repressor with a small synthetic operator (21 or 26 bp) was near the expected diffusion–collision limit ($k_{assoc} \sim 1\text{--}2 \times 10^9 M^{-1} \text{sec}^{-1} \approx k_{dc}$). [Their results exceeds the estimate of k_{dc} for the interaction

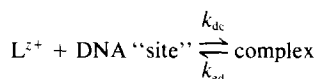
¹⁰⁴ J. C. Lee and S. N. Timasheff, *J. Biol. Chem.* **256**, 7193 (1981).

¹⁰⁵ K. Gekko and S. N. Timasheff, *Biochemistry* **20**, 4667 (1981).

¹⁰⁶ D. V. Goeddel, D. G. Yansura, and M. H. Caruthers, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 3292 (1977).

of *lac* repressor with λ plac DNA ($\sim 10^8 \text{ M}^{-1} \text{ sec}^{-1}$) in part because of the higher diffusion coefficient of the small DNA fragment.] Goeddel *et al.* also found that k_{assoc} decreased slightly with increasing [KCl], with $Sk_{\text{assoc}} \approx -0.5$ for wild-type repressor and $Sk_{\text{assoc}} \approx -1$ for the tight-binding X86 repressor. These results are in semiquantitative agreement with the prediction of Eq. (30) since *lac* repressor behaves in site-specific binding as an oligocation with a valence z in the range $5 \leq z \leq 8$.^{13-15,61,68} Both the magnitude and salt concentration dependence of k_{assoc} in this case appear consistent with a screening affected, diffusion-collision-limited rate constant. Furthermore it appears that the charge on the DNA-binding site of repressor is more important in determining Sk_{dc} than is the net charge on the protein.

Dependence of Elementary Rate Constant for Dissociation of L^{z+} -DNA Complex on Salt Concentration. If association of L^{z+} with DNA is a screening affected, diffusion-collision elementary process and if dissociation of the L^{z+} -DNA complex is also an *elementary* mechanistic step with rate constant k_{ed} (for elementary dissociation step) then the overall association-dissociation process can be represented as



where $K_{\text{obs}} = k_{\text{dc}}/k_{\text{ed}}$. From Eq. (22) for SK_{obs} and Eq. (30) for Sk_{dc} , it follows that

$$Sk_{\text{ed}} = Sk_{\text{dc}} - SK_{\text{obs}} = 0.76z \quad (31)$$

The dependence of k_{ed} on salt concentration [Eq. (31)] for this reaction results from the requirement that $0.76z$ univalent cations recondense on the DNA when L^{z+} dissociates. Hence cations are "reactants" in the dissociation process.^{88,94} This is an example of what appears to be a general principle in analyzing effects of salt concentration on rate constants of ligand-DNA interactions. Rate constants for steps in which cations of the electrolyte associate with DNA exhibit a power dependence on salt concentration, where the exponent is the stoichiometry of participation of ions in the step. This principle even carries over to screening limited association, since the proposed origin of Sk_{dc} [Eq. (30)] is the reassociation of screening ions that accompanies dissociation of the "transition state" of that elementary step.^{88,94} In mixtures of cations, especially of different valence, competitive interactions of the cations must be accounted for in estimating or analyzing salt concentration dependences of dissociation rate constants, as in thermodynamic analyses of SK_{obs} .^{61,62}

Mechanisms Involving Intermediates

Calculated estimates of the magnitude of k_{dc} and of its relatively weak dependences on temperature and salt concentration provide convenient points of reference for comparisons of the behavior of experimentally determined second order rate constants k_{assoc} of site-specific protein–DNA associations. Generally one finds that $k_{assoc} < k_{dc}$, and that the magnitudes of the overall activation energy of association E_{assoc} and salt concentration dependence $S k_{assoc}$ are significantly larger than predicted for a diffusion–collision reaction. Here we consider some simple mechanisms involving intermediates that are consistent with second order kinetics. These demonstrate how temperature and salt concentration can be used to distinguish mechanistic alternatives.

Intermediates Subsequent to Elementary Bimolecular Step. Mechanism of open complex formation between RNA polymerase $E\sigma^{70}$ and promoter DNA: A moderately well-characterized example of a site-specific protein–DNA interaction in which conformational changes of the initial specific complex are mechanistically important is the association of *E. coli* RNA polymerase ($E\sigma^{70}$) with promoter DNA sequences. In excess RNA polymerase, the kinetics of association fit the form of Eq. (26) with $(R_T[RS]) \approx R_T$, so that first order kinetics are observed:

$$\frac{d(S_T - [RS])}{dt} = -k_{assoc} R_T (S_T - [RS]) \quad (32)$$

where $k_{assoc} R_T \equiv k_{obs} = 1/\tau_{obs}$, and where τ_{obs} is a composite first order time constant. Kinetic studies demonstrate that τ_{obs} is a linear function of R_T , with in general a positive (nonzero) intercept.^{107,108} In other words, the kinetic order of association changes from second order (pseudofirst order) in the limit of low (excess) R_T to first order in the limit of very high R_T .

Interpretation of this behavior and of the dependencies of the rate constants on temperature and salt concentration led to the proposal of a three-step minimal mechanism.^{44–46,48,83,109,110} For association of $E\sigma^{70}$ with the λP_R promoter, the kinetically significant steps of the association mechanism under the conditions examined are found to be



¹⁰⁷ W. R. McClure, *Proc. Natl. Acad. Sci. U.S.A.* **77**, 5634 (1980).

¹⁰⁸ W. R. McClure, *Annu. Rev. Biochem.* **54**, 171 (1985).

¹⁰⁹ S. Rosenberg, T. R. Kadesch, and M. J. Chamberlin, *J. Mol. Biol.* **155**, 1 (1982).

¹¹⁰ H. Buc and W. R. McClure, *Biochemistry* **24**, 2712 (1985).

where (to be consistent with the general notation of this chapter) **R** is the polymerase, **S** is the specific DNA site (promoter), **RS** is the functional ("open") complex (generally designated as RP_0) and I_1 and I_2 are intermediate closed complexes (generally designated RP_c and RP_i or RP_{c1} and RP_{c2}).

In excess $E\sigma^{70}$, the steady state solution of this mechanism (which assumes that the concentrations of the intermediate closed complexes are independent of time after an initial transient phase) is

$$k_{\text{assoc}} = \frac{k_{\text{obs}}}{\mathbf{R}_T} = \frac{k_1 k_2 k_3}{k_1 \mathbf{R}_T (k_2 + k_3) + k_3 (k_{-1} + k_2)} \quad (34)$$

$$\tau_{\text{obs}} = \frac{k_{-1} + k_2}{k_1 k_2 \mathbf{R}_T} + \frac{1}{k_2} + \frac{1}{k_3}$$

In the limit of low concentration of $E\sigma^{70}$, k_{assoc} [Eq. (34)] approaches second order behavior:

$$\lim_{\mathbf{R}_T \rightarrow 0} k_{\text{assoc}} = \frac{k_1 k_2}{k_{-1} + k_2} \equiv k_a \quad (35)$$

where k_a is a composite second order rate constant.

In the limit of high concentration of $E\sigma^{70}$, from Eq. (34),

$$\lim_{\mathbf{R}_T \rightarrow \infty} (k_{\text{assoc}} \mathbf{R}_T) = \left(\frac{1}{k_2} + \frac{1}{k_3} \right)^{-1} \equiv k_i \quad (36)$$

and the association kinetics approach a true first order process, with the composite first order isomerization rate constant k_i . Expressed in terms of k_a and k_i , the time constant τ_{obs} [Eq. (34)] is

$$\tau_{\text{obs}} = \frac{1}{k_a \mathbf{R}_T} + \frac{1}{k_i} \quad (37)$$

This functional form is consistent with experimental observations on a wide variety of promoters.¹⁰⁸ [Mechanisms analogous to Eq. (33) with two or more steps yield the functional form of Eq. (37).]

Further information regarding intermediates on the pathway to formation of the functional specific complex may be deduced from measurement of the dissociation rate constant k_{dissoc} . For dissociation of $E\sigma^{70}$ from the λP_R promoter, the kinetically significant steps of dissociation under the conditions examined are



Steady state analysis of mechanism (38), subject also to the assumption $k_3 \gg k_{-2}$ used to define the kinetically significant steps in the association mechanism (33), yields

$$k_{\text{dissoc}} = \frac{k_{-2}k_{-1}}{K_3(k_{-1} + k_2)} \quad (39)$$

where $K_3 \equiv k_3/k_{-3}$.

Consequently, from Eqs. (35) and (39),

$$\frac{k_a}{k_{\text{dissoc}}} = K_1 K_2 K_3 \equiv K_{\text{obs}} \quad (40)$$

where $K_1 \equiv k_1/k_{-1}$, $K_2 \equiv k_2/k_{-2}$, and where K_{obs} is the observed equilibrium constant defined in Eq. (5). The approximate equality $K_1 K_2 K_3 \equiv K_{\text{obs}}$ results from assuming that I_1 and I_2 are relatively unstable intermediates (in comparison to the open complex **RS**) that do not exist at significant concentrations at equilibrium.

Effects of temperature and salt concentration on composite rate constants for association and dissociation of $E\sigma^{70}$ -promoter complexes:

1. Second Order Rate Constant k_a : The steady state solution for the second order rate constant k_a [Eq. (35)] for the $E\sigma^{70}$ -promoter association process is in fact a general expression applicable to any situation in which one or more mechanistically important reversible bimolecular step(s) is followed by one or more mechanistically important irreversible unimolecular step(s) (e.g., Michaelis–Menten enzyme–substrate kinetics). Hence the following discussion is of more general applicability than to $E\sigma^{70}$ -promoter interactions. The steady state expression for k_a encompasses a range of kinetic behavior, determined by the relative magnitudes of k_{-1} and k_2 . The so-called rapid equilibrium limit of the steady state solution is obtained when $k_{-1} \gg k_2$, so that dissociation of the initial complex occurs frequently on the time scale of its isomerization. In this case,

$$\text{Rapid equilibrium } (k_{-1} \gg k_2): \quad k_a = \frac{k_1 k_2}{k_{-1}} = K_1 k_2 \quad (41)$$

The other limit is the sequential limit, in which $k_2 \gg k_{-1}$, so that formation of the initial complex is essentially irreversible on the time scale of its isomerization. In this case,

$$\text{Sequential } (k_2 \gg k_{-1}): \quad k_a = k_1 \quad (42)$$

Here the initial bimolecular step determines the association rate k_a . The

steady state approximation is less accurate when the kinetics approach sequential behavior than when the rapid equilibrium limit is approached.

Temperature and salt concentration are useful variables to determine whether a second order rate constant k_a is closer to the rapid equilibrium or to the sequential limit of the steady state, and hence to determine relative values of k_{-1} and k_2 . However, since k_{-1} and k_2 will generally exhibit different (and, in some cases, large) dependencies on these variables, one must recognize that both absolute and relative values of k_{-1} and k_2 may change over the ranges of temperatures and salt concentrations examined (see Activation Free-Energy Analysis of Association and Dissociation, below).

The activation energy E_a obtained from the temperature dependence of the steady state expression of k_a [Eq. (35)] is

$$E_a = RT^2 \frac{d \ln k_a}{dT} = \Delta H_1^\circ + \left(\frac{1}{k_{-1} + k_2} \right) (k_{-1}E_2 + k_2E_{-1}) \quad (43)$$

where ΔH_1° is the van't Hoff enthalpy change in the process of step 1 [$\Delta H_1^\circ = RT^2(d \ln K_1/dT)$] and E_2 and E_{-1} are the Arrhenius activation energies of the corresponding elementary steps [$E_2 = RT^2(d \ln k_2/dT)$ and $E_{-1} = RT^2(d \ln k_{-1}/dT)$]. In the rapid equilibrium limit, $E_a = \Delta H_1^\circ + E_2$; in the sequential limit $E_a = E_1$.

Analogously, the salt concentration dependence of k_a , Sk_a , has the form

$$Sk_a = SK_1 + \left(\frac{1}{k_{-1} + k_2} \right) (k_{-1}Sk_2 + k_2Sk_{-1}) \quad (44)$$

In the rapid equilibrium limit, $Sk_a = SK_1 + Sk_2$; in the sequential limit $Sk_a = Sk_1$.

For the interaction of RNA polymerase with the λP_R promoter, $Sk_a = -12 \pm 1$,⁴⁶ independent of salt concentration over the range examined (cf. Fig. 4), but E_a is a function of temperature, ranging from 20 ± 5 kcal in the range 25–37° to 40 ± 15 kcal in the range 13–15° (cf. Fig. 5).⁴⁵ Since the magnitudes of k_a are less than the diffusion-collision limit over the range of conditions examined and since the magnitudes of Sk_a and E_a are much larger than expected for a diffusion-limited reaction, it is clear that the rapid equilibrium limit of the steady state is the appropriate one to consider, rather than the sequential limit. The decrease in E_a with increasing temperature could result from an intrinsic negative heat capacity of activation or from a large increase in the ratio k_2/k_{-1} with increasing temperature. In the latter possibility the kinetics would shift from near the

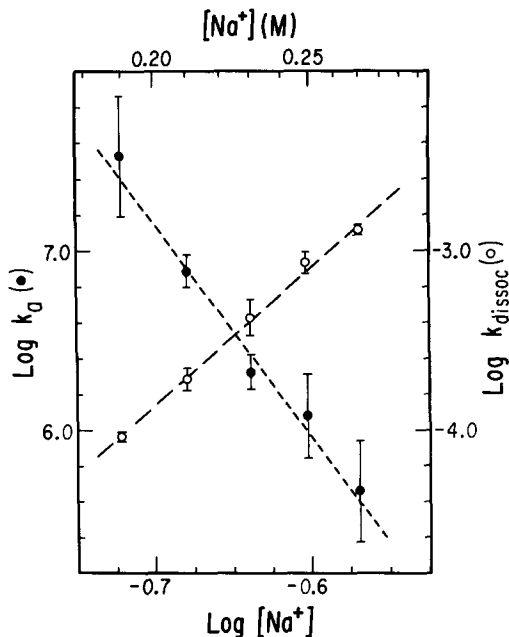


FIG. 4. Log-log plot of association (k_a) and dissociation (k_{dissoc}) rate constants for open complexes between $E\sigma^{70}$ RNA polymerase and the λP_R promoter as functions of NaCl concentration. The dashed lines are weighted linear least-squares fit to the data: $\log k_a = -(1.2 \pm 0.7) - (11.9 \pm 1.1)\log[Na^+]$; $\log k_{dissoc} = (1.5 \pm 0.1) + (7.7 \pm 0.2)\log[Na^+]$. [From J.-H. Roe and M. T. Record, Jr., *Biochemistry* **24**, 4721 (1985).]

rapid equilibrium limit at lower temperature to near the sequential limit at higher temperature. This possibility is inconsistent with the large magnitudes of Sk_a and E_a and with the analysis of dissociation kinetic data (below). Consequently, Roe *et al.*⁴⁵ concluded that the temperature dependence of E_a was the result of an intrinsic large negative $\Delta C_p^{\ddagger}$, which they attributed to a protein conformational change that reduced the amount of nonpolar surface exposed to water in forming the transition state in isomerization of the initial closed complex ($I_1 \rightarrow I_2$). (ΔH_1° is relatively small and thought to be temperature independent.) The large magnitude of Sk_a appears to result from SK_1 , which is the stoichiometry of ion release in forming the initial closed complex.^{27,46,83}

2. First Order Dissociation Rate Constant k_{dissoc} : Analysis of k_a , Sk_a , and E_a for the interaction of $E\sigma^{70}$ with the λP_R promoter indicates that I_1 (the initial closed complex) is in rapid equilibrium with free $E\sigma^{70}$ and promoter DNA on the time scale of isomerization to I_2 ($k_{-1} \gg k_2$) over the range of conditions examined. In this limit,

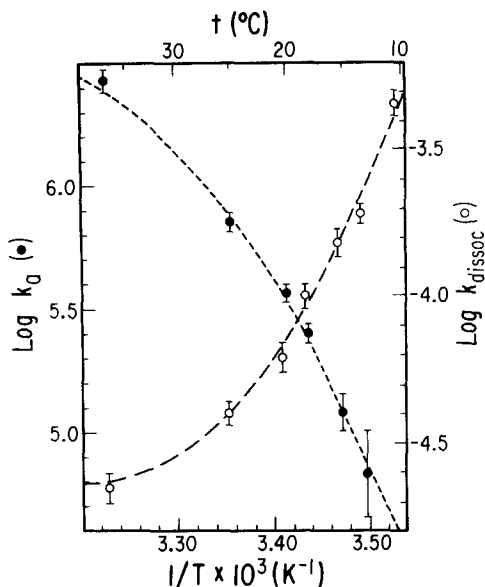
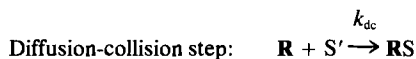
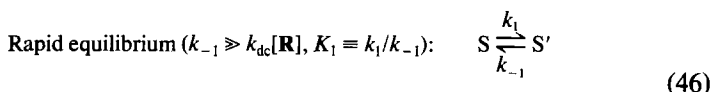


FIG. 5. Arrhenius plot of the temperature dependences of k_a and k_{dissoc} for the $E\sigma^{70}\text{-}\lambda P_R$ promoter interaction. Dashed curves are fits obtained using activation heat capacities $\Delta C_{p,a}^\ddagger = -1.13 \text{ kcal K}^{-1}$ and $\Delta C_{p,\text{dissoc}}^\ddagger = +1.28 \text{ kcal K}^{-1}$, respectively. [From J.-H. Roe, R. R. Burgess, and M. T. Record, Jr., *J. Mol. Biol.* **184**, 441 (1985).]

$$\begin{aligned}
 k_{\text{dissoc}} &= K_3^{-1} k_{-2} \\
 E_{\text{dissoc}} &= E_{-2} - \Delta H_3^\circ \\
 S k_{\text{dissoc}} &= S k_{-2} - S K_3
 \end{aligned}
 \tag{45}$$

Roe and Record found that $S k_{\text{dissoc}} = 8 \pm 1$ (cf. Fig. 4)⁴⁶ and that E_{dissoc} was negative and highly temperature dependent, varying from -9 ± 4 kcal in the range 25–37° to -30 ± 10 kcal in the range 13–15 (cf. Fig. 5).⁴⁵ The negative E_{dissoc} is thought to result from the negative enthalpy change ($\Delta H_{-3}^\circ = -\Delta H_3^\circ < 0$) accompanying the conversion of the open complex (RS) to the intermediate closed complex (I_2), which is interpreted as the enthalpy of helix formation. (The fact that E_{dissoc} is negative demonstrates that the dissociation rate constant is a composite rate constant and not an elementary rate constant, since activation energies of elementary steps are positive.) The positive temperature dependence of E_{dissoc} appears to be an intrinsic positive heat capacity of activation, which is thought to result from the exposure of nonpolar surface in forming the transition state in the isomerization $I_2 \rightarrow I_1$. The large $S k_{\text{dissoc}}$ is thought to result primarily from $-S K_3$, which is interpreted as the stoichiometry of cation uptake that accompanies helix formation.

Intermediates Prior to Elementary Bimolecular Step. Conformational changes in the protein or DNA sites that occur prior to binding are consistent with overall second order kinetics of association, providing the conformational change is in rapid equilibrium on the time scale of association. Consider the hypothetical mechanism



For this situation, the second order rate constant [cf. Eq. (26)] is

$$k_{\text{assoc}} = k_{dc} (1 + K_1^{-1})^{-1} \quad (47)$$

and therefore

$$E_{\text{assoc}} = E_{dc} - (1 + K_1)\Delta H_1^\circ \quad (48)$$

and

$$SK_{\text{assoc}} = SK_{dc} - (1 + K_1)^{-1}SK_1 \quad (49)$$

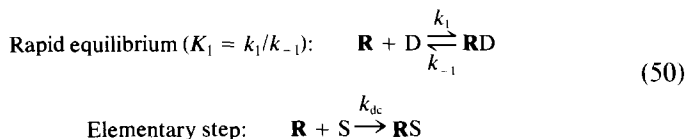
(A rapid conformational equilibrium in the protein would of course produce an analogous result. On the other hand, mechanisms involving intermediates in which the state of oligomerization of the protein changes prior to binding are generally inconsistent with overall second order kinetics of formation of $\mathbf{RS}$.)

In general, a conformational preequilibrium causes k_{assoc} to be significantly smaller than k_{dc} , and yields values of E_{assoc} and SK_{assoc} that also should differ from the values expected for the diffusion-collision step alone. If intermediates also occur subsequent to the diffusion-collision step, clearly both will contribute to the behavior of k_{assoc} and its temperature and salt derivatives.

Nonspecific Binding as Competitor and/or as Facilitating Mechanism. All site-specific DNA-binding proteins exhibit a relatively weak and non-specific binding mode driven principally by the polyelectrolyte effect. Equilibrium specificity ratios $K_{\text{RS}}/K_{\text{RD}}$ for different binding proteins cover a wide range ($\sim 10^1$ – 10^8), and depend on ion concentrations, pH, temperature, and other environmental variables. Under conditions where nonspecific binding to a DNA molecule containing a specific site occurs to a significant extent, it may either retard or facilitate the rate of forming the site-specific complex.

Competition from nonspecific binding: Assume that nonspecific DNA sites D are present in excess, either on the same DNA molecule or on

different DNA molecules from the specific sites, and that nonspecific binding equilibrates rapidly on the time scale of forming a specific complex. Then the competitive effect of nonspecific binding on the observed second order association rate constant for site-specific binding may be summarized by the mechanism



(Conformational changes after formation of the initial site-specific complex may be included if they occur.)

If nonspecific sites are in excess, then $D_T = [\mathbf{D}] + [\mathbf{RD}] \approx [\mathbf{D}]$ and the rapid equilibrium approximation for nonspecific binding yields the following expression for k_{assoc} [Eq. (26)]:

$$k_{\text{assoc}} = k_{dc}(1 + K_1 D_T)^{-1} \quad (51)$$

Competitive effects of DNA molecules not bearing the specific site are readily examined by determining the dependence of k_{assoc} on D_T .

Facilitation from nonspecific binding to a DNA molecule: Riggs *et al.*⁹⁹ made the striking observation that the second order rate constant k_{assoc} [Eq. (26)] for formation of a specific complex between *lac* repressor and the *lac* operator on phage λ DNA ($M_r \sim 3 \times 10^7$; 45 kbp) exceeded the calculated diffusion-collision limit ($\sim 10^8 \text{ M}^{-1} \text{ sec}^{-1}$) at all salt concentrations examined, and exceeded $10^{10} \text{ M}^{-1} \text{ sec}^{-1}$ at low salt concentration. Winter *et al.*¹¹¹ and Barkley¹¹² confirmed and extended these findings (cf. Fig. 6A). Near 0.1 M [KCl], k_{assoc} attains a maximum value of approximately $2 \times 10^{10} \text{ M}^{-1} \text{ sec}^{-1}$; above 0.1 M KCl, $\log k_a$ decreases linearly with $\log[\text{K}^+]$,¹¹¹ as predicted from analysis⁹⁴ of the kinetic data of Riggs *et al.*⁹⁹ in a mixed $\text{Mg}^{2+}/\text{K}^+$ buffer. Below 0.1 NaCl, k_a decreases with decreasing $[\text{K}^+]$ to a low-salt plateau of approximately $3 \times 10^9 \text{ M}^{-1} \text{ sec}^{-1}$ for the repressor-operator interaction on λ plac DNA (Fig. 6A).¹¹¹

To explain these observations, Richter and Eigen¹¹³ and Berg, von Hippel, and co-workers^{42,102,103,111} applied the general principle of Adam and Delbruck¹¹⁴ that a reduction in dimensionality of the search will result in facilitation of a diffusional process. von Hippel and Berg¹⁰³ discuss the

¹¹¹ R. B. Winter, O. G. Berg, and P. H. von Hippel, *Biochemistry* **20**, 6961 (1981).

¹¹² M. D. Barkley, *Biochemistry* **20**, 3833 (1981).

¹¹³ P. Richter and M. Eigen, *Biophys. Chem.* **2**, 255 (1974).

¹¹⁴ G. Adam and M. Delbruck, in "Structural Chemistry and Molecular Biology" (A. Rich and N. Davidson, eds.), p. 198. Freeman, New York, 1968.

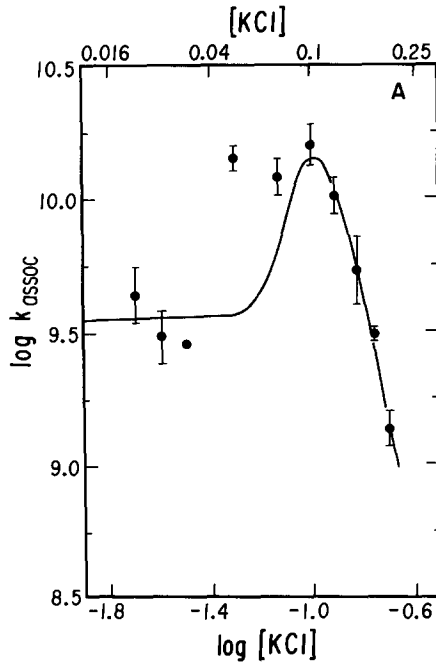


FIG. 6. Plot of $\log k_{\text{assoc}}$ vs $\log [\text{KCl}]$ for the interaction of *lac* repressor with the *lac* operator region contained on (A) λ_{plac} DNA (~ 45 kbp above) and (B) a 6.7-kbp *EcoRI* fragment (p. 337). The theoretical predictions of O. G. Berg, R. W. Winter, and P. H. von Hippel [*Biochemistry* **20**, 6929 (1981)] for a diffusional model of facilitation involving nonspecific binding (with the experimental value of $SK_{\text{obs}}^{\text{RD}}$) and sliding are shown by the solid lines. [Reprinted with permission from R. B. Winter, O. G. Berg, and P. H. von Hippel, *Biochemistry* **20**, 6961 (1981).]

possible random diffusional mechanisms by which nonspecific DNA sites on a molecule carrying a specific site may participate in the mechanism by which a protein locates the specific site. The domain of the DNA flexible coil provides a large target; the protein rapidly associates in a nonspecific manner with DNA sites in the domain by three-dimensional diffusion. Transfer within the domain to the specific site may involve one or more of the following diffusional processes: one-dimensional diffusion along the DNA chain ("sliding"), intersegment transfer (transient "looping"), and relatively long-lived inelastic collisions ("hopping"). None of these transfer processes is adequately expressed by a reaction mechanism with a small number of elementary steps.

For the situation in which sliding is the dominant transfer mechanism, Berg *et al.*⁴² derive the result

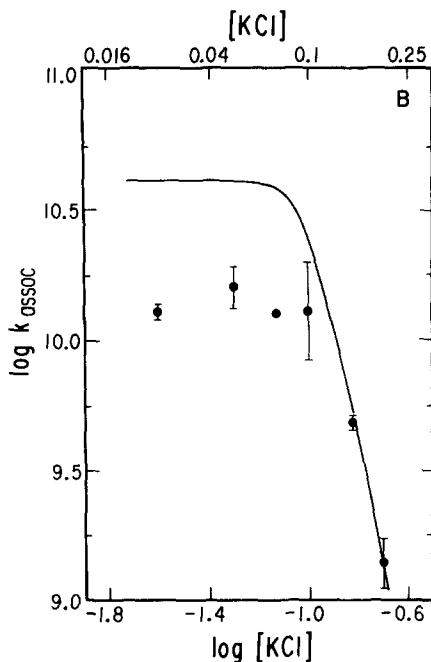


FIG. 6 (continued)

$$k_{\text{assoc}} = k_{\text{dc}}(1 + K_1 D_T)^{-1} F \quad (52)$$

where F is the factor by which the competition-adjusted diffusion-collision rate constant [Eq. (51)] is increased by sliding. At high salt concentration, where nonspecific binding is relatively weak ($K_1 D_T \ll 1$), F is proportional to the square root of the nonspecific binding constant ($F \propto K_1^{0.5}$), because in this limit (where competitive effects of nonspecific binding are negligible) the difference between k_{assoc} and k_{dc} results from the effective extension of the operator site to include adjacent regions of nonspecific DNA from which the repressor may diffuse to the operator target before dissociating. In this limit, $Sk_{\text{assoc}} \approx 0.5 SK_1$. The predicted value of Sk_{assoc} is therefore -6 ± 1 , based on $SK_1 = -12 \pm 2$. This compares well with the value $Sk_{\text{assoc}} = -5.3$ obtained by analysis⁹⁴ of the data of Riggs *et al.*,⁹⁹ and with that estimated from the data of Winter *et al.*¹¹¹ ($Sk_{\text{assoc}} \approx -5$). At lower salt concentrations, the increasingly important competitive effect of nonspecific sites on a very large DNA molecule causes k_{assoc} to pass through a maximum (at the salt concentration where $K_1 D_T = 1$) and then decrease; this behavior is not observed on a 6.7-kbp DNA molecule, where the rate constant reaches a plateau at low salt concentration where the

sliding range encompasses the entire molecule (cf. Fig. 6B).¹¹¹ At all salt concentrations (below 0.2 *M*) and all lengths of DNA molecules investigated, the facilitation factor *F* is much greater than unity; experimental rate constants k_{assoc} generally exceed $10^9 \text{ M}^{-1} \text{ sec}^{-1}$. The level of agreement between the experimental dependences of k_{assoc} on salt concentration and DNA length¹¹¹ and the predictions of the sliding facilitated mechanism of Berg *et al.*⁴² provide strong support for this mechanism, although the ability of *lac* repressor to form stable looped complexes between distant pairs of specific DNA sites^{115,116} suggests that the direct transfer (transient looping) mechanism may also play a role in facilitation of specific binding under some solution conditions.⁴²

Whereas a value of k_{assoc} that is significantly greater than the calculated diffusion-collision limit (k_{dc}) provides strong evidence for a facilitating mechanism involving nonspecific sites, this does not imply that a value of k_{assoc} that is less than k_{dc} provides evidence against the existence of such mechanisms. For example, in a multistep mechanism such as Eq. (33), where an initial specific complex (I_1) rapidly equilibrates with free reactants on the time scale of subsequent conformational changes to form the functional specific complex **RS**, the magnitude and salt dependence of k_{assoc} provide no information regarding whether the formation of I_1 is facilitated. If formation of I_1 involves transfer from nonspecific sites, dissociation of I_1 will also be facilitated by this transfer step, and the equilibrium constant K_1 will be unaffected. In this case, facilitating mechanisms are not kinetically significant even if they do occur. For sequences or solution conditions where the association kinetics are described by the sequential limit of the steady state, then facilitating mechanisms will be kinetically significant if they do occur.

Activation Free Energy Analysis of Association and Dissociation

The energetics of the process of site-specific complex formation by a DNA-binding protein may be represented schematically on a "progress" diagram that plots the relative standard free energies of reactants and the various transition states and intermediates on the pathway. Standard free energy differences (ΔG_i°) between stable states (reactants, intermediates, products) are related to equilibrium constants (K_i) by $\Delta G_i^\circ = -RT \ln K_i$. Standard activation free energy differences ($\Delta G_i^{\ddagger}$) between transi-

¹¹⁵ H. Kramer, N. Niemöller, M. Amouyal, B. Revet, B. von Wilcken-Bergmann, and B. Müller-Hill, *EMBO J.* **6**, 1481 (1987).

¹¹⁶ G. R. Bellomy and M. T. Record, Jr., in "Progress in Nucleic Acid Research and Molecular Biology" (W. Cohn and K. Moldave, eds.), Vol. 39, p. 81. Academic Press, New York, 1990.

tion states and initial (stable) states are related to the rate constants (k_i) of the corresponding elementary steps by an extension of the Eyring equation to noncovalent processes: $\Delta G_i^{\circ\ddagger} = -RT \ln(k_i/A_i)$, where A_i represents the various frequency and nonideality factors for formation and dissociation of the transition state of the i th reaction step.^{117,118} Estimates of the magnitude of A_i require assumptions regarding these factors, and are available only for simple covalent transition states. Values of $\Delta G_i^{\circ\ddagger}$ cannot at present be calculated from rate constants of noncovalent processes. The temperature dependence of a rate constant provides relatively direct information about $\Delta H_i^{\circ\ddagger}$. However, $\Delta S_i^{\circ\ddagger}$ cannot be evaluated, for the same reasons as discussed above for $\Delta G_i^{\circ\ddagger}$. An additional complication in processes involving large changes in water-accessible nonpolar surface and/or vibrational modes is that the existence of a nonzero $\Delta C_{p,i}^{\circ\ddagger}$ results in $\Delta H_i^{\circ\ddagger}$ and $T\Delta S_i^{\circ\ddagger}$ being strong functions of temperature, while $\Delta G_i^{\circ\ddagger}$ is relatively temperature invariant.

Often one is interested in interpreting a composite rate constant in terms of cumulative free energy barriers along a reaction pathway. For example, in the multistep mechanism of Eq. (33) (and in classical Michaelis–Menten kinetics for a one-substrate reaction) the composite second order rate constant k_a [from Eq. (35)] is given by

$$k_a = \frac{k_1 k_2}{k_{-1} + k_2} = K_1 k_2 \left(1 + \frac{k_2}{k_{-1}}\right)^{-1} \quad (53)$$

Therefore

$$\begin{aligned} -RT \ln k_a &= \Delta G_1^{\circ} + \Delta G_2^{\circ\ddagger} - RT \ln A_2 + RT \ln(1 + k_2/k_{-1}) \\ \Delta G_a^{\circ\ddagger} &\equiv \Delta G_1^{\circ} + \Delta G_2^{\circ\ddagger} = -RT \ln k_a/A_2 - RT \ln(1 + k_2/k_{-1}) \end{aligned} \quad (54)$$

where $\Delta G_1^{\circ} = -RT \ln K_1$ and $\Delta G_2^{\circ\ddagger} = -RT \ln k_2/A_2$. Only in the rapid equilibrium limit ($k_{-1} \gg k_2$) is the composite activation free energy barrier $\Delta G_a^{\circ\ddagger}$ directly related to the composite experimental second order rate constant k_a , although the correction term $RT \ln(1 + k_2/k_{-1})$ is not large unless $k_2 \gg k_{-1}$ (the sequential limit). Current investigations of effects of protein and DNA sequence variants on kinetic parameters seek to interpret ratios of rate constants k_a in terms of differences in activation free energy barriers $\Delta \Delta G^{\circ\ddagger}$.^{118–120} For these interpretations to be quantitatively valid, the terms A_2 and k_2/k_{-1} must be the same for all variants.

Figure 7A is a schematic diagram of the relative free energy levels of

¹¹⁷ S. L. Leirmo, Ph.D. Thesis, University of Wisconsin-Madison (1989).

¹¹⁸ B. Beutel, Ph.D. Thesis, University of Wisconsin-Madison (1990).

¹¹⁹ A. Fersht, *Biochemistry* **26**, 8031 (1987).

¹²⁰ J. A. Wells, *Biochemistry* **29**, 8509 (1990).

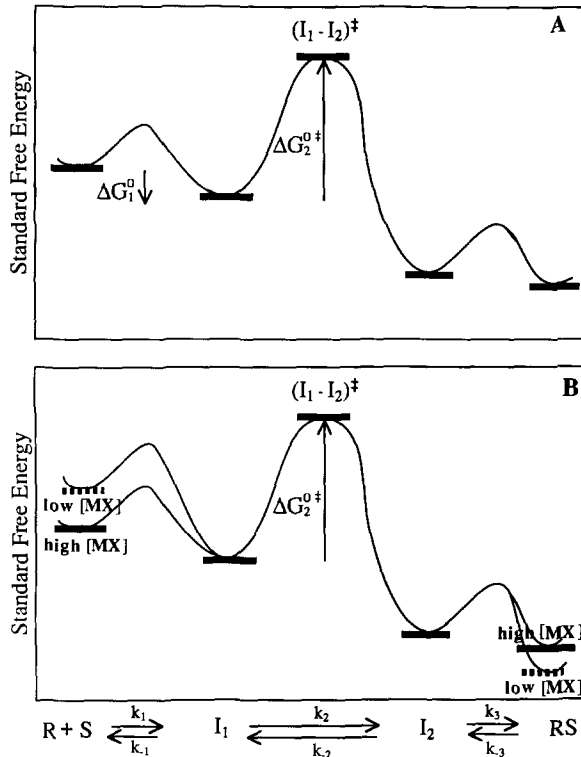


FIG. 7. Schematic free energy diagrams for the two-intermediate mechanism of interaction of $E\sigma^{70}$ (or $E\sigma^{32}$) RNA polymerase with a promoter to form an open complex. (A) The two-intermediate mechanism under conditions (e.g., high salt concentration, $T \approx 37^\circ$) where rapid equilibrium approximations ($k_{-1} \gg k_2$; $k_3 \gg k_{-2}$) are applicable in analysis of both k_{assoc} and k_{dissoc} . [An example of this behavior is the $E\sigma^{70}$ - λP_R promoter interaction (see text).] ΔG_1° is the standard free energy difference between the initial closed complex (I_1) and reactants (R , S). ΔG_2° is the height of the activation barrier between I_1 and the transition state ($I_1 - I_2$) ‡ . This step $I_1 \rightarrow I_2$ is thought to involve a large conformational change in $E\sigma^{70}$. In the rapid equilibrium approximation, the composite second order rate constant k_a is related to the net activation barrier ($\Delta G_a^\circ = \Delta G_1^\circ + \Delta G_2^\circ$) between reactants (R , S) and transition state ($I_1 - I_2$) ‡ by

$$k_a = A_2 e^{-\Delta G_a^\circ / RT}$$

where A_2 is the combination of frequency and nonideality factors involved in passing from intermediate I_1 to the transition state ($I_1 - I_2$) ‡ [cf. Eq. (54)]. (B) Schematic demonstrating the effect of a reduction in univalent salt concentration $[MX]$ on the standard free energies of promoter DNA and the open complex, and the resulting effect of $[MX]$ on the rate constants k_{-1} and k_{-3} for the steps of the mechanism that involve cation uptake. The curve labeled "high $[MX]$ " is the same as in Fig. 7A. At low $[MX]$, the free energy barriers for steps requiring cation uptake (ΔG_1° , ΔG_3°) are increased, so the rate constants k_{-1} and k_{-3} are reduced from their high $[MX]$ values. Consequently the kinetics of association at

the various transition states, stable intermediates, reactants, and products for the prototypic mechanism of open complex formation by RNA polymerase at a promoter [Eq. (33)] at physiological temperature ($\sim 37^\circ$) and cation concentration ($\sim 0.2\text{ M K}^+$). At this temperature and relatively high salt concentration, the highest activation free energy barrier in each direction appears to be that separating the two intermediate closed complexes.^{44–46,83,117} Because the surrounding barriers are not as high, rapid equilibrium approximations ($k_{-1} \gg k_2$; $k_3 \gg k_{-2}$) appear appropriate to analyze both association and dissociation kinetic data (cf. Fig. 7A).

Changes in salt concentration have major effects on the relative heights of these barriers. As the salt concentration is reduced, the height of the free energy barrier to dissociation of the intermediate I_1 to free DNA increases, because of the additional free energy cost at low salt concentration of accumulating cations in the transition state by which I_1 dissociates (cf. Fig. 7B). At low salt concentration the free energy of the free DNA site also increases. The association kinetics are predicted to shift from rapid equilibrium toward sequential as the salt concentration is reduced, because the barrier for $I_1 \rightarrow R + S$ increases with decreasing salt concentration, whereas the barrier for isomerization ($I_1 \rightarrow I_2$) is thought to involve a conformational change in RNA polymerase and therefore be relatively salt concentration independent. The kinetics of association of $E\sigma^{32}$ RNA polymerase with the *groE* promoter on a 450-bp restriction fragment appear to shift in this manner when the [NaCl] is reduced.²⁸ At low salt concentration, the second order association rate constant k_a approaches the estimated diffusion–collision limit ($k_{dc} \approx 10^8\text{ M}^{-1}\text{ sec}^{-1}$) and the association process is sequential; at higher salt concentration k_a is drastically reduced because of the increase in the rate constant for dissociation of the initial closed complex (k_{-1}).²⁸ The kinetics of *lac* repressor–*lac* operator association may also exhibit a shift from rapid equilibrium toward sequential with decreasing salt concentration.¹¹¹ At the higher salt concentrations examined ($>0.1\text{ M KCl}$), the facilitating nonspecific binding step appears to be in rapid equilibrium on the time scale of specific complex formation. At lower salt concentration, the coupled processes of nonspecific association and transfer become more sequential, until in the low salt limit the lifetime of the nonspecific complex is sufficiently long that dissociation

sufficiently low salt concentration may no longer conform to the rapid equilibrium approximation, since k_{-1} may no longer be much larger than k_2 . [This situation appears to apply to the interaction of $E\sigma^{32}$ with the *groE* promoter (see text).] Since the location of the progress diagram on the free energy axis is arbitrary, the diagram designated “low [MX]” has been normalized so that the [MX]-independent step $I_1 \rightarrow I_2$ is at the same free energy level as at “high [MX].”

apparently does not occur on the time scale of transfer to the specific site.¹¹¹

Summary

The concentration and nature of the electrolyte are key factors determining (1) the equilibrium extent of binding of oligocations or proteins to DNA, (2) the distribution of bound protein between specific and nonspecific sites, and (3) the kinetics of association and dissociation of both specific and nonspecific complexes. Salt concentration may therefore be used to great advantage to probe the thermodynamic basis of stability and specificity of protein–DNA complexes, and the mechanisms of association and dissociation. Cation concentration serves as a thermodynamic probe of the contributions to stability and specificity from neutralization of DNA phosphate charges and/or reduction in phosphate charge density. Cation concentration also serves as a mechanistic probe of the kinetically significant steps in association and dissociation that involve cation uptake. In general, effects of electrolyte *concentration* on equilibrium constants (quantified by SK_{obs}) and rate constants (quantified by Sk_{obs}) are primarily cation effects that result from the cation-exchange character of the interactions of proteins and oligocations with polyanionic DNA. The competitive effects of Mg^{2+} or polyamines on the equilibria and kinetics of protein–DNA interactions are interpretable in the context of the cation-exchange model. The nature of the anion often has a major effect on the magnitude of the equilibrium constant (K_{obs}) and rate constant (k_{obs}) of protein–DNA interactions, but a minor effect on SK_{obs} and Sk_{obs} , which are dominated by the cation stoichiometry. The order of effects of different anions generally follows the Hofmeister series and presumably reflects the relative extent of preferential accumulation or exclusion of these anions from the relevant surface regions of DNA-binding proteins. The question of which anion is most inert (i.e., neither accumulated nor excluded from the relevant regions of these proteins) remains unanswered.

The characteristic effects of temperature on equilibrium constants and rate constants for protein–DNA interactions also serve as diagnostic probes of the thermodynamic origins of stability and specificity and of the mechanism of the interaction, since large changes in thermodynamic and activation heat capacities accompany processes with large changes in the amount of water-accessible nonpolar surface area. Although more work remains to be done to verify the generality and the quantitative implications of these large heat capacity effects, one may anticipate that they will serve to detect and quantify the contributions of removal of nonpolar surface from water (the hydrophobic effect) to the steps of site-specific

binding, in much the same way that cation concentration effects have served to detect and quantify the contributions of neutralization of DNA phosphates and/or reduction in phosphate charge density (the polyelectrolyte effect).

Acknowledgments

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[17] Detecting Cooperative Protein–DNA Interactions and DNA Loop Formation by Footprinting

By ANN HOCHSCHILD

Introduction

Weak interactions between adjacently and nonadjacently bound transcriptional regulators play an essential role in gene regulation in both prokaryotes and eukaryotes. In particular, many examples of “action at a distance” (i.e. regulatory events mediated by proteins bound far from the start point of transcription) involve long-range interactions between proteins bound at separated sites on the DNA. Such interactions induce the formation of loops in the DNA (for reviews, see Refs. 1 and 2).

Long-range protein–protein interactions were first implicated in the regulation of transcription by studies of the arabinose and galactose operons of *Escherichia coli*. The classical mechanism of transcriptional repression in prokaryotes is steric occlusion; the promoter and operator are overlapping sites and bound repressor prevents RNA polymerase from gaining access to the promoter. However, transcriptional repression of the arabinose operon was found to depend on an operator site located far upstream of the promoter.³ This repression depends on an interaction between a molecule bound at the upstream site (located at position –280

¹ S. Adhya, *Annu. Rev. Genet.* **23**, 227 (1989).

² A. Hochschild, in “DNA Topology and Its Biological Effects” (N. R. Cozzarelli and J. C. Wang, eds.), p. 107. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1990.

³ T. M. Dunn, S. Hahn, S. Ogden, and R. F. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 5017 (1984).

relative to the start point of transcription) and a second molecule bound near the start point of transcription.⁴ Repression of the galactose operon also depends on an interaction between nonadjacently bound repressors, one bound upstream of the promoter (at position -61) and the other bound within the first structural gene (at position +53).⁵⁻⁸ These are examples of two different mechanisms for transcriptional repression, both of which involve the formation of a DNA loop.

Transcriptional regulators of other prokaryotic operons also participate in long-range interactions,⁹⁻¹⁵ as do DNA-binding proteins involved in other processes, such as site-specific recombination and DNA replication.¹⁶⁻²⁰ The function of these interactions differs from example to example. In some cases the interaction may serve simply to increase site occupancy. At the *lac* operon, for example, occupancy of the primary operator (which overlaps the promoter) is enhanced by the presence of two pseudo-operators, located 93 bp upstream and 401 bp downstream of the primary operator, respectively.¹² In other cases, the interaction may serve to impose a structural constraint on the intervening DNA. At the *gal* operon, for example, site occupancy alone is not sufficient to mediate repression; rather the looped structure contributes directly to repression.⁸ Finally, long-range interactions may serve to bring together noncontiguous segments of DNA (such as in site-specific recombination) or to promote the formation of a well-defined higher order structure.

Several methodologies, including DNase footprinting, the gel-binding assay, and electron microscopy, have been exploited to demonstrate protein-mediated DNA loop formation *in vitro*. This chapter will focus on

⁴ K. Martin, L. Huo, and R. F. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 3654 (1986).

⁵ M. H. Irani, L. Orosz, and S. Adhya, *Cell (Cambridge, Mass.)* **32**, 783 (1983).

⁶ H.-J. Fritz, H. Bicknase, B. Gleumes, C. Heibach, S. Rosahl, and R. Ehrling, *EMBO J.* **2**, 2129 (1983).

⁷ A. Majumdar and S. Adhya, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 6100 (1984).

⁸ R. Haber and S. Adhya, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 9683 (1988).

⁹ M. C. Mossing and M. T. Record, *Science* **233**, 889 (1986).

¹⁰ H. Kramer, M. Niemoller, M. Amouyal, B. Revet, B. V. Wilcken-Bergmann, and B. Muller-Hill, *EMBO J.* **6**, 1481 (1987).

¹¹ S. Sasse-Dwight and J. D. Gralla, *J. Mol. Biol.* **202**, 107 (1988).

¹² S. Oehler, E. R. Eismann, H. Kramer, and B. Muller-Hill, *EMBO J.* **9**, 973 (1990).

¹³ G. Dandanell and K. Hammer, *EMBO J.* **4**, 3333 (1985).

¹⁴ L. J. Reitzer and B. Magasanik, *Cell (Cambridge, Mass.)* **45**, 785 (1986).

¹⁵ A. J. Ninfa, L. J. Reitzer, and B. Magasanik, *Cell (Cambridge, Mass.)* **50**, 1039 (1987).

¹⁶ H. Echols, *Science* **133**, 1050 (1986).

¹⁷ M. Gellert and H. Nash, *Nature (London)* **325**, 401 (1987).

¹⁸ N. L. Craig, *Annu. Rev. Genet.* **22**, 77 (1988).

¹⁹ R. C. Johnson and M. I. Simon, *Trends Genet.* **3**, 262 (1987).

²⁰ S. Mukherjee, H. Erickson, and D. Bastia, *Cell (Cambridge, Mass.)* **52**, 375 (1988).

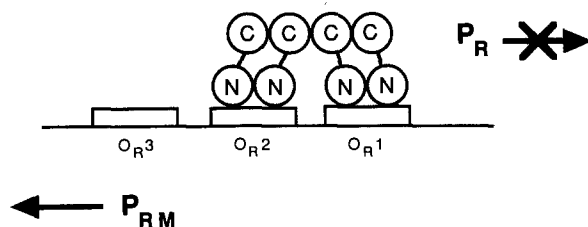


FIG. 1. Cooperative binding of λ repressor to adjacent sites in the right operator (O_R). In a λ lysogen repressor dimers bind cooperatively to O_{R1} and O_{R2} , thereby repressing transcription from promoter P_R and activating transcription from promoter P_{RM} . O_{R1} is a high-affinity site, whereas O_{R2} and O_{R3} are lower affinity sites. Thus, the dimer at O_{R1} stabilizes the association of the second dimer with O_{R2} and, at physiological concentrations of repressor, O_{R3} remains mostly unoccupied because only two dimers cooperate at one time. Note that the carboxyl domain of repressor mediates the cooperative interaction.

DNase footprinting, as applied to complexes formed by the binding of λ repressor to artificially separated operator sites.^{21,22}

Repressor Structure

λ repressor is a two-domain protein that binds its 17-bp operator as a dimer. Repressor dimers bind cooperatively to adjacent operator sites on the phage chromosome (see Fig. 1).²³ The amino domain of the repressor contacts the DNA, whereas its carboxyl domain mediates both dimerization and the cooperative interaction between adjacently bound dimers.^{24,25}

Cooperative Binding

The DNase footprinting assay for protein-mediated DNA loop formation is based on the fact that an interaction between DNA-bound proteins will be reflected in cooperative DNA binding. This cooperativity should be detectable *in vitro* whether or not it is physiologically relevant *in vivo*. (Depending on the intracellular concentration of protein, site occupancy *in vivo* may or may not be enhanced appreciably by cooperative binding.)

²¹ A. Hochschild and M. Ptashne, *Cell (Cambridge, Mass.)* **44**, 681 (1986).

²² A. Hochschild and M. Ptashne, *Nature (London)* **336**, 353 (1988).

²³ A. D. Johnson, B. J. Meyer, and M. Ptashne, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 5061 (1979).

²⁴ C. O. Pabo, R. T. Sauer, J. M. Sturtevant, and M. Ptashne, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 1608 (1979).

²⁵ The amino domain itself dimerizes weakly and binds DNA even when separated from the carboxyl domain.

Cooperative binding is detected as follows. A template is prepared carrying two protein-binding sites (A and B). A second template is prepared carrying only one of the two sites (B, for example). Footprints are performed under equilibrium conditions to determine whether or not the affinity of the protein for site B is higher when site A is present on the same DNA fragment (see Fig. 2).

More specifically, we wish to determine the equilibrium dissociation constant for the binding of a repressor dimer to site B in the presence and absence of site A on the same DNA fragment. This equilibrium can be represented as follows



where O represents free operator, R_2 represents free dimer, and R_2O represents bound operator.²⁶ The equilibrium dissociation constant for the repressor dimer/operator interaction is defined

$$K_D = [R_2][O]/[R_2O] \quad (2)$$

At half-maximal site occupancy $[O] = [R_2O]$, and $K_D = [R_2]$. The quantity that we can actually measure in a footprint experiment is the total repressor concentration at which half-maximal site occupancy is achieved $[R_T]_{1/2}$. To determine K_D we must, therefore, be able to derive $[R_2]_{1/2}$ from $[R_T]_{1/2}$. This calculation is complicated by the fact that repressor exists as a mixture of monomers and dimers at equilibrium.²⁷ The monomer/dimer equilibrium



is governed by the equilibrium dissociation constant

$$K_1 = [R]^2/[R_2] = 2 \times 10^{-8} M \quad (4)$$

If the operator concentration is low compared with the total repressor concentration at which half-maximal site occupancy is achieved, the presence of the operator-bearing DNA will not significantly perturb the monomer/dimer equilibrium, and we may calculate $[R_2]_{1/2}$ as follows. Using the stoichiometric relation

$$[R_T] = [R] + 2[R_2] + 2[R_2O] \quad (5)$$

we replace $[R_2]$ in Eq. (4), obtaining

$$K_1 = 2[R]^2/([R_T] - [R] - 2[R_2O])$$

or

²⁶ A. D. Johnson, C. O. Pabo, and R. T. Sauer, this series, Vol. 65, p. 839.

²⁷ R. T. Sauer, Ph.D. Dissertation, Harvard University, Cambridge, Massachusetts (1979).

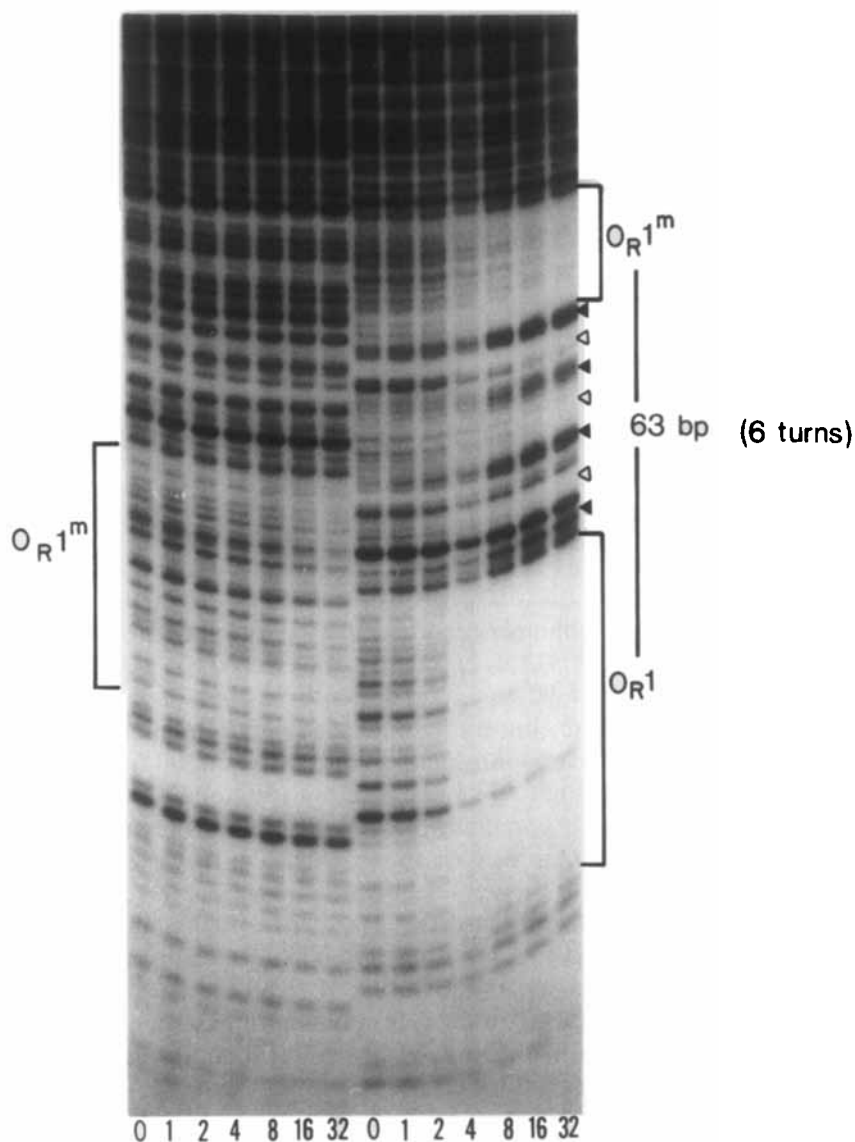


FIG. 2. DNase I footprinting assay showing that λ repressor binds cooperatively to operator sites separated by six turns of the DNA helix. (Reprinted from Ref. 21; copyright held by Cell Press, Cambridge, MA.) The binding of repressor to operator site O_{R1}^m is measured in the absence (left panel) and presence (right panel) of a high-affinity site (O_{R1}) six turns away on the same DNA template. Reactions were incubated with increasing concentrations of repressor, a concentration of 1 corresponding to 13.5 nM repressor monomer. The experiment shows that between 16 and 32 relative units of repressor was required to occupy O_{R1}^m half-maximally on the single-site template, whereas only 4 relative units was required on the two-site template. Changes in the DNase I sensitivity of the region spanning the two sites are indicated by arrowheads; the black arrowheads indicate enhanced cleavages and the open arrowheads indicate diminished cleavages.

$$[R]^2 + (K_1/2)[R] = (K_1/2)Q \quad (6)$$

where

$$Q = [R_T] - 2[R_2O]$$

Solving Eq. (6) for $[R]$, we find

$$[R] = -(K_1/4) + \sqrt{(K_1/4)^2 + (K_1/2)Q} \quad (7)$$

Since $[R_T] \gg [O_T]$, we may replace Q with $[R_T]$, giving

$$[R] = -(K_1/4) + \sqrt{(K_1/4)^2 + (K_1/2)[R_T]} \quad (8)$$

We thus obtain $[R]$ as a function of the known quantities K_1 and $[R_T]_{1/2}$. We then use Eq. (5) to solve for $[R_2]$, i.e., K_D . Again, since $[R_T] \gg [O_T]$, we can simplify Eq. (5):

$$2[R_2] = [R_T] - [R] \quad (9)$$

Consider a numerical example. Assume that half-maximal occupancy of a given site occurs at a total repressor concentration of $10^{-8} M$ (expressed as repressor monomer) and that the concentration of operator containing DNA is $< 10^{-10} M$. Using Eq. (8), $[R] = 6 \times 10^{-9} M$, and from Eq. (9) $[R_2] = 2 \times 10^{-9} M$. Note that under these conditions we do not need to know the precise concentration of the operator containing DNA and, further, that small variations in operator concentration from experiment to experiment will not affect our determination of K_D .

The extra binding energy coming from the cooperative interaction will be reflected primarily in an increase in the affinity of the protein for the weaker of the two sites if they differ in strength (see Table I and legend for some sample values and derivation). This is because among the singly bound DNA molecules the strong site is more likely to be occupied than the weak, and the presence of the strong site will increase binding to the weak site more than the other way around. In practice, therefore, cooperativity is most easily detected by examining binding to a relatively weak site in the presence and absence of a strong site.

The Effect of Inter-operator Spacing on DNA Loop Formation

In our experiments with λ repressor we observed a periodic dependence of cooperativity on the spacing between the two binding sites. We detected cooperative binding when the center-to-center distance between the two operators was an integral number (n) of turns of the DNA helix (5, 6, or 7), assuming 10.5 bp/turn, but not when the inter-operator distance was close to $n + 1/2$ turns (4.6, 5.5, or 6.4). Since repressor binds primarily to one side of the DNA helix, the bound repressor molecules are located

TABLE I
THE EFFECT OF COOPERATIVITY ON SITE OCCUPANCY^a

$K_1 (M)$	$K_2 (M)$	K_{coop}	$K_2^* (M)$	K_2/K_2^*
3.3×10^{-9}	3.3×10^{-9}	3.6×10^{-2}	6.3×10^{-10}	5
3.3×10^{-9}	5.0×10^{-8}	3.6×10^{-2}	3.5×10^{-9}	14
3.3×10^{-9}	5.0×10^{-7}	3.6×10^{-2}	2.1×10^{-8}	24
3.3×10^{-9}	5.0×10^{-6}	3.6×10^{-2}	1.8×10^{-7}	28

^a Consider two operators, O_1 and O_2 , located on the same DNA molecule. K_1 and K_2 are dissociation constants that describe the intrinsic affinities of O_1 and O_2 for repressor and K_{coop} is a constant that describes the cooperative term (defined below). The values chosen for K_{coop} , $^{28} K_1$ (all lines), and for K_2 (line 2) are based on the λ system; 3.3×10^{-9} and $5 \times 10^{-8} M$ are the intrinsic dissociation constants for the binding of repressor to O_{R1} and O_{R2} , respectively. K_2^* is a dissociation constant for the binding of repressor to O_2 when the binding is cooperative, and the ratio K_2/K_2^* is thus the factor by which binding to O_2 increases due to cooperativity. The table shows that for fixed values of K_1 and K_{coop} , the ratio K_2/K_2^* increases as O_2 is weakened. That is, the weaker O_2 is compared with O_1 , the more the energy of cooperativity is reflected in an increase in the affinity of O_2 for repressor. For the given value of K_{coop} , the ratio K_2/K_2^* approaches a limiting value of approximately 28 when essentially all of the energy of cooperativity contributes to an increase in the affinity of O_2 for repressor.

To understand the meaning of K_{coop} , consider the following equilibrium relations: $K_1 = [R][O]/[RO_1]$ and $K_2 = [R][O]/[RO_2]$, where $[R]$ denotes the concentration of free repressor, $[O]$ the concentration of free DNA molecules, $[RO_1]$ the concentration of DNA molecules on which O_1 alone is occupied, and $[RO_2]$ the concentration of DNA molecules on which O_2 alone is occupied. If the binding to the two sites is noncooperative, then the dissociation of R_2O (a DNA molecule on which both O_1 and O_2 are occupied) to free repressor and free operator is described by the following equilibrium relation: $K_1 K_2 = [R]^2 [O]/[R_2O]$. If binding to the two sites is cooperative, we define K_{coop} such that $[R]^2 [O]/[R_2O] = K_1 K_2 K_{\text{coop}}$. K_{coop} can be determined experimentally by comparing the intrinsic affinity of each operator site for repressor with the affinity of that site when the binding is cooperative. That is, K_1 and K_2^* can be measured in separate experiments, as can K_2 and K_2^* . $K_{\text{coop}} = K_2^* K_2 / K_1 K_2$.

The following calculation was performed to compute K_2/K_2^* for each value of K_2 (see Ackers *et al.*²⁸ for a detailed discussion of the λ system). Let fO_2 stand for the fractional occupancy of O_2 . We can write:

$$fO_2 = \frac{[RO_2] + [R_2O]}{[O] + [RO_1] + [RO_2] + [R_2O]}$$

Dividing top and bottom by $[O]$ and using the equilibrium relations, $K_1 = [R][O]/[RO_1]$, $K_2 = [R][O]/[RO_2]$, and $K_1 K_2 K_{\text{coop}} = [R]^2 [O]/[R_2O]$, we can rewrite the equation:

$$fO_2 = \frac{[R]/K_2 + [R]^2/K_1 K_2 K_{\text{coop}}}{1 + [R]/K_1 + [R]/K_2 + [R]^2/K_1 K_2 K_{\text{coop}}}$$

We can now set $fO_2 = 0.5$ and solve for $[R]$, which equals K_2^* . Performing the appropriate algebra gives the simplified equation:

$$[R]^2/K_1 K_2 K_{\text{coop}} + \{(K_1 - K_2)/K_1 K_2\}[R] - 1 = 0$$

on the same side of the helix when the operators are separated by an integral number of turns and on opposite sides when the sites are separated by $n + 1/2$ turns. To bring together two molecules that are bound on opposite sides of the DNA helix, bending of the DNA is insufficient; twisting and/or writhing is also required.

The amount of energy required to twist the DNA and bring sites that are on opposite sides of the helix into alignment for loop formation can be derived from the free energy of supercoiling, $\Delta G\tau$, which has been measured as a function of DNA length by Horowitz and Wang.²⁹ According to their data the cost of misalignment by half a turn is less than 0.1 kcal/mol for separations greater than ~ 2500 bp and ~ 3 kcal/mol for separations ~ 200 bp in length. With separations that are less than 100 bp, the cost of misalignment is predicted to be very high: ~ 7 kcal for 100 bp and ~ 13 kcal for 50 bp.³⁰ Therefore, the fact that we fail to observe cooperative binding to nonintegrally spaced sites separated by less than 100 bp is not surprising.

To strengthen the argument that the observed cooperative binding was mediated by a direct protein-protein interaction and that the apparent side-of-the-helix effect was due to the difficulty of bringing together molecules bound on opposite sides of the helix, we introduced a small gap into one strand of the DNA between two operators separated by a nonintegral number of turns (see Fig. 3A). Under these circumstances the DNA is

²⁸ G. K. Ackers, A. D. Johnson, and M. Shea, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 1129 (1982).

²⁹ D. Horowitz and J. C. Wang, *J. Mol. Biol.* **173**, 75 (1984).

³⁰ J. C. Wang and G. N. Giaever, *Science* **240**, 300 (1988).

or

$$[R]^2 + (K_1 - K_2)K_{\text{coop}}[R] - K_1K_2K_{\text{coop}} = 0$$

Solving for $[R]$, we find

$$[R] = \frac{(K_2 - K_1)K_{\text{coop}} + \sqrt{(K_1 - K_2)^2K_{\text{coop}}^2 + 4K_1K_2K_{\text{coop}}}}{2}$$

Note that when $K_2 = K_1$, $[R] = K_1\sqrt{K_{\text{coop}}}$, and when $K_2 \gg K_1$ (O_2 significantly weaker than O_1),

$$\begin{aligned} [R] &= \frac{K_2K_{\text{coop}} + \sqrt{K_2^2K_{\text{coop}}^2 + 4K_1K_2K_{\text{coop}}}}{2} \\ &= \frac{K_2K_{\text{coop}} + \sqrt{K_2K_{\text{coop}}(K_2K_{\text{coop}} + 4K_1)}}{2} \end{aligned}$$

Finally, when K_2 is several orders of magnitude greater than K_1 , $K_2K_{\text{coop}} \gg 4K_1$, $[R]$ approaches the limiting value K_2K_{coop} , and the ratio K_2/K_2^* approaches $1/K_{\text{coop}}$.

expected to turn freely around the single-stranded region, and therefore the energetic barrier that would otherwise prevent cooperative binding should be removed. This prediction was borne out and repressor bound cooperatively to operators separated by 6.4 turns when a four-nucleotide gap was introduced into one of the DNA strands roughly midway between the operators (Fig. 3B).²¹

It should be noted that there may be proteins that bind cooperatively to nonadjacent operator sites irrespective of their angular alignment. For example, consider a hypothetical protein that binds its operator as a dimer and wraps around the DNA in such a way that the monomers lie on opposite sides of the DNA helix. Given such a configuration, one can imagine interactions involving either two or four monomers that would occur regardless of the spacing between the operators. It should also be noted that flexibility in the protein might reduce the amount of twisting that would be required to allow molecules that are not optimally aligned to interact. Finally, as the distance between binding sites is increased, any periodic effect of spacing is expected to diminish. Thus, although the observation of a periodic effect of spacing on cooperativity argues in favor of a direct protein-protein interaction, the absence of such an effect does not rule out the interaction.

Changes in the Susceptibility of Bent DNA to DNase Cleavage

The distortion of the DNA backbone that is induced by the cooperative binding of λ repressor to separated operators results in a change in the sensitivity of the DNA to DNase I cleavage. The binding of repressor to integrally spaced operators is accompanied by the appearance of an alternating set of enhanced and diminished cleavages affecting the DNA between the two operators (see Fig. 2). The enhanced and diminished cleavages are separated by roughly 5 bp; assuming that the interaction between the bound repressors induces a smooth bend in the DNA, the enhanced cleavages lie on the outside of the protein/DNA complex, whereas the diminished cleavages lie on the inside of this complex. A similar pattern of enhanced and diminished cleavages was previously observed with a small circular DNA molecule,³¹ and it has also been observed in other cases of protein-mediated loop formation.³²⁻³⁵

³¹ H. R. Drew and A. A. Travers, *J. Mol. Biol.* **186**, 773 (1985).

³² H. Kramer, M. Niemoller, M. Amouyal, B. Revet, B. V. Wilcken-Bergmann, and B. Muller-Hill, *EMBO J.* **6**, 1481 (1987).

³³ J. J. Salvo and N. D. F. Grindley, *EMBO J.* **7**, 3609 (1988).

³⁴ D. Valenzuela and M. Ptashne, *EMBO J.* **8**, 4345 (1989).

³⁵ P. A. Beachy, unpublished observations with a Ubx protein from *Drosophila*.

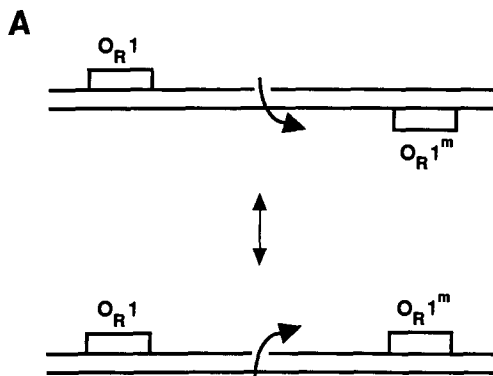


FIG. 3. The binding of repressor to a gapped duplex carrying nonintegrally spaced operator sites. (A) Schematic illustration of the gapped duplex, indicating free rotation around the single-stranded region. (B) DNase footprinting assay showing that repressor binds cooperatively to operator sites separated by 6.4 turns of the DNA helix on a gapped template. (Reprinted from Ref. 21; copyright held by Cell Press, Cambridge, MA.) The DNA template in the left six lanes is an intact duplex carrying O_R1 and O_R1^m separated by 6.4 turns of the DNA helix, and the DNA template in the right six lanes is the same except that it bears a four-nucleotide gap (on the unlabeled strand) in the region spanning the two operators. A repressor concentration of 1 corresponds to 13.5 nM repressor monomer. The experiment shows that ~ 16 relative units of repressor was required to occupy O_R1^m half-maximally on the intact duplex, whereas only about 2 relative units was required on the gapped duplex. Note that the presence of the gap changes the cleavage pattern between the two operators. However, the binding of repressor to the gapped template does not induce a pattern of enhanced and diminished cleavages, as is observed with the ungapped template.

These periodic changes in the susceptibility to DNase I cleavage of the DNA spanning two binding sites are a very sensitive indicator of loop formation. Under conditions where the stability of the looped and unlooped complexes are similar and cooperative binding may not be detectable, one may nevertheless detect changes in the cleavage pattern arising from a subpopulation of protein/DNA complexes that is looped.

The enhanced and diminished cleavages that are observed with bent DNA presumably reflect the narrowing and widening of the minor groove on the inside and outside of the bend, respectively. In addition, the DNA on the inside of the bent protein/DNA complex may be less accessible to DNase I. The presence of these anomalies in the cleavage pattern suggests that there is a single preferred conformation of the bent DNA in the looped complex. In fact, these anomalies are not seen when repressor binds to gapped DNA; presumably this is because the gapped DNA adopts a variety of conformations, or because the gapped DNA is kinked rather than bent smoothly. Likewise, when the protein-binding sites are located relatively far apart we would not necessarily expect to observe the anomalous cleav-

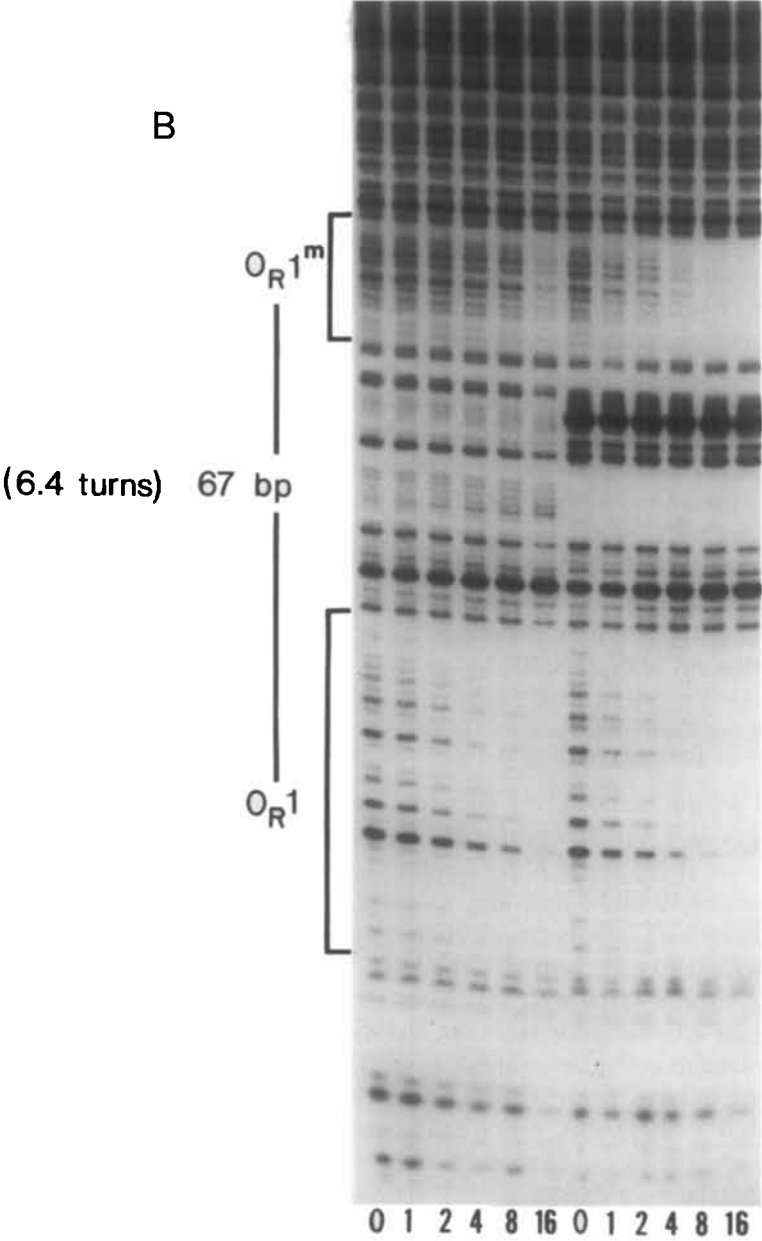


FIG. 3. (continued)

ages, because the conformation of the intervening DNA is apt to be more variable.

Protein Fragments and Fusion Proteins

In the case of λ repressor, cooperative binding to separated operator sites, like cooperative binding to adjacent operators, depends on the carboxyl domain; the purified amino domain binds noncooperatively to operators separated by both integral and nonintegral numbers of turns. Furthermore, a single amino acid substitution in the carboxyl domain of the repressor eliminates cooperative binding to both nonadjacent and adjacent operators.²²

When carrying out experiments with protein fragments or with fusion proteins one may, of course, fail to observe cooperative binding even though the intact protein binds cooperatively. However, it is also possible that a fusion protein may bind cooperatively although the native protein does not. The foreign protein may, for example, mediate the formation of oligomers that can bind simultaneously to two binding sites. This situation would be analogous to that of *lac* repressor, which forms stable tetramers that can mediate loop formation by binding simultaneously to a pair of nonadjacent operators.^{10,11}

Effect of Distance on Cooperative Binding

It would be useful to know how the probability of loop formation changes as a function of the distance separating the two protein-binding sites (see Refs. 2 and 30 for a more detailed discussion). When the sites are relatively close together, as are the λ operators on the constructs described here, this is a very complex problem. However, when the sites are sufficiently far apart on the DNA so that the size of the protein is negligible compared with the distance separating the binding sites, we can consider an analogous reaction, the cyclization of DNA fragments with complementary single-stranded ends. The cyclization probabilities, called *j* factors, have been measured previously as a function of fragment length.³⁶⁻³⁸ The *j* value, which can be defined as the ratio of two equilibrium constants, K_c/K_a , where K_c is the cyclization constant and K_a is the bimolecular equilibrium association constant for joining two linear molecules, is a measure of the effective concentration of one end of the molecule

³⁶ J. C. Wang and N. R. Davidson, *J. Mol. Biol.* **19**, 469 (1966).

³⁷ J. E. Mertz and R. W. Davis, *Proc. Natl. Acad. Sci. U.S.A.* **69**, 3370 (1972).

³⁸ D. Shore, J. Langowski, and R. L. Baldwin, *Proc. Natl. Acad. Sci. U.S.A.* **78**, 4833 (1981).

in the vicinity of the other. Both theoretical and experimental analyses indicate that j reaches a maximum of $\sim 10^{-7} M$ for DNA fragments of ~ 700 bp. With decreasing fragment length, DNA flexibility becomes a critical parameter and the j value is determined primarily by an enthalpic term that describes the cost of deforming and bending the DNA. With increasing fragment length, on the other hand, the DNA more nearly resembles a flexible coil and the j value is determined primarily by an entropic term that describes the loss in configurational entropy of the DNA molecule when its ends are held in proximity to one another. This loss increases with fragment length; that is, the probability of one end finding the other decreases with increasing fragment length.

Based on this analogy with ring formation, the effective concentration of one DNA-bound protein in the vicinity of the other is not expected to exceed $10^{-7} M$, at least *in vitro*. We can therefore expect to observe loop formation over some range of intersite separations provided that the K_a for the bimolecular reaction of two DNA bound proteins exceeds $10^7 M^{-1}$. This follows since K_c , the equilibrium constant for ring formation, equals $j(K_a)$, which is >1 if $K_a > 10^7 M^{-1}$ (see Fig. 4). If we know the equilibrium association constant for the relevant protein-protein interaction in solution ($\sim 10^6 M^{-1}$ for the association of free λ repressor dimers to form tetramers)²⁷ it is reasonable to assume that the K_a for the association of proteins bound to separate DNA molecules will be at least as strong. If there is an allosteric effect of DNA binding on the oligomerization reaction, the DNA-bound molecules may interact more strongly than free molecules in solution. It is also possible that an entropic effect may favor the interaction of DNA-bound molecules; for example, their immobilization on the DNA may serve to orient the interacting surfaces. In the case of λ repressor we do not observe loop formation at large separations, suggesting that the K_a for the bimolecular association of the DNA-bound dimers is less than $10^7 M^{-1}$, or less than 10 times the value measured for free dimers ($10^6 M^{-1}$).

Now let us consider relatively short separations. If the size of the protein is significant compared with the distance separating the two binding sites, the looped complex will not in fact resemble a DNA ring. In the case of interacting λ repressor molecules separated by about six turns of the helix the complex can be modeled as a semicircle bridged by the interacting protein molecules. Under these circumstances both DNA flexibility and protein flexibility are expected to be important parameters. Treating the protein as a rigid body one might expect that increasing the length of DNA over which the bend is distributed would facilitate complex formation. However, our observations with λ repressor are inconsistent with this expectation. The magnitude of the cooperative effect is relatively constant

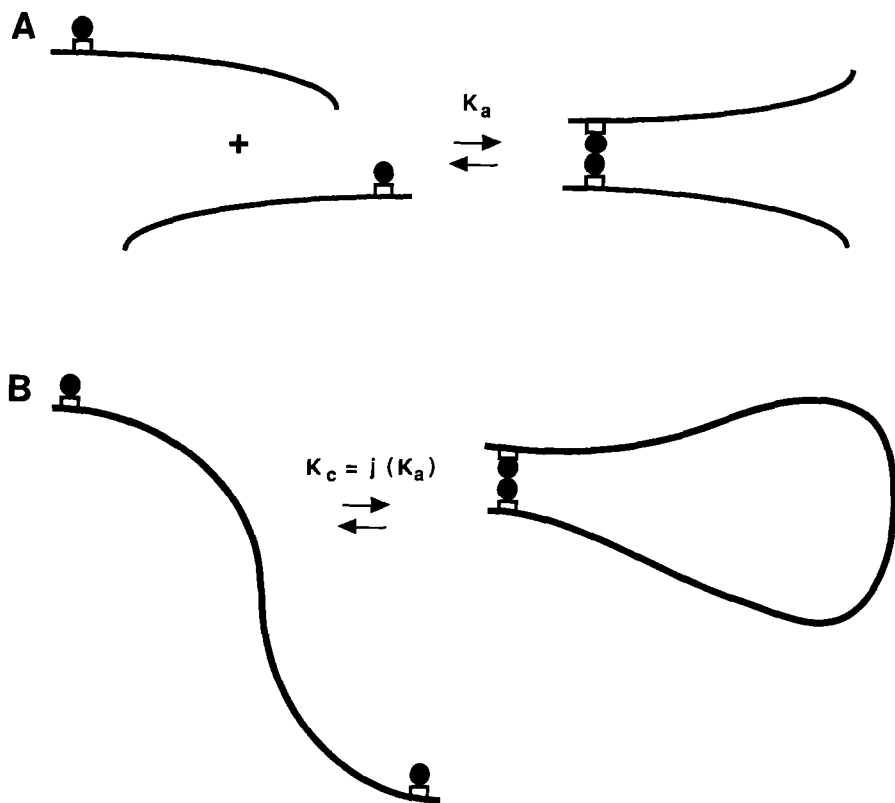


FIG. 4. Using the j value to estimate the probability of protein-induced loop formation. (A) The bimolecular association of two DNA-bound protein molecules. K_a is the bimolecular equilibrium constant for the association of two protein molecules that are prebound to separate DNA fragments. (B) The unimolecular association of two DNA-bound protein molecules. K_c is the unimolecular equilibrium constant for the association of two protein molecules that are prebound to a single DNA molecule. The j value is defined as the ratio K_c/K_a (see text), and so $K_c = j(K_a)$. Loop formation is favored provided that $K_c > 1$.

at 5, 6, 7, and 8 turns (within the sensitivity of our assay), and then diminishes gradually until 20 turns, at which point little or no cooperativity can be detected. These observations suggest that the detailed conformation of the looped protein-DNA complex may not be constant over this range of separations and that factors other than DNA flexibility are playing a significant role in determining the overall energetics of loop formation.

Construction of DNA Templates Carrying One or Two Operators

Several related templates were constructed carrying one or both of two repressor-binding sites, O_{R1} and a mutant derivative of O_{R1} (O_{R1}^m) to

which repressor binds more weakly ($K_D O_{R1}^m = 8[K_D O_{R1}]$). To construct this series of templates we inserted a 45-bp DNA fragment carrying λO_{R1} at various positions within the polylinker of pEMBL8+, as outlined in Fig. 5.

To generate the gapped DNA template we prepared heteroduplexes by annealing an excess of single-stranded circles derived from the 6.4-turn plasmid to the 6.0-turn parent plasmid linearized at the *Bam*HI site (see Fig. 5). The single-stranded circles were prepared by superinfecting plasmid-containing cells with phage F1 as described by Dente and Cortese.³⁹ The two DNA samples were mixed so that the single-stranded circles were present in an approximate fivefold molar excess over the double-stranded linears. The DNA in this mixture was denatured and renatured according to a procedure described by Kirkegaard and Wang⁴⁰ to produce the gapped circles.

Preparation of Labeled Restriction Fragments

The plasmid DNAs ($\sim 3 \mu\text{g}$) were digested with *Hind*III and *Pvu*II (see map) to generate an operator-containing fragment of ~ 250 bp in the case of the single-operator construction and fragments of ~ 300 bp in the case of the two-operator constructions. The gapped circles were treated in the same manner as the intact plasmids. The fragments were 3' end labeled at the *Hind*III end with [α -³²P]dATP and reverse transcriptase (purchased from Life Sciences, St. Petersburg, FL). On the gapped template this protocol results in the incorporation of the label at the end of the intact strand. The radiolabeled fragments were purified after electrophoresis through a 5% polyacrylamide gel. In each case the appropriate band was excised from the gel. The following protocol was used to elute the DNA: The gel slice is placed in a 15-ml tube and crushed with a glass rod. After the addition of 2.5 ml elution buffer (500 mM NH_4Ac , 10 mM MgAc , 1 mM EDTA, 0.1% SDS),⁴¹ the tubes are rocked overnight at 37°. The samples are then centrifuged for several minutes in a clinical centrifuge to pellet the acrylamide. The supernatant containing the labeled DNA is poured into a clean tube. After the addition of $\sim 15 \mu\text{g}$ of carrier tRNA, it is dripped through a 0.45- μm Millex filter (Millipore, Bedford, MA) into a 30-ml centrifuge tube. Ethanol (2.5 vol) is added and the labeled DNA is precipitated. The DNA pellet is redissolved in 400 μl of 0.3 M NaOAc, transferred to an Eppendorf tube, and reprecipitated.

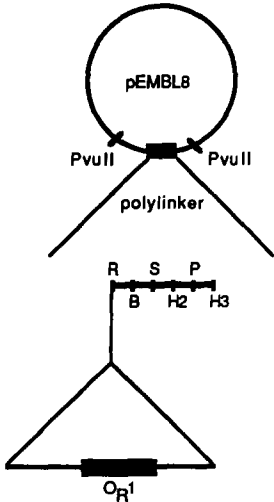
To obtain an estimate of the specific activity of each DNA fragment, the amount of radioactivity in the dried DNA pellet is determined by

³⁹ L. Dente and R. Cortese, this series, Vol. 155, p. 111.

⁴⁰ K. Kirkegaard and J. C. Wang, *J. Mol. Biol.* **185**, 625 (1985).

⁴¹ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

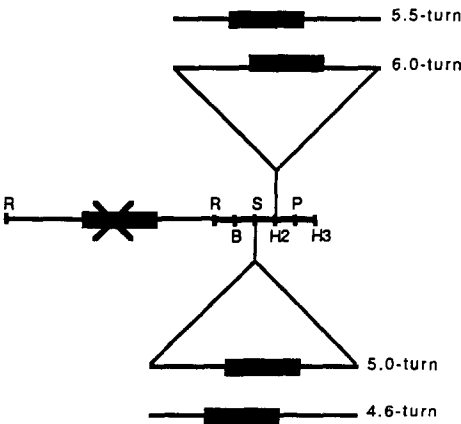
A



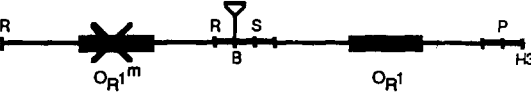
B



C



D



Cerenkov counting. For the purpose of these estimates, we assume 100% recovery of each gel-purified DNA fragment (so as not to underestimate the concentration of operator in the final reaction). Ideally it is desirable to load 10,000 cpm of labeled fragment in each lane so that a suitable footprint can be obtained after overnight exposure of the gel. In our case we wished to keep the concentration of DNA fragment below 0.2 nM, well below the repressor concentration required for half-maximal occupancy of our strongest site, ~20 nM. We generally obtain $1-2 \times 10^6$ cpm/labeled restriction fragment, or 20–40,000 cpm/ 2.4×10^{-14} mol fragment. The reaction volume is 200 μ l (see below), and therefore we can normally use 20,000–40,000 cpm/reaction without exceeding a concentration of 0.12 nM in DNA fragment.

DNase Footprinting Reactions

Our footprinting protocol is essentially that of Johnson *et al.*²³ The reactions are performed in a buffer consisting of 10 mM Tris-HCl (pH 7.0), 2.5 mM MgCl₂, 1 mM CaCl₂, 0.1 mM EDTA, 200 mM KCl, bovine serum albumin at 100 μ l/ml, and chick blood DNA at 2.5 μ l/ml. After the addition of the labeled DNA fragment to an appropriate volume of protection buffer, the mixture is added to microcentrifuge tubes in 200- μ l aliquots. Various amounts of repressor are added to these reaction tubes and the mixtures are allowed to come to equilibrium for 20 min at 37°. Fresh dilutions of the repressor stock solution are prepared for each experiment. The reactions are initiated by the addition of a predetermined amount of DNase I (see below) at 30-sec intervals. Each tube is quickly mixed by gentle vortexing after the addition of the DNase. The mixtures are incubated at 37° for 12 min, and then stopped by the addition of 50 μ l of cold

FIG. 5. Construction of plasmid DNAs carrying one or two λ operators. The single-operator and two-operator constructions were generated as follows: (1) An *Eco*RI fragment carrying λ O_R1 was inserted into the unique *Eco*RI site of pEMBL8+; (2) a single base-pair substitution was introduced into O_R1 by site-directed mutagenesis to generate O_R1^m. The resulting plasmid was the source of the single-operator template; (3) the O_R1 fragment was reintroduced into either the *Sma*I site or the *Hind*II site. O_R1 is not centered on the *Eco*RI fragment, and thus insertion of the fragment in one or the other orientation generates different spacing between O_R1 and O_R1^m. This set of plasmids was the source of four of the two-operator templates; (4) 4 bp was inserted between O_R1 and O_R1^m on the 6.0-turn construction by cutting the plasmid at the unique *Bam*HI site, filling out the 3' ends, and religating the blunt ends, thus generating the 6.4-turn construction. The gapped duplex was prepared by annealing single-stranded circular DNA carrying the 4-bp insertion at the *Bam*HI site to the 6.0-turn parent plasmid linearized at the *Bam*HI site. Restriction sites are abbreviated as follows: *Eco*RI, R; *Bam*HI, B; *Sma*I, S; *Hinc*II, H2; *Pst*I, P; and *Hind*III, H3.

8 M NH_4Ac containing 250 $\mu\text{g/ml}$ tRNA and 1 ml of cold ethanol. The DNA is precipitated by immersing the reaction tubes into a dry ice-ethanol bath for ~ 5 min. The tubes are then centrifuged for 5 min in a microcentrifuge. The pellets should be clearly visible. The supernatant is aspirated; if care is taken to remove all visible droplets it is not necessary to rinse the pellets. After drying the pellets for several minutes under vacuum, the amount of radioactivity in each tube is again determined by Cerenkov counting.

In some cases it is necessary to subject the samples to a phenol/chloroform extraction prior to the precipitation (when using protein extracts, for example). This step was needed when examining the binding of mutant proteins that, because of a low affinity for operator DNA, had to be added at high concentrations. If the final DNA sample does not enter the gel properly, this may be an indication that the phenol/chloroform step is needed. In that case, the reactions are stopped by the addition of 200 μl of phenol/chloroform, the aqueous phase is transferred to a clean tube, and the DNA is precipitated as described above.

Finally the pellets are resuspended in 6–10 μl of loading dye consisting of 80% (v/v) deionized formamide/50 mM Tris-borate (pH 8.3)/1 mM EDTA/0.1% xylene cyanol/0.1% bromphenol blue. Prior to electrophoresis through an appropriate percentage acrylamide sequencing gel (in our case 8%), the samples are heated at 90° for 90 sec and then chilled on ice. If the samples are stored prior to electrophoresis it is preferable to freeze the dry pellets. However, we generally found that samples could be saved and rerun, if necessary, without significant fragment breakage.

Before carrying out a set of footprint experiments, a pilot experiment is performed to determine the optimum concentration of DNase I. The concentration should be adjusted so that each DNA molecule is cleaved only once between the labeled end and a point beyond the region of interest. That is, the probability of obtaining a second cut somewhere between the labeled end and a primary cut within the region of interest should be reduced as much as possible. Thus, the further the region of interest is from the labeled end, the lower the appropriate concentration of DNase I. The pilot experiment is performed over a range of DNase I concentrations to find a concentration at which the DNA is neither underreacted (too much uncleaved fragment and ladder too faint) nor overreacted (little or no uncleaved fragment and smaller fragments more abundant than larger fragments). For the DNA fragments described here we found that adding DNase to a final concentration of ~ 5 ng/ml gave a suitable amount of cleavage. We store our DNase I (purchased from Worthington, Freehold, NJ) in frozen aliquots at 1 mg/ml in H_2O . Fresh dilutions should be prepared for each experiment.

Before performing a series of experiments with a given protein preparation, the DNA mixture should be incubated with protein in the absence of DNase to ensure that there is no contaminating nuclease activity. In our experience, preparations of partially purified λ repressor have a significant amount of endogenous nuclease activity.

Experimental Reproducibility

To facilitate direct comparisons in these experiments, it is advisable to treat the single-site template as a standard and to measure binding to that template along with the two-site templates in any given experiment. If the K_D measurements are not consistent from experiment to experiment, possible sources of the variability include the following: (1) the protein preparation may be unstable; (2) the binding of some proteins (such as λ repressor) is quite temperature dependent and consequently small variations in the incubation temperature may have a significant effect on the K_D ; (3) the concentration of DNA fragment may be too high so that the measured binding constant is not actually independent of DNA concentration. A good test of the latter possibility is to vary the DNA concentration intentionally over a small range and check that the K_D is not affected.

[18] DNA Contacts Probed by Modification Protection and Interference Studies

By ANDREAS WISSMANN and WOLFGANG HILLEN

Introduction

Many methods for studying the sequence-specific recognition of DNA by proteins are available to date. One approach is to obtain cocrystals of the complex of interest and solve its X-ray structure at high resolution. This yields detailed information about the chemical nature of the contacts. To achieve this goal will usually take several years and the results reflect a static description of a dynamic complex. Another approach is to study protein-DNA complexes in solution using chemical methods.¹ This yields information about the complex much more quickly and is also very valuable for the confirmation and extension of crystallographic results to solution conditions.

The use of chemical methods relies on the fact that some chemicals react specifically with DNA and lead to breakage of the strand at the site of modification. Using mild reaction conditions only a single hit per molecule of DNA needs to be achieved, leading to a nucleotide-specific cleavage pattern. This may be influenced by the presence of a protein bound tightly to a specific sequence in the fragment. The result can either be protection of a nucleotide covered by the protein or enhanced reaction of another nucleotide in close proximity to the protein. This approach is outlined schematically in Fig. 1. Some examples of protection experiments are covered by other chapters in this volume. Here, modification protection protocols using dimethyl sulfate for purine N-7 methylation² and diethyl pyrocarbonate for carbethoxylation of purine N-7³ will be presented.

Another particularly quick and informative chemical method to probe DNA-protein contacts is to determine sites where chemical modifications interfere with binding of the protein. In the case of methylation this is complementary to the protection experiment. The general approach of this experiment is outlined in Fig. 2. The crucial step is to separate modified DNA fragments according to their affinity for the binding protein. One portion should contain modifications only at positions where recognition

¹ U. Siebenlist, R. B. Simpson, and W. Gilbert, *Cell (Cambridge, Mass.)* **20**, 269 (1980).

² W. Hillen, K. Schollmeier, and C. Gatz, *J. Mol. Biol.* **172**, 185 (1984).

³ L. Runkel and A. Nordheim, *J. Mol. Biol.* **189**, 487 (1986).

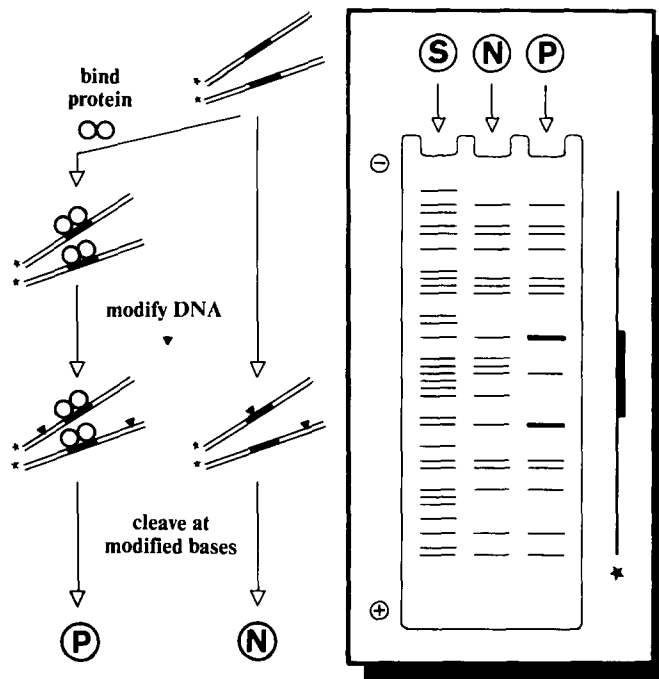


FIG. 1. Schematic outline of a modification-protection experiment. The left-hand side displays the experimental approach. DNA fragments are shown as bars with the protein-binding site in black. Asterisks denote a radioactive label at one end of a DNA strand. Protein is represented by a circle and is shown as a dimer. The modification is indicated by the black triangle. For the protection experiment the protein-DNA complex is modified and the DNA cleaved at the modified bases (P). In parallel, the same procedure is done for free DNA (N). At the right of the gel a schematic representation of the radioactively labeled DNA strand is given, the protein-binding site being indicated by the black box. (S) denotes a Maxam and Gilbert G + A reaction of the same DNA fragment as used in (P) and (N). In lane P residues protected by the protein can be identified as missing bands and residues with enhanced reactivity by increased intensities.

is not affected. The other portion contains modifications in sites preventing protein binding. This goal can either be achieved by polyacrylamide gel electrophoresis or by nitrocellulose filtration. In this chapter, interference experiments using dimethyl sulfate (purine N-7 methylation), *N*-ethyl-*N*-nitrosourea (phosphate ethylation), and diethyl pyrocarbonate (purine N-7 carboxymethylation) will be described.

Both experiments, modification protection (Fig. 1) and interference (Fig. 2), identify specific sites of the DNA that are located close to the bound protein. Experimental conditions for both approaches will be detailed and examples are presented for Z-DNA-antibody³ and Tet repres-

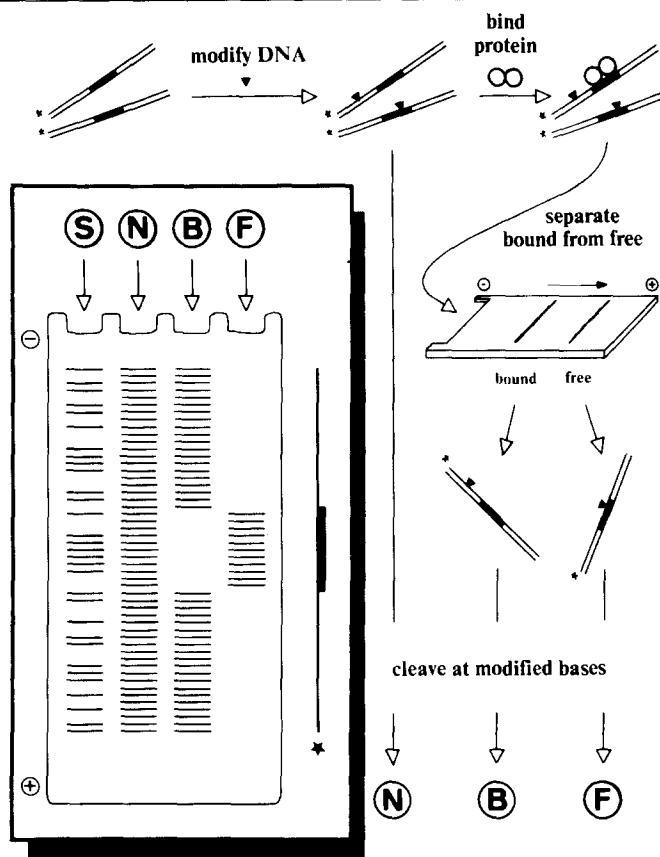


FIG. 2. Schematic outline of a modification interference experiment. The upper and right-hand side display the experimental approach. DNA, protein, and reagent are symbolized as in Fig. 1. The DNA is modified and a part of the sample is directly cleaved at the modified bases (N). The remaining portion of the modified DNA is incubated with the protein to allow binding. The bound and nonbound (free) fractions of the DNA are separated by gel electrophoresis, eluted, cleaved, and analyzed on sequencing gels (B, bound DNA; F, free DNA). On the left-hand side the anticipated results are presented. At the right of the gel a schematic representation of the radioactively labeled DNA strand is depicted with the protein-binding site indicated by a black box. (S) denotes a Maxam and Gilbert G + A reaction of the DNA fragment. The three other lanes show typical results of an ethylation interference experiment. Lane N shows all possible lengths of cleaved DNA fragments indicating the non-base-specific modification. DNA fragments analyzed in lane B define those sites of the DNA, where modification does not interfere with protein binding. The opposite holds true for sample F. Here, DNA fragments indicate that the modification resulted in interference with binding of the protein.

sor-*tet* operator complexes.^{2,4} The technical aspects will be presented first, followed by a description of the results. Finally, the interpretation of the results will be discussed.

Modification of DNA

The DNA of interest needs to be labeled at one end of a single strand. This is achieved by published procedures using either [γ -³²P]dATP and polynucleotide kinase or [α -³²P]dATP and Klenow DNA polymerase. The labeled, redigested DNA fragment can finally be eluted from a polyacrylamide gel and purified by standard procedures.⁵ The radioactivity needed at this point should be between 10⁵ and 10⁶ cpm; however, it is possible to obtain results with less.

Afterward the DNA is subjected to the modification reaction. It should be specific with respect to the attacked function on the DNA and, in particular for the interference analysis, rather small sizes of the introduced residues are desirable, because this allows a more specific interpretation in terms of protein-DNA contacts. The two mainly used chemicals for DNA modification fulfilling these requirements are dimethyl sulfate and *N*-ethyl-*N*-nitrosourea. Diethyl pyrocarbonate shows a structure-dependent reaction specificity⁶ and has successfully been used to map protein binding to Z-DNA^{3,7} while the application to B-DNA has been of only moderate success.⁴ The products of their specific reactions with DNA are depicted in Fig. 3.

Methylation of DNA⁵

The most widely used modification to study protein-DNA interaction is the methylation with dimethyl sulfate, which attacks the N-7 of guanines (G) and the N-3 of adenines (A), as is well known from the chemical nucleotide sequencing method. With regard to the analysis of protein-DNA interactions the methylation of the guanine N-7 in double-stranded DNA is of greater practical importance. This reaction can either be performed as a protection or an interference experiment.

Between 0.1 and 1 pmol DNA fragment in 200 μ l of 50 mM sodium cacodylate, pH 8.0, and 1 mM EDTA are mixed with 1 μ l of a carrier DNA solution containing 1 mg/ml sonified pWH802 DNA (any DNA that does not contain the binding site of interest can be used here), 2 μ l tRNA

⁴ C. Heuer and W. Hillen, *J. Mol. Biol.* **202**, 407 (1988).

⁵ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

⁶ W. Herr, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 8009 (1985).

⁷ B. Johnston and A. Rich, *Cell (Cambridge, Mass.)* **42**, 713 (1985).

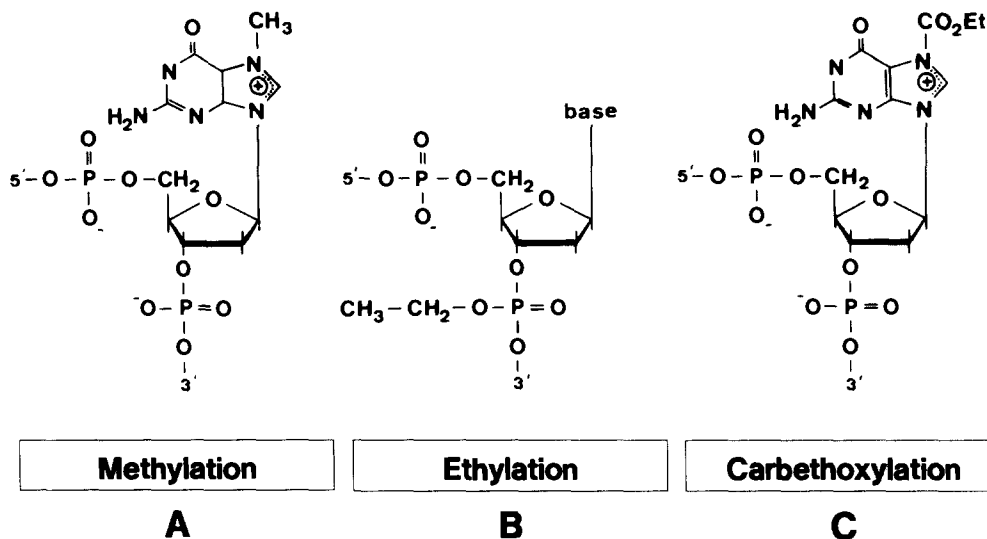


FIG. 3. Chemical structures of modified nucleotides. The reaction products of DNA and dimethyl sulfate (A), *N*-ethyl-*N*-nitrosourea (B), and diethyl pyrocarbonate (C) are shown. Dimethyl sulfate leads to the methylation of the N-7 in guanines and to a minor extent to methylation of the N-3 in adenines (not shown). *N*-ethyl-*N*-nitrosourea gives rise to ethylated phosphates and diethyl pyrocarbonate causes a carbethoxylation of the N-7 of purines (guanosine is shown as an example here).

solution containing 10 mg/ml yeast tRNA, and 0.5 μ l dimethyl sulfate (New England Nuclear, Boston, MA) in that order and then incubated at ambient temperatures for 2 min. The methylation reaction is terminated by adding 50 μ l of 1.5 *M* sodium acetate, 1 *M* 2-mercaptoethanol, 100 μ g/ml tRNA, followed by the addition of 750 μ l of ethanol. The mixture is kept in liquid nitrogen for 2 min and centrifuged for 15 min at 13,000 rpm in a Biofuge A (Heraeus, Osterode, Germany). The pellet is dissolved in 250 μ l of 0.3 *M* sodium acetate, 0.1 mM EDTA, 25 μ g/ml tRNA and reprecipitated by adding 750 μ l of ethanol and treatment as described above. The precipitate is dissolved in 50 μ l of 0.4 *M* sodium chloride and reprecipitated with 1 ml of ethanol as described above. Finally, the pellet is washed three times with 0.1 ml ethanol each, dried for 2 min *in vacuo*, and stored for further use.

For the protection from methylation by dimethyl sulfate the solution of the DNA fragment is mixed with an equimolar or up to 10⁴-fold excess of DNA-binding protein (Tet repressor in the example shown below) in 1 to 8 μ l of protein storage buffer [5 mM potassium phosphate, pH 8.0, 4 mM 2-mercaptoethanol, 50% glycerol (v/v) in this case]. The complex is

allowed to form for 10 min at ambient temperatures. Then the carrier DNA, tRNA, and dimethyl sulfate in the amounts described above are added. The following steps are performed as detailed above.

It should be noted that the amount of dimethyl sulfate, the reaction temperature, and time can be varied to yield the optimal methylation of one hit per fragment. Usually, however, this is not necessary due to the large excess of nonspecific competitor DNA. The strand cleavage reaction at the methylated positions is done exactly as described by Maxam and Gilbert.⁵

Ethylation of DNA^{8,9}

Between 0.1 and 1 pmol DNA fragment in 100 μ l of 50 mM sodium cacodylate, pH 7.0, 1 mM EDTA is mixed with 1 μ l sonicated carrier DNA (1 mg/ml, pWH802 in the examples shown here, but any DNA may be used) and 100 μ l of a freshly prepared saturated solution of *N*-ethyl-*N*-nitrosourea (Sigma, St. Louis, MO) in ethanol is added. The mixture is kept at 50° for 1 hr. After addition of 20 μ l of 3 M sodium acetate the DNA is precipitated with 0.5 ml of ethanol as described above. The pellet is washed with ethanol, dried *in vacuo* for 2 min, and stored for further use.

For strand cleavage the ethylated DNA samples are dissolved in 30 μ l of 10 mM sodium phosphate, pH 7.0, 1 mM EDTA. Five microliters of 1 M sodium hydroxide is added and the mixture is incubated at 90° for 30 min. Then 5 μ l of 1 M HCl is added, followed by 1 μ l of a glycogen solution (10 mg/ml) and 100 μ l of ethanol. After precipitation for 5 min in liquid nitrogen and centrifugation the pellet is washed with ethanol, dried *in vacuo*, and redissolved in 98% formamide containing 0.1% bromphenol blue and xylene cyanol. The sample is incubated at 100° for 2 min, immediately placed in ice water, and applied to a sequencing gel.

Carbethoxylation of DNA^{6,10}

DNA (0.1 to 1 pmol) in 200 μ l of 50 mM sodium cacodylate, pH 7.0, 1 mM EDTA is mixed with 1 μ l of carrier DNA solution (sonified pWH802 or any other DNA at 1 mg/ml) and 3 μ l of ice-cold diethyl pyrocarbonate (obtained from Sigma). Since diethyl pyrocarbonate is not dissolved in water the mixture is vigorously vortexed for 15 min at ambient temperatures. The DNA is then precipitated with ethanol, washed with ethanol, and briefly dried *in vacuo*. Strand cleavage at the modified positions is done exactly as described.⁶

⁸ U. Siebenlist and W. Gilbert, *Proc. Natl. Acad. Sci. U.S.A.* **77**, 122 (1980).

⁹ W. Hendrickson and R. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 3129 (1985).

¹⁰ D. A. Peattie and W. Gilbert, *Proc. Natl. Acad. Sci. U.S.A.* **77**, 4679 (1980).

For protection from ethoxylation³ 1.5 μg of plasmid DNA is incubated with 10 μg monoclonal antibody (corresponding to a 90-fold molar excess) in 100 μl of 100 mM sodium chloride, 60 mM Tris-HCl, pH 7.4 for 20 min at 37°. The tubes are transferred to ice and 100 μl of cold, 50 mM sodium cacodylate, pH 7.1 is added followed by 3 μl of diethyl pyrocarbonate. After thorough vortexing the mixture is incubated at 20° for 10 min. Then the vortexing is repeated, followed by another 10-min incubation at 20°. The samples are further processed according to Herr⁶ with a phenol extraction prior to the final precipitation. The DNA is then digested with a suitable restriction endonuclease, labeled at the 5' or 3'-end, and redigested with a second endonuclease. The products are separated on 5% (w/v) polyacrylamide gels, eluted, cleaved, and analyzed as described.⁶

Separation of Modified DNA Fragments According to Protein-Binding Capacity

The gel mobility shifts observed for most protein-DNA complexes provide a very convenient method to separate the bound fraction of modified DNA from those species in which the modification prevents DNA binding.⁹ A second method is the filtration over nitrocellulose, which works well for complexes with high-affinity constants and a large protein portion, e.g., a strong promoter with RNA polymerase.⁸ The experimental goal is, in any case, to obtain the most complete separation of these DNAs.

For the formation of Tet repressor-*tet* operator complexes the modified DNA is adjusted to a concentration of about 10^{-8} M in 10 to 30 μl of 10 mM Tris-HCl, pH 7.4, 10 mM potassium chloride, 10 mM magnesium chloride, 1 mM dithiothreitol, 0.1 mM EDTA, 5% glycerol (v/v), and 50 $\mu\text{g}/\text{ml}$ bovine serum albumin. Then Tet repressor is added to reach a 4- to 10-fold molar excess over DNA and the mixture is incubated at 37° for 20 min. Immediately afterward the samples are applied to carefully rinsed slots of a 5% polyacrylamide gel (0.25% bisacrylamide) and electrophoresed at 40 V overnight in 60 mM Tris-borate, pH 8.3, 1 mM EDTA. The bands are visualized by autoradiography and eluted from the gel by a standard procedure.⁵ The DNA is then cleaved and analyzed on sequencing gels.⁵

Modification Protection Analysis

Methylation Protection

Protection from methylation by dimethyl sulfate has been very widely used according to the protocol described above to determine protein-DNA

contacts *in vitro*¹ and, in a few cases, *in vivo*.¹¹ It is not within the scope of this chapter to provide a list of references completely covering this topic; instead, only references to the original protocols are given.

The great popularity of this experiment is due to two main reasons. First, it is very easily done because it involves only the standard Maxam and Gilbert sequencing reaction for guanosine. It can also be used to study contacts to adenine (A); however, this requires more drastic reaction conditions, making it more difficult to observe A protection in addition to G protection. Second, dimethyl sulfate is a small reagent specific for the guanosine modification at position N-7 located in the major groove of the B-form structure of DNA (see Fig. 3). Thus, a protection can be interpreted as a result of a very close contact of the studied protein to the major groove of the bound DNA in the near vicinity or even at the N-7 of the guanine moiety. As shown schematically in Fig. 1, formation of a protein–DNA complex may also lead to an enhanced reactivity of residues. This is explained either by protein binding-induced structural alterations of the DNA leading to a more exposed reaction site, or by the formation of a so-called hydrophobic pocket by the protein on the DNA attracting the reagent and increasing its local concentration. In this interpretation the sites of increased reactivity may indicate the boundary of the protein on the DNA. A result of this experiment is not presented here. The data for Tet repressor–*tet* operator interaction discussed below are taken from Ref. 2.

Carbethoxylation Protection

Treatment of DNA with diethyl pyrocarbonate leads to carbethoxylation of purines at N-7 (see Fig. 3). It has been demonstrated that the efficiency of this reaction depends on the secondary structure of the DNA. The reactivity of purines in Z-DNA, e.g., is so much greater compared to B-DNA that modification by diethyl pyrocarbonate has been used as an indication for Z-DNA formation under conditions of negative superhelicity.^{6,7} Consequently, the protection from this modification by binding of Z-DNA-specific monoclonal antibodies has been used to determine close contacts of the antibody to the DNA.³ A typical result is displayed in Fig. 4. The optical impression of altered intensities are usually quantified by densitometry based on the intensity of a reference band located outside of the area of interest. In the case of antibody binding it was possible to combine the results of carbethoxylation protection with the global structure of antibodies known from X-ray analysis to propose a detailed model describing the orientation of the protein on the DNA.

¹¹ H. Nick and W. Gilbert, *Nature (London)* **313**, 795 (1985).

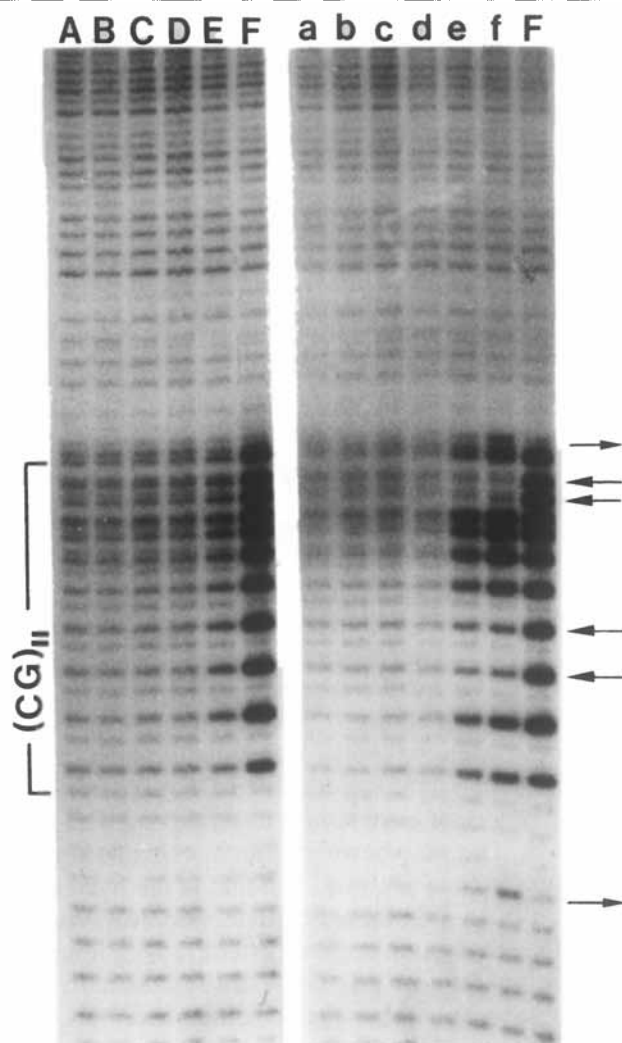


FIG. 4. Hyperreactivity of Z-DNA to diethyl pyrocarbonate and protection by an antibody. Preparations of a plasmid containing a $(C-G)_{11}$ repeat sequence were adjusted to six different negative superhelical densities, $-\sigma$ (A, 0; B, 0.012; C, 0.023; D, 0.035; E, 0.046; F, 0.058) and treated with diethyl pyrocarbonate in either the absence (lanes A to F) or presence (lanes a to f) of a monoclonal antibody. The bracket indicates the region of 22 bp of alternating d(C-G). Arrows indicate guanine residues with either a reduced (arrows pointing left) or enhanced (arrows pointing right) hyperreactivity toward diethyl pyrocarbonate on complex formation with a Z-DNA-specific monoclonal antibody. [Reproduced by permission of Academic Press, Inc., London (Runkel and Nordheim³).]

While carbethoxylation is particularly suitable for Z-DNA modification its use for protein-DNA complexes with B-type structures is rather limited. In the case of *tet* operator-Tet repressor recognition the modification of the pure *tet* regulatory sequence turned out to be quite inhomogeneous and the interpretation of modification interference was hampered by the lack of consistency within the palindromic *tet* operator sequences. However, although the interpretation of the results obtained for single base pairs was not straightforward, the points of interference are located in the center of the operator palindrome.⁴ This is in good agreement with other protection⁴ and genetic data¹² concerning Tet repressor-*tet* operator interaction.

Modification Interference Analysis

Separation of Bound from Nonbound Fraction of Modified DNA

For complexes with large protein portions showing complete binding to nitrocellulose the filtration technique is very suitable.⁸ Repressor-operator complexes usually contain smaller proteins, which in some cases show only very poor retention on nitrocellulose.^{13,14} For these cases in particular and for many other cases for the sake of simplicity the gel mobility shift technique is the method of choice.¹⁵⁻¹⁷ An example is shown in Fig. 5 for the Tet repressor-*tet* operator complex with native and partially ethylated DNA. It is clearly visible that the ethylated DNA cannot be completely complexed by the repressor even though a sixfold molar excess of protein over DNA is used. The gel shown in Fig. 5 contains three regions of interest for the interference analysis. The first is the band corresponding to the complexed DNA. This should contain modifications only at those positions where the ethyl group does not interfere with repressor binding. In our hands, the analysis of this DNA always leads to clear results. The second band corresponds to the remaining free DNA. It contains modifications at sites where the ethyl residue prevents complex formation. The analysis of this band leads to results that are less clear. We believe that this fraction is always contaminated with some residual nonbound DNA, because binding may never be quantitative. This could lead to

¹² A. Wissmann, I. Meier, and W. Hillen, *J. Mol. Biol.* **202**, 397 (1988).

¹³ A. Joachimiak, R. L. Kelley, R. P. Gunsalus, C. Yanofsky, and P. B. Sigler, *Proc. Natl. Acad. Sci. U.S.A.* **80**, 668 (1983).

¹⁴ L. S. Klig, I. P. Crawford, and C. Yanofsky, *Nucleic Acids Res.* **15**, 5339 (1987).

¹⁵ M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

¹⁶ M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **11**, 141 (1983).

¹⁷ A. Revzin, J. A. Ceglarek, and M. M. Garner, *Anal. Biochem.* **152**, 172 (1986).

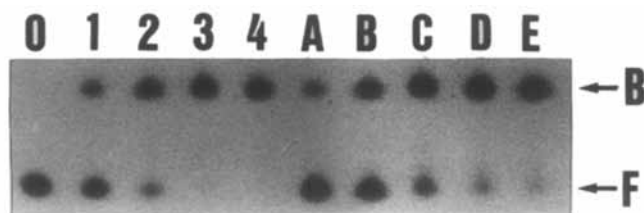


FIG. 5. Gel mobility of native and partially ethylated *tet* operator DNA in the presence of Tet repressor. A 64-bp *tet* operator containing DNA fragment was electrophoresed on a 5% polyacrylamide gel. Lanes 0 to 4 contain native DNA. In lane 0 no repressor was added to the DNA, whereas in lane 1 a 0.25-fold, in lane 2 a 0.75-fold, in lane 3 an equimolar, and in lane 4 a 4-fold molar excess of Tet repressor over *tet* operator was added. Lanes A to E contain the same DNA fragment after ethylation. The molar ratios of Tet repressor over *tet* operator added to the DNA in these lanes were as follows: lane A, 0.25-fold; lane B, 0.75-fold; lane C, equimolar; lane D, 4-fold; lane E, 6-fold. The arrows on the right denote the positions of the bound (B) and the free (F) DNA. [Reproduced by permission of Academic Press, Inc., London (Heuer and Hillen⁶).]

the increased background seen in those experiments. The third region is located between the two bands. It contains DNA from complexes that have undergone dissociation during electrophoresis and are, therefore, of reduced stability. An example of this analysis for methylation interference is discussed below.

Analysis of Interference

The electrophoretic analysis of cleavage products from the modified DNA fragments yields the positions where modification has interfered with protein binding. Ideally, these positions should be blank in the bound fraction of the modified DNA. When a gel separation is used this goal can generally be achieved for the positions showing the strongest interference for both methylated and ethylated DNA. An example from our own work on *tet* operator–Tet repressor recognition is shown in Fig. 6. The data indicate clearly that the interference signals are of different intensities. The methylation interference of the plus strand yields a single, strong signal for the guanosine at position 2, the same which is protected from methylation. For the minus strand, on the other hand, the corresponding strong G signal is accompanied by a weaker interference band for adenosine at position 3. However, since this is observed only on one strand of the two *tet* operators we are not sure that this signal indeed represents a contact. At operator O₁ an adenosine interference is found at position 4 of the minus strand (not shown). It has been shown that this methylation is less destructive for complex formation than the respective modification

OPERATOR 2

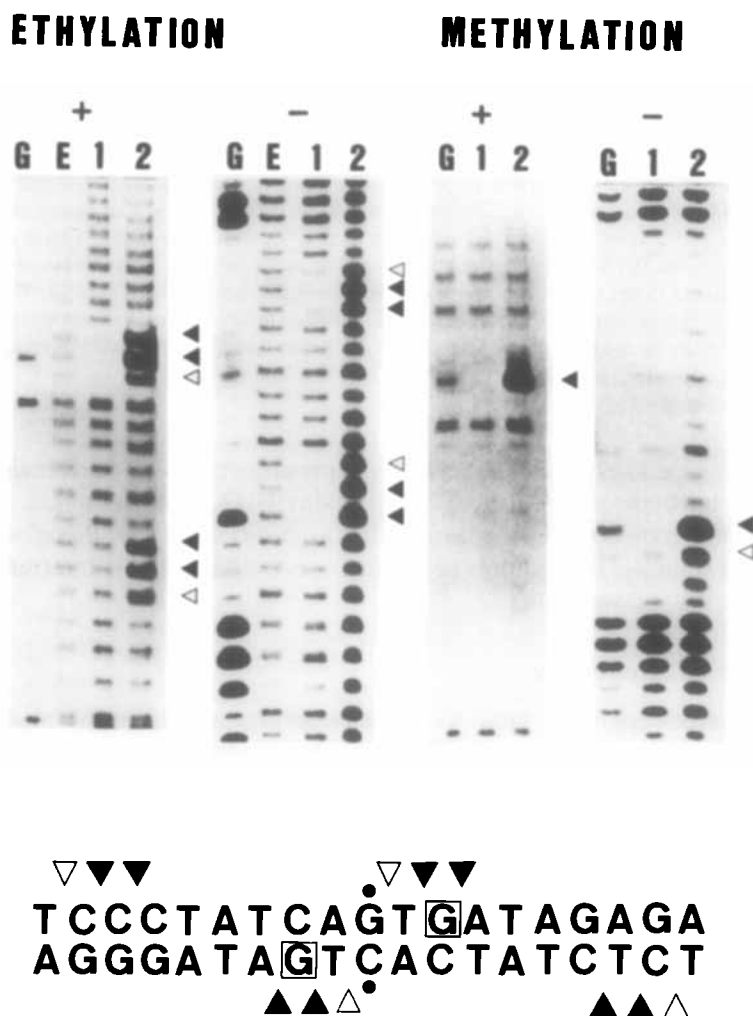


FIG. 6. Results of ethylation and methylation interference studies obtained for *tet* operator O_2 . The + denotes the upper (5'-end-labeled) and the - the lower (3'-end-labeled) strand of the DNA sequence of *tet* operator O_2 shown at the bottom. In lanes G the products of the Maxam and Gilbert G reaction were loaded, whereas lanes E show the cleavage products of the ethylated DNA. Lanes 1 contain the bound and lanes 2 the nonbound fractions of the ethylated DNA. Stronger contacts are indicated by filled triangles, whereas weaker contacts are marked by open triangles. At the bottom the deduced contact sites are indicated in the sequence of *tet* operator O_2 . The phosphate contacts are indicated as triangles as described

at the guanosine by comparing the signals obtained from the nonbound DNA with the ones from the region between the two bands on the polyacrylamide gel (result not shown). Very clearly, the G methylation prevents complex formation completely while the A methylation at position 4 yields a strong signal in the interband region. This has been interpreted as a weaker interference since the complex can be formed, but dissociates partially during gel electrophoresis.

The ethylation interference results in Fig. 6 reveal six positions on each strand of *tet* operator, where the modification reduces Tet repressor binding. They fall into two sets of three interference signals each, one of which is located close to the palindromic center of the *tet* operator, while the other is located at the outside. For the identification of the interference site it needs to be considered that strand cleavages in the G reaction and after ethylation lead to different products.⁸ Since strand breakage can occur on both sides of the ethylated phosphate, the corresponding band are doublets, which are only resolved for short molecules. As a result, they trail the marker bands produced by G cleavage slightly. On the basis of the different intensities of these interference bands it is concluded that the modifications exhibit quantitatively different effects on Tet repressor binding. Inspection of the bound DNA reveals that the center positions are clearly blank while the outer positions show an increased background. This is interpreted as partial binding of the DNA modified at the outside while ethylation at the center prevents repressor binding completely under these conditions. It is, thus, concluded that the contacts at the inside of *tet* operator are more important for Tet repressor binding than the ones at the outside. Similar differences in intensity are found for the bands within each set of three contacted phosphates. A densitometric analysis of the data in Fig. 6 (not shown) indicates that the innermost interference signals of the inside contacts and the outermost signals of the outside contacts are somewhat weaker than the respective other two signals. It is, therefore concluded that the contacts of Tet repressor to these phosphates are less important for complex formation than the others. For this kind of analysis it is usually necessary to study a variety of modification conditions to yield the optimal undermodification of the sites on the DNA, and to vary the exposition times of the gels to yield the proper low intensities of bands for quantification. Usually, the experiments yielding the best pictures for publication are not the ones allowing the most reliable quantification.

above and contacts to the N-7 of G are marked by boxed nucleotides. The dots indicate the palindromic center of the operator. By definition this base pair is localized at position 0 of the operator. From here base pairs are numbered to the outside of the palindromic sequence (1 to 9). [Reproduced by permission of Academic Press, Inc., London (Heuer and Hillen⁴).]

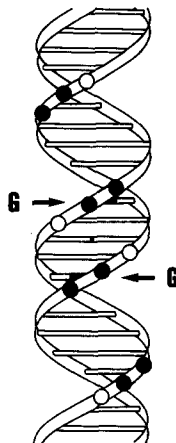


FIG. 7. Projection of the *tet* operator B-DNA structure showing Tet repressor contacts as deduced from modification protection and interference studies. The strong phosphate contacts of Tet repressor to *tet* operator DNA are indicated as filled circles, whereas the weaker contacts are marked by open circles. The G residues showing methylation protection and methylation interference are indicated by the filled bars at the respective base pairs. The palindromic center of the operator is shown as a black dot in the middle of the bar. [Reproduced by permission of Academic Press, Inc., London (Heuer and Hillen⁴).]

An important caveat relates to the interpretation of the results. An interference is observed, when the binding protein in the complex with DNA approaches the modified function so closely that the ethyl or methyl moiety cannot be accommodated. Furthermore, the energy required for a potential accommodation in the complex must reduce the association constant below the inverse concentrations of the binding experiment used to separate free from bound DNA. Thus, the interference results may depend on the methodology used. In addition, conclusions can only be derived in terms of contacts in the sense of distance and not in the sense of possible interactions that may or may not occur at the sites identified by interference analysis.

Interpretation of Results

The interpretation of protection and interference results is aided by either a three-dimensional computer drawing or a two-dimensional projection of the anticipated DNA structure in which the identified contacts are marked. For the examples described in this chapter, this has been done in the case of Z-DNA antibody binding to Z-DNA and Tet repressor binding to B-form *tet* operator DNA. In both cases the location of protein contacts

on the surface of the DNA structure yields hints about the potential protein structures involved in complex formation. Assumptions about the structure of the Z-DNA-specific monoclonal antibodies have been derived from the crystal structures of other antibodies and indicate a good structural fit of the proteins to the contact sites on the DNA.

The DNA-binding domain of Tet repressor shows strong homologies to many other prokaryotic DNA-binding proteins having an α helix–turn– α helix supersecondary structure in this domain.¹⁸ The ethylation interference results look indeed very similar to the ones obtained with these other proteins and their respective cognate DNAs.¹⁹ However, the methylation protection and interference does not fit very well into this picture because it appears to be displaced from the center between the contacted phosphates. This interpretation is shown in Fig. 7. Furthermore, a quantitative saturating mutational analysis of the *tet* operator sequence for binding of Tet repressor has indicated that base pairs at positions 3 of the two half-sides are most important, followed by base pairs 2 and 4, while 1, 5, and 6 are less important.²⁰ Thus, the most important sequence as well as the protected guanosine are not located in the center between the ethylation contacts, as was proposed for other bacterial DNA-binding proteins.¹⁹ Whether this pattern can be achieved by alterations in the position of the recognition α helix compared to other known structures or whether other parts of the Tet repressor contribute to *tet* operator binding cannot be distinguished by these results. The two examples presented here demonstrate clearly that DNA modification-protection and modification-interference analysis yields valuable information about the structure of protein–DNA complexes contributing one piece to the tertiary structure puzzle. In addition, the necessary experimental setup is commonly available and the experimental procedures involved are not difficult. For these reasons they will most likely be even more extensively used in the future as they have been in the past.

¹⁸ C. O. Pabo and R. T. Sauer, *Annu. Rev. Biochem.* **53**, 293 (1984).

¹⁹ R. H. Ebright, in "Protein Structure, Folding and Design" (D. Oxender, ed.), p. 207. Alan R. Liss, New York, 1986.

²⁰ C. Sizemore, A. Wissmann, U. Gülland, and W. Hillen, *Nucleic Acids Res.* **18**, 2875 (1990).

[19] Hydroxyl Radical Footprinting

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MARGARET F. WEIDNER, BETH A. DOMBROSKI,
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Introduction

Advances in cloning and purification of biological molecules have made available for study an ever-increasing number of proteins that bind to DNA. With this large number of new proteins has come the recognition that there are several strategies in protein design that Nature has employed for producing sequence-specific protein-DNA complexes. The helix-turn-helix and the zinc finger are the best understood at the moment, but other protein structures are also known that mediate contact with DNA. Along with the variety in protein structural motifs that bind to DNA, one might expect to find a corresponding variety in the structures of protein-DNA complexes.

Footprinting¹ is one experimental approach that can provide detailed information on how a protein binds to DNA. Since footprinting was first devised, with its beginning in the use of the enzyme deoxyribonuclease I (DNase I) to digest a DNA-protein complex,²⁻⁴ much progress has been made in employing other cleavage reagents that are capable of revealing higher resolution views of a protein-DNA complex. Many of these new reagents are small molecules, not enzymes. The smallest chemical species that has been used for footprinting, and therefore the one likely to provide the highest resolution structural information, is the hydroxyl radical ($\cdot\text{OH}$).⁵

In this chapter we discuss the use of the hydroxyl radical to make footprints of protein-DNA complexes. In an earlier volume of "Methods in Enzymology" we presented our original experimental protocol for hydroxyl radical footprinting.⁶ Since then we have refined the method, applied it to a wide variety of systems, and developed new experimental

¹ T. D. Tullius, *Annu. Rev. Biophys. Biophys. Chem.* **18**, 213 (1989).

² D. J. Galas and A. Schmitz, *Nucleic Acids Res.* **5**, 3157 (1978).

³ A. D. Johnson, B. J. Meyer, and M. Ptashne, *Proc. Natl. Acad. Sci. U.S.A.* **176**, 5061 (1979).

⁴ M. Noll, *Nucleic Acids Res.* **1**, 1573 (1974).

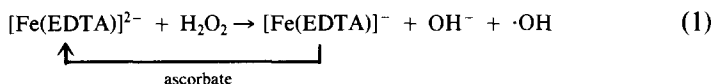
⁵ T. D. Tullius, *Trends Biochem. Sci.* **12**, 297 (1987).

⁶ T. D. Tullius, B. A. Dombroski, M. E. A. Churchill, and L. Kam, this series, Vol. 155, p. 537.

methods, based on the chemistry of the hydroxyl radical with DNA, that give additional information on the structure of a protein–DNA complex. The enhancements to the method that we will cover here include modifications to the composition of the cleavage reagent to give footprints in media containing scavengers of the hydroxyl radical, quantitative treatment of data from hydroxyl radical footprinting, the use of mobility-shift gel electrophoresis to “amplify” footprints, and a new method, the missing nucleoside experiment, which provides information on contacts with DNA that are important for stability in a protein–DNA complex.

Chemistry behind Hydroxyl Radical Footprinting

Three compounds are combined in the footprinting reaction mixture to produce the hydroxyl radical: $[\text{Fe}(\text{EDTA})]^{2-}$, hydrogen peroxide, and sodium ascorbate. The reaction by which the hydroxyl radical is generated, called the Fenton reaction, is shown in Eq. (1):



In this reaction an electron from iron(II)-EDTA serves to reduce and break the O–O bond in hydrogen peroxide, giving as products iron(III)-EDTA, the hydroxide ion, and the neutral hydroxyl radical. Sodium ascorbate is present to reduce the iron(III) product to iron(II), thereby establishing a catalytic cycle and permitting low (micromolar) concentrations of iron(II)-EDTA to be effective in cleaving DNA. A consequence of this scheme is that the concentrations of the three chemical species [iron(II)-EDTA, hydrogen peroxide, and sodium ascorbate] may be varied to optimize the generation of the hydroxyl radical under different solution conditions, e.g., to compensate for the presence of radical scavengers in the binding buffer of a protein–DNA complex. Specific examples of concentrations of each of the three reagents that are used in particular footprinting experiments are discussed in later sections.

The Fenton reaction [Eq. (1)] is not the only way to generate the hydroxyl radical for footprinting. We have published a protocol for making a footprint of the bacteriophage λ repressor using ionizing (γ) radiation to produce the hydroxyl radical in aqueous buffered solution.⁷ This technique, which requires no additional chemical reagents, may be useful in some situations (for example, *in vivo*) where it is difficult to perform the Fenton reaction.

⁷ J. J. Hayes, L. Kam, and T. D. Tullius, this series, Vol. 186, p. 545.

Updated Version of Basic Hydroxyl Radical Footprinting Technique

Reagents

The quality of the water used in buffers for hydroxyl radical footprinting is critical. Organic contaminants in particular can interfere with the reaction, for example by reacting with iron(II), hydrogen peroxide, or the hydroxyl radical. For this reason all solutions used in experiments involving the hydroxyl radical are prepared with water purified by a Milli-Q system (Millipore, Bedford, MA), which includes cartridges for ion exchange and removal of organics, and a 0.44- μ m filter for removal of bacteria.

To prevent nicking of the DNA by radiation products, radioactively end-labeled DNA is stored at an activity of 50,000 disintegrations per minute (dpm)/ μ l or lower in a dilute buffer, such as 10 mM Tris-Cl, 0.1 mM EDTA (pH 8.0) (which we refer to hereafter as TE buffer), at 4°.

An aqueous solution of iron(II) is prepared by dissolution of ferrous ammonium sulfate $[(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}]$ (99+%; Aldrich, Milwaukee, WI) in water at a concentration of 10–100 mM. The complex of iron(II) with EDTA is prepared by mixing equal volumes of the solution of ferrous ammonium sulfate with a solution of EDTA (Gold Label; Aldrich) that is twice the concentration of the iron solution. These solutions are prepared in acid-washed glassware to prevent the introduction of extraneous iron. We segregate glassware used to prepare iron solutions from other glassware in the laboratory to obviate the possibility of iron contamination of DNA solutions. A solution of ascorbate ion is prepared by dissolving sodium ascorbate (Sigma, St. Louis, MO) in Milli-Q-purified water. Hydrogen peroxide solutions are made by dilution of a 30% (v/v) solution (J. T. Baker, Phillipsburg, NJ). The three reagents can be made up fresh prior to use. However, a more convenient approach is to freeze the ascorbate and $[\text{Fe}(\text{EDTA})]^{2-}$ stock solutions in small aliquots. An alternative is to make stock solutions of iron(II) and EDTA separately. Iron(II) stock solutions can be stored as frozen aliquots, and EDTA solutions can be stored at 4°. These stock solutions can then be combined just prior to performing the experiment.

General Technique for Hydroxyl Radical Footprinting

A typical starting mixture for a hydroxyl radical footprinting experiment contains the binding buffer, 0 to 0.5 μ g of nonspecific DNA, 50,000 to 200,000 dpm of singly end-labeled DNA containing the protein-binding site, and the desired amount of protein, in a volume of 70 μ l. This solution should be incubated at the appropriate temperature to allow protein to

bind to the DNA. Then 10 μ l each of the three stock solutions (each at 10 times the final concentration desired in the reaction mixture) of ascorbate, hydrogen peroxide, and iron(II)-EDTA are placed as drops on the inside of the 1.5-ml Eppendorf reaction tube that contains the protein-DNA complex. To initiate the cleavage reaction, the reagents are mixed together on the side of the reaction tube and then added to the protein-DNA mixture. The reaction is allowed to proceed for 2 min at room temperature. The reaction can be performed at 4° if necessary, but this often requires modification of the reaction conditions to increase the rate of the cutting reaction. The reaction is quenched by adding 0.1 M thiourea (5 to 30 μ l) (a hydroxyl radical scavenger), and 0.2 M EDTA (2 μ l). To remove protein, samples are extracted with a volume of phenol equal to the volume of the reaction mixture, and then the phenol layer is back extracted with TE buffer. The aqueous solution is extracted with ether three times to remove any residual phenol. The DNA is then precipitated twice by addition of ethanol. The DNA pellet is rinsed with ethanol and dried in a SpeedVac concentrator (Savant, Farmingdale, NY).

Effect of Experimental Conditions on Hydroxyl Radical Cleavage of DNA

The concentrations of reagents in the hydroxyl radical footprinting method originally developed by Tullius and Dombroski⁸ were 1 mM ascorbate, 0.03% hydrogen peroxide, 10 μ M iron(II), and 20 μ M EDTA. This mixture was found to generate sufficient hydroxyl radical for introduction of not more than one gap in the backbone of a DNA molecule.⁹ These concentrations are adequate for DNA solutions that contain small amounts of scavengers of the hydroxyl radical, such as DNA in TE buffer. Such a cleavage reagent often is suitable for structural studies of free DNA molecules, and for some footprinting experiments. However, sometimes the reagents and buffers used for a protein-DNA complex reduce the rate of cleavage of DNA by the hydroxyl radical. The most avid scavenger of hydroxyl radical commonly encountered in protein-DNA systems is glycerol. We have found that a concentration of >0.5% (v/v) glycerol significantly inhibits cleavage of DNA by the hydroxyl radical. To a lesser degree, common buffers such as Tris and HEPES also reduce DNA cleavage. Thus DNA in a 10 mM Tris solution will be more efficiently cleaved than DNA in a 50 mM solution of Tris.⁶ It is also possible that high concentrations of bovine serum albumin (BSA) or nonspecific DNA will

⁸ T. D. Tullius and B. A. Dombroski, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 5469 (1986).

⁹ T. D. Tullius and B. A. Dombroski, *Science* **230**, 679 (1985).

decrease the hydroxyl radical cutting of the radioactively labeled DNA by competing for reaction with the hydroxyl radical.

There are two ways to increase the rate of cleavage of DNA. One is to lower the concentration of scavengers of the hydroxyl radical that are present in the reaction mixture. Since these reagents often are necessary for protein stability and binding to DNA, usually another means must be found for increasing DNA cleavage. This can be accomplished by increasing the concentrations of some or all of the reagents that generate the hydroxyl radical. Because some proteins are sensitive to hydrogen peroxide,¹⁰ increasing its concentration is not advised. The best method for augmenting the cleavage reagent is to increase the concentrations of both $[\text{Fe}(\text{EDTA})]^{2-}$ and ascorbate. Raising the concentration of $[\text{Fe}(\text{EDTA})]^{2-}$ increases cutting by increasing the production of the hydroxyl radical by Fenton chemistry. Increasing the concentration of ascorbate is important because ascorbate reduces one of the products of the reaction, $\text{Fe}(\text{III})\text{EDTA}$, to $\text{Fe}(\text{II})\text{EDTA}$. Therefore it is preferable to have a concentration of ascorbate several times that of iron to maintain a catalytic cycle.

More cleavage is not always better, though. Reaction conditions are desired that produce no more than one cut per DNA molecule. Since the DNA is labeled at one end of one strand of the duplex, only the fragment resulting from the cut closest to the label will be seen on the autoradiograph of the gel. If the reaction conditions produce more than one cut per molecule, it is possible that an initial cut could occur far from the label and a second cut would be made between the label and the first cut. In this case, the fragment resulting from the first cut would be lost and only the smaller fragment containing the label would be observed.¹¹ Thus overcutting gives a biased picture of the cleavage frequency along the DNA molecule. To ensure that only one cut per molecule is produced, a simple Poisson calculation shows that the percentage of uncut DNA in the sample should be greater than 70%.¹² Conversely, a sufficient amount of cleavage is needed for a good signal-to-noise ratio in the experiment. This is why careful regulation of the length of the reaction and the concentrations of the cleavage reagents is necessary.

Effects of Hydroxyl Radical Cutting Reagents on Protein Binding

Previously it was demonstrated that DNase I footprinting can be used to test whether a particular component of the hydroxyl radical-generating system affects the ability of the protein of interest to bind to DNA.⁶ This

¹⁰ K. E. Vrana, M. E. A. Churchill, T. D. Tullius, and D. D. Brown, *Mol. Cell. Biol.* **8**, 1684 (1988).

¹¹ L. C. Lutter, *J. Mol. Biol.* **124**, 391 (1978).

¹² M. Brenowitz, D. F. Senear, M. A. Shea, and G. K. Ackers, this series, Vol. 130, p. 132.

control experiment is useful in cases where cleavage of DNA occurs in footprinting samples, but samples with protein give the same cleavage pattern as free DNA. In this procedure, each of the three components of the hydroxyl radical cleavage reaction ($[\text{Fe}(\text{EDTA})]^{2-}$, ascorbate, and hydrogen peroxide) are added independently to samples of the DNA-protein complex. Then DNase I digestion is performed to determine whether protein remains bound in the presence of each reagent. Lack of a DNase I footprint in the presence of one of the reagents indicates sensitivity of the protein to that reagent.

So far the only one of the three reagents that we have found to which protein-DNA binding is sensitive is hydrogen peroxide. Both the zinc finger protein TFIIA⁶ and the "copper fist" protein CUP2¹³ do not bind to DNA in the presence of the standard concentration of hydrogen peroxide, which may be due to the oxidation of sulfhydryl groups that are necessary for metal binding in these proteins. In order to footprint these proteins it was necessary to use a concentration of hydrogen peroxide of only 0.003%. To compensate, a concentration of 100 μM $[\text{Fe}(\text{EDTA})]^{2-}$ was used for footprinting TFIIA. For CUP2 footprinting, 1 mM $[\text{Fe}(\text{EDTA})]^{2-}$ and 20 mM ascorbate were needed to provide sufficient cutting.

Sequencing Gel Electrophoresis and Gel Drying

The DNA pellet is dissolved in formamide-dye mixture, and denatured by heating to 90° for 3–5 min. Bromphenol blue is often omitted from the dye mixture for all but the marker lanes since it can interfere with the resolution of particular bands. Electrophoresis is performed on a denaturing polyacrylamide gel. Gels are made according to the procedure described by Maxam and Gilbert.¹⁴ The gel and running buffer is 1 $\times$ TBE [100 mM Tris-Cl, 100 mM sodium borate, 2 mM EDTA (pH 8.3)]. Denaturing polyacrylamide gels are 50% by weight urea with a ratio of 19:1 acrylamide: bisacrylamide. For electrophoresis we use Hoefer (San Francisco, CA) Poker-face sequencing gel equipment. The products of a Maxam and Gilbert guanine-specific sequencing reaction¹⁴ performed on the labeled DNA fragment serve as size markers for the gel.

The percentage of acrylamide used in the gel depends on the distance from the region of interest to the radioactively labeled end of the DNA. Table I lists the percentages of acrylamide required to resolve DNA fragments of particular lengths. Percentages between the values in Table I are used to optimize the resolution of a particular range of fragments. The

¹³ C. Buchman, P. Skroch, W. Dixon, T. D. Tullis, and M. Karin, *Mol. Cell. Biol.* **10**, 4778 (1990).

¹⁴ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

TABLE I
PERCENTAGE OF ACRYLAMIDE FOR SEPARATING
DNA MOLECULES OF VARIOUS LENGTHS

Length of DNA fragment (bp)	Acrylamide (%)
3-14	25
5-30	20
10-50	15
15-120	10
50-200	5

ability to resolve the smallest fragments depends mostly on the salt content of the sample.

A larger number of fragments can be resolved on one gel by using wedge spacers, which are thin at the top of the gel and thick at the bottom. Gels are prerun for at least 1 hr at a constant power of 65-75 W, until the gel temperature is between 45 and 50°. Samples are loaded on the gel as quickly as possible. During electrophoresis the gel temperature should return to the original prerunning temperature and then remain at this temperature. This is generally achieved by electrophoresis at a constant power of 55-65 W.

After electrophoresis the gel is dried onto Whatman (Clifton, NJ) 3MM paper, with a Hoefer slab gel dryer (model SE 1160). Our technique for transfer of a sequencing gel to filter paper has been described.¹⁵ This procedure is particularly well suited for transfer of high-percentage acrylamide gels, which are difficult to dry by the standard method.

Autoradiography

The dried gel is autoradiographed with preflashed Kodak (Rochester, NY) XAR-5 film, either at -70° with a Du Pont (Wilmington, DE) Cronex Lightning Plus intensifying screen, or at room temperature without a screen. A gel with a total of 100,000 dpm/lane should be exposed at -70° with an intensifying screen for 15 hr, or at room temperature without a screen for 6 days (150 hr). We generally make a first exposure at -70° to allow an early look at the gel and to determine the time needed for the room temperature exposure. Exposure at room temperature eliminates the parallax effect of both the screen and the gel contributing to darkening of the film, and gives sharper bands. To prevent exposure of the film by static electricity from gel cracking we place a piece of paper between the

¹⁵ G. E. Shafer, M. A. Price, and T. D. Tullius, *Electrophoresis* **10**, 397 (1989).

gel and the film during the first exposure. If hydroxyl radical cutting is weak, exposing the film longer is advisable.

Data Analysis

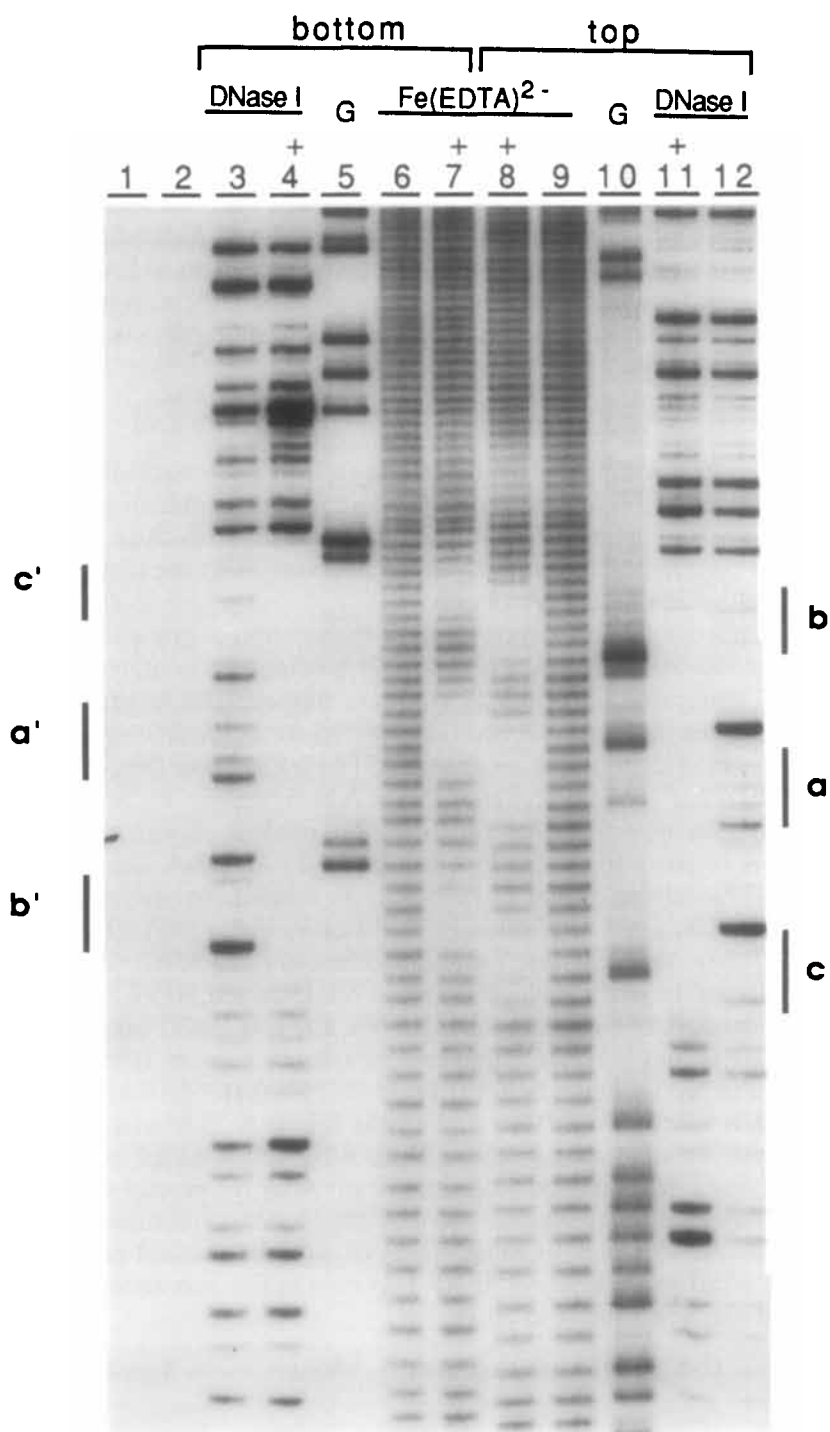
Qualitative analysis of one-dimensional densitometer scans is adequate for determination of protections at the nucleotide level. A one-dimensional scan is produced by scanning the autoradiograph with a Joyce Loebel Chromoscan 3 densitometer at an aperture width of 0.05 cm. A full discussion of methods for quantitative analysis of footprinting data is presented in the next section.

Footprinting of NF I-DNA Complex with Hydroxyl Radical

As a specific example of a footprinting experiment, we present a protocol developed in this laboratory for footprinting nuclear factor I (NF I), a protein involved in initiation of replication by adenovirus (Ad). (Independent hydroxyl radical footprinting experiments on NF I have recently been published by another laboratory.¹⁶)

A complication of this system is that the solution of protein and DNA found necessary to produce a saturated binding site contained a final glycerol concentration of 0.5%. We were unsuccessful in producing a footprint using the standard conditions for hydroxyl radical footprinting.^{6,8} On performing the footprinting experiment with a series of concentrations of $[\text{Fe}(\text{EDTA})]^{2-}$, we discovered that it was necessary to increase the concentration of $[\text{Fe}(\text{EDTA})]^{2-}$ to 4.5 mM to produce enough cleavage of the DNA to see a footprint. In a volume of 25 μl , DNA samples were prepared by mixing 5 fmol of radioactively labeled Ad5 DNA (100,000 dpm) with 0.5 μl of 5 M NaCl in $2\times$ NF I assay buffer [100 mM HEPES (pH 7.0), 50 mM MgCl_2 , and 1 mM dithiothreitol (DTT)]. A 1 : 20 dilution of NF I from the stock solution [20 ng/ μl NF I, 25 mM HEPES (pH 7.5), 20% glycerol, 0.01% Nonidet P-40 (NP-40), 100 mM NaCl, 1 mM EDTA, 1 mM DTT, and 0.1 mM phenylmethylsulfonyl fluoride (PMSF)] was prepared in NF I dilution buffer [10 mM HEPES (pH 7.0), 0.01% NP-40, 100 mM NaCl, 1 mM EDTA, and 1 mM DTT]. A 25- μl aliquot of the diluted protein solution, containing 25 ng of NF I, was added to the DNA mixture, for a final reaction volume of 50 μl . After the protein was added to the DNA solution, the sample was incubated for 20 min at room temperature to allow for protein binding. To DNA samples that did not contain protein, 25 μl of 1% glycerol in NF I dilution buffer was added to bring the final reaction volume to 50 μl .

¹⁶ H. Zorbas, L. Rogge, M. Meisterernst, and E.-L. Winnaker, *Nucleic Acids Res.* **17**, 7735 (1989).



Immediately before the cutting reactions were initiated, equal volumes of Fe(II) (250 mM) and EDTA (500 mM) solutions were mixed. A solution of $[\text{Fe}(\text{EDTA})]^{2-}$ that is 50 mM or greater in concentration is initially light green, but under these conditions Fe(II) is readily oxidized and the solution begins to turn rust in color. It is advised that, at these concentrations, a fresh solution of $[\text{Fe}(\text{EDTA})]^{2-}$ be prepared for every three to four reaction samples.

The reaction was begun by the addition of 2 μl each of $[\text{Fe}(\text{EDTA})]^{2-}$ [125 mM Fe(II), 250 mM EDTA], ascorbate (28 mM), and hydrogen peroxide (0.84%). After 2 min at room temperature the reaction was stopped with 150 μl of stop reagent [112 μl TE buffer, 10 μl 0.1 M thiourea, 25 μl 3 M sodium acetate, 2 μl 0.2 M EDTA, and 1 μl t-RNA (5 mg/ml)]. Protein was removed by phenol extraction. The DNA left in the aqueous solution was then subjected to two ether extractions, two ethanol precipitations, and rinsed and dried.¹⁷

An autoradiograph of the resulting NF I footprint is shown in Fig. 1. Three regions of protection are visible. The individual protections cover three to four nucleotides. The protected regions on each strand are offset by 3 base pairs (bp) in the 3' direction. This offset indicates equivalent protein interactions across the minor groove of DNA, since the backbone positions closest to each other across the minor groove are 3 bp apart in the sequence.¹⁸ The individual protected regions are separated by 8.5 to 9.5 bp. In many aspects, the hydroxyl radical protection pattern of NF I resembles that of the prokaryotic bacteriophage λ repressor.⁸ Using the protection results from λ repressor for comparison, it appears that NF I binds to one face of the Ad DNA helix as a dimer, with monomer units

¹⁷ B. A. Dombroski, Ph.D. Dissertation, Johns Hopkins University, Baltimore, Maryland (1988).

¹⁸ H. R. Drew and A. A. Travers, *Cell (Cambridge, Mass.)* **37**, 491 (1984).

FIG. 1. Hydroxyl radical footprints of nuclear factor I on both strands of the 326-bp Ad DNA fragment. Information for the bottom or the top strand was obtained by labeling the 5' or the 3' end of the *HindIII/RsaI* fragment, respectively. Lane 1, untreated Ad DNA labeled on the top strand. Lane 2, Ad DNA, labeled on the top strand, subjected to the conditions of the hydroxyl radical cleavage reaction but with water substituted for the cutting reagents. Lanes 3 and 12, products of DNase I digestion of the Ad DNA fragment alone. Lanes 4 and 11, products of DNase I digestion of the nuclear factor I-Ad DNA complexes (25 ng NF I). Lanes 5 and 10, products from Maxam-Gilbert guanine-specific sequencing reactions. Lanes 6 and 9, products of hydroxyl radical digestion of free Ad DNA. Lanes 7 and 8, products of hydroxyl radical cleavage of the nuclear factor I-Ad DNA complex (25 ng NF I). The labels a, b, c, a', b', and c', and the lines associated with them identify the hydroxyl radical footprints.

contacting the DNA in adjacent major grooves, and contacting each other across the minor groove at the dyad axis of symmetry of the consensus sequence. Mutational analysis of the NF I consensus sequence is consistent with dimer binding. A mutation in one of the half-sites led to a decrease (but not complete loss) in affinity of NF I.¹⁹

Quantitative Treatment of Footprinting Data

As discussed in other sections of this chapter, the nucleotide-level resolution of hydroxyl radical footprinting allows the description of binding site topography at a level of detail afforded by few other techniques. While in many cases a qualitative treatment of footprinting data is sufficient to address the questions at hand, there are situations in which precise quantitation of reactivity at each nucleotide position is desirable. The relative lack of sequence specificity of hydroxyl radical cleavage of DNA makes it especially suitable for such applications. For example, the smoothly periodic pattern of hydroxyl radical cleavage observed with nucleosomal DNA, as opposed to the irregular pattern produced by digestion with DNase I, has allowed accurate calculation of the helical periodicity of specific DNA sequences when bound on the surface of a nucleosome.²⁰ The ability to examine structural periodicity at nucleotide resolution improves significantly over other techniques,^{21,22} which detect the global conformation of the DNA but give no indication of local structural variations.

Another potential application of quantitative hydroxyl radical footprinting data is in the measurement of thermodynamic parameters of protein binding or DNA structural transitions. Ackers and co-workers have developed a footprint titration technique¹² in which the binding isotherm for the interaction of a protein with its specific recognition sequence on DNA is generated by quantitative analysis of DNase I footprints produced at various concentrations of protein. This technique, while powerful, is limited by the low resolution of DNase I as a probe of DNA structure. By using hydroxyl radical as the probe in the footprint titration experiment, it should be possible to examine both DNA structure and the thermodynamics of protein-DNA interactions at the nucleotide level.

Finally, quantitative analysis of hydroxyl radical reactivity at each position along the DNA backbone could give detailed structural informa-

¹⁹ K. A. Jones, J. T. Kadonaga, P. J. Rosenfeld, T. J. Kelly, and R. Tijan, *Cell (Cambridge, Mass.)* **48**, 79 (1987).

²⁰ J. J. Hayes, T. D. Tullius, and A. Wolffe, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 7405 (1990).

²¹ J. C. Wang, *Proc. Natl. Acad. Sci. U.S.A.* **176**, 200 (1979).

²² H.-M. Wu and D. M. Crothers, *Nature (London)* **308**, 509 (1984).

tion not easily obtainable by other methods. The measured rate of cleavage reflects the reactivity or accessibility of the target atom or bond in each nucleotide toward attack by the hydroxyl radical. This information could be used as an aid in modeling torsion angles in the DNA, and in positioning a protein domain on its binding site, in such a way that the observed pattern of reactivity is generated.

Since the mechanism of hydroxyl radical cleavage of DNA is as yet poorly understood, we do not know which atom(s) or bond(s) in a nucleotide to include in such a calculation. Instead, we have begun to approach this kind of quantitative application of hydroxyl radical footprinting from the reverse direction: that is, by comparing hydroxyl radical reactivity with calculated structural parameters from a protein–DNA complex for which the three-dimensional structure has been elucidated by X-ray crystallography. For this work we have studied the complex of λ repressor with the O_L1 operator site, for which the X-ray cocrystal structure was reported by Jordan and Pabo.²³ The goals of our work are twofold: in addition to evaluating the feasibility of deriving quantitative structural information from hydroxyl radical footprinting data, we hope to obtain important clues as to the mechanism of hydroxyl radical cleavage of DNA.

Methods

Autoradiographic data from footprinting experiments can be quantitated using either one-dimensional (1D) or 2D scanning densitometry. In the case of 1D densitometry the scanner beam is set to run down the length of a gel lane, detecting the optical density at each position along its path. This results in a series of peaks, each corresponding to a band on the gel (see Fig. 2). Visual inspection and comparison of such scans yields qualitative information about footprints; integration of the area under each peak is performed when quantitative data are desired. In theory, since electrophoresis is a 1D process, a 1D scan should accurately represent the data. However, 1D scans have inherent inaccuracies. For instance, the optical density across the width of the gel lane in a given band is not usually constant, and a scan with a narrow beam may not sample the same amount of all the bands. This problem is particularly obvious if the gel lane is not absolutely straight. Scanning with a beam as wide as the gel lane can reduce this problem. However, if the bands are not exactly perpendicular to the path of the scanning beam, resolution between peaks is decreased.

Because of the inherent tradeoff between accuracy and resolution in

²³ S. R. Jordan and C. O. Pabo, *Science* **242**, 893 (1988).

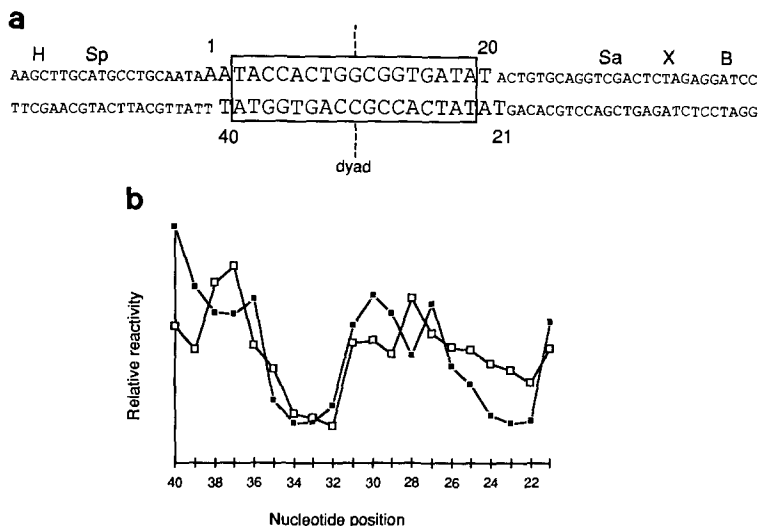


FIG. 2. Comparison of solvent-accessible surface area with hydroxyl radical cleavage frequency in the λ repressor- O_L1 complex. (a) DNA sequence of the insert region of plasmid pOL1-1. Larger letters indicate the sequence corresponding to the 20-mer oligonucleotide used in the cocrystal structure reported by Jordan and Pabo,²³ with numbering according to the scheme of Jordan and Pabo. The 17-bp binding site is boxed, with the pseudopalindrome dyad indicated. Restriction sites convenient for labeling are shown (H, *Hind*III; Sp, *Sph*I; Sa, *Sal*I; X, *Xba*I; B, *Bam*HI). (b) Relative hydroxyl radical cleavage frequencies (open squares) for the nucleotides on the bottom strand of the sequence shown in (a) are plotted along with the calculated accessible surface (probe radius 1.4 Å) for the sugar 3' hydrogens (closed squares). See text for experimental and theoretical details. The values have been adjusted to an arbitrary scale for purposes of comparison.

1D scanning, it is often preferable to scan in two dimensions. A 2D scan in effect transfers an image of the gel into a computer, for subsequent manipulation by image analysis methods. For the experiments described here, we used a photographic scanning system developed by the group of Professor Gary Ackers (Washington University, St. Louis, MO). The system consists of an Eikonix (Ektron Applied Imaging, Bedford, MA) model 1412 camera, interfaced to a Hewlett-Packard Vectra computer, which digitizes the optical density for each position on the autoradiograph and creates a computer file consisting of a series of pixels.¹² The digitized data can then be displayed on a computer and analyzed interactively using an image analysis program.¹² For these studies we used the program Image, which was written for the Macintosh II by W. Rasband (National Institutes of Health, Bethesda, MD). Using this software a box is drawn on the computer screen around each individual band to define the area for integral calculation. A pseudocolor feature, available in many image analysis programs, colors each pixel according to its optical density, and is useful in

visualizing where the baseline between two adjacent bands occurs. Care must be taken to leave enough space between lanes when samples are loaded on the gel for the image analysis software to be able to determine an accurate background optical density level for the autoradiograph.¹²

We have begun to analyze gels with laser densitometers²⁴ that are capable of scanning autoradiographic films or imaging phosphor plates. The PhosphorImager²⁴ in particular allows for a much larger linear range for data acquisition than film-based systems, since imaging phosphor plates have a dynamic range of nearly 10^5 .

With both 1D and 2D scans, there often is uncertainty as to where to place the border between adjacent bands. This can be caused by poor gel resolution or by electrophoretic artifacts, such as trailing of bands, or by the presence of minor bands between nucleotide positions, which are most likely side products of the hydroxyl radical cleavage reaction. Inaccurate division of bands can introduce error into the analysis of the data. In the case where quantitation is for the purpose of calculating helical periodicity,²⁰ this error can be greatly reduced by using a smoothing algorithm that averages the optical density of each band with those of the two adjacent bands. This three-band averaging results in a general dampening of the data, and so is useful only in cases where accurate quantitation of each band is not the goal, or where dramatic changes in optical density from one band to the next are not observed.

In most cases of footprinting proteins, three-band averaging would likely distort the footprint. Here, the best way to avoid error due to improper division between bands is to quantitate the footprints from several independent experiments and average the data. Comparison of data for the same strand of DNA labeled at the 5' end versus the 3' end can be a useful check for artifact. While the sequence itself will be inverted on the gel, and therefore contributions of electrophoretic anomaly to error should differ, the pattern of hydroxyl radical cleavage frequency should be independent of which end of the DNA is labeled.

Example: Quantitation of λ Repressor Footprint

Although hydroxyl radical footprinting of the λ repressor had already been reported by our laboratory,⁸ the published experiments were not done under conditions designed for quantitative analysis of the footprinting data. We wished to repeat these experiments in a way that would optimize the resolution of the DNA cleavage products in the region of the repressor-

²⁴ Molecular Dynamics model 300 Computing Densitometer, and model 400 PhosphorImager, available from Molecular Dynamics, 240 Santa Ana Court, Sunnyvale, CA 94086.

²⁵ R. T. Sauer, C. O. Pabo, B. J. Meyer, M. Ptashne, and K. C. Backman, *Nature (London)* **279**, 396 (1979).

binding site, and which would generate data that would be directly comparable to the cocrystal structure.²³ Since the cocrystal contained an amino-terminal fragment of λ repressor²⁵ bound to the O_L1 -binding site, we set out to duplicate this complex in our footprinting experiments.

A synthetic duplex oligonucleotide with the precise nucleotide sequence of the O_L1 operator, flanked by *Pst*I-compatible 3' overhangs, is cloned into the *Pst*I site of plasmid pUC18 (Fig. 2a). Use of a cloning site in the interior of the polylinker region provides us with multiple potential labeling sites on either side of the repressor binding site, so that either strand of DNA can be labeled at either the 5' or the 3' end, at distances ranging from 13 to 28 nucleotides from the first nucleotide in the binding site. Uniquely end-labeled DNA fragments are generated by digesting at one of the polylinker restriction sites, labeling the 5' or 3' ends by standard methods,²⁶ performing a second restriction digest with *Pvu*II to generate two labeled fragments, and isolating the O_L1 -containing fragment from a polyacrylamide gel.

Fragments labeled at several of the possible labeling sites are treated with the hydroxyl radical and electrophoresed through sequencing gels containing various concentrations of polyacrylamide to determine the optimum conditions for resolving the region of the O_L1 -binding site. We obtain the best results when the DNA is labeled approximately 20 nucleotides from the first nucleotide in the binding site and the footprinting products are run on a 15% (w/v) acrylamide sequencing gel (19:1 acrylamide:bis-acrylamide) with electrophoresis continuing until the bromophenol blue tracking dye has migrated 25–26 cm from the origin. These conditions place the DNA fragment representing the first nucleotide in the O_L1 -binding site (i.e., the end of the site closest to the labeled end of the DNA) very near the bottom of the gel, and array the 17 nucleotides comprising the binding site over a distance of roughly 13 cm. While this combination of labeling distance and electrophoresis conditions works well for these experiments, other combinations likely exist that would yield comparable results. Important considerations in designing this type of experiment are to maximize gel resolution by labeling as close as possible to the region of interest, keeping in mind that the closer the binding site is to the end of the DNA, the more problems of inefficient ethanol precipitation of small DNA fragments after hydroxyl radical cleavage, and potentially inefficient binding of protein to a site near the end of the DNA, come into play.

After empirically determining the optimum concentrations of footprinting reagents to use in the presence of λ repressor-binding buffer

²⁶ T. Maniatis, E. F. Fritsch, and J. Sambrook, "Molecular Cloning: A Laboratory Manual." Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1982.

(see above), several footprinting experiments are carried out with the λ repressor 92-amino acid amino-terminal fragment used in the cocrystal study. (The repressor fragment was kindly provided by Dr. Carl Pabo of M.I.T.) Binding of the protein to the DNA is performed essentially as previously described, using a protein concentration that gives fully saturated footprints (i.e., complete protection) with DNase I in parallel reactions. Footprints are obtained for both DNA strands, and with the label at both ends of each strand. After gel drying and autoradiography, the films are scanned with the Eikonix camera system (courtesy of Dr. Gary Ackers) described above. The scanned image is imported into the Image program. The bands corresponding to the 17 nucleotides of the O_L 1-binding site are individually integrated and the data are normalized to the integral for a band outside the binding site for which the cleavage frequency is known to be unaffected by the presence of λ repressor. Care is taken to ensure that the autoradiographic exposures are within the linear range of both the X-ray film and the densitometry equipment. Data from at least three independent footprinting experiments are averaged to provide a data set for comparison with crystallographic data.

Analysis of the atomic coordinates of the repressor cocrystal (provided by Dr. Carl Pabo) is performed with the QUANTA/CHARMm molecular modeling package from Polygen (Waltham, MA), run on a Silicon Graphics (Mountain View, CA) 4D/20 Personal Iris computer. The program uses the algorithm of Lee and Richards²⁷ to compute the solvent-accessible surface area for each atom of the DNA, using a variety of probe radii. A spreadsheet program (Microsoft Excel, run on a Macintosh II) is then used to sort the data by atom type and plot data for atoms of interest versus nucleotide position for comparison with hydroxyl radical cleavage frequency data.

For several of the deoxyribose hydrogens the pattern of surface accessibility to a probe sphere the size of a water molecule shows similarities to the hydroxyl radical footprinting pattern. As an example of such a comparison, the calculated surface accessibilities of the deoxyribose 3' hydrogens along the bottom strand of the O_L 1-binding site are plotted with the relative hydroxyl radical cleavage rates for this sequence in Fig. 2b. Areas of strong protection from hydroxyl radical cleavage (e.g., positions 22–24 and 32–34) correspond to positions of decreased solvent accessibility, suggesting that reactivity of the sugars with the hydroxyl radical is determined at least in part by accessibility to the reagent. Furthermore, the correlation between reactivity and accessibility of the 3' hydrogen indicates that this hydrogen is one of the targets for abstraction by hy-

²⁷ B. Lee and F. M. Richards, *J. Mol. Biol.* **55**, 379 (1971).

droxyl radical. Other hydrogens may also be targets, since their accessibility patterns also resembled the footprinting pattern to varying degrees (data not shown), and since it is unlikely that solvent accessibility is the sole determinant of reactivity. Experiments in progress in our laboratory are aimed at determining the mechanism of DNA cleavage by the hydroxyl radical by replacing each sugar hydrogen with deuterium and measuring the resulting isotope effect on the rate of cleavage.

Correlation between chemical reactivity and calculated surface accessibility has been noted previously by other workers studying tRNA structure.^{28–30} The observation that such a correlation exists in a protein–DNA complex demonstrates that it is possible to derive quantitative structural information from hydroxyl radical footprinting of macromolecular complexes. The sophistication of the derived structural information should increase as our understanding of the mechanism of hydroxyl radical cleavage of DNA improves.

Increasing Contrast of Hydroxyl Radical Footprint by Purifying Protein–DNA Complex on Mobility Shift Gel

Many proteins produce strong hydroxyl radical footprints when studied using the methods described above. However, proteins that bind to DNA weakly often produce very weak hydroxyl radical footprints, most likely due to a high level of signal from unbound DNA. Since the data from these weak footprints are hard to interpret, it is desirable to increase the footprint signal compared to the signal from unbound DNA. The footprint signal can be increased over the noise of the cleavage pattern of naked DNA by separating the protein–DNA complex from naked DNA by electrophoresis on a native (mobility shift^{31,32}) polyacrylamide gel.

The protein is bound to the DNA, and the mixture is treated with the hydroxyl radical cleavage reagent as in a conventional footprinting experiment. After 2 min the reaction is stopped and the mixture is loaded immediately on a native polyacrylamide gel to separate the protein–DNA complex from the unbound DNA. The DNA from the band containing the protein–DNA complex is eluted from the gel, denatured by heating, and electrophoresed on a sequencing gel to produce the footprint pattern.

In this section, we apply our protocol for hydroxyl radical footprinting

²⁸ S. R. Holbrook and S.-H. Kim, *Biopolymers* **22**, 1145 (1983).

²⁹ D. Romby, D. Moras, M. Bergdoll, P. Dumas, V. V. Vlassov, E. Westhof, J. P. Ebel, and R. Giege, *J. Mol. Biol.* **184**, 455 (1985).

³⁰ J. A. Latham and T. R. Cech, *Science* **245**, 276 (1989).

³¹ M. M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).

³² M. Fried and D. M. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

with gel isolation of complexes to two proteins, the restriction endonuclease *EcoRI* and CUP2, a yeast transcriptional activator protein. While the footprint of *EcoRI* is clear in a standard footprinting experiment, a stronger footprint can be produced at a lower protein : DNA ratio by separating bound from free DNA on a mobility shift gel. With CUP2 we had difficulty observing any regions of decreased cutting using the standard procedure for hydroxyl radical footprinting. However, a clear CUP2 footprint was obtained by performing hydroxyl radical footprinting with gel isolation of the CUP2–DNA complex.

Hydroxyl Radical Footprinting of EcoRI–DNA Complex

Protocol for Footprinting without Mobility Shift Gel Electrophoresis.

Binding conditions for *EcoRI*–DNA complex: The radioactively labeled DNA restriction fragment containing the *EcoRI* binding site (2 nM) is incubated with *EcoRI* endonuclease in binding buffer [0.1 M KCl, 10 mM Tris (pH 7.4), 1 mM EDTA] at room temperature for 20–30 min. [Since magnesium ion is necessary for cleavage of DNA by *EcoRI*, the lack of magnesium in this buffer permits binding of the enzyme but not cleavage of DNA.] The volume of the solution is 70 μ l. Concentrations of *EcoRI* ranging from 3 to 80 nM are used for footprint titration experiments.

Hydroxyl radical cleavage reaction: Samples are subjected to hydroxyl radical cleavage by the addition of 10 μ l of 2.5 mM $[\text{Fe}(\text{EDTA})]^{2-}$, 10 μ l of 10 mM sodium ascorbate, and 10 μ l of 0.3% hydrogen peroxide. Samples are allowed to react for 2 min. The cleavage reaction is terminated by the addition of 90 μ l of stop buffer (13 mM EDTA, 13 mM thiourea, 13 mg/ml tRNA, 0.1 M KCl). Samples are extracted with phenol and ether. The DNA is precipitated twice with ethanol. The pellet is rinsed with ethanol and dried. DNA samples are prepared for loading and electrophoresis on an 8% sequencing gel as described above.

Protocol for Footprinting with Mobility Shift Gel Electrophoresis.

Binding conditions for *EcoRI*–DNA complex: Binding solutions contain binding buffer [20 mM KCl, 10 mM Tris (pH 7.4), 2.5 mM EDTA], and either 102 nM DNA [2 nM radioactively labeled restriction fragment, and 100 nM unlabeled self-complementary oligonucleotide, d(CGCGAATTCGCG), which contains the *EcoRI* recognition sequence] and 100 nM *EcoRI*, or 2 nM labeled DNA and 50 nM *EcoRI*. The total volume of the solution is 35 μ l.

Hydroxyl radical cleavage reaction: Samples are subjected to hydroxyl radical cleavage in the same manner as described above, except that the reaction is quenched using 5 μ l of 0.1 M thiourea. Native dye mixture [50% (v/v) glycerol, 2% (w/v) xylene cyanol] (10 μ l) is added to samples

prior to nondenaturing polyacrylamide gel electrophoresis, bringing the volume of the sample to 80 μ l.

Nondenaturing gel electrophoresis: The *Eco*RI-DNA complex is separated from free DNA on a 16 cm $\times$ 19.5 cm $\times$ 1.5 mm gel consisting of 9.5% (w/v) acrylamide, 0.5% (w/v) bisacrylamide in 1 $\times$ TBE (pH 7.4). The gel is preelectrophoresed for 1 hr at 60 V. Samples are electrophoresed at 60 V at room temperature overnight. Autoradiography is used to identify radioactively labeled DNA in the gel. The gel slices containing radioactive DNA are crushed and incubated in 0.5 M ammonium acetate, 1 mM EDTA to elute the DNA. Small portions of the gel slices are crushed and eluted with 0.5 M ammonium acetate, 1 mM EDTA, and 10 mM $MgCl_2$ to test for DNA cleavage activity at the *Eco*RI sequence. Gel slices are incubated in elution buffer for 1.5 hr at 37°. The eluants are extracted with phenol and ether and made 0.3 M in sodium acetate. The DNA is precipitated twice with ethanol, rinsed, and dried. DNA is prepared for electrophoresis on an 8% sequencing gel as described above.

Results: We have produced hydroxyl radical footprints of *Eco*RI, bound to its recognition sequence GAATTC, with or without the additional step of separating the *Eco*RI-DNA complex from unbound DNA by mobility shift gel electrophoresis. The mobility shift gel shows that even at a ratio of 25 : 1 of *Eco*RI to its DNA recognition sequence a sizeable amount of unbound DNA is present in the reaction mixture (Fig. 3). From the autoradiographs and the densitometer scans of these footprints (Fig. 4) it is apparent that the protection patterns that result from the two procedures are quite similar. Both show a sharp decrease in hydroxyl radical cleavage two nucleotides prior to the G in the recognition sequence. This low level of cleavage is maintained throughout the recognition sequence. Past the C at the 3' end of the recognition sequence a gradual increase in cleavage occurs for about four nucleotides, at which point the cleavage frequency is again equal to that of the flanking sequences. The footprint pattern is stronger when unbound DNA has been separated by mobility shift gel electrophoresis. This is most apparent in the densitometer scans. Removal of free DNA by gel electrophoretic separation reveals that cleavage within the recognition sequence GAATTC is almost completely blocked by bound *Eco*RI. When the free DNA is not removed from the sample, weak bands are observed in the region of the recognition sequence.

In the X-ray crystal structure of the complex of *Eco*RI with its DNA-binding site,³³ the endonuclease is seen to interact with the DNA backbone

³³ J. A. McClarin, C. A. Frederick, B.-C. Wang, P. Greene, H. W. Boyer, J. Grable, and J. M. Rosenberg, *Science* **234**, 1526 (1986).

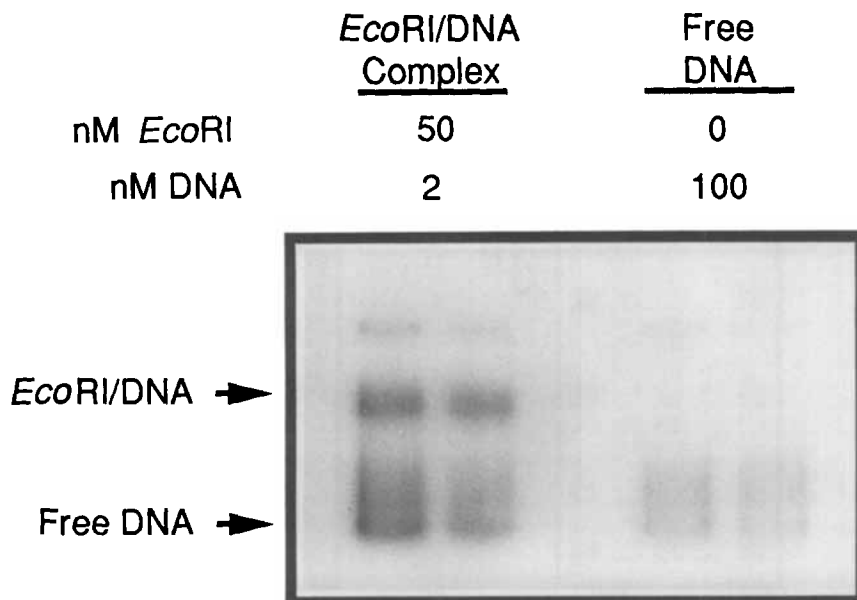


FIG. 3. Separation of the *Eco*RI–DNA complex from free DNA on a mobility shift gel. All samples contained a singly end-labeled 207-bp DNA restriction fragment containing the *Eco*RI recognition sequence GAATTC. Each sample was loaded in two lanes. Unbound DNA is labeled as free DNA, and migrates farther on the gel than DNA bound by protein.

of eight nucleotides, six that encompass the recognition sequence (GCGAATTC) and two nucleotides to the 5' side. The DNA backbone of those same eight nucleotides plus four others are protected from hydroxyl radical cleavage. The extra backbone protections are probably due to the presence of the protein at nearby nucleotides which results in a decrease in reaction with hydroxyl radical outside the recognition sequence. The hydroxyl radical footprint of the *Eco*RI–DNA complex corresponds in general to expectations based on the crystal structure.

Hydroxyl Radical Footprint of CUP2 Bound to Upstream Activation Site C of Yeast Metallothionein Gene

Binding Conditions for the CUP2–DNA Complex. Binding solutions contain 23.5 ng CUP2, 0.25 ng singly end-labeled DNA (500,000 dpm), 1 mg BSA, 1% poly(vinyl alcohol), 20 ng poly(dI) · poly(dC), 12.5 mM HEPES–NaOH buffer (pH 8.0), 50 mM KCl, 6.25 mM MgCl₂, 0.5 mM EDTA, and

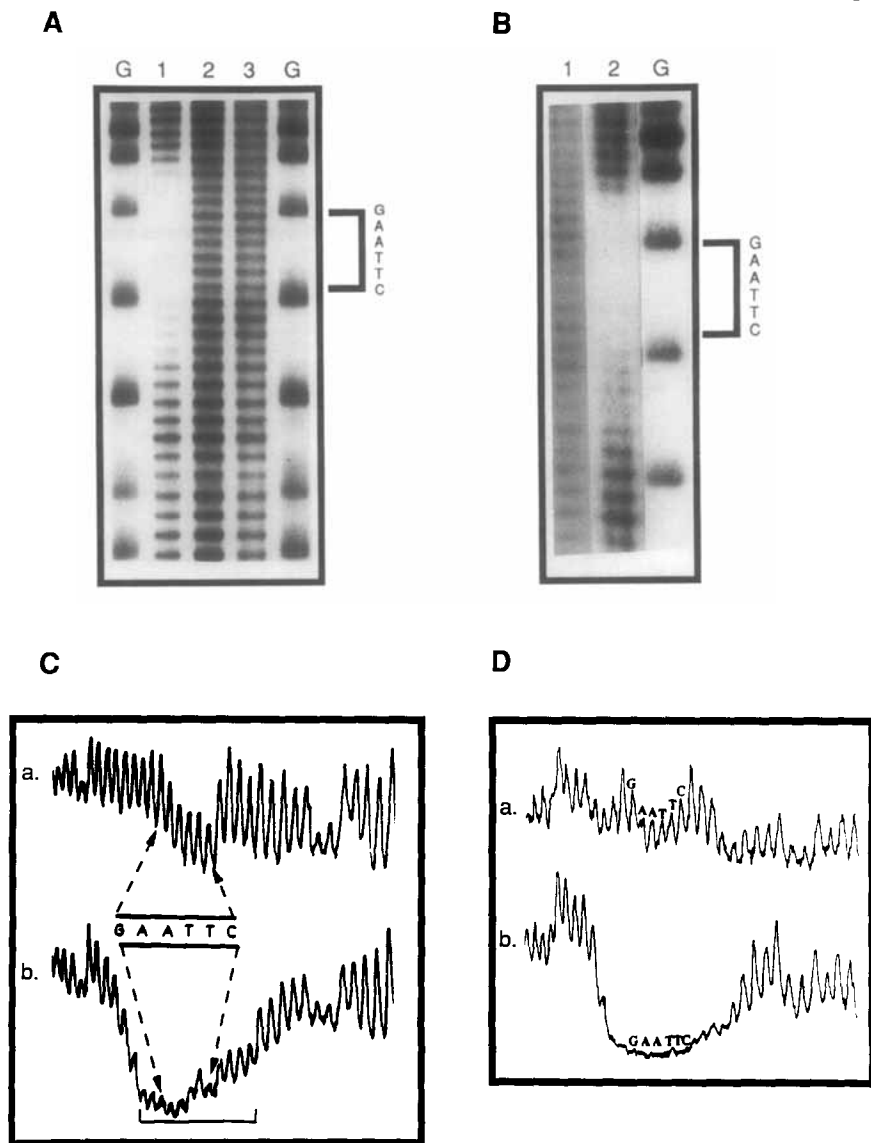


FIG. 4. Comparison of hydroxyl radical footprints of *EcoRI* produced with or without the additional step of gel isolation of the *EcoRI*-DNA complex. (A) and (C) show an autoradiograph and corresponding densitometer scan of the hydroxyl radical footprint of *EcoRI* produced by the standard footprinting technique (without gel isolation of the *EcoRI*-DNA complex). The autoradiograph and densitometer scan of a hydroxyl radical footprint of *EcoRI* produced by the standard footprinting technique followed by gel isolation of the *EcoRI*-DNA complex are shown in (B) and (D), respectively. In (A) are shown hydroxyl radical cleavage

0.5 mM DTT, in a total volume of 10 μ l. To ensure that equilibrium is reached for binding of the protein to DNA, the solution is incubated on ice for 15 min.

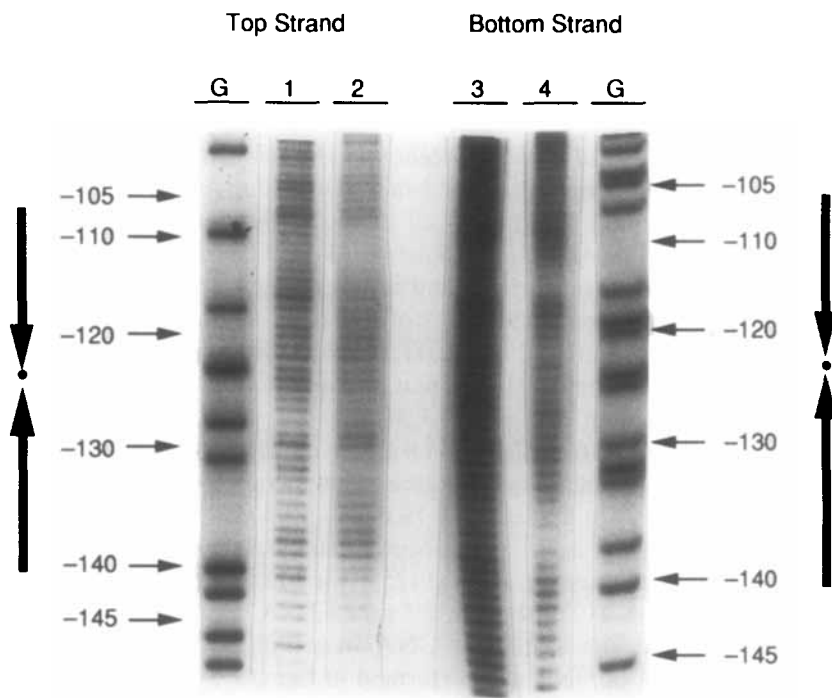
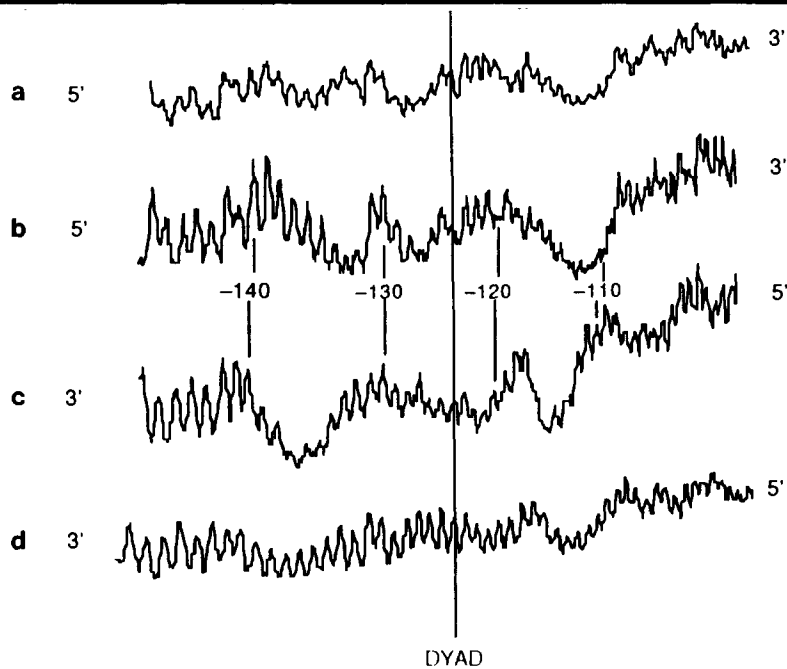
Hydroxyl Radical Cleavage Reaction. After warming the CUP2–DNA mixture to room temperature for 1 min, the cutting reaction is initiated by mixing together on the inner wall of the Eppendorf tube 2 μ l each of [Fe(EDTA)]²⁻ [8 mM iron(II), 16 mM EDTA], 0.024% hydrogen peroxide, and 160 mM sodium ascorbate, and adding this reagent to the CUP2–DNA mixture. The final concentrations of the constituents of the cleavage reagent are 1 mM iron(II), 2 mM EDTA, 0.003% hydrogen peroxide, and 20 mM sodium ascorbate. After 2 min at room temperature the reaction is stopped by addition of 5 μ l of 0.1 M thiourea.

Nondenaturing Gel Electrophoresis. To each sample, 4 μ l of loading dye (0.25% bromphenol blue, 0.25% xylene cyanol, and 30% glycerol) is added. Samples are loaded on a 16 cm $\times$ 19.5 cm $\times$ 1.5 mm 6% native polyacrylamide gel (acrylamide : bisacrylamide, 80 : 1). Electrophoresis is performed at room temperature at 125 V for 3 hr in a buffer consisting of 25 mM Tris–borate–HCl (pH 8.3), 2.5 mM EDTA. Detection of radioactive DNA, excision and elution of DNA from the gel, and sequencing gel electrophoresis of DNA are performed in the same manner as described above for the *Eco*RI–DNA complexes.

Results. The CUP2 protein was bound to a restriction fragment containing the UASc- and UASd-binding sites.¹³ Under the conditions used a single protein–DNA complex was observed via electrophoresis on a mobility shift gel. The best separation was achieved with the protein–DNA complex running slightly slower than the unbound DNA.

The resulting hydroxyl radical footprint of CUP2 is shown in Fig. 5. The footprint is highly symmetrical around the pseudodyad axis of symmetry of

products of a singly end-labeled 207-bp DNA restriction fragment (2 nM) in the absence (lane 3) and the presence (lane 2, 20 nM; lane 1, 80 nM) of *Eco*RI endonuclease. Lanes marked G are products of a Maxam–Gilbert guanine-specific sequencing reaction. The *Eco*RI recognition sequence, GAATTC, is indicated by a bracket. In (C), the upper scan (a) shows the cleavage of DNA in the absence of *Eco*RI and corresponds to lane 3 in (A). The lower scan (b) shows the cleavage pattern of DNA in the presence of *Eco*RI and corresponds to lane 1 in (A). The sequence reads 5' to 3', left to right, for each scan. The *Eco*RI recognition sequence is marked by arrows. In (B), lanes 1 and 2 show products of hydroxyl radical cleavage of a DNA restriction fragment that was isolated from the nondenaturing gel shown in Fig. 3. Lane 1, free DNA; lane 2, DNA isolated from the upper band on the mobility shift gel (the band containing the *Eco*RI–DNA complex). Lane marked G contains products of a Maxam–Gilbert guanine-specific sequencing reaction. The *Eco*RI recognition sequence, GAATTC, is indicated by a bracket. In (D), the upper scan (a) corresponds to lane 1 and the lower scan (b) corresponds to lane 2 in (B).

A**B**

the binding site. CUP2 protects three regions of each strand. Each set of protections at the ends is offset across the DNA strands from each other by three bases in the 3' direction. The protected region on each strand in the center of the site is offset by six bases in the 5' direction from the corresponding protected region on the other strand.

These results indicate that in each half of the UASc, CUP2 crosses the minor groove at the ends of the binding site between the two major grooves where methylation interference detects contacts with guanines.¹³ The protection at the center indicates that CUP2 interacts with the major groove at the center of the binding site.

Comments on Technique of Hydroxyl Radical Footprinting Followed by Mobility Shift Gel Electrophoresis

As discussed above, purification of a protein–DNA complex on a nondenaturing gel prior to sequencing gel electrophoresis can significantly enhance the hydroxyl radical footprint signal by elimination of bands that derive from cleavage of free DNA. However, to use this technique three technical pitfalls must be overcome. First, separation of a protein–DNA complex from free DNA by gel electrophoresis must be achieved. The conditions necessary to effect a separation depend on the protein, the DNA fragment length, and the number of binding sites. A general technique for separating free from bound DNA by mobility shift gel electrophoresis has been described.³⁴

³⁴ W. Hendrickson and R. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 3129 (1985).

FIG. 5. Hydroxyl radical footprinting of the CUP2–UASc complex. (A) shows the autoradiograph of a gel containing the hydroxyl radical cleavage pattern for the UASc in the absence of protein (lanes 1 and 3), and with bound CUP2 (lanes 2 and 4). Lanes 1 and 2 are hydroxyl radical cleavage products derived from the top strand, and lanes 3 and 4 are cleavage products derived from the bottom strand. Lanes marked G are products of a Maxam–Gilbert G reaction performed on the respective labeled strands. Nucleotides are numbered relative to the start site of transcription. The vertical lines indicate the region of almost perfect dyad symmetry. The dots are placed at position –124, the center of the pseudodyad. The corresponding densitometer scans are shown in (B). Scans a and b were made from lanes 1 and 2, and scans c and d were made from lanes 4 and 3, respectively. The vertical line marks the pseudodyad axis of symmetry. The footprint is clearly symmetrical from one strand to the other about the pseudodyad, with symmetry-related protections shaped identically. On the top strand the strongest protections occur at positions –133, –127, and –112, while on the bottom strand the symmetry-related protections are at positions –136, –121, and –115, respectively. From Buchman *et al.* (1990).¹³

Second, a substantial amount of DNA must be shifted into the protein–DNA complex and successfully eluted from the gel. Two means can be used to improve the amount of DNA that is shifted. One is to increase the amount of protein in the sample. The other is to omit the addition of marker dyes (bromphenol blue and xylene cyanol) to the protein–DNA sample. We usually realize a low recovery (about 50%) of DNA by elution from the gel. In order to have a sufficient amount of DNA to load on the sequencing gel, we have found it necessary to start the experiments with 5 to 10 times the desired amount.

Third, a protein–DNA complex must be stable during footprinting and loading onto the gel. Instability of the protein–DNA complex could lead to the following problems in interpreting the footprinting pattern of a protein–DNA complex that was gel isolated after the footprinting reaction. First, dissociation of the protein–DNA complex during gel loading or electrophoresis will diminish the protection signal, since some DNA that should have run in the band with complexed DNA instead runs as free DNA. Second, and more problematic, reassociation of protein with gapped DNA will be biased due to the effect of gaps on the affinity of protein for DNA, and will lead to a decreased cutting frequency observed at the nucleosides involved in specific contacts with the DNA. A gap at a critical place in the DNA sequence interferes with protein binding, as we have shown in the missing nucleoside technique³⁵ for studying protein–DNA complexes, which is described in the next section. In this situation, a decrease in signal in the data from the protein-bound DNA could therefore indicate either a protein protection or a protein contact.

The analysis becomes more complicated if there is more than one form of DNA–protein complex separated by the mobility shift gel, since a gap in the DNA will inhibit formation of the fully saturated complex, but may favor formation of one of the partially saturated complexes, resulting in an enhancement of signal for particular nucleosides in those complexes. Two control experiments may be performed to test for instability of the protein–DNA complex. One involves adding unlabeled DNA containing the binding site to the sample after hydroxyl radical treatment to see if exchange takes place. In the second control, comparison of the results of the experiment with conventional hydroxyl radical footprints and missing nucleoside data will show whether the pattern on the gel is more consistent with a footprint or a missing nucleoside signal.

³⁵ J. J. Hayes and T. D. Tullius, *Biochemistry* **28**, 9521 (1989).

Missing Nucleoside Experiment: Direct Information on Energetically Important Contacts of Protein with DNA

The set of all interactions between a DNA-binding protein and its recognition site can be divided conceptually into two types based on the architecture of DNA. One group of interactions is composed of the contacts between the protein and the sugar-phosphate backbone of DNA. The other group includes the interactions between the base moieties of DNA and the protein. The hydroxyl radical footprinting experiment yields high-resolution structural information about protein-DNA contacts of the first type. This technique shows where along a DNA molecule the sugar-phosphate backbone of DNA comes into close contact with the bound protein. However, little is learned about the second type of interaction, that between the protein and the bases of DNA, from the footprinting experiment.

We have described a new technique that can be used to obtain information about contacts between DNA and protein that fall into the second group. This technique, called the missing nucleoside experiment,³⁵ can be used to quickly determine at which positions in a DNA molecule the bases are in contact with a sequence-specific DNA-binding protein. This method is related to the missing contact experiment.³⁶ The missing nucleoside experiment uses hydroxyl radical cleavage chemistry to randomly remove nucleosides from DNA. The ability of the gapped DNA to bind to protein is then tested by gel mobility shift. Bases that make important contacts to protein are identified by sequencing gel electrophoresis of the bound and unbound fractions of DNA recovered from the mobility shift gel. The missing nucleoside experiment can be used to scan a DNA molecule for specific contacts at single-nucleotide resolution in just one experiment.

Overview of Missing Nucleoside Method

The hydroxyl radical cleavage reaction is used to generate DNA fragments that contain, on average, fewer than one randomly placed single-nucleoside gap per fragment. The gapped DNA molecules are then mixed with the protein, and DNA bound to protein is separated from free DNA by electrophoresis on a native polyacrylamide gel. Bands containing bound and free DNA are excised from the native gel, and the DNA is eluted and run on a sequencing electrophoresis gel for determination of the cleavage patterns in the bound and free samples. Since the DNA strand is already broken in the gapping reaction, no other treatment is necessary to develop the pattern. Although this procedure might seem very similar to that discussed in the previous section, for the missing nucleoside analysis

³⁶ A. Brunelle and R. F. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6673 (1987).

naked DNA is treated by the hydroxyl radical before the protein is introduced. In contrast, for gel purification of DNA-protein complexes before sequencing gel electrophoresis, the hydroxyl radical reaction is performed on the DNA-protein complex in the usual way.

The missing nucleoside experiment detects gaps in the DNA molecule that interfere with binding enough that the protein and DNA no longer migrate as a complex on the gel. A nucleoside that is important to formation of the protein-DNA complex yields a weak or missing band on the sequencing gel in the lane containing DNA that was bound to protein, or a high-intensity band in the lane in which free DNA is run. Nucleosides that do not make energetically important contacts also give a positive experimental signal, but with the opposite pattern of band intensity. Complementary information is therefore obtained from the two lanes (containing bound and free DNA) that result from the original sample. In contrast, protection (footprinting) methods give a negative result for bases not involved in protein contact.

Advantages of Missing Nucleoside Experiment

Site-directed mutagenesis is a powerful adjunct to direct structural studies and can be used to investigate contacts with protein at each base pair in a recognition sequence. Many time-consuming manipulations are needed to scan even a small binding site using mutagenesis. For example, a saturation mutagenesis analysis of λ repressor binding to its operator required the synthesis of over 50 new versions of the DNA recognition site.³⁷ The binding affinity of repressor for each of the mutant operators was then determined. While the results from the mutation experiments and the missing nucleoside experiment generally agree,³⁵ the latter procedure took only a few days to complete—certainly a much shorter time than that required to complete the former experiment. A second advantage of the missing nucleoside experiment over mutational methods arises because often only one member of a base pair is involved in specific recognition.²³ Since the identities of both bases in a base pair are changed by mutagenesis, it is not easy to determine which is the important contact. The missing nucleoside experiment, on the other hand, can yield information about contacts to a particular base that is independent of the contribution of the base on the opposite strand.³⁵ Moreover, not all base substitutions at a particular position result in a reduction in binding affinity.^{38,39}

Another experimental approach, methylation interference, involves

³⁷ A. Sarai and Y. Takeda, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 6513 (1989).

³⁸ T. Pieler, J. Hamm, and R. G. Roeder, *Cell (Cambridge, Mass.)* **48**, 91 (1987).

³⁹ Y. Takeda, A. Sarai, and V. M. Rivera, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 439 (1989).

assessing the effect of alkylation of the DNA bases on protein binding. Modified positions that interfere with protein binding are revealed on induction of backbone cleavage at the alkylated sites of the DNA molecules that were unable to bind to protein. However, information is available for only a subset of base-specific contacts and, unfortunately, only contacts with purines can generally be detected.

As mentioned above, the missing nucleoside experiment is related to the "missing contact" method of Brunelle and Schleif.³⁶ This experimental approach can also be used to assess the contribution to protein binding of each member of a base pair independently. However, in the missing contact method Maxam–Gilbert sequencing chemistry is used to remove the bases, so three separate experiments are needed to determine the bases important for protein binding. With the missing nucleoside technique a single chemical reaction, and thus a single lane on a sequencing gel, is all that is needed for the determination. The missing nucleoside experiment thus reduces the amount of work and material required for an analysis compared to the missing contact method, and since all of the signal appears in just one sequencing gel lane, the results are somewhat easier to interpret.

Methods

Proteins and DNA Fragments. Any protein–DNA complex can be analyzed by this method provided that the complexed DNA can be isolated from unbound DNA in some manner, and that enough sample is recovered to generate a signal on a sequencing gel. In our laboratory gel mobility shift is used to effect the separation, so the radioactively labeled DNA fragment must be of the appropriate size (see Troubleshooting, below). We also found that the ends of the fragment must be further than 15–20 bp from the binding site to be assayed, in order to prevent dissociation of short oligomers from the gapped DNA molecule.⁴⁰

Hydroxyl Radical Cleavage. The labeled DNA molecule is randomly gapped by reaction with the hydroxyl radical. Typically, a relatively large quantity of labeled fragment, 250–500 fmol, enough for two or three missing nucleoside experiments, is dissolved in 70 μ l TE buffer. For comparison, a typical hydroxyl radical footprinting experiment contains about 10–20 fmol of labeled fragment. The DNA is treated with a total of 30 μ l of the cleavage reagent (10 μ l of each of the individual components) in the manner described above. The cleavage reaction is terminated by the addition of 30 μ l of 0.1 M thiourea, 10 μ l of 0.2 M EDTA (pH 8.0), 20 μ l of 3 M sodium acetate, and 40 μ l of TE buffer, to bring the volume of the

⁴⁰ V. Rimphanitchayakit, G. F. Hatfull, and N. D. F. Grindley, *Nucleic Acids Res.* **17**, 1035 (1989).

reaction mixture to 200 μ l. (These solutions can be added to the cleavage reaction together, as a cocktail.) The DNA is then precipitated by the addition of 500 μ l of ethanol at -20° . The sample is reprecipitated by first dissolving the pellet in 200 μ l of 0.3 M sodium acetate, 0.2 mM EDTA to remove traces of iron that may remain in the sample. The final pellet is rinsed with 70% ethanol (-20°), dried under vacuum, and then dissolved in a storage buffer consisting of 10 mM Tris-Cl, 0.1 mM EDTA (pH 8.0). In the case of proteins requiring metal cofactors, such as *Xenopus* transcription factor IIIA, EDTA is omitted from the final storage buffer.

Formation of Protein–DNA Complexes. Buffers for protein binding are prepared as in the usual procedure. Note that buffer systems that contain a high level of glycerol (or any other radical scavenger), which would be unsuitable for use in a hydroxyl radical footprinting experiment, are fully compatible with the missing nucleoside experiment. Protein is added to labeled, gapped DNA that is dissolved in binding buffer, and the system is allowed to come to equilibrium. A typical binding solution contains 125–250 fmol of gapped, labeled DNA. The amount of protein to be added is determined empirically and can be estimated by titrating small quantities of the labeled DNA with protein. (The amount of protein required to saturate a binding site on DNA in a footprinting experiment generally was found to be less than that required to saturate the DNA when measured by gel mobility shift.) The amount of protein or of unlabeled competitor DNA in the binding mixture is adjusted to give approximately 95% formation of complex as judged by the intensities of the bands on the mobility shift gel. This amount of protein generally results in good bound and unbound samples, but some proteins may require small adjustments in the amount of complex formed to optimize the signal observed on the sequencing gel (see Troubleshooting, below).

Fractionation and Analysis of Bound and Unbound DNA. DNA bound to protein is separated from free DNA by electrophoresis on a native polyacrylamide (mobility shift) gel as described.⁴¹ For further comments on gel mobility shift techniques, see Troubleshooting, below. Radioactive bands containing bound and free DNA are excised from the gel and the DNA is eluted. DNA is precipitated by addition of ethanol, rinsed with ethanol, dried under vacuum, and dissolved in 10–20 μ l of TE buffer. Note that typically in the unbound sample little or no radioactivity is detectable, while a majority of the original counts will appear in the bound sample. About one-fourth of each sample is then analyzed by sequencing gel electrophoresis. The amount of the bound or unbound sample that is loaded may be adjusted slightly on later sequencing gels to better match

⁴¹ A. Wolffe, *EMBO J.* 7, 1071 (1988).

the band intensities of the two samples, or to match the intensities of bands from corresponding footprinting sample.

Applications of Missing Nucleoside Experiment

The missing nucleoside method has been used to analyze two well-understood protein–DNA complexes, the complexes of the bacteriophage λ repressor and Cro proteins with the O_R1 operator site.³⁵ A detailed comparison of the missing nucleoside results with the protein–DNA contacts detected crystallographically and genetically has been presented.³⁵

λ Repressor– O_R1 Complex. In the case of the λ repressor–operator complex, an example of the correlation between the cocrystal structure and the missing nucleoside experiment is most easily observed at position 2 in the operator. The crystal structure reveals an intricate network of contacts with the adenine member of the base pair at this position, while no contact is detected with the thymine. The missing nucleoside experiment³⁵ indicates exactly the same circumstance. A large contact signal is detected at the adenosine at position 2, while the signal corresponding to the thymidine indicates that this nucleoside is uncontacted by the protein.

Another interesting example of the data available from the missing nucleoside experiment involves the central 5 bp of the operator and the structural unit of the λ repressor that contacts them. The repressor has an amino-terminal arm that wraps around the center of the operator and contacts the backside of the DNA in the major groove.⁴² The supposition that the arms contact the central bases of the operator is borne out by the cocrystal structure.²³ The missing nucleoside data⁴⁵ show that the arms make extensive and energetically important contacts with positions 7, 8, and the central dyad base. These contacts agree well with those detected by mutagenesis methods in the center of the operator.³⁷ The data also indicate that the structure of the amino-terminal arm is not the same in each half of the dimer. Specifically, the results suggest that the amino-terminal arm that extends from the repressor monomer that contacts the consensus half of the operator binds much more tightly to the DNA than does the arm that is associated with the nonconsensus half of the operator. This proposal correlates well with the cocrystal structure, which shows that the arm that contacts the consensus half-site is the more ordered.²³

Cro– O_R1 Complex. We also have used the missing nucleoside technique to analyze a related system, the complex of the λ Cro protein with the O_R1 operator sequence. Cro and λ repressor bind to the same DNA sites, but with different affinities. The two proteins might thus be expected

⁴² C. O. Pabo, W. Krovatin, A. Jeffrey, and R. T. Sauer, *Nature (London)* **298**, 441 (1982).

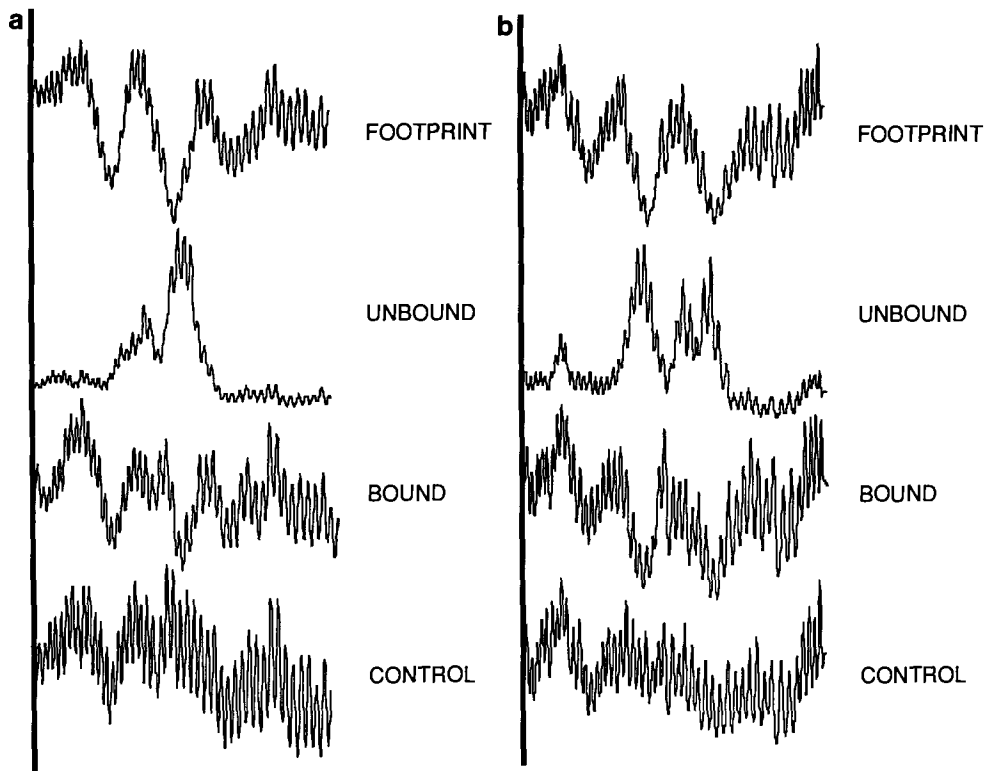


FIG. 6. Missing nucleoside analysis of the λ Cro- O_R1 complex. Shown are densitometer scans of the autoradiograph of the denaturing gel on which was separated DNA that was subjected to a standard hydroxyl radical footprinting reaction (footprint), DNA extracted from a mobility shift gel that was complexed with Cro (bound) and which ran as free DNA (unbound), and DNA that was subjected to cleavage by the hydroxyl radical in the absence of protein (control). Scans for the two DNA strands [(a) top strand, (b) bottom strand] are shown.

to make different sets of contacts with DNA. Indeed, despite marked similarities in the DNase I and hydroxyl radical footprints of Cro and repressor,⁸ the missing nucleoside experiment yields different patterns for the two proteins.

One particular difference occurs at the center of the operator site. Cro protein does not have the same amino terminal arm that repressor has. Takeda *et al.*³⁹ demonstrated by mutagenesis experiments that the central 3 bp of O_R1 contribute little to the energetics of Cro binding. The missing nucleoside analysis shows that the center of the O_R1 site is largely devoid of contacts with Cro (Fig. 6). With λ repressor, by contrast, large contact

signals are observed in this region,³⁵ which we attribute to the interaction of the arms of repressor with the center of the binding site.

Another Possible Use of the Method. The missing nucleoside experiment can be used to locate the DNA-binding site for a protein that is available only in a complicated extract, and for which the protein–DNA complex is observed only as a band on a mobility shift gel. Such an extract might be unsuitable for normal hydroxyl radical footprinting, because it contains high concentrations of radical scavengers (glycerol, for example). However, a sort of reverse footprint can be obtained by applying the missing nucleoside technique to this system. Protein-bound, mobility-shifted DNA fragments from such an experiment, even if the bound band represents only a small fraction of the total labeled DNA in the sample, should yield an obvious bound signal at the location of the recognition sequence.

Troubleshooting Missing Nucleoside Experiment

We have made several observations that might aid attempts to employ this technique with other systems. Each of these points may or may not apply to a particular protein–DNA complex. Some fine tuning of the parameters may be required before an optimal result is obtained. Most of the suggestions concern a process that is to date poorly understood: the separation of free DNA from protein–DNA complexes by mobility shift gel electrophoresis.^{31,32}

It also should be noted that introducing gaps in a DNA molecule might affect the binding of a protein because of structural changes induced in the DNA backbone⁴³ as a result of the missing nucleoside. Comparison of our results for λ repressor with the cocrystal structure and with mutagenesis experiments showed that nearly all of the missing nucleoside signals are consistent with the loss of a contact with a nucleoside.³⁵ However, with other protein–DNA systems changes in DNA structure should also be considered.

1. *Gel.* Other gel systems than the one employed for this study have been used for mobility shift. We have obtained a quantitative shift for the TFI_{II}A–5S RNA gene complex using a 0.7% agarose gel with 0.5 $\times$ TB buffer [50 mM Tris-Cl (pH 8.3), 50 mM boric acid] at room temperature. However, all of the proteins that we have investigated with the missing nucleoside technique worked well with the gel system described by Wolffe⁴¹: 4% acrylamide, 0.08% bisacrylamide, in a buffer consisting of

⁴³ G. B. Koudelka, P. Harbury, S. C. Harrison, and M. Ptashne, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 4633 (1988).

20 mM HEPES (pH 8.3), 0.1 mM EDTA, 5% glycerol. The same solution without glycerol is used as the running buffer. The gel has the thickness of a sequencing gel, which conserves gel material and minimizes the volume of gel from which a sample must later be recovered. This thickness does, however, make it difficult to produce good wells and requires that the gel be loaded with a 0.2-mm diameter syringe. It is absolutely necessary for the large plate to be liberally siliconized before every gel is poured. Care must be taken when plastic wrap is put on and taken off since these gels are very fragile.

2. *Gel buffers.* Buffers such as $0.5 \times$ TB or TBE can be used in this system, depending on the effect of EDTA on protein binding. However, the HEPES buffer yielded superior results compared to any buffer containing Tris, especially in the experiments with TFI_{IIA}. Although high concentrations of EDTA can interfere with the binding of metal-containing proteins, when a small amount of EDTA (0.1 mM) was included in the gel buffer the sharpness of bands was improved. This might be due to the scavenging of excess metal ions, such as free zinc that remains from TFI_{IIA}-DNA complex formation. A low level of EDTA does not seem to affect the TFI_{IIA}-DNA complex since the exchange rate of zinc in the protein-DNA complex is probably much slower than it is in the free protein.

3. *Gel running conditions.* Once a protein-DNA complex enters the gel, the complex seems to be very stable and migrates without dissociation. However, the equilibrium can be disturbed by the process of transferring the sample from a tube to the gel matrix. An interesting example involves λ repressor. With a gel run at room temperature, only a minute fraction of the labeled DNA sample can be made to run as a complex with repressor, regardless of protein concentration. Footprinting experiments show that this is true for concentrations of repressor well above that required to saturate the binding site *in vitro*. In contrast, Cro protein easily shifts the mobility of the DNA molecule under these conditions. However, when the repressor sample is cooled and run on a gel at 4°, a quantitative shift of DNA into the bound band results. The on/off rates for repressor might be much slower at 4°, essentially freezing the equilibrium of the sample and thus allowing the complex to enter the gel.

4. *Loading sample.* Artifacts can be introduced during the preparation and loading of the sample. For this reason the gel loading solution (1 $\times$ HEPES/EDTA buffer, 40% (v/v) glycerol, 0.025% dye) is added just before loading. The sample is carefully layered into the well without mixing with the gel running buffer.

5. *Sample manipulations.* The strength of the contact signal observed can be improved by the above suggestions. In addition, certain manipula-

tions of the sample before loading, such as the addition of unlabeled competitor DNA, can help. A typical sample is prepared under conditions that maximize binding to a labeled DNA fragment. Then an excess of unlabeled, ungapped DNA, which usually contains the protein-binding site, is added. The amount of competitor DNA is adjusted so as to compete for only the most weakly bound protein. Alternatively, a large excess of competitor is added and the bound complexes are challenged for a specific length of time.³⁶ The time is controlled either by directly loading the sample on the gel or by cooling the sample to 0° on ice to slow the exchange process before loading.

6. *Miscellaneous.* The length of the DNA fragment is important, since a long fragment makes it difficult to separate bound from free DNA. As mentioned above, the binding site cannot be located too close to the end of the fragment, because dissociation of short oligonucleotides from the duplex after the gapping reaction will leave single-stranded regions in the binding site that are incapable of binding protein.⁴⁰ It might be necessary to do two separate experiments to get good bound and unbound samples. Since only a few bands occur in the unbound lane on the sequencing gel (and thus there is little radioactivity associated with unbound DNA), excellent unbound samples can be obtained by cutting out the place on the native gel where the free DNA is expected to run, even if there is no apparent radioactive band there.

Acknowledgments

We thank Carl Pabo for providing us with the oligonucleotide containing the O_L1 operator site and for samples of the λ Cro protein and the 1–92 fragment of λ repressor, Gary Ackers for supplying λ repressor and for making available to us the 2D gel scanning system, and Irina Russu for generous provision of *EcoRI*. We are grateful to Michael Karin for providing samples of *CUP2* and a clone containing the *UASc*, and Thomas J. Kelly for giving us samples of NF I and clones of Ad DNA. This work was supported by PHS Grant GM 41930. T.D.T. is a fellow of the Alfred P. Sloan Foundation, a Camille and Henry Dreyfus Teacher-Scholar, and the recipient of a Research Career Development Award (CA 01208) from the PHS. J.R.L. acknowledges the support of a postdoctoral fellowship from the Institute for Biophysical Research on Macromolecular Assemblies at Johns Hopkins University (an NSF Biological Center). W.J.D. is the recipient of a National Research Service Award from the PHS.

[20] Nuclease Activity of 1,10-Phenanthroline–Copper in Study of Protein–DNA Interactions

By DAVID S. SIGMAN, MICHIO D. KUWABARA, CHING-HONG B. CHEN,
and THOMAS W. BRUCE

Introduction

The nuclease activity of the 1,10-phenanthroline–copper complex efficiently nicks DNA and RNA under physiological conditions.^{1,2} Two reactants are essential for the cleavage reaction: the 1,10-phenanthroline–cuprous complex and hydrogen peroxide. The chemistry of B-DNA scission has been studied in detail, and proceeds by the kinetic mechanism summarized in Fig. 1.^{3–9} The reaction is funneled through a noncovalent intermediate that is responsible for the structural and sequence-dependent variability of the nucleolytic activity. DNA scission is achieved by the oxidative species formed by the one-electron oxidation of the 1,10-phenanthroline–cuprous complex by hydrogen peroxide when the coordination complex is bound to the surface of DNA.

With the 2 : 1 1,10-phenanthroline–cuprous complex, reaction pathway (a) (Fig. 2) accounts for over 70–90% of the scission events. This pathway involves the C-1 hydrogen of the deoxyribose as the initial site of oxidative attack. Pathway (b) results from oxidative attack on the C-4 hydrogen.⁵ Either reaction pathway requires that the initial sites of reaction be in the minor groove of B-DNA.

The nuclease activity of 1,10-phenanthroline–copper has been used in two distinct ways in the study of protein–nucleic acid interactions. In the first, copper complexes prepared from 1,10-phenanthroline (OP–Cu) or

¹ D. S. Sigman, D. R. Graham, V. D'Aurora, and A. M. Stern, *J. Biol. Chem.* **254**, 12269 (1979).

² D. S. Sigman, *Acc. Chem. Res.* **19**, 180 (1986).

³ L. M. Pope, K. A. Reich, D. R. Graham, and D. S. Sigman, *J. Biol. Chem.* **257**, 12121 (1982).

⁴ M. D. Kuwabara, C. Yoon, T. E. Goyne, T. B. Thederahn, and D. S. Sigman, *Biochemistry* **25**, 7401 (1986).

⁵ T. E. Goyne and D. S. Sigman, *J. Am. Chem. Soc.* **109**, 2846 (1987).

⁶ L. E. Pope and D. S. Sigman, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 3 (1984).

⁷ G. J. Murakawa, C.-H. B. Chen, M. D. Kuwabara, D. Nierlich, and D. S. Sigman, *Nucleic Acids Res.* **17**, 5361 (1989).

⁸ D. S. Sigman, A. Spassky, S. Rimsky, and H. Buc, *Biopolymers* **24**, 183 (1985).

⁹ T. B. Thederahn, M. D. Kuwabara, T. A. Larsen, and D. S. Sigman, *J. Am. Chem. Soc.* **111**, 4941 (1989).

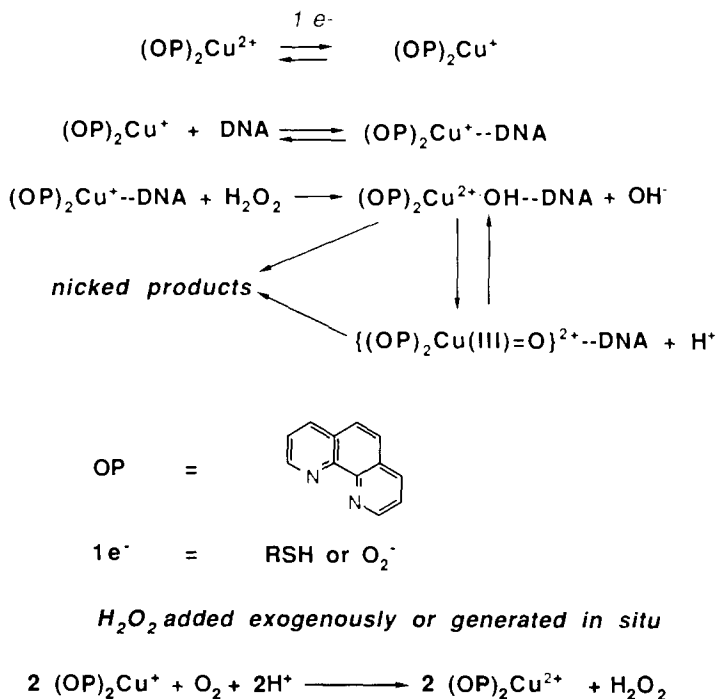


FIG. 1. Kinetic mechanism of the nuclease activity of 1,10-phenanthroline-copper. The reaction pathway is obligatory because the 1,10-phenanthroline-cupric ion DNA complex is not reducible.

5-phenyl-1,10-phenanthroline (5- ϕ -OP-Cu) provides footprints of protein binding and can detect protein-induced structural changes in DNA.^{10,11} In the second approach, 1,10-phenanthroline has been attached to DNA-binding proteins.¹² The orientation of the binding protein with respect to the DNA recognition sequence can be inferred by examining the positions of cleavage.

OP-Cu Footprinting for Studying Protein-DNA Interactions

The nuclease activity of 1,10-phenanthroline-copper makes single-strand nicks in double-stranded DNAs under physiological conditions. Piperidine treatment of the modified DNA is not necessary to cleave the

¹⁰ A. Spassky and D. S. Sigman, *Biochemistry* **24**, 8050 (1985).

¹¹ M. D. Kuwabara and D. S. Sigman, *Biochemistry* **26**, 7234 (1987).

¹² C.-H. B. Chen and D. S. Sigman, *Science* **237**, 1197 (1987).

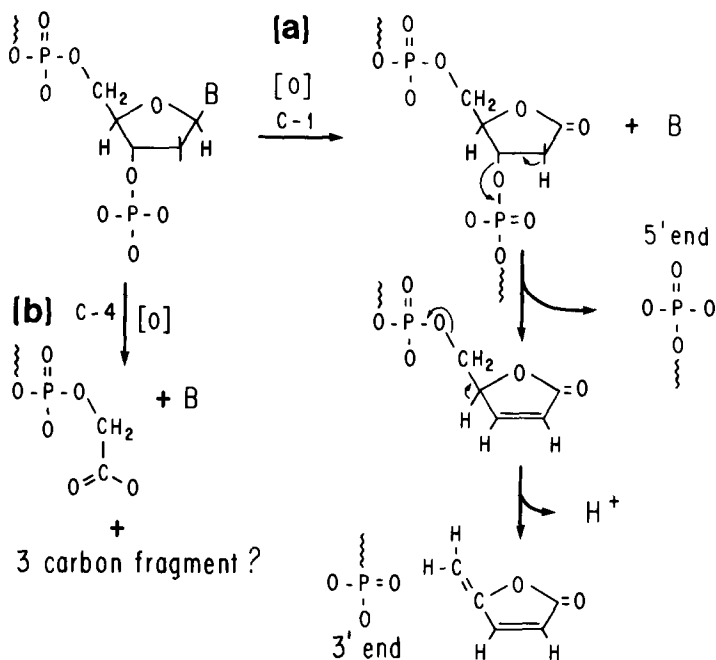


FIG. 2. Postulated reaction mechanism for the scission of DNA by the 2:1 1,10-phenanthroline-copper complex. Reaction pathway (a) accounts for greater than 70% of the scission events at any sequence position. The three-carbon fragment derived from ribose in pathway (b) has not been isolated although it is known that base propenals are not formed.

phosphodiester backbone. In contrast to DNase, which interacts with DNA both in the major and minor groove, the chemistry of the nuclease activity of OP-Cu is restricted to the minor groove. This aspect of the mechanism of OP-Cu is relevant to the interpretation of footprinting results. A ligand will protect scission at a given sequence position if it sterically blocks access to the C-1 hydrogen either directly or indirectly, or if it alters the minor groove geometry so that the tetrahedral coordination complex binds poorly. Conversely, if the ligand binds exclusively in the major groove without sterically blocking access to, or grossly perturbing, the dimensions of the minor groove of the recognition sequence, then no footprint will be observed. For example, *EcoRI* makes sequence-specific contacts in the major groove.^{13,14} According to the X-ray structure of Rosenberg and colleagues, the minor groove is accessible to solvent even

¹³ C. A. Frederick, J. Grable, M. Melia, C. Samudzi, L. Jen-Jacobson, B. C. Wang, P. Greene, H. W. Boyer, and J. M. Rosenberg, *Nature (London)* **309**, 327 (1984).

¹⁴ J. A. McClarin, C. A. Frederick, B. C. Wang, P. Greene, H. W. Boyer, and J. M. Rosenberg, *Science* **234**, 1526 (1986).

though the DNA structure is slightly kinked. Although DNase I is blocked from cutting within the recognition sequence, the OP-Cu scission reaction is not inhibited.⁴

A unique feature of OP-Cu as a footprinting reagent is its ability to detect protein-induced changes in DNA.¹⁰ This unanticipated feature of the reactivity of OP-Cu may arise from the affinity of this hydrophobic cation for protein surfaces and/or its intercalation into underwound DNA. For example, single-stranded DNA formed at the active site of *Escherichia coli* RNA polymerase of the transcriptionally competent open complex is efficiently cut on the template strand by OP-Cu and even more effectively cleaved by the copper complex of 5-phenyl-1,10-phenanthroline (5- ϕ -OP-Cu) (Fig. 3).¹⁵ The same scission pattern is observed with the wild-type promoter in the presence of the cyclic AMP-binding protein as is observed in the strong cAMP-independent *lacUV5* promoter.¹⁰ Presumably the DNA in both open complexes attains the same structures. Both reagents can be used to detect the series of intermediates formed during the initiation of transcription with the *lacUV5* promoter. In fact, 5- ϕ -OP-Cu can detect double-stranded DNA at the leading edge of a transcription bubble both in the open and elongation complexes (Fig. 3, lanes b and e).¹⁵ The inference that DNA downstream of the positions of nucleotide triphosphate incorporation is double stranded relies on the observation of nicking on both strands. Scission sites on the template strand upstream of the site of nucleotide triphosphate incorporation are not associated with any cutting on the nontemplate strand. Magnesium ion is essential for the enzyme-induced melting of the double-stranded DNA at the active site of RNA polymerase.¹¹

Detection of these strained intermediates appears to be a unique feature of the phenanthroline-copper complexes. Neither MPE nor DNase I can detect polymerase-induced changes at the start of the transcription. In addition to detecting single-stranded regions in the *lac* system, the OP-Cu reagent also can detect parallel sites in the *gal*, *mal*, *ara*, and *mer* operons.^{16,17} 5- ϕ -OP-Cu also detects strain in the DNA induced by merR and RNA polymerase upstream of the *MerT* transcription start site.¹⁷

Footprinting within Gel Matrix

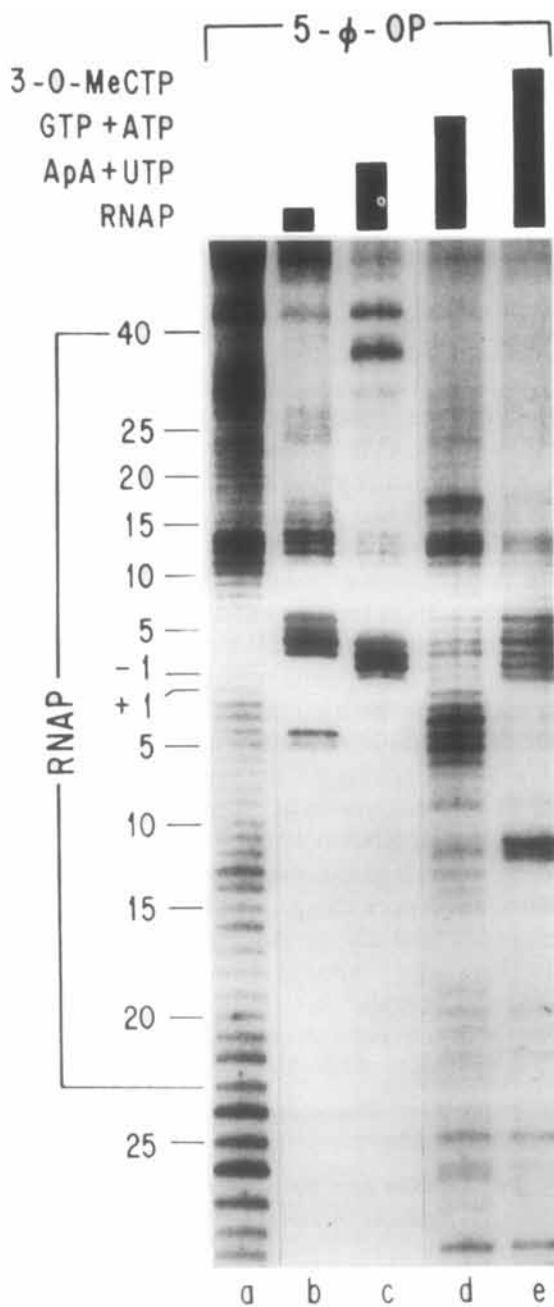
Experimentally, OP-Cu can be used as a footprinting reagent in a completely analogous fashion as DNase.¹⁸ The protein-DNA complex is

¹⁵ T. B. Thederahn, A. Spassky, M. D. Kuwabara, and D. S. Sigman, *Biochem. Biophys. Res. Commun.* **168**, 756 (1990).

¹⁶ A. Spassky, S. Rimsky, H. Buc, and S. Busby, *EMBO J.* **7**, 1871 (1988).

¹⁷ B. Frantz and T. V. O'Halloran, *Biochemistry* **29**, 4747 (1990).

¹⁸ D. J. Galas and A. Schmitz, *Nucleic Acids Res.* **5**, 3157 (1978).



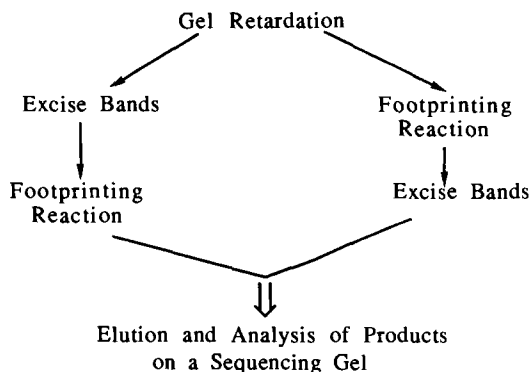


FIG. 4. Footprinting protein-DNA complexes using OP-Cu following separation by gel retardation. Two possible protocols to identify DNA sequences recognized by DNA-binding protein.

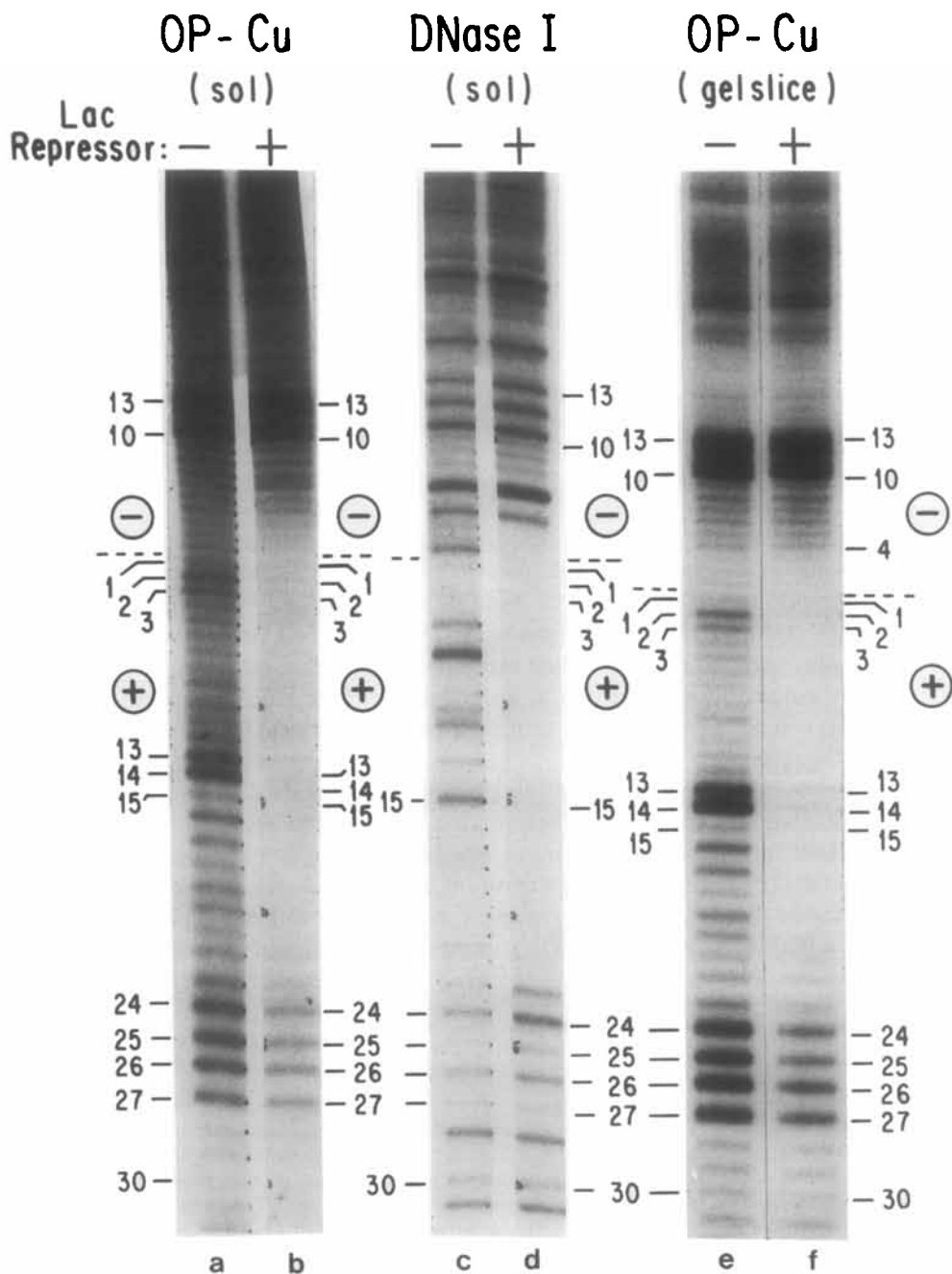
formed and attacked with the nucleolytic agents and the digested products analyzed on a sequencing gel. Alternatively, in procedures parallel to methylation interference assays, the DNA can be lightly digested either with DNase or OP-Cu (or ferrous EDTA or MPE), used to form a DNA-protein complex, and then separated from unbound DNA with the widely employed gel-retardation procedure.^{19,20} The DNA bound to the protein will not be modified at sequence positions crucial for binding (see [19], this volume).

However, a novel footprinting procedure has been developed using OP-Cu that incorporates gel retardation *prior* to the nuclease reaction (Fig. 4).¹¹ In gel retardation assays, the binding of a protein to a labeled restriction fragment alters the migration of the DNA in a nondenaturing acrylamide gel.^{19,20} Even in a complex mixture of proteins, separating a protein that binds to a labeled DNA is possible. To identify the sequence-

¹⁹ D. M. Crothers, *Nature (London)* **325**, 464 (1987).

²⁰ M. M. Garner and A. Revzin, *Trends Biochem. Sci.* **11**, 395 (1986).

FIG. 3. Footprints by 5- ϕ -OP-Cu of the binding of RNA polymerase as a function of added nucleotides. The footprinting reagents used are indicated at the top. *lacUV5* fragment labeled on the 5' end of the template strand was used for all experiments. Five different reaction conditions were used. The components of each reaction mixture are reflected by the size of the bar on top of each lane: (a) DNA scission pattern in absence of any addition; (b) in the presence of RNA polymerase; (c) ApA and UTP added; (d) nucleotides ApA, UTP, GTP, and ATP are added; (e) chain-terminating 3-*O*-methyl-CTP. Brackets indicate the limits of footprint observed with DNase I.



specific contacts between the protein and DNA, the footprinting reaction with OP-Cu can be carried out directly in the gel. The small size and ready diffusibility of the reactants of the nuclease activity—1,10-phenanthroline-copper ion, 3-mercaptopropionic acid, and hydrogen peroxide (molecular oxygen)—and the insensitivity of the reaction to inhibition by organic radical traps permit the reaction to proceed readily within the gel matrix. Deoxyoligonucleotide products are directly eluted from the gel and analyzed on sequencing gels. For initial calibration of this method, the binding of the *lac* repressor to the *lac* operator was examined. The same footprints were obtained within the gel as were observed in solution using OP-Cu. In turn, these patterns were consistent with those obtained in the solution using DNase I (Fig. 5).

In addition to convenience, combining these two methodologies has additional advantages. They ensure that the footprinting reaction is carried out on a discrete electrophoretic species. Moreover, they allow the footprinting of unstable complexes that are difficult to detect in solution for two reasons. First, unbound DNA is separated from the DNA-protein complex and, therefore, the background cutting of unbound DNA is greatly reduced. Second, the gel matrix inhibits the dissociation rate of the protein from its binding site on DNA. Unlike procedures in which the DNA is modified first and then used to form complex, there is no possibility that structural changes induced by the prior modification might influence the footprint obtained. Furthermore, the DNA-protein complex on which the footprints reaction is carried out is fully functional. As a result, molecules can be diffused in the gel matrix (e.g., ATP or inorganic ions) to interact with the protein-DNA complex prior to the footprinting reaction. It is then possible to determine their impact on the structure of the protein-DNA complex. For example, it was possible in this way to show that the magnesium ion was essential for forming single-stranded regions at the active site of RNA polymerase.²¹

²¹ M. D. Kuwabara and D. S. Sigman, *Biochemistry* **26**, 7234 (1987).

FIG. 5. Comparison of the OP-Cu and DNase I footprints of the *lac* repressor-*lac* operator binding in solution with that obtained of the 1:1 complex isolated by gel retardation. The footprinting experiment was carried out in an acrylamide plug that had been excised from the gel. The 1:1 complex was isolated from an incubation mixture containing 9.4 nM L8-UV-5 DNA (2.5×10^6 cpm) and 19 nM *lac* repressor. Lanes a and b, OP-Cu digestion in solution: (a) free DNA; (b) repressor-DNA complex; lanes c and d: (c) free DNA; (d) repressor-DNA complex; lanes e and f, OP-Cu digestion within the gel slice: (e) unbound DNA from gel retardation assay; (f) 1:1 repressor-DNA complex from gel retardation assay.

Experimental Procedures for Footprinting

Footprinting in Solution

5'-Labeled L8-UV5 186-bp DNA (10^5 cpm; 0.03 pmol) is incubated with 50 ng of *lac* repressor in 50 mM Tris-HCl, pH 8.0, 10 mM $MgCl_2$, and 0.2 mg/ml bovine serum albumin (BSA) in 10 μ l for 10 min at 37°. For OP-Cu footprinting, 1 μ l of 1 mM OP/0.23 mM $CuSO_4$ and 1 μ l of 58 mM 3-mercaptopropionic acid (MPA) are added to the 10 μ l mixture and incubated for 5.0 min at 37°. Digests are quenched by first adding 1 μ l of 28 mM 2,9-dimethyl-OP and then adding 10 μ l of the reaction mixture to the sucrose-urea mixture described in the paragraph below. The quenched solutions containing the sample are then heated at 90° for 5 min, and then loaded directly onto 10% sequencing gels that were run at 45 W for 3 hr. Gels are exposed to X-ray film at -20°.

The same footprinting experiment is carried out using DNase I in the following way. DNase I (1 μ l of 40 U/ml) is added and incubated for 2.0 min at 37°. Digests are stopped by the addition of a 10- μ l aliquot of the digestion solution to 10 μ l of a mixture composed of the following: 300 mg of sucrose, 840 mg of urea, 40 μ l of 0.5 M EDTA, 100 μ l of 0.1% (w/v) bromphenol blue, and 100 μ l of 0.1% (w/v) xylene cyanol. In order to obtain 10 ml of a homogeneous mixture, the latter is liquefied by heating to 90°.

In Situ OP-Cu Footprinting following Gel Retardation Assays

Slab gels are used for all gel retardation assays.^{19,20}

Footprinting Reaction in the Acrylamide Matrix. The same four reaction components are necessary for the OP-Cu footprinting reaction in the acrylamide matrix as in solution: 1,10-phenanthroline (OP), 2,9-dimethyl-1,10-phenanthroline (2,9-dimethyl-OP), cupric sulfate, and MPA. Prepare the following solutions just prior to use: mix equal volumes of 40 mM OP (in 100% ethanol) with 9.0 mM $CuSO_4$ (in water); dilute 1/10 with water to 2.0 mM OP/0.45 mM $CuSO_4$ (solution A); dilute neat MPA 1/200 with water to 58 mM MPA (solution B).

Two methods of carrying out the footprinting reaction within the acrylamide matrix are used. In the first, the wet retardation gel is exposed to X-ray film for 30 min at room temperature until the retarded bands are visible. The bands of interest are then excised with a razor and immersed in 100 μ l of 50 mM Tris-HCl, pH 8.0. Then, 10 μ l of solution A is added followed by 10 μ l of 28 mM 2,9-dimethyl-OP (in 100% ethanol) and 270 μ l of a solution containing both 0.5 M ammonium acetate and 1 mM EDTA. The DNA is then eluted overnight at 37°. The eluted DNA is ethanol

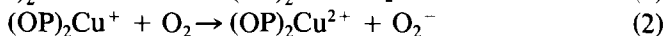
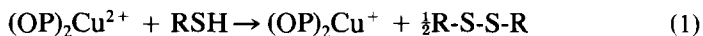
precipitated (carrier DNA is unnecessary), resuspended in 80% (v/v) formamide, 10 mM NaOH, 1.0 mM EDTA, 0.1% bromphenol blue, and 0.1% xylene cyanol, then loaded on a 10% sequencing gel.

An alternate method of carrying out the footprinting within the gel takes advantage of the ready availability of the reagents and involves immersion of the whole gel in a 1,10-phenanthroline-copper solution without the prior identification of the retarded bands. The first step of this alternative procedure is the immersion of the slab gel into 200 ml of Tris-HCl, pH 8.0 buffer. The gel is removed from its glass plate, then 20 ml of solution A and 20 ml of solution B are added; the digestion is allowed to proceed for 10 min at room temperature. To quench the reaction, 20 ml of 28 mM 2,9-dimethyl-OP is added, and the resulting solution is allowed to stand for 2.0 min. The gel is rinsed with distilled water and exposed to X-ray film for 30 min at room temperature. Bands of interest are cut from the gel and eluted overnight at 37° in 0.5 M ammonium acetate/1 mM EDTA as above. Eluted DNA is ethanol precipitated, resuspended in 80% (v/v) formamide, 10 mM NaOH, 1.0 mM EDTA, 0.1% bromphenol blue, and 0.1% xylene cyanol, and loaded on a 10% sequencing gel. The significant advantage of this procedure is that if a long exposure is essential to identify weak bands, the footprint will not be perturbed by protein denaturation. Reagents are not reusable because thiol is consumed and the 1,10-phenanthroline is oxidatively destroyed.

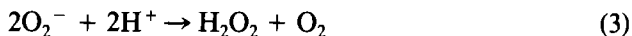
Reactivity Features of OP-Cu Relevant to Footprinting Applications

In this section, practical features of the reactivity of the nuclease activity of OP-Cu are reviewed.

Reducing Agent and Hydrogen Peroxide. Two species serve as coreactants of the nuclease activity: the 1,10-phenanthroline-cuprous complex and hydrogen peroxide. Typically the reaction has been carried out by reducing cupric ion *in situ* with thiol. The resulting cuprous complex is then oxidized by molecular oxygen via superoxide reforming the cupric complex and generating hydrogen peroxide through the spontaneous dismutation of superoxide.²² Given the complexity of the system, it has not been possible to determine the equivalent of a K_m for H_2O_2 . Although the addition of 3 mM H_2O_2 ensures that this coreactant will not be rate limiting, it has usually not been added because it causes background scission.



²² D. R. Graham, L. E. Marshall, K. A. Reich, and D. S. Sigman, *J. Am. Chem. Soc.* **102**, 5419 (1980).



The thiol generally used to activate the chemistry is 3-mercaptopropionic acid. It has been chosen because it is not volatile and is a poorer chelating agent than mercaptoethanol, mercaptoethylamine, or dithiothreitol. Early studies with these thiols caused complexities that can be avoided with 3-mercaptopropionic acid.^{23,24} NADH, in the presence of H_2O_2 ,²⁵ can also be used to activate the scission chemistry. However, the multistep pathway essential to the accomplishment of the one-electron reduction of cupric ion makes these reaction conditions very susceptible to inhibition. Ascorbic acid has also been used to activate the chemistry with apparently satisfactory results.²⁶

The addition of thiol (or ascorbic acid) provides the most convenient way to activate the scission chemistry. Reaction mixtures containing 1,10-phenanthroline, cupric ion, and thiol should not be prepared and allowed to stand because thiol will be consumed [Eqs. (1)–(3)]. Activation of the scission chemistry by cupric ion addition is not advisable. Cupric ion is a ubiquitous contaminant, and will be present at sufficient concentrations to support measurable levels of reaction prior to the desired initiation time of the reaction.

Quenching Nuclease Activity. In footprinting applications, reactions must be carried out under single-hit conditions. As a result, the nuclease activity must be efficiently quenched to avoid overdigestion. This can be achieved with the ligand 2,9-dimethyl-1,10-phenanthroline, which is an analytical reagent (i.e., neocuproin) for copper. It forms a stable cuprous complex with a characteristic absorption maximum at 430 nm. It blocks the nuclease activity by sequestering all the available copper in an inert form. Because no other metal ion can substitute for copper, the reaction is efficiently inhibited.

The *o*-methyl substituents stabilize the cuprous complex relative to the cupric complex through steric effects. In the tetrahedral cuprous complex, there is no interference between the methyl groups in the 2:1 complex. However, in the cupric complex a planar geometry is energetically disfavored due to steric hindrance of the methyl groups. In fact, a single ortho substituent on a phenanthroline is sufficient to destroy the ability of its

²³ V. D'Aurora, A. M. Stern, and D. S. Sigman, *Biochem. Biophys. Res. Commun.* **78**, 170 (1977).

²⁴ V. D'Aurora, A. M. Stern, and D. S. Sigman, *Biochem. Biophys. Res. Commun.* **80**, 1025 (1978).

²⁵ K. A. Reich, L. E. Marshall, D. R. Graham, and D. S. Sigman, *J. Am. Chem. Soc.* **103**, 3582 (1981).

²⁶ Q. Guo, M. Lu, N. C. Seeman, and N. R. Kallenbach, *Biochemistry* **29**, 570 (1990).

copper complex to catalyze the oxidation of thiol and to nick DNA. The 1,10-phenanthroline-copper complex must be able to cycle between the oxidized and reduced forms in both of these reactions.

Sensitivity of Nuclease Activity to Reaction Conditions. Reaction mixtures used to study protein-DNA interactions often contain multiple components. The advantage of the nuclease activity of OP-Cu is its hardness. It proceeds readily under conditions where biological catalysts are usually inactivated and other chemical nucleases can be inhibited.

Influence of Buffer Components

Glycerol/ethylene glycol: These frequently added components that are used to stabilize protein preparations have little effect on scission chemistry up to 1 *M*. Ferrous-EDTA, in contrast, is inhibited by these reagents in the millimolar range.²⁷

EDTA: High concentrations of EDTA will inhibit the reaction by decreasing the copper concentration. The effect of EDTA will be decreased if magnesium ions are present.

Magnesium ions: 1,10-Phenanthroline is not a good ligand for magnesium ions (or other alkaline earth ions). The scission reaction will therefore not be inhibited by magnesium concentrations as high as 30 mM. However, a possible mechanism of inhibition at very high concentrations (0.1 *M* and above) is the displacement of the cupric ion by a magnesium ion. This can be overcome by the addition of cupric ions.

Chloride ions: At high concentrations (>0.1 *M*), chloride ions can sequester copper ions. The addition of cupric ions should counteract this problem.

Protein denaturants: A strong advantage of the nuclease activity is that it is functional even in the presence of high concentrations of protein denaturants (e.g., guanidine hydrochloride). Although these compounds may sequester copper ions or consume thiol, the reaction frequently will proceed if increased amounts of reactants are added. The addition of exogenous hydrogen peroxide (e.g., 7 mM) is particularly helpful if the reaction is sluggish.

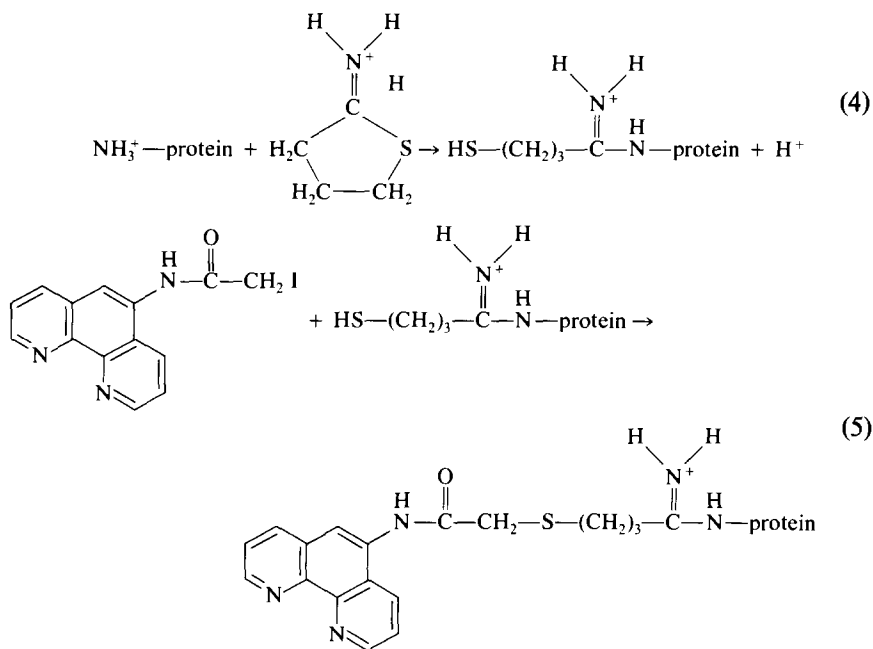
Derivatization of DNA-Binding Proteins with 1,10-Phenanthroline-Copper

In addition to suggesting a method for the design of artificial restriction enzymes, the chemical modification of DNA-binding proteins with 1,10-phenanthroline provides a novel approach for studying the interaction

²⁷ T. D. Tullius and B. A. Dombroski, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 5469 (1986).

of a protein to its recognition sequence. Presently, three proteins, the *Escherichia coli trp* repressor,¹² cAMP-binding protein,²⁸ and the bacteriophage λ Cro protein,²⁹ have been derivatized with 1,10-phenanthroline. In each case, the pattern of scission is consistent with modes of binding that have been inferred from crystallographic studies or other chemical approaches. An important component of all targeted cleavage reactions involving 1,10-phenanthroline-copper is that scission will occur only if the C-1 hydrogen of the deoxyribose in the minor groove is accessible to the scission reagent.

A general method involving lysine derivatization has been applied to the *E. coli trp* repressor. Iminothiolane reacts with primary amine groups to generate sulfhydryl groups that are then alkylated with 5-iodoacetyl-1,10-phenanthroline.¹² In the case of the *trp* repressor, four phenanthrolines are attached per *trp* repressor monomer.



A possible concern with any chemical modification procedure is the potential loss of sequence-specific binding. In the case of *trp* repressor, DNase

²⁸ R. H. Ebright, Y. W. Ebright, P. S. Pendergrast, and A. Gunasekera, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 2882 (1990).

²⁹ T. W. Bruice, J. Wise, and D. S. Sigman, *Biochemistry* **29**, 2185 (1990).

footprinting indicated that this was not a problem; the modified protein bound with comparable affinity as the wild-type protein. Scission was achieved by the addition of both thiol and copper ion (Fig. 6). It is also dependent on the presence of the corepressor, L-tryptophan, which is essential for the sequence-specific binding of the protein.³⁰

Two operators, *aroH* and *trpEDCBA*, were assayed as substrates for the OP-derivatized *trp* repressor. Both were nicked within the recognition sequence. The pattern of scission, however, with the two operators was distinct. These results are consistent with previous suggestions, based on DNase I footprinting and methylation studies, that the repressor binds to four successive major grooves on the *trpEDCBA* operator and three successive minor grooves on the *aroH* operator.³¹ Thus, the OP-*trp* repressor provides chemical evidence that there is not a canonical orientation between a DNA-binding protein and a recognition sequence.

Ebright and colleagues have derivatized the *E. coli* cAMP-binding protein (CAP) at a unique cysteine residue using 5-iodoacetyl-1,10-phenanthroline.²⁸ This cysteine residue is at position 10 of the helix-turn-helix domain of this transcription activator. Scission on both strands at the center of the 22-nucleotide long symmetric recognition sequence was observed (Fig. 7), and was absolutely dependent on the addition of copper, thiol, and cAMP. Thus, both in the case of the *trp* repressor and CAP, the ligands essential for sequence-specific binding are also required for the scission chemistry. The observed sites of scission are fully consistent with the postulated model for CAP binding.

The helix-turn-helix format in DNA-binding proteins was first identified in the λ phage Cro protein by Matthews and colleagues.^{32,33} In their pioneering studies, Matthews and co-workers suggested that the C-terminal end of the protein could extend into the minor groove of the recognition sequence and enhance the stability of the complex. Because the wild-type Cro protein contains no cysteine residue, a unique site of alkylation of the protein by 5-iodoacetamido-1,10-phenanthroline^{33a} could be generated by changing the C-terminal alanine residue into a cysteine residue (mutant protein designated as Cro-Cys) using site-directed mutagenesis.

Gel retardation assays provided a convenient method to demonstrate that the site-directed mutagenesis of the Cro to generate Cro-Cys was achieved as planned and that Cro-Cys alkylated by 5-iodoacetamido-1,10-phenanthroline still could bind to the recognition sequence with high affin-

³⁰ D. N. Arvidson, C. Bruce, and R. P. Gunsalus, *J. Biol. Chem.* **261**, 238 (1986).

³¹ A. A. Kumamoto, W. G. Miller, and R. P. Gunsalus, *Genes Dev.* **1**, 556 (1987).

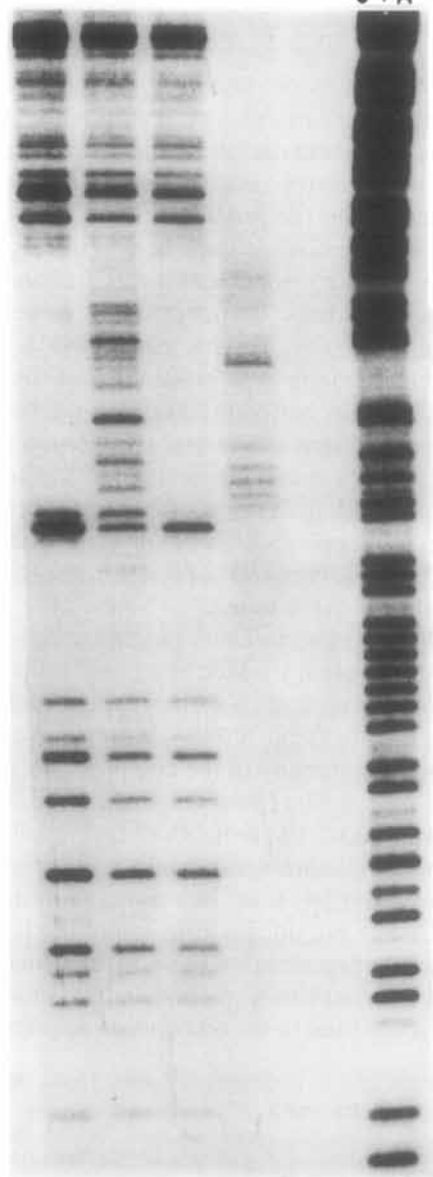
³² R. G. Brennan and B. W. Matthews, *Trends Biochem. Sci.* **13**, 286 (1989).

³³ R. G. Brennan and B. W. Matthews, *J. Biol. Chem.* **264**, 1903 (1989).

^{33a} For synthesis of reagents, see end of chapter.

DNase I	+	+	+	-	-
trp rep	+	-	-	-	-
OP-trp rep	-	-	+	+	+
trp	+	-	+	+	-

G+A



41
40
32
30
29
28
27

⊖
⊕

a b c d e f

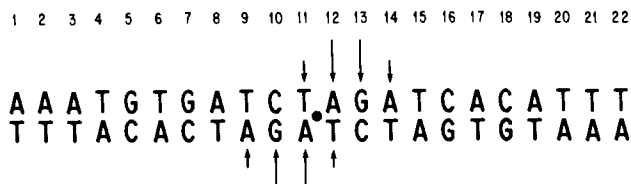


FIG. 7. Site-specific DNA scission of recognition sequence by cAMP-binding protein-alkylated cysteine-10 of the helix-turn-helix domain. (Adapted from Ebright *et al.*²⁸)

ity (Fig. 8A). Extracts of *E. coli* transformed with pUC119 containing the gene for Cro or Cro-Cys were used as the source of protein for the gel retardation assay. Chemical modification was carried out on the extracts directly.

To determine if sequence-specific scission could be achieved, copper ion and thiol were added to the gel slice containing the complex of OR-3 operator and Cro-Cys-OP (Fig. 8A). Scission is observed within the binding site identified by OP-Cu footprinting. These experiments provide direct chemical support for the model proposed by Matthews and colleagues. An important feature of the 1,10-phenanthroline-derivatized Cro protein is that it retains high affinity for the target sequence. The Cro protein, therefore, may serve as a convenient starting point for generating a family of semisynthetic restriction endonucleases. Possibly stable mutants of Cro-Cys can be generated in which the DNA-binding domain has been altered to bind other recognition sequences.

Methods for Preparation of 1,10-Phenanthroline-Modified Proteins

trp Repressor. The procedures used for derivatizing the *E. coli trp* repressor are summarized below. The protein is reacted with 2-iminothiolane (Pierce Chemical Company, Rockford, IL), according to a procedure adapted from a method first applied to the modification of ribosomal subunits.³⁴ From a freshly prepared 2-iminothiolane hydrochloride solution [0.5 M in 0.1 M (pH 8.0) phosphate buffer], 2.4 μ l is added to 0.021

³⁴ R. Jue, J. M. Lambert, L. R. Pierce, and R. R. Traut, *Biochemistry* **17**, 5399 (1978).

FIG. 6. Binding and scission of the *aroH* operator with OP-*trp* repressor. Template strand 5' labeled. Lanes a to c: DNase footprints of the binding of the *trp* repressor and OP-*trp* repressor to the 5'-labeled template strand of *aroH*. Lane a: native *trp* repressor; lane b: control; lane c: OP-*trp* repressor; lane d: OP-*trp* repressor with L-tryptophan, Cu²⁺, and 3-mercaptopropionic acid; lane e: same as lane d but lacking L-tryptophan; and lane f: G + A sequencing lane. Temperature = 25°.

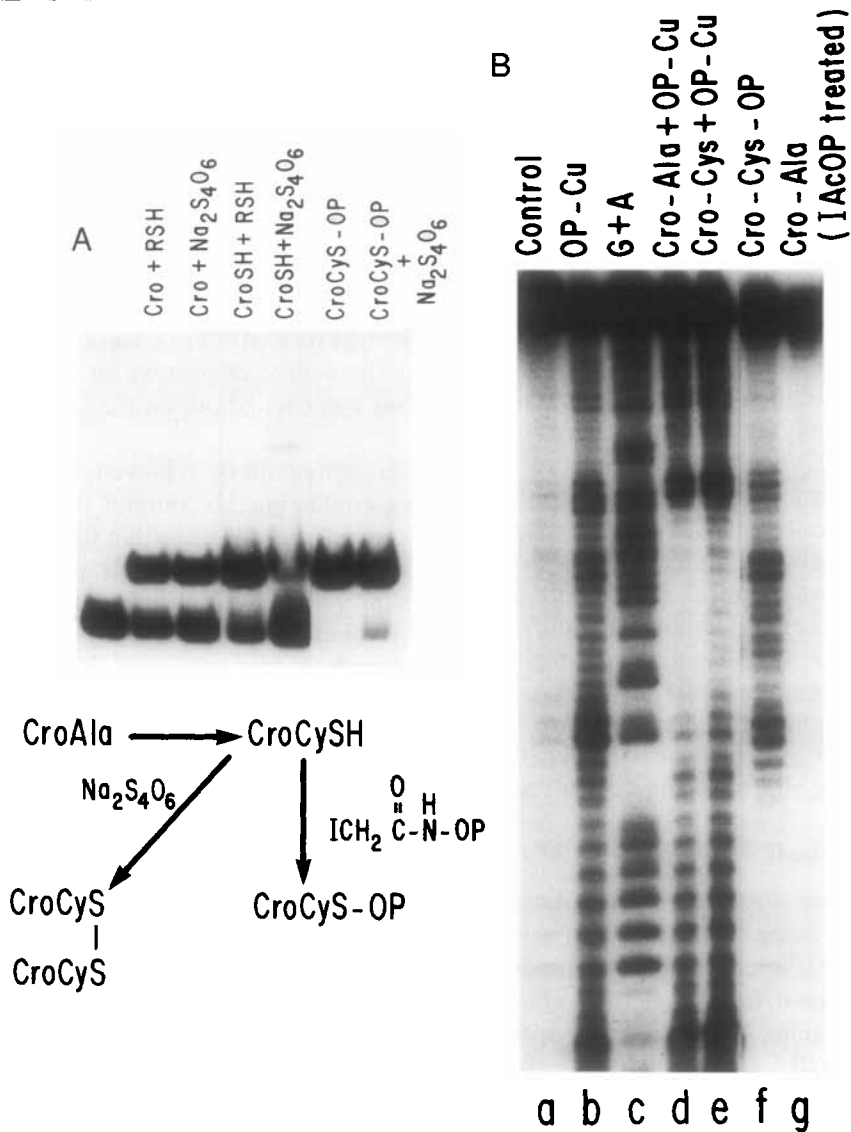


FIG. 8. Interaction of Cro protein with OR-3. (A). Gel retardation assay: (a) no protein added; (b) *E. coli* extract containing *cro* in the pUC119 plasmid. (B). Scission of retarded band within the gel. (a) Control DNA; (b) free DNA cleaved by OP-Cu; (c) G + A sequencing lane; (d) Cro-DNA complex cleaved by OP-Cu. e, f, g.

μmol of *trp* repressor in 250 μl of phosphate buffer containing 1% 2-mercaptoethanol and 1 mg of L-tryptophan at 4°. After overnight incubation at 4°, the mixture is passed through a Sephadex G-25 (coarse) spin column to remove the unreacted 2-iminothiolane. The spin column is equilibrated in 0.1 M phosphate buffer, pH 8.0, prior to use. Then, 1.6 mmol of 5-iodo[³H]acetamido-OP (specific activity, 3 mCi/mmol) in *N,N*-dimethylformamide (4 ml) is added to the eluant collected from the spin column. The alkylation reaction proceeds at 4° overnight. The product is isolated by passing the mixture through a Sephadex G-25 spin column.

The *aroH* operator region is used to assay binding and the scission activity of the modified protein. The operator derives from a 257-bp *Bam*HI-*Eco*RI fragment, which had been cloned into pBR327.³¹ The control region of *trpEDCBA* is contained in a *Bam*HI-*Eco*RI restriction fragment that had also been cloned into pBR327. These fragments are 5' and 3' labeled. For the footprinting and cleavage reactions, a solution of labeled *aroH* operator fragment (100 nM, 10,000 cpm), *trp* repressor or OP-*trp* repressor (100 to 200 nM), and L-tryptophan (10 mM) is incubated at room temperature for 5 min in 10 μl of Tris-HCl (30 mM, pH 8.0), KCl (100 mM), and MgCl₂ (3 mM). Footprinting reactions are carried out with 0.04 U of DNase I for 5 min at room temperature. The reactions are quenched by the addition of "stop" solution [transfer RNA (200 mg/ml), 2 M sodium acetate, and 10 mM EDTA]. The DNA is then precipitated with ethanol, redissolved in Maxam-Gilbert loading buffer, and analyzed by the use of a 10% polyacrylamide gel that contains 8.34 M urea, and is cross-linked at a ratio of 19:1. The chemical cleavage reaction is initiated by adding 1 μl of 20 μM cupric sulfate and 1 μl of 58 mM mercaptopropionic acid. After 30 min at room temperature or at 37°, the reaction is quenched by adding 2,9-dimethyl-1,10-phenanthroline followed by 1 μl of stop solution. The DNA is then precipitated with ethanol, redissolved in Maxam-Gilbert loading buffer, and analyzed by 10% polyacrylamide (19:1 acrylamide:bisacrylamide cross-linked gel) containing 8.34 M urea.

Cyclic AMP-Binding Protein. As described by Ebright and colleagues,²⁸ the cAMP-binding protein is derivatized with 5-iodoacetamido-1,10-phenanthroline in the following way. Reaction mixtures (50 μl) contain 2.7 nmol CAP, 210 nmol 5-iodoacetamido-1,10-phenanthroline, 20 mM Tris-HCl (pH 8.0), 200 mM KCl, 0.1 M EDTA, 5% glycerol, and 1.7% dimethylformamide. Reactions are carried out in the dark and proceeded for 3 hr at 23°, followed by 15 hr at 4°. The product is purified by chromatography on BioGel P-6DG (Bio-Rad, Richmond, CA), and stored at -70° in 20 mM Tris-HCl (pH 8.0), 200 mM KCl, 5% glycerol. All solutions are treated with Chelex 100 (Bio-Rad) to remove trace metals.

Ebright and colleagues assay the scission reaction in a solution containing 1 nM ^{32}P -labeled target fragment, 80 nM OP-modified CAP, 7.5 μM cupric sulfate, 2.5 mM 3-mercaptopropionic acid, 0.2 mM cAMP, 10 mM MOPS-NaOH (pH 7.3), 20 mM NaCl, 50 mg/ml bovine serum albumin, and 2.2% ethanol. Reaction time is 1 hr at 37°; the reaction is quenched by the addition of 2,9-dimethyl-1,10-phenanthroline to 3.0 mM. Reaction mixtures are extracted twice with phenol and twice with chloroform prior to ethanol precipitation.

λ Phage Cro Protein Gel retardation techniques provide a powerful method for purifying DNA-binding proteins. It is therefore possible to derivatize DNA-binding proteins with 1,10-phenanthroline in crude reaction mixtures and then purify the derivatized protein by gel retardation. The scission reaction can then be activated within the gel matrix. This experimental protocol has been used in studies with Cro protein in which both the wild-type and Cro-Cys protein are expressed in low yield in the pUC119 plasmid.

In situ modification of proteins containing sulfhydryl groups can be carried out in extracts by first ensuring that the sulfhydryl group is reduced by adding a thiol (e.g., mercaptoethanol or 3-mercaptopropionic acid) to attain a concentration of 2.0 mM. Then 5-iodoacetamido-1,10-phenanthroline in dimethylformamide (DMF) is added to a final concentration of 5.0 mM in 10% DMF. The reaction mixture is incubated on ice for 60 min, then 3-mercapto-propionic acid (to 10 mM) is added to inactivate the 5-iodoacetyl derivative. The resulting preparation can then be used in gel retardation experiments. The protein is separated from excess 5-iodoacetamido-1,10-phenanthroline by the gel retardation separation. This procedure allows an initial screen to determine if the modification procedure influences binding and whether a protein-targeted reaction can be achieved.

Synthesis of Reagents

5-Amino-1,10-phenanthroline. 5-Amino-1,10-phenanthroline is a key intermediate for derivatizing proteins and nucleic acids. A convenient high-yield synthesis of this known compound is presented below. Ten grams (0.044 mol) of 5-nitro-1,10-phenanthroline (available from G. F. Smith Co., Columbus, OH) is dissolved in 400 ml of absolute ethanol by boiling, and 475 ml of ammonium sulfide (~22%) is heated on a steam bath. The 5-nitro-1,10-phenanthroline solution is added dropwise to the ammonium sulfide with a continual flow of nitrogen. When the addition is completed, 250 ml more of ammonium sulfide is added and the reaction mixture is allowed to reflux for 1 hr. The solution is cooled to room

temperature, and then is extracted with chloroform. The chloroform extracts are dried and distilled at atmospheric pressure. The residue is crystallized from ethanol/water and dried under vacuum. Yield: 5.5 g (63.5%); mp, 268–270°; thin-layer chromatography (TLC; silica, 5% ammonia in methanol) 1 spot, R_f 0.45, running slower than the starting material.

5-Iodoacetamido-1,10-phenanthroline. Five grams (0.024 mol) of dicyclohexylcarbodiimide is added to a solution of 10 g (0.054 mol) of iodoacetic acid in 150 ml of ethyl acetate. Dicyclohexylurea forms immediately, but stirring is continued for 2 hr. The urea is removed by filtration, and the solution is concentrated to dryness by rotatory evaporation.

5-Amino-1,10-phenanthroline (1.6 g, 0.0082 mol) is dissolved by heating in 200 ml of acetonitrile. The solution is allowed to cool at room temperature. Then the iodoacetic anhydride dissolved in 75 ml of acetonitrile is added. After the solution is stirred overnight at room temperature, a crystalline product is obtained. The product is washed with cold 5% sodium bicarbonate, washed with water, and finally dried under vacuum. Yield: 2 g (70%). Analytical data: TLC (silica, 5% ammonia in methanol) 1 spot; R_f 0.24. Elemental analysis: Calculated C 46.20%, H 2.75%, N 11.60%, I 35.00%; found C 46.41%, H 2.91%, N 11.74%, I 34.72%.

[21] Base Analogs in Study of Restriction Enzyme–DNA Interactions

By CHRISTOPHER R. AIKEN and RICHARD I. GUMPORT

Introduction

Enzymologists have long sought to discern the topography of the active sites of enzymes by examining substrate analogs for their ability to serve as reactants. Such investigations aim to contribute to our understanding of the kinetic and chemical mechanisms as well as the stereochemistry and stereoselectivity of a reaction. The interpretation of results from studies using analogs is based on the assumption that steric and electronic variations in the substrate will reveal contacts or interactions with the enzyme. The effects of substrate analogs are exerted on the enzyme–substrate complex as reflected in its equilibrium dissociation constant or the K_m value, or on catalysis itself as manifested by changes in the overall reaction rate (k_{cat}) or the rate of an individual step in the mechanism. Although the paradigm of enzyme and substrate as a rigid, lock-and-key-

like system is incorrect because both partners are flexible to varying extents, the analog approach remains a valuable means to study the geometric and electronic interfaces between an enzyme and its substrate during a reaction. A complementary approach has more recently been developed through protein engineering techniques that create enzyme analogs in which one amino acid is substituted for another in order to monitor directly the role of specific amino acid side chains in the interactions. The combined use of mutant enzymes with substrate analogs should be a powerful means to elucidate mechanisms.

In this chapter we discuss the principles and assumptions of the substrate-analog approach, evaluate the merits, explain the limitations, and mention artifacts sometimes encountered when it is used to assess site-specific enzyme-DNA interactions. The aim of such studies with enzymes that react with specific sites on DNA has often been to determine the molecular basis of the "recognition" of the target nucleotide sequence. In this context, recognition means the capability of the enzyme to select the targeted sequence from relatively higher concentrations of similar sequences and to catalyze a specific reaction with the DNA. We will confine our detailed considerations to the type II restriction endonucleases. For a review of DNA recognition by restriction enzymes see Ref. 1.

Nucleic acid analogs have also been used with proteins that bind to specific sites on DNA but do not catalyze reactions with it, e.g., repressors. It should be noted that enzymes, in contrast to simple DNA-binding proteins, are afforded added opportunities to recognize their targets because chemistry occurs; covalent bonds are broken and formed. The natures of the substrates change during the course of the catalysis as a transition state develops from them and is tightly bound by the enzyme. Functional groups, possibly newly formed, on the developing transition state may interact with the enzyme. Such groups may be unavailable for contact with the enzyme in the initial enzyme-substrate complex. Thus, an enzyme may bind to DNA with some degree of site specificity yet fail to catalyze a reaction efficiently because prerequisites for the formation of the transition state are absent. With enzymes, some recognition can occur during the chemical steps as well as in the initial binding of the substrate.

The effects of analog substitutions can arise directly from lost interactions because of missing or added groups on the modified bases themselves

¹ S. P. Bennett and S. E. Halford, in "Current Topics in Cellular Regulation" (B. L. Horecker, E. R. Stadtman, P. B. Chock, and A. Levitzki, eds.), Vol. 30, p. 57. Academic Press, San Diego, California, 1989.

or from indirect effects as a result of conformational changes in the DNA introduced by the presence of the analogs. Whereas high-resolution structural analyses of specific DNA-protein complexes often definitively indicate the atomic interactions between the macromolecules, they cannot yet do so for the changing structures in an enzyme-catalyzed reaction with DNA. In addition, structural analyses of enzyme-substrate complexes are sometimes conducted in the absence of cofactors necessary for reaction; consequently, such structures may not represent the biologically active complexes. Kinetic studies using analogs and other methods that monitor the dynamics of a reaction under quasiphysiological conditions can sometimes uniquely reveal important interactions. For these reasons there is a continuing role for analog studies of protein-nucleic acid interactions despite the considerable ambiguities that exist in their interpretation (see below).

Analogs

DNA and oligodeoxyribonucleotides containing base analogs, i.e., bases different from Ade,² Cyt, Gua, and Thy, can be obtained in several ways. Natural base analogs exist in the genomes of some organisms. For example, methylated (Me) derivatives of Ade and Cyt occur in some bacterial genomes,³ and HmUra and glycosylated derivatives of HmCyt are found in the DNAs of some bacteriophages.⁴ Analogs can also be formed in DNA and oligonucleotides through chemical modification reactions, e.g., 7-MeGua and 3-MeAde can be created by reaction with Me₂SO₄.⁵ Such sources of analog-containing DNA are unsuitable for systematic studies because the variety of bases attainable is limited, and it is difficult to assure complete modification at a unique site.

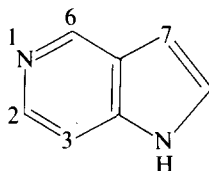
Synthetic organic chemistry greatly enlarges the repertoire of structural variants available and, coupled with methods for their site-specific introduction into oligodeoxyribonucleotides, enables a large number of analogs to be examined. Variants at most of the sites available for potential interactions on each of the bases have been made. Tables I and II show some of the purine and pyrimidine compounds that have been incorporated into oligonucleotides, and list some of the restriction enzymes tested with

² Abbreviations follow the IUPAC suggestions in *Biochemistry* 9, 4022 (1970), except that Pur stands for purine itself and not an unknown purine.

³ H. A. Sober, ed., "CRC Handbook of Biochemistry, Selected Data for Molecular Biology," 2nd Ed., p. H-117. The Chemical Rubber Co., Cleveland, Ohio, 1970.

⁴ H. A. Sober, ed., "CRC Handbook of Biochemistry, Selected data for Molecular Biology," 2nd Ed., p. H-4. The Chemical Rubber Co., Cleveland, Ohio, 1970.

⁵ A. M. Maxam and W. Gilbert, this series, Vol. 65, p. 499.

TABLE I
PURINE ANALOGS

2,3-Positions	1,6-Positions	7-Position	Base	Enzymes examined
H ₂ NC=N	HN—CO	N	Guanine	
HC=N	HN—CO	N	Hypoxanthine	<i>EcoRI</i> , ^{a,b} <i>EcoRV</i> ^c
H ₂ NC=N	N=CH	N	2-Aminopurine	<i>EcoRI</i> , ^{a,b,d} <i>EcoRV</i> ^c
H ₂ NC=N	HN—CO	CH	7-Deazaguanine	<i>EcoRI</i> ^{d,e}
HC=N	HN—CO	CH	7-Deazahypoxanthine	<i>HindII</i> , <i>SalI</i> , <i>TaqI</i> ^f
OC—NH	HN—CO	N	Xanthine	None ^g
H ₂ NC=N	N=COCH ₃	N	O ⁶ -Methylguanine	<i>HpaII</i> , <i>HhaI</i> , ^h and others
CH ₃ NHC=N	HN—CO	N	N ² -Methylguanine	<i>BglII</i> , <i>Sau3AI</i> , <i>MboI</i> ⁱ
HC=N	N=CNH ₂	N	Adenine	
HC=N	N=CH	N	Purine	<i>EcoRI</i> , ^b <i>EcoRV</i> ^c
H ₂ NC=N	N=CNH ₂	N	2,6-Diaminopurine	<i>EcoRI</i> ^{a,b}
HC=N	N=CNHOCH ₃	N	N ⁶ -Methoxyadenine	None ^j
HC=N	N=CNHCH ₃	N	N ⁶ -Methyladenine	<i>EcoRI</i> , ^{a,d} <i>EcoRV</i> , ^k <i>Sau3AI</i> , <i>MboI</i> , <i>MflI</i> ^l
HC=CH	N=CNH ₂	N	3-Deazaadenine	<i>EcoRV</i> , ^{m,n} <i>BglIII</i> , <i>Sau3AI</i> , <i>MboI</i> ⁱ
HC=N	N=CNH ₂	CH	7-Deazaadenine	<i>EcoRI</i> , ^o <i>EcoRV</i> , ^{k,m} <i>HindII</i> , <i>SalI</i> , <i>TaqI</i> , ^f <i>BglII</i> , <i>Sau3AI</i> ^p
HC=N	N=CH	CH	7-Deazapurine	<i>HindII</i> , <i>SalI</i> , <i>TaqI</i> ^f

^a C. A. Brennan, M. D. Van Cleve, and R. I. Gumpert, *J. Biol. Chem.* **261**, 7270 (1986).

^b L. W. McLaughlin, F. Benseler, E. Graeser, N. Piel, and S. Scholtissek, *Biochemistry* **26**, 7238 (1987).

^c J. Mazzairelli, S. Scholtissek, and L. W. McLaughlin, *Biochemistry* **28**, 4616 (1989).

^d D. R. Lesser, M. R. Kurpiewski, and L. Jen-Jacobson, *Science* **250**, 776 (1990).

^e F. Seela and H. Driller, *Nucleic Acids Res.* **14**, 2319 (1986).

^f J. Jiricny, S. G. Wood, D. Martin, and A. Ubasawa, *Nucleic Acids Res.* **14**, 6579 (1986).

^g R. Eritja, D. M. Horowitz, P. A. Walker, J. P. Ziehler-Martin, M. S. Boosalis, M. F. Goodman, K. Itakura, and B. E. Kaplan, *Nucleic Acids Res.* **14**, 8135 (1986).

^h J. M. Voigt and M. D. Topal, *Biochemistry* **29**, 1632 (1990).

ⁱ A. Ono and T. Ueda, *Nucleic Acids Res.* **15**, 3059 (1987).

^j H. Nishino, A. Ono, A. Matsuda, and T. Ueda, *Nucleic Acids Res. Symp. Ser. No. 21*, 123 (1989).

^k A. Fliess, H. Wolfes, F. Seela, and A. Pingoud, *Nucleic Acids Res.* **16**, 11781 (1988).

^l A. Ono and T. Ueda, *Nucleic Acids Res.* **15**, 219 (1987).

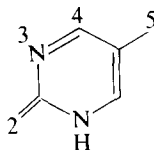
^m P. C. Newman, V. U. Nwosu, D. M. Williams, R. Cosstick, F. Seela, and B. A. Connolly, *Biochemistry* **29**, 9891 (1990).

ⁿ R. Cosstick, X. Li, D. K. Tuli, D. M. Williams, B. A. Connolly, and P. C. Newman, *Nucleic Acids Res.* **18**, 4771 (1990).

^o F. Seela and A. Kehne, *Biochemistry* **26**, 2232 (1987).

^p A. Ono, M. Sato, Y. Ohtani, and T. Ueda, *Nucleic Acids Res.* **12**, 8939.

TABLE II
PYRIMIDINE ANALOGS



2-Position	3,4-Positions	5-Position	Base	Enzymes examined
O	HN—CO	CH ₃	Thymine	
O	HN—CO	H	Uracil	<i>EcoRI</i> , ^{a,b} <i>EcoRV</i> ^{c,d}
O	HN—CS	CH ₃	4-Thiothymine	<i>EcoRV</i> ^{d,e}
S	HN—CO	CH ₃	2-Thiothymine	<i>EcoRV</i> ^d
O	N=CH	CH ₃	5-Methyl-2-pyrimidinone	<i>EcoRV</i> ^{d,f}
O	HN—CO	(H)CH ₃	5,6-Dihydrothymine ^g	None ^h
O	HN—CO	F	5-Fluorouracil	<i>MvaI</i> ⁱ
O	HN—CO	Br	5-Bromouracil	<i>EcoRI</i> , ^a <i>EcoRII</i> , ^j <i>EcoRV</i> , ^k <i>BglII</i> , <i>Sau3AI</i> , <i>MboI</i> ^l
O	HN—CO	CN	5-Cyanouracil	<i>BglII</i> , <i>Sau3AI</i> , <i>MboI</i> ^l
O	HN—CO	C ₂ H ₅	5-Ethyluracil	<i>BglII</i> , <i>Sau3AI</i> , <i>MboI</i> ^l
O	N=CNH ₂	H	Cytosine	
O	N=CNH ₂	CH ₃	5-Methylcytosine	<i>EcoRI</i> , ^{a,b} <i>EcoRII</i> , ^j <i>EcoRV</i> , ^{c,m} <i>MvaI</i> ⁱ
O	N=CNH ₂	Br	5-Bromocytosine	<i>EcoRV</i> ^m
O	N=CNHCH ₃	H	N ⁴ -Methylcytosine	<i>Sau3AI</i> , <i>MboI</i> , <i>MflI</i> ⁿ

^a C. A. Brennan, M. D. Van Cleve, and R. I. Gumport, *J. Biol. Chem.* **261**, 7270 (1986).

^b L. W. McLaughlin, F. Benseler, E. Graeser, N. Piel, and S. Scholtissek, *Biochemistry* **26**, 7238 (1987).

^c J. Mazzearelli, S. Scholtissek, and L. W. McLaughlin, *Biochemistry* **28**, 4616 (1989).

^d P. C. Newman, V. U. Nwosu, D. M. Williams, R. Cosstick, F. Seela, and B. A. Connolly, *Biochemistry* **29**, 9891 (1990).

^e B. A. Connolly and P. C. Newman, *Nucleic Acids Res.* **17**, 4957 (1989).

^f H. P. Rappaport, *Nucleic Acids Res.* **16**, 7253 (1988).

^g The 5,6-double bond is reduced in dihydrothymine and there is an additional hydrogen at position 6.

^h D. Molko, A. M. Delort, and R. Teoule, *Biochemie* **67**, 801 (1985).

ⁱ E. A. Kubareva, C.-D. Pein, E. S. Gromova, S. A. Kuznezova, V. N. Tashlitzki, D. Cech, and Z. A. Shabarova, *Eur. J. Biochem.* **175**, 615 (1988).

^j A. A. Yolov, M. N. Vinogradova, E. S. Gromova, A. Rosenthal, D. Cech, V. P. Veiko, V. G. Metelev, V. G. Kosykh, Y. I. Buryanov, A. A. Bayev, and Z. A. Shabarova, *Nucleic Acids Res.* **13**, 8983 (1985).

^k A. Fliess, H. Wolfes, A. Rosenthal, K. Schwellnus, H. Blocker, R. Frank, and A. Pingoud, *Nucleic Acids Res.* **14**, 3463 (1986).

^l T. Hayakawa, A. Ono and T. Ueda, *Nucleic Acids Res.* **16**, 4761 (1988).

^m A. Fliess, H. Wolfes, F. Seela, and A. Pingoud, *Nucleic Acids Res.* **16**, 11781 (1988).

ⁿ A. Ono and T. Ueda, *Nucleic Acids Res.* **5**, 219 (1987).

them. These lists are not exhaustive with respect to either analogs or enzymes. In this chapter, we have selected representative studies with analogs that illustrate useful approaches or methods.

For an ideal analysis, one would remove or alter each of the functional groups of the heterocyclic base that has the potential to interact specifically with the enzyme, namely, the O or N atoms on accessible surfaces of the DNA that could act as hydrogen bond donors or acceptors and the methyl group of Thy that could be involved in hydrophobic or van der Waals interactions.⁶ The perfect analog would have only one of its specificity determinants removed without added new atoms that might introduce steric blockages or electronic incompatibilities that interfered with any other interactions near the site of modification. Examples of analogs approaching this ideal are purines (Pur²) for Ade, where the exocyclic amino group of Ade is replaced with a hydrogen atom,⁷ or Ura, where the methyl group of Thy is similarly replaced.^{7,8} In practice, one functional group has been substituted for another, for example, bromine replaced a methyl group when BrUra was substituted for Thy.⁸ Sometimes a single, larger group has been added in place of a hydrogen atom, e.g., when 5-McCyt replaces Cyt.⁸ The relative positions of the amino and carbonyl groups on a base pair have also been interchanged, as when a 2,6-(NH₂)₂Pur : Ura base pair was substituted for a Gua : Cyt base pair.⁷

Ideally, the introduction of the analog should not decrease the thermal stability of the duplex under the conditions of the assay so that the substrate is significantly denatured. The analog should also not cause conformational changes in the DNA structure that could affect nearby interactions with the normal base pairs by altering their relative locations. Adverse conformational effects could exert themselves on the direct readout of adjacent bases or the indirect readout of sequence-specific contacts with the ribose-phosphodiester backbone. Since some site-specific DNA-binding proteins significantly distort the DNA on binding, an additional, related requirement is that the analog should not alter the deformability of the DNA. Since all these qualities are unlikely to be present with any analog, the extent to which a particular structural variant lacks them will determine the degree to which the interpretation of the results attained with it will be compromised. These difficulties often necessitate ancillary information for a more complete understanding of the molecular effects of the substitutions. For example, the effects of several analogs on the bend-

⁶ N. C. Seeman, J. M. Rosenberg, and A. Rich, *Proc. Natl. Acad. Sci. U.S.A.* **73**, 804 (1976).

⁷ L. W. McLaughlin, F. Benseler, E. Graeser, N. Piel, and S. Scholtissek, *Biochemistry* **26**, 7238 (1987).

⁸ C. A. Brennan, M. D. van Cleve, and R. I. Gumport, *J. Biol. Chem.* **261**, 7270 (1986).

ing of the *EcoRI* recognition sequence have been correlated with their effects on the activity of the enzyme.⁹ The location of the enzyme with respect to the major and minor grooves of the bound DNA target, as determined by X-ray crystallography,¹⁰ has also been useful in explaining the effects of analog substitution experiments.^{7,8} Just as with enzyme or protein analogs produced by mutagenesis procedures, the more structural information available, the easier it is to design and interpret substrate-analog experiments.

Preparation of Substrates for Type II Restriction Enzymes

If properly protected and activated nucleotide analogs are available, an oligodeoxyribonucleotide can be synthesized by current automated chemical methods. Analog synthons containing 7-deazaAde, *N*⁶-MeAde, 5-BrCyt, 5-IoCyt, 5-MeCyt, 7-deazaGua, *O*⁶-MeGua, hypoxanthine (Hpx), Pur, 2,6-(NH₂)₂Pur, Ura, 5-BrUra, 5-FIUra, and 5-IoUra are currently available commercially. The preparations of many other analog-containing oligonucleotides have been described (see Tables I and II for references) but the protected, activated nucleotides are not commercially available. Alternatively, T4 RNA ligase¹¹ may be used to prepare sufficient quantities of modified oligonucleotides for the kinetic analyses used in most studies of restriction endonucleases. If the 2'-deoxyribonucleoside of the analog is available, it is converted to the 2'-deoxyribonucleoside 3',5'-bisphosphate derivative by reaction with pyrophosphoryl chloride.¹² The deoxyribonucleoside bisphosphate serves as a substrate for RNA ligase,¹³ and is joined by the enzyme to the 3'-hydroxyl of an oligodeoxyribonucleotide at least three nucleotides long. After removal of the 3' phosphate from this initial product, the extended oligonucleotide bearing the analog is similarly added to a second oligomer to form the analog-containing product.¹¹ When annealed to a complementary strand, a duplex containing an analog at a unique site is formed. Because chemical synthesis provides more material and thus facilitates the characterization and structural studies that can aid in interpreting results, it is the preferred method.

Oligodeoxyribonucleotides, eight or more base pairs long, serve as

⁹ S. Diekmann and L. W. McLaughlin, *J. Mol. Biol.* **202**, 823 (1988).

¹⁰ J. A. McClarin, C. A. Frederick, B.-C. Wang, P. Greene, H. W. Boyer, J. Grable, and J. M. Rosenberg, *Science* **234**, 1526 (1986).

¹¹ C. A. Brennan and R. I. Gumpert, *Nucleic Acids Res.* **13**, 8665 (1985).

¹² J. R. Barrio, M. D. Barrio, N. J. Leonard, T. E. England, and O. C. Uhlenbeck, *Biochemistry* **10**, 2077 (1987).

¹³ O. C. Uhlenbeck and R. I. Gumpert, in "The Enzymes" (P. D. Boyer, ed.), Vol. 15, p. 31. Academic Press, New York, 1982.

substrates for many restriction enzymes.^{1,14} Since the sequence recognized by most type II enzymes is palindromic, the easiest synthetic approach is to make a self-complementary oligonucleotide. Self-complementary oligonucleotides can sometimes form intramolecular hairpin structures that complicate their use as restriction enzyme substrates.¹⁵ Duplexes formed by self-complementary oligomers that contain an analog will have symmetrically disposed substitutions in each strand. If the effect of an analog in one strand only is to be tested, then complementary oligonucleotides with unique sequences must be synthesized. For enzymes recognizing palindromic sequences, this requires synthesizing two oligonucleotides with sufficiently long, nonself-complementary flanking sequences to suppress intramolecular hairpin formation by either strand.

At a minimum, the melting temperature (T_m) of the duplex under the conditions of the assays must be determined to ensure that the substrate is a duplex. Usually, this is impossible to measure directly because of the low concentrations of oligomer used in the assays. As a result, the T_m must be measured at several higher concentrations, and the value interpolated to the lower concentration.^{11,16} Other information, e.g., structural analyses of oligomers containing analogs, can facilitate the interpretation of the results obtained.^{9,17-19} Careful consideration should also be given to the base pairs adjacent to the recognition sequence, because they can affect the activity of the enzyme. Flanking base pairs influence the stability and structure of the oligonucleotide,²⁰ and with *EcoRI*, by contacting the enzyme,²¹ can directly affect its activity.²²⁻²⁴

One must also ascertain whether the decreased ability of a substrate analog to be cleaved is due to the modified base or to the presence of an adventitious enzyme inhibitor. Careful purification and characterization

¹⁴ P. J. Greene, M. S. Poonian, A. L. Nussbaum, L. Tobias, D. E. Garfin, H. W. Boyer, and H. M. Goodman, *J. Mol. Biol.* **99**, 237 (1975).

¹⁵ F. Seela and A. Kehne, *Biochemistry* **26**, 2232 (1987).

¹⁶ O. C. Uhlenbeck, P. N. Borer, B. Dengler, and I. Tinoco, *J. Mol. Biol.* **73**, 483 (1973).

¹⁷ A.-M. Delort, J. M. Neumann, D. Molko, M. Herve, R. Teoule, and S. T. Dinh, *Nucleic Acids Res.* **13**, 3343 (1985).

¹⁸ G. M. Clore, H. Oschkinat, L. W. McLaughlin, F. Benseler, C. S. Happ, E. Happ, and A. M. Gronenborn, *Biochemistry* **27**, 4185 (1988).

¹⁹ P. C. Newman, V. U. Nwosu, D. M. Williams, R. Cosstick, F. Seela, and B. A. Connolly, *Biochemistry* **29**, 9891 (1990).

²⁰ B. A. Connolly and F. Eckstein, *Biochemistry* **23**, 5523 (1984).

²¹ A.-L. Lu, W. E. Jack, and P. Modrich, *J. Biol. Chem.* **256**, 13200 (1981).

²² M. Thomas and R. W. Davis, *J. Mol. Biol.* **91**, 315 (1975).

²³ J. Alves, A. Pingoud, W. Haupt, J. Langowski, F. Peters, G. Maass, and C. Wolff, *Eur. J. Biochem.* **140**, 83 (1984).

²⁴ M. D. Van Cleve, Ph.D. Thesis, University of Illinois, Urbana, Illinois (1989).

of the oligonucleotide reduces the likelihood of this potential artifact. Experiments in which the effect of the modified oligonucleotide on the cleavage of DNA or another oligonucleotide containing the unmodified recognition sequence are useful for differentiating these phenomena.^{7,8,25} Inhibition due to binding of the modified, duplex oligonucleotide to the enzyme can be clearly distinguished from nonspecific general inhibition as follows.⁷ If the T_m of a modified oligonucleotide is lower than that of an unmodified substrate (and both T_m values are within the range of temperatures at which the enzyme is active), the reaction is performed at a temperature between the two T_m values. If the modified substrate is denatured while the normal substrate remains a duplex and the inhibition of cleavage of the unmodified substrate disappears, it can be concluded that the analog itself is binding to the enzyme and that there is no thermally stable, general inhibitor present.⁷ This method depends on the endonuclease using only duplex DNA as a substrate. Analyses of the mode of inhibition by analog-containing oligonucleotides have revealed examples of competitive inhibition and allowed the determination of the dissociation constant (K_I) of the enzyme-analog complex.^{7,26}

Assays of Type II Restriction Endonucleases with Analog-Containing Substrates

Early studies with analog-containing substrates often reported only qualitative results using preparations of enzymes that were not homogeneous. Some studies simply reported whether the DNA was cleaved, while others determined the rate of cleavage at a single concentration of substrate with excess enzyme. Quantitative interpretations of the effects of analog substitutions require the determination of the K_m and k_{cat} values from steady state studies or of individual rate constants from single-turnover experiments and of equilibrium association constants (K_A).

Analog-containing substrates are usually synthesized with 5'-hydroxyl groups that are subsequently labeled with T4 polynucleotide kinase and [γ -³²P]ATP. Cleavage can be followed by separating the products from the substrates by thin-layer chromatography²⁷ or gel electrophoresis.²⁸ Cleavage has also been followed by elution of the products from a DEAE-cellulose filter and quantifying the disappearance of substrate.⁷ For the

²⁵ J. Mazzarelli, S. Scholtissek, and L. W. McLaughlin, *Biochemistry* **28**, 4616 (1989).

²⁶ C. R. Aiken, L. W. McLaughlin, and R. I. Gumport, *J. Biol. Chem.* **266**, (1991), in press.

^{26a} C. R. Aiken, E. W. Fisher, and R. I. Gumport, *J. Biol. Chem.* **266**, (1991), in press.

²⁷ E. Jay, R. Bambara, R. Padmanabhan, and R. Wu, *Nucleic Acids Res.* **1**, 331 (1974).

²⁸ A. A. YOLOV, M. N. Vinogradova, E. S. Gromova, A. Rosenthal, D. Cech, V. P. Veiko, V. G. Metelev, V. G. Kosykh, Y. I. Buryanov, A. A. Bayev, and Z. A. Shabarova, *Nucleic Acids Res.* **13**, 8983 (1985).

first two methods, the locations of substrate and product are determined by autoradiography, and the extent of reaction determined by excising the labeled components and quantifying them by liquid scintillation counting. The filter-binding method measures only the amount of substrate remaining after hydrolysis, and is less reliable for determining small amounts of product. Alternatively, the hydrolysis reaction of unlabeled oligonucleotides can be monitored by UV absorbance after high-performance liquid chromatography²⁹ (HPLC) to follow the disappearance of substrate or appearance of product.^{7,19,30,31} HPLC assays do not readily lend themselves to the analysis of reaction mixtures containing low concentrations of oligonucleotides because they rely on UV absorbance for analyte detection. Since no continuous assay has yet been developed for oligonucleotide substrates, aliquots of a reaction mixture must be analyzed individually.

For steady state analyses, the assays are manipulated so that the kinetics fit the Michaelis–Menten equation. The Mg^{2+} cofactor must be saturating so that the measured velocities are a function of only the oligonucleotide concentrations. The following conditions must be met for the assays to be valid. The initial velocity (v_0) of the reaction must be linear with respect to both the times of reaction and the enzyme concentrations used. The DNA concentration should be sufficiently greater (usually by at least 10-fold) than that of the enzyme, and the assay times and enzyme concentrations chosen so that enzyme binding or the reaction itself does not consume so much (>5 to 10%) of the oligonucleotide that the v_0 changes as a consequence of free substrate depletion. With very poor substrates, it is sometimes difficult to measure the reaction without using relatively high concentrations of enzyme.³⁰ For the best estimates of K_m , the substrate concentrations should range from about three to eight times below and above the ultimately determined K_m value, and curve-fitting techniques (see Methods of Procedure, below) rather than plots of the various linear forms of the Michaelis–Menten equation should be used to determine the kinetic values and their associated errors of measurement. In practice, the data obtained tend to be erratic, and several measurements at each substrate concentration are required to obtain reliable values of k_{cat} and K_m . In addition, one should attempt to ensure that reaction components other than the DNA do not influence the reaction. For example, the K_m values for Mg^{2+} with an analog-containing substrate and the unsubstituted control were different with *EcoRV*.³⁰

²⁹ L. W. McLaughlin, *Chem. Rev.* **89**, 309 (1989).

³⁰ P. C. Newman, D. Williams, R. Cosstick, F. Seela, and B. A. Connolly, *Biochemistry* **29**, 9902 (1990).

³¹ J. Alves, A. Pingoud, W. Haupt, J. Langowski, F. Peters, G. Maass, and C. Wolff, *Eur. J. Biochem.* **140**, 83 (1984).

For reactions conducted under single-turnover conditions, two criteria need be met to ensure that all the substrate is enzyme bound. The concentration of enzyme should be greater than that of the substrate, and the concentrations of enzyme and substrate should each be greater than the K_A value for the formation of the enzyme–substrate complex. The enzyme excess is proved by showing that the observed initial velocity is not increased by the further addition of enzyme.³² Additional confirmation that all the substrate is enzyme bound by a physical technique such as mobility retardation during gel electrophoresis has also been reported.³² The assay conditions should be those that favor cleavage at canonical sites in spite of high enzyme concentrations, which often favor “star” activity.³³ This consideration also applies to steady state studies, i.e., avoidance of cleavage at noncanonical sites by maintenance of high-specificity conditions. The reactions are usually initiated by the addition of Mg^{2+} , and the progress of the reaction is monitored by the withdrawal of aliquots with time. The rate of product release from the enzyme does not influence the results, for it is eliminated as a consideration by the strongly denaturing conditions used to quench the reaction.³² The substrates can be designed so that the cleavage products of the two strands have different sizes or, alternatively, either strand can be labeled individually so that hydrolysis of each can be monitored independently. Equations that describe the appearance of the radiolabeled cleavage products from each strand of a nonself-complementary oligonucleotide duplex in terms of the four first-order rate constants that together describe the complete cleavage reaction of an asymmetrically located *EcoRI* site have been reported.³²

Equilibrium association constants (K_A) for specific binding to a target sequence have been determined by nitrocellulose filter-binding assays.^{32,34,35} These measurements are carried out in the absence of Mg^{2+} to prevent hydrolysis of the substrates.³⁴ Direct determinations under equilibrium conditions^{34,35} or indirect assays using equilibrium–competition techniques^{34,36} have been reported. Equations for the solution of equilibrium–competition data from experiments using oligonucleotides with noncanonical sites or mismatched base pairs in the recognition sequence of *EcoRI* that are applicable to analog-containing oligonucleotides

³² D. R. Lesser, M. R. Kurpiewski, and L. Jen-Jacobson, *Science* **250**, 776 (1990).

³³ B. Polisky, P. Greene, D. Garfin, B. McCarthy, H. Goodman, and H. Boyer, *Proc. Natl. Acad. Sci. U.S.A.* **72**, 3310 (1975).

³⁴ B. J. Terry, W. E. Jack, R. A. Rubin, and P. Modrich, *J. Biol. Chem.* **258**, 9820 (1983).

³⁵ L. Jen-Jacobson, M. Kurpiewski, D. Lesser, J. Grable, H. W. Boyer, J. M. Rosenberg, and P. J. Greene, *J. Biol. Chem.* **258**, 14638 (1983).

³⁶ L. Jen-Jacobson, D. Lesser, and M. Kurpiewski, *Cell (Cambridge, Mass.)* **45**, 619 (1986).

have been published.³⁷ For assays under equilibrium conditions, the K_A values determined with excess enzyme should equal those obtained with excess oligonucleotide. Efforts were made with *EcoRI* to show that the absence of Mg^{2+} does not yield a K_A value different from the association constant under conditions allowing hydrolysis.³² Some restriction endonucleases, e.g., *EcoRV* and *TaqI*, do not appear to bind specifically to their target sequences in the absence of Mg^{2+} .^{38,39}

Since the endonucleases have affinities for DNA lacking their recognition sequences, i.e., they bind nonspecifically to DNA,³⁴ the effects of the concentration of potential nonspecific sites on the oligonucleotide should be considered in calculating the K_A value for specific binding. If the ratios of K_A (specific) to K_A (nonspecific) are similar to those for *EcoRI* binding to plasmid DNA ($\sim 10^4$ to 10^5),³⁴ the correction for nonspecific binding on a short oligonucleotide would be negligible because there are relatively few nonspecific sites. The effects on the binding resulting from the proximity of termini to the target sequence should also be kept in mind. In addition to the structure of DNA at duplex ends being different than that located within the helix,²⁰ the enzymes probably contact more base pairs than the canonical recognition sequence^{10,21} and specific targets near a terminus may lack one or more of these nucleotides. K_m values are higher with short oligonucleotides than with polymeric DNA, supporting the view that the flanking sequences contribute to the interactions between enzyme and substrate.^{1,14}

Interpretation of Results

The effects of the analog substitutions are commonly assessed by determining the kinetics of the hydrolysis reaction under either steady state or single-turnover conditions. The most easily interpretable result is the lack of an effect on the reaction metrics. In this case, one can safely conclude that the determinant under consideration does not interact with the enzyme and, additionally, that the analog substitution does not alter the DNA conformation such that any nearby interactions are affected. Usually, however, an analog substitution affects the reaction in at least one of its kinetic or thermodynamic constants. In a study of *EcoRI* with a battery of analog substrates, this finding has been interpreted to indicate a close interfacial contact between the enzyme and substrate.⁸

For analog substitutions that have kinetic consequences, the straight-

³⁷ V. Thielking, J. Alves, A. Fliess, G. Maass, and A. Pingoud, *Biochemistry* **29**, 4682 (1990).

³⁸ J. D. Taylor, I. G. Badcoe, A. R. Clarke, and S. E. Halford, *Biochemistry* **30**, (1991) in press.

³⁹ J. Zebala and F. Barany, personal communication.

forward interpretation is that the effect is due to the loss of an interaction between the enzyme and a group that has been deleted, or to a steric interference between the enzyme and the region of the base to which a new group was added. The direct inference is that the enzyme and the substrate normally "touch" one another at this site at some time during the reaction. Such an unambiguous conclusion is seldom warranted in the absence of other information that eliminates the indirect causal effects previously mentioned.^{8,19,25,30} Given this general caveat, analog experiments have been interpreted as follows.

Effects on both the initial velocity of the reaction and substrate binding are determined. The specificity constant, k_{cat}/K_m , is calculated from steady state results.⁴⁰ For reactions measuring single turnovers, the counterpart of the specificity constant, k_1K_A , is calculated.³² In this expression, k_1 is the first-order rate constant for the irreversible hydrolysis of the first phosphodiester bond and K_A is the equilibrium association constant of the enzyme and its recognition sequence in the absence of Mg^{2+} .³² Both the k_{cat}/K_m and k_1K_A values reflect the recognition of the substrate by the enzyme. For example, for two substrates A and B under steady state conditions, the ratio of the velocities $v_A/v_B = \{[A](k_{\text{cat}}/K_m)_A\}/\{[B](k_{\text{cat}}/K_m)_B\}$ expresses the relative selectivity of the enzyme for the two substrates.⁴⁰ For equimolar concentrations of A and B, the ratio of the specificity constants directly reflects the selectivity of the enzyme for one substrate with respect to the other. Because the values are derived from steady state experiments, the specificity constant, k_{cat}/K_m , measures the selectivity of the enzyme as a function of the appearance of the free, cleaved products of the reaction. If the "nicked" product after cleavage of the first strand of the duplex dissociates from the enzyme, steady state studies can measure the rate of nicking rather than of double-strand cleavage.

Analogously, the expression $v_A/v_B = \{[A](k_1K_A)_A\}/\{[B](k_1K_A)_B\}$ provides the relative selectivity of an enzyme under single-turnover conditions. The k_1K_A ratio reflects the selectivity as a function of the hydrolysis of the first bond cleaved.³² The specificity constant ratio may be more physiologically relevant than the ratio of k_1K_A values because DNA that has been cleaved in both strands at a target sequence may have more of a biological consequence than does nicked DNA. Conversely, the value (k_1K_A) is valuable for determining the energetics of the atomic interactions of recognition because it is a more direct measure of the events at a discrete step in the reaction—the first bond hydrolysis.³² This value is not subject to ambiguities introduced by such mechanistic considerations as whether

⁴⁰ A. Fersht, "Enzyme Structure and Mechanism." Freeman, New York, 1985.

the rate-limiting step of the reaction under steady state conditions occurs after bond hydrolysis.

In steady state studies, effects predominantly on k_{cat} have been regarded as indicative of alterations in interactions between the enzyme and the transition state.^{3,30,40} Since the rate-limiting step with polymeric DNA substrates is related to product release, i.e., is at some point after the initial phosphodiester bond hydrolysis with *EcoRI*⁴¹ and *EcoRV*,⁴² the interpretation of the steady state values may be complicated because the "recognition" occurs before the rate-determining step of the reaction. However, with some short oligonucleotide substrates, k_{cat} values approach those seen for the bond hydrolysis step determined under single-turnover conditions with DNA.^{8,30,43} This finding suggests that product release may not be rate limiting with these oligodeoxyribonucleotides and that k_{cat} may approximate k_1 , the first-order rate constant for bond hydrolysis. This hypothesis has not been confirmed.³⁰ Although K_m is likely not a true equilibrium dissociation constant (K_s) for the enzyme–substrate complex because of the complexity of the mechanisms of the endonucleases, changes in this value have been interpreted as arising primarily from effects on macroscopic binding phenomena involving all of the enzyme-bound forms of the substrate.⁴⁰ Keeping in mind that $K_m \neq K_s$, K_m values have been considered to be indicative of total binding effects.

Both k_{cat}/K_m and k_1K_A ratios can be used to estimate the binding energy provided by a single group on the base in the transition state given appropriate circumstances. If the determinant under consideration is not chemically involved in the bond-making and bond-breaking reactions and there are no inductive effects, the ratios are the binding energy ratios.^{40,44–46} By comparing the canonical sequence with the analog-containing sequence using *EcoRV*, the general equation $\Delta G_{\text{app}} = RT \ln[(k_{\text{cat}}/K_m)_{\text{analog}}/(k_{\text{cat}}/K_m)_{\text{canonic}}]$ has been used to estimate the contribution to the interaction energy of a group that has been deleted in the analog.³⁰ Likewise, with *EcoRI*, $\Delta\Delta G_1^\ddagger = -RT \ln[(k_1K_A)_{\text{noncanonic or analog}}/(k_1K_A)_{\text{canonic}}]$ has been used to estimate the energy cost in reaching the transition state of deleting a group.³² The former method depends on product appearance under steady state conditions, whereas the latter measures the first-order rate of the initial bond hydrolysis under single-turnover conditions. In spite of their

⁴¹ P. Modrich and D. Zabel, *J. Biol. Chem.* **251**, 5866 (1976).

⁴² S. E. Halford and A. J. Goodall, *Biochemistry* **27**, 1771 (1988).

⁴³ B. J. Terry, W. E. Jack, and P. Modrich, in "Gene Amplification and Analysis" (J. G. Chirikjian, ed.), Vol. 5, p. 103. Elsevier Science Publishing, New York, 1987.

⁴⁴ A. R. Fersht, *Trends Biochem. Sci.* **12**, 301 (1987).

⁴⁵ A. R. Fersht, *Biochemistry* **26**, 8031 (1987).

⁴⁶ A. R. Fersht, *Biochemistry* **27**, 1577 (1988).

differences, both methods have yielded anticipated values for the energies of some interactions.^{44,47} Unanticipated values have required interpretations that invoke indirect effects such as steric clashes and impaired conformational adjustments of enzyme or substrate to account for the energy discrepancies observed.^{30,32} These elegant studies represent the first attempts to quantify the energetics of the interactions between a submolecular determinant on the oligonucleotide in the transition state and the amino acid side chains of restriction endonucleases.

Methods of Procedure

Oligodeoxyribonucleotides containing modified bases are now most commonly synthesized using automated methods with β -cyanoethyl phosphoramidite synthons (see references in Tables I and II). After synthesis, the deprotected oligonucleotides are dried *in vacuo*, and purified by polyacrylamide gel electrophoresis in the presence of 7 M urea or by reversed-phase HPLC. We have found that HPLC is convenient for the isolation of oligonucleotides shorter than 20 bases.

Procedure 1. Purification of Oligonucleotides by Reversed-Phase HPLC

Prepare 1 liter of each of the following solutions using HPLC-grade methanol:

Buffer A: 50 mM potassium phosphate, pH 5.9

Buffer B: 50 mM potassium phosphate, pH 5.9,
50% (v/v) methanol

Equilibrate an octylsilica (C_8) reversed-phase HPLC column (4.6×150 or 250 mm) at 1 ml/min with buffer A. Inject 5 to 10 optical density (OD) units (measured at 260 nm) of oligomer in 200 μ l of buffer A, and separate with a linear gradient of increasing buffer B. Perform a trial run using 0.1 OD unit of material to determine the conditions for optimal separation. In our hands, a decanucleotide eluted at a concentration of approximately 60% buffer B ($\sim 30\%$ methanol). Collect the product peak, as detected by absorbance at 254 nm, and reinject a small amount (0.05 OD units) with the detector set at high sensitivity to analyze for purity. Oligonucleotides do not necessarily elute in order of their increasing lengths during this procedure, so it can be difficult to identify the desired product unless it is the major product of the synthesis. An analysis using polyacrylamide gel

⁴⁷ J. A. Wells, *Biochemistry* **29**, 8509 (1990).

electrophoresis may be performed to check the length of the product. It is necessary to equilibrate the column completely using at least 20 column volumes of solvent before each injection to ensure reproducible results. Dry the pooled fractions containing the product *in vacuo*, dissolve in H₂O, and desalt by passage over a Sephadex G-25 column (NAP-5 Sephadex from Pharmacia, Inc., Piscataway, NJ) and elution with H₂O. Store the product frozen.

Procedure 2. Purification of Oligonucleotides by Polyacrylamide Gel Electrophoresis

Polyacrylamide gel electrophoresis is the method of choice for purification of longer oligonucleotides (>20 bases), but it can also be employed in the purification of shorter products.

Pour a 20% (w/v) polyacrylamide-7 M urea sequencing gel.⁵ Dry the oligonucleotide *in vacuo* and dissolve it in the loading buffer [10 mM NaOH, 80% (v/v) formamide, 1 mM EDTA, 0.1% (w/v) xylene cyanol, 0.1% (w/v) bromphenol blue]. Load approximately 0.5 OD units (A_{260}) of oligomer in 5 μ l/cm of well width. Separate the mixture at 1000 V until the dyes have migrated a sufficient distance to indicate movement of the product to approximately half the length of the gel. Remove the gel and transfer it to a polyethylene film (Saran wrap). Remove one glass plate, overlay the gel with the plastic wrap, invert the plate, and tease the gel away from the glass with a spatula while simultaneously pulling on the plastic wrap.

Visualize the bands under reflected, short-wave UV light by placing the plastic-wrapped gel on a support that fluoresces on UV illumination, e.g., an X-ray film intensifying screen or a thin-layer chromatography plate impregnated with a fluorophore. Excise the desired products from the gel, and isolate the oligonucleotides by electroelution or by a crush-and-soak method.⁴⁸ Our laboratory routinely uses the EpiGene device (EpiGene, Baltimore, MD) for electroeluting the oligonucleotides onto columns of DEAE-cellulose (5 mm diameter $\times$ 3 mm) in the bicarbonate form. The DNA is recovered (~50% yield) by washing and eluting the columns with 0.5 ml 2 M triethylammonium bicarbonate (TEABC) [prepared by neutralization with CO₂ (gas) of triethylamine that has been distilled in the presence of chlorosulfonic acid to eliminate primary and secondary amines and to remove nonvolatile impurities]. Remove the TEABC *in vacuo* by repeated coevaporation with 50% (v/v) methanol. Dissolve the product in H₂O.

⁴⁸ T. Maniatis, E. F. Fritsch, and J. Sambrook, "Molecular Cloning: A Laboratory Manual," 2nd Ed. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1989.

Procedure 3. Nucleoside Analysis of Oligodeoxyribonucleotides Containing Modified Bases¹¹

After purification, the presence of the modified base is confirmed by HPLC analysis of the products of snake venom phosphodiesterase and alkaline phosphatase digestion of the oligomer.

Prepare a reaction mixture (50 μ l) containing:

Purified oligonucleotide: 0.2 OD units (A_{260})
Snake venom phosphodiesterase (5 mg/ml): 5 μ l
Bacterial alkaline phosphatase (1 mg/ml): 5 μ l
Tris-HCl (0.1 M, pH 7.5), 20 mM $MgCl_2$: 25 μ l
Sufficient H_2O to bring to a final volume of 50 μ l

Combine the solutions and incubate at 37° for 2 hr. Inject a sample of the digested oligonucleotide onto a reversed-phase column (C_8 or C_{18}) equilibrated in buffer A as described in procedure 1. We advise the use of a guard column to protect the primary column from contamination. Elute the column with a linear gradient of 0–30% buffer B in 20 min, followed by a 30 to 50% elution over 10 min. Wash the column for 3 min in buffer B before reequilibrating it. The deoxyribonucleosides are detected at 254 nm. The areas of the peaks are determined with an integrator that is connected to the spectrophotometer of the HPLC apparatus. Deoxynucleoside standards, including the analog deoxynucleoside, are individually injected to determine their retention times and relative recoveries. The mole fraction of each nucleoside is determined by dividing the area of each peak by the extinction coefficient at 254 nm of the corresponding deoxyribonucleoside, and dividing the area of the normalized peak by the lowest value obtained.

Procedure 4. Partial Snake Venom Digestion of Oligonucleotides to Determine Length

Prepare the reaction as in procedure 3 but use a 5'-³²P-labeled oligonucleotide (10^5 to 10^6 cpm) and omit the alkaline phosphatase. Start the reaction by adding the phosphodiesterase. Withdraw 5- μ l aliquots at intervals of 1, 2, 5, 10, 20, 30, 40, 50, and 60 min and spot them onto a plastic-backed, DEAE-cellulose thin-layer plate (20 cm long). Develop the plate as described in procedure 7. After autoradiography, the number of components in the nested set of products will correspond to the length of the oligonucleotide.

If desired, the oligonucleotides can be sequenced by either the wander-

ing-spot²⁷ or the Maxam-Gilbert^{5,49} methods. In practice, this is usually unnecessary if the length is correct, and the expected 5'-nucleoside and the modified base are present. The 5'-terminal nucleoside is determined by complete digestion of the 5'-³²P-labeled oligonucleotide as described previously followed by thin-layer chromatography of the deoxynucleotides and standards on cellulose TLC plates using saturated (NH₂)₂SO₄/1 M sodium acetate/2-propanol [80/18/2 (v/v/v)] as the eluant.

Assay of Type II Restriction Endonucleases with Oligodeoxyribonucleotides

Assays of type II restriction endonucleases using oligonucleotides containing base analogs as substrates are most commonly performed using radiolabeled oligomers followed by the separation of substrates and products by homochromatography²⁷ or polyacrylamide gel electrophoresis.²⁸

Procedure 5. Labeling of Oligonucleotides Using Polynucleotide Kinase and [γ -³²P]ATP

Evaporate to dryness 1 nmol of purified oligonucleotide in a siliconized microfuge tube. Redissolve in 12.5 μ l of a solution containing:

Tris-HCl, pH 7.5, 100 mM

MgCl₂, 20 mM

Dithiothreitol, 20 mM

Bovine serum albumin, 0.1 mg/ml

Spermidine (neutralized to pH 7.5), 2 mM

Add 9.5 μ l of distilled water, and 2 μ l of [γ -³²P]ATP (6000 Ci/mmol, 140 mCi/ml). Add 1 μ l (10 units) of T4 polynucleotide kinase, and incubate at 37° for 30 min. If possible, the ATP concentration should be >1 μ M to allow its efficient utilization by polynucleotide kinase. Add 1 μ l of 1 mM ATP and an additional 1 μ l of the kinase, and return the reaction to 37° for 30 min. The second addition of unlabeled ATP and enzyme ensures that all the oligonucleotide is phosphorylated. In endonuclease assays using short oligomers, it is necessary for all the molecules to be phosphorylated to ensure that the enzyme has only one possible substrate. In addition, the concentration of the substrate may be more accurately estimated if only one species is present. An alternative labeling procedure involves decreasing the specific radioactivity of the commercial [γ -³²P]ATP with ATP, and using a molar excess over the oligonucleotide of total ATP

⁴⁹ A. M. Banaszuk, K. V. Deugau, J. Sherwood, M. Michalak, and B. R. Glick, *Anal. Biochem.* **128**, 281 (1983).

with a lower specific radioactivity in the phosphorylation reaction. This procedure allows one to control more easily the specific radioactivity of the products.

Purification of Labeled Oligonucleotides and Determination of Concentrations

After labeling, it is sometimes necessary to purify the oligonucleotides before using them in the enzyme assays. In assays of the *RsrI* and *EcoRI* endonucleases using decanucleotides, contaminating [γ - ^{32}P]ATP, remaining from the polynucleotide kinase reaction, comigrated with the trimer product of hydrolysis during homochromatography. Chromatography of the labeled oligonucleotide mixtures using Nensorb 20 (Du Pont, Inc., Wilmington, DE) eliminated virtually all of the unreacted ATP. The columns were used as described by the manufacturer. However, the oligonucleotides obtained were not free of UV-absorbing impurities. Therefore, we employed chromatography using NAP-5 Sephadex columns (see procedure 1) as a final purification step. The resulting products were free of enzyme inhibitors and ATP, and yielded spectra typical of pure oligonucleotides. The oligonucleotide concentrations were determined spectrophotometrically using extinction coefficients calculated by summing the published values for the individual mononucleotides.⁵⁰ A correction in the molar extinction coefficients to account for hypochromicity can be introduced by measuring the increase in absorbance of a sample of the *duplex* oligonucleotide on digestion to completion with snake venom phosphodiesterase and alkaline phosphatase.¹⁹

Procedure 6. Kinetic Assays of Restriction Endonucleases Using Labeled Oligonucleotides as Substrates

Dispense aliquots containing the desired amounts of the labeled oligonucleotides necessary to form 50- μl reactions into siliconized microfuge tubes. Evaporate to dryness *in vacuo*. Redissolve in 25 μl restriction endonuclease assay buffer. The buffer should be one that gives maximal activity and high recognition specificity (no star activity) with the enzyme under study. Heat to 65° when using decanucleotides,⁷ or to higher temperatures for longer oligonucleotides, for 5 min, and cool to room temperature. This protocol ensures the formation of a duplex oligomer. Equilibrate the tubes at the desired temperature (20° for decanucleotides). The temperature is chosen to be low enough to

⁵⁰ B. Janik, "Physicochemical Characteristics of Oligonucleotides." IFI/Plenum, New York, 1971.

maintain the oligonucleotides in the duplex form and high enough to allow enzyme activity. Incubate the mixture for at least 5 min to dissolve the oligonucleotide. Start the reactions by adding 25 μ l of the assay buffer containing an appropriate dilution of the restriction endonuclease. Withdraw 5- μ l aliquots at intervals, and stop the reaction by either spotting directly onto DEAE-cellulose thin-layer plates (for homochromatography) or by pipetting the aliquot into 5 μ l of a solution containing a fivefold molar excess of EDTA over the Mg^{2+} in the reaction mixture (for gel electrophoresis). We have found that aliquots can be withdrawn every 2 min from six separate reactions, if the starting times are staggered. Normally, five aliquots are taken from each reaction and analyzed for product formation. The assays must be replicated enough times to provide precise results.

Procedure 7. Homochromatography for Separation of Substrates and Products

Prepare homochromatography solvent (Homomix 6).²⁷ Mix together 80 ml H_2O and 20 g yeast RNA (Sigma Chemical Co., St. Louis, MO). With cooling and stirring, add 20 ml 5 M KOH. Incubate at 37° with gentle agitation for 48 hr. Remove, add 100 ml H_2O , and neutralize with 1 M HCl. Add 420 g urea and H_2O to bring to a volume of 1000 ml. Stir the suspension until the urea is dissolved. Filter through Whatman (Clifton, NJ) #1 paper, and store at 4°. This protocol prepares Homomix 6, which is useful for separating oligonucleotides up to 12 nucleotides long. For longer oligonucleotides, other homomixes can be prepared by decreasing the amount of KOH or the incubation time, or both.²⁷ In a thin-layer chromatography tank, place 50–100 ml of Homomix (enough to cover the bottom to the depth of 0.5 cm). Place the entire tank in a 65° oven or incubator, and allow the tank to equilibrate. Spray the bottom 3 cm of the 20-cm long DEAE thin-layer plate onto which reaction aliquots were spotted with a fine mist of water from an atomizer. The chromatogram should be wet but not dripping. Without allowing the plate to dry, place it in the chromatography tank and develop the solvent to the top of the chromatogram. Remove the plate, allow it to dry, and mark the corners with radioactive ink. The Homomix solvent may be stored in a sealed container and reused. Cover with polyethylene film and subject the chromatogram to autoradiography on X-ray film. Develop the film and mark the positions of the spots corresponding to the substrate and product on the chromatogram. Excise these regions of the chromatogram, and determine their radioactivity by Cerenkov or

liquid scintillation counting. For each time point, the extent of reaction is determined using the following equation:

$$\text{Product (\%)} = \frac{\text{radioactivity in product}}{\text{radioactivity in product} + \text{radioactivity in substrate}}$$

The initial rate of each reaction is determined by plotting the product formation, expressed as moles, as a function of time and determining the slope of the straight line by regression analysis.

Procedure 8. Gel Electrophoresis for Separation of Substrates and Products

If the oligonucleotides are too long for the products to be separated from the substrate by homochromatography, electrophoresis in polyacrylamide gels as described in procedure 2 is used. Dry each aliquot, dissolve it in gel-loading buffer, and load the sample onto a 20% polyacrylamide gel containing 7 M urea.^{5,28} The electrophoresis is performed as described above, and the gel dried and subjected to autoradiography. Excise the product and substrate bands, quantify their radioactivities, and determine the extent of the reaction as described in procedure 7.

Procedure 9. Analysis of Data and Calculation of Kinetic Constants

The kinetic assays should be performed so that their initial velocities are directly proportional to the times of reaction and the concentrations of enzyme used. After measuring the initial rates of reaction for various concentrations of modified and unmodified substrates, the kinetic constants may be calculated using one of the standard linearization equations (Lineweaver–Burk⁷ or Eadie–Hofstee⁸); or, preferably, they are determined by computer-assisted direct fitting to the Michaelis–Menten equation. Our laboratory has employed a Fortran version of the kinetics program⁵¹ of W. W. Cleland (courtesy of B. V. Plapp, University of Iowa) to calculate k_{cat} and K_m values from the kinetic data. The computer-assisted analysis has the advantage of providing an estimate of the precision of the measurements. In practice, the data obtained using the procedures given here require several measurements of the reaction rate at each concentration of substrate to give reliable values.

⁵¹ W. W. Cleland, this series, Vol. 63, p. 103.

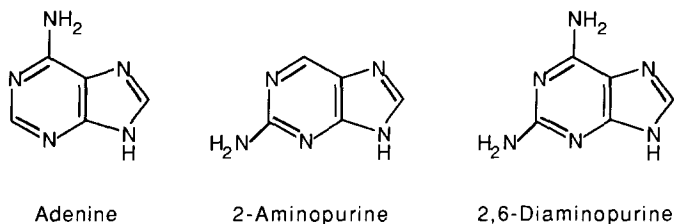


FIG. 1. Adenine and two analogs used in studies of the *EcoRI* and *RsrI* endonucleases. In 2-aminopurine, the amino group of adenine, which is found in the major groove of B-form DNA, has been removed and placed at C-2, where it lies in the minor groove. The analog 2,6-diaminopurine contains amino groups in both grooves.

Examples with *EcoRI* and *RsrI* Endonucleases

RsrI and *EcoRI* endonucleases catalyze the identical reaction of cleaving the duplex sequence d(GAATTC) at the phosphodiester bonds between the G and A residues.⁵²⁻⁵⁴ Both enzymes have been purified to homogeneity and their genes sequenced.^{41,53-57} The gene sequences reveal that the proteins have >50% amino acid identity, and that the amino acids postulated to be involved in DNA recognition and catalysis as determined by crystallographic and mutational analysis of *EcoRI* are present in corresponding positions in both amino acid sequences.^{54,55} The enzymes have similar molecular weights and both function as homodimers.^{41,53} They display similar turnover numbers, Michaelis constants, and specificity constants when assayed on plasmid DNA substrates. Both enzymes cleave DNA at noncanonical sites under appropriate reaction conditions.^{33,53} The two enzymes differ with respect to their isoelectric points, optimal reaction conditions, and sensitivity to *N*-ethylmaleimide.⁵³ Unlike *EcoRI*, *RsrI* endonuclease is essentially inactive at 37° and has reduced activity at NaCl concentrations above 10 mM.^{53,54}

An experiment was performed with *EcoRI* endonuclease using the duplex octadeoxyribonucleotide d(GGAATTCC)⁸ with 2-NH₂Pur or 2,6-(NH₂)₂Pur (2NH₂-Ade) (Fig. 1) at the second adenine position. The results

⁵² J. Hedgepeth, H. M. Goodman, and H. W. Boyer, *Proc. Natl. Acad. Sci. U.S.A.* **69**, 3448 (1972).

⁵³ C. Aiken and R. I. Gumport, *Nucleic Acids Res.* **16**, 7901 (1988).

⁵⁴ P. J. Greene, B. T. Ballard, F. Stephenson, W. J. Kohr, H. Rodriguez, J. M. Rosenberg, and H. W. Boyer, *Gene* **68**, 43 (1988).

⁵⁵ F. H. Stephenson, B. T. Ballard, H. W. Boyer, J. M. Rosenberg, and P. J. Greene, *Gene* **85**, 1 (1989).

⁵⁶ P. J. Greene, M. Gupta, H. W. Boyer, W. E. Brown, and J. M. Rosenberg, *J. Biol. Chem.* **256**, 2143 (1981).

⁵⁷ A. K. Newman, R. A. Rubin, S.-H. Kim, and P. Modrich, *J. Biol. Chem.* **256**, 2131 (1981).

of this experiment indicate the importance of having ancillary information to aid in interpreting results. The experiment was performed to determine if there was a contact between the enzyme and the C-6 amino group of the central Ade base. The analog 2-NH₂Pur lacks this amino group in the major groove of B-form DNA but has an added amino group in the minor groove that forms a hydrogen bond with the paired Thy base in the opposite strand. With this analog, the amino group on the purine has been switched from the major to the minor groove of the DNA. When measured under steady state conditions, the specificity constant, k_{cat}/K_m , of the oligomer containing the 2-NH₂Pur base was decreased 11-fold with respect to the unmodified octamer.⁸ This result suggested that the amino group in the major groove of the DNA made an important contact with the enzyme.

The analog 2,6-(NH₂)₂Pur was used as a control to test for the effect of the addition of an amino group to the minor groove of the DNA. When this analog, containing 2,6-(NH₂)₂Pur, was tested, the specificity constant remained depressed ninefold.⁸ The result with this substrate indicated that the major effect caused by the analog substitutions was due to the insertion of the amino group into the minor groove and not to its deletion from the major groove. In the absence of any other information, one might have concluded that the addition of the amino group in the minor groove interfered with some interaction with the enzyme in that groove of the DNA. Fortunately, an X-ray structural of the cocrystal of the enzyme and oligonucleotide containing the recognition site was available¹⁰ and showed that protein was absent from the minor groove. The structure also revealed that the DNA in the complex was not in a normal B form, i.e., it was kinked with the minor groove compressed. Thus, we were able to conclude that the effect of the addition of the amino group to the minor groove was probably due to its interference with the kinking of the DNA observed when it is bound to the enzyme.^{10,58} A subsequent experiment, in which the C-6 amino group was simply deleted by substitution of Pur for Ade at the central position of the sequence, indicated that the removal of the amino group decreased the specificity constant twofold.⁷ Taken together, these results suggest that although there is likely a contact between the amino group and the enzyme, its loss is not as disruptive as is the introduction of an amino group into the minor groove. Thus, these experiments were more revealing of the role of DNA conformational factors than of amino acid-base interactions. In general it has been observed that the removal of a functional group from a base is less disruptive of the interaction between the enzyme and DNA than is addition of a new group.^{7,8}

⁵⁸ Y. Kim, J. C. Grable, R. Love, P. J. Greene, and J. M. Rosenberg, *Science* **249**, 1307 (1990).

Another example from studies with *EcoRI* illustrates the value of experiments with analogs in the absence of confirmatory structural information. Removal of the methyl groups from either of the Thy bases in the GAATTC recognition sequence yielded decreased specificity constants with *EcoRI*, with the effect being larger when the second Thy residue was converted to Ura.^{7,8} The activities of the substrates were restored on the substitution of BrUra for the Ura.⁸ These results suggested that the methyl groups of these two pyrimidines were involved in interactions with the enzyme, although the structure of the enzyme-DNA complex failed to reveal electron density in this region and recognition was assigned to interactions solely with the purines.¹⁰ Genetic studies⁵⁹ have implicated, and the determination of the *EcoRI*-DNA structure in several new crystal forms⁵⁸ now accommodate, a role for interactions of the enzyme with the pyrimidines.

The analog 2-NH₂Pur has also been useful in a comparative study of the *RsrI* and *EcoRI* endonucleases.²⁶ Because the two enzymes catalyze the identical reaction and have a high degree of amino acid identity,⁵⁵ we anticipated that they would interact with their duplex target sequence, d(GAATTC), similarly. They appear to do so in the complex that is formed during site-specific binding in the absence of Mg²⁺. If the enzyme is bound site specifically to covalently closed circular DNA and the complex relaxed with a type I topoisomerase, both enzymes induce the same unwinding of the helix (~25°) as determined by the change in the linking number of the DNA.^{26a,60} Similarly, when both enzymes are tested for site-specific binding to DNA fragments in which the target site is located at different distances from the ends of the DNA, they bend the DNA to the same extent (~50°) as determined by gel mobility shift assays.^{26a,61} Finally, the site-specific and nonspecific equilibrium thermodynamic binding constants for the two enzymes are similar and they thus show nearly identical ratios of specific to nonspecific binding (~10⁵) despite the different solvent and temperatures at which the values were determined.^{26a} All these results suggest similar modes of interaction between the enzymes and the target sequence.

However, when experiments with analog-containing substrates were performed in order to study the cleavage reactions, in contrast to the binding events, substantial differences between the enzymes were observed. Decameric oligodeoxyribonucleotides containing 2-NH₂Pur at the central Ade position of the recognition sequence (GAATTC) were assayed

⁵⁹ J. Heitman and P. Model, *Proteins* **2**, 185 (1990).

⁶⁰ R. Kim, P. Modrich, and S.-H. Kim, *Nucleic Acids Res.* **12**, 7285 (1984).

⁶¹ J. Thompson and A. Landy, *Nucleic Acids Res.* **16**, 9687 (1989).

with both enzymes. *EcoRI* cut the substrate with a k_{cat}/K_m ratio ninefold lower than that with the unmodified control as a result of a relatively unchanged k_{cat} value and an increased K_m value.^{7,8,26} In contrast, *RsrI* failed to cut the analog-containing sequence, and the analog was not an inhibitor of the cleavage of the unmodified sequence, suggesting that the analog did not bind to the enzyme.²⁶ Thus, *EcoRI* binds and cleaves the substrate containing 2-NH₂Pur whereas *RsrI* fails even to bind the compound.

In order to insert an amino group into the minor groove, 2,6-(NH₂)₂Pur was substituted at this position for Ade, and the two enzymes compared. *EcoRI* cleaved the substrate well, whereas *RsrI* hydrolyzed it at a rate just above the level of detection.²⁶ For *RsrI*, the analog containing 2,6-(NH₂)₂Pur was a strong competitive inhibitor ($K_i \sim 10^{-9} M$) of the cleavage of the unmodified sequence.²⁶ Thus, in spite of the high degree of amino acid identity and the similar interactions with the target sequence in the absence of Mg²⁺, the two enzymes react with the same substrate analogs differently, i.e., they exhibit selectivity or recognition in the chemistry of the cleavage and not just in the binding. Because this difference in reactivity occurs under conditions *in vitro* in which each enzyme exhibits high-specificity cleavage, it is likely that it is due to a property of the enzymes themselves rather than to the reaction conditions.

Studies using substrate analogs and restriction endonucleases have provided information about interactions between the enzymes and their DNA substrates and, additionally, between the enzymes and transition states of the reactants. The ability to monitor interactions between the enzyme and DNA at stages of the hydrolysis reaction beyond the initial ground-state, enzyme-substrate complex is a valuable attribute of this approach. Such studies have directly indicated that recongition may occur during the chemistry of the reaction as well as at the level of substrate binding. Studies with substrate analogs should continue to serve as useful partners with genetic and structural approaches in our attempts to understand the molecular nature of sequence-specific protein-DNA interactions.

Acknowledgments

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[22] Probing Information Content of DNA-Binding Sites

By GARY D. STORMO

Introduction

Information content is a convenient way to describe the specificity of a DNA-binding protein.¹ In its simplest form information content is equivalent to the common description of a restriction enzyme as a "6-base cutter" or a "4-base cutter." That is, information content is a measure of the amount of specificity required for recognition, independent of the mechanism of recognition. Equivalently, it is a measure of the frequency of sites expected in a random sequence. We usually measure information content in *bits*, rather than bases. Since the choice of 1 base requires two bits, a "6-base cutter" has an information content of 12 bits. A restriction enzyme with multiple recognition sites, such as *HincII*, which cuts at GTYRAC (Y = C or T, R, = A or G),² recognizes 10 bits, or 5 bases, of information, even though its recognition site is 6 bases wide, because of the ambiguity in some of the positions.

The utility of information content as a measure of specificity is that it is applicable even to cases of high ambiguity, which is a typical characteristic of regulatory sites and which distinguishes them from restriction sites. Even though restriction sites can contain ambiguity they have essentially an all-or-none activity. That is, *HincII* cuts at every site that matches the sequence GTYRAC, and only such sites are cut. Regulatory sites, on the other hand, are usually similar to a "consensus sequence," but may differ at almost any position and at more than one position in any particular example. Furthermore, some positions within the consensus sequence may be more variable than others. The most studied example is *Escherichia coli* promoter sequences. They contain a two-part consensus sequence: a "–35" region of TTGACA and a "–10" region of TATAAT. In a compilation of over 200 promoters each of the 4 bases has been found at least once at each position of the consensus sequence, and none of the examples is a perfect match to it.³ But promoters are not random, because any particular promoter will likely match at most of the consensus positions, and some of the positions are highly, although not absolutely, conserved.

¹ T. D. Schneider, G. D. Stormo, L. Gold, and A. Ehrenfeucht, *J. Mol. Biol.* **188**, 415 (1986).

² G. D. Stormo, *Annu. Rev. Biophys. Biophys. Chem.* **17**, 241 (1988).

³ C. B. Harley and R. P. Reynolds, *Nucleic Acids Res.* **15**, 2343 (1987).

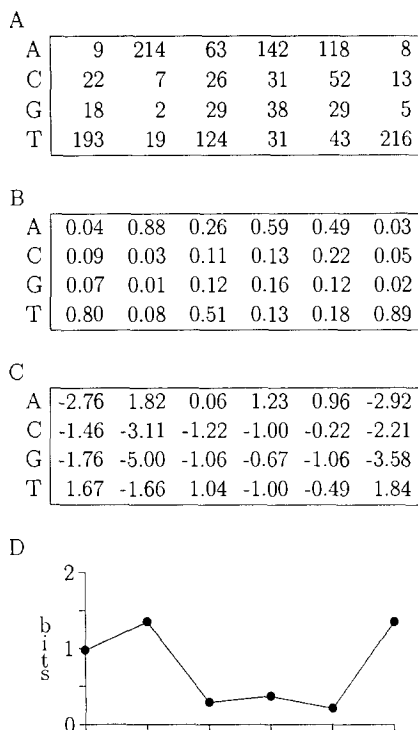


FIG. 1. Matrices from *E. coli* -10 promoter sequences.³ (A) The number of occurrences of each base at each position of the 242 nonmutant promoters. (B) The fraction of each base at each position, from the above data. (C) The logarithms (to the base 2) of those fractions divided by 0.25, the genomic frequency of each base in *E. coli*. (D) The information content¹ of each position. This is the dot product of the vectors in each column of the matrices in (B) and (C), as calculated in Eq. (1). (Reprinted with permission.⁴)

The simplistic notion of specificity used above for restriction sites is not easily adapted to the highly variable sequences in regulatory sites, but information content is. Figure 1⁴ illustrates the calculation of information content for the -10 region of *E. coli* promoters. The basic formula for the information content at any position in a binding site is

$$I_{\text{seq}} = \sum_{b=A}^T f_b \log_2(f_b/p_b) \quad (1)$$

where f_b is the fraction of each base b in the aligned sites, and p_b is the genomic proportion of each base.^{1,2,4} When the genomic proportions of

⁴ G. D. Stormo, this series, Vol. 183, p. 211.

each base are equal ($p_b = 0.25$ for all b) the formula reduces to $I_{\text{seq}} = 2 + \sum f_b \log_2 f_b$, which can vary from 0, for a random position, to 2, for a position that is absolutely conserved.¹ The DNA-binding sites for several regulatory proteins have been examined.^{1,5} The information content of each is very close to that expected based on the number of sites in the genome. For highly ambiguous sites information content determined assuming independent positions, as in Eq. (1), is a lower bound on the value that would be determined using higher order analyses.⁶ Therefore, the result of near equality between expected and observed frequencies¹ is not inconsistent with the idea that regulatory sites should be "overspecified."^{7,8} That result does allow one to estimate the frequency expected for a particular type of site, and to gauge whether or not the pattern obtained for the specificity of a particular protein is appropriate. For example, the large discrepancy we originally observed for the T7 promoters led to the experiments described below, from which we determined that there were additional constraints on phage promoters than just transcription initiation activity.⁵

Experimental Determination of Information Content

While the information content determined from several examples of the binding sites of a protein can be a valuable tool in understanding its specificity, often a more detailed analysis is needed. For example, the known binding sites may have other constraints that contribute to their information content, but which are not required for function by the protein. It is also likely that not enough natural sites are known to have a reliable picture of the specificity of the protein from those data alone.

A variety of experimental procedures can also be used to determine the information content of a protein. The basic strategy is to randomize the binding sites and to select from the entire population those that retain functional activity. The definition of a functional site may be qualitative; either a particular site works or it does not, or it may involve a quantitative assay that lets one use the relative affinity of the protein for different sites in the calculation of information content. Such experiments can either be carried out *in vitro* or by putting the randomized sites back into cells and determining function *in vivo*.

⁵ T. D. Schneider and G. D. Stormo, *Nucleic Acids Res.* **17**, 659 (1989).

⁶ G. Z. Hertz and G. D. Stormo, submitted.

⁷ P. H. von Hippel, in "Biological Regulation and Development" (R. F. Goldberger, ed.), p. 279. Plenum, New York, 1979.

⁸ O. G. Berg and P. H. von Hippel, *J. Mol. Biol.* **193**, 723 (1987).

In Vivo Experiments

In our early studies of information content in regulatory sites the T7 promoters were an exception to the observation that information content is approximately equal to that expected from the frequency of sites.¹ In fact, T7 promoters had about twice the expected information content. We hypothesized then that perhaps only part of the information was required for promoter function, polymerase binding and transcription initiation, and that the remaining information was conserved for other, unknown, reasons. In order to test this hypothesis we synthesized randomized variants of a 27-bp region that contained the promoter consensus sequence.⁵ At each position of this synthesized promoter we used a mixture of nucleotides designed to give the consensus base 85% of the time and each of the other bases 5% of the time. In so doing we expected to get the complete consensus sequence about 1% of the time, assuring us that we would have a positive control of promoters that had to function. We also expected that there would be an average of about four changes per promoter. These randomized promoter sequences were then cloned into a plasmid that would serve to screen for functional promoters in cells that could be induced to make T7 RNA polymerase. The screen was that functional promoters killed the cells when induced, whereas nonfunctional promoters did not. Comparing the sequences that were recovered by the screen to analyses done by Chapman and Burgess,⁹ we concluded that promoters had to be within two- or threefold the activity of the consensus promoter to pass the screen.

Two hundred clones were screened and 58 (29%) of them were found to be functional. Five had multiple inserts, leaving 53 (including 3 with the consensus sequence) that were functional promoters. Figure 2 shows the composition by position of the functional promoters and of the T7 phage promoters for comparison. In determining the information content from these data two factors are important. One is that the population proportions of the bases, p_b , are not equimolar, but rather they have been biased to be 85% the consensus base, which varies from position to position. When these numbers are used in Eq. (1), the information content comes out at about 2.9 bits, which is not too far removed from the 1.8 bits expected from the 29% functional promoters. But the other important consideration to remember is that this is the information relative to the biased synthetic promoters. What we really want to know is the amount of information relative to a T7 infection, in which case the genomic proportions of the bases are approximately equal. We could have obtained that

⁹ K. A. Chapman and R. R. Burgess, *Nucleic Acids Res.* **15**, 5413 (1987).

L	$\phi 10$	Experiment T7-W				Phage Promoters			
		A	C	G	T	A	C	G	T
-21	a	43	5	4	1	8	3	4	2
-20	a	49	2	1	1	12	0	2	3
-19	a	41	0	12	0	10	1	3	3
-18	t	3	2	2	46	3	1	3	10
-17	t	3	1	1	48	0	3	0	14
-16	a	48	0	3	2	16	0	0	1
-15	a	51	1	0	1	17	0	0	0
-14	t	2	0	0	51	0	0	0	17
-13	a	47	1	1	4	14	1	0	2
-12	c	0	53	0	0	0	16	1	0
-11	g	7	0	46	0	2	0	15	0
-10	a	47	2	1	3	17	0	0	0
-9	c	0	53	0	0	0	17	0	0
-8	t	0	0	0	53	0	0	0	17
-7	c	0	53	0	0	0	17	0	0
-6	a	48	1	2	2	17	0	0	0
-5	c	0	53	0	0	0	16	1	0
-4	t	1	3	1	48	0	0	0	17
-3	a	42	4	4	3	17	0	0	0
-2	t	2	0	0	51	5	0	0	12
-1	a	42	4	5	2	16	0	0	1
0	g	2	0	50	1	2	0	15	0
1	g	1	1	50	1	2	0	15	0
2	g	2	4	46	1	5	0	12	0
3	a	45	4	2	1	10	0	7	0
4	g	3	4	42	4	5	1	11	0
5	a	47	3	1	2	13	2	0	2

FIG. 2. Tabulated results for strong T7 promoters. The numbers of each kind of base at each position (L) are given for both the functional promoters recovered by the screen and for sequences at promoters in wild-type T7 phage. The column labeled $\phi 10$ refers to a particular T7 promoter, the sequence of which is identical to the phage consensus. A symmetry element is indicated by boxed letters. (Reprinted with permission.⁵)

number by completely randomizing the promoters instead of leaving a bias toward the consensus sequence. However, since we had only a screen, not a selection, for activity, that would have required screening many thousands of clones to get a few that were functional. Instead, we can use the data we collected and "normalize" it to see what we would have obtained had we done the complete randomization. The assumption is that the data obtained accurately reflect the selectivity of the polymerase. We assume that the ratios of f_b/p_b are equal to the ratios of the activities for each variant base.^{2,4,5,8} That is equivalent to assuming that the data reflect the differences in binding energies to each possible base, and that those

energy differences are independent of the base proportions in the population of possible binding sites.

The data we have are of f_b and p_b values. The normalization procedure is to find the f'_b values for new values of p'_b .⁵ The constraints that must be satisfied are that the sums of f_b and p_b must equal one, and that the ratios of all the f_b/p_b values must be unchanged. For the case of $p'_b = 0.25$ this is straightforward, because then $\sum f'_b/p'_b = 4$. The fraction of each base calculated to be in binding sites if the nonsite bases are equimolar is

$$f_b^s = (f_b/p_b)/(\sum f_b/p_b) \quad (2)$$

We consider this situation, of equimolar nonsite bases, to be a "standard state," and the binding constant determined at this state, $K_b^s = 4f_b^s$, to be the specific binding constant. It can vary between 0 and 4, with a value of 1 for nonspecific binding. The formula for information content then becomes

$$I_{\text{seq}} = \sum_{b=A}^T f_b^s \log_2 K_b^s \quad (3)$$

which is easily seen to be proportional to the average specific binding energy.^{2,4,13}

When our randomized T7 promoter data are treated in this manner, the resulting information content at each position is as shown in Fig. 3. Also shown is the information content as calculated from the phage promoters.^{1,5} The most important result is that much of the information within the phage promoters is not required for promoter activity. In fact, only about half of the total conserved information is required, placing T7 promoters into the same class as other regulatory sites where the information content observed is about as expected from the frequency of sites.¹ Careful examination of the data in Figs. 2 and 3 reveals several interesting features. First, there are some positions, such as -7, -8, and -9, that are absolutely conserved in both the phage and the functional promoters. There are several other positions, such as -3, -4, -6, and -10, that are absolutely conserved in the phage promoters, but at which any base can exist in a functional promoter. This finding alone confirms our hypothesis that there are constraints at T7 promoters other than those required by the polymerase for transcription activity. At several other positions there is information in the functional promoters, but less than in the phage promoters. The results at positions -5 and -12, where it looks like functional promoters contain more information than phage promoters, are not significant, but due to the small sample size. If we had seen even one nonconsensus base at those positions the information content would

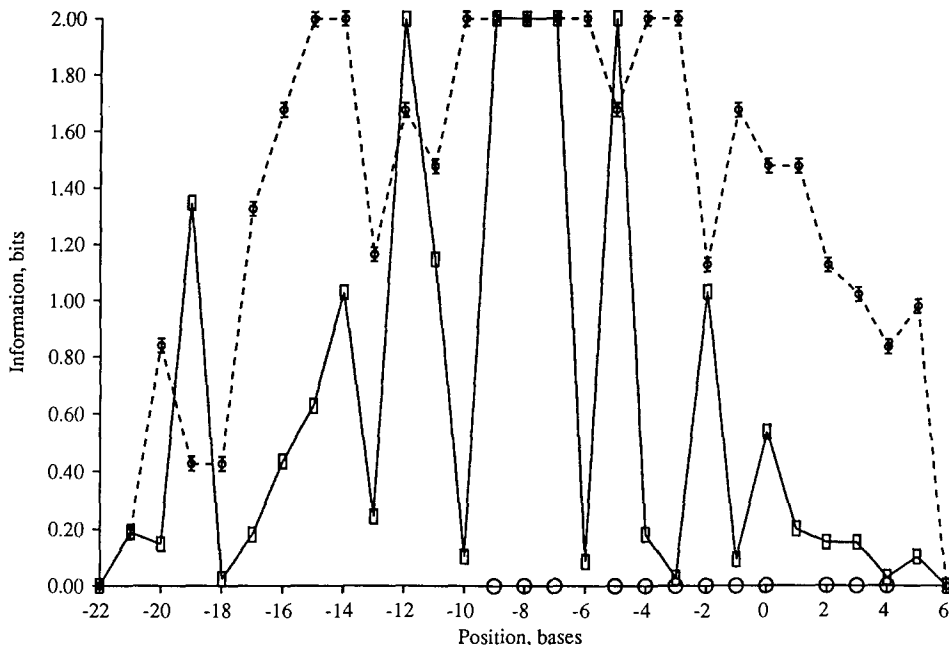


FIG. 3. Information at T7 promoters. The abscissa is the position along the sequence, in bases, while the ordinate shows the amount of information at each position in bits. (Φ — Φ), Information from the phage promoters; ($\square$ — $\square$), normalized information in strong T7 promoters recovered by the screen. ($\circ$), Symmetry elements on the abscissa. (Reprinted with permission.⁵)

have been lower than the phage value. That is, given the bias in the randomized promoters, the phage data lead us to expect between zero and one nonconsensus base in a sample size of 53, and the observation of zero cannot be considered significantly different than that expectation. At position -19 the result is significant and indicates that a G at that position may lead to more promoter activity than the consensus A, which must be conserved for other reasons. Overall the information content analysis of *in vivo* functioning T7 promoters gives us a very different picture of the sequence requirements than is obtained from the phage examples.

If we had quantitative measurements of how strong each promoter is we could have incorporated those data into the analysis. For example, several years ago we reanalyzed data about the context effects on nonsense suppression in *E. coli*.¹⁰ The efficiency of amber suppression by *su2* had been determined for many different contexts, and we analyzed the data

¹⁰ G. D. Stormo, T. D. Schneider, and L. Gold, *Nucleic Acids Res.* **14**, 6661 (1986).

by multiple regression using several alternative models. As shown in Fig. 4, the simple model that the context effect is determined by the 6 bases surrounding the amber codon, three on each side, and that the effects of each base are independent and additive does quite well in describing the data. Figure 4b shows the fit by this model between the predicted and observed values, which has a correlation coefficient of about 0.93. Figure 4a shows the values of relative $\ln K_b$ that were determined by the regression, in which we had arbitrarily set $K_T = 1$ at each position. (There are only three independent parameters per position, so we set one and determine the others relative to that one.)¹⁰ We can normalize those values to a standard activity constant, as described above, and then calculate the information content for suppression contexts, as in Eq. (3). The result is a total of 0.70 bits, with 0.58 coming from the two positions 3' to the amber codon, positions 3 and 4. While it was clear in our original paper that these were the two most important positions for determining suppression efficiency, this information content analysis lets us quantitate the relative importance of each position.

In Vitro Experiments

One can also determine information content from *in vitro* binding experiments. In fact, these make it easy to use quantitative assays of binding affinity in the determination. One method is to perform extensive binding analysis on mutant binding sites, as has been done for the phage λ proteins Cro and Repressor.^{11,12} In these experiments every single base change was made to the consensus λ operator and the binding constant was determined for each protein to each variant operator. The results were presented as $\Delta\Delta G$ values, which can easily be converted to specific binding constants, as described above. The information content determined by these calculations is about 15.5 and 14 bits for Cro and Repressor, respectively. These numbers are low compared to that determined from the known binding site sequences or from the frequency of known operators.¹ These calculations, based on quantitative measurements of independent changes, tend to underestimate the complete specificity of the proteins.⁶ Nonetheless, these values are consistent with many experiments of determining the affinity to sites with multiple changes, and serve to constrain the models of how the proteins obtain their required specificity.^{11,12}

Information content can also be determined from experiments in which the protein is bound to many different sites at once, providing there is a

¹¹ Y. Takeda, A. Sarai, and V. M. Rivera, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 439 (1989).

¹² A. Sarai and Y. Takeda, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 6513 (1989).

a

position:	-3	-2	-1	0	+1	+2	+3	+4	+5
A	+0.02	-0.20	+0.30		a		+1.72	-1.15	-0.22
C	+0.35	+0.28	+0.16				+0.54	-0.98	-0.45
base G	-0.03	+0.07	-0.53			g	+1.78	-1.55	-0.30
T	0.00	0.00	0.00	t			0.00	0.00	0.00

+ 1.85 = ln(suppression activity)

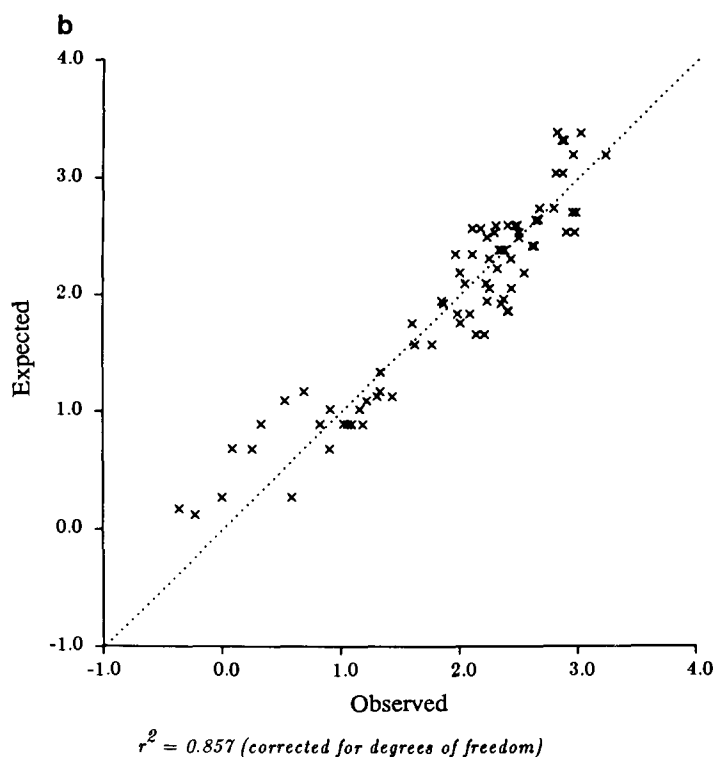


FIG. 4. Regression results for six mononucleotides vs suppression activity. (a) Matrix of relative ln(suppression activity) constants. (b) Plot of the observed ln(activities) vs those predicted from the matrix of (a). (Reprinted with permission.¹⁰)

means of determining the partitioning of each site between bound and unbound fractions. We have been developing such a method for the Mnt repressor of phage P22.¹³ Mnt is a repressor that binds as a tetramer to a symmetric operator sequence to regulate expression from the P_{ant} promoter.¹⁴ We synthesized an operator with one randomized position, which could be cut with a different restriction enzyme for each different base in the variable position: . . . AGGTCCACGGTGGAnCTagt . . . The upper-case letters are the wild-type operator sequence, and the n is the randomized position. Depending on whether the variable base is an A, C, G, or T, the oligonucleotide can be cut by *Spe*I, *Sau*96I, *Alu*I, or *Mbo*I, respectively. Mnt repressor was added to the randomized synthetic operator in sufficient quantity to bind approximately half of the DNA, and the bound and unbound fractions were separated in a gel mobility shift assay.¹⁵ Both bands were eluted from the gel and the proportion of each base in each fraction was determined by cutting with the restriction enzymes. The results of this experiment were that the specific binding constants, K_b^s , were 0.71, 2.36, 0.28, and 0.64 for A, C, G, and T, respectively. The information content for this position is 0.41 bits.

The partitioning of each site into the two fractions could be determined by quantitative sequencing of each fraction. Then an equivalent experiment could be done at any position, regardless of whether an appropriate set of restriction enzymes could be found. It would also be possible to randomize several positions at once and determine the information content of each in a single experiment. Then only a few binding experiments might be sufficient to determine the complete specificity of the protein. So as to minimize the effects of nonindependent binding interactions, such an experiment should probably be done with nonadjacent positions randomized in a particular experiment.

Summary

An information content analysis of protein-binding sites gives a quantitative description of the specificity of the protein, independent of the mechanism of specificity. It gives useful information about the total specificity of the protein and about the individual positions within the binding sites. Information content is consistent with both thermodynamic and statistical analyses of specificity.^{2,4} When applied to a collection of known binding sites, the description provided may be limited by the sample size

¹³ G. D. Stormo and M. Yoshioka, *Proc. Natl. Acad. Sci.* **88**, 5699 (1991).

¹⁴ A. K. Vershon, S.-M. Liao, W. R. McClure, and R. T. Sauer, *J. Mol. Biol.* **195**, 311 (1987).

¹⁵ K. L. Knight and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 797 (1989).

or by unknown constraints on those sites. Experimental procedures to determine the information content can give much more reliable measures. A large number of functional sites can be obtained from a much larger pool of randomized potential sites. Quantitative assays for the activity of different sites can be easily incorporated into the analysis, thereby increasing its sensitivity. Both *in vitro* and *in vivo* experiments are amenable to information content analysis.

Acknowledgments

The experiments on the Mnt-binding specificity were performed using protein kindly provided by Ken Knight and Robert Sauer. The work described in this chapter has been supported by NIH Grant GM28755.

[23] Protein Chemical Modification as Probe of Structure-Function Relationships

By KATHLEEN S. MATTHEWS, ARTEMIS E. CHAKERIAN, and
JOSEPH A. GARDNER

Chemical modification is a useful tool for examining the participation of specific amino acid side chains in the structure and function of proteins. In addition, chemical modification provides the potential for cross-linking and, in the case of enzymes, for design of reagents that are activated by the enzymatic mechanism. A number of reviews and books are available for consultation in designing an experimental approach for chemical modification of DNA-binding proteins.¹⁻⁷ Figure 1 provides a flow chart for developing an experimental design to modify a protein. In the ideal situation, a specific reagent will react with only one type of side chain; in

¹ R. L. Lundblad and C. M. Noyes, "Chemical Reagents for Protein Modification, Volumes I and II." CRC Press, Boca Raton, Florida, 1984.

² C. H. W. Hirs, ed., this series, Vol. 11.

³ C. H. W. Hirs and S. N. Timasheff, eds., this series, Vol. 25.

⁴ C. H. W. Hirs and S. N. Timasheff, eds., this series, Vol. 47.

⁵ A. N. Glazer, R. J. DeLange, and D. S. Sigman, "Chemical Modification of Proteins." American Elsevier, New York, 1975.

⁶ G. E. Means and R. E. Feeney, "Chemical Modification of Proteins." Holden-Day, San Francisco, California, 1971.

⁷ R. E. Feeney, *Int. J. Pept. Protein Res.* **29**, 145 (1987).

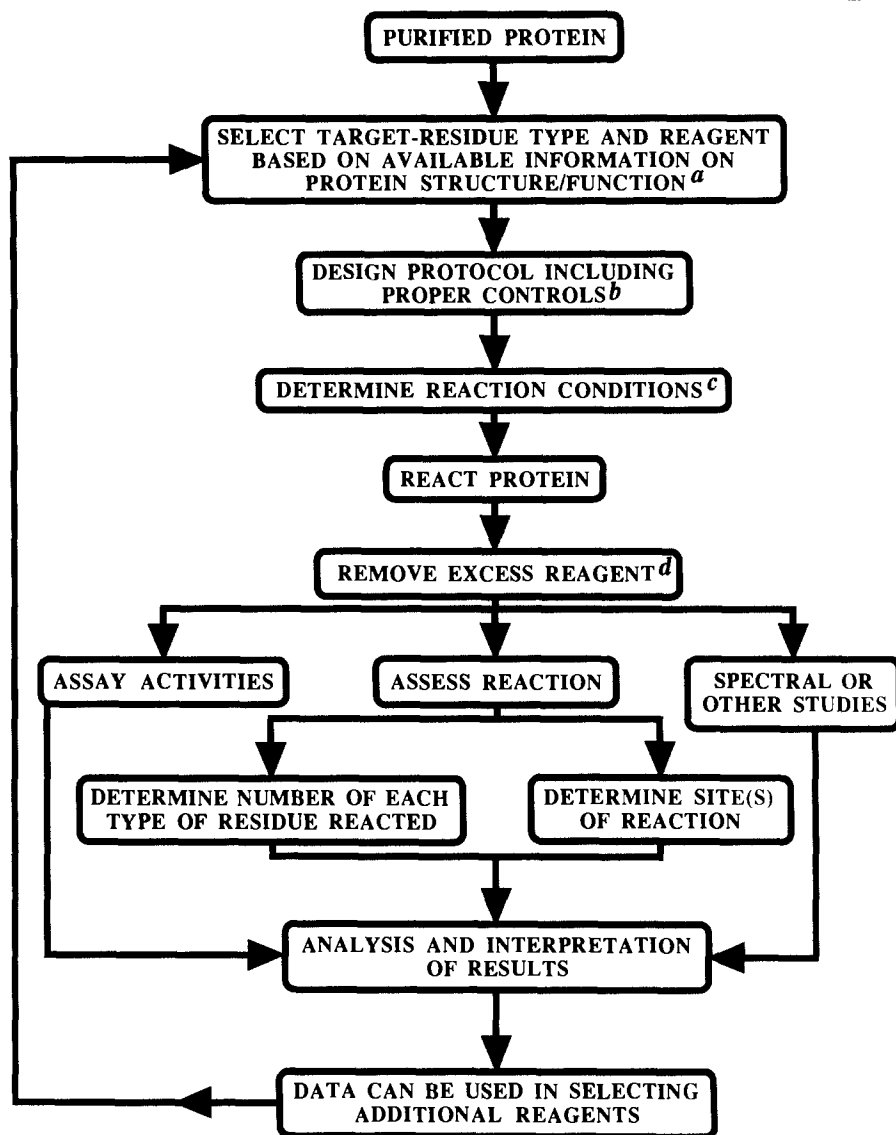


FIG. 1. Flow chart for design of chemical modification experiments. (a) For example, DNA-binding proteins may have lysine or arginine residues in or near the binding site. (b) Controls may include no reagent, preblocked reagent, with/without ligands, etc. (c) Reaction conditions to be considered include buffer, ionic strength, pH, temperature, reaction time, reagent concentration (ratio or absolute), presence or absence of sulfhydryl reagents, metal ions, reaction stop procedure, etc. (d) Excess reagent may be removed by dialysis, chromatography, etc.

reality, however, multiple residues are frequently affected, and the effects of modification of each type must be assessed carefully to interpret the results obtained. By adjusting the reaction conditions (e.g., molar ratio of reagent to protomer, pH, salt concentration, temperature, and time of reaction), it may be possible to elicit greater selectivity for residue type. It should be noted that in some cases it is the ratio between reagent and protein that determines the extent of modification, while in others, the absolute concentration of the reagent is the determining factor. Thus, care in selecting reaction conditions as well as reagent and protein concentrations is essential. Once modification has been achieved, identification of the sites of reaction and assessment of the effects on the structural and functional properties of the protein must be accomplished.

Determination of the effects of modification on activity is specific for each protein under consideration; all the activities must be evaluated (DNA binding, RNA binding, modulator ligand binding, enzymatic activity, oligomer formation, etc.). Where possible, kinetic as well as equilibrium parameters should be measured. Modification of specific amino acid residues can affect the conformation of the protein and thereby exert an indirect effect on binding or enzymatic properties; reaction can also alter residues that are directly involved in these functions. Distinguishing conformational from direct effects can be difficult; detailed analyses of the reaction pattern and functional properties under a variety of experimental conditions may provide sufficient information to make the distinction. In addition, other methods to determine conformational effects (e.g., circular dichroism spectra, proteolytic susceptibility, and NMR studies) can be used to detect alterations in structure consequent to modification. The ability of ligands/substrates to protect residues against modification is frequently a useful method to confirm the participation of specific side chains in the binding of these molecules. However, it is essential to confirm that the ligands (particularly DNA and RNA) do not themselves react with the reagent and do not interfere in nonpassive ways with protein reaction.

The purity of the protein is a consideration in designing chemical modification experiments, since the relative reactivity of different proteins varies, and a contaminant may alter the concentration of reagent. Selection of control conditions is exceptionally important in order to have a standard for comparison; in some cases, exposure to reaction conditions alone can cause alterations in functional capacity of the protein. Thus, it is essential to include protein processed identically, but without reagent, for comparison to the corresponding sample actually exposed to reagent. In some cases, a "stop" compound is added to prevent further reaction (e.g., dithiothreitol to sulfhydryl reagents, lysine to reagents specific for the ϵ -amino group); where this type of protocol is followed, the control sample

includes the "stop" compound added to modifying reagent prior to exposure to the protein. Thoughtful design of the experimental approach and inclusion of multiple controls can allow unambiguous interpretation of results and prevent unnecessary experimental repetition.

Analysis of the modified protein to determine all residues affected by the reaction is essential. In some cases, assessment can be made by spectrophotometric or fluorometric methods, while in others a secondary reaction is required. These will be detailed below for each of the residue types and different reagents. Extent of modification of a residue type provides useful information, but identification of the specific amino acid(s) modified is of greatest utility in interpreting the effects of a reaction on the structure and the functional capacity of a protein. This analysis can be complex and sometimes requires careful experimental design; a few specific examples will be provided in the following sections. The microtechniques now available for amino acid analysis, peptide separation, and peptide sequencing diminish significantly the quantities necessary for this type of study and make the task of determining the site(s) of reaction within the structure more feasible on proteins that are available only in small amounts.

Selectivity of a reagent is a major consideration, both with respect to residue type and to specific reaction at given sites within the protein. Reaction is determined by a combination of the inherent reactivity of the side chain, the influence of the surrounding protein environment on reaction (inhibitory or facilitative), and the conditions selected for the reaction. An excellent example of increased reactivity and specificity is Lys-7 reaction in ribonuclease with reagents that have an acidic moiety that mimicks the phosphate on RNA.⁸ The selectivity of a particular modification can be increased in some cases by conducting the reaction at low molar ratios of reagent to protomer or by limited time exposure of the protein to the reagent. In most cases, experimental assessment is required to determine the optimal reaction conditions for a specific protein with a given reagent. However, some guidelines and examples of previous modification reactions with DNA-binding proteins will be provided below for each reagent.

Two methods that have been widely used to determine the number of a specific side-chain type that are essential for protein function are based on the kinetics of the reaction⁹ and on statistical analysis.^{10,11} For the

⁸ G. E. Means and R. E. Feeney, *J. Biol. Chem.* **246**, 5532 (1971).

⁹ D. E. Koshland, Jr., W. J. Ray, Jr., and M. J. Erwin, *Fed. Proc.* **17**, 1145 (1958).

¹⁰ C.-L. Tsou, *Sci. Sin., Ser. B (Engl. Ed.)* **11**, 1535 (1962).

¹¹ R. B. Yamasaki, A. Vega, and R. E. Feeney, *Anal. Biochem.* **109**, 32 (1980).

kinetic method, the pseudo-first order rate constant for the reaction of a given side-chain type and the associated loss of functional activity are compared to yield data on the number of essential residues. In the statistical approach (or graphical method), the number of essential residues is determined from measurement of the number of residues modified and the remaining functional capacity in samples varying in total extent of modification. Although useful in estimating the number of essential residues, these methods assume reactivity of only one residue type and are compromised by the problem of whether previous modification affects subsequent reactivity. In general, utilization of multiple reagents for a given side chain combined with careful analysis of the protein at low levels of modification provide more interpretable data.

Classification of reagents with respect to the primary amino acid residue modified provides a means to select specific side chains and reagents for a particular study. Based on the reagent, a set of conditions can be identified for initiating an experimental program to determine the effects of modification on structure and function of a DNA-binding protein (Fig. 1 and Tables I–III). It should be noted that selection of both side chain and reagent is important, and previous efforts using specific reagents may be helpful in making this choice. Useful reference material can be found in Refs. 1–7 for selecting reagent(s) as well as the initial set of conditions to be explored and precautions to be exercised. For example, some reagents and products are photolabile or thermolabile or unstable in water; thus, familiarity with the requirements specific to working with a given reagent will prevent expensive and potentially time-consuming errors in experimental design. The sections that follow describe reagents for modification of sulfhydryl residues (cysteine), amino groups (lysine), guanidino groups (arginine), phenols (tyrosine), indoles (tryptophan), and imidazoles (histidine). By far, the most specific reactions can be obtained with sulfhydryl residues, and these will be discussed first and in the most detail.

Cysteine Residues

Cysteine side chains are generally the most reactive in proteins, and a wide variety of reagents are available with reactivities that are influenced significantly by the relative surface exposure and surrounding environment of the residues. For modification of cysteine residues, it is essential that all potentially reactive agents (e.g., dithiothreitol and 2-mercaptoethanol) be removed from the sample and that the cysteine residues be protected from oxidation. The most straightforward way to achieve this is dialysis or gel filtration into the buffer to be utilized for modification. The buffer must be purged of oxygen to prevent sulfhydryl oxidation using either an

inert gas or nitrogen (10–20 min of bubbling gas through the buffer solution from the bottom using a gas dispersion tube is normally sufficient). In evaluating the extent of reaction and effects on functional activities, it is necessary to handle the modified protein in such a manner as to ensure that the product is not exposed to conditions that might reverse the reaction.

There are several important categories of cysteine reagents listed in Table I with examples of application to DNA-binding proteins and an indication of the conditions used in these experiments. A brief summary of the different reactions and more information on specific examples follow.

α -Ketoalkyl Halides

α -Ketoalkyl halides (e.g., iodoacetic acid and iodoacetamide) have been used to modify cysteines in a variety of proteins, and derivatives of these reagents provide an even greater range of modification. T7 RNA polymerase modification with iodoacetamide with incorporation of ~ 1.2 mol/mol protein led to inactivation of the protein.¹² The reagent excess utilized was 10- to 20-fold over protein at pH 7.7 in 20 mM sodium phosphate, 100 mM NaCl, 1 mM EDTA or 100 mM NH_4HCO_3 , 1 mM EDTA. The reaction was stopped by addition of 2-mercaptoethanol, and peptide analysis indicated the sites of reaction. Although the highest reactivity is with cysteine residues and careful control of conditions can ensure selective reaction, these reagents will also react with histidine and lysine moieties, particularly at higher reagent concentrations. Reaction occurs primarily with exposed residues for the parent compounds, but less available sulfhydryls can be modified with derivatives substituted with apolar moieties. The apolar moieties apparently facilitate reaction at hydrophobic sites that are removed from the surface of the protein. For example, in the native tetrameric lactose repressor protein, none of the three cysteines in each monomer is reactive to iodoacetic acid or iodoacetamide, but at least two of these side chains can be modified with 2-bromoacetamido-4-nitrophenol (BNP) and one with at least partial selectivity with *N*-[(iodoacetyl)aminoethyl]-5-naphthylamine 1-sulfonate (IAEDANS).^{13–15}

General reaction conditions include protein at approximately 1–10 μM with reagent in excess at a 10- to 200-fold molar ratio. High pH can facilitate the reaction, and a usual buffer would be Tris-HCl with the pH adjusted to ~ 8 –9. The reaction can be stopped with excess sulfhydryl-

¹² G. C. King, C. T. Martin, T. T. Pham, and J. E. Coleman, *Biochemistry* **25**, 36 (1986).

¹³ C. F. Sams, B. E. Friedman, A. A. Burgum, D. S. Yang, and K. S. Matthews, *J. Biol. Chem.* **252**, 3153 (1977).

¹⁴ J. M. Schneider, C. I. Barrett, and S. S. York, *Biochemistry* **23**, 2221 (1984).

¹⁵ J. A. Gardner and K. S. Matthews, *Biochemistry* **30**, 2707 (1991).

TABLE I
REACTION OF CYSTEINE RESIDUES

Reagent	Protein	[Reagent]	[Protein]	Buffer	Additives	Ref.
α -Ketoalkyl halides Iodoacetamide	T7 RNA poly- merase	10–20 <i>M</i> excess	1–10 μ <i>M</i>	20 mM sodium phosphate, pH 7.7 100 mM NH ₄ HCO ₃ , pH 7.7	100 mM NaCl/1 mM EDTA 1 mM EDTA	<i>a</i> <i>a</i>
	BNP LAEDANS	10–200 <i>M</i> excess ~8 mM	0.2–0.7 mg/ml 1–3 mg/ml	0.1 <i>M</i> Tris-HCl, pH 9.0 0.2 <i>M</i> Tris-HCl, pH 8.0–8.6	1.0 <i>M</i> NaCl 0.5 <i>M</i> KCl/0.1 mM EDTA	<i>b</i> <i>c, d</i>
Mercurials <i>p</i> -CMB	Yeast RNA polymerase I	1–20 μ <i>M</i>	0.04–0.06 mg/ml	50 mM Tris-HCl, pH 8	0.1 <i>M</i> KCl/0.5 mM EDTA/10% glycerol	<i>e</i>
	<i>lac</i> repressor <i>lac</i> repressor	1–5 <i>M</i> excess 0–5 <i>M</i> excess	0.1–0.5 mg/ml 0.5 mg/ml	1.0 <i>M</i> Tris-HCl, pH 9.0 0.2 <i>M</i> Tris-HCl, pH 8.0		<i>f</i> <i>g</i>
Disulfide exchangers DTNB	RecA	0.5–1 mM	6–20 μ <i>M</i>	0.01 <i>M</i> potassium phos- phate, pH variable	1.0 mM EDTA/0.5 <i>M</i> KCl	<i>h, i</i>
	CAP	5–40 μ <i>M</i>	0.2 mg/ml	0.1 <i>M</i> sodium phosphate, pH 8.0	0.1 <i>M</i> NaCl	<i>j</i>
MMTS	Histone H3	78 μ <i>M</i>	50 μ <i>M</i>	2.4 mM sodium phos- phate, pH 5.6	Varied	<i>k</i>
	Glucocorticoid receptor	0.25 mM	Cytosol extract	50 mM HEPES, pH 7.9–8.0		<i>l</i>
DFDS	Glucocorticoid receptor	0.25 mM	Cytosol extract	50 mM HEPES, pH 7.9–8.0		<i>l</i>
	<i>lac</i> repressor Histone H3	1–100 <i>M</i> excess ~3 mg/ml	12.5 μ <i>M</i> ~33 mg/ml	0.1 <i>M</i> Tris-HCl, pH 7.6 10 mM sodium phosphate, pH 7.5		<i>m</i> <i>n</i>
Maleimides <i>N</i> -Ethylmaleimide	<i>lac</i> repressor	1–100 <i>M</i> excess	20–50 μ <i>M</i>	0.24 <i>M</i> potassium phos- phate, pH 7.0	5% glycerol	<i>o</i>
	Integrase pro- tein from λ	5 mM	40 μ g/ml	50 mM Tris-HCl, pH 7.4	0.9 <i>M</i> KCl/0.1 mM EDTA/1 mM mercapto- ethanol	<i>p</i>

RNA polymerase σ	5–20 mM	0.5–0.6 mg/ml	50 mM TEA-sulfate, pH 7	<i>q</i>
Glucocorticoid receptor TFIIIA	5 mM 10 mM	Crude extract, 20 mg/ml	10 mM Tris-HCl, pH 7.5 10 mM HEPES, pH 7.5	<i>r</i> <i>s</i>
			65 mM KCl/5 mM MgCl ₂ /0.5 mM DTT/ 10% glycerol 5% glycerol	<i>o</i>
N-Pyrenemaleimide	1–25 M excess	20 μ M	0.24 M potassium phosphate, pH 7.0	<i>t</i>
<i>o</i> -PDM	1–10 M excess	0.2–1.0 mg/ml	50–100 mM HEPES, pH 8	<i>u</i>
Oxidation				<i>u</i>
NBS	1–5 M excess	10–70 μ M	0.1 M Tris-HCl, pH 7.8 0.24 M potassium phosphate, pH 7.0	

- ^a G. C. King, C. T. Martin, T. T. Pham, and J. E. Coleman, *Biochemistry* **25**, 36 (1986).
^b C. F. Sams, B. E. Friedman, A. A. Burgum, D. S. Yang, and K. S. Matthews, *J. Biol. Chem.* **252**, 3153 (1977).
^c J. M. Schneider, C. I. Barrett, and S. S. York, *Biochemistry* **23**, 2221 (1984).
^d D. E. Kelsey, T. C. Rounds, and S. S. York, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 2649 (1979).
^e P. Bull, U. Wyneken, and P. Valenzuela, *Nucleic Acids Res.* **10**, 5149 (1982).
^f D. S. Yang, A. A. Burgum, and K. S. Matthews, *Biochim. Biophys. Acta* **493**, 24 (1977).
^g A. A. Burgum and K. S. Matthews, *J. Biol. Chem.* **253**, 4279 (1978).
^h S. Kuramitsu, K. Hamaguchi, T. Ogawa, and H. Ogawa, *J. Biochem. (Tokyo)* **90**, 1033 (1981).
ⁱ S. Kuramitsu, K. Hamaguchi, H. Tachibana, T. Horii, T. Ogawa, and H. Ogawa, *Biochemistry* **23**, 2363 (1984).
^j E. Eilen and J. S. Krakow, *J. Mol. Biol.* **114**, 47 (1977).
^k J. Palau and J. R. Daban, *Arch. Biochem. Biophys.* **191**, 82 (1978).
^l J. E. Bodwell, N. J. Holbrook, and A. Munck, *Biochemistry* **23**, 1392 (1984).
^m T. J. Daly, J. S. Olson, and K. S. Matthews, *Biochemistry* **25**, 5468 (1986).
ⁿ E. Wingender and A. Arellano, *Anal. Biochem.* **127**, 351 (1982).
^o R. D. Brown and K. S. Matthews, *J. Biol. Chem.* **254**, 5128 (1979).
^p W. Ross and A. Landy, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 7724 (1982).
^q C. S. Narayanan and J. S. Krakow, *Nucleic Acids Res.* **11**, 2701 (1983).
^r M. Kalimi and K. Love, *J. Biol. Chem.* **255**, 4687 (1980).
^s J. J. Bieker and R. G. Roeder, *J. Biol. Chem.* **259**, 6158 (1984).
^t C. Pampeno and J. S. Krakow, *Biochemistry* **18**, 1519 (1979).
^u S. P. Manly and K. S. Matthews, *J. Biol. Chem.* **254**, 3341 (1979).

containing compounds (e.g., dithiothreitol, 2-mercaptoethanol), and remaining reagent is removed by dialysis or gel filtration. The protein can then be analyzed spectrophotometrically or by peptide mapping techniques. One advantage of many iodoacetamide derivatives is the presence of a spectroscopically detectable moiety, using either standard spectrophotometry in the visible range or fluorometry. These probes can be used for spectral studies and yield information regarding the effects of modification. A number of probes with a variety of fluorophores and chromophores are available (e.g., 2-bromoacetamido-4-nitrophenol, IAEDANS).

Mercurials

Several different types of mercurials are available, including *p*-chloromercuribenzoate (*p*-CMB), fluorescein mercuric acetate, and 2-chloromercuri-4-nitrophenol. Reaction results in altered spectral characteristics of the reagent in most cases and thus can be monitored spectrophotometrically. These modifications are normally executed at pH 8–9 in Tris-HCl buffers with very low molar ratios of reagent, as reaction is nearly stoichiometric.^{16–18} Titration using small aliquots of reagent is an effective means to determine the total number of reactive cysteines in the protein. An advantage of the mercurial reagents and the disulfide exchange reagents to be discussed next is the ability to reverse the reaction using excess sulfhydryl-containing compounds (2-mercaptoethanol or dithiothreitol); while this reversibility is useful in determining the effects of reaction and assuring that no irreversible changes have occurred, it may complicate mapping procedures and require development of specific strategies for identifying sites of reaction.

Disulfide Exchangers

Free sulfhydryls will exchange with disulfide-containing reagents to produce a mixed disulfide at the site of reaction. Several different reagents are available, including 5,5'-dithiobis(2-nitrobenzoic acid) (DTNB), methyl methane thiosulfonate (MMTS), and difluoroscein disulfide. DTNB reaction with sulfhydryl residues results in the release of TNB anion, which has maximum absorption at ~410 nm, so that reaction can be followed spectrophotometrically. Reaction of exposed sulfhydryl residues with disulfide exchange reagents can be stoichiometric, but buried residues or those in an unusual environment may require higher molar ratios of

¹⁶ P. Bull, U. Wyneken, and P. Valenzuela, *Nucleic Acids Res.* **10**, 5149 (1982).

¹⁷ D. S. Yang, A. A. Burgum, and K. S. Matthews, *Biochim. Biophys. Acta* **493**, 24 (1977).

¹⁸ A. A. Burgum and K. S. Matthews, *J. Biol. Chem.* **253**, 4279 (1978).

reagent to achieve reaction. All three cysteine residues in the *Escherichia coli* RecA protein are reactive with DTNB; however, in the presence of ATP or ADP, only one cysteine residue reacts.^{19,20} Binding of single-stranded (ss) DNA to the RecA protein does not affect the reactivity with DTNB.²⁰ For these studies, potassium phosphate buffer was utilized at pH 7.9 at varying salt concentration with 1.0 mM DTNB and 6 μ M protein. With the cAMP-binding protein (CAP) from *E. coli*, it has been observed that two cysteines will react in the absence of cAMP, whereas in its presence, the formation of a disulfide linkage occurs between the two sulfhydryls.²¹ This result was interpreted as indicating the close proximity of two antiparallel β -sheet structures when the CAP protein is in its DNA-binding conformation.

Local environment may be an important factor in reactivity. For example, it has been shown that reaction of DTNB with histones can be facilitated by electrostatic effects,²² but this phenomenon is suppressed by the presence of high salt or guanidine hydrochloride. In eukaryotic systems, DTNB has been used to treat cellular extracts to determine the effects on glucocorticoid receptor complexes; the results indicate that sulfhydryl modification inhibits the binding of activated dexamethasone-receptor complexes to DNA cellulose.²³ The conditions for these modifications were pH 8 with a concentration of 0.25 mM reagent.

MMTS does not contain a chromophoric probe to allow spectrophotometric assessment, but its small size and uncharged -S-Me addition to a free sulfhydryl make it a very effective reagent for assessment of the role of cysteine residues in protein function. Reaction of MMTS with exposed and partially buried sulfhydryl residues appears to be stoichiometric; however, in the lactose repressor protein this reagent can also react at high molar ratios with residues that are involved in a subunit interface and are unavailable to other sulfhydryl reagents.²⁴ Generally, MMTS reaction is carried out at pH 7.5 to 8.0 in Tris-HCl or HEPES buffer; following modification, detection of modified residues requires irreversible reaction with a second reagent to provide a chromophoric signal for unreacted cysteine sites.²⁵ In this fashion, the cysteines that were not modified

¹⁹ S. Kuramitsu, K. Hamaguchi, T. Ogawa, and H. Ogawa, *J. Biochem. (Tokyo)* **90**, 1033 (1981).

²⁰ S. Kuramitsu, K. Hamaguchi, H. Tachibana, T. Horii, T. Ogawa, and H. Ogawa, *Biochemistry* **23**, 2363 (1984).

²¹ E. Eilen and J. S. Krakow, *J. Mol. Biol.* **114**, 47 (1977).

²² J. Palau and J. R. Daban, *Arch. Biochem. Biophys.* **191**, 82 (1978).

²³ J. E. Bodwell, N. J. Holbrook, and A. Munck, *Biochemistry* **23**, 1392 (1984).

²⁴ T. J. Daly, J. S. Olson, and K. S. Matthews, *Biochemistry* **25**, 5468 (1986).

²⁵ J. A. Gardner and K. S. Matthews, *Anal. Biochem.* **167**, 140 (1987).

by MMTS can be identified by conventional peptide mapping methods. Because of its small size, this reagent is very useful for determining the significance of an ionizing sulfhydryl moiety to function, and the effects are readily reversible by addition of sulfhydryl reagents.

Difluorescein disulfide has been utilized to examine cysteine residues in histone H3; this fluorophore allows assessment of conformational changes that influence the fluorescence parameters associated with the probe.²⁶ Reaction occurred at pH 7.5 in a sodium phosphate buffer; in this case, high concentrations of histone were utilized (~33 mg/ml) with ~3 mg/ml reagent. Prior to reaction, a low level of dithiothreitol was utilized to reduce the sulfide linkage between the two cysteines in histone H3; reagent was then added in excess over the combined protein and added dithiothreitol concentrations. This protocol illustrates the ability to modify cysteine residues that are normally linked by disulfide bonds in the protein structure.

Maleimide Derivatives

N-Ethylmaleimide and other *N*-substituted maleimides have been used widely to modify cysteine residues. This reagent is generally cysteine specific and requires high molar ratios for reaction; with the lactose repressor, *N*-substituted maleimide molar ratios of 1- to 100-fold were utilized in a potassium phosphate buffer at pH 7 with protein concentrations of 20 to 50 μ M.²⁷ For *N*-ethylmaleimide, there is no chromophore for assessment of extent of reaction; for other *N*-substituted maleimides (e.g., *N*-pyrenemaleimide), spectrophotometric or fluorometric methods can be used to assess the extent of reaction. Spectral probes introduced by this reaction can be used to assess environmental changes consequent to ligand binding.²⁸ The maleimide reaction is irreversible, a feature that facilitates mapping procedures to identify specific sites of modification. Differential effects on functional properties have been observed in some studies. For example, for the integrase protein (int) from bacteriophage λ , reaction with *N*-ethylmaleimide (40 μ g/ml protein in 0.9 *M* KCl, 50 mM Tris-HCl, pH 7.4, 1 mM 2-mercaptoethanol, 0.1 mM EDTA, 2 mg/ml bovine serum albumin) resulted in selective loss of int interaction with one of the consensus recognition sequences for the protein²⁹; this result was consistent with previous reports that this modification abolished the ability of int to execute recombination and to form a heparin-resistant complex with DNA

²⁶ E. Wingender and A. Arellano, *Anal. Biochem.* **127**, 351 (1982).

²⁷ R. D. Brown and K. S. Matthews, *J. Biol. Chem.* **254**, 5128 (1979).

²⁸ R. D. Brown and K. S. Matthews, *J. Biol. Chem.* **254**, 5135 (1979).

²⁹ W. Ross and A. Landy, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 7724 (1982).

without effect on its topoisomerase activity.³⁰ This type of differential effect on separate activities can be utilized to decipher roles of various subdomains of protein sequence/structure in these functional properties.

RNA polymerase modification with *N*-ethylmaleimide resulted in reaction of three cysteine residues with accompanying loss of σ subunit activity³¹; kinetic analysis of the inactivation data indicated that a single cysteine was essential for activity. Reaction was carried out in 50 mM triethylamine-sulfate at pH 7 with 5–20 mM *N*-ethylmaleimide and 0.6 mg/ml σ subunit. The reaction can also be carried out in crude fractions; for example, the cytosol fraction from rat liver homogenates was modified in 10 mM Tris-HCl, 0.25 M sucrose, pH 7.5 at a total protein concentration of 20 mg/ml and 5 mM reagent concentration.³² The results of this study indicated that *N*-ethylmaleimide treatment yielded a glucocorticoid receptor complex that was frozen in its intact, unactivated form. Similarly, reaction of transcription factor IIIA (TFIIIA) with 10 mM *N*-ethylmaleimide (15 min, 30°) either in the absence or presence of 5S RNA or the 5S RNA gene results in abolition of its ability to promote transcription.³³

An example of using specific chemical modification for cross-linking is the use of *o*-phenylenedimaleimide to cross-link the subunits of the cAMP-binding protein (CAP). The cross-linking reaction is modulated by the presence of cAMP, and the cross-linked protein loses DNA-binding activity.³⁴ A 10-fold molar excess of reagent was sufficient to generate the cross-linked species in HEPES at pH 8; addition of excess dithiothreitol was used to stop the reaction prior to analysis.

Oxidation

Controlled oxidation of cysteine residues can also be of utility in deciphering the role specific sulfhydryls play in the activity of proteins. Although most oxidizing agents affect a broader range of side chains, careful control of reaction conditions can result in specific cysteine modification. *N*-Bromosuccinimide has been used with the lactose repressor protein, and the effects of cysteine and methionine oxidation were distinguished by ligand protection measurements.³⁵ Reaction with *N*-bromosuccinimide can be carried out between pH 7 and 8 in either phosphate or Tris-HCl

³⁰ Y. Kikuchi and H. A. Nash, *Proc. Natl. Acad. Sci. U.S.A.* **76**, 3760 (1979).

³¹ C. S. Narayanan and J. S. Krakow, *Nucleic Acids Res.* **11**, 2701 (1983).

³² M. Kalimi and K. Love, *J. Biol. Chem.* **255**, 4687 (1980).

³³ J. J. Bieker and R. G. Roeder, *J. Biol. Chem.* **259**, 6158 (1984).

³⁴ C. Pampeno and J. S. Krakow, *Biochemistry* **18**, 1519 (1979).

³⁵ S. P. Manly and K. S. Matthews, *J. Biol. Chem.* **254**, 3341 (1979).

buffer; the reagent can be inactivated by the addition of excess dithiothreitol. By varying the molar ratio, it may be possible to control the extent of cysteine oxidation. Other protocols and reagents can be utilized for these types of experiments, and the choice of oxidizing reagents is broad. Care must be exerted to determine all residues affected by the reaction and to assess, where possible, whether the product is sulfenic (-SOH), sulfinic (-SO₂H), or sulfonic (-SO₃H) acid at the cysteine sites. The sulfonic acid derivative is not affected by addition of sulfhydryl reagents, while the sulfenic and sulfinic acid products can revert at least partially to free sulfhydryl under reducing conditions. In addition, disulfide bonds can form under oxidizing conditions.

Lysine Residues

The α - and ϵ -amino groups in a protein are targets for reactions with a variety of reagents with relatively high specificity. A list of reagents with examples of DNA-binding proteins examined and conditions utilized is given in Table II. Care must be taken to evaluate the reaction of cysteine residues in a protein when lysine side chains are the targeted amino acid due to the high reactivity of the sulfhydryl group. The unprotonated form of the amino group is usually the reactive species; thus, elevated pH (i.e., above 8) to generate a significant proportion of the unprotonated species is generally required for reaction to occur at a reasonable rate. It should be noted that many buffer systems can be used for reacting lysine residues, but in general Tris and other compounds which contain a primary or secondary amine should be avoided, as they will react with reagent and lower the concentration available for protein modification. As with other amino acid side chains, the specific reactivity of a residue is influenced by the surrounding environment. In DNA-binding proteins, lysine and arginine residues are involved frequently in nonspecific ionic interactions with the sugar phosphate backbone, and it is important to assess the degree to which changes in specific affinity may be correlated with alterations in the nonspecific binding properties of the modified protein.

Aryl and Sulfonyl Halides

These classes of compounds have been useful in a number of contexts, particularly the introduction of spectroscopic probes into protein structure (e.g., dansyl chloride, fluorodinitrobenzene). Low levels of modification with concomitant introduction of chromophore/fluorophore frequently can be achieved with minimal effects on the functional properties of the protein to generate a species that can be used in a variety of experimental designs

to measure physical and functional parameters. Dansyl chloride modification of the lactose repressor protein resulted in loss of specific DNA binding at low molar ratios of reagent without effect on inducer-binding properties.³⁶ The presence of nonspecific DNA results in protection of two lysine residues and concomitant protection of operator DNA-binding activity. The dansyl group can be used for fluorescence measurements in addition to determining lysine participation in functional activity.

Trinitrobenzene Sulfonate (TNBS)

This reagent reacts with relatively high specificity with amino groups, and the reaction can be followed spectrophotometrically.³⁷ Reaction of the σ subunit of RNA polymerase with TNBS required modification of five lysine residues for complete inactivation; using kinetic analysis, one lysine was implicated as critical for function.³⁸ The modified σ subunit was able to form holoenzyme with binding affinity similar to the unmodified protein; however, the modification compromised the ability to form a tight complex with promoter and to stimulate RNA chain initiation. It is noteworthy that proteolytic digestion analysis indicated a conformational change in the σ subunit following trinitrophenylation; this example illustrates the importance of complete and careful analysis of the reaction product.

Lactose repressor protein modified with TNBS exhibited differential effects on its three binding activities: inducer binding was increased, operator and nonspecific DNA binding were decreased, although to differing degrees.³⁹ The presence of ligands influenced the extent of reaction and the degree of binding perturbation caused by the modification. The trinitrophenyl chromophore is useful for mapping the sites of reaction, as it absorbs in the visible region; this absorbance can also be used to estimate the extent of reaction of TNBS with protein, although ancillary analysis is required to confirm these results.

Pyridoxal Phosphate

This reagent selectively produces a Schiff's base with α - and ϵ -amino groups in proteins; in addition, pyridoxal phosphate (PLP) reaction introduces a chromophore, thereby facilitating analysis, and the reaction is reversible (although reduction can be utilized to generate an irreversible

³⁶ W.-T. Hsieh and K. S. Matthews, *Biochemistry* **24**, 3043 (1985).

³⁷ A. F. S. A. Habeeb, *Anal. Biochem.* **14**, 328 (1966).

³⁸ C. S. Narayanan and J. S. Krakow, *Biochemistry* **24**, 6103 (1982).

³⁹ P. A. Whitson, A. A. Burgum, and K. S. Matthews, *Biochemistry* **23**, 6046 (1984).

TABLE II
REACTION OF LYSINE RESIDUES

Reagent	Protein	[Reagent]	[Protein]	Buffer	Additives	Ref.
Sulfonyl halides and sulfonates						
Dansyl chloride	<i>lac</i> repressor	1–32 <i>M</i> excess	1 mg/ml	0.24 <i>M</i> potassium phosphate, pH 8.4	5% glucose	<i>a</i>
TNBS	RNA polymerase σ	1–10 mM	0.3–1.0 mg/ml	80 mM Tris, pH 9.0	80 mM borate/2.5 mM EDTA	<i>b</i>
	<i>lac</i> repressor	2–16 <i>M</i> excess	0.6–2.0 mg/ml	2.5% triethanolamine, pH 9.0		<i>c</i>
Pyridoxal phosphate	DNA polymerase I	1 mM	1 μ M	50 mM HEPES, pH 8.0	10 mM NaCl/20% glycerol/5 mM Mg(OAc) ₂ /1 mM DTT	<i>d</i>
	DNA polymerase β	0–500 μ M	200 μ g/ml	50 mM HEPES, pH 7.8	1 mM MnCl ₂ /10 mM NaCl/20% glycerol	<i>e</i>
Acid anhydrides	Chromatin	0–0.5 mg/ml	0.19 mg DNA/ml	10 mM HEPES, pH 8.2		<i>f</i>
	Nucleosomes	0.5–0.7 mg/ml	0.14–0.35 mg DNA/ml	10 mM HEPES, pH 8.2	5 mM EDTA 0.1 <i>M</i> PMSF	<i>g</i>
Reductive alkylators	Sodium cyanoborohydride	10 mM	0.5–1.0 mg/ml	20 mM HEPES, pH 7.5		<i>h</i>
	Formaldehyde	10-fold [Lys]				
	Formaldehyde	6 mM				
	Sodium borohydride	~0.4 mg/ml	1.6 mg/ml	45 mM potassium phosphate, pH 8.0	4.5 mM EDTA/10 μ M cAMP/10% glycerol	<i>i</i>

product). Inhibition of modified *E. coli* DNA polymerase I large fragment with PLP required two molecules of PLP, although reduction of the modified enzyme demonstrated incorporation of 3 mol PLP/mol of enzyme.⁴⁰ Interestingly, the presence of dNTPs during modification resulted in a decrease of 1 mol PLP incorporated/mol enzyme with a product that exhibited DNA polymerase activity but was compromised in the elongation function. The site of PLP attachment is Lys-758, and the inactivation is dependent on the presence of divalent metal ions.⁴¹ Similar results have been observed with mammalian DNA polymerase β , although the complexity of the reaction appears greater with this enzyme.⁴² The site protected by dNTPs in DNA polymerase β is Lys-71, presumed to participate in the nucleotide substrate-binding pocket.

Carboxylic Acid Anhydrides

A variety of anhydrides will react with amino groups in proteins at alkaline pH, although modification of tyrosine can complicate interpretation of the results with this type of modification. Charge neutralization or reversal of charge at reacted lysines can be achieved with this reagent class (e.g., acetylation or succinylation). Chromatin dissociation has been obtained using dimethylmaleic anhydride (citraconic anhydride).⁴³ Incubation at pH 6 reverses the reaction, and the proteins released from chromatin by this method are able to reform nucleosome-like structures. Similar results have been obtained with nucleosomal particles and with isolated core-histone octamers; dimethylmaleic anhydride elicits a biphasic dissociation of H2A:H2B dimers, while acetic anhydride does not have this effect.^{44,45} Use of both reagents at higher molar excesses yields dissociation of nucleosomal particles, a result that was interpreted to signify the participation of lysine groups in binding of histones to the DNA. These studies illustrate the specific effects that can be elicited by modification and the necessity for careful planning and exploration of multiple reagents for a specific purpose.

Reductive Alkylation

This reaction has the advantage of preserving the charge characteristic of the lysyl side chain, although side reactions can occur. Either mono- or disubstituted products are observed, depending on the reagent and

⁴⁰ A. K. Hazra, S. Detera-Wadleigh, and S. H. Wilson, *Biochemistry* **23**, 2073 (1984).

⁴¹ A. Basu and M. J. Modak, *Biochemistry* **26**, 1704 (1987).

⁴² A. Basu, P. Kedar, S. H. Wilson, and M. J. Modak, *Biochemistry* **28**, 6305 (1989).

⁴³ E. Palacián, A. López-Rivas, J. A. Pintor-Toro, and F. Hernández, *Mol. Cell. Biochem.* **36**, 163 (1981).

⁴⁴ J. Jordano, M. A. Nieto, and E. Palacián, *J. Biol. Chem.* **260**, 9382 (1985).

⁴⁵ M. A. Nieto and E. Palacián, *Biochemistry* **27**, 5635 (1988).

conditions. Although larger alkyl groups can be introduced, methyl is most commonly used in proteins to minimize perturbation of structure. Formaldehyde is utilized as the source of the methyl group, and sodium cyanoborohydride serves as the reducing agent; the use of the latter is important, as it is stable in aqueous solution at neutral pH, is specific for Schiff's base reduction, and does not affect aldehydes or disulfide bonds. Several authors have noted the necessity of recrystallizing the sodium cyanoborohydride immediately prior to use and problems with interference from other components in the reaction solution (e.g., sulfhydryl reagents, metal ions, ammonium ion).^{46,47} This method has found application in the area of DNA-binding proteins in the introduction of ¹³C or ¹⁴C into the structure of proteins. For example, the lysine residues in gene 5 protein from fd bacteriophage, a ssDNA-binding protein, have been modified using sodium cyanoborohydride (10 mM) and formaldehyde (10-fold lysyl side-chain concentration) in a 20 mM HEPES buffer at pH 7.5.⁴⁸ Of the seven lysyl residues, six react in the free protein, while only three are available in the presence of ssDNA sequences; identification of the three protected residues was possible by using radiolabeled formaldehyde for the reaction, and the results suggested that these lysine side chains may be involved in binding to DNA. Reductive methylation has also been utilized to radiolabel the CAP protein for determination of protein-DNA stoichiometry.⁴⁹ The reaction with CAP protein utilized sodium borohydride as the reductant following reaction with formaldehyde.

Imido Esters

This class of compounds shares the advantage of reductive alkylation in maintaining the charge on the lysine residues. Cross-linking reactions are a specific application of imido ester modification, and a broad variety of reagents, some cleavable, to react amino functions is available (e.g., dimethyl suberimidate). Many types of imido esters have been utilized in protein modification, although relatively minimal application to DNA-binding proteins has occurred. One reason may be marginal effects on the functional properties of DNA-binding proteins due to the maintenance of charge in the reaction product; this situation has been observed using reaction of several imido esters with the lactose repressor.⁵⁰ ssDNA-binding proteins have been examined using ethyl acetimidate, and the

⁴⁶ N. Jentoft and D. G. Dearborn, *J. Biol. Chem.* **254**, 4359 (1979).

⁴⁷ N. Jentoft and D. G. Dearborn, *Anal. Biochem.* **106**, 186 (1980).

⁴⁸ L. R. Dick, A. D. Sherry, M. M. Newkirk, and D. M. Gray, *J. Biol. Chem.* **263**, 18864 (1988).

⁴⁹ M. G. Fried and D. M. Crothers, *Nucleic Acids Res.* **11**, 141 (1983).

⁵⁰ K. S. Matthews and J. Rex, unpublished data (1980).

data indicate that modification of lysines may alter functional properties and that the presence of ssDNA protects specific residues from acetylation.^{51,52} An example of a specialized imido ester reaction is the biotinylation of CAP protein from *E. coli* using sulfosuccinimidyl 6-(biotinamido)hexanoate; the biotin introduced by this reaction was utilized in identifying and separating subunits involved in exchange reactions with unmodified CAP dimer.⁵³ Modification was carried out on a DNA cellulose substrate to protect binding activity in a 20 mM HEPES buffer at pH 8 with 0.6 mM reagent; reaction was stopped with a Tris-HCl buffer at similar pH.

Iminothiolane (an imido thioester) has been used to convert the amino function to a sulfhydryl group in the *trp* repressor from *E. coli* using a phosphate buffer, pH 8, ~5 mM reagent, ~0.1 mM protein.⁵⁴ The resulting sulfhydryl was alkylated by 5-iodoacetamido-1,10-phenanthroline to generate a site-specific endonuclease based on the affinity of the *trp* repressor for its cognate operator sequences. Such combined reactions can be used to manufacture altered proteins with unusual properties and with desired specificities.

Other Reagents

A broad range of additional reagents is utilized for lysine modification, but these have found limited application in the study of DNA-binding proteins. As with all of the amino acids, careful investigation of the large number of available reagents is required to determine the best choice for a particular application.

Arginine

A limited number of reagents is available for arginine modification, in part due to the high pK of the guanidino moiety. Those widely used are phenylglyoxal (and its derivatives), 2,3-butanedione, and 1,2-cyclohexanedione. Examples of reactions with DNA-binding proteins and conditions used are summarized in Table III(A).

Phenylglyoxal

Phenylglyoxal will react with amino groups, particularly the α -amino group, and this side reaction must be assessed in determining the effects of modification. Takahashi⁵⁵ has demonstrated that two phenylglyoxal

⁵¹ A. Tsugita and G. G. Kneale, *Biochem. J.* **228**, 193 (1985).

⁵² S. Bayne and I. Rasched, *Biosci. Rep.* **3**, 469 (1983).

⁵³ A. M. Brown and D. M. Crothers, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 7387 (1989).

⁵⁴ C.-H. B. Chen and D. S. Sigman, *Science* **237**, 1197 (1987).

⁵⁵ K. Takahashi, *J. Biol. Chem.* **243**, 6171 (1968).

molecules react with a single guanidino group. RNA polymerase holoenzyme from *E. coli* has been modified using this reagent, and kinetic analysis of the inactivation data indicates that reaction of a single arginine is responsible for the loss in activity.⁵⁶ Inhibition of DNA polymerases from eukaryotic, prokaryotic, and RNA tumor viruses with phenylglyoxal has been demonstrated; the modification appears to interfere with template binding of the DNA polymerases and suggests arginine interaction with the template as well as a common mechanism for these enzymes.⁵⁷ The presence of template during reaction protected against activity loss. The reaction was carried out in a HEPES buffer at pH 7.8, with phenylglyoxal concentrations up to 0.5 mM. Phenylglyoxal reaction with the lactose repressor protein did not affect inducer binding, but diminished operator and nonspecific DNA binding with modification of one to two equivalents of arginine per monomer.⁵⁸ Protection of activity was observed with reaction in the presence of operator DNA concomitant with diminished reactivity of arginine residues. Reaction was carried out in this study using sodium bicarbonate buffer, pH 8.3, with protein concentration $\sim 25 \mu\text{M}$ and reagent concentrations ranging to 2.5 mM. Bicarbonate/carbonate buffers have been shown to increase the rate of reaction with arginine and to minimize reaction with the α - and ϵ -amino groups.⁵⁹ It is possible to obtain (Amersham, Arlington Heights, IL) or synthesize radiolabeled phenylglyoxal to facilitate determination of the extent of reaction and sites of modification. Several derivatives of phenylglyoxal (*p*-nitro-, *p*-hydroxy-, and *p*-azido-) are available, but utilization of these reagents with DNA-binding proteins has been minimal.

2,3-Butanedione

Borate stabilizes the reaction product between butanedione and arginine, and this buffer should be utilized for experiments using butanedione. Optimal pH and borate concentration for reaction should be determined for each protein. Removal of borate or the presence of excess free arginine can reverse the reaction and regenerate the side chain in the protein. In addition, this reagent can polymerize, and the polymeric species display diminished reactivity relative to the monomer⁶⁰; thus, distillation of the reagent on a routine basis is requisite as well as storage in the dark to prevent photoreactions.^{60,61} Side reactions have been observed with

⁵⁶ V. W. Armstrong, H. Sternbach, and F. Eckstein, *FEBS Lett.* **70**, 48 (1976).

⁵⁷ A. Srivastava and M. J. Modak, *J. Biol. Chem.* **255**, 917 (1980).

⁵⁸ P. A. Whitson and K. S. Matthews, *Biochemistry* **26**, 6502 (1987).

⁵⁹ S.-T. Cheung and M. L. Fonda, *Biochem. Biophys. Res. Commun.* **90**, 940 (1979).

⁶⁰ J. F. Riordan, *Biochemistry* **12**, 3915 (1973).

⁶¹ D. Petz, H.-G. Löffler, and F. Schneider, *Z. Naturforsch. C: Biosci.* **34**, 742 (1979).

TABLE III
REACTION OF OTHER RESIDUES

Reagent	Protein	[Reagent]	[Protein]	Buffer	Additives	Ref.
Phenylglyoxal	RNA polymerase holoenzyme	0.36 mM	0.45 mg/ml	100 mM bicine, pH 8.0	10 mM MgCl ₂	a
	DNA and RNA polymerases	Up to 0.5 mM		50–80 mM HEPES, pH 7.8	Variable	b
	<i>lac</i> repressor	0–2.5 mM	2.7×10^{-5} M	0.1 M NaHCO ₃ , pH 8.3		c
2,3-Butanedione	<i>lac</i> repressor	0–25 mM	2.4×10^{-5} M	50 mM borate, pH 8.0		c
	<i>Bam</i> HI methylase	0–40 mM	6 units	20–50 mM borate, pH 8.0		d
	<i>Bam</i> HI endonuclease	0–30 mM	50 units	0–200 mM borate, pH 8.0		e
	BS-NS	10 mM	0.77 mg/ml	100 mM HEPES, pH 8.0	50 mM borate	f
Diethyl pyrocar-bonate	RNA polymerase	0.11–1.0 mM	1–4 μ M	0.1 M phosphate, pH 6.0		g
	<i>lac</i> repressor	0–3 mM	0.5 mg/ml	0.24 M potassium phosphate, pH 7.5	5% glycerol	h
	fd gene 5 protein	64 M excess	0.8 mM	0.05 M Tris-HCl, pH 8	0.15 M NaCl	i
TNM	<i>lac</i> repressor	2–50 M excess	0.1–1.0 mg/ml	0.1 M Tris-HCl, pH 8.0	5% glucose	j
	<i>lac</i> repressor	2–50 M excess	0.1–1.0 mg/ml	0.24 M potassium phosphate, pH 8.0		j
	Topoisomerase I and II	1–3 mM	10–35 μ g/ml	50 mM Tris-HCl, pH 8.0		k

I ₂ /KI	<i>lac</i> repressor	0–60 M excess I ₂ 0–240 M excess KI	0.6–0.8 mg/ml	0.04 M Tris-HCl, pH 7.5	0.01 M Mg(OAc) ₂ /0.2 M KCl/5% glycerol	<i>l</i>
D. Reaction of tryptophan residues						
NBS	<i>lac</i> repressor	1–20 M excess	25–100 μM	1.0 M Tris-HCl, pH 7.8		<i>m</i>
	<i>lac</i> repressor	1–20 M excess	25–100 μM	0.2 M potassium phosphate, pH 7.6		<i>m</i>
Oxidation	RNA polymerase	25–30 M excess		pH 7.9		<i>n</i>
	T4 gene 32 protein	0.1 M trichloroethanol/UV irradiation	~1–10 μM	10 mM cacodylate, pH 7.6	10 mM NaCl/0.2 mM EDTA	<i>o</i>
UV irradiation	<i>lac</i> repressor	UV irradiation	0.15 mg/ml	0.1% NH ₄ HCO ₃ , pH 8.7		<i>p</i>

^a V. W. Armstrong, H. Sternbach, and F. Eckstein, *FEBS Lett.* **70**, 48 (1976).

^b A. Srivastava and M. J. Modak, *J. Biol. Chem.* **255**, 917 (1980).

^c P. A. Whitson and K. S. Matthews, *Biochemistry* **26**, 6502 (1987).

^d G. Nardone, J. George, and J. G. Chirikjian, *J. Biol. Chem.* **259**, 10357 (1984).

^e J. George, G. Nardone, and J. G. Chirikjian, *J. Biol. Chem.* **260**, 14387 (1985).

^f M. Lammi, M. Paci, and C. O. Gualerzi, *FEBS Lett.* **170**, 99 (1984).

^g A. W. Abdulwajid and F. Y.-H. Wu, *Biochemistry* **25**, 8167 (1986).

^h C. F. Sams and K. S. Matthews, *Biochemistry* **27**, 2277 (1988).

ⁱ R. A. Anderson, Y. Nakashima, and J. E. Coleman, *Biochemistry* **14**, 907 (1975).

^j W.-T. Hsieh and K. S. Matthews, *J. Biol. Chem.* **256**, 4856 (1981).

^k L. Klevan and Y.-C. Tse, *Biochim. Biophys. Acta* **745**, 175 (1983).

^l T. G. Fanning, *Biochemistry* **14**, 2512 (1975).

^m R. B. O'Gorman and K. S. Matthews, *J. Biol. Chem.* **252**, 3565 (1977).

ⁿ B. Wasyluk and A. D. B. Malcolin, *Biochem. Soc. Trans.* **3**, 654 (1975).

^o J.-J. Toulmé, T. LeDoan, and C. Hélène, *Biochemistry* **23**, 1195 (1984).

^p M. Spodheim-Maurizot, M. Charlier, and C. Hélène, *Photochem. Photobiol.* **42**, 353 (1985).

histidine.⁶¹ Reaction of 2,3-butanedione with the lactose repressor protein resulted in loss of arginine with no evident modification of lysine residues.⁵⁸ Reaction of one to two arginine residues was sufficient to inhibit operator DNA binding, but no effect was observed on inducer binding; these data are consistent with the results obtained using phenylglyoxal to modify the arginines in the lactose repressor protein.

*Bam*HI methylase is inhibited by treatment with 2,3-butanedione with protection against inactivation by the presence of DNA but not by *S*-adenosylmethionine. Borate at 50 mM was used in these studies with 20–40 mM butanedione; the extent of enzyme inactivation was maximal at 50 mM borate.⁶² In like manner, *Bam*HI endonuclease is inhibited by reaction with 2,3-butanedione, with the dinucleotide pdGpdG providing maximum protection for the enzyme against inactivation by this reagent; however, in this case the optimum concentration of borate was 100 mM.⁶³ The effects of arginine modification by 2,3-butanedione on DNA binding of the prokaryotic DNA-scaffolding protein BS-NS from *Bacillus stearothermophilis* has indicated that at least one arginine residue in this protein may be required for binding to DNA.⁶⁴ The buffer used in this study was 100 mM HEPES, pH 8.0, 50 mM borate with 10 mM 2,3-butanedione. It should be emphasized again that the presence of borate stabilizes the adduct formed with arginine and 2,3-butanedione,⁶⁵ and the optimal concentration for each system must be determined.

1,2-Cyclohexanedione has also been used to modify arginine, and an adduct is stabilized by borate; however, there are side reactions that are irreversible as well. This and other complications have presumably prevented widespread use of this reagent with DNA-binding proteins.

Histidine

Although a number of reagents will modify the imidazole moiety of histidine, specificity of reaction is obtained with only a few of these compounds. When attempting to modify histidine residues, particular care must be exercised to assess reactions with all other potential targets of reaction.

⁶² G. Nardone, J. George, and J. G. Chirikjian, *J. Biol. Chem.* **259**, 10357 (1984).

⁶³ J. George, G. Nardone, and J. G. Chirikjian, *J. Biol. Chem.* **260**, 14387 (1985).

⁶⁴ M. Lammi, M. Paci, and C. O. Gualerzi, *FEBS Lett.* **170**, 99 (1984).

⁶⁵ J. F. Riordan, *Mol. Cell. Biochem.* **26**, 71 (1979).

General Reagents

Early studies of DNase II from porcine spleen utilized iodoacetamide reaction to demonstrate alkylation at a single histidine to inactivate the protein.⁶⁶ Substrate DNA partially protected the enzyme against this inactivation using 0.2 M sodium acetate buffer, pH 4.6, with up to 0.2 mM iodoacetate. Photooxidation was another method used to target histidine, but difficulties with specificity have precluded widespread application with DNA-binding proteins. While a number of reagents have been applied to other proteins, none has achieved the specificity observed with diethyl pyrocarbonate [Table III(B)].

Diethyl Pyrocarbonate

Diethyl pyrocarbonate (ethoxyformic anhydride) reacts with high specificity with histidine in the pH range from 5.5 to 7.5, although side reactions with lysine and tyrosine, and in some cases cysteine, must be monitored carefully. Diethyl pyrocarbonate reaction with histidines is a function of the concentration of the reagent rather than the molar ratio to the protein in solution. The reagent itself is sensitive to hydrolysis in aqueous solutions, with a half-life of approximately 10 min at pH 7 and shorter times at higher pH.⁶⁷ Stock solutions of reagent can be prepared in anhydrous ethanol and standardized by reaction with imidazole following the reaction spectrophotometrically. Monosubstitution of the imidazole ring can be reversed at alkaline or acidic pH or at neutral pH with hydroxylamine, while dicarbethoxylated imidazole is stable. Reaction is monitored by the increase in absorbance at 240 nm ($\Delta\epsilon \sim 3200 M^{-1} \text{ cm}^{-1}$),⁶⁸ and reaction with tyrosine can be spectrophotometrically determined by a decrease in absorbance at 280 nm.

RNA polymerase undergoes rapid inactivation on exposure to diethyl pyrocarbonate with formation of carbethoxyhistidine and no tyrosine, lysine, or cysteine reaction.⁶⁹ A large number of histidines were modified (six to nine), although substrate ATP (but not template DNA) provided protection against inactivation. Using the statistical method of analysis, it was determined that a single histidine was critical for enzyme activity with a reaction rate seven times greater than other histidines. Phosphate buffer, pH 6, was utilized for the reaction with 0.1 to 1.0 mM reagent. Reaction

⁶⁶ R. G. Oshima and P. A. Price, *J. Biol. Chem.* **248**, 7522 (1973).

⁶⁷ E. W. Miles, this series, Vol. 47, p. 431.

⁶⁸ J. Ovadi, S. Libor, and P. Elodi, *Acta Biochim. Biophys. Acad. Sci. Hung.* **2**, 455 (1967).

⁶⁹ A. W. Abdulwajid and F. Y.-H. Wu, *Biochemistry* **25**, 8167 (1986).

of lactose repressor protein with diethyl pyrocarbonate resulted in diminished inducer, operator, and nonspecific DNA binding.⁷⁰ A maximum of three histidines per subunit were modified along with a single lysine residue; the loss of DNA binding was correlated with histidine modification, and the binding could be restored by the addition of hydroxylamine. The presence of inducer sugars had no effect on histidine modification or loss of DNA-binding activity; however, inducer protected against lysine reaction and prevented loss of inducer binding. An interesting aspect of this study was the absence of evidence for a difference in incorporation of radiolabel with/without inducer present during reaction; diethyl pyrocarbonate can apparently function as a catalyst to cross-linking between carboxylate and amino functions, and the reagent may catalyze internal linkages as well as cross-links between molecules (this type of reaction may occur more readily at lower pHs).⁶⁷

Tyrosine

The modification of the phenolic side chain can be achieved with a measure of specificity using several different types of reagents [Table III(C)]. Reaction in some cases alters the UV/visible absorption properties of the tyrosine and results in a chromophore useful for spectral studies.

Tetranitromethane (TNM)

Nitration of tyrosine in proteins significantly alters the pK of the phenolic group, and reduction to aminotyrosine further alters the pK . Spectral properties are also influenced by this reaction, with nitrotyrosine absorbing in the visible range (~ 380 – 430 nm). Alkaline pH favors tyrosine nitration, and it is essential to monitor oxidation of sulfhydryl groups under these conditions. Although rare, histidine, methionine, and tryptophan modification have been observed on occasion. Another concern with tetranitromethane modification is cross-linking between tyrosine residues (both intra- and intermolecular cross-links have been observed); this reaction tends to occur more readily at low pH. For this reason, low concentrations of protein are desirable to minimize the possibility of intermolecular reactions. Determination of cross-linking is required to interpret the effects of tetranitromethane reaction on functional activities. The stability of nitrotyrosine to hydrolysis enables ready assessment of the extent of reaction by amino acid analysis of the product. Reduction of nitrotyrosine to aminotyrosine can be accomplished using $Na_2S_2O_4$ at slightly elevated

⁷⁰ C. F. Sams and K. S. Matthews, *Biochemistry* **27**, 2277 (1988).

pH. Tertiary reactions can be targeted at the *p*-amino group. Tetranitromethane is soluble in apolar solvents, and reaction cannot be assumed to relate to solvent exposure, as "buried" groups may react due to concentration of the reagent in an apolar region of the molecule.

Gene 5 protein of fd phage has been reacted with tetranitromethane with modification of three of the five tyrosines in the protein.⁷¹ Reaction results in ~100-fold reduced binding affinity for fd DNA. The presence of fd DNA during the reaction protects all tyrosyl residues from nitration. Similarly, nitration of gene 32 protein from bacteriophage T4 results in modification of five of the eight tyrosines with complete loss of DNA-binding activity and protection from nitration in the presence of DNA.⁷² The effects of reaction on cysteine residues were not determined in these studies.

In studies of the reaction of the lactose repressor protein, the effects of nitration were obscured by oxidation of cysteine residues.⁷³ Reaction of cysteine residues with *N*-ethylmaleimide (which does not affect the functional properties of the lactose repressor) followed by tetranitromethane treatment resulted in loss of DNA-binding activities with modification of primarily two tyrosine residues.⁷⁴ Cross-linking observed in this system was minimized by reducing the concentration of the protein. Reduction of the nitrotyrosine to aminotyrosine with Na₂S₂O₄ restored nonspecific DNA binding and partially restored operator DNA binding. The presence of nonspecific DNA during the modification with tetranitromethane reduced nitration and protected DNA-binding capabilities. Topoisomerases I and II from *Micrococcus luteus* are inactivated by treatment with tetranitromethane, and activity is protected by the presence of DNA in the reaction mixture.⁷⁵ Unfortunately, this study did not determine effects on other amino acid side chains.

Iodination

A variety of methods are available for iodination of proteins, although side reactions may occur due to the oxidizing capacity of the reagents. This reaction is particularly useful for the introduction of radiolabel at specific sites in proteins. Iodine/iodide solutions at alkaline pH are non-

⁷¹ R. A. Anderson, Y. Nakashima, and J. E. Coleman, *Biochemistry* **14**, 907 (1975).

⁷² J. E. Coleman, K. R. Williams, G. C. King, R. V. Prigodich, Y. Shamoo, and W. H. Konigsberg, *J. Cell. Biochem.* **32**, 305 (1986).

⁷³ M. E. Alexander, A. A. Burgum, R. A. Noall, M. D. Shaw, and K. S. Matthews, *Biochim. Biophys. Acta* **493**, 367 (1977).

⁷⁴ W.-T. Hsieh and K. S. Matthews, *J. Biol. Chem.* **256**, 4856 (1981).

⁷⁵ L. Klevan and Y.-C. Tse, *Biochim. Biophys. Acta* **745**, 175 (1983).

mally utilized; it is important to remove sulfhydryl reagents from the buffer prior to reaction. It is also possible to iodinate using iodine monochloride (ICl) at slightly alkaline pH or by peroxidase/hydrogen peroxide/sodium iodide. Cross-linking has been observed in iodination reactions, and this by-product should be monitored carefully in the analysis of the reaction product. The lactose repressor protein has been iodinated with I_2/KI in Tris-HCl at pH 7.5; this reaction resulted in loss of operator-binding activity with minimal effect on inducer binding.⁷⁶ The presence of operator DNA during reaction provided partial protection of binding activity and diminished modification; it should be noted that in this case oxidation of cysteine residues was observed, an occurrence that complicates interpretation of the effects of tyrosine iodination on binding activity.

Tryptophan

Tryptophan is sensitive to oxidation by a variety of agents, including *N*-bromosuccinimide (NBS), UV irradiation, and photosensitized oxidation [Table III(D)]. Unfortunately, most of these methods are not specific, and other amino acids are affected. Thus, it is essential to determine the effect of modification on residues other than tryptophan (e.g., cysteine, methionine, tyrosine). Assessment of tryptophan modification is complicated by the instability of this amino acid to many protocols for acid hydrolysis preparatory to amino acid analysis.

N-Bromosuccinimide

Modification with NBS requires recrystallization of the reagent from water immediately prior to each use, and halide ions should be avoided in the buffer system. Low molar ratios of reagent to tryptophan are sufficient to achieve modification, and greater specificity for tryptophan is observed at acid pH values. Reaction can be followed by observing the decrease in absorbance at 280 nm; although the extent of modification can be estimated spectrophotometrically, the decrease in absorbance can undergo reversal in the presence of excess reagent and other products of the reaction. Thus, it is essential to evaluate the extent of tryptophan oxidation by alternative means, e.g., amino acid analysis of protein hydrolyzed under conditions to preserve tryptophan. The reaction of NBS with lactose repressor protein illustrates the problems encountered at neutral to basic pH (required for stability of this protein); cysteine is as reactive as tryptophan, and methionine and tyrosine modification was also observed at molar ratios up to 20-

⁷⁶ T. G. Fanning, *Biochemistry* **14**, 2512 (1975).

fold.⁷⁷ Absorption and fluorescence spectra of modified protein can provide information on the contribution of specific tryptophans to the spectral properties of the parent protein.⁷⁸ Similarly, RNA polymerase is completely inactivated by modification with NBS (25-fold molar ratio, pH 7.9), even when the sulfhydryl residues in the protein are protected by DTNB.⁷⁹ Spectral measurements indicated oxidation of approximately two tryptophan residues, and the affinity for ATP was decreased fourfold by the reaction. However, protection against inactivation by NBS was not observed in the presence of DNA/ATP/GTP. Circular dichroism spectra indicated no differences between modified and unmodified RNA polymerase. These two studies indicate the significant problems in achieving specificity for this reaction, and any efforts to modify tryptophan must include careful attention to potential side reactions.

Oxidation

Direct photooxidation of tryptophan, sensitized photochemical reaction of this side chain, and selective free-radical oxidation have all been utilized to examine the role of tryptophan in DNA-binding proteins. Trichloroethanol has been used as a sensitizing agent for oxidizing gene 32 protein from phage T4 (gp32); a decrease in absorbance at 280 nm was observed, with an accompanying increase at wavelengths less than 260 nm and greater than 290 nm.⁸⁰ Only one type of photoproduct appears to be produced in this reaction.⁸¹ Three tryptophan residues were modified, and the two remaining intact tryptophan residues exhibit minimal fluorescence emission (presumably due to energy transfer to the photoproduct). Irradiation in the presence of trichloroethanol resulted in diminished affinity for DNA; protection of tryptophan by the presence of DNA was not investigated in this study. Similar experiments on gp32 have been executed using oxidation by selective free-radical anions at slightly acidic pH. The radical anions I_2^- , Br_2^- , and SCN_2^- were produced by steady state γ radiolysis.⁸² Low irradiation doses were sufficient to decrease binding to DNA; in addition to tryptophan oxidation, cysteine was also affected by this reaction. As cysteine oxidation also partially inhibits DNA-binding activity, differentiation of the effect of modification of sulfhydryl and

⁷⁷ R. B. O'Gorman and K. S. Matthews, *J. Biol. Chem.* **252**, 3565 (1977).

⁷⁸ R. B. O'Gorman and K. S. Matthews, *J. Biol. Chem.* **252**, 3572 (1977).

⁷⁹ B. Wasylyk and A. D. B. Malcolm, *Biochem. Soc. Trans.* **3**, 654 (1975).

⁸⁰ J.-J. Toulmé, in "Progress in Tryptophan and Serotonin Research," p. 853. de Gruyter, Berlin, 1984.

⁸¹ J.-J. Toulmé, T. LeDoan, and C. Hélène, *Biochemistry* **23**, 1195 (1984).

⁸² J. R. Casas-Finet, J.-J. Toulmé, C. Cazenave, and R. Santus, *Biochemistry* **23**, 1208 (1984).

indole groups is complex. However, the presence of ssDNA protects the tryptophans and one of the sulfhydryl residues from oxidation, consistent with the interpretation that the tryptophans are required for gp32 activity.

Ultraviolet irradiation of the lactose repressor protein (without sensitizer) results in photooxidation of 1 Eq of the two tryptophan residues per monomer with the formation of *N*-formylkynurenine.⁸³ The formation of this photoproduct results in quenching of fluorescence of the remaining tryptophan by energy transfer and precludes photoreaction at the second site. The photooxidation occurs with equal frequency at each of the two tryptophan residues without ligand present, but in the presence of inducer, one tryptophan is protected while the other is slightly more vulnerable to oxidation.

Conclusion

Chemical modification is an effective tool for evaluating the participation of specific types of side chains in the functional properties of a DNA-binding protein. Examples of a variety of reactions have been presented to provide an overview of the range of reagents and their specificity. Careful selection of reagent and reaction conditions as well as controls utilized is essential for interpretation of results. In addition, it is imperative that assessment of reaction products include not only the targeted amino acid based on apparent reagent specificity, but also all potential by-products of reactions with other amino acids. Finally, the effects of reaction on all activities and on the structural integrity of the protein must be determined; coupled with this measurement, data on ligand protection from reaction may be useful in interpreting the role of specific side chains in the functional properties of the protein. Application of this technique without thorough preparation and careful experimental design can result in misinterpretation of results that in turn misleads other investigators. Despite the limits inherent in chemical modification of proteins and the pitfalls of crafting the appropriate protocol for a given protein, this method coupled with other approaches, genetic studies and physical measurements in particular, has yielded significant information about the contribution of specific amino acids to the structure and function of DNA-binding proteins and has a place in the experimental repertoire applied to investigations of this class of proteins.

Acknowledgments

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⁸³ M. Spodheim-Maurizot, M. Charlier, and C. Hélène, *Photochem. Photobiol.* **42**, 353 (1985).

[24] Characterization of Protein–DNA Complexes by Affinity Cleaving

By PETER B. DERVAN

High-resolution crystallographic views of protein–DNA complexes reveal the structural complexity of protein–DNA interactions.^{1–5} The combination of direct protein–DNA contacts mediated by multiple hydrogen bonds and sequence-dependent DNA conformational effects limits our ability to make detailed structural predictions, even if a new DNA-binding protein can be assigned to a structural class such as helix–turn–helix,^{1–4} double-barreled helix,⁵ zinc-binding finger,⁶ or scissor grip-leucine zipper.⁷ In the absence of high-resolution crystallographic and nuclear magnetic resonance (NMR) data, solution methods such as affinity cleaving can be used to characterize the topology of protein–DNA complexes and correlate sequence similarities with known structural classes.

The conversion of a sequence-specific DNA-binding protein into a sequence-specific DNA-cleaving protein by covalent attachment of the iron chelator, ethylenediaminetetraacetic acid (EDTA), to a specific amino acid residue creates a class of hybrid affinity-cleaving proteins that are available through chemical synthesis. Moreover, a structural domain consisting of naturally occurring amino acids that binds transition metals and oxidatively cleaves DNA extends this method to recombinant methods for protein synthesis.

¹ A. K. Aggarwal, D. W. Rodgers, M. Drottar, M. Ptashne, and S. C. Harrison, *Science* **242**, 899 (1988).

² J. E. Anderson, M. Ptashne, and S. C. Harrison, *Science* **326**, 846 (1987).

³ Z. Otwinowski, R. W. Schevitz, R. G. Zhang, C. L. Lawson, A. Joahimiak, R. Q. Marmorstein, B. F. Luisi, and P. B. Sigler, *Nature (London)* **335**, 321 (1988).

⁴ S. R. Jordan and C. O. Pabo, *Science* **242**, 893 (1988).

⁵ J. A. McClarin, C. A. Frederick, B. C. Wang, P. Greene, H. W. Boyer, J. Grable, and J. M. Rosenberg, *Science* **234**, 1526 (1986).

⁶ J. Miller, A. D. McLachlan, and A. Klug, *EMBO J.* **4**, 1609 (1985); J. M. Berg, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 99 (1988); G. Párraga, S. J. Horvath, A. Eisen, W. E. Taylor, L. E. Hood, E. T. Young, and R. E. Klevit, *Science* **241**, 1489 (1988); M. S. Lea, G. P. Gippert, R. V. Saman, D. A. Case, and P. Wright, *Science* **245**, 635 (1989).

⁷ W. H. Landschulz, P. F. Johnson, and S. L. McKnight, *Science* **240**, 1759 (1988); C. R. Vinson, P. B. Sigler, and S. L. McKnight, *Science* **246**, 911 (1989).

Affinity Cleaving

Attachment of EDTA · Fe to a DNA-binding moiety creates a DNA-cleaving molecule that functions under physiologically relevant pH, temperature, and salt conditions.⁸ The cleavage reaction can be initiated by addition of a reducing agent such as dithiothreitol or sodium ascorbate.⁸ If the DNA-binding molecule is sequence specific, the EDTA · Fe cleaves at highly localized sites on DNA restriction fragments and plasmids.⁹⁻¹⁶ Because the EDTA · Fe-cleaving moiety is not sequence specific, the cleavage specificity is determined only by the binding specificity of the molecule being investigated. EDTA · Fe-equipped DNA-binding molecules cleave DNA by oxidation of the deoxyribose backbone via a diffusible oxidant, presumably hydroxyl radical.⁹⁻¹⁶ Cleavage of both DNA strands is observed and typically extends over 4 to 6 bp.⁹⁻¹⁶ Due to the right-handed nature of double-helical DNA, the groove in which the EDTA · Fe is located can be identified by cleavage pattern analysis.

Cleavage Pattern Analysis

Affinity cleaving has been used to study the sequence-specific recognition of double-helical DNA by naturally occurring DNA-binding antibiotics^{9,10}, peptide analogs that bind in the minor groove,¹¹ and oligonucleotide-

⁸ R. P. Hertzberg and P. B. Dervan, *J. Am. Chem. Soc.* **104**, 313 (1982); R. P. Hertzberg and P. B. Dervan, *Biochemistry* **23**, 3934 (1984).

⁹ P. G. Schultz, J. S. Taylor, and P. B. Dervan, *J. Am. Chem. Soc.* **104**, 6861 (1982); J. S. Taylor, P. G. Schultz, and P. B. Dervan, *Tetrahedron* **40**, 457 (1984); P. G. Schultz and P. B. Dervan, *J. Am. Chem. Soc.* **105**, 7748 (1983).

¹⁰ P. B. Dervan, *Science* **232**, 464 (1986).

¹¹ R. S. Youngquist and P. B. Dervan, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 2565 (1985); R. S. Youngquist and P. B. Dervan, *J. Am. Chem. Soc.* **107**, 5528 (1985); J. H. Griffin and P. B. Dervan, *J. Am. Chem. Soc.* **108**, 5008 (1986); J. P. Sluka and P. B. Dervan, in "New Synthetic Methodology and Functionally Interesting Compounds, Proceedings of the 3rd International Kyoto Conference on New Aspects of Organic Chemistry" (Z. I. Yoshida, ed.), Vol. 25, p. 307. Elsevier, New York, 1986; R. S. Youngquist and P. B. Dervan, *J. Am. Chem. Soc.* **109**, 7564 (1987); W. S. Wade and P. B. Dervan, *J. Am. Chem. Soc.* **109**, 1574 (1987); J. H. Griffin and P. B. Dervan, *J. Am. Chem. Soc.* **109**, 6840 (1987).

¹² H. E. Moser and P. B. Dervan, *Science* **238**, 645 (1987); L. C. Griffin and P. B. Dervan, *Science* **245**, 967 (1989).

¹³ J. P. Sluka, S. J. Horvath, M. F. Bruist, M. I. Simon, and P. B. Dervan, *Science* **238**, 1129 (1987); J. P. Sluka, S. J. Horvath, A. C. Glasgow, M. I. Simon, and P. B. Dervan, *Biochemistry* **29**, 6551 (1990); D. P. Mack, J. A. Shin, J. H. Griffin, M. I. Simon, and P. B. Dervan, *Biochemistry* **29**, 6561 (1990).

¹⁴ M. G. Oakley and P. B. Dervan, *Science* **248**, 847 (1990).

¹⁵ K. Graham and P. B. Dervan, *J. Biol. Chem.* **265**, 16534 (1990).

¹⁶ J. P. Sluka, J. H. Griffin, D. P. Mack, and P. B. Dervan, *J. Am. Chem. Soc.* **112**, 6369 (1990).

directed recognition of the major groove by triple-strand formation.¹² From these studies, which involve less complicated DNA-binding motifs, numerous cleavage patterns caused by a diffusible oxidant produced in either the minor or the major groove of duplex DNA have been analyzed. EDTA · Fe located in the minor groove generates an asymmetric cleavage pattern with maximal cleavage loci shifted to the 3' side on opposite strands. When the EDTA · Fe is located in the major groove, the maximal cleavage loci are 5' shifted; in addition, cleavage of lower efficiency occurs on the distal strands of the adjacent minor grooves.¹³ This results in an overall pattern of a *pair* of 3'-shifted asymmetric cleavage loci of *unequal* intensity on opposite strands. These patterns can be explained if the diffusible radical generated from the localized EDTA · Fe reacts with the major and minor grooves of DNA with unequal rates and preferentially (although not necessarily exclusively) in the minor groove.¹⁷ This database allows interpretation of the affinity-cleaving results from conformationally more complex proteins (Fig. 1).

Characterization of Protein–DNA Structures by Affinity Cleaving

Incorporation of EDTA · Fe at discrete amino acid residues within a protein allows the position and groove location of the modified residues to be mapped to nucleotide resolution.^{13–16} The secondary and tertiary structures of DNA–protein complexes can be analyzed by affinity cleaving using two approaches. First, the amino acid location of the EDTA on a polypeptide chain of constant length may be varied in order to reveal key topological features of the protein–DNA complex (e.g., location of NH₂ versus COOH terminus).^{13–15} Moreover, the length of the polypeptide may be incrementally changed while the EDTA is kept at the same terminus to examine the influence of substructures on the binding affinity of the protein and the conformational flexibility of a specific peptide fragment.^{13,14}

Sequence Specificity Mapping

Protein–EDTA · Fe can cause sequence-specific double-strand breaks on DNA.¹³ The locations of site-specific protein binding on large DNA such as plasmids can be mapped by nondenaturing agarose gel electrophoresis. This provides information about DNA sites of similar sequence that

¹⁷ T. D. Tullius and B. A. Dombrowski, *Science* **230**, 679 (1985); T. D. Tullius and B. A. Dombrowski, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 5469 (1986); T. D. Tullius, B. A. Dombrowski, M. E. A. Churchhill, and L. Kam, this series. Vol. 155, p. 537.

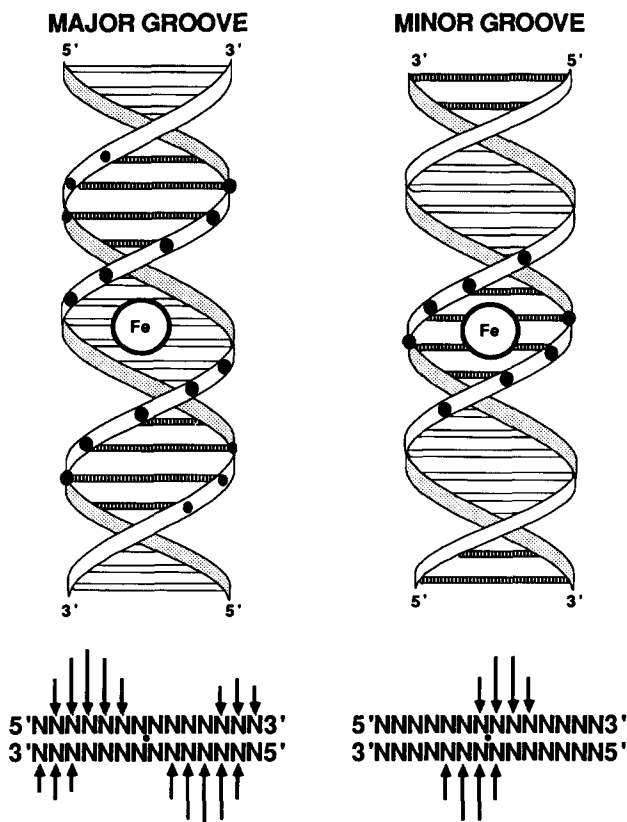


FIG. 1. Cleavage pattern analysis. Filled circles represent points of cleavage along the phosphodiester backbone. Sizes of circles represent extent of cleavage.

have affinity for the protein, which bears on the assignment of DNA base pair-protein interactions critical for sequence-specific recognition.

Methods

Synthetic procedures for the introduction of the metal chelator, EDTA, at unique amino acid positions of proteins by solid-phase methods have been described.¹⁶ Two protected derivatives of EDTA compatible with Merrifield solid-phase protein synthesis employing *N*-tert-butyloxycarbonyl (Boc)-protected amino acids were developed. The first reagent is a dipeptide with three of four carboxyl groups of EDTA protected as benzyl

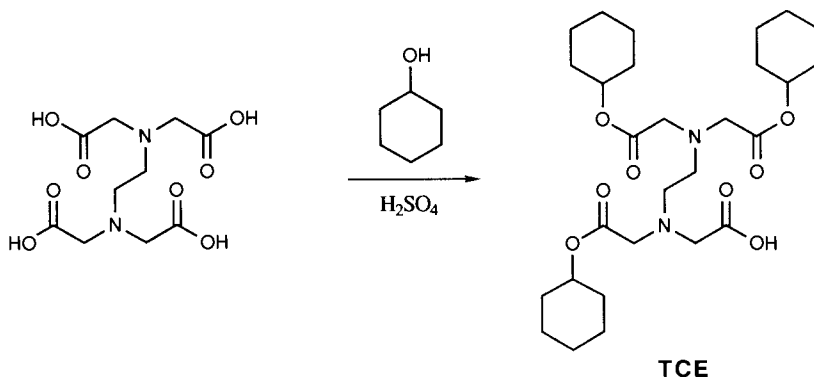


FIG. 2. Synthesis of tricyclohexyl ester of EDTA (TCE).¹⁶

esters and the fourth coupled to a γ -aminobutanoic acid linker, referred to as tribenzyl-EDTA-GABA (BEG).^{13,16} A second reagent is the tricyclohexyl ester of EDTA, TCE.¹⁶ BEG and TCE allow the modification of the NH₂ terminus and/or lysine side chains of resin-bound peptides and proteins. Due to the ease of synthesis, stability, coupling efficiency, and flexibility in choice of linker, TCE is the likely reagent for most applications. On deprotection and cleavage from the resin, a protein is produced with EDTA at a defined amino acid position. The availability of protein-EDTA conjugates extends the affinity-cleaving method to the study of protein-DNA complexes in solution.

*Synthesis of TCE*¹⁶

A stirred slurry of EDTA (10 g, 34 mmol) in cyclohexanol (70 ml) containing concentrated sulfuric acid (2 ml) is heated at 115° for 10 hr, at which time nearly all of the EDTA has dissolved. The mixture is cooled to room temperature, poured into saturated aqueous sodium bicarbonate solution (500 ml), and extracted with ether (twice with 200 ml). The combined extracts are dried (Na₂SO₄) and concentrated under reduced pressure and then *in vacuo* to afford a thick gum. The product is purified by flash chromatography [3% (v/v) CH₃OH in CH₂Cl₂ followed by 30% CH₃OH in CH₂Cl₂] to afford TCE as a brittle foam (3.9 g, 21%). ¹H NMR (deuterium-labeled dimethylsulfoxide, Me₂SO-*d*₆): δ 4.68 (m, 3H), 3.50 (s, 4H), 3.49 (s, 2H), 3.37 (s, 2H), 2.71 (s, 4H), 1.78–1.21 (m, 30H) ppm; infrared (IR; film) 3448, 2937, 2858, 1735, 1634, 1452, 1198 cm⁻¹ (see Fig. 2).

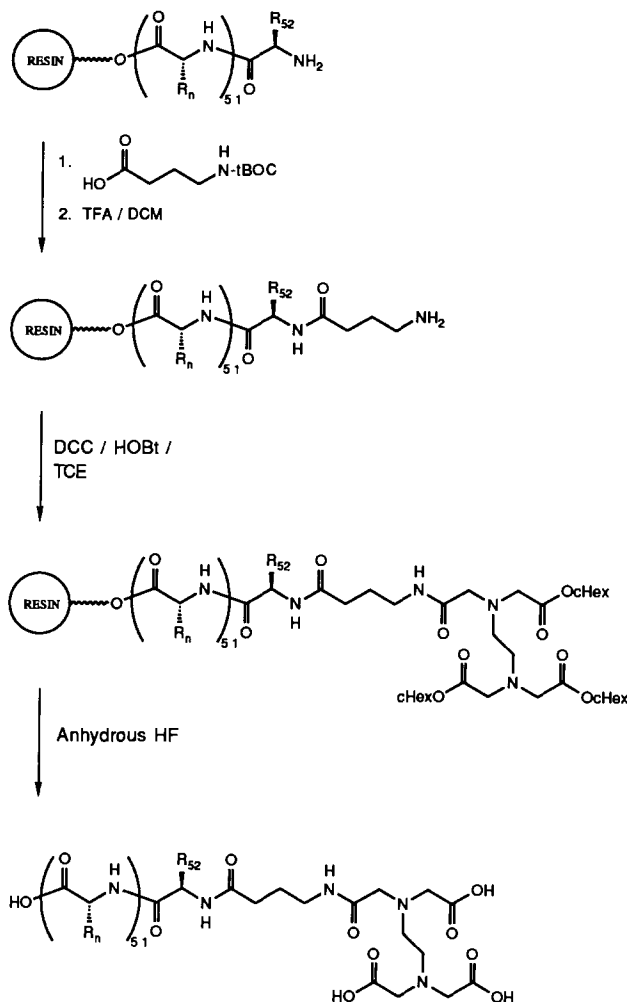


FIG. 3. Scheme for the attachment and deprotection of TCE to the NH_2 terminus of a protein using solid-phase methods.¹⁶

Peptide Syntheses¹⁶

Peptides and proteins are prepared manually or by automated methods employing *N-tert*-butoxycarbonyl (Boc) amino acid derivatives for Merrifield solid-phase synthesis.¹⁸ *N-α*-Boc-L-amino acids are used with the following side-chain protecting groups: Arg(Tos), Asp(OBzl), Glu(OBzl), His-dinitrophenol (DNP), Lys(Cl-Z), Ser(Bzl), Thr(Bzl), Trp-carbohydrate (CHO), and Tyr(Br-Z). Coupling yields are determined by quantita-

tive ninhydrin monitoring, with acceptable values being $\geq 99.7\%$ in the beginning of the synthesis and gradually decreasing to 99% near the end.

TCE Coupling to NH₂ Terminus of Resin-Bound Peptides (Fig. 3)

A typical procedure for the coupling of TCE to the NH₂ terminus of resin-bound peptide is as follows.¹⁶ An ~ 100 -mg sample of peptide/resin (ca. 100 $\mu\text{mol/g}$, total peptide ca. 10 μmol) is placed in a 12 $\times$ 80 mm reactor, and the resin is swollen in CH₂Cl₂ for 15 min. The resin is washed five times with CH₂Cl₂, the terminal Boc-protecting group is removed, and the resin neutralized using standard procedures. TCE (125 mg, 194 μmol) and HOBt (50 mg, 330 μmol) are dissolved in dimethylformamide (DMF, 2 ml) and DCC (42 mg, 203 μmol) is added. The solution is stirred 30 min at room temperature and added to the peptide/resin along with sufficient DMF to fill the reactor two-thirds full. Ninhydrin analysis indicates $\geq 98\%$ reaction yield after 2–3 hr. The peptide/resin is washed four times with DMF and four times with CH₂Cl₂.

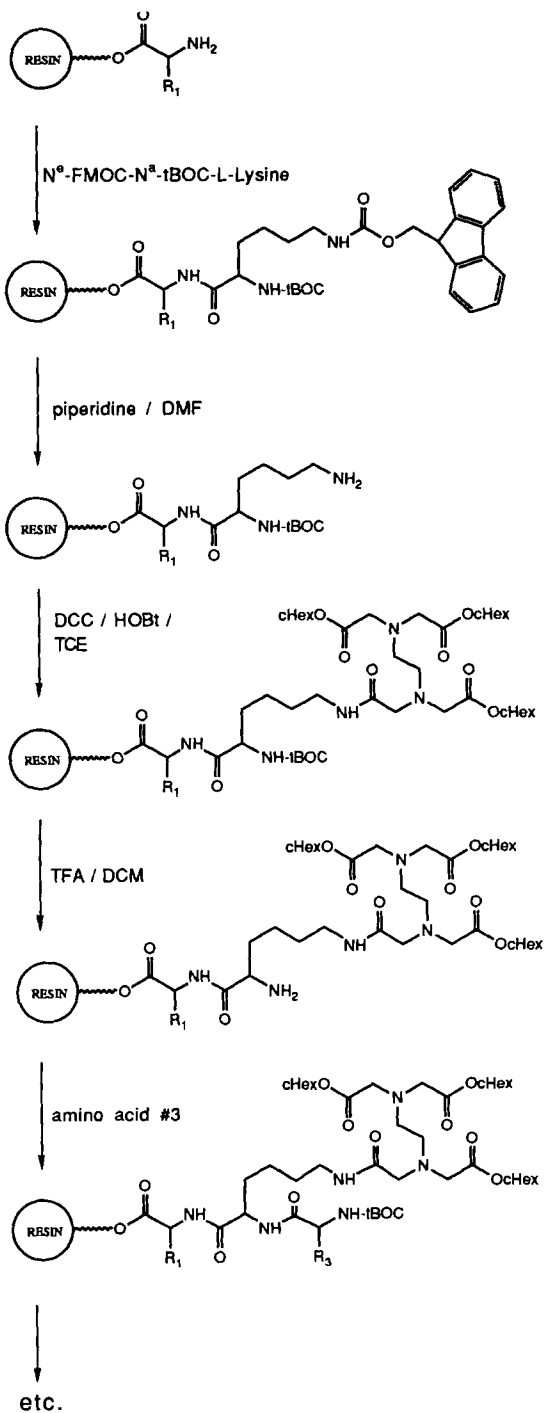
TCE Coupling to ϵ -NH₂ Side Chain of Lysine Residue (Fig. 4). *N*- α -Boc-*N*- ϵ -Fmoc-L-lysine is coupled to the growing peptide as an HOBt ester.¹⁶ The resin is washed five times with CH₂Cl₂, followed by five washes with DMF. The *N*- ϵ -Fmoc group is selectively removed using 20% piperidine in DMF (a 1-min reaction step followed by a 10-min step) and the resin is washed five times with DMF. TCE (1.08 g, 2 mmol, 4 Eq based on a 0.5-mmol synthesis) and HOBt (0.46 g, 3.4 mmol, 6.8 Eq) are dissolved in DMF (2 ml) and DCC (0.41 g, 2 mmol, 4 Eq) is added. The solution is stirred 30 min at room temperature and added to the peptide resin along with sufficient DMF to fill the reactor vessel two-thirds full. Ninhydrin analysis indicates 99.9% reaction after 1 hr. The peptide resin is washed five times with DMF and five times with CH₂Cl₂. The *N*- α -Boc is removed as usual with trifluoroacetic acid (TFA) and the synthesis continued with the standard protocols.

Deprotection and Purification of Protein-EDTA

Deprotection of the resin-bound peptide is carried out as described by Kent.¹⁹ The histidine-protecting group, DNP, is removed at 25° using 20% (v/v) 2-mercaptoethanol and 10% (v/v) DIEA in DMF; this treatment is repeated twice (30 min each). After removal of the *N*- α -Boc group with TFA and drying of the resin, all other side chain-protecting groups as well as the peptide–resin bond are cleaved using anhydrous HF, in the presence

¹⁸ R. B. Merrifield, *Adv. Enzymol.* **32**, 221 (1969).

¹⁹ S. B. H. Kent, *Annu. Rev. Biochem.* **57**, 957 (1988).



of *p*-cresol and *p*-thiocresol as scavengers, for 60 min at 0°. For peptides containing a Trp residue, the scavengers used are anisole and 1,4-butanedithiol. The HF is removed under vacuum. The crude protein is precipitated with diethyl ether, collected on a fritted funnel, dissolved in water, and filtered through, leaving the resin on the frit. A small sample is then removed, microfiltered, and subjected to analytical high-performance liquid chromatography (HPLC) [Vydac, 25 × 4.6 mm C₄ column (Separations Group, Hesperia, CA), 0 to 60% acetonitrile/0.1% TFA over 60 min]. The remaining solution is frozen and lyophilized. Residual DNP groups are removed from the crude protein by treatment in 4 M guanidine hydrochloride, 50 mM Tris, pH 8.5, and 20% 2-mercaptoethanol for 1 hr at 50°. ²⁰ This solution is injected onto a semipreparative C₄ HPLC column (25 × 1 cm) and developed with H₂O/0.1% TFA until the guanidine and 2-mercaptoethanol have eluted. A gradient of 0–60% acetonitrile/0.1% TFA is then run over 240 min. Fractions are collected and assayed for the presence of the desired protein by analytical HPLC.

Cleavage Reactions by Protein-EDTA · Fe on DNA Restriction Fragments

Protein-EDTA · Fe cleavage reactions on DNA restriction fragments labeled at the 5' (and 3') ends are run in a total volume at 15 μl. Final concentrations are 20 mM phosphate, pH 7.5, 20 mM NaCl, 100 μM in base pairs of calf thymus DNA, ~15,000 cpm of ³²P-end-labeled DNA restriction fragment, 5 mM dithiothreitol (DTT), and 5 μM protein-EDTA · Fe. The protein-EDTA · Fe is allowed to equilibrate with the DNA for 10 min at 25°; cleavage is then initiated by the addition of DTT and allowed to proceed for 60 min at 25°. The reactions are terminated by ethanol precipitation, dried, and resuspended in 5 ml of 100 mM Tris-borate-EDTA, 80% formamide solution. The ³²P-labeled DNA products are analyzed by denaturing polyacrylamide gel electrophoresis followed by autoradiography. Densitometric analysis of the gel autoradiogram and comparison of individual lanes with sequence marker lanes allows assignment of DNA cleavage to nucleotide resolution.

²⁰ N. Camerman, A. Camerman, and B. Sarkar, *J. Chem.* **110**, 7572 (1988).

FIG. 4. Synthetic scheme for the attachment of TCE to the ε-NH₂ group of a lysine in the second amino acid position on a resin-bound peptide. [D. P. Mack, J. A. Shin, J. H. Griffin, M. I. Simon, and P. B. Dervan, *Biochemistry* **29**, 6561 (1990).]

Double-Strand Cleavage of Plasmid DNA by Protein-EDTA · Fe

Protein-EDTA · Fe cleavage reactions on linearized plasmid DNA ^{32}P -labeled at one (or both) ends are run in a total volume of 45 μl . Final concentrations are 20 mM phosphate, pH 7.5, 20 mM NaCl, 100 μM in base pairs of calf thymus DNA, $\sim 30,000$ cpm of ^{32}P -end-labeled linearized plasmid DNA, 5 mM DTT, and 2.5 μM protein-EDTA · Fe. The protein-EDTA · Fe is allowed to equilibrate with the DNA for 10 min at 25°; cleavage is then initiated by the addition of DTT and allowed to proceed for 60 min at 25°. The reaction is terminated by the addition of 5 μl of 10 $\times$ Ficoll solution (25% in water, type 400; Sigma, St. Louis, MO). The ^{32}P -labeled DNA products are analyzed on a 4 mm $\times$ 20 cm 1% (w/v) agarose gel. Densitometric analysis of the gel autoradiogram and comparison of individual lanes with molecular weight markers allow assignment of cleavage loci to within 20 bp.

Cleavage Reactions by Ni · GGH-Protein on DNA Restriction Fragments

Nickel-mediated Gly-Gly-His (GGH)-protein cleavage reactions on DNA restriction fragments labeled with ^{32}P at the 5' (and 3') ends are run in a total volume of 20 μl . Final concentrations are 20 mM phosphate, pH 7.5, 20 mM NaCl, 100 μM in base pairs of calf thymus DNA, $\sim 15,000$ dpm of ^{32}P -end-labeled DNA restriction fragment, 5 μM monoperoxyphthalic acid, magnesium salt, and 5 μM Ni · GGH-protein. The Ni · GGH-protein is allowed to equilibrate with the DNA for 10 min at 25°; cleavage is then initiated by the addition of monoperoxyphthalic acid and allowed to proceed for 15 min at 25°. The reactions are terminated by ethanol precipitation, dried, and resuspended in 50 μl of 0.1 M *n*-butylamine and heated to 90° for 30 min. The reactions are then dried by lyophilization and resuspended in 5 μl of 100 mM Tris-borate-EDTA, 80% formamide solution. The ^{32}P -labeled products are analyzed by denaturing polyacrylamide gel electrophoresis, followed by autoradiography. Densitometric analysis of the gel autoradiogram and comparison of individual lanes with sequence marker lanes allow assignment of DNA cleavage to nucleotide resolution.

*Discussion**Synthetic Procedure and Experimental Design*

Protecting groups for the EDTA carboxylic acids are chosen based on standard side chain-protection methods used for glutamic and aspartic acids in Merrifield solid-phase protein synthesis employing *N*-tert-butyl-

oxycarbonyl (Boc) amino acid derivatives. In BEG, three of the four carboxyl groups of EDTA are protected as benzyl esters. The fourth carboxyl is coupled via an amide bond to a γ -aminobutanoic acid (GABA) linker designed to minimize disruptions of protein structure by the attached EDTA $\cdot$ Fe chelates (Fig. 2). The protected dipeptide, tribenzyl-EDTA-GABA (BEG), can be synthesized in four steps.¹⁶ The utility of BEG was demonstrated by the introduction of EDTA at the NH_2 terminus of the DNA-binding domain from *Hin* recombinase, residues 139–190.¹³

The tricyclohexyl ester of EDTA (TCE) has been developed as an improved alternative to BEG.¹⁶ TCE may be prepared on a larger scale than BEG, is more stable than BEG, and offers more flexibility than BEG in that a variety of amino acid linkers (or no linker at all) may be used. TCE was synthesized in a single step by incomplete esterification of EDTA with cyclohexanol (Fig. 3). TCE couples with high efficiency to the NH_2 terminus or ϵ - NH_2 group on the side chain of a lysine residue at the COOH terminus of resin-bound peptides (>98%) in the presence of DCC and HOBT in DMF.

Although BEG and TCE are formally amino acids, they are suitable only for capping an amino terminus or lysine side chain since they do not have a primary or secondary amino group for further chain extension. Current methodology only allows attachment of EDTA to the NH_2 residues of resin-bound protected peptides. In addition, the approach of transient protection of the ϵ - NH_2 side chain of lysine with Fmoc, for subsequent coupling with TCE, may be limited to the first few residues at the COOH terminus at the beginning of the peptide chain synthesis. EDTA modification of internal ϵ -lysine side chains of peptides in the *tert*-Boc-based synthesis has not yet been documented. Thus, BEG and TCE represent only the first steps in the development of a class of modified amino acids compatible with Merrifield synthesis for the construction of hybrid proteins with novel functions.

Cleavage Pattern Analysis of Helix–Turn–Helix Motif Protein–DNA Complexes

This methodology has been used to investigate the tertiary structures of two putative helix–turn–helix protein–DNA complexes and a leucine zipper–DNA complex. The DNA-binding domains of *Hin* recombinase and $\gamma\delta$ resolvase are thought to bind DNA by a helix–turn–helix motif. A synthetic 52-residue protein based on the sequence-specific DNA-binding domain of *Hin* recombinase (residues 139–190) with EDTA $\cdot$ Fe at the

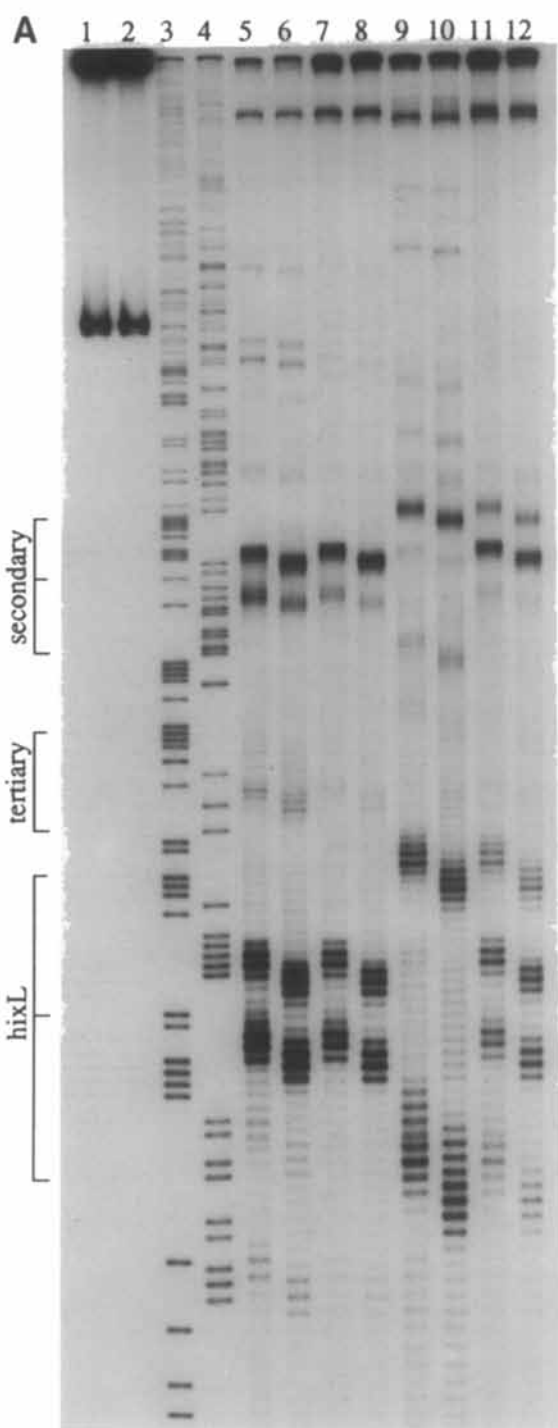
²¹ K. Kim, S. G. Rhee, and E. R. Stadtman, *J. Biol. Chem.* **260**, 15394 (1985).

NH₂ terminus, Fe · EDTA–*Hin*(139–190), produced two cleavage patterns on the symmetry axis side of the 13-bp inverted half-sites (see Fig. 5).¹³ The cleavage patterns on opposite strands of DNA for Fe · EDTA–*Hin*(139–190) were shifted to the 3' side at all binding sites. This asymmetry reveals that the NH₂ terminus of *Hin*(139–190) is bound in the minor groove of DNA near the symmetry axis of *Hin*-binding sites.¹³ A binding model put forward for *Hin*(139–190) includes a helix–turn–helix–turn–helix structure in the major groove with residues at the NH₂ terminus extending across the DNA phosphodiester backbone, making specific contacts to the adjacent minor groove.¹³ Attachment of EDTA · Fe to a lysine side chain (Ser-183 → Lys-183) at the COOH terminus of *Hin*(139–184) affords an asymmetric cleavage pattern on opposite strands that is also shifted to the 3' side. Within the context of the helix–turn–helix model, these data are consistent with the putative recognition helix oriented toward the symmetry axis of the site (Fig. 6).¹³

From sequence comparison, the 43-residue DNA-binding domain of $\gamma\delta$ resolvase is also thought to bind DNA by a helix–turn–helix motif. Incorporation of EDTA · Fe at the NH₂ and COOH termini of $\gamma\delta$ (141–183), respectively, affords cleavage patterns similar to *Hin*(139–190). We have assigned the NH₂ terminus bound in the minor groove of DNA near the symmetry axis of $\gamma\delta$ -binding sites and the recognition helix oriented toward the symmetry axis of the site.¹⁵

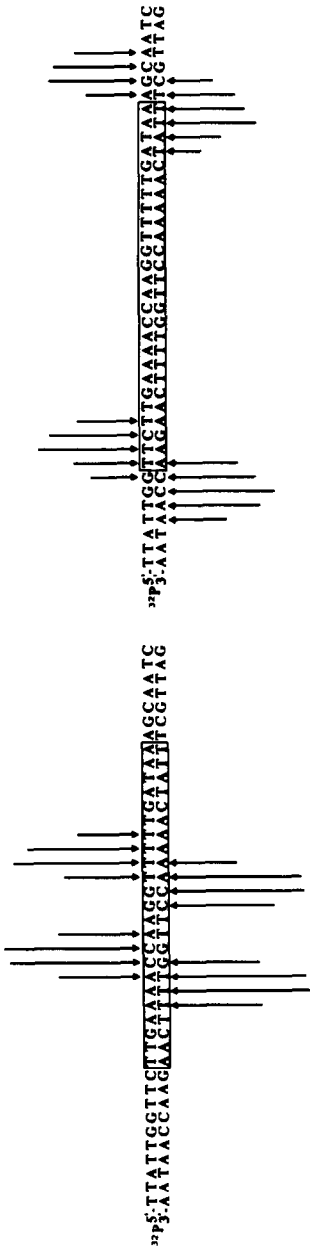
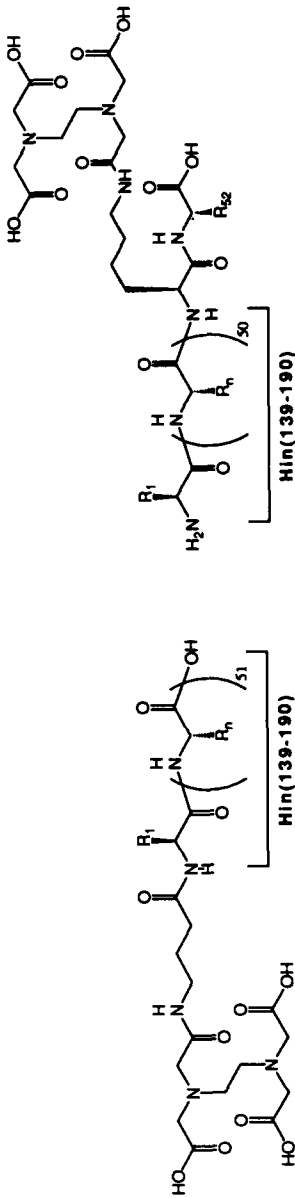
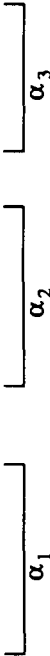
Cleavage Pattern Analysis of Y-Shape Motif Protein–DNA Complexes

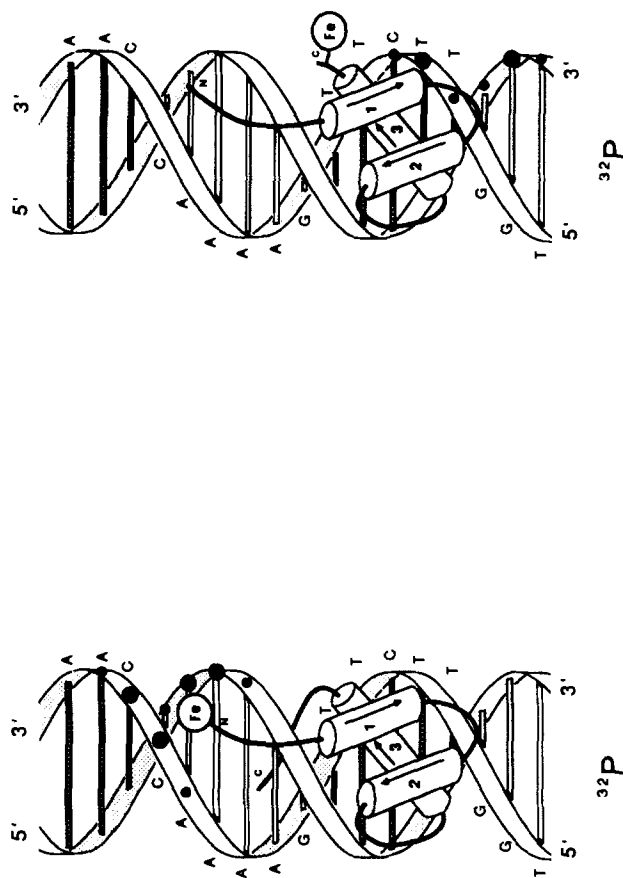
The DNA-binding domain of the transcriptional activator GCN4 is thought to bind DNA by a dimeric Y-shaped leucine zipper motif.⁷ EDTA was attached to the NH₂ terminus of the DNA-binding domain of GCN4 (residues 222–281).¹⁴ Because both halves of the Fe · EDTA 60-mer dimers contain Fe · EDTA at the NH₂ terminus, the affinity-cleavage pattern should result from the sum of two localized Fe · EDTA moieties. The cleavage pattern observed consists of three cleavage loci of unequal intensity. With regard to the cleavage model in Fig. 1, the simplest interpretation of this cleavage pattern is that it results from the superimposition of two adjacent major groove cleavage patterns along one face of the DNA. Cleavage occurs on both strands in three adjacent minor grooves, with the most efficient cleavage in the central minor groove proximal to both Fe · EDTA moieties. These data strongly suggest that the two Fe · EDTA moieties are located in adjacent major grooves. The positions of the Fe · EDTA moieties may be assigned by assuming that they lie in the center of the two 5'-shifted patterns. The Fe · EDTA moieties, and hence the NH₂ termini of the dimer, are 9 to 10 bp apart and are located 4 to 5 bp on



B

GRPRAINKHEQEQISRLLLEKGGHPRQQLAIIFGIGVSTLYRYFFPASSIKKRMN



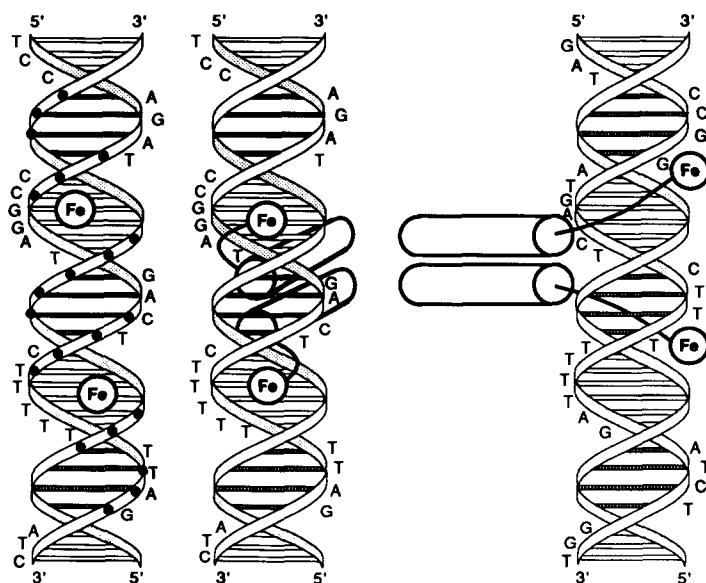
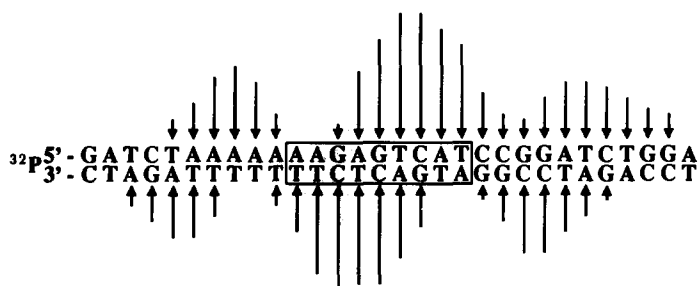
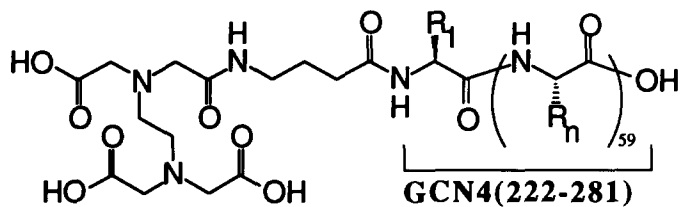


PESSDPAALKRARNTAARRSRARKLQRMK

Basic Region

QLEDKVEELLSKNYHLENEVARLKKLVGER

Leucine Zipper Region



either side of the central C in the binding sites 5'-CTGACTAAT-3' and 5'-ATGACTCTT-3'¹⁴ (see Fig. 6). Although not a proof, this is consistent with a Y-shaped model for a class of DNA-binding proteins important in the regulation of gene expression.

Design of Sequence-Specific DNA-Cleaving Proteins Using Naturally Occurring α -Amino Acids

From X-ray diffraction analysis, the tripeptide, GGH, is known to bind Cu(II) in a square planar complex with coordination from an imidazole nitrogen, two deprotonated peptide nitrogens, and the terminal amino group.²⁰ Although a crystal structure of GGH · Ni(II) is not available, the Ni(II) complex of GGH has been studied by other techniques.²² Crystal structures of tetraglycine with Cu(II) or Ni(II) indicate that the metal ions are bound by peptide ligands in a similar fashion.²³

GGH was attached to the NH₂ terminus of the DNA-binding domain of *Hin* recombinase (residues 139–190) to afford a new 55-residue protein, GGH[*Hin*(139–190)].²⁴ GGH[*Hin*(139–190)] at 1.0 μ M concentrations (pH 7.5, 25°) in the presence of 1 Eq of Ni(OAc)₂ and monoperoxyphthalic acid cleaves double-helical DNA at four 13-bp sites (termed *hixL* and secondary) on base workup (see Fig. 7²⁵). The cleavage patterns observed at the *hixL*-binding sites are strong and occur predominantly on one strand of each DNA site with single base specificity, while the cleavage at the secondary sites is modest and covers one to two base positions on both DNA strands. Maximal cleavage on opposite strands of the DNA is asymmetric to the 3' side, consistent with the known location of the NH₂ terminus of *Hin*(139–190) in the minor groove. On base workup, the

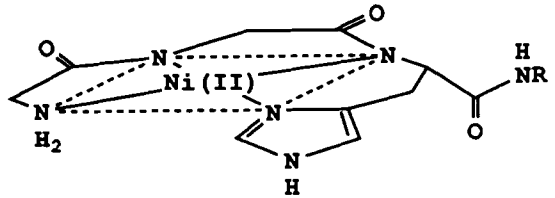
²² F. P. Bossu and D. W. Margerum, *Inorg. Chem.* **16**, 1210 (1977); C. E. Bannister, J. M. T. Raycheba, and D. W. Margerum, *Inorg. Chem.* **21**, 1106 (1982); T. Sakurai and A. Nakahara, *Inorg. Chim. Acta* **34**, L243 (1979).

²³ H. C. Freeman and M. R. Taylor, *Acta Crystallogr.* **18**, 939 (1965); H. C. Freeman, J. M. Guss, and R. L. Sinclair, *Chem. Commun.* (1968); F. P. Bossu, E. B. Paniago, D. W. Margerum, S. T. Kirsey, and J. L. Kurtz, *Inorg. Chem.* **17**, 1034 (1978).

²⁴ D. P. Mack, B. L. Iverson, and P. B. Dervan, *J. Am. Chem. Soc.* **110**, 7572 (1988).

²⁵ D. P. Mack and P. B. Dervan, *J. Am. Chem. Soc.* **112**, 4604 (1990).

FIG. 6. Sequence is the DNA-binding domain of GCN4 (residues 222–281). Putative dimerization domain and DNA contact basic region are indicated by brackets. Fe · EDTA-GCN4(222–281) reveals that the location of the NH₂ termini of a dimer of this DNA-binding protein are separated by 9 to 10 bp in the major groove on the same face of the DNA. (See Oakley and Dervan.¹⁴)

$$\alpha_1 \quad \alpha_2 \quad \alpha_3$$


DNA termini at the cleavage site are 3'- and 5'-phosphate consistent with oxidative degradation of the deoxyribose backbone.

The small number of deoxyribose cleavage positions within each binding site implies that a nondiffusible oxidant is generated by the Ni(II)·GGH[(*Hin*(139–190)]–peracid reactions. One explanation for the difference in reactivity observed for the sites is that the structure of the DNA is different at each site, and the nondiffusible reactive moiety on the metalloprotein is sensitive to these sequence-dependent conformational differences. The highly reactive nature of the oxidizing species, the precision of the cleavage reaction, and the identity of the DNA termini created on treatment with butylamine imply that the species may be a high-valent nickel-bound oxygen that abstracts a specific hydrogen atom on the deoxyribose backbone.²⁵

Future Applications: Protein Affinity Cleaving

Because the reactive oxidant generated from EDTA·Fe can cause cleavage of proteins,^{21,26} protein–EDTA·Fe molecules can be used for affinity-cleaving studies of protein structure in the presence and absence of DNA. In principle, analyses of protein fragments caused by protein autocleavage of protein–EDTA·Fe complexes should reveal the proximal location of the amino acid equipped with EDTA·Fe and its neighbor amino acid sequences in three-dimensional space.

Acknowledgments

This work was supported by grants from the DARPA University Research Initiative Program and the National Foundation for Cancer Research. I am indebted to my students and co-workers who made this work possible; in particular, J. P. Sluka, D. P. Mack, J. H. Griffin, K. Graham, J. A. Shin, and M. G. Oakley. We are grateful for helpful discussions and collaborations with the M. J. Simon research group and for generous technical advice from S. B. H. Kent and S. J. Horvath.

²⁶ T. M. Rana and C. F. Meares, *J. Am. Chem. Soc.* **112**, 2457 (1990).

FIG. 7. Sequence of designed protein with GGH at the NH₂ termini of the DNA-binding domain of *Hin* recombinase. Nickel-mediated oxidative cleavage occurs in the minor groove predominantly on one strand of the *Hin*-binding site by a nondiffusible oxidant (see Mack *et al.*²⁴; Mack and Dervan,²⁵). Circle represents point of cleavage.

[25] Identification of Amino Acid Residues at Interface of Protein–Nucleic Acid Complexes by Photochemical Cross-Linking

By KENNETH R. WILLIAMS and WILLIAM H. KONIGSBERG

Introduction

This chapter will focus on providing a general approach for identifying amino acid residue(s) that have been cross-linked to nucleic acids using light. Ultraviolet light can be employed to detect the presence of specific protein–nucleic acid complexes. Because irradiation of these complexes with ultraviolet light will produce covalent linkages between amino acid residues and nucleic acid bases with minimal perturbation of the system being studied, this technique can be used to identify residues at the interface between these species. The reaction is thought to occur via a free radical mechanism where the radical is generated by photoexcitation of a nucleic acid base followed by abstraction of a hydrogen atom from a favorably positioned amino acid residue to produce a purinyl or, much more likely, a pyrimidinyl radical along with a radical on the side chain of the amino acid residue that serves as the hydrogen atom donor. The formation of the covalent bond results from the geminate recombination of the purinyl or pyrimidinyl radical with the corresponding radical on the proximate amino acid residue.¹ For this reason, ultraviolet photolysis produces zero-length cross-links, in contrast to chemical cross-linking agents, which interpose spacers of varying distance between the protein and the nucleic acid. Since model studies suggest that virtually any amino acid residue can participate in photolysis, the promiscuous nature of the ultraviolet induced reaction makes it very useful for probing the interface of these complexes (for reviews see Refs. 1 and 1a). In addition to the generality of its application (it can be used *in vitro* or *in vivo*) the simplicity of the procedure is also an advantage. For example, the only equipment required is an inexpensive germicidal lamp that is available commercially.

Photochemical cross-linking has also been employed to detect protein bound to specific sites on double-stranded DNA by a modification of the method that permits transfer of ³²P from specific phosphodiester bonds at

¹ M. D. Shetlar, *Photochem. Photobiol. Rev.* **5**, 105 (1980).

^{1a} J. W. Hockensmith, W. L. Kubasek, W. R. Vorachek, E. M. Everts, and P. H. Von Hippel, this volume [13].

the binding site, to the protein.^{1b} This approach allows proteins that are present in only small amounts to be identified. By using ³²P-labeled site-specific probes, it may provide a valuable way of estimating pool sizes of specific DNA-binding proteins that cannot be done by DNase I, free radical footprinting, or electrophoretic mobility shift assays.^{1b}

While any nucleotide residue in DNA or RNA might, in principle, take part in the photo-cross-linking reaction, it appears that, in those systems that have been most extensively studied, thymine is by far the most reactive base. Hence, at 252.7 nm, a wavelength that is close to the maximum emission (253.4 nm) for a germicidal lamp, no cross-linking of (dG)₁₀ or (dA)₁₀ to the bacteriophage gene 32 single-stranded DNA-binding protein (gp32) could be observed while the ratio of (dT)₁₀ to (dC)₁₀ cross-linking to gp32 was 200:1.^{1a}

To test the potential of any protein to undergo photo-cross-linking, the simplest approach involves the use of an oligonucleotide such as (dT)₈ that has been 5'-end labeled with [³²P]ATP. After photolysis most of the non-cross-linked (dT)₈ can be removed by trichloroacetic acid precipitation prior to digestion of the remaining mixture of free and cross-linked protein with trypsin. By taking advantage of the large negative charge on tryptic peptides that are cross-linked to (dT)₈, it is possible to use anion-exchange high-performance liquid chromatography (HPLC) to rapidly separate cross-linked from non-cross-linked tryptic peptides. If there are multiple cross-linking sites on the protein then reversed-phase ion-pairing HPLC can be used for the final step in purification. Surprisingly, recoveries of oligonucleotide-peptide adducts from reversed-phase supports are very poor unless an ion pairing reagent such as triethylammonium acetate is present.^{2,3} The amino acid residue that has been cross-linked can then be identified by amino acid sequencing. If a solid-phase sequencer is used, then the site of attachment can be identified by scintillation counting of an aliquot of each of the resulting phenylthiohydantoin amino acid derivatives. If a gas or liquid phase instrument is used, the phenylthiazolinone derivative of the [³²P](dT)₁₀-labeled amino acid will not be extracted from the Polybrene-coated support disk and so a direct identification cannot be made on the basis of scintillation counting. However, there will be a "hole" at that position in the sequence that, if the primary structure of the protein is known, can be used to assign the amino acid residue involved in cross-linking.

^{1b} B. Safer, R. B. Cohen, S. Garfinkel, and J. A. Thompson, *Mol. Cell. Biol.* **8**, 105 (1980).

² B. M. Merrill, K. R. Williams, J. W. Chase, and W. H. Konigsberg, *J. Biol. Chem.* **259**, 10850 (1984).

³ B. M. Merrill, K. L. Stone, F. Cbianchi, S. H. Wilson, and K. R. Williams, *J. Biol. Chem.* **263**, 3307 (1988).

An alternative approach to that just described involves the use of a long polynucleotide or naturally occurring nucleic acid such as bacteriophage fd DNA in place of an oligonucleotide of defined length. In this instance sedimentation⁴ or gel filtration can be used to separate the free and cross-linked nucleic acid from the non-cross-linked protein. Following extensive micrococcal nuclease digestion, the cross-linked protein–oligonucleotide complex can then be treated with trypsin and the cross-linked peptide–oligonucleotide complex isolated by electrophoresis at low pH⁴ or by anion-exchange and reversed-phase HPLC in the presence of an ion pairing reagent. The nucleic acid can either be uniformly labeled *in vivo* prior to cross-linking or, alternatively, it can be 5'-end labeled with bacteriophage T4 polynucleotide kinase and [γ -³²P]ATP after the nuclease digestion. While the use of a high-molecular-weight nucleic acid permits the facile separation of free and cross-linked protein, it introduces a significant level of heterogeneity that complicates the task of purifying the final peptide–oligonucleotide complex. Unless the micrococcal nuclease digestion is complete, leaving only a single nucleotide residue linked to the protein, there will be a population of cross-linked protein (and ultimately peptide) molecules that will be covalently linked to a series of oligonucleotides of different lengths and base composition with the potential to separate on anion-exchange and reversed-phase ion-pairing HPLC. For example, despite extensive micrococcal nuclease digestion of the bacteriophage fd gene 5 protein–fd single-stranded (ss) DNA cross-linked complex, the cross-linked gene 5 protein still migrated on sodium dodecyl sulfate (SDS)-polyacrylamide gel electrophoresis as a diffuse band corresponding to a population of gene 5 monomers linked to a series of oligonucleotides ranging in length from 3 to 10.⁴ This heterogeneity substantially decreased the yield of any single oligonucleotide–peptide species. If, on the other hand, conditions could be found to obtain complete nuclease digestion this would substantially decrease the ability of the anion-exchange step to separate the cross-linked peptide from all of the other non-cross-linked tryptic peptides.

Experimental Procedures

Optimization of Photocross-Linking Conditions

To use photochemical cross-linking for identifying amino acid residues at the interface of protein–nucleic acid complexes sufficient amounts of the purified oligonucleotide–peptide covalent complex must be isolated to

⁴ P. R. Paradiso and W. H. Konigsberg, *J. Biol. Chem.* **257**, 1462 (1982).

permit detailed chemical characterization. Since the yield of the cross-linked product is dependent on the extent of cross-linking that can be achieved, it is necessary to determine the dose response. This information can be easily obtained using a short oligonucleotide such as (dT)₈ that has been 5'-³²P end labeled using polynucleotide kinase as described by Maniatis *et al.*⁵ The resulting [³²P](dT)₈ can then be separated from unincorporated [γ -³²P]ATP using a Du Pont (Newton, CT) Neksorb 20 column following the manufacturer's recommended procedures and buffers. The [³²P](dT)₈ and the protein of interest are then mixed under conditions (low salt, high protein concentration, and, if necessary, excess oligonucleotide) that are known to result in near stoichiometric binding based on the amount of protein present. The irradiation is usually carried out in a cold room to minimize sample heating using a 15-W germicidal lamp (General Electric Co.). The sample (20–150 μ l) is placed on a Parafilm sheet at a distance of 5–15 cm away from the light source. At a distance of 5 cm, irradiation times typically range from 1 to 30 min and the actual photon flux can be measured with a dosimeter (S. I. Schumberger Co.). Maximal extent of cross-linking is usually reached with an exposure that will provide photon fluxes in the range of 0.2 to 2.0×10^5 ergs/mm² (see Refs. 2–4, 6). The amount of protein that has been cross-linked to the labeled oligonucleotide can then be estimated by SDS–polyacrylamide gel electrophoresis followed by autoradiography. More quantitative measurements of the extent to which the protein has been cross-linked can be made by using HPLC gel filtration in the presence of a denaturant such as 6 M guanidine hydrochloride² or, if the cross-linked complex is acid stable, by a simple filter assay.⁶ In the latter case, irradiated samples can be blotted onto Whatman (Clifton, NJ) gf/c41 filter paper that has been presoaked in 15% trichloroacetic acid (TCA) and then dried prior to use. After spotting several samples onto a grid that has been penciled onto the gf/c41 filter paper, it is placed on a Büchner funnel, rinsed once under vacuum with 50 ml cold 10% TCA, three times with 50 ml cold H₂O, and three times with 20 ml cold ethanol. The filters are then baked at 67° for 15 min prior to immersing in 10 ml Hydrofluor scintillation fluid. After scintillation counting, the maximum extent of protein cross-linking can then be determined based on the specific activity of the [³²P](dT)₈ and the initial protein concentration.

In the case of three single-stranded nucleic acid-binding proteins whose cross-linking sites have been determined, using the procedure outlined

⁵ T. Maniatis, E. F. Fritsch, and J. Sambrook, in "Molecular Cloning: A Laboratory Manual," p. 125. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1982.

⁶ Y. Shamoo, K. R. Williams, and W. H. Konigsberg, *Proteins: Struct. Funct. Genet.* **4**, 1 (1988).

TABLE I
PHOTOINACTIVATION OF PROTEINS UNDER CONDITIONS USED
FOR IDENTIFYING CROSS-LINKING SITES

Nucleic acid-binding protein	Dose (ergs/mm ²)	Inactivation ^a (%)	Ref.
Bacteriophage fd gene 5	1.5×10^5	0	7
<i>E. coli</i> SSB	1.3×10^5	50	2
Bacteriophage T4 gene 32	0.9×10^5	33	6
HeLa A1 hnRNP	0.6×10^5	0	3

^a As determined either by stoichiometric fluorescence titrations with nucleic acid before and after irradiation, or in the case of the fd gene 5 protein by comparing cross-linking efficiencies before and after irradiation.

above (which will be described in more detail below)^{2,3,6} the percentage cross-linking ranged from 11.8% for the *Escherichia coli* SSB ssDNA-binding protein² to 33% for the A1 hnRNP protein.³ The amount of protein used in these studies ranged from 39 nmol for the bacteriophage T4 ssDNA-binding protein encoded by gene 32 (gp32, Ref. 6) to 250 nmol for *E. coli* SSB.² Based on these three studies^{2,3,6} it is reasonable to expect an overall average recovery of about 50% for the cross-linked peptide(s) following tryptic digestion and anion-exchange HPLC.

Controls Required to Determine Specificity of Labeling

Prior to identifying a site of photochemical cross-linking it is essential to show that cross-linking is limited to the nucleic acid-binding site of the native protein. Since exposure to UV light can lead to protein denaturation it is important to determine the extent of denaturation that is occurring during photolysis and to demonstrate that only the native protein can be cross-linked to the nucleic acid. As shown in Table I,⁷ the extent of protein denaturation that has been presumed to occur under the conditions used for identifying cross-linking sites in four nucleic acid-binding proteins ranges from 0 to 50%. Since the data in Table I show that similar doses of UV light lead to 50% inactivation of *E. coli* SSB² compared to no detectable inactivation of the fd gene 5 protein,⁷ it is apparent that there is considerable variation in the photosensitivity of proteins. Thus, an essential control requires determination of the extent of protein nucleic acid cross-linking that can be obtained before and after the protein alone has been subjected

⁷ P. R. Paradiso, Y. Nakashima, and W. H. Konigsberg, *J. Biol Chem.* **254**, 4739 (1979).

to the same photon flux as that used in the cross-linking reaction. Hence Merrill *et al.*² demonstrated that prior irradiation of SSB with 1.3×10^5 ergs/mm² of UV light in the absence of [³²P](dT)₈ led to a 38% decrease in the extent of cross-linking compared to an identical sample of SSB that had not been preirradiated.² These data argue strongly that it is the native rather than the photoinactivated SSB that is involved in cross-linking. Another important control is the demonstration that disruption of the complex, for example by adding a sufficiently high NaCl concentration to result in at least a 50% dissociation of the protein–nucleic acid complex, directly correlated with a decreased extent of cross-linking.^{2,3,6} Finally, advantage has also been taken of the finding^{1,7} that thymine is the most photoreactive base. Hence, Merrill *et al.*³ found that a 20:1 molar ratio of (dA)₈ to [³²P](dT)₈ completely prevented cross-linking as estimated by SDS polyacrylamide gel electrophoresis followed either by Coomassie Blue staining or autoradiography. These data are best explained by assuming that [³²P](dT)₈ can cross-link only at the A1 oligonucleotide-binding site(s) and, hence, when the [³²P](dT)₈ is effectively removed from the binding site by (dA)₈ competition, the extent of A1 : [³²P](dT)₈ cross-linking is decreased. The final proof that cross-linking is nonrandom comes from actually identifying the site of cross-linking, as will be described below.

Measurement of Extent of Photodegradation of the Cross-Linked Protein–Nucleic Acid Complex

In addition to photoinactivation of the protein^{2,3,6} and photolysis reactions involving the oligonucleotide,¹ another factor that accounts for the inability to obtain 100% cross-linking of the protein is reversal or photodegradation of the cross-linked protein–nucleic acid complex. In the case of the bacteriophage fd gene 5 protein, it appears that the cross-linking reaction can be reversed. Thus, a maximum of 22% cross-linking of the gene 5 protein–poly(dT) complex was observed when either a mixture of the individual components or the covalently linked gene 5–poly(dT) complex itself was exposed to 1.5×10^5 ergs/mm² of UV light.⁷ To obtain this result with the covalent gene 5–poly(dT) complex, the complex was first isolated by gel filtration in 1.0 M NaCl, and then, after desalting, it was exposed to the same dosage of UV light that was used to form the complex initially. Gel filtration in SDS of the complex after reexposure to UV light indicated that reirradiation resulted in breakage of nearly 80% of the covalent bonds in the gene 5 protein–poly(dT) complex.⁷ Most importantly, it was also shown that the free gene 5 protein and poly(dT), which had failed to cross-link during the initial UV exposure, and which had been separated by gel filtration in 1.0 M NaCl, gave the same cross-linking efficiency of about

22% when desalted, recombined, and exposed again to the same dosage of UV light.⁷ From these results it was concluded that in the case of gene 5 protein, no detectable photodenaturation occurred on exposure of the protein to 1.5×10^5 ergs/mm² of UV light.⁷ Paradiso *et al.*⁷ demonstrated a significant difference in the action spectrum for the forward and reverse photochemical reaction. They found that the action spectrum for covalent bond formation between the gene 5 protein and [³²P](dT)₄ was similar to the UV absorption spectrum of thymine, which has a maximum near 265 nm. This result suggested that absorption of UV light by thymine results in its excitation and that this is the primary event leading to covalent bond formation with the gene 5 protein.⁷ In contrast, the extent of bond breakage in the reverse reaction was inversely related to the incident wavelength (at a constant dosage of 1.5×10^5 ergs/mm²) down to 230 nm, which was the lowest wavelength tested.⁷ It appeared therefore that in the gene 5-[³²P](dT)₄ system, the nature of the absorbing species in the product is different from that in the reactant. In addition to the gene 5-[³²P](dT)₄ system, photodissociation of the cross-linked protein-nucleic acid complex has also been demonstrated in the case of the *E. coli* SSB protein-(dT)₈ complex.² In this instance, exposure of the purified SSB-[³²P](dT)₈ cross-linked complex to 1.3×10^5 ergs/mm² of UV light at 257 nm resulted in bond breakage of about 40% of the covalent complex.²

Chemical Stability of Photochemically Cross-Linked Protein-Nucleic Acid Complexes

Because of the numerous possible cross-linking reactions that may occur when protein-nucleic acid complexes are exposed to UV light, the resulting covalent complexes may differ substantially in their stability to acids, bases, or denaturants, depending on the protein being studied. In all instances examined so far in detail, the resulting complexes are stable to high concentrations of salts and denaturants, such as 6 M guanidine hydrochloride and 8 M urea, as well as to boiling in SDS at pH 8. Similarly, the covalent complex formed with the gene 5 protein is stable to 7% perchloric acid⁷ and those formed with bacteriophage T4 gp32,⁶ *E. coli* SSB,² and the A1 hnRNP protein³ are resistant to exposure to 5–10% trichloroacetic acid at 25°. In contrast, the covalent gene 5 protein-DNA complex is disrupted by 50% (v/v) formic acid⁴ or by treatment with 1.0 M acetic or 6 N HCl for 15 min at 25°. Similarly, Hockensmith *et al.*^{1a} found that a 30-min incubation at room temperature in 10% (v/v) acetic acid (1.6 M) and 50% (v/v) methanol led to 75% disruption of the covalent gp32-DNA complex that had been formed using a laser UV light source. These conditions are similar to those routinely used for fixing and Coomas-

sie Blue staining of SDS–polyacrylamide gels. While the *E. coli* SSB–DNA complex is stable at 25° to anhydrous trifluoroacetic acid (TFA) it is disrupted in this acid if the temperature is raised to 45°. ² Although the covalent gene 5 protein–DNA complex has been shown to be stable when 15-hr incubations were carried out in 0.1 *N* NaOH or in concentrated NH₄OH at 25°, ⁷ the alkali stability of the other protein–nucleic acid complexes mentioned above has not yet been studied.

Isolation of Peptides Cross-Linked to ³²P-Labeled Oligonucleotides or Nucleic Acids

The approach that has most often been used to identify nucleic acid cross-linking sites in proteins is to cross-link the protein to a ³²P-labeled oligonucleotide such as (dT)₈, digest the cross-linked complex with trypsin, and then use anion-exchange HPLC to rapidly isolate the corresponding [³²P](dT)₈-labeled tryptic peptide. Using this approach the non-cross-linked [³²P](dT)₈ can be removed by a trichloroacetic acid precipitation (TCA) step that immediately follows the photolysis. ^{2,3,6} After washing the TCA pellet three times with cold acetone, the mixture of cross-linked and non-cross-linked protein is then suspended in a volume of 8 *M* urea, 0.4 *M* NH₄HCO₃ to give a final protein concentration in the range of 0.1 to 5 mg/ml. If the protein contains disulfide bonds, the sample can be reduced and carboxamidomethylated, to ensure complete denaturation, and digested as described previously. ^{8,9} Briefly, the protein is reduced by adding dithiothreitol to a final concentration of 4.5 mM and incubating it at 50° for 15 min. After cooling to room temperature, iodoacetamide is added to a final concentration of 10 mM and the sample is incubated for an additional 15 min at room temperature. Prior to adding a 1:25 (w/w) ratio of trypsin, the sample is diluted with water to give a final urea concentration of 2 *M*. After 24 hr at 37° the resulting peptide mixture can be directly injected onto an AX300 (SynChrom, Lafayette, IN) anion-exchange HPLC column that had been equilibrated at a flow rate of 1.0 ml/min with buffer A [20 mM KH₂PO₄, pH 6.6, 4.8% (v/v) ethanol, 40 mM KCl]. The column is then eluted over a period of 100 min or more with a linear gradient extending to 1.0 *M* KCl in buffer A. ^{2,3,6} Peptides are detected by their absorbance

⁸ K. L. Stone, M. B. LoPresti, N. D. Williams, J. Myron Crawford, R. DeAngelis, and K. R. Williams, in "Techniques in Protein Chemistry" (T. E. Hugli, ed.), p. 377. Academic Press, New York, 1989.

⁹ K. L. Stone, M. B. LoPresti, J. Myron Crawford, R. DeAngelis, and K. R. Williams, in "A Practical Guide to Protein and Peptide Purification for Microsequencing" (P. T. Matsudaira, ed.), p. 33. Academic Press, New York, 1989.

at 220 nm² or 254 nm^{3,6} and by Cerenkov counting of the resulting fractions, which can conveniently be collected in 1.5-ml Eppendorf tubes that are then capped prior to storage. At this point individual peaks can either be desalted and then analyzed and sequenced or can be further purified by directly injecting them onto a reversed-phase HPLC column in the presence of an ion-pairing reagent. A Du Pont Nensorb 20 nucleic acid purification cartridge can be used to desalt cross-linked peptides rapidly that have been isolated by anion-exchange HPLC. As described previously,^{3,6} the Nensorb 20 cartridge is equilibrated with 5 mM triethylammonium acetate at pH 7.0 prior to sample loading. After washing the cartridge with the same buffer, the cross-linked peptide(s) can be eluted with 1 ml 75% methanol and then reduced in volume using a Savant (Farmingdale, NY) Speed-Vac. Fractions selected for further purification can be injected directly onto a Vydac (Hesperia, CA; 300-Å pore size, 5-μm particle size) or similar C₁₈ column that had been equilibrated at a flow rate of 1 ml/min with buffer B [10% (v/v) acetonitrile, 90% (v/v) 0.1 M triethylammonium acetate at pH 6.8]. After washing the column with the same solvent, peptides can be eluted with a linear gradient extending over about 100 min to 80% (v/v) acetonitrile, 20% (v/v) 0.1 M triethylammonium acetate. Peptides are detected as previously^{2,3} by their absorbance at 254 nm and by Cerenkov counting of the collected fraction. If necessary, they can be individually desalted as described above prior to amino acid analysis or sequencing.

If the protein of interest has been cross-linked to a high-molecular-weight nucleic acid such as fd ssDNA^{4,7} then the non-cross-linked protein can be removed by high-speed centrifugation⁴ or gel filtration in the presence of a high concentration of salt. At this point it may be possible to carry out an extensive chemical and/or proteolytic degradation of the protein followed by high-speed centrifugation or gel filtration, now in the presence of a low concentration of salt, to isolate the cross-linked peptide(s). Alternatively, the mixture of cross-linked and non-cross-linked DNA may be digested with micrococcal nuclease as previously described⁴ and the free oligonucleotides removed by dialysis. If necessary, the cross-linked protein:oligonucleotide complex can be 5'-end labeled using [γ -³²P]ATP and bacteriophage T4 polynucleotide kinase.⁴ The peptide-oligonucleotide complex, which because of the expected variability in the length and base composition of the resulting oligonucleotide may be heterogeneous, can then be purified by anion-exchange and reversed-phase HPLC as described above or by paper electrophoresis at pH 1.9.^{4,7}

Identification of Amino Acid Residues that Participate in Cross-Linked Adduct

Providing that the complete primary structure of the protein being studied is known, it may be possible to identify the cross-linked peptide by amino acid analysis following acid hydrolysis. Having done this, Paradiso *et al.*⁷ then used extensive aminopeptidase M and pronase digestion to release all non-cross-linked amino acids from the cross-linked oligonucleotide. Following performic acid oxidation and hydrolysis, the cross-linked amino acid was then identified as cysteine, based on amino acid analysis.⁷ This may not represent a general approach, however, since it depends on the ability of acid hydrolysis to regenerate the free amino acid from the amino acid–nucleotide adduct. Differences in the extent of regeneration of the original amino acid may be expected, depending on the exact chemical nature of the cross-link. In most instances, the identification of the cross-linked peptide and amino acid has been made on the basis of amino acid sequencing. In the case of the bacteriophage T4 gene 32, *E. coli* SSB² and A1 hnRNP³ proteins, the site of cross-linking was inferred from the absence of an identifiable phenylthiohydantoin derivative at a particular position in the peptide sequence. In all three of these proteins only the amino acid side chain (as opposed to the peptide bond) appeared to be involved in cross-linking because in each case amino acid sequencing proceeded as expected before and after the cross-linked residue was reached. In the case of *E. coli* SSB, the cross-linked peptide was subjected to solid-phase sequencing so that a sufficiently polar solvent, such as trifluoroacetic acid, could be used that would extract the oligonucleotide-labeled phenylthiazolinone derivative of the cross-linked amino acid.² Since the oligonucleotide had been ³²P-labeled it was possible using this solid-phase sequencing approach to directly confirm the cycle position corresponding to the site of cross-linking.

Results

Relative Efficiency and Extent of Cross-Linking of Several Single-Stranded Binding Proteins to Nucleic Acids

In terms of nucleotide base specificity, results obtained using a germicidal lamp source^{3,7} seem to parallel similar experiments carried out with laser cross-linking.^{1a} In both cases thymine appears to be the most reactive base. As shown in Table II for the bacteriophage fd gene 5 protein, no cross-linking was detected with either poly(dA) or poly(dG) and poly(dT)

TABLE II
PHOTOCROSS-LINKING OF BACTERIOPHAGE fd
GENE 5 PROTEIN TO NUCLEIC ACIDS^a

Nucleic acid	Gene 5 protein cross-linked (%)
Poly(dT)	21.4
Poly(dA-dT)	11.4
fd ssDNA	7.0
Poly(dC)	4.4
Poly(dA)	0
Poly(dG)	0

^a Data are taken from Ref. 7 and were determined by BioGel A-5 M chromatography (Bio-Rad, Richmond, CA) that was carried out in 0.5% SDS following exposure of mixtures of ¹⁴C-labeled gene 5 protein and the above nucleic acids to 1.5×10^5 ergs/mm² ultraviolet light. The irradiation was carried out at 4° in 10 mM Tris-HCl, pH 7.2, 1 mM EDTA, and 0.1% 2-mercaptoethanol.

was five times more reactive in this regard than poly(dC). When poly(dA-dT) and fd ssDNA was used, there was a clear correlation between the content of thymine and the extent of cross-linking. Thus, poly(dA-dT) had 50% efficiency and fd ssDNA, which contains 24% thymine, had 33% cross-linking efficiency relative to poly(dT).⁷ Although an extensive study was not carried out, the A1 hnRNP protein appears similar to the gene 5 protein in that A1 cross-links with a maximum efficiency of about 35% to oligo(dT)₈ (Fig. 1) compared with no detectable cross-linking to (dA)₈.

The maximum extent of cross-linking depends not only on the specific protein as well as the UV dose but also on the fraction of the protein that is actually complexed to the nucleic acid under the conditions of irradiation. Table III gives relative cross-linking efficiencies for a number of single-stranded nucleic acid-binding proteins. Although 1.3×10^5 ergs/mm² appeared to provide near maximum cross-linking for F plasmid SSB (data not shown), the data in Table III indicate that with the homologous *E. coli* SSB protein, optimum cross-linking requires at least twice this dose. With the calf thymus UP1 protein, which is actually a proteolytic fragment corresponding to the NH₂-terminal two-thirds of A1,^{10,11} binding

¹⁰ A. Kumar, K. R. Williams, and V. Szer, *J. Biol. Chem.* **261**, 11266 (1986).

¹¹ S. Riva, C. Morandi, P. Tsoulfas, M. Pandolfo, G. Biamonti, B. Merrill, K. R. Williams, G. Multhaup, and K. Beyereuther, *EMBO J.* **5**, 2267 (1986).

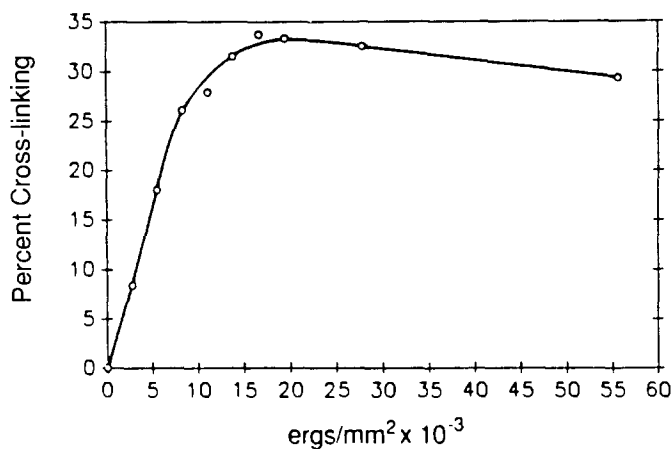


FIG. 1. Dose-response curve for the photocross-linking of A1 to $[^{32}\text{P}](\text{dT})_8$. A1 was mixed with $[^{32}\text{P}](\text{dT})_8$ in a 1 : 40 ratio of protein monomer to phosphate, exposed for varying lengths of time to UV light, then chromatographed on an HPLC gel-filtration column equilibrated in 6 *M* guanidine hydrochloride as described in Ref. 3. The results indicate that the maximum amount of A1 cross-linked is approximately 33%, on exposure of the A1-oligodeoxynucleotide complex to 16,700 ergs/mm² of UV light.

TABLE III
EFFICIENCY OF CROSS-LINKING OF VARIOUS SINGLE-STRANDED NUCLEIC ACID-BINDING PROTEINS TO $[^{32}\text{P}](\text{dT})_8^a$

Protein	Monomer protein concentration (μM)	Molar ratio of oligonucleotide to protein monomer	Dose (ergs/mm ²)	Protein cross-linked (%)
Bacteriophage T4 gene 32 protein	20	2.5 : 1	2.0×10^5	20.0
F plasmid SSB	20	2.5 : 1	1.3×10^5	27.3
<i>E. coli</i> SSB	20	2.5 : 1	1.3×10^5	11.8
	20	2.5 : 1	2.0×10^5	22.0
Rat liver lactate dehydrogenase	20	2.5 : 1	2.0×10^5	3.8
Calf thymus UPI	5	2.5 : 1	2.0×10^5	37.8
	5	8.0 : 1	2.0×10^5	85.6

^a Data are taken from Ref. 2 and were determined by HPLC gel filtration on a Waters (Milford, MA) I-125 column that had been equilibrated with 6 *M* guanidine hydrochloride. The $[^{32}\text{P}](\text{dT})_8$: protein mixtures were exposed to the indicated dose of ultraviolet light at 4° in 0.1 ml of 10 *mM* Tris-HCl, pH 8.0, 1 *mM* 2-mercaptoethanol, 10% glycerol, 310 *mM* NaCl.

to (dT)₈ in 310 mM NaCl is sufficiently weak that even a 2.5-fold molar excess of this oligonucleotide was insufficient to drive most of the protein into the complex. Hence, the molar ratio of oligonucleotide to protein monomer must be increased further, to about 8 : 1, to reach a near maximal extent of UP1 cross-linking (Table III). The unusually large percentage of UP1 cross-linking may result partly from the existence of four different cross-linking sites in this protein.³ Similarly, the seemingly low percentage of cross-linking of the rat liver lactate dehydrogenase (LDH) may result from its relatively low affinity for ssDNA. Although this enzyme binds sufficiently tightly to ssDNA that it was mistaken for a new rat liver helix-destabilizing protein, it binds less tightly than the other proteins listed in Table III to (dT)₈. It is likely therefore that the percentage cross-linking efficiency for LDH might be considerably improved by increasing the oligonucleotide to protein ratio above 2.5, the value that was used and reported in Table III.

Cross-Linked Amino Acid Residues Identified in Several ssDNA- and RNA-Binding Proteins

Identification of the phenylalanine residue in the *E. coli* SSB protein photo-cross-linked to [³²P](dT)₈ provides a good example of how the approach outlined above can be applied to a specific protein. As shown in Fig. 2, anion-exchange HPLC readily separates even the unusually acidic T-14,15 tryptic peptide, which has a net negative charge of −5, from the cross-linked peptide, which elutes at about 45 min and has a net negative charge of at least −7. Under the conditions used in Fig. 2 all other SSB tryptic peptides elute in the void volume of the column and a small amount of residual [³²P](dT)₈ elutes as shown at about 66 min. All four A1 tryptic peptides that were cross-linked to [³²P](dT)₈ also eluted from an anion-exchange HPLC column prior to free [³²P](dT)₈³ whereas, in the case of the T4 gene 32 protein, the cross-linked tryptic peptide actually coeluted with free [³²P](dT)₈.⁶ As shown in Fig. 3 further purification of the [³²P]dT₈ cross-linked, SSB tryptic peptide was achieved by reversed phase HPLC in the presence of an ion-pairing reagent. Solid-phase amino acid sequencing of the two major peaks in Fig. 3 indicated that both contained the same 6-residue tryptic peptide that corresponds to residues 57 to 62 in *E. coli* SSB. Since only a single SSB tryptic peptide was found cross-linked to [³²P](dT)₈, the multiple peaks resolved on ion-pairing HPLC (Fig. 3) may result from photochemical changes such as thymine dimer formation occurring within the oligonucleotide. As was also the case with the bacteriophage T4 gene 32 and the A1 hnRNP protein,³ the actual site of SSB cross-linking was indirectly identified by the absence of an expected phe-

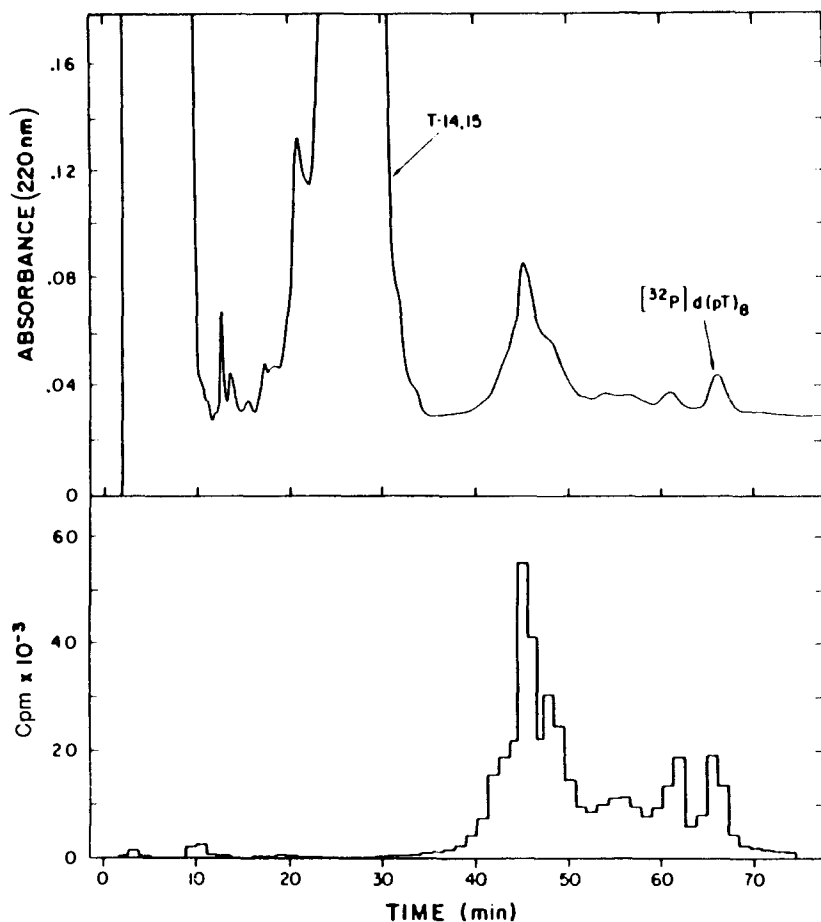


FIG. 2. Anion-exchange HPLC of [³²P](dT)₈ cross-linked tryptic peptides from SSB. SSB (250 nmol of monomer) and [³²P](dT)₈ (2 mmol of phosphate) were mixed together, irradiated with 1.3×10^5 ergs/mm² of UV light, precipitated with trichloroacetic acid, and then digested with trypsin as described in Ref. 2. The resulting peptide mixture was then injected onto a Synchropak AX300 anion-exchange HPLC column equilibrated with 20 mM potassium phosphate, pH 6.6, 4.8% ethanol, 40 mM KCl. The column was eluted at a flow rate of 1.0 ml min⁻¹ with linear gradients of 0.8 M KCl in 20 mM potassium phosphate, 4.8% ethanol (buffer B) into 20 mM potassium phosphate, 48% ethanol (buffer A), 0–30 min (5–50% buffer B), 30–63 min (50–70% buffer B), and 63–99 min (70–100% buffer B). The elution positions of the SSB carboxyl-terminal tryptic peptide (T-14,15, residues 155–177) and of free [³²P](dT)₈ are indicated above.

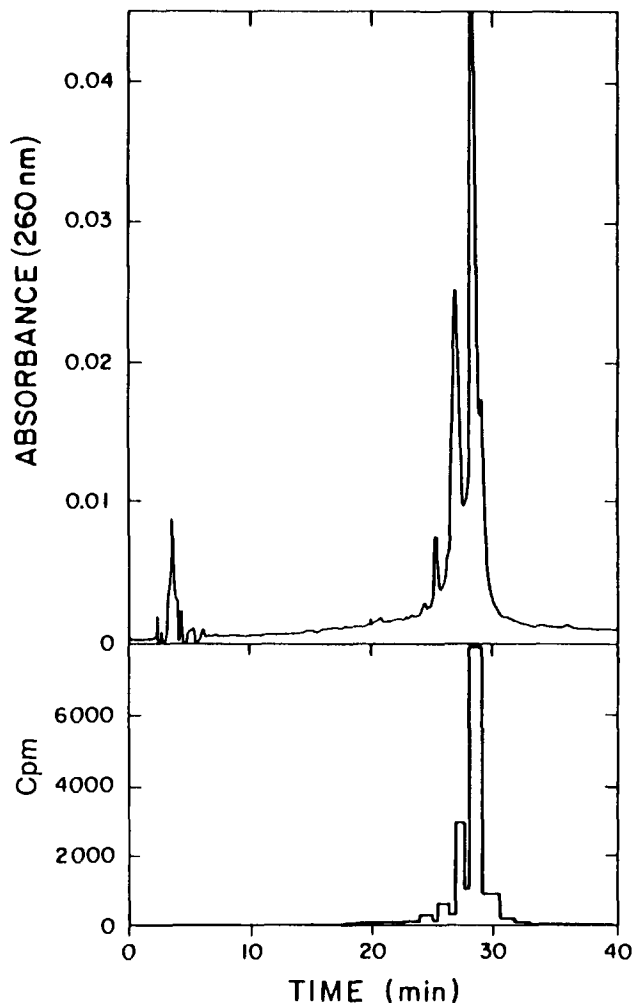


FIG. 3. Ion-pairing HPLC of the $[^{32}\text{P}](\text{dT})_8$ cross-linked tryptic peptide from SSB. The cross-linked peptide isolated via anion-exchange HPLC (see Fig. 2) was further purified as described in Ref. 2 by ion-pairing HPLC on a Waters C_{18} column equilibrated in 90% 0.1 M triethylammonium acetate (buffer A) and 10% acetonitrile (buffer B). The column was eluted with a linear gradient from 10 to 25% buffer B over 40 min at a flow rate of 1 ml/min. Both major peaks eluted contain the same cross-linked peptide, which corresponds to SSB tryptic peptide T-8 (residues 57–62).

nylthiohydantoin derivative of a particular amino acid residue following sequencing of the cross-linked peptide. As shown in Fig. 4, in the case of the *E. coli* SSB cross-linked peptide, the phenylthiohydantoin derivative of Phe-60, which corresponds to cycle 4 in Fig. 4, was missing. Further evidence suggesting Phe-60 as the site of cross-linking was provided by subjecting a portion of the trifluoroacetic acid wash from each sequence cycle to scintillation counting. As expected, most of the ^{32}P label was released at cycle 4.²

As summarized in Table IV,¹²⁻¹⁴ at least four different amino acids (cysteine, phenylalanine, tyrosine, and methionine) have been identified as photochemical cross-linking sites in nucleic acid-binding proteins. In half of the examples cited in Table IV, oligo(dT)₈ was the nucleic acid whereas in the other half a naturally occurring, high-molecular-weight nucleic acid was cross-linked. Although the A1 hnRNP protein binds RNA *in vivo*, a deoxyoligonucleotide was used for the cross-linking experiment because of its high efficiency of cross-linking. In general, single-stranded RNA versus single-stranded DNA-binding proteins have only about an order of magnitude difference in specificity with respect to sugar type¹⁵ and hence it seems reasonable in these instances to take advantage of whichever oligonucleotide has the higher cross-linking efficiency. An appropriate control would be to determine whether an oligonucleotide such as (U)₈ could effectively compete in a [^{32}P](dT)₈ photo-cross-linking assay. Of the proteins listed in Table IV the A1 hnRNP was the only one in which multiple cross-linking sites were found. Phe-16 is the major A1 cross-linking site since it accounted for nearly 65% of the bound [^{32}P](dT)₈. Approximately 10–15% cross-linking occurred at each of the remaining three sites in A1, which together actually account for two analogous pairs of cross-linking sites. That is, A1 contains an approximately 90-amino acid internal region of sequence similarity such that when residues 3–93 are aligned with residues 94–194, 32% of the amino acid residues in these two regions are identical.¹⁶ Hence, Phe-16 and Phe-107 as well as Phe-58 and Phe-149 occupy analogous positions in terms of this 90-amino acid internal region of sequence similarity. The finding that both of these 90-amino acid domains can be cross-linked to [^{32}P](dT)₈ greatly strengthened the notion that each of these domains corresponds to an independently folded nucleic acid-binding domain.³ In the case of the L4 and S7 ribosomal proteins the

¹² P. Maly, J. Rinke, E. Ulmer, C. Zwieb, and R. Brimacombe, *Biochemistry* **19**, 4179. (1980).

¹³ K. Moller, C. Zwieb, and R. Brimacombe, *J. Mol. Biol.* **129**, 489 (1978).

¹⁴ C. Zwieb and R. Brimacombe, *Nucleic Acids Res.* **6**, 1775 (1979).

¹⁵ J. Chase and K. R. Williams, *Annu. Rev. Biochem.* **55**, 103 (1986).

¹⁶ B. M. Merrill, M. B. LoPresti, K. L. Stone, and K. R. Williams, *J. Biol. Chem.* **261**, 878 (1986).

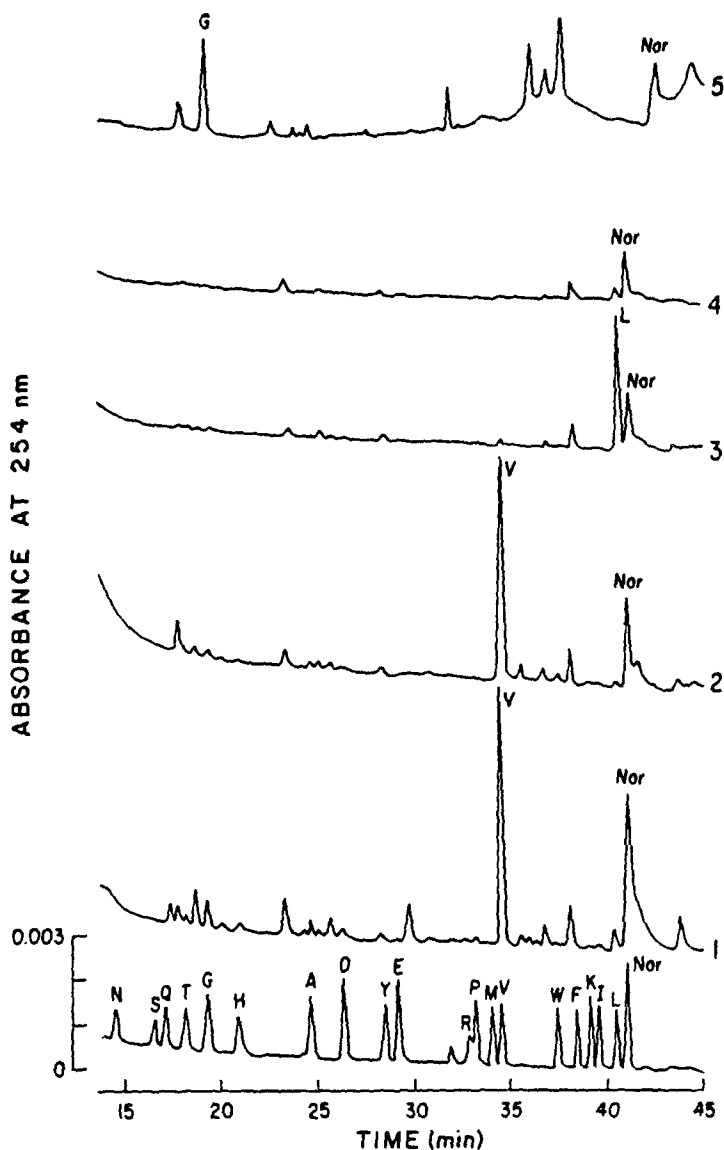


FIG. 4. HPLC traces of phenylthiohydantoin derivatives from cycles 1–5 of the solid-phase sequencing of the SSB- ^{32}P (dT) $_8$ cross-linked peptide coupled to an aminopolystyrene resin. An internal standard of 200 pmol norleucine (Nor) was added to each sequenator cycle. Samples were run on a Waters HPLC system as described in Ref. 2. The sequence is indicated by the one-letter amino acid symbol adjacent to the corresponding peak. The bottom tracing represents 100 pmol of a standard phenylthiohydantoin derivative mixture. The norleucine (Nor) peak in cycle 5 elutes slightly later than those in cycles 1–4 because cycle 5 was run on the HPLC on a different date.

TABLE IV
IDENTIFICATION OF PHOTOCHEMICAL CROSS-LINKING SITES
IN NUCLEIC ACID-BINDING PROTEINS

Protein	Nucleic acid	Cross-linked amino acid	Ref.
Bacteriophage fd gene 5	fd ssDNA	Cys-33	4,7
Bacteriophage T4 gene 32	(dT) ₈	Phe-183	6
<i>E. coli</i> SSB	(dT) ₈	Phe-60	2
Human A1 hnRNP	(dT) ₈	Phe-16	3
		Phe-58	
		Phe-107	
		Phe-149	
<i>E. coli</i> 50S ribosomal protein S7	23S RNA	Tyr-35	12
<i>E. coli</i> 30S ribosomal protein S7	16S RNA	Met-114	13,14

exact site of cross-linking was located both with respect to the protein as well as the nucleic acid. In both cases a single uridine residue was involved that was contained within the sequence A-A-U-G-G in 16S RNA^{13,14} and within the sequence A-A-U-A-G- in 23S RNA.¹² Despite the observation that with the T4 gene 32 protein (rU)₁₀ has less than 0.5% of the laser cross-linking efficiency of (dT)₁₀,^{1a} uridine does provide a high degree of cross-linking in the case of the L4 and S7 ribosomal proteins.

Use of Nucleotide Analogs to Increase Photocross-Linking Efficiency of Nucleic Acid-Protein Complexes

Incorporation of photosensitive analogs into nucleic acids may greatly increase their cross-linking efficiency, thus enabling the study of protein-nucleic acid complexes that would not normally form cross-links. An added benefit of this approach is that the photon flux required to obtain cross-linking is greatly reduced and, hence, problems due to photodenaturation of the protein are minimized. In the case of the bacteriophage T4 gene 32 protein, 20% cross-linking to (dT)₈ required 9×10^4 ergs/mm², which resulted in inactivating nearly 33% of the free protein that was initially present.⁶ In contrast, by using poly(adenylic-8-azidoadenylic) acid, which had been prepared with 8-azidoadenosine 5'-diphosphate and polynucleotide phosphorylase, nearly 100% cross-linking efficiency was achieved¹⁷ with a dose (9×10^2 ergs/mm²) that was equal to only 1% of the photon flux previously used for (dT)₈.⁶ Similarly, 50% cross-linking of the *lac* repressor to bromodeoxyuridine-substituted *lac* operator required

¹⁷ R. L. Karpel, V. Y. Levin, and B. E. Haley, *J. Biol. Chem.* **262**, 9359 (1987).

8×10^3 ergs/mm² (Ref. 18), which is considerably less than the dosages that are typically used (Table III) in the absence of a photosensitive nucleotide analog. Other photosensitive analogs that have been used include 5-azido-2'-deoxyuridine 5'-triphosphate, which can be incorporated into DNA by DNA polymerase I,¹⁹ and 4-thio-UTP.²⁰ In the latter case the 4-thio-UTP was incorporated into RNA transcripts by RNA polymerase III transcription complexes that could then be photo-cross-linked to the nascent RNA transcripts by irradiating with ultraviolet light above 280 nm.²⁰

Azido derivatives of nucleotides have also been cross-linked to proteins in high yield. In the case of the complex formed between *E. coli* RecA protein and 8-azido-ATP, a maximum of about 15% cross-linking was achieved²¹ and a single tyrosine residue was found to be labeled.²² Another system that has been examined in detail is 8-azido-dATP cross-linking to the Klenow fragment of DNA polymerase. A maximum of 50% cross-linking was achieved and covalent bonds were formed at five adjacent chemical groups centered around Tyr-766²³: that is, the peptide carbonyl of Ile-765, the α -carbon of Tyr-766, and the peptide amide, β -carbon, and γ -carbon of Tyr-766. Cross-linking to either the peptide carbonyl or α -carbon was manifested by an inability to sequence past Leu-764 and Ile-765, respectively.²³ One potential disadvantage of the use of the azido photoaffinity label is that, as in the case of the Klenow fragment, the high reactivity of the azido group may result in cross-linking to multiple sites around a single amino acid residue, which then may result in heterogeneity on reversed-phase HPLC peptide mapping. Thus, HPLC fractionation of a tryptic digest of the 8-azido-[³²P]dATP-labelled Klenow fragment resulted in five different radioactive peaks (corresponding to each of the five proposed cross-linking sites around Tyr-766), all of which separated and contained the same tryptic peptide.²³

Another approach that has been used to decrease the extent of photoinactivation while cross-linking proteins to nucleotides is the use of a photosensitizer such as acetone. Whereas irradiation of ribonuclease with ultraviolet light of wavelength equal to 260 nm or above resulted in 80% inactivation after 1 hr, similar irradiation with ultraviolet light of wavelength equal to 300 nm or above in the presence of 5% acetone resulted in

¹⁸ S.-Y. Lin and A. D. Riggs, *Proc. Natl. Acad. Sci. U.S.A.* **71**, 947 (1974).

¹⁹ R. K. Evans, J. D. Johnson, and B. E. Haley, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 5382 (1986).

²⁰ B. Bartholomew, C. F. Meares, and M.E. Dahmus, *J. Biol. Chem.* **265**, 3731 (1990).

²¹ K. L. Knight and K. McEntee, *J. Biol. Chem.* **260**, 867 (1985).

²² K. L. Knight and K. McEntee, *J. Biol. Chem.* **260**, 10185 (1985).

²³ J. Rush and W. H. Konigsberg, *J. Biol. Chem.* **265**, 4821 (1990).

less than 60% inactivation.²⁴ Under the latter conditions the extent of cross-linking of ribonuclease to pUp was about 15%.²⁴ The primary sites of pUp cross-linking to ribonuclease appeared to be Ser-80 and Thr-82.²⁵

DNA Affinity Labeling of Proteins Binding to Specific Sequences in Double-Stranded DNA

Numerous sequence elements in double-stranded DNA have been identified that exhibit specificity for the binding of certain proteins. Usually these sequence specific DNA-binding proteins are involved in the regulation of gene expression. To aid in the identification and characterization of these proteins, Safer *et al.*^{1b} have developed a method that allows transfer of ³²P from specifically labeled phosphodiester bonds at or near the center of the binding site to the protein by ultraviolet irradiation of a mixture of proteins (usually present in a cellular extract or in an eluate from chromatographic columns used to fractionate the extract) with a double-stranded deoxynucleotide probe that has been labeled with ³²P in one phosphodiester bond in both the coding and in the noncoding strands. Safer *et al.*^{1b} used three different methods to introduce the ³²P label at single internal phosphodiester bonds on each DNA strand (the Safer *et al.* reference should be consulted for the experimental details of probe labeling). Incubations with the probe were carried out at 24° for 10 min with about 2 μ l of concentrated protein fraction in a total of 20 μ l of a buffer [containing 40 mM(NH₄)₂SO₄, 1 mM EDTA, 20 mM Tris-HCl (pH 7.9), and 1 mM dithiothreitol]. Generally 10 fmol of the ³²P-labeled DNA probe was incubated with a solution that contained between 1 to 10 μ g of total protein. Photolysis was performed with UV light (260 to 280 nm) at 4° with an average intensity of 600 μ W/cm². After a specified time of irradiation, CaCl₂ was added to a final concentration of 10 mM followed by addition of 50 units of micrococcal nuclease. After a 30-min digestion at 37° a portion of each sample was subjected to SDS-PAGE followed by autoradiography.

In this situation described here, sequence-specific proteins bind stably at the site into which ³²P-labeled phosphodiester bonds have been introduced while nonspecific protein-DNA interactions are randomly distributed across the fragment and are weaker. The extensive nuclease digestion leaves only small oligonucleotides covalently linked to the proteins. Some of these oligonucleotides contain the ³²P-labeled specific binding sequence. The SDS-PAGE separation followed by autoradiography allows the visualization of only those protein(s) that have been photo-cross-linked to the

²⁴ J. Sperling and A. Havron, *Biochemistry* **15**, 1489 (1976).

²⁵ A. Havron and J. Sperling, *Biochemistry* **16**, 5631 (1977).

labeled region of the probe. It also permits the estimation of the apparent molecular weight of the protein, since the small oligonucleotide remaining attached to the protein would not significantly alter its mobility. Using this technique Safer *et al.*^{1b} identified two proteins having molecular weights of 116,000 and 45,000 from a HeLa whole-cell extract that could be covalently linked by this procedure to ³²P-labeled probes containing the adenovirus type 2 major late upstream promoter sequence (UPS). One of these probes was a 63-base pair (bp) fragment and another contained 357 bp. Unlabeled competitor DNA fragments containing the same internal UPS-binding sequence were used as competitors and, as expected, the extent of UV cross-links diminished with increasing concentration of the competitor DNA. A DNA fragment without the UPS failed to diminish the extent of ³²P-labeled cross-link product when the competitor fragment was incubated at high molar excess in the presence of the ³²P-labeled probe containing the UPS. When a shorter, 38-bp probe containing within it the UPS ³²P labeled at position 19 was used, the specificity of labeling was reduced because of the formation of covalent links with end-binding proteins that extend far enough along the DNA oligomers to allow them to be cross-linked. This problem could be overcome by labeling in the presence of higher concentrations of nonspecific unlabeled DNA competitors that increase the signal-to-noise ratio. One of the requirements for this method to succeed is the presence of ³²P phosphodiester bonds adjacent to pyrimidines within the binding site. Most often the sequence of these binding sites are such that this requirement is met. The fact that the probes contain only normal nucleotides rather than analogs such as bromodeoxyuridine, which has been used in other cross-linking studies, is an advantage since UV-induced breaks at bromodeoxyuridine sites in the DNA probes are avoided and thus the background of nonspecific end binding proteins that might otherwise be visualized in this procedure is minimized. Finally Safer *et al.*^{1b} point out that if an excess of the labeled probe is used and the duration and/or intensity of the photolysis is determined experimentally by the methods described, then this technique may be useful for estimating the pool size of these DNA-binding proteins in a crude extract. They state that, because of its sensitivity, it is possible to detect proteins that would not ordinarily be found with a DNase I footprint assay.

Discussion

Factors that Influence Rate and Extent of Photochemical Cross-Linking

The extent of protein-nucleic acid cross-linking that can be achieved is a complex function of several factors that relate both to the experimental

conditions as well as to the intrinsic nature of the nucleic acid, the protein, and the three-dimensional structure of the resulting complex. An important variable often overlooked is the fraction of the protein in the reaction mixture that is actually complexed with the nucleic acid. That is, is the final salt concentration low enough and the final protein and nucleic acid (binding site) concentration high enough to ensure that most of the protein is actually in a protein–nucleic acid complex as opposed to being free in solution? If the K_{app} for the complex is not known then preliminary experiments should be carried out where the extent of protein cross-linking is measured as a function of an increasing concentration of nucleic acid. The value of this approach is illustrated in Table III, which shows that increasing the molar ratio of oligonucleotide to UPI from 2.5 to 8 resulted in increasing the extent of cross-linking from about 40% to nearly 86%. Obviously, at the UPI concentration used in this experiment (Table III), the 2.5 molar ratio of oligonucleotide/UPI was not sufficiently high to drive most of the protein into a complex. Another important variable is the dose of ultraviolet light that is used. While it is best to generate a dose–response curve similar to that in Fig. 1 for each system that is being studied, a total dose of about 2.0×10^5 ergs/mm², based on the data in Table III, has so far proved to be near optimum. With a General Electric 15-W germicidal lamp this does corresponds to about a 20-min exposure at a distance of about 7.5 cm. Once an optimum dose has been found for the system under study then it is important to determine the extent of photoinactivation of the protein that is occurring with this dose and, using one or more of the methods discussed in the text, to demonstrate that only the native, as opposed to the denatured protein, can be cross-linked and then only when it is complexed with nucleic acid. By carrying out a full dose–response study it is possible to determine rapidly the maximum dose of ultraviolet light that is consistent with minimal photoinactivation of the protein and photodegradation of the resulting amino acid–base adduct.

While there appears to be agreement that thymine is the most photoreactive deoxynucleotide and uridine the ribonucleotide base yielding the greatest extent of photochemical cross-linking to proteins, no data are currently available on the relative tendencies of the 20 different amino acids (when they are in proteins as opposed to being free in solution) to participate in adduct formation with a photochemically excited base. In general, model studies carried out on amino acid–nucleotide mixtures seem to be of little use in this regard. For example, while Schott and Shetlar²⁶ failed to observe any phenylalanine–thymine cross-linking (whereas they observed significant thymine cross-linking to cysteine, lysine, and arginine), phenylalanine accounts for six of the nine cross-linking

²⁶ H. N. Schott and M. D. Shetlar, *Biochem. Biophys. Res. Commun.* **59**, 1112 (1974).

sites that have been identified in the six nucleic acid-binding proteins listed in Table IV. The high propensity for phenylalanine to be cross-linked may reflect an unusually high reactivity for this amino acid when it is held in a certain configuration in proteins or it may simply reflect the essential role that this amino acid may play in the overall mechanism of binding of proteins to single-stranded nucleic acids. There is considerable biochemical and physicochemical data that suggest that the bacteriophage T4 gene 32, the *E. coli* SSB, and the A1 hnRNP proteins (which together account for all of the phenylalanine cross-linking sites identified in Table IV) share a common mechanism of binding involving the close approach of aromatic amino acid side chains with the bases of the single-stranded polynucleotide (see Ref. 15 for a review). In the absence of any relevant data, it seems prudent to assume that any of the 20 different amino acids found in proteins can in principal be photo-cross-linked to nucleic acids.

Because ultraviolet light is a "zero-length" cross-linker, an amino acid residue would have to be in extremely close proximity to cross-link to an excited base. In addition, studies on the bacteriophage fd gene 5 protein suggest that topology must also play an essential role in determining the ability of amino acids to cross-link to nucleic acids. Hence, even though ¹H nuclear magnetic resonance (NMR) data requires that Tyr-26 and Phe-73 form part of the DNA-binding domain of the bacteriophage fd gene 5 protein²⁷ and the data in Table IV demonstrate that both of these amino acids can be cross-linked to nucleic acids, neither of these residues can in fact be cross-linked in the gene 5-ssDNA complex.^{4,7} Despite these studies on the gene 5 protein, which suggest that a relatively specific topological arrangement between an amino acid and a nucleotide base must be reached in order to achieve photochemical cross-linking, the fact remains that all five of the single-stranded nucleic acid-binding proteins (bacteriophage T4 gp32, *E. coli* and F plasmid SSB, calf thymus UPI, and in Refs. 4 and 7 the bacteriophage fd gene 5 protein) that have been subjected to UV photolysis in the presence of oligonucleotides can in fact be cross-linked to oligo(dT) or poly(dT) with an efficiency above 20%.

Photochemical Cross-Linking Limited to Those Amino Acid Residues in Native Protein that Are at Interface of Protein-Nucleic Acid Complex

Without exception, all of the numerous controls that have been used, including prior denaturation of the protein and the use of competing nucleic acids and high salt concentrations, confirm that only the *native* protein

²⁷ G. C. King and J. E. Coleman, *Biochemistry* **27**, 6947 (1988).

can be cross-linked and only when it is complexed with nucleic acid. That cross-linking is restricted to amino acid residues that are at the interface of the protein–nucleic acid complex is suggested by studies on four proteins whose three-dimensional structures have been solved and whose cross-linking sites have been identified. In the case of the bacteriophage fd gene 5 protein, the cysteine residue that was cross-linked to fd DNA (Cys-33) is located within a predicted DNA-binding groove that is composed of a three-stranded antiparallel β sheet encompassing approximately residues 11–40.^{7,28} The three amino acids that are cross-linked in the ribonuclease A–(pUp) complex (Ser-80, Ile-81, and Thr-82) are known to constitute the bottom of the binding site for the pyrimidine ring.^{25,29} Based on the recently published crystal structure of the U1 small ribonucleoprotein A, which contains a eukaryotic RNA binding motif that is homologous to the two, 90-residue domains found in the A1 hnRNP protein, the two phenylalanine residues that were cross-linked in each of the A1 RNA binding domains (3) are located exactly next to each other on adjacent β -strands that form the proposed RNA binding surface.³⁰ Similarly, 8-azido-dATP cross-linking to Tyr-766 in the Klenow fragment of DNA polymerase I is consistent with the location of the dNTP-binding site as determined by crystallography, chemical modification, footprinting, and NMR spectroscopy (see Ref. 23 for a summary of these studies). Although a three-dimensional structure is not yet available for the *E. coli* SSB ssDNA-binding protein, the involvement of Phe-60 [which was the residue found cross-linked to (dT)₈ (Ref. 2)] in binding has been confirmed by site-directed mutagenesis. Substitution of Phe-60 by several other amino acid residues revealed a direct correlation between the hydrophobicity of the amino acid at this position and the relative *E. coli* SSB affinity for single-stranded nucleic acids.³¹ Since NMR spectroscopy revealed little or no conformational change as a result of these mutations, the conclusion was reached that Phe-60 in SSB is involved with a hydrophobic interaction with ssDNA.³¹

In summary, photo-cross-linking provides a useful approach for identifying one or more amino acid residues at the interface of protein–nucleic acid complexes. This technique requires minimal instrumentation, is equally applicable *in vivo* and *in vitro*, is highly specific, and proceeds with minimal perturbation of the system being studied.

²⁸ G. D. Brayer and A. McPherson, *J. Mol. Biol.* **169** 565 (1983).

²⁹ F. M. Richards and H. W. Wyckoff, in "Atlas of Molecular Structures in Biology" (F. M. Richards and H. W. Wyckoff, eds.), p. 1. Oxford Univ. Press (Clarendon), London and New York, 1973.

³⁰ K. Ngai, C. Oubridge, T. Jessen, J. Li, and P. Evans, *Nature (London)* **348**, 515 (1990).

³¹ I. Bayer, A. Fliess, J. Greipel, C. Urbanke, and G. Maass, *Eur. J. Biochem.* **179**, 399 (1989).

[26] Use of Nonsense Suppression to Generate Altered Proteins

By JEFFREY H. MILLER

Introduction

The study of protein structure–function relationships relies heavily on the ability to generate altered proteins with known sequence changes. The first investigations of altered proteins involved variants that were recognized in the population.^{1,2} Later, mutagens were employed to create altered proteins (e.g., Refs. 3–5). These *in vivo* techniques were responsible for much of the knowledge of the effects of altered proteins, until the onset of the recombinant DNA era in the mid-1970s. At that point, a variety of methods became available that allowed the investigator to program specific changes into cloned genes and create the exact altered protein that was desired. These procedures, some of which are reviewed in this volume, include site-directed mutagenesis,⁶ gene synthesis,⁷ Kunkel mutagenesis,⁸ cassette mutagenesis,⁹ and combinatorial cassette mutagenesis.¹⁰ Each technique can be used to great advantage in the appropriate situation. This chapter describes an additional method for generating altered proteins that is based on the suppression of nonsense mutations. This latter method greatly simplifies the task of examining a large number of amino acid replacements in the protein under study.

¹ V. M. Ingram, *Nature (London)* **180**, 326 (1957).

² M. F. Perutz and H. Lehmann, *Nature (London)* **219**, 902 (1968).

³ C. Yanofsky, B. C. Carlton, J. R. Guest, D. R. Helinski, and U. Henning, *Proc. Natl. Acad. Sci. U.S.A.* **51**, 266 (1966).

⁴ G. Streisinger, F. Mukai, W. J. Dreyer, B. Miller, and S. Horiuchi, *Cold Spring Harbor Symp. Quant. Biol.* **26**, 25 (1961).

⁵ G. Streisinger, Y. Okada, J. Emrich, J. Newton, A. Tsugita, E. Terzaghi, and M. Inouye, *Cold Spring Harbor Symp. Quant. Biol.* **31**, 77 (1966).

⁶ M. Smith, *Annu. Rev. Genet.* **19**, 423 (1985).

⁷ H. G. Khorana, *Science* **203**, 614 (1979).

⁸ T. A. Kunkel, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 488 (1985).

⁹ J. A. Wells, M. Vasser, and D. B. Powers, *Gene* **34**, 315 (1985).

¹⁰ J. F. Reidhaar-Olson and R. T. Sauer, *Science* **241**, 53 (1988).

The Principle of the Method

Nonsense mutations signal chain termination during protein synthesis (see reviews by Eggertsson and Soll¹¹ and Gorini¹²). In bacteria, and in most other organisms, the triplets UAG, UAA, and UGA serve as nonsense codons. In the absence of tRNAs that recognize these codons (the normal situation), protein release factors mediate chain termination and release from the ribosome. In *Escherichia coli*, there are two release factors, RF1, which recognizes UAG and UAA, and RF2, which recognizes UAA and UGA. The effects of nonsense mutations can be reversed by suppressor mutations, which allow some translation through the nonsense codon. The nonsense codons have alternate names. UAG is termed "amber," UAA "ochre," and UGA "opal." The initial suppressors were found to result in altered tRNAs, most of which carried a change at the anticodon of a dispensible copy of a tRNA, now enabling it to read one of the three nonsense codons. Each suppressor tRNA inserted a specific amino acid, namely the same amino acid inserted by the unaltered tRNA. Subsequent work, however, showed that at least one suppressor tRNA inserted a different amino acid than the original tRNA. When the tRNA^{Trp} is converted to a UAG suppressor by a change at the anticodon, glutamine is inserted at a UAG site 95% of the time.^{13,14} For a summary of many of the suppressors found *in vivo*, see the reviews by Gorini¹² and by Eggertsson and Söll.¹¹

The initial set of suppressors allowed the insertion of up to five amino acids at a UAG codon. These included glutamine, tyrosine, serine, and leucine, at efficiencies greater than 10%, and lysine at efficiencies from 1 to 2%. Lysine, glutamine, and tyrosine could also be inserted at UAA codons, and tryptophan could be inserted at a UGA codon. In 1979, these suppressors were used on a collection of 90 nonsense mutations,¹⁵ most of which were UAG mutations, in the *lacI* gene of *E. coli*, which encodes the *lac* repressor protein. The approximately 300 altered repressors that were generated by this technique, each with a known amino acid replacement, did represent an extensive collection of mutant proteins with interesting properties. However, the limitations of this procedure were evident. Namely, only certain amino acids could be inserted, because of the very small set of nonsense suppressors available. Also, nonsense mutations

¹¹ G. Eggertsson and D. Söll, *Microbiol. Rev.* **52**, 354 (1988).

¹² L. Gorini, *Annu. Rev. Genet.* **4**, 107 (1970).

¹³ M. Yaniv, W. R. Folk, P. Berg, and L. Soll, *J. Mol. Biol.* **86**, 245 (1974).

¹⁴ J. Celis, C. Coulondre, and J. H. Miller, *J. Mol. Biol.* **104**, 729 (1976).

¹⁵ J. H. Miller, C. Coulondre, M. Hofer, U. Schmeissner, H. Sommer, A. Schmitz, and P. Lu, *J. Mol. Biol.* **131**, 191 (1979).

constructed as a set of four to six oligonucleotides, as shown in Fig. 2. When the synthetic genes were reintroduced into *E. coli* cells, suppressor activity was monitored by measuring the level of β -galactosidase produced by the *lacZ* gene carrying a UAG mutation. Figure 3 summarizes the methodology. The nature of the amino acid inserted was determined by purifying dihydrofolate reductase produced in a strain carrying a UAG mutation early in the plasmid-encoded *fol* gene, which encodes dihydrofolate reductase, depicted in Fig. 4. The amino-terminal end of the suppressed protein was then sequenced.

Some of the newly constructed suppressors operated at high efficiency, whereas others did not. Therefore attempts were made to improve the level of suppression.¹⁷ In some cases, changing the bases directly following the anticodon to -A-A- improved the efficiency of suppression (Fig. 5), as expected from previous work.^{19,20} In one case, a hybrid tRNA consisting of the anticodon stem and loop of the phenylalanine tRNA UAG suppressor and the remainder of the molecule from the proline tRNA was constructed (Fig. 6) and found to operate at high efficiency,¹⁷ in contrast to the parent proline tRNA suppressor, which did not function as a suppressor at all. In a second case, McClain and Foss²¹ used a variant of a hybrid tRNA^{Phe}/tRNA^{Arg} suppressor to construct an arginine-inserting tRNA amber suppressor.

The determinants for correct recognition of tRNAs by their cognate synthetases are under intense study (see review by Schulman and Abelson²²). At least two classes of tRNAs can be identified. Some tRNAs do not appear to use the anticodon as part of the recognition, whereas others rely heavily on the anticodon for synthase recognition. The first category can be converted into nonsense suppressors and still retain recognition by the correct synthase, but the second cannot, at least not without additional constructions. The sequencing results of the suppressed dihydrofolate reductases bear out these predictions, since in some cases the amino acid inserted by the suppressor is the correct one, and in other cases it is either a mixture of amino acids, or else is predominantly or exclusively either glutamine or lysine.

Table I summarizes the information on the efficiency and nature of the amino acid inserted for a series of suppressors, including both the naturally occurring suppressors and those derived by *in vitro* constructions.

¹⁹ M. Yarus, *Science* **218**, 646 (1982).

²⁰ D. Bradley, J. V. Park, and L. Soll, *J. Bacteriol.* **145**, 704 (1981).

²¹ W. H. McClain and K. Foss, *Science* **241**, 1804 (1988).

²² L. H. Schulman and J. Abelson, *Science* **240**, 1591 (1988).

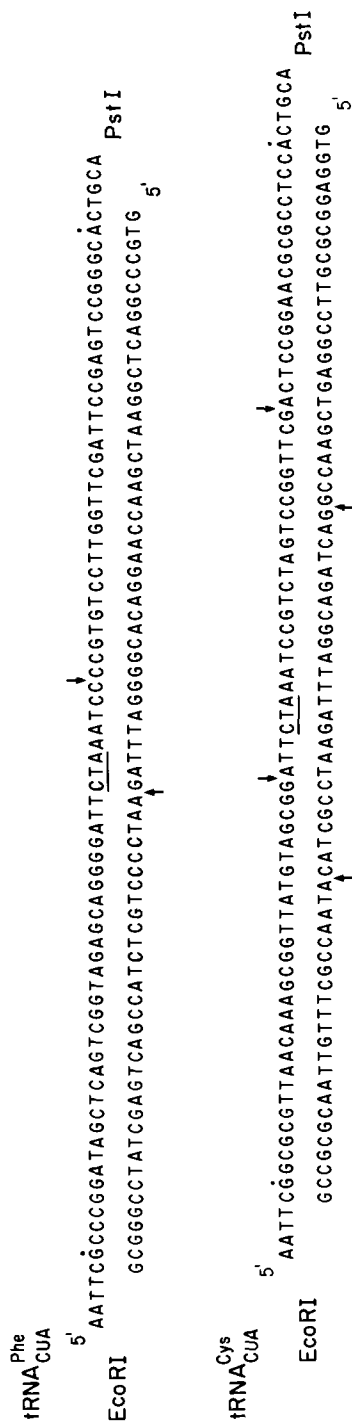


Fig. 2. Synthetic genes for tRNA^{Phe} and tRNA^{Cys} . The first and last nucleotides encoding each tRNA (indicated by a dot) are immediately flanked by the cohesive ends of *EcoRI* and *PstI* restriction endonuclease sites, respectively. Arrows indicate the oligonucleotide junctions, and the altered anticodon is underlined. Note that the tRNA^{Phe} gene does not encode the 3'-terminal C and A residues. These residues are added *in vivo* by the tRNA nucleotidyltransferase [M. P. Deutscher, *Crit. Rev. Biochem.* **17**, 45 (1984)].

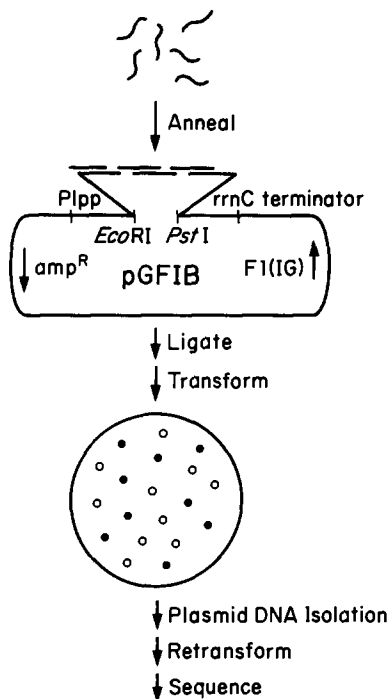


FIG. 3. The steps involved in constructing a synthetic tRNA suppressor gene that suppresses a *lacZ* amber mutation (see text).

Advantages and Disadvantages

Let us examine some of the advantages and disadvantages of using nonsense suppressors. The strong point of the technique is that it makes possible the generation of a large number of variants with a relatively small amount of work. If mutations are being generated by site-directed mutagenesis, then by constructing an amber (UAG) mutation at a given site, 12–13 different mutants are created simply by employing different suppressors. It becomes a feasible project to examine 200 altered proteins with known sequence changes simply by constructing 16 site-directed mutations. As will be seen below, systematic studies of amino acid exchanges can be carried out that would not otherwise be attempted. This method thus serves as an important extension to simple site-directed mutagenesis, since it offers a 12- to 13-fold payoff for each amber mutation constructed by site-directed mutagenesis. In addition, in cases where it is important to control for unwanted secondary mutations resulting from site-directed mutagenesis, only the amber construct needs to be sequenced in its entirety, rather than each individual missense construct.

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DHFR wt: Ile Ser Leu Ile Ala Ala Leu Ala Val Asp Arg
 ATC AGT CTG ATT GCG GCG TTA GCG GTA GAT GCG

DHFR am: am Asn . . .
 TAG AAT . . .

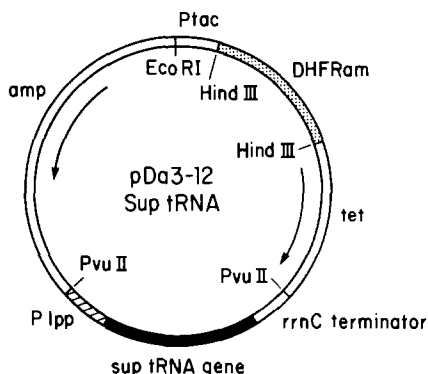


FIG. 4. Plasmid used to determine the amino acid inserted by different amber suppressors. The *E. coli fol* gene was mutated to an amber at the tenth coding position, and the following codon was also changed from GAT to AAT. [For details of the construction see J. Normanly, J.-M. Masson, L. G. Kleina, J. Abelson, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 6548 (1986).]

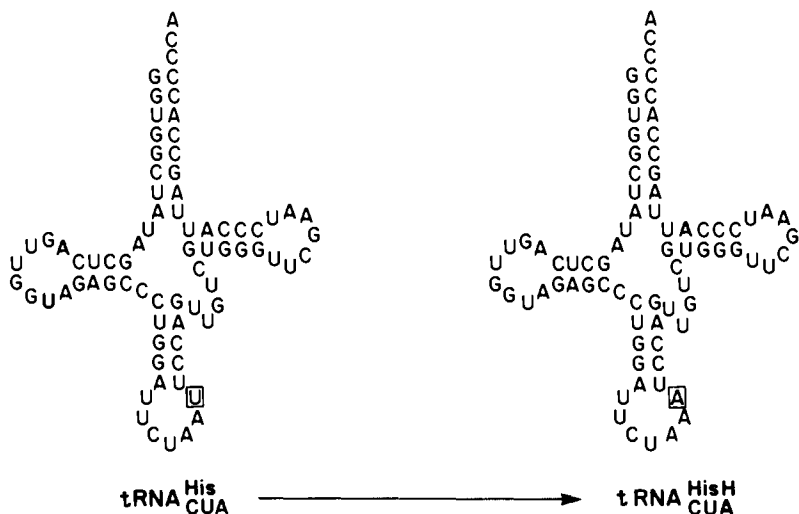


FIG. 5. Alterations of tRNA^{His}. The suppressor tRNA derived from a histidine tRNA, and resulting from a synthetic tRNA gene is shown, as is the change made near the anticodon that improves suppression efficiency.

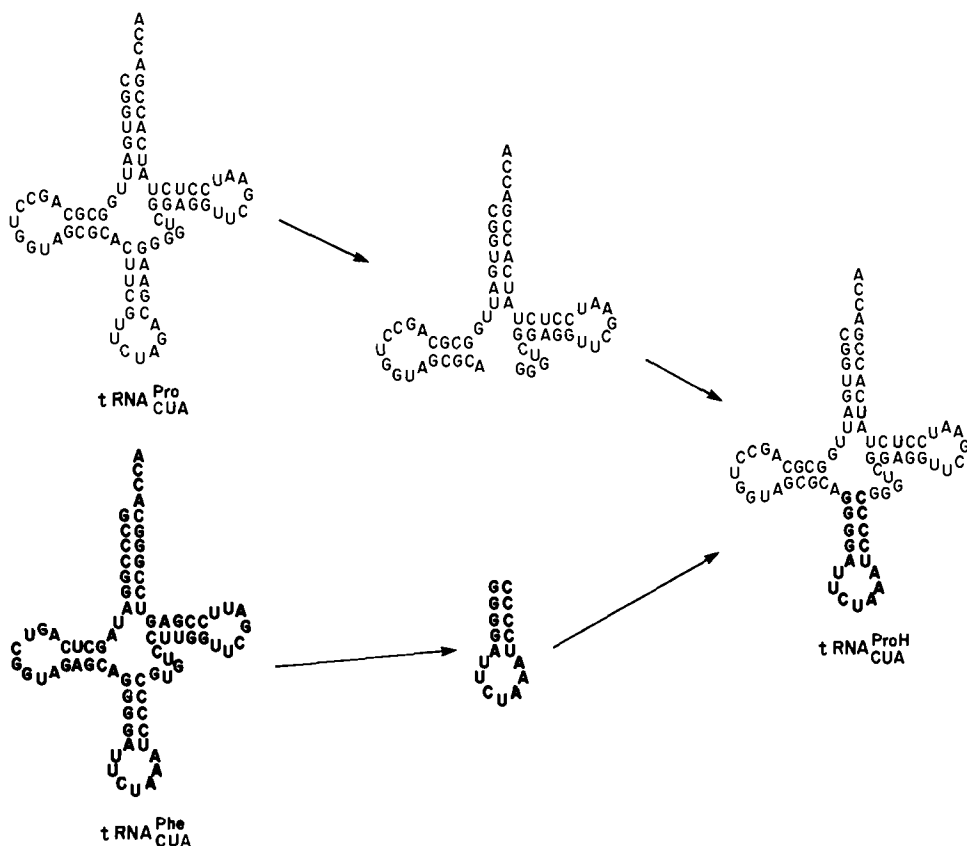


FIG. 6. Construction of tRNA^{ProH}. The sequences used to produce a hybrid tRNA via gene synthesis, derived from sequences of tRNA^{Phe} and tRNA^{Pro}, are shown.

There are potential pitfalls, however, some of which can be overcome with additional experiments. First, there are still seven amino acids that are not inserted efficiently via nonsense suppression. If one of these residues (valine, isoleucine, aspartic acid, asparagine, methionine, tryptophan, or threonine) is to be examined, then additional mutations must be generated. It is anticipated that at least some of the remaining seven amino acids will also be able to be inserted via amber suppression, since efforts to convert the altered recognition by tRNA synthases are underway in several laboratories. However, as will be seen below, the consequences of 12 or 13 amino acid replacements reveal an enormous amount of information regarding the tolerance to substitutions of a specific amino acid residue. Second, the efficiency varies for each suppressor, and for each amber site. Although

TABLE I
EFFICIENCY OF SUPPRESSION AND SPECIFICITY OF INSERTION OF tRNA SUPPRESSORS^a

Suppressor	Gene	Amino acid inserted	Efficiency of suppression at UAG codons (%)
Su1	<i>supD</i>	Serine	6–54
Su2–89	<i>supE</i>	Glutamine	32–60
Su3	<i>supF</i>	Tyrosine	11–100
Su5	<i>supG</i>	Lysine	0.6–3; 5–29 ^b
Su6	<i>supP</i>	Leucine	30–100
tRNA ^{Ala} _{CUA}	Synthetic	Alanine	8–83
tRNA ^{Cys} _{CUA}	Synthetic	Cysteine	17–54
tRNA ^{Glu} _{CUA}	Synthetic	Glutamic acid (80%), glutamine (20%)	8–100
tRNA ^{Gly} _{CUA}	Synthetic	Glycine	24–100
tRNA ^{His} _{CUA}	Synthetic	Histidine	16–100
tRNA ^{Lys} _{CUA}	Synthetic	Lysine	9–29
tRNA ^{Phe} _{CUA}	Synthetic	Phenylalanine	48–100
tRNA ^{Pro} _{CUA}	Synthetic	Proline	9–60
FTOIΔ26	Synthetic	Arginine	4–28; 4–47 ^b

^a Taken from L. G. Kleina, J.-M. Masson, J. Normanly, J. Abelson, and J. H. Miller, *J. Mol. Biol.* **213**, 705 (1990).

^b Assayed in a strain carrying the *mar* mutation (S. M. Ryden and L. A. Isaksson, *Mol. Gen. Genet.* **193**, 38 (1984).

for the most part the efficiency range is only severalfold, there are cases where a greater range is exhibited. If threshold levels of a protein need to be achieved for activity, or if the actual specific activity of a suppressed protein is being determined, the variation of levels is not a significant factor. Third, there is the possibility that certain suppressors add a second amino acid at undetectable (less than 2–3%) levels. This could complicate certain analyses. The best use of the method is to screen a large number of altered proteins to find the few very interesting ones, and then to reconstruct these interesting mutants by designing missense mutations. This would solve both the variation in amount problem, and also the potential problem from minute amounts of a second amino acid being inserted at an amber site. Finally, there are potential complications arising from fragments resulting from the chain-terminating codon. These fragments can occur by two different mechanisms. Since nonsense suppression is rarely 100%, there will always be some level of the polypeptide containing the amino-terminal portion of the protein up until the point corresponding to the amber fragment. If this fragment is not degraded and can interfere with the wild-type protein (negative complementation), then difficulties arise with the nonsense suppression method. Also, sometimes

reinitiation of protein synthesis after chain termination can occur at internal reinitiation sites.²³⁻²⁵ In the *lacI* gene, this occurs at several positions in the beginning of the mRNA, resulting in fragments that negatively complement wild-type repressor.²⁶ Since reinitiation is rarely efficient, and since only the unsuppressed nonsense codons are subject to subsequent reinitiation, this problem can be overcome by very efficient suppression, or by converting the reinitiation codons to other codons still specifying the same amino acid.²⁷

Types and Sources of Suppressors Available

In order to employ nonsense suppressors, it is necessary to be able to place each suppressor in combination with each nonsense mutation in the same strain. If the constructions for achieving this are very laborious, then it defeats the purpose of using nonsense suppression. Suppressors are currently available on the *E. coli*¹¹ and *Salmonella typhimurium* chromosome,²⁸ on different plasmids, and on specialized transducing phage such as λ and $\phi 80$. Although the simplest situation is to have the suppressor on the chromosome, this is not always possible. Also, some of the synthetic suppressors have greatly reduced efficiency when encoded by single-copy genes on the chromosome. On the other hand, greatly increased copy number of some of the suppressor tRNAs inhibits cell growth. Thus, several tRNA suppressors are virtually lethal when cloned onto the very high copy number pUC series of vectors. This effect varies with the suppressor. The serine-inserting Su1 suppressor is very efficient when cloned onto pBR322 and also allows normal growth, whereas the tyrosine-inserting Su3 suppressor causes slow growth and operates inefficiently on the same type of plasmid. Since many genes that will be subjected to suppression of nonsense mutations are carried on plasmids, it is important to employ compatible plasmids for the cloned gene and the cloned suppressor tRNA genes. The synthetic suppressors operate efficiently on a pBR322-based plasmid derived from pEMBL, pGFIB,²⁹ shown in Fig.

²³ A. Sarabhai and S. Brenner, *J. Mol. Biol.* **27**, 145 (1967).

²⁴ T. Grodzicker and D. Zipser, *J. Mol. Biol.* **38**, 305 (1968).

²⁵ T. Platt, K. Weber, D. Ganem, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **69**, 897 (1972).

²⁶ J. G. Files, K. Weber, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **71**, 667 (1974).

²⁷ L. G. Kleina and J. H. Miller, *J. Mol. Biol.* **212**, 295 (1990).

²⁸ F. Winston, D. Botstein, and J. H. Miller, *J. Bacteriol.* **137**, 443 (1979).

²⁹ J.-M. Masson and J. H. Miller, *Gene* **47**, 179 (1986).

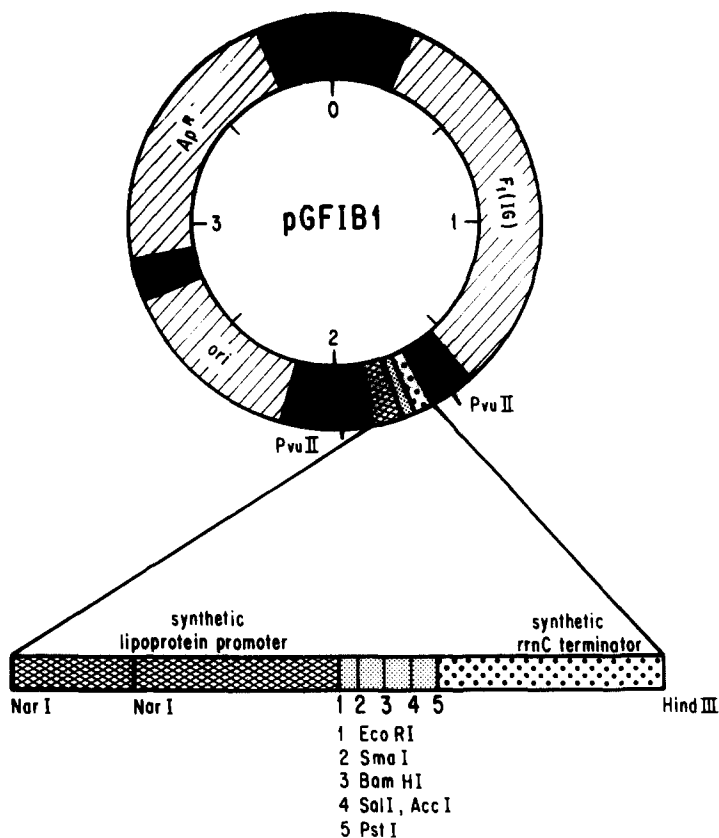


FIG. 7. Plasmid pGFIB-1. This vector [J.-M. Masson and J. H. Miller, *Gene* **47**, 179 (1986)] is a derivative of pEMBL-8+ [L. Dente, G. Cesareni, and R. Cortese, *Nucleic Acids Res.* **11**, 1645 (1983)]. The *lac* promoter was replaced with a synthetically constructed promoter based on the promoter sequence of the *E. coli* lipoprotein gene *lpp* [K. Nakamura and M. Inouye, *Cell (Cambridge, Mass.)* **18**, 1109 (1979)].

7, which would necessitate using a plasmid such as pACYC for the cloned gene carrying the amber mutations. However, different pACYC derivatives of most of the synthetic suppressors have been constructed (R. Cowl, M. Michaels, and C. W. Kim, unpublished, 1991), and although the efficiency of the suppressors is reduced severalfold, strains carrying these plasmids grow much better than those carrying the suppressors on the pEMBL plasmid. The use of pACYC plasmids carrying the synthetic tRNA suppressor genes creates the opportunity to employ pBR322-based plasmids for the cloned gene carrying the amber mutations.

TABLE II
SUPPRESSOR EFFICIENCY IN DIFFERENT CONTEXTS

tRNA suppressor	β -Galactosidase activity (% wild type) ^a			
	A26 ^b /AUG	A16/ACA	O17/CGC	O13/CUG
Su1 (Ser)	25.4	21.1	6.3	26.0
Su2 (Gln)	17.8	9.6	0.8	11.3
Su3 (Tyr)	100	42.6	10.6	46.4
Su6 (Leu)	100	61.7	30.4	57.8
tRNA ^{His} _{CUA}	23.7	17.5	0.8	11.7
tRNA ^{Phe} _{CUA}	100	75.2	48.0	80.8

^a β -Galactosidase activity was determined in a *lacI*-Z fusion system as described by J. H. Miller and A. M. Albertini, *J. Mol. Biol.* **164**, 59 (1983). See also L. G. Kleina, J.-M. Masson, J. Normanly, J. Abelson, and J. H. Miller, *J. Mol. Biol.* **213**, 705 (1990).

^b Site/following codon.

Technical Factors: Procedures and Problems

Context Effects on the Efficiency of Suppression

The efficiency of amber (as well as other nonsense) suppression is affected by the sequence surrounding the amber triplet in the mRNA.^{17,30-32} Studies using *lac* fusions indicate that the most important factor in the mRNA sequence is the two bases following the UAG.^{31,32} The codons fit into several patterns. In general, UAG triplets followed by codons beginning with A are suppressed efficiently, those followed by G are moderately well suppressed, and those followed by U and C are poorly suppressed. One exception is the CUX family of codons, which allows very efficient suppression when following an amber codon. Also, the AUX family of codons allows the most efficient suppression of all, when following an amber codon. Table II gives some examples of different suppressors operating on UAG codons followed by different sequences. Although this information offers the possibility of constructing each nonsense mutation with the most favorable sequence context, in most cases the amino acid following the one corresponding to the UAG codon would be substituted, a situation one clearly wishes to avoid. Therefore, in order to maintain the same surrounding amino acid sequence, one is limited to completely silent changes. The three amino acids encoded by six different triplets (leucine,

³⁰ L. Bossi and J. R. Roth, *Nature (London)* **286**, 123 (1980).

³¹ L. Bossi, *J. Mol. Biol.* **164**, 73 (1983).

³² J. H. Miller and A. M. Albertini, *J. Mol. Biol.* **164**, 59 (1983).

TABLE III
EFFECT ON SUPPRESSION EFFICIENCY OF
CODONS FOLLOWING UAG TRIPLET

Codon following UAG triplet	Effect on UAG suppression
Leucine	
CUX	Favorable
UUA, UUG	Unfavorable
Arginine	
CGX	Unfavorable
AGA, AGG	Favorable
Serine	
UCX	Unfavorable
AGU, AGC	Favorable

serine, and arginine) offer the only possibilities for improving the context without changing the amino acid sequence. Thus, when an amber mutation is followed by a Leu, Ser, or Arg codon, it is feasible to change that to a better codon. Table III shows the preferred codons for each of the groups of triplets that comprise the sixfold degenerate codons. In each case one set of triplets is favorable and the other unfavorable. For instance, the CGX family of arginine codons is extremely unfavorable for suppression, whereas the AGA and AGG family is favorable. Likewise, the UUA and UUG codons for leucine are unfavorable for suppression when they follow the amber codon, but the CUX family of triplets is very favorable for suppression.

Instability of Suppressors

Maintaining tRNA suppressor genes on moderate or high-copy-number plasmids requires constant selection to prevent cells that have lost the suppressor from overtaking the population, which they can do quite readily, since they can grow more rapidly. Selection to maintain the plasmid itself is usually carried out by using relevant antibiotic resistance markers. For instance, the pGFIB-based constructs are maintained in the presence of ampicillin, and the pACYC constructs are maintained in the presence of chloramphenicol. However, the suppressor tRNA gene insert can be lost from the plasmid, so it is necessary to have a suppressible amber mutation in the strain. It is convenient to use amber mutations in required genes. The original strains carrying the plasmid-encoded suppressor tRNA genes^{16,17} also carry an *argE* amber mutation that requires at least 2–3% suppression to allow normal growth in the absence of arginine.

Therefore, cells are usually grown and stored on minimal medium with ampicillin (or chloramphenicol). They can grow on rich medium (with antibiotic) overnight if inoculated from these minimal plates without suffering a severe loss of the plasmid insert.

Applications

Let us examine some applications of the amber suppressors in generating altered proteins. Two studies serve to illustrate the typical uses of this technology.

Studies with the lac Repressor

In the *lacI* gene of *E. coli*, nonsense mutations have been constructed at over 300 of the 360 coding positions in the gene (P. Markiewicz, L. G. Kleina, and J. H. Miller, unpublished, 1991). The initial 90 of these were obtained by *in vivo* mutagenesis,¹⁵ and the remainder added by site-directed mutagenesis *in vitro* (Kleina and Miller²⁷, P. Markiewicz and J. H. Miller, unpublished, 1991). In designing the UAG mutations, unfavorable leucine, arginine, or serine codons that followed the UAG triplet were converted to more favorable ones if the need arose. This is illustrated in Fig. 8. In the case of the *lacI* gene, the amber mutations were carried on an F' *lac proAB* eipsome that could be transferred into each suppressor strain by conjugation. In practice, 50 amber mutants were gridded onto nutrient agar in a Petri dish and replica mated into the suppressor strains. The newly created suppressor derivatives were then further replicated onto different indicator media to allow examination of the different LacI phenotypes. The fortunate use of a conjugative plasmid and the availability of sensitive indicator plates permit the examination of a very large number of altered *lac* repressors.

Figure 9 represents the effects of more than 3300 amino acid replacements created by suppressing nonsense mutations at 270 sites in the *lac* repressor (see also Kleina and Miller²⁷). This quasi-systematic study of amino acid replacements reveals much about the structure-function relationships of the protein. The distribution of residues that are sensitive to replacement is certainly not even. In the amino-terminal end (the first 59 residues), which has been shown to bind to DNA and operator,³³⁻³⁶ many

³³ T. Platt, J. G. Files, and K. Weber, *J. Biol. Chem.* **248**, 110 (1973).

³⁴ B. Mueller-Hill, *Prog. Biophys. Mol. Biol.* **30**, 227 (1975).

³⁵ N. Geisler and K. Weber, *Biochemistry* **16**, 938 (1977).

³⁶ R. Ogata and W. Gilbert, *Proc. Natl. Acad. Sci. U.S.A.* **75**, 5851 (1978).

Amino Acid Position of Amber	Sequence			Amino Acid Position of Amber	Sequence		
4	PRO	VAL	THR	20	THR	VAL	SER
	CCA	GTA	ACG		ACC	GTT	TCC
		↓↓↓			ACC	↓↓↓	
	CCA	TAG	ACG			TAG	AGC
5	VAL	THR	LEU	21	VAL	SER	ARG
	GTA	ACG	TTA		GTT	TCC	CGC
		↓↓	↓↓			↓↓	↓↓
	GTA	TAG	CTG		GTT	TAG	AGA
8	TYR	ASP	VAL	22	SER	ARG	VAL
	TAC	GAT	GTC		TCC	CGC	GTG
		↓↓				↓↓↓	↓
	TAC	TAG	GTC		TCC	TAG	GTC
9	ASP	VAL	ALA	23	ARG	VAL	VAL
	GAT	GTC	GCA		CGC	GTG	GTG
		↓↓↓				↓↓	↓
	GAT	TAG	GCA		CGC	TAG	GTC
10	VAL	ALA	GLU	24	VAL	VAL	ASN
	GTC	GCA	GAG		GTG	GTG	AAC
		↓↓↓			↓	↓↓	
	GTC	TAG	GAG		GTC	TAG	AAC
14	ALA	GLY	VAL	25	VAL	ASN	GLN
	GCC	GGT	GTC		GTG	AAC	CAG
		↓↓↓			↓	↓↓	
	GCC	TAG	GTC		GTC	TAG	CAG
15	GLY	VAL	SER	28	ALA	SER	HIS
	GGT	GTC	TCT		GCC	AGC	CAC
		↓↓↓	↓↓			↓↓↓	
	GGT	TAG	AGT		GCC	TAG	CAC
16	VAL	SER	TYR	29	SER	HIS	VAL
	GTC	TCT	TAT		AGC	CAC	GTT
		↓↓				↓↓	
	GTC	TAG	TAT		AGC	TAG	GTT
19	GLN	THR	VAL	30	HIS	VAL	SER
	CAG	ACC	GTT		CAC	GTT	TCT
		↓↓↓				↓↓↓	↓↓
	CAG	TAG	GTT		CAC	TAG	AGT

FIG. 8. Construction of new amber sites in the *lacI* gene constructed *in vitro*. The number given is that of the amino acid specified by the codon converted to amber. Arrows indicate the sequence changes made from the wild type to create the mutant. In some cases the bases in the following coding position were altered so as to allow more efficient suppression without changing the following amino acid. [See also L. G. Kleina and J. H. Miller, *J. Mol. Biol.* **212**, 295 (1990).]

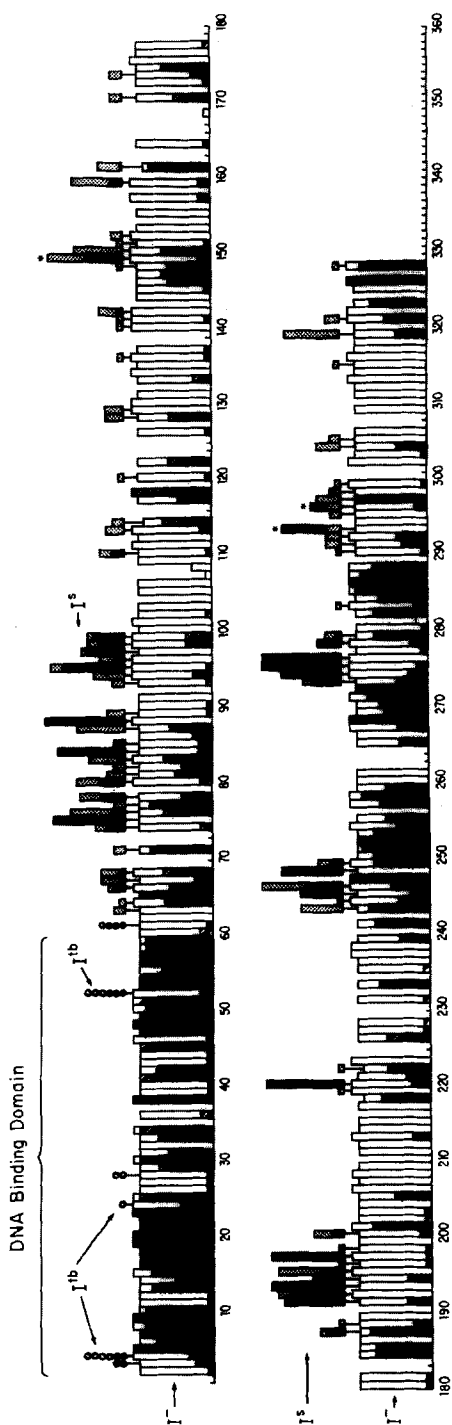


FIG. 9. Effects of amino acid replacements in the *lac* repressor. The consequences of over 3300 single amino acid replacements are depicted here. The repressor protein is shown as an array of 360 sites (amino acids). At each position where amino acids can be exchanged through nonsense suppression a bar has been drawn on the linear axis of the repressor polypeptide. The height of the bar represents the number of exchanges made (12 or 13 in the case of UAG sites, 3 to 4 in the case of UAA sites, and 1 in the case of UGA sites). Each bar is divided into invisible segments, the number of segments corresponding to the number of replacements made. When a replacement generates the I^- phenotype, a corresponding portion of the bar is filled in. If the replacement does not alter the I^- phenotype, then the segment is left open or blank. Intermediate effects are represented by a diagonal stripe. Thus, if all of the replacements destroy repressor function, then the whole bar is filled in (see position 22 or 38). If none of the replacements creates the I^- phenotype, then the whole bar is left open (see positions 153 and 189). Above the first set of bars the replacements that create I^s repressors are shown. Here the bars indicate only those replacements causing this change in phenotype, the height of the bar indicating the number of replacements that generate the I^s repressor. The black segments are strong effects, and the open or dotted segments are weaker effects. The I^s phenotype can sometimes result from tighter binding to operator, as occurs for certain replacements in the amino-terminal portion (residues 1 to 61 of the repressor). These repressors are depicted as I^{lb} (tight binding) and are represented by circles above the main set of bars. Filled-in circles indicate strong effects, and open circles weaker effects (P. Markiewicz, L. G. Kleina, and J. H. Miller, unpublished, 1991).

residues are sensitive to substitutions. This is in contrast to the remaining amino acids, many of which can be freely substituted, as is evident in the bottom portion of Fig. 9. Even so, many of these later residues do not tolerate proline, which shows up in this figure as the solitary black segment in an otherwise open box (see the legend to Fig. 9 for more details). There are, of course, residues in the latter part of the protein that are essential, and an inspection of Fig. 9 reveals that the segment between amino acids 239 and 287 has a high density of residues sensitive to substitution.

The upper part of Fig. 9 displays the position of residues involved in inducer binding and/or the allosteric change in response to inducer binding, which causes the repressor to lose affinity for operator DNA. The resulting I^s phenotype is easy to score. While some type of I^s effect results from replacing many of the residues from positions 70–100, in the latter half of the protein I^s repressors are generated by substitutions at five almost evenly spaced clusters. These can be seen in Fig. 9 at residues centered around positions 195, 220, 246, 274, and 296. This spacing has prompted speculation that some of these clusters may represent the ends of β turns in the repressor structure.³⁷

The circles shown above the line in the amino-terminal portion of the repressor represent repressors that bind operator and DNA more tightly. Several examples of such tight-binding repressors have been detected before,^{38–40} resulting from both missense and suppressed nonsense mutations.

The diagram in Fig. 9 compiles much detailed information into a single perspective. Many of the individual altered repressors have been analyzed more quantitatively.²⁷ The use of nonsense mutations facilitates spectroscopic studies of proteins, as has been shown for both fluorescence and NMR studies of the repressor.^{41–43} Together with the emerging three-dimensional structure of the repressor,⁴⁴ the data represented in Fig. 9

³⁷ J. H. Miller, *J. Mol. Biol.* **131**, 249 (1979).

³⁸ A. Jobe and S. Bourgeois, *J. Mol. Biol.* **72**, 139 (1972).

³⁹ A. Schmitz, C. Coulondre, and J. H. Miller, *J. Mol. Biol.* **123**, 431 (1978).

⁴⁰ J. L. Betz, *J. Mol. Biol.* **195**, 495 (1987).

⁴¹ H. Sommer, P. Lu, and J. H. Miller, *J. Biol. Chem.* **251**, 3774 (1976).

⁴² M. A. C. Jarema, K. T. Arndt, M. Savage, P. Lu, and J. H. Miller, *J. Biol. Chem.* **256**, 6544 (1981).

⁴³ M. A. C. Jarema, P. Lu, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **78**, 2702 (1981).

⁴⁴ H. C. Pace, P. Lu, and M. Lewis, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 1870 (1990).

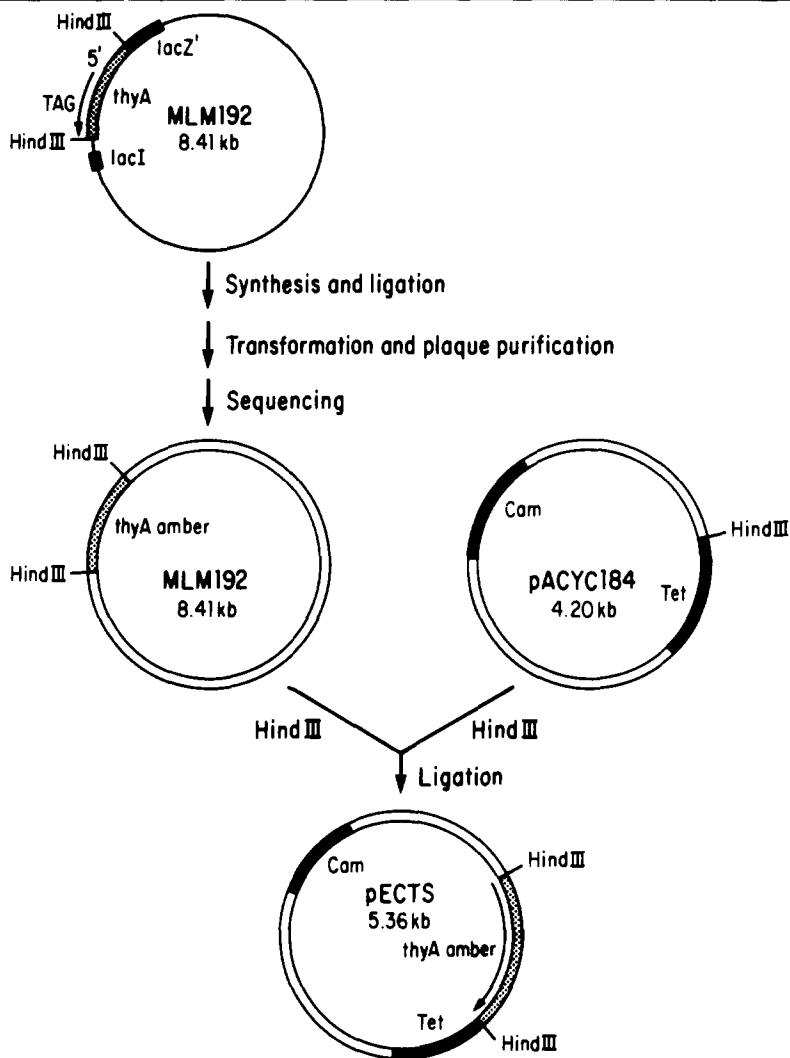


FIG. 10. Construction of pECTS amber mutants. The *thyA*-containing *Hind*III fragment from plasmid pATAH was subcloned into M13mp19 to create MLM192. Oligonucleotides containing amber stop codons were annealed to the template and extended and ligated with T4 DNA polymerase and ligase. Mutants were identified by sequencing and were subcloned into pACYC 184 to create the pECTS amber mutant series. [For further details see M. L. Michaels, C. W. Kim, D. A. Matthews, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 3957 (1990).]

TABLE IV
SUPPRESSION PATTERNS OF THYMIDYLATE SYNTHASE^a

Residue	Nonpolar					Polar				Basic			Acidic
	Gly	Ala	Pro	Phe	Leu	Cys	Ser	Tyr	Gln	His	Lys	Arg	Glu
Glu-14	+	+	+	+	+	+	+	+	+	+	+	+	+
Arg-21	+/-	-/+	+/-	-	-	-	+/-	-	-	+/-	-	+	-
Phe-30	-	-	-	+	+	-	-	-	-	-	-	-	-
Gln-33	+	+	+	+	+	+	+	+	+	+	+	+	+
Arg-35	+	+	+	+	+	+	+	+	+	+	+	+	+/-
Trp-80	-/+	-/+	-	+/-	-	-	-	-	-	-/+	-	-	-/+
Asp-81	+	+	+	+	+	+	+	+	+	+	+	+	+
Asp-105	+	+	+	+	+	+	+	+	+	+	+	+	+
Asp-110	-	-	-	-	-	+/-	+/-	+/-	-	-	-	-	-
Asn-121	+	+	+	+	+	+	+	+	+	+	+	+	+
Arg-126	-/+	-/+	+/-	-	-	-	+/-	+	-	+	+	+	-
Arg-127	+	+	+	+	+	+	+	+	+	+	+	+	+
Cys-146	-	-	-	-	-	+	+/-	-	-	-	-	-	-
His-147	+	+	+	-	+/-	+	+	-	+/-	+	-	+	-
Gln-151	-	-	-/+	-	-	-/+	-	-/+	+	+	-	-	+
Arg-166	-/+	-	-	-	-	-	-	-	-	-/+	-	+	-
Asp-169	-	-	-	-	-	+/-	-	-	-	-	-	-	-
Asn-177	+/-	-	-	-	+/-	-	+/-	-	+/-	-	-	-/+	-
Gly-204	+	+	-	-	-	-	+/-	-	-	-	-	-	-
Glu-223	+	+	+	+	+	+	+	+	+	+	+	+	+

^a Purified colonies of 20 *thyA* amber mutations transformed into 13 amber suppressor strains were streaked onto media with or without thymidine and growth of individual colonies was compared at 37°: +, 50–100%; +/-, 10–50%; -/+, less than 10%; -, no individual colonies observed. [From M. L. Michaels, C. W. Kim, D. A. Matthews, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 3957 (1990).]

should allow a detailed understanding of the structure–function relationships in the repressor.

Studies with Thymidylate Synthase

Escherichia coli thymidylate synthase is required for the biosynthesis of dTMP (thymidine monophosphate) from dUMP (deoxyuridine monophosphate), in the presence of the cofactor 5,10-methylenetetrahydrofolate.⁴⁵ We used site-directed mutagenesis to introduce 20 different amber mutations into the *E. coli thyA* gene.⁴⁶ The sites were chosen based on the

⁴⁵ D. V. Santi and P. V. Danenberg, in "Folates and Pterines" (R. L. Blakely and S. J. Benkovic, eds.), Vol. 1, p. 345. Wiley, New York, 1984.

⁴⁶ M. L. Michaels, C. W. Kim, D. A. Matthews, and J. H. Miller, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 3957 (1990).

TABLE V
ACTIVITIES OF SUPPRESSED THYMIDYLATE SYNTHASE (TS) VARIANTS

Wild-type amino acid	Amino acid substituted	Growth phenotype	Proposed role	Total activity (per milligram extract protein per hour)	[TS] _{μg/ml} in the extract	Specific activity (per milligram TS protein per hour)	Percent of wild type
Phe-30	Ala-30	-	Crystal structure	0.80 (±0.73)	5.12 (±1.75)	0.16 (±0.20)	0.32
	Phe-30	+	suggests an	973.52 (±142.38)	32.51 (±1.80)	23.36 (±4.71)	46.46
	Ser-30	-	important role	0.37 (±0.13)	4.96 (±1.11)	0.08 (±0.05)	0.16
	Tyr-30	+	in tertiary	77.03 (±17.26)	1.88 (±0.59)	42.58 (±22.90)	84.69
	Leu-30	+	structure of the binding pocket	246.82 (±18.46)	3.94 (±0.73)	61.51 (±16.00)	122.33
Gln-33	Ala-33	+	Conserved amino	38.28 (±5.74)	1.16 (±0.04)	26.09 (±4.81)	51.89
	Phe-33	+	acid among all	409.02 (±169.86)	6.06 (±2.22)	36.35 (±28.41)	72.30
	Ser-33	+	known TS's.	519.72 (±39.88)	7.12 (±0.71)	49.72 (±8.78)	98.89
	Tyr-33	+	Crystal structure	485.25 (±16.25)	16.34 (±10.44)	48.67 (±32.72)	96.80
	Leu-33	+	suggests no role	675.62 (±63.32)	31.30 (±8.92)	22.26 (±8.43)	49.27
Arg-127	Gln-33	+	in catalysis	602.52 (±54.22)	16.15 (±0.91)	49.25 (±7.21)	97.95
	Ala-127	+	Crystal structure	27.56 (±14.12)	0.69 (±0.02)	28.78 (±15.63)	57.24
	Phe-127	+	suggests an	54.10 (±1.56)	10.51 (±2.37)	4.61 (±1.17)	9.17
	Ser-127	+	important role	54.94 (±5.27)	4.22 (±0.35)	10.60 (±1.90)	21.08
	Tyr-127	+	in dUMP binding	48.93 (±8.42)	8.76 (±1.19)	8.48 (±2.61)	16.86
His-147	Leu-127	+		550.26 (±77.75)	27.74 (±5.58)	20.05 (±6.87)	39.88
	Arg-127	+		361.54 (±21.40)	13.24 (±2.29)	31.82 (±7.39)	63.29
	Ala-147	+	Crystal structure	93.01 (±11.82)	38.43 (±3.31)	3.45 (±0.74)	6.86
	Phe-147	-	suggests an	0.73 (±0.03)	49.30 (±3.11)	0.03 (±0.01)	0.06
	Ser-147	+	important role	21.19 (±1.67)	57.18 (±3.51)	0.63 (±0.09)	1.25
Wild-type plasmid	Tyr-147	-	during catalysis	1.02 (±1.70)	9.96 (±3.26)	0.12 (±0.20)	0.24
	Leu-147	±		5.71 (±1.97)	19.14 (±2.96)	0.49 (±0.24)	0.97
	His-147	+		1436.61 (±23.89)	46.69 (±4.40)	43.08 (±4.78)	85.68
	None	+		790.05 (±151.35)	26.40 (±7.22)	50.28 (±23.38)	100.00

three-dimensional structure of the enzyme.⁴⁷ In this case, we employed a plasmid that expressed the *thyA* gene in a strain that was *ThyA*⁻. Figure 10 depicts the use of this plasmid to construct the mutations. The *thyA*-encoded plasmid was derived from pACYC,^{48,49} which is compatible with the pBR322-based plasmids carrying the synthetic superrepressor genes. Although the activity of thymidylate synthase can be roughly estimated by scoring for *Thy*⁺ and *Thy*⁻, as shown in Table IV, it is more informative to assay directly for thymidylate synthase activity in crude extracts, and then to calculate specific activities after determining the concentration of thymidylate synthase by quantitative Western blotting. Table V depicts a sample of the results obtained.

Summary

The use of suppressed nonsense mutations to generate altered proteins can greatly simplify studies in which a large number of defined mutant proteins are sought. If site-directed mutagenesis is used to generate specific mutations, then for every amber (UAG) mutation constructed, as many as 13 different amino acids can be inserted at the corresponding site in the protein. This allows a rapid screening of many altered proteins for those with interesting properties. Once identified, the interesting substitutions can be regenerated by missense changes, to avoid some of the potential problems of the method. Nonsense suppression has been used to generate more than 3300 amino acid replacements in the *E. coli lac* repressor, and close to 250 amino acid substitutions in *E. coli* thymidylate synthase.

Acknowledgment

The work described here was partly supported by a grant from the National Science Foundation (DMB-8417353).

⁴⁷ D. A. Matthews, K. Appelt, and S. J. Oatley, *J. Mol. Biol.* **205**, 449 (1989).

⁴⁸ A. Y. C. Chang and S. N. Cohen, *J. Bacteriol.* **148**, 1141 (1978).

⁴⁹ M. Belfort and J. Pederson-Lane, *J. Bacteriol.* **160**, 371 (1984).

[27] Random Mutagenesis of Protein Sequences Using Oligonucleotide Cassettes

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Introduction

Investigations of protein structure and function often rely on the analysis of mutant proteins. With the advent of methods for rapid and economical chemical synthesis of DNA, there has been a steadily increasing use of oligonucleotide-directed mutagenesis¹ and cassette mutagenesis^{2,3} to create specific mutations at particular sites. These synthetic methods can also be used to create random mutations at one or more codons in a gene. In this chapter, we describe and evaluate several methods of cassette mutagenesis that allow random mutations to be generated, and discuss the application of these methods to the analysis of the structure and function of DNA-binding proteins. In addition to the techniques described here, other methods are available for the efficient generation of protein variants. These include approaches based on the suppression of amber mutations⁴ and oligonucleotide-directed methods that generate either random changes^{5,6} or specific changes.⁷

A simple form of cassette-mediated random mutagenesis involves mutating a single codon to encode all 20 naturally occurring amino acids. Individual clones are then isolated from the resulting population of mutants, sequenced, and screened for neutral, conditional, or defective phenotypes. This method can be somewhat labor intensive, but because changes are limited to a single position, the interpretation of the resulting phenotypes is relatively straightforward. As described below, this ap-

¹ M. J. Zoller and M. Smith, this series, Vol. 100, p. 468.

² S. J. Eisenbeis, M. S. Nasoff, S. A. Noble, L. P. Bracco, D. R. Dodds, and M. H. Caruthers, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 1084 (1985).

³ J. A. Wells, M. Vasser, and D. B. Powers, *Gene* **34**, 315 (1985).

⁴ J. H. Miller, this volume [26].

⁵ D. D. Loeb, R. Swanstrom, L. Everitt, M. Manchester, S. E. Stamper, and C. A. I. Hutchison, *Nature (London)* **340**, 397 (1989).

⁶ J. D. Hermes, S. C. Blacklow, and J. R. Knowles, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 696 (1990).

⁷ B. C. Cunningham and J. A. Wells, *Science* **244**, 1081 (1989).

proach is easily extended to the simultaneous mutagenesis of several codons, either in contiguous blocks or in noncontiguous residue positions. In either of these multiple mutagenesis experiments, a biological selection is generally applied to identify active clones, which are then sequenced. Such experiments generate lists of functional sequences, from which one can determine the spectrum of substitutions that are tolerated. An analysis of the resulting substitution patterns is then used to determine the importance of the mutagenized positions and to check for possible combinatorial effects.

All cassette mutagenesis procedures involve the synthesis of a small, double-stranded DNA molecule that can be ligated into a larger vector fragment to reconstruct the gene of interest (Fig. 1). The cassette is created by chemical synthesis or a combination of chemical and enzymatic synthesis. The backbone molecule is generated by digestion of plasmid or viral DNA with restriction enzymes. To introduce a cassette into a particular region of a gene requires that appropriate restriction sites occur in the gene sequence. If such sites are not present in the wild-type gene, it is usually possible to modify the sequence in such a way that restriction sites are introduced every 30–40 bases without altering the encoded protein sequence.^{8,9}

In the following sections, we discuss mutagenic strategies and issues that arise in the interpretation of results from random mutagenesis experiments. We then describe techniques for preparation of oligonucleotide cassettes, for preparation of plasmid backbones, for transformation, and for rapid sequencing of mutant genes. Finally, we discuss problems that can arise in cassette mutagenesis experiments.

Mutagenic Strategies

Genetic Selections and Screens. Efficient functional selections or screens are extremely important in most randomization experiments, especially when multiple codons are mutagenized. For sequence-specific DNA-binding proteins that can be expressed in *Escherichia coli*, it is often possible to develop selections *in vivo* based on antibiotic resistance.^{10,11}

⁸ M. Nassal, T. Mogi, S. S. Karnik, and H. G. Khorana, *J. Biol. Chem.* **262**, 9264 (1987).

⁹ A number of computer programs are available that identify sites in gene sequences at which recognition sites for restriction endonucleases may be introduced without altering the protein sequence. One such program, SeqSearcher for the Apple Macintosh computer, is available from Jim Hu, Dept. of Biology, Room 16-833, Massachusetts Institute of Technology, Cambridge, MA 02139.

¹⁰ M. C. Mossing, J. U. Bowie, and R. T. Sauer, this volume [29].

¹¹ S. J. Elledge, P. Sugiono, L. Guarente, and R. W. Davis, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 3689 (1989).

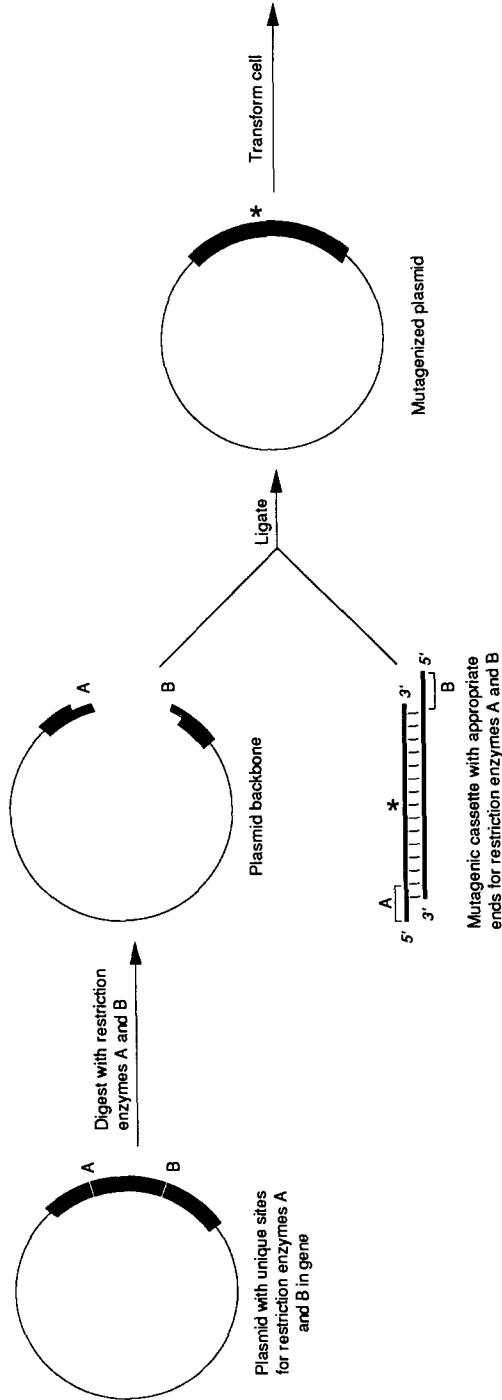


FIG. 1. Cassette mutagenesis procedure. A plasmid or viral vector is digested with restriction enzymes A and B to generate a backbone fragment. An oligonucleotide cassette corresponding to the region between sites A and B is synthesized with mutations at one or more positions. The cassette is then ligated to the backbone fragment to generate the mutagenized vector, which is transformed into the appropriate cells.

	T		C		A		G	
T	TTC	Phe	TCC	Ser	TAC	Tyr	TGC	Cys
	TTG	Leu	TCG		TAG	Stop	TGG	Trp
C	CTC	Leu	CCC	Pro	CAC	His	CGC	Arg
	CTG		CCG		CAG	Gln	CGG	
A	ATC	Ile	ACC	Thr	AAC	Asn	AGC	Ser
	ATG	Met	ACG		AAG	Lys	AGG	Arg
G	GTC	Val	GCC	Ala	GAC	Asp	GGC	Gly
	GTG		GCG		GAG	Glu	GGG	

FIG. 2. Genetic code with third codon position restricted to G or C. All of the codons shown are accessible in an NN_C^G randomization. Further restrictions are also possible. For example, an NT_C^G randomization would restrict the available codons to the first column.

Colony screens based on repression of *lacZ* or *galK* expression are also straightforward in many cases. In interpreting phenotypes based on any selection or screen, it is important to know what level of activity is required. If the stringency of the selection or screen is very high, then small reductions in activity or level may result in a conditional or even defective phenotype, and fewer residue substitutions will be tolerated. If the stringency is lower, then more substitutions would be expected to be classified as functionally neutral. In our experience, a selection that requires approximately 10% of wild-type activity provides a useful metric that allows the importance of residue positions to be established.

Codon Restrictions. In codon randomization experiments, the position(s) of interest can be changed to either a limited set or a complete set of the 20 naturally occurring amino acids. Complete randomization at the codon level can be achieved by an NNN randomization (where N indicates a mixture of A, C, G, and T). However, as shown in Fig. 2, the third codon position can be restricted to G or C without eliminating any of the amino acids, and thus an NN_C^G randomization is sufficient to achieve complete randomization with respect to possible amino acid sequences. This restriction is useful both because it reduces the overall DNA sequence complexity and because it reduces the coding discrepancy between residues like Met and Trp, which have a single codon, and residues like Leu, Arg, and Ser, which have six codons in an NNN randomization but only three codons in an NN_C^G randomization. Randomizations can also be restricted to specific regions of the genetic code. For example, a randomization using

NT_C^G would be restricted to the five hydrophobic amino acids Phe, Leu, Ile, Met, and Val, whereas an NA_C^G randomization would be restricted to the seven relatively hydrophilic side chains Tyr, His, Gln, Asn, Lys, Asp, and Glu. In either complete or limited randomizations of this type, equal quantities of each desired base are generally included. Thus in unselected populations, one would expect to recover each residue at frequencies that were roughly proportional to the number of codons for that residue. However, random mutagenesis can also be performed using unequal quantities of each base,^{12,13} in order to bias recoveries toward a specific residue, which is often the wild type. For example, a "biased" randomization might include 70% of the wild-type base at each codon position and 10% of each of the other bases.

Single-Codon Randomizations. Mutagenesis of single-codon positions is the most basic kind of cassette randomization experiment. In general, one uses an unbiased NN_C^G randomization to generate all of the possible amino acids and then analyzes the sequences and phenotypes of clones from the resulting population of variants. A good mixture of different sequences can generally be obtained simply by sequencing unselected clones. The phenotypes conferred by each different sequence can then be determined. Alternatively, it may be desirable in some cases to screen clones for phenotype first and then to sequence several candidates from each phenotypic class.

How many unselected clones from an NN_C^G randomization need to be sequenced to recover a given number of different amino acids? Monte Carlo simulations (assuming a random distribution at the nucleotide level) show that sequencing 30 candidates generally results in the identification of 14–16 residues, whereas sequencing almost 100 candidates is necessary to have a greater than 50% chance of recovering all 20 amino acids.¹⁴ For most studies, there is no need to analyze all possible substitutions; enough different changes are recovered by sequencing 20–30 candidates to evaluate the importance of the residue being studied. When recovery of a complete set of substitutions is desirable, it is often easiest to sequence approximately 50 clones, and then make the remaining 1 or 2 substitutions by conventional site-directed mutagenesis using a cassette.

Randomization of Multiple Codons. Simultaneous randomization of more than one codon is useful because many positions can be examined in a single experiment. Moreover, potential interactions among residues can be tested. However, in randomization experiments involving multiple

¹² A. R. Oliphant, A. L. Nussbaum, and K. Struhl, *Gene* **44**, 177 (1986).

¹³ D. E. Hill, A. R. Oliphant, and K. Struhl, this series, Vol. 155, p. 558.

¹⁴ J. F. Reidhaar-Olson and R. T. Sauer, *Proteins: Struct. Funct. Genet.* **7**, 306 (1990).

codons, it is rarely desirable to try to recover or analyze all of the resulting sequences. This is true both because of numerical complexity and because analysis of multiply mutant sequences that confer a defective phenotype is generally uninformative. As a consequence, the analysis is usually limited to active sequences that are identified by using a biological selection for activity. Screening among unselected candidates can also be used to find active clones, but can be tedious if only a small number of the randomly generated sequences are active.

In multiple codon randomizations, it is often desirable to limit the number of codons being mutated or the extent of randomization to ensure that some active sequences are recovered. If an NN_C^G randomization is applied to 5 codons, then there are 20^5 or 3,200,000 different amino acid sequences generated. The complexity at the level of nucleic acid sequences is even greater. In principle, with a sufficiently powerful selection, one could recover any active sequence (including wild type). In practice, however, the total library of transformants may be smaller (e.g., 10^5 – 10^6) than the sequence complexity and thus there is no guarantee that any active sequences will be recovered. By contrast, if the NN_C^G randomization were restricted to 3 codons, then the amino acid sequence complexity would be only 8000, and there would be an excellent chance of recovering all active sequences from the mutagenized library. If the goal of a study is a comprehensive survey of all sequences, then it is probably best to limit complete randomizations to three or at most four codons.

Sequence complexity in multiple codon randomizations can also be limited by restricting the randomization to a subset of the amino acids or by biasing the randomization toward the wild-type residue (see section on codon restriction). Restricting the randomization is useful if the subset of residues introduced at each position (these subsets can be different for different codons) provide useful information. For example, in one study of hydrophobic packing in the core of λ repressor, a randomization of 3 positions was restricted to 5 hydrophobic residues, allowing a comprehensive survey of the 125 possible residue combinations.¹⁵ In biased randomizations, the extent of mutagenesis can be controlled by adjusting the level of mutant bases. For example, in studies of the Arc repressor,¹⁶ blocks of approximately 10 adjacent codons were mutagenized by contaminating the wild-type nucleotide at each position with 7.5% of each of the other 3 bases. At this level of mutagenesis, each codon has about a 60% chance of encoding the wild-type residue and about a 40% chance of encoding a mutant amino acid. Hence, the mutagenized population should contain

¹⁵ W. A. Lim and R. T. Sauer, *J. Mol. Biol.* **219**, 359 (1991).

¹⁶ J. U. Bowie and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 2152 (1989).

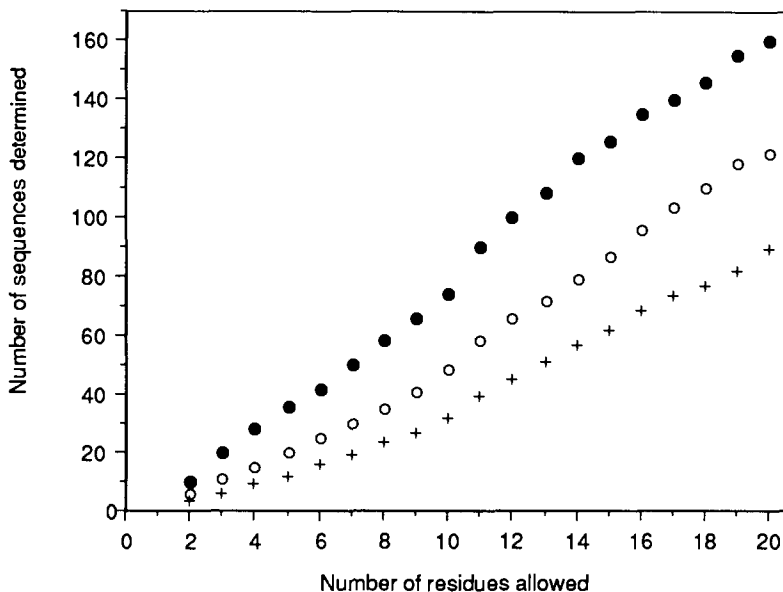


FIG. 3. Number of sequences required to observe all allowed residues. Monte Carlo simulations of NN^G random mutagenesis experiments were performed to find the number of sequences that need to be determined to observe all allowed residues 50% of the time (+), 80% of the time (○), and 95% of the time (●). For example, if one determines 50 sequences and observes 10 different residues at a randomized position, then one knows with about 80% confidence that all allowed residues have been recovered. With 10 observed residues, about 75 sequences would be required to increase the confidence level to 95%.

occasional wild-type sequences (at a frequency of approximately 0.6%) but most genes should contain an average of four amino acid substitutions in the mutagenized region. Substitutions resulting from "biased" randomizations will be somewhat limited because single base changes are more probable than double changes, and triple changes are extremely rare. In practice, however, enough substitutions can generally be recovered to assess the general requirements for function.

In analyzing functional variants, how does one decide that enough sequences have been determined to be confident that all tolerated substitutions have been identified? As shown in Fig. 3, the number of sequences that need to be determined depends both on the number of residues allowed and the desired confidence limits. Nevertheless, several points are clear. For example, if 20 sequences are determined and only 1 or 2 residues are recovered at a given position, then it is quite likely that other residues would be nonfunctional. However, if in the set of 20 sequences, 10 different residues are recovered, then sequencing more candidates is likely to result

in additional allowed residues. In general, if a large number of different side chains are observed to be tolerated at a given position, then it is probably dangerous to assume that the residues that have not been recovered are nonfunctional. However, it is generally not necessary to know with certainty that all allowed changes have been recovered. The simple fact that a large number of chemically dissimilar side chains allow function suggests that the position under study cannot be critical for function.

In any experiment where more than one codon is randomized, it is possible that some of the functional substitutions recovered at one position depend on simultaneous changes at other positions in the sequence. In such cases, the observed substitutions might not be tolerated as single substitutions in otherwise wild-type backgrounds. Such examples are most common in the case of interacting residues. For example, when three or four interacting residues in the hydrophobic core of the N-terminal domain of λ repressor were randomized together, residue substitutions were found to be allowed that were known not to be tolerated as single substitutions.¹⁷

Applications and Analysis

Identification of Allowed Substitutions. Cassette randomization experiments are extremely useful in identifying the importance of residues in a protein sequence. Because the method of mutagenesis ensures that a complete ensemble of sequences is present in the initial randomized population, many different substitutions should be allowed if the chemical identity of the side chain is not important for function. In contrast, if only the wild-type residue or a small set of related side chains is recovered among the functional sequences, then it is reasonable to infer that most other substitutions would result in a defective phenotype.

Figure 4 shows residue substitutions that are functionally allowed in the helix I region of the N-terminal domain of λ repressor.¹⁴ Some positions tolerate many chemically dissimilar side chains, suggesting that these positions play little or no role in either structure or function. Other positions are restricted to a single residue or a small set of chemically similar side chains. In the crystal structure of the N-terminal domain bound to operator DNA,¹⁸ the ϵ -amino group of Lys-19 and the hydroxyl of Tyr-22 make direct contacts with the phosphate backbone of the operator, indicating a direct role of these side chains in DNA binding. In addition to its role in DNA binding, the ring portion of Tyr-22 forms part of the hydrophobic core of the protein and thus serves a structural role as well. Ala-15,

¹⁷ W. A. Lim and R. T. Sauer, *Nature (London)* **339**, 31 (1989).

¹⁸ S. R. Jordan and C. O. Pabo, *Science* **242**, 893 (1988).

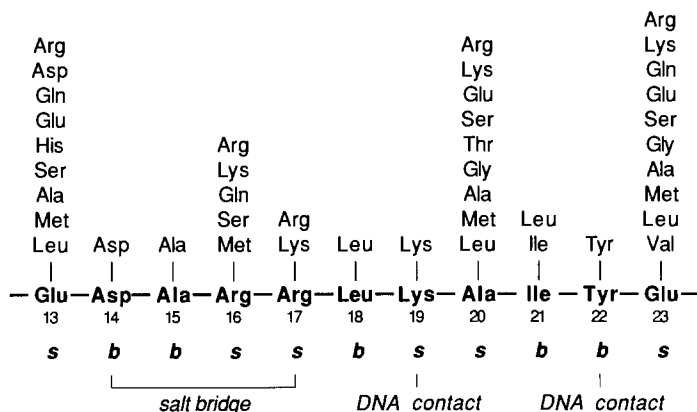


FIG. 4. Allowed substitutions identified from random mutagenesis experiments. The allowed residues in a helical region of the N-terminal domain of λ repressor are shown above the wild-type sequence. The numbers below the wild-type sequence indicate residue numbers. An *s* below the sequence indicates that the wild-type residue is on the surface of the protein; a *b* indicates that the residue is buried (0–25% fractional side chain accessibility). The side chains of Asp-14 and Arg-17 interact through a salt bridge. Lys-19 and Tyr-22 make contacts with the operator DNA. In general, residues that are either structurally or functionally important are conserved.

Leu-18, and Ile-21 also form part of the hydrophobic core, while Asp-14 and Arg-17 form a charge stabilized hydrogen bond. Hence, these residues are likely to be important for maintaining the folded structure of the protein.

Discriminating between Structural and Functional Effects. The examples discussed above show that allowed substitutions can be restricted because the residue mediates contact with the DNA or is important for protein folding or stability. In the absence of structural information, these effects can usually be distinguished by biochemical studies of purified proteins.^{19,20} However, it is sometimes possible to eliminate mutations that cause structural defects if more than one phenotype that depends on structure can be monitored. For example, in studies of the positive control function of λ repressor, randomized candidates were first selected for their ability to mediate repression (thereby ensuring that the mutant proteins could fold and bind DNA) and then screened for a positive control phenotype.²¹ This allowed the identification of a single residue that was critical

¹⁹ M. H. Hecht, J. M. Sturtevant, and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 5685 (1984).

²⁰ H. C. M. Nelson and R. T. Sauer, *J. Mol. Biol.* **192**, 27 (1986).

²¹ F. D. Bushman, C. Shang, and M. Ptashne, *Cell (Cambridge, Mass.)* **58**, 1163 (1989).

for positive control. As discussed below, strategies involving antibody screens or intracellular proteolysis can also be used to distinguish structural and functional effects.

Proteolysis and Structural Defects. Thermally unstable variants of many DNA-binding proteins, including the N-terminal domain of λ repressor, λ Cro, and P22 Arc, are subject to rapid degradation in *E. coli*.^{22–25} As a result, variants bearing destabilizing substitutions are generally present at intracellular levels lower than wild type, and the presence of normal levels of a mutant protein *in vivo* is often indicative of thermal stability. For example, in studies of the Arc repressor, blocks of approximately 10 codons at a time were subject to a “biased” randomization and unselected colonies were screened by sodium dodecyl sulfate (SDS) gel electrophoresis for Arc protein levels.¹⁶ The *arc* genes from colonies displaying moderate to high levels of protein were sequenced, and the corresponding proteins were purified and studied by circular dichroism to confirm that they were stably folded. These studies resulted in a list of “structurally” tolerated substitutions. By comparing this list with that of functionally tolerated substitutions, it was possible to distinguish positions likely to be directly involved in DNA recognition from those likely to be involved in stabilization of structure.

Although proteolysis screens can be used in a preliminary way to identify structurally stable proteins, the absence of a mutant protein in a cell lysate does not always indicate structural instability. For example, in a study of the N-terminal domain of λ repressor, the five C-terminal codons of the gene were subjected to a complete NN_C^G randomization and colonies were screened by SDS gels for protein levels.²⁶ Candidates found to have extremely low levels of the N-terminal domain (and to be rapidly degraded) had hydrophobic C-terminal pentapeptides. By contrast, candidates with high levels of protein had hydrophilic C-terminal pentapeptides. While these results provided clear evidence that hydrophobic residues at the C terminus resulted in proteolytic instability, the T_m of one of the most rapidly degraded variants was found to be identical to wild type. Hence, proteolytic instability and thermal instability are not always correlated. This example illustrates that purification and biochemical analysis of mutant proteins are ultimately required to understand the effects of particular mutations.

²² D. A. Parsell and R. T. Sauer, *J. Biol. Chem.* **264**, 7590 (1989).

²³ J. U. Bowie and R. T. Sauer, *J. Biol. Chem.* **264**, 7596 (1989).

²⁴ A. A. Pakula, V. B. Young, and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 8829 (1986).

²⁵ A. A. Pakula and R. T. Sauer, *Proteins: Struct. Funct. Genet.* **5**, 202 (1989).

²⁶ D. A. Parsell, K. R. Silber, and R. T. Sauer, *Genes Dev.* **4**, 277 (1990).

Observed Densities of Restricted Sites in DNA-Binding Proteins. In the 92-residue N-terminal domain of λ repressor, 60 residues have been studied by random mutagenesis.^{14,17,21,27-30} Of these residues, roughly one-half exhibit highly restricted substitution patterns, and thus play important roles of some kind in repressor structure and function. Many of the important residues are buried in the hydrophobic core or the dimer interface of the protein. Of the remaining restricted positions, some are involved in hydrogen bonds or salt bridges, some are directly required for DNA binding, and one is essential in protecting the protein from intracellular proteolysis.

In the P22 Arc repressor, the entire 53 residues of the protein have been characterized in a "biased" randomization study.¹⁶ Here, the identities of approximately one-third of the residues were found to be functionally important, and one-half were structurally important. The remaining residues could be freely substituted and thus are unimportant for either structure or function.

Interplay of Single-Codon and Multiple-Codon Randomizations. It is often useful to apply single-codon and multiple-codon randomizations in a sequential fashion, as the two methods can provide different types of information. For example, in mapping antibody-binding epitopes on the surface of λ repressor, a region of the gene was subjected to a multiple-codon "biased" randomization and candidates displaying repressor activity and binding to a conformation-specific monoclonal antibody were identified.²⁸ Several surface positions were found to be invariant in the antibody-reactive clones, suggesting that the side chains at these positions played important roles in antibody binding. This was confirmed by performing single-codon NN_C^G randomizations on each of these positions, sorting the resulting mutants into reactive and nonreactive classes, and then purifying representative mutants and measuring antibody affinities directly. In this case, the multiple-codon randomization provided a rapid way to identify important positions, while the single-codon randomizations provided detailed information about the sequence requirements at each of these positions.

Multiple-codon randomizations can also be performed after single-codon studies, as a test of the additivity of allowed substitutions. For example, in studies of leucine zipper function, single-codon randomizations were first performed to identify allowed substitutions at four of the

²⁷ J. F. Reidhaar-Olson and R. T. Sauer, *Science* **241**, 53 (1988).

²⁸ R. M. Breyer and R. T. Sauer, *J. Biol. Chem.* **264**, 13355 (1989).

²⁹ J. F. Reidhaar-Olson, D. A. Parsell, and R. T. Sauer, *Biochemistry* **29**, 7563 (1990).

³⁰ N. D. Clarke and C. O. Pabo, manuscript in preparation.

conserved leucine positions.³¹ In these experiments, other hydrophobic residues such as Val, Ile, and Met were found to be tolerated at each of the four positions. However, when all four of these leucine positions were randomized simultaneously, it was found that each of the functional sequences contained at least two and usually three leucines, suggesting that many of the substitutions that were tolerated singly could not be tolerated together. In this case, the multiple-codon randomization revealed combinatorial effects that could not be identified in the single-codon randomizations.

Random Mutagenesis Techniques

Synthesis of First Strand of Cassette

Oligonucleotides can be conveniently prepared using any of a number of commercially available DNA synthesizers. On some synthesizers, the machine can be programmed to deliver variable amounts of any of the four bases during any coupling step. If this is not the case, then it is generally necessary to have additional bottles with mixtures of bases (e.g., equal amounts of A, C, G, and T). These bottles may then be attached to extra delivery ports on the synthesizer, if available. If the machine has capacity for only four bottles of nucleotide solutions, it will be necessary to interrupt the synthesis and replace one of the bottles with the base mixture at the appropriate step of the synthesis. Following removal of the final product from the solid support, the oligonucleotide may be of sufficient purity to use directly. However, gel purification on a 12–20% (w/v) polyacrylamide/urea gel³² is often desirable in order to remove incomplete DNA fragments that accumulate during the course of the synthesis. Following electrophoresis, the appropriate band is located by UV shadowing or staining with ethidium bromide, and excised. The oligonucleotide is then eluted from the gel slice and can be further purified (e.g., by passage over a reversed-phase C₁₈ Sep-Pak column from Waters Associates, Milford, MA).

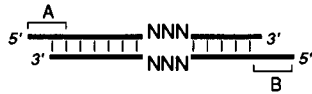
Chemical Synthesis of the Complementary Strand

Figure 5A and B shows two possibilities for chemical synthesis of the second strand of the cassette. In the first, the same NN_G random mixture is included in both strands at the codon or codons being randomized (Fig. 5A). In theory, mismatches generated at the randomized codon could lead

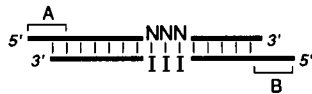
³¹ J. C. Hu, E. K. O'Shea, P. S. Kim, and R. T. Sauer, *Science* **250**, 1400 (1990).

³² F. M. Ausubel, R. Brent, R. E. Kingston, D. D. Moore, J. G. Seidman, J. A. Smith, and K. Struhl, "Current Protocols in Molecular Biology." Wiley, New York, 1989.

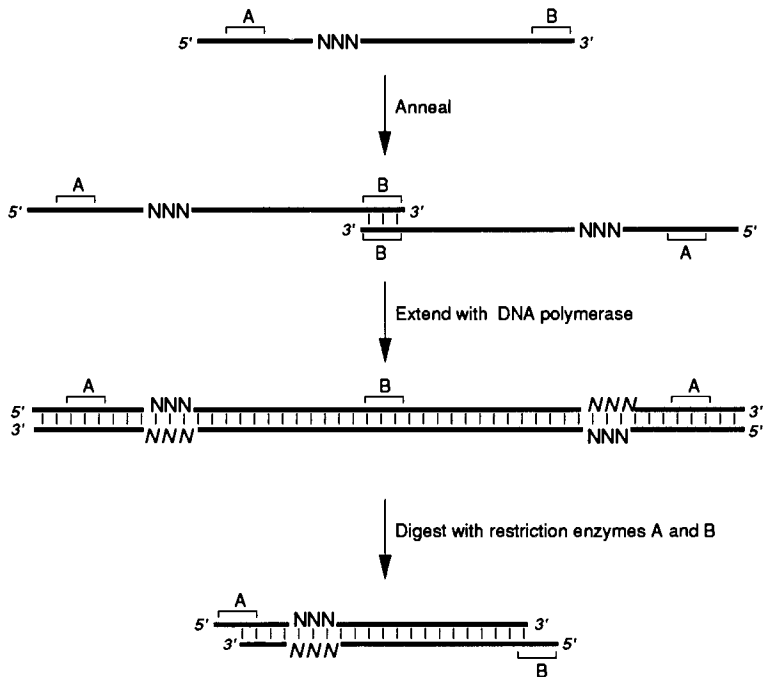
A. Random bases on both strands



B. Inosine pairing



C. Enzymatic second-strand synthesis



to inefficient pairing or problems with heterogeneity following transformation. However, in practice this method appears to be completely satisfactory, in both efficiency and randomness, when codons are randomized individually. We have not tested the technique extensively with multiple randomizations. An alternative method (Fig. 5B) uses inosine in the second strand opposite the randomized positions in the first strand. Since inosine is able to base pair with each of the four standard bases,³³ such cassettes are able to anneal and ligate with high efficiency. As discussed later, this procedure occasionally introduces some bias toward particular bases, but appears to be sufficiently random in most instances. This technique has been used successfully to randomize as many as three residue positions simultaneously.²⁷

Procedure. Following synthesis and purification, each strand of the cassette is diluted to a concentration of $0.5 \mu\text{M}$ in $40 \mu\text{l}$ kinase buffer [50 mM Tris-HCl (pH 7.5), 10 mM MgCl_2 , 5 mM dithiothreitol, 0.1 mM spermidine, 0.1 mM EDTA, $2 \mu\text{M}$ ATP], and phosphorylated using 20 units of T4 polynucleotide kinase (New England Biolabs, Beverly, MA). Phosphorylation is carried out for 1.5 hr at 37° , followed by 10 min at 65° to inactivate the kinase. (As discussed later, phosphorylation of the oligonucleotides is particularly important when using the inosine method.) The two oligonucleotides are then annealed by mixing at $0.2 \mu\text{M}$ in $40 \mu\text{l}$ annealing buffer [10 mM Tris-HCl (pH 8.0), 10 mM MgCl_2], heating at 80° for 10 min, and cooling slowly to room temperature.

Enzymatic Second Strand Synthesis

When both strands of the mutagenic cassette are prepared by chemical synthesis, there is always the potential problem of noncomplementarity due to mismatch formation during the annealing step. This can be avoided

³³ F. H. Martin, M. M. Castro, F. Aboul-ela, and I. Tinoco, Jr., *Nucleic Acids Res.* **13**, 8927 (1985).

FIG. 5. Three strategies for construction of the complementary strand of a mutagenic cassette. (A) Random bases are included on both strands of the cassette during synthesis. (B) Inosines are included on the complementary strand opposite the randomized positions on the first strand. (C) The complementary strand is synthesized enzymatically to prevent mismatches in the cassette. In this case, the first strand is synthesized with a self-complementary region at its 3' end corresponding to restriction site B. Following extension with DNA polymerase, the double-stranded molecule is digested with restriction enzymes A and B to yield the final cassette. An italicized *N* indicates an enzymatically-inserted base that is complementary to the random base on the opposite strand. Throughout this figure, randomized codons are indicated as NNN for simplicity; in practice, complete randomization may be accomplished by using NNN_C^G .

by performing enzymatic second strand synthesis, using the first, randomized strand as a template as shown in Fig. 5C.^{12,13} The first strand is synthesized with a self-complementary 3' end that contains the recognition sequence for one of the two restriction enzymes. The recognition sequence for the other restriction enzyme is included near the 5' end. Extension of the oligonucleotide is performed using the Klenow fragment of DNA polymerase I. The result of the self-primed DNA synthesis is a double-stranded oligonucleotide that contains no mismatches. Digestion with the two restriction enzymes yields a cassette with the appropriate ends for insertion into the plasmid backbone. Efficient digestion requires the presence of several additional base pairs beyond the restriction sites at the ends of the cassette; for most enzymes, three additional bases at the 5' end of the oligonucleotide appear to be sufficient. The enzymatic method is more involved than either of the chemical methods for second strand synthesis, and can lead to higher levels of unwanted mutations at cassette positions that have not been mutagenized (see Bonus Mutations, below). However, the enzymatic method does not appear to skew randomizations toward particular bases, does not introduce potential problems due to mismatched bases, and is almost certainly the best method to use in "biased" randomizations.

Figure 6 shows a variation on the enzymatic approach that allows the synthesis of larger mutagenic cassettes.¹⁵ In this case, two oligonucleotides are synthesized, each covering half the distance between the two restriction sites, with a complementary 9-base overlap at their 3' ends. Each oligonucleotide contains one of the two appropriate restriction sites plus several additional bases at its 5' end. The oligonucleotides are annealed and filled in, each priming second strand synthesis of the other. Digestion with the two enzymes yields the final randomized cassette with the appropriate ends for cloning into the plasmid backbone. Any of the codons between the two sites may be randomized. This method has been used to randomize four codons combinatorially, the most distant of which were 100 bp apart in the sequence.¹⁷

Procedure. For enzymatic second strand synthesis, we use approximately 5 μ g of the template oligonucleotide. Following annealing, extension is performed in 50 μ l of a solution containing 10 mM Tris-HCl (pH 7.5), 10 mM $MgCl_2$, 100 μ g/ml bovine serum albumin, 50 mM NaCl, and 250 μ M dNTPs using 10 units of the Klenow fragment of DNA polymerase I (Boehringer-Mannheim, Mannheim, Germany) or Sequenase (United States Biochemical, Cleveland, OH). The Sequenase enzyme has higher processivity but also exhibits a higher misincorporation rate (see below). After 1 hr at 37°, 5 μ l of 2.5 mM dNTPs, 0.5 μ l of 0.5 M dithiothreitol,

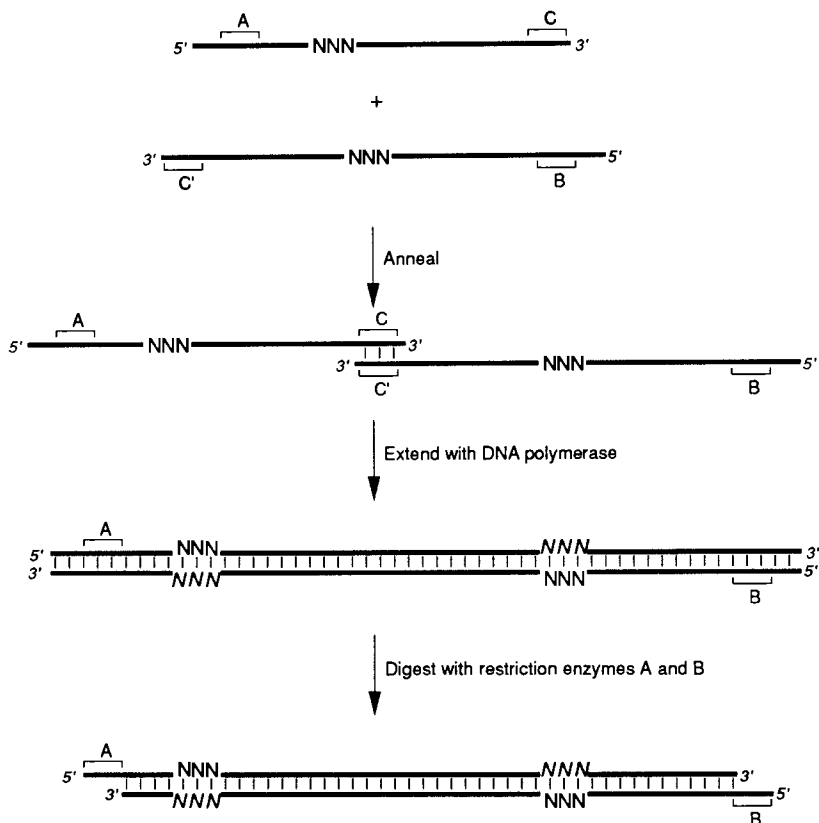


FIG. 6. A strategy for randomizing multiple codons distant in the sequence. Two oligonucleotides are synthesized that together encode the region between restriction sites A and B. The first oligonucleotide corresponds to the top strand, the second oligonucleotide corresponds to the bottom strand. The two oligonucleotides are synthesized with a complementary 9-base overlap at their 3' ends. Each molecule serves as a template for extension of the other by DNA polymerase. Digestion with restriction enzymes A and B yields the final, full-length cassette.

and 10 units of enzyme are added, and the reaction is allowed to proceed at 37° for an additional hour. The DNA is then ethanol precipitated, and approximately 10 μ g of the double-stranded oligonucleotide is digested with 40 units of each restriction enzyme at 37° overnight. Extension and digestion reactions are monitored by running the DNA on 6% (w/v) denaturing polyacrylamide gels. The final extended and digested cassettes are purified on 8% nondenaturing gels.

Preparation of Plasmid Backbone

If plasmids containing the wild-type gene (or any active gene) are used to prepare a backbone fragment, then the main consideration is preventing significant contamination of the backbone fragment by uncut or singly cut plasmid molecules. Such contamination will give rise to an unwanted background of unrandomized clones. This can be an especially serious problem in multiple codon randomizations, where only a small fraction of the randomized sequences may be active. A simple strategy that allows contamination due to uncut or singly cut molecules to be detected is to introduce a silent mutation at one of the nonrandomized positions in the cassette to a degenerate codon. When candidates are sequenced, contaminants can be readily identified because they will have the wild-type DNA sequence.

Contamination with uncut or singly cut molecules can be reduced by overdigestion with restriction enzymes or by careful gel purification of the backbone fragment, but these precautions are not always sufficient. The best solution to this problem is to purify the backbone from a plasmid that contains a large "stuffer" fragment cloned between the restriction sites to be used for insertion of the mutagenic cassette (for an example of the use of "stuffer" fragments, see Ref. 27). The stuffer fragment serves two purposes. First, it disrupts the coding sequence, which should result in inactivation of the gene. This ensures that any functional genes recovered following random mutagenesis result from the insertion of oligonucleotide cassettes rather than from backbone reclosure. Second, if the stuffer fragment is fairly large, its excision following digestion with restriction endonucleases leads to a significant change in mobility during gel electrophoresis. Consequently, the plasmid backbone can be readily purified away from uncut or singly cut plasmid DNA.

Procedure. Plasmid DNA that has been purified by CsCl gradient centrifugation is the most reliable for digestion with restriction enzymes, although DNA purified by miniprep procedures³² is often of sufficient purity. In typical experiments, 5 μ g of plasmid DNA is digested with 5–10 units of each restriction enzyme for 1 hr at 37°. An additional 5–10 units of each enzyme is added, and the incubation is continued for 1 hr. Restriction fragments are separated by electrophoresis on low melting point agarose gels. DNA fragments are isolated from gel slices by any of several methods, including the Qiagen Cartridge (Qiagen, Inc., Studio City, CA), Elutip (Schleicher & Schuell, Heane, NH), and Gene Clean (Bio 101, Inc., La Jolla, CA) protocols. Alternatively, the slice may be melted, phenol extracted, and the DNA purified by ethanol precipitation.³²

Double-stranded oligonucleotide cassettes are ligated to plasmid back-

bone overnight at 4–14° in ligation buffer [50 mM Tris-HCl (pH 7.5), 10 mM MgCl₂, 10 mM dithiothreitol (DTT), 1 mM spermidine, 1 mM ATP, 100 µg/ml bovine serum albumin]. Ligation reactions are performed at roughly equimolar concentrations of insert and plasmid backbone.

Transformation

Since the goal of a random mutagenesis experiment is usually to sample as much of sequence space as possible, the number of residue positions that can be examined in a single experiment rapidly becomes limited by the efficiency of each step in the procedure, and especially by the efficiency of transformation. We have found, using the transformation protocol of Hanahan,³⁴ that it is possible to obtain 10⁵–10⁶ transformants from 100 ng of DNA. This is enough to ensure that most of sequence space is sampled in experiments in which one to three residue positions are randomized at a time. However, randomizing more than three positions generally requires scaling up the transformation procedure. An alternative is to use electroporation,³² which gives transformation efficiencies about 100-fold greater than Hanahan transformation.

Sequencing

Since random mutagenesis generates a large number of mutant genes, the ability to rapidly sequence these genes is essential. We have found it convenient to perform mutagenesis using plasmids bearing an M13 origin of replication to allow production of single-stranded plasmid DNA.³⁵ This DNA is then sequenced using the dideoxy method.³⁶ An alternative is to isolate and sequence the double-stranded plasmid DNA.^{37,38}

We typically perform 48, 72, or 96 sets of dideoxy-sequencing reactions at a time. Sequencing this many candidates at once is facilitated by the use of microtiter dishes for all of the sequencing reactions. Sequencing reagents may be rapidly dispensed into the microtiter wells using a repeating pipette. The labeling and extension reaction is performed in one row of the dish, and then aliquots of each reaction are transferred using a multichannel pipette to separate wells for the A, C, G, and T termination reactions. Sequencing kits are available (e.g., from Amersham, Arlington Heights, IL) with reagents already dispensed into microtiter wells. The

³⁴ D. Hanahan, *J. Mol. Biol.* **166**, 557 (1983).

³⁵ R. J. Zagursky and M. L. Berman, *Gene* **27**, 183 (1984).

³⁶ F. Sanger, S. Nicklen, and A. R. Coulson, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 5463 (1977).

³⁷ E. Y. Chen and P. H. Seeburg, *DNA* **4**, 165 (1985).

³⁸ D. Seto, *Nucleic Acids. Res.* **18**, 5905 (1990).

use of double-fine sharktooth combs on 34-cm wide gels allows 24 sets of sequencing reactions to be loaded per sequencing gel.

Procedure. To prepare single-stranded DNA for sequencing, a 1.5-ml aliquot of $2 \times \text{YT}^{39}$ containing 1.5×10^7 M13 RV-1 helper phage⁴⁰ is inoculated with 30 μl of a fresh overnight culture. The cells are grown aerobically in a roller drum and infection is allowed to proceed for 6 hr at 37°. The cultures are then transferred to 1.5-ml Eppendorf tubes and centrifuged for 10 min. The full 10-min centrifugation is necessary to pellet cell debris that can otherwise interfere with the DNA sequencing. A 1.2-ml portion of the supernatant, containing the phage, is transferred to an Eppendorf tube containing 300 μl of 2.5 M NaCl in 20% polyethylene glycol (M_r 8000). After 30 min at room temperature, this solution is centrifuged for 10 min at 4° to pellet the phage. The supernatant is poured off, the tubes are briefly centrifuged again, and the residual supernatant is removed by aspiration. It is crucial that all the polyethylene glycol be removed, since it will inhibit the sequencing reactions. The phage pellet is suspended in 100 μl TES [20 mM Tris-HCl (pH 7.5), 10 mM NaCl, 0.1 mM EDTA], a 50- μl aliquot of phenol is added, and the tubes are vortexed for 30 sec. After 5 min at room temperature, the tubes are again vortexed for 30 sec, and then centrifuged for 10 min at 4°. An 80- μl portion of the upper, aqueous supernatant is removed and added to 4 μl of 3 M sodium acetate. The single-stranded DNA is precipitated by addition of 200 μl ethanol. The DNA is pelleted by centrifugation for 15 min at 4° and rinsed with 200 μl 70% ethanol. Following another 5-min centrifugation, the DNA is dried in a Savant (Farmingdale, NY) Speed-Vac and dissolved in 25 μl TES. Eight microliters of this DNA is used for sequencing according to the Sequenase (United States Biochemical) protocol. Using this amount of DNA, 200–300 bases can be sequenced. However, when sequencing very close to the sequencing primer (<50 bases), better results are obtained if the purified single-stranded DNA is dissolved in 12 μl of TES, and an 8- μl aliquot is used for sequencing.

To prepare double-stranded plasmid DNA for sequencing, 20 μl of a fresh overnight culture is added to 2 ml of L broth (LB),³⁹ and the cells are grown at 37° for 5 hr (it is important to use a freshly saturated culture). A 1.5 ml portion of culture is transferred to a 1.5-ml Eppendorf tube, and the cells are pelleted by centrifugation for 30 sec. The cell pellet is suspended in 300 μl STET [8% sucrose, 0.5% (v/v) Triton X-100, 50 mM EDTA, 50 mM Tris-HCl (pH 8)], and a 20- μl aliquot of 10 mg/ml lysozyme

³⁹ J. Miller, "Experiments in Molecular Genetics." Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1972.

⁴⁰ A. Levinson, D. Silver, and B. Seed, *J. Mol. Appl. Genet.* **2**, 507 (1984).

is added. After 5 min at room temperature, the solution is heated in a boiling water bath for 2 min and then centrifuged for 5 min at room temperature. The pellet is removed with a toothpick, and 200 μ l of 2.5 M ammonium acetate/75% 2-propanol is added. After 5 min at room temperature, the solution is centrifuged for 5 min at room temperature, and the pellet is washed with 200 μ l 70% ethanol. Following another 5-min centrifugation, the pellet is dried in a Savant Speed-Vac, and then resuspended in 20 μ l TE [10 mM Tris-HCl (pH 8), 1 mM EDTA].

To sequence using the double-stranded DNA as a template, the entire sample prepared as described above is used. To the DNA are added 4 μ l of 16 ng/ μ l sequencing primer and 6 μ l of 2 M NaOH. The solution is incubated at 37° for 15 min. The DNA is precipitated by addition of 10 μ l of 3 M sodium acetate (pH 5.2) and 200 μ l of cold ethanol, and is pelleted by centrifugation for 15 min at 4°. The pellet is washed with 200 μ l of 70% ethanol. Following a 5-min centrifugation, the supernatant is removed and the pellet is dried in a Savant Speed-Vac. The dried pellet is resuspended in 12 μ l of sequencing buffer [prepared by mixing 2 μ l of 5 $\times$ Sequenase buffer (United States Biochemical), 1 μ l of 0.1 M DTT, and 9 μ l of water] and the sequencing is continued according to the Sequenase protocol, beginning with the labeling step.

Methodological Considerations

Sources of Nonrandomness. In using random mutagenesis techniques to study protein structure and function, it is important that the distribution of amino acids at the mutagenized positions be as near to truly random as possible. The nature of the genetic code dictates that the distribution cannot be perfectly random, since different amino acids are encoded by different numbers of codons. However, at the nucleotide level, a random distribution should be possible. In practice, there are several potential sources of nonrandomness to consider. Some of these sources are difficult or impossible to control experimentally. For example, the stability or translation of an mRNA may be sensitive to the presence of particular codons at some positions, causing certain protein sequences to be under-represented among the functional sequences. Deviations from a random distribution of bases may also arise in the construction of the cassette. If the coupling efficiency during oligonucleotide synthesis is not the same for each of the four bases, there will be an unequal distribution of bases in the final cassette. Bias may also be introduced in the annealing of the two oligonucleotide strands. For example, in the inosine pairing method, randomized sequences rich in C may pair better than other sequences

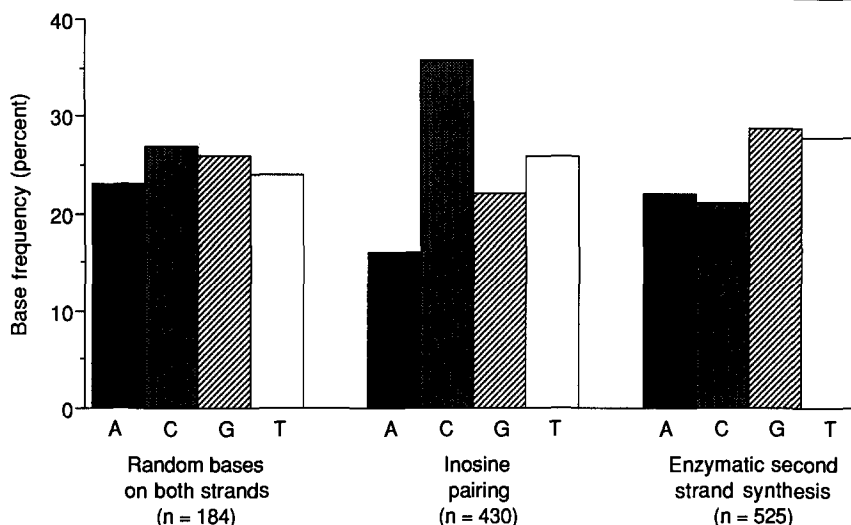


FIG. 7. Base frequencies observed in randomization experiments. The frequency with which each base was observed on the randomized strand is shown, using three different strategies for construction of the complementary strand. In each case, sequences from unselected populations were used, and only data from fully randomized positions were included (i.e., the third position in NN^G randomizations was omitted). The total number of base sequences determined for each strategy is indicated.

to the inosine-containing strand.³³ As a result, such sequences may be overrepresented in the mutagenized pool.

Evaluations of Base Distributions. To estimate the nucleotide distribution within pools of mutagenized plasmids generated by different methods, we sequenced a number of randomized genes from transformants that were not subjected to a functional selection. Figure 7 shows the frequency of each nucleotide observed at randomized positions, mutated using (1) random bases on both strands of the cassette, (2) the inosine pairing method, or (3) enzymatic second strand synthesis. This figure shows data from fully randomized bases only; that is, the third base in NN^G randomizations is omitted. All three methods give fairly even distributions of the four bases, although the inosine-pairing method overrepresents C and underrepresents A. The bias toward C in the inosine method probably reflects a slight base-pairing preference.³³ If the 5' ends of the oligonucleotide cassette are not phosphorylated prior to ligating into the plasmid backbone (thereby leaving a nick), then this bias toward C can be extreme at positions near the end of the cassette. For example, in one such experiment, C was recovered 100% of the time at the terminal codon, 95% of the time at the penultimate codon, and approximately 70% of the time at

the next two codons. This may result from exonuclease digestion of the randomized bases near the nick, followed by preferential insertion of C opposite the inosines in the gapped duplex during repair synthesis *in vivo*.

Bonus Mutations. Another problem that may be encountered during construction of the mutagenic cassette is the incorporation of the wrong base at nonrandomized positions. When randomizing positions by including NN_C^C on both strands or by using the inosine-pairing method, "bonus" mutations are observed at a frequency of about 0.03%/base throughout the cassette region. As a consequence, for cassettes of 30–60 bp, roughly 1–2% of the sequences are found to contain mutations at nonrandomized positions. These mutations may arise from misincorporation of nucleotide bases during synthesis of the oligonucleotides on the DNA synthesizer. Alternatively, they could reflect chemical modification of bases or failure to deprotect bases completely, with subsequent errors arising during replication in the cell. Another problem is the occurrence of single-base deletions in the cassette region. Such deletions appear in roughly 10% of the sequences from unselected populations.

A somewhat higher frequency of bonus mutations is observed when the second strand of the cassette is synthesized enzymatically. In one set of experiments in which the Sequenase enzyme was used to extend the oligonucleotides as shown in Fig. 5C, 30% of the candidates analyzed contained additional mutations (including deletions) at nonrandomized positions. Since the cassettes in these experiments were roughly 100 bp in length, this frequency represents a 0.3% misincorporation rate at each base. Many of these errors probably arise during the extension reaction, both as a consequence of the lack of an editing function in the Sequenase enzyme and because the extension reaction is usually performed at high nucleotide concentrations. Although the Sequenase enzyme is more error prone than the Klenow fragment of DNA polymerase I, it can be used to extend some oligonucleotides that cannot be extended with the Klenow fragment. Hence, there is a trade-off of efficiency versus fidelity when choosing between these two enzymes. Native T7 polymerase has many desirable properties that may help to overcome some of these problems,⁴¹ although we have not had extensive experience with this enzyme.

Heterogeneity. In cassette randomization experiments, there is always heterogeneity among the plasmid molecules, prior to transformation into recipient cells. Single plasmids will encode different sequences as a consequence of the deliberate introduction of random bases during construction of the plasmids, and some or even all of the plasmids may bear mismatches as a consequence of the method of second strand synthesis of the muta-

⁴¹ K. Bebenek and T. A. Kunkel, *Nucleic Acids Res.* **17**, 5408 (1989).

genic cassette. Heterogeneity may persist after transformation if a single cell is transformed with plasmids bearing different sequences or if DNA replication of mismatched strands yields more than one sequence. Fortunately, such examples are rare in our experience. Presumably, multiple transformation events are uncommon and repair *in vivo* corrects most mismatches before replication can occur. As a consequence, it is possible to use primary transformants for selections, screens, and sequencing in most cases. Nevertheless, the possibility of mixed populations in cells must be kept in mind. It is good practice to restreak candidates to single colonies before single-stranded template DNA is prepared for sequencing, and to retest activity phenotypes after colony purification. Heterogeneity may be indicated as a potential problem if phenotypes are not stable, if independent isolates of the same sequence appear to confer different phenotypes, or if sequencing suggests the presence of more than one base at a given position. In these cases, plasmid DNA should be purified and used to retransform cells before further experiments.

[28] Linker Insertion Mutagenesis as Probe of Structure-Function Relationships

By STEPHEN P. GOFF and VINAYAKA R. PRASAD

The *in vitro* modification of cloned genes is one of the most powerful methods available for the localization of functional domains of a given gene product. The analysis of the function of mutant gene products, and correlation of the position of mutations with their effects, can quickly permit a determination of the essential regions encoding that function. A variety of mutagenesis techniques, variously generating substitutions, insertions, and deletions, can be used to reveal the structural organization of domains in proteins. Amino acid substitution, readily achieved with oligonucleotides, is perhaps the most commonly used form of mutagenesis. When the structure of a gene is only poorly understood, however, it is difficult and expensive to use this method to make a large library of mutants with changes scattered across the gene. In contrast, linker insertion mutagenesis offers a rapid means of structure-function analysis in the absence of clues suggesting specific site-directed mutations.

In linker insertion mutagenesis, short palindromic oligonucleotides containing the recognition sequence for a particular restriction enzyme are inserted at known locations throughout the length of the gene, and the effects of these insertions on protein function and stability are studied.

Insight into the localization of functional domains of several proteins has been obtained by the procedure (for examples, see Refs. 1–9). Mutagenesis by linker insertion is attractive because the insertions are genetically stable, the mutations are easily mapped and monitored, they facilitate subsequent manipulations (such as the generation of terminal or internal deletions) employing the new restriction sites inserted, and the effects of the mutations on protein function are significant but often restricted to a relatively local part of the protein. In a surprising number of cases, linker insertion mutations have been found to confer a temperature-sensitive phenotype on the target gene.^{10–15} This property is one of the most powerful features of the method.

We have been interested in understanding the structure and function of reverse transcriptase (RT), a key enzyme in the replication of retroviruses. It is a multifunctional protein, displaying RNA-directed DNA polymerase, DNA-directed DNA polymerase, and ribonuclease H activities. Understanding the manner in which these enzymatic activities, residing on the same polypeptide, are coordinated during replication requires knowledge of the spatial organization of domains encoding them. To this end, we have used linker mutagenesis for mapping the two major domains of retroviral reverse transcriptases. The availability of bacterially expressed functional proteins allowed us to carry out analysis of a series of linker insertion and deletion mutants of two different viral RTs.^{4,9} We describe below linker mutagenesis methods, listing the advantages and disadvantages for each method, and briefly summarize the results of the analysis of RTs as examples of applications.

- ¹ J. C. Stone, T. Atkinson, M. Smith, and T. Pawson, *Cell (Cambridge, Mass.)* **37**, 549 (1984).
- ² M. Ng and M. L. Privalsky, *J. Virol.* **58**, 542 (1986).
- ³ S. D. Lyman and L. R. Rohrschneider, *Mol. Cell. Biol.* **7**, 3287 (1987).
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Generation of Conventional Linker Insertion Libraries

It is straightforward to generate a single insertion, or a small number of insertions, at specific locations in a cloned gene. Briefly, a circular plasmid DNA is cleaved by an enzyme that cuts only once, linkers are joined to the ends of the linear DNA, and the DNA is recircularized. But the real power of the method is in its capability of generating larger libraries of mutants, members of which contain insertions at single sites scattered along the entire coding region. The procedure for creating insertions at multiple sites involves the generation of a pool of permuted linear plasmid DNA molecules obtained by cleavage at one of a series of positions along the length of the gene, adding linkers to the ends of the linear DNAs, recircularizing the DNA and transforming bacteria, and screening the colonies for the presence of the newly created restriction site in their plasmids (Fig. 1). We will describe each step in turn.

Generation of Permuted Linear DNA Molecules

Several different methods can be used to generate a random pool of linear molecules derived from the cleavage of supercoiled plasmid containing the gene of interest. One can either use controlled digestion of the plasmid DNA with DNase I to generate double-stranded breaks,^{16,17} or partial digestion with a restriction enzyme that cuts the gene of interest at multiple sites. Restriction enzymes are preferable, because the precise site of insertion and the altered amino acid sequence can usually be deduced from limited mapping data and a known nucleotide sequence. When DNase I is used to create sites for insertion, the resulting mutations must generally be examined very closely—often by DNA sequence analysis—to precisely map the position and structure of the mutations. DNase I does not often make blunt ends, and the resulting insertions are often “dirty,” i.e., associated with loss or duplication of parental nucleotides.

In choosing a restriction enzyme for linearizing the plasmid, it is best to use enzymes with the maximum number of sites in the sequence of interest, and the minimum number of sites in the vector. In general this requirement demands that the vector be as small as possible; the pUC series of plasmids are eminently suitable. Simplest to use are enzymes that produce blunt ends; our preferred enzymes include *Hae*III, *Alu*I, *Fnu*DII, *Rsa*I, and *Nla*IV. Enzymes that produce staggered cuts yield cohesive termini that must be blunted with the Klenow fragment of DNA

¹⁶ F. Heffron, M. So, and B. J. McCarthy, *Proc. Natl. Acad. Sci. U.S.A.* **75**, 6012 (1978).

¹⁷ K. Tatchell, K. A. Nasmyth, B. D. Hall, C. Astell, and M. Smith, *Cell (Cambridge, Mass.)* **27**, 25 (1981).

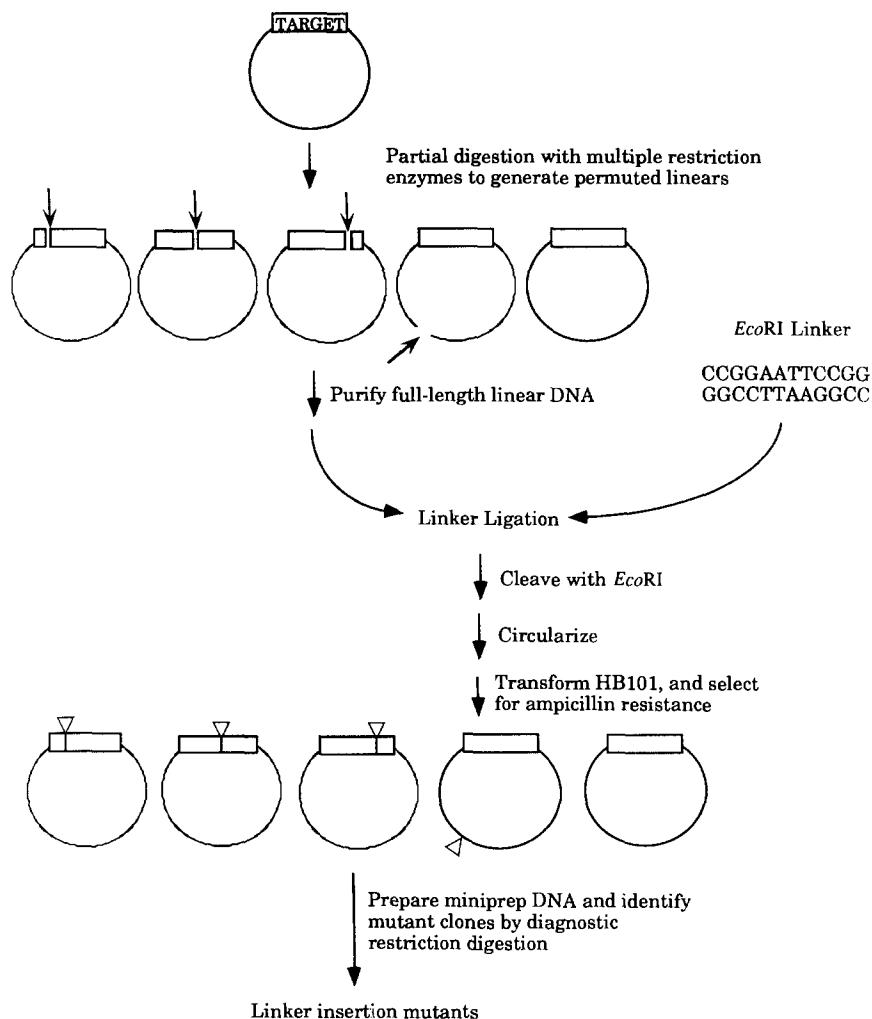


FIG. 1. Construction of a library of linker insertion mutants. A plasmid DNA carrying the target gene is exposed to limited digestion by any one of a variety of restriction enzymes to produce a collection of permuted linear molecules. The full-length linear DNA is purified, and synthetic oligonucleotides containing a restriction enzyme recognition site are added to the termini. The DNA is treated with the cognate enzyme (in this example, *EcoRI*) and recircularized by ligation at low DNA concentrations. The DNA is used to transform bacteria, and individual clones are screened for the presence and position of insertions.

polymerase I (see below) before linkers are added; suitable examples of this class are *Dde*I, *Hinf*I, *Sau*3AI, and *Sau*96I.

A partial digestion of the plasmid DNA containing the target gene is carried out in one of the following ways.

Time-Course Method. This is the most common method for producing partial digests of plasmid DNA. One sets up a digestion of about 5–20 μ g of DNA with 1–2 units of the restriction enzyme of choice, and aliquots of 2–5 μ g are withdrawn into an equal volume of 20 mM EDTA at various times. Digestion times are often very short, in the range of 1–10 min. With some enzymes, digestion must be limited by carrying out the reaction at reduced temperatures of 20° or even 0°. The products of digestion are displayed by electrophoresis on a 0.8% (w/v) agarose gel to determine the time point at which most DNA is in the full-length, linear form. Ideally, the products of partial digestion should contain very few molecules that are uncut or multiply cut and should contain the maximum amount of full-length linears. Conditions of electrophoresis that maximize the separation of full-length linears from contaminating form I and form II DNAs should be used whenever possible.

While this method is generally easy to use, it is often hard to hit on the time point that yields an appropriate level of digestion due to the fact that some enzymes act too rapidly. With some enzymes there can be serious difficulties with “hot spots,” i.e., preferential cleavage at one or a few sites.

Ethidium Bromide Method. Slowing the rate of restriction enzyme reaction by including inhibitors increases the chances of obtaining an appropriate level of digestion. If the inhibitor acts specifically or selectively on those DNAs that have already been cleaved, the effect is to increase the yield of singly cut molecules. One way of achieving this is to include ethidium bromide in the reaction mixture.¹⁸ Ethidium will bind to a limited extent in supercoiled DNA, inhibiting digestion to a similarly limited extent; once the DNA has been cleaved (or nicked), much more ethidium will intercalate and so inhibit further digestion. In our laboratory, we use the following method.

Set up digestions as follows:

Supercoiled DNA	2–5 μ g
Recommended buffer (10 $\times$)	10 μ l
Ethidium bromide (0.5 mg/ml)	0, 5, 10, 20, or 50 μ l
Water	to 100- μ l total volume
Restriction enzyme	25 units (U)

¹⁸ R. C. Parker, R. M. Watson, and J. Vinograd, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 851 (1977).

The digestion is allowed to proceed for an extended period (1–3 hr, depending on the restriction enzyme being used). At the end of the reaction, a small aliquot of the reaction is electrophoresed on a 0.8% agarose gel alongside an equal quantity of uncut plasmid DNA to determine the extent of digestion. If an appropriate level of digestion has been obtained in one of them, then the rest of that reaction is fractionated on a preparative 0.8% agarose gel.

In all cases, once a set of permuted linear DNAs has been generated, the full-length DNA is isolated after electrophoresis. If the termini produced by the chosen restriction enzyme are not blunt, or if DNase I is used to linearize the DNA, the ends must be blunted with the Klenow fragment of DNA polymerase I. A typical incubation includes

DNA	~1 μ g
Tris-HCl, pH7.5	50 mM
MgCl ₂	10 mM
DTT	0.1 mM
dXTP	100 μ M each
Water	to 50- μ l total volume
Klenow	2–5 U

Reactions are carried out for 15–30 min at 20°; the DNA is recovered after phenol extraction and precipitation with ethanol.

Addition of Linkers to DNA Termini and Circularization

First, a linker of the correct size and sequence must be chosen. With phosphorylated linkers, the sequence must include a recognition site for a restriction enzyme that does not cut the parental DNA. Linkers containing recognition sites for a large array of enzymes are now available, and usually a suitable linker is readily identified. There are some additional considerations in choosing a linker. The sequence should generally not include a terminator triplet (TAG, TAA, or TGA) in any of the three possible reading frames; thus, *Spe*I linkers (sequence GACTAGTC) or *Bcl*I linkers (sequence CTGATCAG) would be poor choices. In addition, the insertion of the linker into a particular restriction site should not create a terminator triplet. Thus, the insertion of some *Cla*I linkers (those of sequence GATCGATC) into *Rsa*I sites (sequence GT'AC) would be ill-advised. There is not a great deal of information concerning the effects of insertions of different sequence. Because random insertions of a given linker occur in all three translational reading frames of the target gene, three different sets of inserted amino acids are generated; different novel condons are also created at the joints between linker and site. Presumably

some inserted amino acids have much more serious consequences than others, but no systematic studies of these ideas have been made.

We have had most experience with linkers 8, 10, or 12 nucleotides (nt) in length. Although 6-nt linkers can be used,¹⁹ these shorter linkers are generally added very inefficiently; thus, if it is desirable to add a 6-mer, a 12-mer consisting of a tandem repeat of a 6-mer may be easier. For example, in the case of *EcoRI*, a linker of the sequence GAATTCGAATTC is first added; the eventual digestion with *EcoRI* and religation will reduce the size of the final insertion to only 6 bp (see below). There is some evidence that short insertions (say, 6 bp or less) are more often silent than larger insertions, but studies of this notion are limited. It is probably also true that 3- and 6-bp insertions generate temperature-sensitive phenotypes more often than larger insertions.

There are two ways to add linkers to DNA termini.

Phosphorylated Linkers. The chemistry used in the synthesis of linkers generally produces nonphosphorylated 5' termini. Many commercial firms now provide prephosphorylated linkers, and we have used these preparations with good success. If phosphorylated linkers are not available, linkers can be efficiently and easily phosphorylated with polynucleotide kinase and ATP:

Linker	100–1000 ng
10 × Buffer L (500 mM Tris-HCl, pH 7.6; 100 mM MgCl ₂ ; 100 mM dithiothreitol (DTT); 5 mM ATP)	1 μl
Water	to 10-μl total volume
T4 polynucleotide kinase	2–10 U

Incubation is for 30 min at 37°.

In adding linkers to DNA, we have found that the use of a 40- to 50-fold molar excess of linkers over DNA termini is sufficient to allow ligation of more than one linker per end. It is helpful to keep the reaction volume small to promote addition. A ligation reaction is set up as follows:

Permuted linear plasmid DNA	100 ng
Phosphorylated <i>EcoRI</i> linkers	40- to 50-fold molar excess (100 ng, depending on the DNA length)
10 × Buffer L (see above)	1 μl
Water	to 10-μl total volume
T4 DNA ligase	2–10 U

¹⁹ F. Barany, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 4202 (1985).

The reaction is allowed to proceed for 12–18 hr at 20°. It is helpful to replenish the ATP once after 8 hr. Add 0.5 μ l of 10 mM ATP and continue the incubation.

At the conclusion of the reaction, the ligase is inactivated by heating the ligation mix at 65° for 5 min and the DNA is digested with the restriction enzyme whose recognition sequence is present on the linker. This digestion removes the majority of the linkers appended to each terminus, and leaves on each end only one-half of the total duplex linker; each terminus now displays a cohesive end. It is necessary to use a large excess of the enzyme in order to obtain complete digestion of all the linker molecules that are ligated, in the presence of a large excess of free linkers. Alternatively, one can first electrophoretically separate the unligated linkers from the DNA containing the linkers, and then digest it.

In the final step, the digested DNA is recircularized in a simple reaction.

DNA	50–100 ng
10 $\times$ buffer L	5 μ l
Water	to 50- μ l total volume
T4 DNA ligase	2–10 U

The optimal DNA concentration for promoting self-ligation is low, in the range of 1–10 μ g/ml. The reaction is allowed to proceed for 10–20 hr at 15°.

Nonphosphorylated Linkers. Occasionally, one needs to insert linkers containing a restriction site that is also present on the gene being mutated. This will lead to deletions when the linkers on the DNA termini are subjected to digestion. In such an instance, the use of nonphosphorylated linkers will obviate the need for digestion with the restriction enzyme during the cloning step. The procedure is essentially similar to the one described above except that the linkers are not phosphorylated. The ligation of the 3'-OH ends of the double-stranded linker to the PO₄ at the 5' ends of the linear DNA proceeds as expected, while the 5'-OH ends of the linker cannot be ligated to the 3'-OH ends of the linear DNA molecule. After one linker is added to the DNA, other linkers cannot be added to the modified termini. Thus, there is no need to remove oligomerized linkers by cleavage with the cognate restriction enzyme. In these reactions, it must be noted that there is often a significant background of circularization of the DNA without linkers.

At the end of the ligation, the DNA is heated at 65° to melt the unligated linker strand from the DNA, and then the linear DNA is separated from circles and free linkers by electrophoresis. After recovery, the DNA can generally be used directly to transform bacteria; the presence of the single-stranded linker tails on the termini is usually sufficient to induce circular-

ization by annealing. The nicks in the DNA are repaired by the bacteria *in vivo*. Alternatively, the linear DNA can be phosphorylated with fresh ATP and polynucleotide kinase, and circularized *in vitro* using ligase reactions set up as described above.

Recovery of Clones and Mapping Insertions

After circular molecules containing inserts are generated, the DNA is used to transform bacteria. Competent cells can generally be transformed directly with a portion of the ligation mix without any purification of the DNA. The resulting colonies should each contain plasmid molecules with a single linker inserted at a particular position.

In the next step, the insertion mutations in individual clones must be localized. Insertions in the target gene being analyzed as well as in nonessential regions of the plasmid vector will be recovered. The most straightforward approach is to prepare minipreparations of DNAs and to digest each preparation with a pair of restriction enzymes: the enzyme whose recognition sequence is on the linker, and an enzyme cutting once on the parental plasmid. Such a digest will give the distance from the marker site to the linker, but it will leave symmetry ambiguities. Subsequent double digestion with an enzyme recognizing a different marker site can then be used to resolve these ambiguities.

In many cases it will be important to determine the exact site of insertion of a linker. When there are many potential target sites (i.e., when the sites for cleavage by the linearizing restriction enzyme are closely spaced) fine mapping is required. We usually examine each mutant DNA by digestion with a frequent-cutting enzyme, with or without the linker enzyme, and examine the digests by electrophoresis on polyacrylamide gels. Very occasionally it will be necessary to determine the DNA sequence in the area of the insertion to resolve ambiguities.

It is worth noting that sometimes short deletions are generated in the procedure. These result from joining of linkers to linear DNAs that have been cleaved not once but twice by the partial digestion; if two cuts are very close together the resulting linear will be nearly full length and will not be resolved from the singly cut linears. Even more rarely, Klenow DNA polymerase I used to fill in the cohesive ends can remove bases unexpectedly before linkers are added. It should also be noted that sometimes larger insertions than expected can be formed. These often arise when tandem insertions of two or more copies of the linker are formed at a given site; usually the cause is a failure by the linker restriction enzyme to remove the extra linkers added to the termini. In all these cases, frameshift and other more drastic mutations can be generated. It is very

helpful to have an independent check that the mutants still encode a full-length protein before firmly assigning a particular phenotype to a given, supposedly simple, insertion mutation.

Mutagenesis of Human Immunodeficiency Virus Reverse Transcriptase

We have applied the simple linker insertion method described above to probe the structure of the reverse transcriptase enzyme (RT) encoded by the human immunodeficiency virus (HIV-1).⁹ This enzyme is a heterodimeric protein in the virion particle, consisting of 66- and 51-kDa subunits that are closely related; the smaller subunit is a proteolytic fragment of the larger. The two share the same N-terminal sequence. For mutagenesis of HIV RT sequences, we employed an expression construct termed pHRTRX2 that encodes the p66 version of HIV RT as a fusion protein with a portion of the bacterial TrpE protein.²⁰

We cleaved pHRTRX2 DNA with a variety of enzymes, controlling the digestions with time, isolated full-length linears, and when necessary blunted the termini with Klenow. We added phosphorylated *EcoRI* linkers followed by digestion with *EcoRI*. The inserts were mapped, and were found to be broadly scattered along the gene (Fig. 2). In virtually every case, the mutant protein was still synthesized and was stably retained in the cell. The analysis of the mutants for the RNA-dependent DNA polymerase activity showed that the polymerase activity was localized to the N-terminal two-thirds of the RT molecule; most mutants with insertions in the N-terminal region were inactive, while most in the C-terminal region were still active (Fig. 2). To define the polymerase domain more firmly, we then generated directional deletions from each end, employing the *EcoRI* sites in the inserted linkers. This deletion analysis helped further localize the minimal domain encoding the polymerase function to the N-terminal p51 region. Interestingly, rare insertions in HIV RT were highly disruptive of polymerase activity even though they mapped in the C terminus. For example, insertion mutant A5 in the C-terminal region affects the polymerase function even though many deletions that span this mutation do not (e.g., mutant R8). This insertion, therefore, may interfere with the proper folding of the molecule.

It should be added that this type of domain mapping would not yield simple results for a protein without discrete functional domains. For example, a linker insertion analysis of adenovirus DNA polymerase similar to that described above for RT showed that functional residues of that polymerase were distributed across the length of the molecule.⁷

²⁰ N. Tanese, V. R. Prasad, and S. P. Goff, *DNA* 7, 407 (1988).

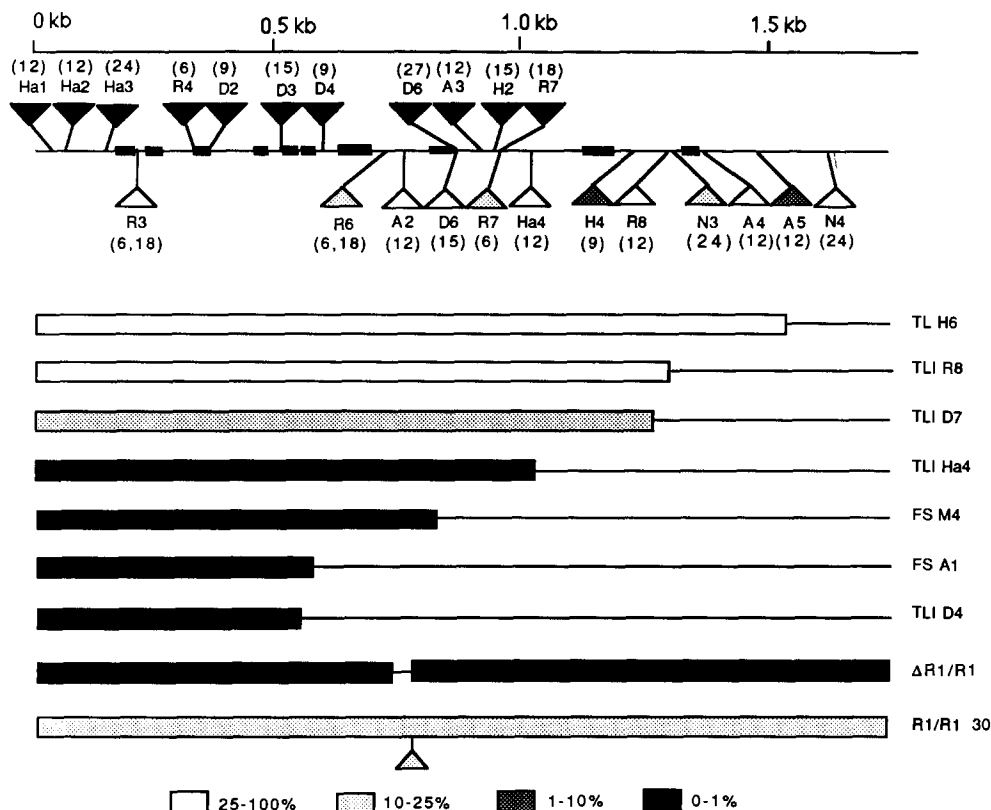


FIG. 2. An application of linker insertion mutagenesis: analysis of HIV reverse transcriptase. A library of mutants was generated, and members of the library were assayed for the levels of DNA polymerase activity in the encoded protein. Top line: Triangles indicate the position of each insertion; the numbers in parentheses indicate the size of the insertion (in bp). The enzyme activity of each mutant is indicated by the shading from black to white (see key at bottom). Lower lines: a series of C-terminally truncated enzymes were similarly assayed. (From Ref. 9.)

Cassette-Mediated Linker Insertions

In some cases the addition of linkers to linearized DNA is inefficient, as when the linearizing restriction enzyme leaves cohesive or heterogeneous termini; large numbers of colonies need to be screened to find those DNAs containing linkers. There are methods that help eliminate the burden of screening a large number of colonies in search of the rare mutant clones.

Such methods rely on the use of cassettes that contain selectable or screenable genetic markers, symmetrically flanked by multiple restriction sites. Markers used in cassette-mediated linker insertions include genes for resistance to kanamycin²¹ and streptomycin/spectinomycin²² or a gene for suppressor tRNA, *supF*, in combination with *lacZam* host bacteria.²³ The general idea is a two-step procedure: first the cassette is inserted at random positions along the target gene; then the bulk of the cassette is excised, leaving only a few nucleotides at the site of the original insertion. The final result is similar to that achieved with conventional linker insertion.

The method differs in several ways from the conventional method described above. We will outline the procedure for the *supF* cassette ("sup-link"), discussing each difference from the simple linker method in turn.

Design and Preparation of Linker Insertion Cassette

The cassette consists of a small fragment of DNA, carrying an autonomous marker; for example, we built a cassette beginning with a 220-bp fragment containing the *supF* gene from the plasmid π VX.²³ Restriction sites are first engineered to flank the gene on both termini. The *supF* gene is flanked by *Eco*RI sites and was recovered by simple cleavage with *Eco*RI. The choice of the enzyme site proximally flanking the marker gene is important because this is the site that will ultimately be present in the insertion mutation in the target gene, and in particular the site must not appear in the target plasmid. A second site is then appended to the fragment, usually in the form of linkers. In the *supF* case, the termini were filled in by treatment with Klenow polymerase I, and the *Eco*RI sites were reconstructed by addition of *Pvu*II linkers to the blunted termini (Fig. 3). The resulting DNA thus consists of the *supF* gene, flanked first by *Eco*RI sites, and then in turn flanked by *Pvu*II sites. Cleavage with *Pvu*II generates the final cassette.

The cassette used for insertional mutagenesis is generally built only once and cloned into a suitable vector. To clone the *supF* cassette, we simply inserted the fragment into the single *Pvu*II site of pBR328. Once formed, the cassette can be prepared whenever needed simply by excision of the cassette from a stock of plasmid DNA.

²¹ M. L. Smith and G. F. Crouse, *Gene* **84**, 159 (1989).

²² P. Prentki and H. M. Krisch, *Gene* **29**, 303 (1984).

²³ L. I. Lobel and S. P. Goff, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 4149 (1984).

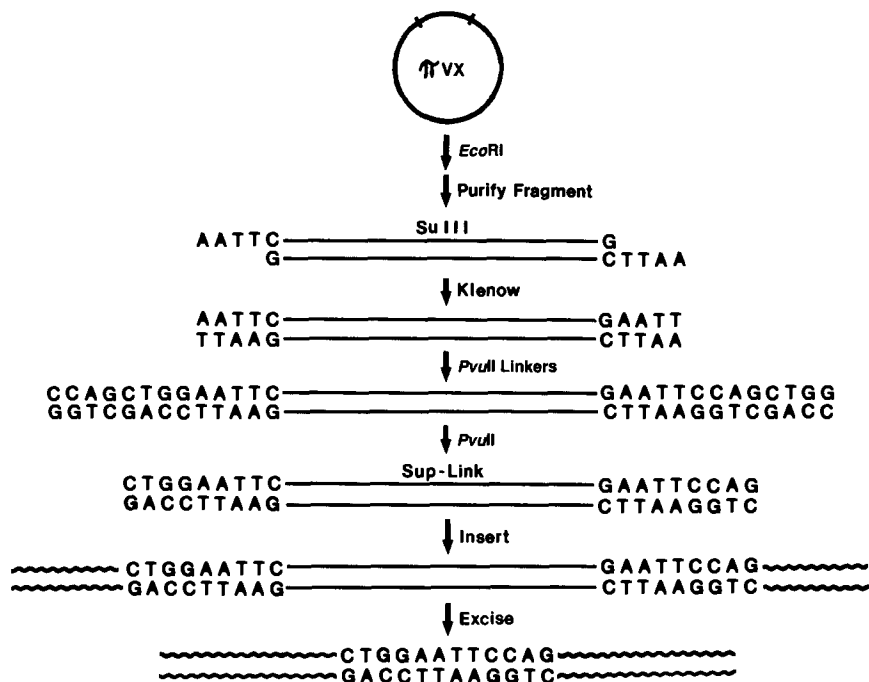


FIG. 3. Construction and use of the suppressor-linker cassette. A DNA fragment carrying the *supF* marker was isolated after *EcoRI* cleavage, the overhanging termini were filled in, and *PvuII* linkers were added. Cleavage with *PvuII* generates the "sup-link" DNA, a blunt-ended DNA cassette suitable for insertional mutagenesis. Successive insertion of the cassette onto a blunt-ended linear target DNA, followed by excision with *EcoRI*, generates a net 12-bp insertion (bottom line). (From Ref. 23.)

Insertion of Cassette into Target Gene

The target DNA is linearized as before and the termini are blunted if necessary. The full-length linear DNA is simply ligated to the cassette (produced by cleavage with *PvuII*) in a small volume; we generally use a large molar excess of the cassette to drive recovery of mutant clones. Typically 0.1–0.5 μ g of target DNA is joined to 100 ng of the cassette, in a total volume of perhaps 10 μ l. This ligation mix is then used directly to transform bacteria to drug resistance. In our experiments, we use the host strain CC114 [Δ (*ara leu*)7697 LacZ⁻Y14(Am) *galU galK HsdR⁻ hsdM⁺ strAr rif^r argE(Am) srl::Tn10, recA1*] as best able to stably propagate many plasmids and to serve as an indicator of the *supF* gene. Recipient cells can be directly plated on MacConkey lactose plates containing ampicillin, and the presence of a *supF* insertion is directly visible as a red colony.

The overall efficiency of the insertion process may vary, but the proportion of the selected (red) colonies that contain an insertion mutant is very high (essentially 100%).

In some cases, it is preferable to pick individual red colonies and map the insertions in the resident plasmids of each clone. Desirable clones can be chosen and used in the step to follow. Alternatively, it is often preferable to pool large numbers of red colonies and prepare bulk DNA from the pooled colonies. This pool of DNA can then be used in the next step.

Excision of Cassette: Generation of Final Insertion

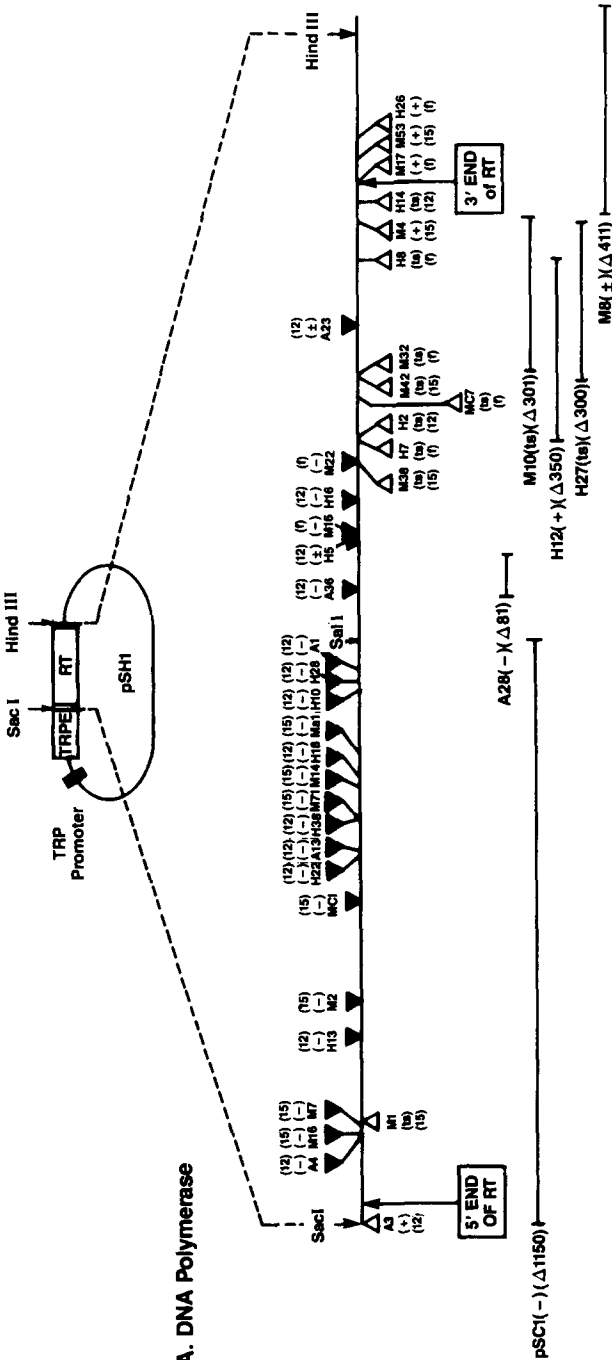
In a second step, the bulk of the cassette is excised. Here the design of the cassette makes this trivial. The plasmid DNA is simply cut with the enzyme whose sites proximally flank the selectable marker of the cassette; this cut removes the selectable marker and leaves only the outer bases of the original cassette appended to the termini of the linear target DNA (Fig. 3). The linear DNA is then recircularized by ligation at low concentrations, and the resulting DNA is used to transform bacteria. Clones from which the cassette has been successfully excised can readily be detected by screening or selecting for loss of the selectable marker. If a single mapped cassette insertion was subjected to the excision procedure, the position of the resulting small insertion is of course known. If a pool of cassette insertion clones was originally made, individual mutants can now be recovered and mapped as for the conventional linker insertion method.

Insertions in Moloney Murine Leukemia Virus Reverse Transcriptase

We made use of the *supF* cassette in our own experiments to generate insertions in reverse transcriptase of the Moloney murine leukemia virus.⁴ The parental plasmid for the procedure was pSH1, expressing the monomeric 80-kDa enzyme as a TrpE fusion protein.²⁴ The enzyme exhibits both activities normally associated with RTs: DNA polymerase activity and RNase H. Our goal was to localize both activities if possible, and determine whether they could be separated by mutation.

We used a variety of restriction enzymes to generate full-length linears of the target DNA, joined the linears to the *supF* cassette, and recovered a large number of clones carrying the suppressor tRNA marker. The efficiency was high even when rather poor enzymes like *MnI* were used for the partial digestion. We excised the marker gene from the clones with *EcoRI* and recovered the final mutants (Fig. 4). The inserts ranged in size from 9 to 15 bp.

²⁴ N. Tanese, M. Roth, and S. P. Goff, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 4944 (1985).



A. DNA Polymerase

B. RNase H

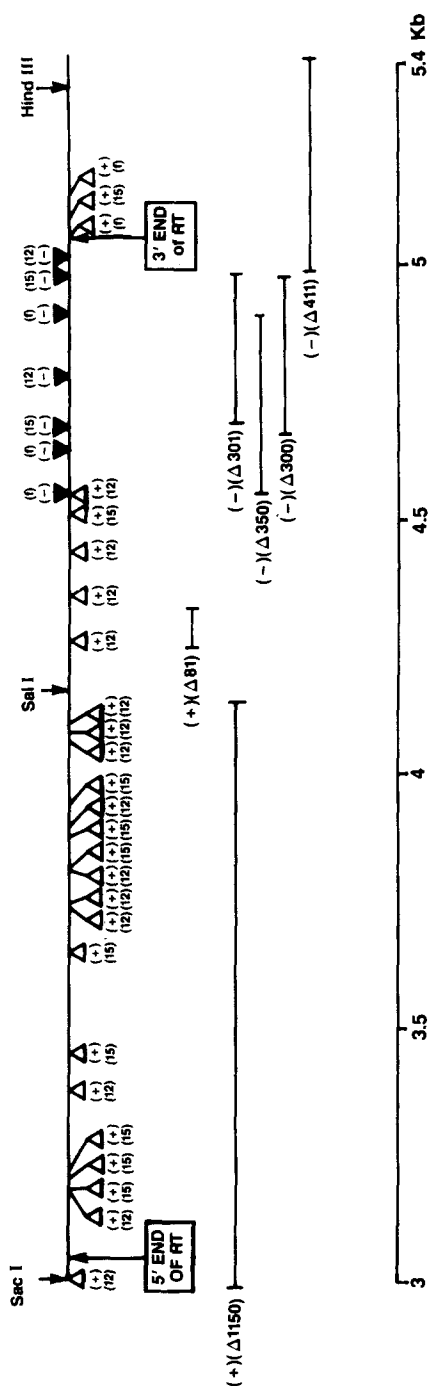


FIG. 4. An application of cassette-mediated insertional mutagenesis: analysis of Moloney murine leukemia virus reverse transcriptase. A plasmid encoding an enzymatically active fusion protein was used to generate a library of insertion mutants with the *supF* cassette, and individual mutants were assayed for both DNA polymerase and RNase H activity. The parental plasmid is drawn at the top; expanded views of the target gene are shown below. (A) DNA polymerase activity; (B) RNase H activity. The position of each linker insertion is indicated by the position of a triangle. Solid triangles denote little ($\pm$) or no (-) activity, and open triangles denote wild-type (+) or substantial levels of temperature-sensitive (ts) activity. Numbers in parentheses indicate the number of base pairs inserted at each site. Frameshift mutations are indicated by (f). A number of deletion mutants are included; the position and extent of the deleted sequences are indicated by the bars below each map. (From Ref. 4.)

Each bacterial clone was induced to express the enzyme, and lysates were tested for both the RNA-dependent DNA polymerase activity and RNase H activity characteristic of the murine viral enzyme. In the case of this enzyme the results were extremely clear: insertions in the N-terminal two-thirds of the protein abolished the polymerase activity with little or no effect on the RNase H activity; insertions in the C-terminal one-third abolished the RNase H activity with little or no effect on the polymerase. The conclusion was inescapable that the enzyme was a two-domain protein, with an N-terminal polymerase and a C-terminal RNase H.

The presence of the *EcoRI* site at each mutation made it very simple to build more variants using the insertion mutants as parents. Because the boundary between the two domains was apparent from the analysis, we generated new clones that separately expressed either the polymerase or RNase H domains alone. The assays of these constructs confirmed the existence of two domains. Fully separable, nonoverlapping sequences of amino acids were able to express either one or the other of the two activities.

Short Insertions Created without Linkers

While the insertion of short sequences via linker insertion is broadly useful, there are other, less frequently employed approaches for the formation of small insertions.

Insertion by Fill-In Reaction

This is a simple method of insertion mutagenesis, especially useful in introducing one-codon insertions into a gene, and therefore one-residue insertions into the gene product.^{12,25} A target plasmid DNA is treated with a frequently cutting restriction enzyme that generates a 3-base, 5'-protruding end; digestion is limited such that full-length linears are produced. A number of such enzymes are available, namely, *AvaII*, *DdeI*, *HinfI* and *Sau96I*. After the linear molecules are generated, the single-stranded projections are filled in by treatment with the Klenow fragment of DNA polymerase I, and the flush-ended, linear molecules are recircularized in a simple ligation reaction at low DNA concentration. The result is the net insertion of 3 bp at the original site of cleavage. In general no new restriction site is created; the mutation must be detected as the loss of a restriction site at that position.

²⁵ J. D. Boeke, *Mol. Gen. Genet.* **181**, 288 (1981).

Although the method of insertion per se is easy, confirming the presence of the insertion and mapping it can be time consuming. As the process of insertion by filling in destroys the existing restriction site and does not necessarily create new sites, no further manipulations at that site can be carried out without resorting to subsequent oligonucleotide-directed mutagenesis.

Insertion by Oligonucleotide-Directed Mutagenesis

All the methods described above are dependent on the presence of a restriction site at the position of the insertion. For initial random mutagenesis, no single site is a particularly important target for mutagenesis. But as information about a protein is acquired, it may be desirable to place an insertion at a very critical site, and because the probability of finding a restriction site at any given position is not high, no restriction enzyme site may be available in the vicinity. In such cases one must employ oligonucleotide-directed linker insertion mutagenesis. An oligonucleotide is synthesized with a short insertion relative to the wild-type sequence, and is used to prime synthesis on single-stranded clones in M13. The procedure allows insertion of a desired sequences (encoding any desired amino acids) at any site.²⁶

Conclusions

Linker insertion is a broadly useful procedure for the generation of libraries of mutations scattered along the coding region of a target gene. The mutations often provide the most rapid definition of the regions of a protein required for a given enzymatic activity.

Acknowledgments

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²⁶ T. A. Kunkel, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 488 (1985).

[29] A Streptomycin Selection for DNA-Binding Activity

By MICHAEL C. MOSSING, JAMES U. BOWIE, and ROBERT T. SAUER

Introduction

Any DNA-binding protein that can bind to a specific target site and block access of RNA polymerase or accessory proteins to nearby promoter sites can regulate gene expression. With the appropriate configuration of promoters, binding sites, and reporter genes, it is possible to monitor the activity of DNA-binding proteins in *Escherichia coli* and also to require such activity for survival of the bacterium. The chief advantage of performing such screens or selections in *E. coli* is that many candidate clones can be processed in parallel and thus rare active genes can be identified. This is true whether the candidate genes are members of a cDNA library and the goal is to identify the source of a particular binding activity, or whether the candidates are heavily mutagenized variants of a well-characterized protein and the goal is to probe the range of amino acid substitutions that are compatible with function.

Screens have been described for the activities of particular DNA-binding proteins based on transcriptional fusions of heterologous promoter-operator elements to easily assayed reporter genes such as *lacZ*,¹ *galK*,² *trpE*,³ and chloramphenicol acetyltransferase.⁴ Biological selections for the activities of specific repressor proteins based on nutritional pathways⁵ or phage resistance^{6,7} have also been reported. Several more general selection schemes have also been developed.⁸⁻¹⁰ Each of these

¹ K. Bertrand, K. Postle, L. V. Wray, and W. S. Reznikoff, *J. Bacteriol.* **158**, 910 (1978).

² K. McKenney, H. Shimatake, D. Court, U. Schmeissner, C. Brady, and M. Rosenberg, "Gene Amplification and Analysis" (J. G. Chirikjian and T. S. Papas, eds.), pp. 383-415. Elsevier/North-Holland, New York, 1981.

³ K.-O. Cho and C. Yanofsky, *J. Mol. Biol.* **204**, 41 (1988).

⁴ A. K. Vershon, J. U. Bowie, T. M. Karplus, and R. T. Sauer, *Proteins: Struct. Funct. Genet.* **1**, 302 (1986).

⁵ J. H. Miller, "Experiments in Molecular Genetics." Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1972.

⁶ M. H. Hecht and R. T. Sauer, *J. Mol. Biol.* **186**, 53 (1985).

⁷ P. Youderian, A. Vershon, S. Bouvier, R. T. Sauer, and M. M. Susskind, *Cell (Cambridge, Mass.)* **35**, 777 (1983).

⁸ N. P. Benson, P. Sugiono, D. N. Arvidson, R. P. Gunsalus, and P. Youderian, *Genetics* **114**, 1 (1987).

methods involves the construction of hybrid promoter-operator regions that directly or indirectly control a gene whose expression can be conditionally required for cell survival. In this chapter, we describe the use of a positive selection involving streptomycin resistance to select for the DNA-binding activities of Lac repressor, λ Cro, and P22 Arc repressor. We also discuss several applications of this selection, including the isolation of neutral mutations, the isolation of second site revertants, and attempts to select proteins with altered DNA-binding specificity.

In *E. coli*, the wild-type ribosomal protein S12 is the target of the antibiotic streptomycin. Streptomycin-resistant alleles of the S12 gene are recessive to the wild-type, sensitive allele.¹¹ Thus, a diploid expressing both the resistant and sensitive genes is killed by the antibiotic. In a strain in which the streptomycin-resistant gene is expressed constitutively from its natural promoter, and the dominant, streptomycin-sensitive gene is expressed from a promoter that is repressible by a DNA-binding activity of interest, resistance to streptomycin is dependent on repressor activity.¹⁰ This is illustrated schematically in Fig. 1. The stringency of a selection of this type can be altered in a number of ways. First, the concentration of streptomycin to which resistance is required can be increased or decreased. Second, expression of the DNA-binding protein can be regulated so that the level of activity subject to selection can be varied. Finally, the expression level of the S12 gene can be adjusted by varying gene dosage, promoter strength, operator sequences, or translational signals.

Methods

In the following sections, we describe the construction of vectors that bring the S12 gene under control of the P22 Arc, Lac, or λ Cro repressors. In some cases, these vectors are described in some detail, since they may prove useful as starting points for construction of vectors that could be used in positive selections for other DNA-binding proteins. We also describe control experiments that address potential problems in using the streptomycin selection.

To bring the wild-type S12 gene under Arc control,¹⁰ the S12 gene was separated from its promoter and translational signals by restriction cleavage with *Hpa*I, which cuts after the fourth codon of the S12 gene (see

⁹ S. J. Elledge, P. Sugiono, L. Guarente, and R. W. Davis, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 3689 (1989).

¹⁰ J. U. Bowie and R. T. Sauer, *J. Biol. Chem.* **264**, 7596 (1989).

¹¹ J. Lederberg, *J. Bacteriol.* **61**, 549 (1951).

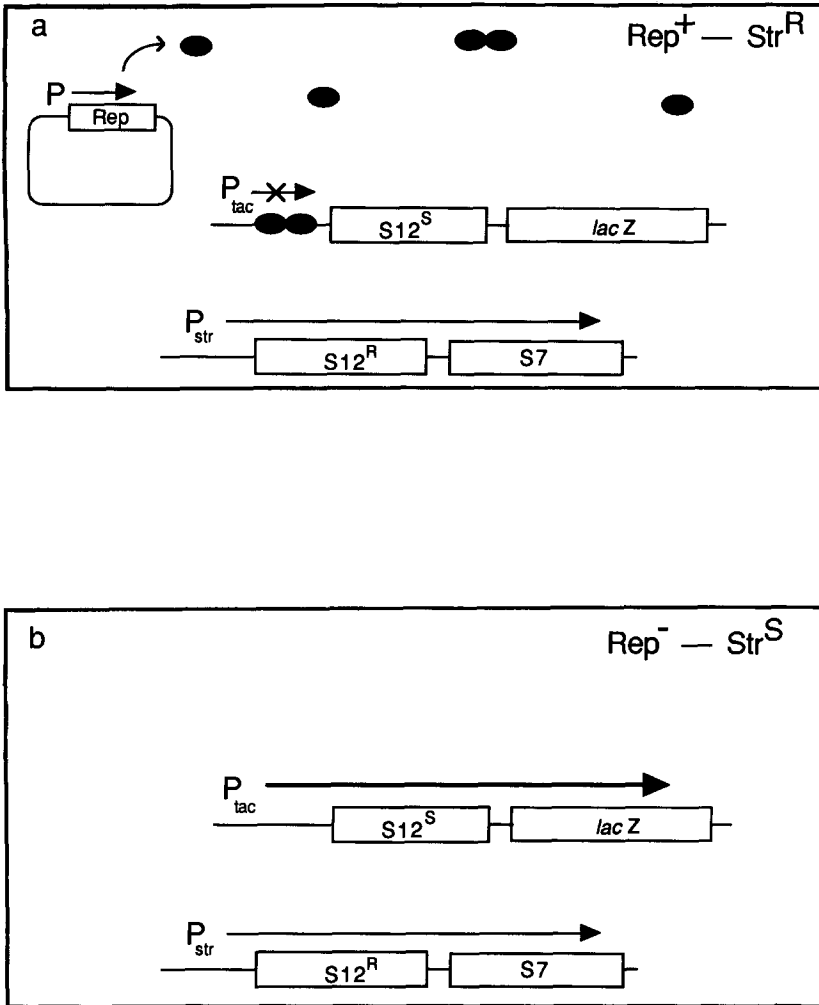


FIG. 1. Principle of the streptomycin selection for DNA-binding activity. Both cells contain a constitutively expressed copy of a streptomycin-resistance gene (S12^R) and a copy of a dominant, streptomycin-sensitive gene (S12^S) under control of a *tac* promoter. The cell shown in (a) also contains a plasmid that expresses a DNA-binding protein (Rep), which represses transcription of the S12^S gene initiated at P_{tac} . As a consequence, cell a is resistant to streptomycin whereas cell b (which contains no DNA-binding protein) is sensitive to streptomycin. Note that a cell that expressed an inactive form of Rep would also be streptomycin sensitive.

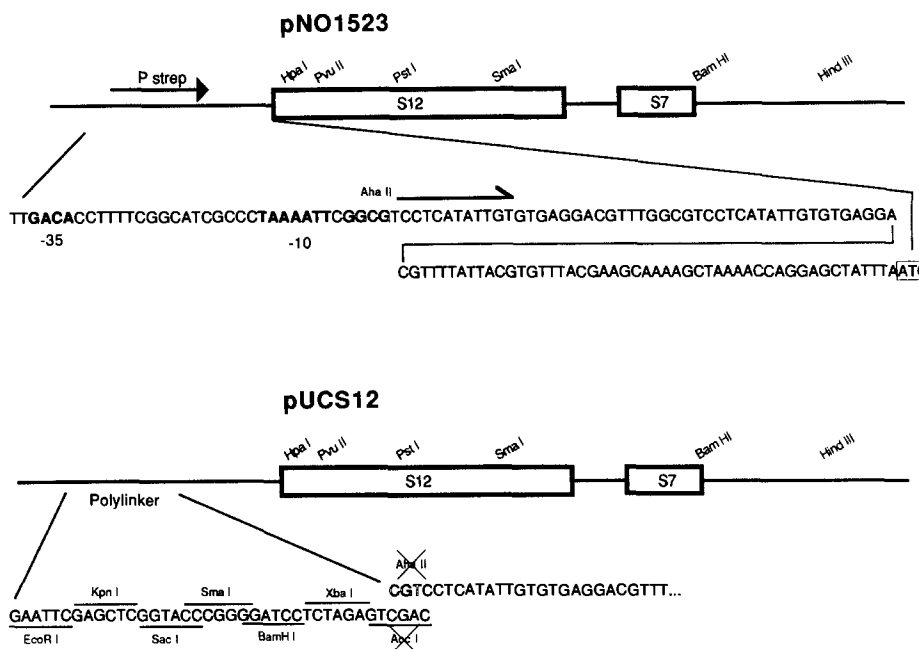


FIG. 2. S12 subcloning. Plasmid pNO1523¹² contains the wild-type S12 gene of *E. coli*. The nucleotide sequence of the S12 promoter and leader mRNA region are shown. The transcription start point is indicated by an arrow and the translation initiation site is boxed. The *AhaII* restriction site overlaps the transcription start site. Plasmid pUCS12 contains the *AhaII*-*HindIII* fragment of pNO1523, cloned into the *AccI*-*HindIII* backbone of plasmid pUC118 (J. Vieira and J. Messing, this series, Vol. 153, p. 3). The *AccI*-*AhaII* junction destroys the sites for both restriction enzymes.

Fig. 2).¹² The 5' end of the gene was then replaced with a synthetic cassette that recreated the S12-coding sequence and ribosome binding site and introduced a unique *BstEII* restriction site upstream of the gene. This *BstEII* site was used to introduce synthetic sequences containing the wild-type *arc* operator and the Arc-repressible promoter P_{ant} (bearing a mild down mutation T→A at position -7 of the promoter). The sequence of the promoter-operator region is shown in Fig. 3A. The P_{ant} -S12 fusion was then cloned into a pBR322-derived plasmid containing the wild-type *arc* gene (under *tac* promoter control) to generate plasmid pUS405. Otherwise identical plasmids containing defective alleles of the *arc* gene were also constructed. *Escherichia coli* strain UA2F is normally streptomycin resistant because it contains the *strA* allele of the S12 gene (see Table I

¹² D. Dean, *Gene* **15**, 99 (1981).

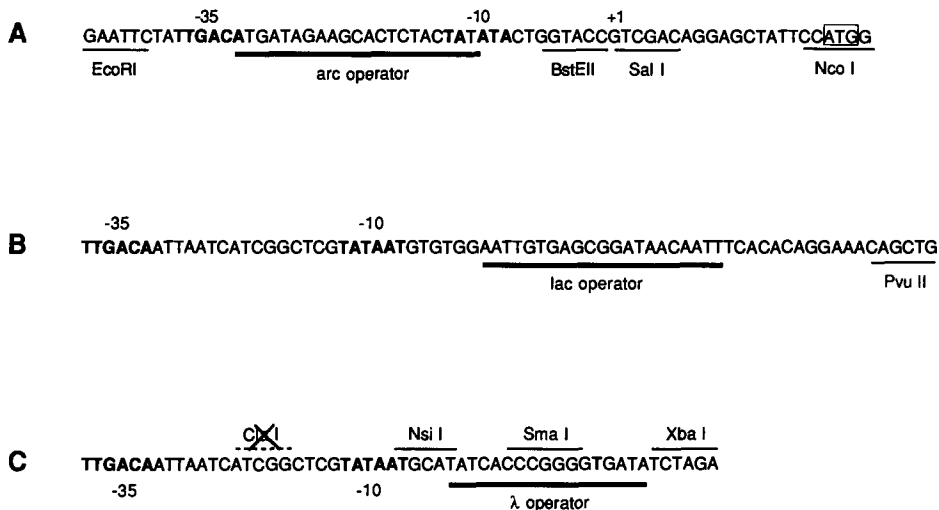


FIG. 3. Sequences of promoter-operator regions. (A) Sequence of the *ant* promoter and *arc* operator. The operator is positioned between the -10 and -35 regions of the promoter. The *Nco*I site overlaps the start point of translation of the wild-type S12 gene. (B) A portion of the *Eco*RI-*Pvu*II fragment, containing the *tac* promoter and *lac* operator from plasmid *ptac*12.¹³ The *Eco*RI site is approximately 200 bases to the left of the -35 region. The translational start of the S12 gene is approximately 115 bases to the right of the *Pvu*II site. (C) A portion of the *Eco*RI-*Xba*I fragment containing a *tac* promoter and the λ 1 operator. The *Eco*RI site is approximately 200 bases to the left of the -35 region. The translational start of the S12 gene is approximately 100 bases to the right of the *Xba*I site.

for complete genotype). When this strain was transformed with pUS405, transformants were found to be streptomycin resistant if the plasmid contained an active *arc* gene and streptomycin sensitive if the plasmid contained a defective *arc* gene.

In selections for Arc activity, strain UA2F was transformed with pUS405 or related plasmids bearing defective or highly mutagenized *arc* genes. Following the heat shock step of the transformation protocol, cells were allowed to grow at 37° in 5 ml of LB broth plus 20 mM MgSO_4 for 1 hr. The cells were then collected by centrifugation and resuspended in 0.5 ml of LB broth containing 20 mM MgSO_4 and 100 $\mu\text{g/ml}$ ampicillin. Selections for Arc⁺ transformants were performed by plating 100- μl portions of cells on LB plates supplemented with 100 $\mu\text{g/ml}$ ampicillin, 50 $\mu\text{g/ml}$ streptomycin, and 2 mg/ml IPTG (isopropyl- β -D-thiogalactopyranoside). In strain UA2F/pUS405, Arc synthesis from the *tac* promoter is repressed by Lac repressor. The inducer IPTG is included in the selection plates because it inactivates Lac repressor, thereby resulting in higher

TABLE 1
BACTERIAL STRAINS, PHAGES, AND PLASMIDS

	Comments	Source/Ref.
Strains		
GW5100	$\Delta(lac\ pro)\ thi\ strA\ SupE\ endA\ sbcB$ with F' <i>traD36 ProAB lacI^Q lacZ</i> $\Delta M15$	G. Walker (MIT)
GW5180	<i>recA::cml</i> derivative of GW5100	G. Walker (MIT)
GW5100F ⁻	F ⁻ derivative of GW5100	G. Walker (MIT)
GW5180F ⁻	F ⁻ derivative of GW5180	G. Walker (MIT)
UA2F	<i>strA thi⁻ his⁻ lacZ⁻ lacY⁺ sup^o recA⁻</i> (λ AC201) F' <i>lacI^Q lacZ::Tn5(kanR) pro⁺</i>	(4)
Phages		
λ I12	<i>imm²¹ λP_R-trpA-lacZ</i> fusion	(a)
λ tac-S12	<i>imm²¹ P_{tac}-lac</i> operator-S12 fusion	This study
λ tac/ λ I-S12 ^b	<i>imm²¹ P_{tac}-λI</i> operator-S12 fusion	This study
λ AC201	<i>imm²¹ P_{ant}-cat-lacZ</i> fusion	(4)
Plasmids		
pNO1523	Contains wild-type S12 gene (see Fig. 2)	Pharmacia ¹²
pAP119	pBR322 derivative, P _{tac} - λ Cro fusion	(c)
pI ^Q	pACYC184 derivative, contains <i>lacI^Q</i> gene on ~1-kb fragment inserted at <i>EcoRI</i> site	J. Wang (Harvard)
pUCroRS	pUC18 derivative, contains P _{tac} - λ Cro fusion from pAP119 inserted in polylinker	This study
pUCS12	pUC118 derivative, contains S12 sequences from pNO1523 (see Fig. 2)	This study
pUCTacS12	pUCS12 derivative, P _{tac} - <i>lac</i> operator-S12 fusion (see Fig. 3)	This study
pUCTac/ λ I	pUC118 derivative, contains P _{tac} - λ operator construct (see Fig. 3)	This study
pUS405	pBR322 derivative, P _{ant} - <i>arc</i> operator-S12 fusion, P _{tac} - <i>arc</i> fusion	(10)

^a R. Maurer, B. J. Meyer, and M. Ptashne, *J. Mol. Biol.* **139**, 147 (1980).

^b Phage with the λ 2, λ 3, and λ 4 operators are designated λ tac/ λ 2-S12, etc.

^c A. A. Pakula and R. T. Sauer, *Proteins: Struct. Funct. Genet.* **5**, 202 (1989).

levels of Arc expression. The streptomycin selection for Arc activity was generally found to be reliable as long as it was performed on freshly transformed cells. However, to ensure that streptomycin-resistant candidates were indeed Arc⁺, an independent screen for Arc activity was performed. This screen relies on the fact that strain UA2F contains a gene for chloramphenicol acetyltransferase under control of an Arc-repressible promoter.⁴ As a result, a cell that is Arc⁺, such as UA2F/pUS405, is both streptomycin resistant and chloramphenicol sensitive.

In constructing the P_{ant}-S12 fusion used for the Arc selection, 5' sequences apparently required for stability or high-level translation of the

S12 mRNA were removed. While expression of the S12 gene from the multicopy plasmid pUS405 was sufficient to confer a streptomycin-sensitive phenotype, we worried that this reduced expression might preclude the construction of single-copy selection vectors. To facilitate the construction of selection strains in which a single copy S12 gene would be expressed at high levels, the S12 gene in plasmid pNO1523¹² was separated from its natural promoter near the start of transcription by restriction cleavage with *AhaII* (see Fig. 2). The resulting S12 fragment, which includes all of the natural leader sequences, was cloned into the *AccI* site of the polylinker in plasmid pUC118 to generate pUCS12, as shown in Fig. 2. The S12 gene was then brought under *lac* operator control by cloning the *EcoRI*–*PvuII* restriction fragment bearing the *tac* promoter and *lac* operator from plasmid ptac12¹³ between the *EcoRI* and *SmaI* sites in the polylinker of pUCS12. We refer to the resulting fusion as the *tac*–S12 fusion and the plasmid as pUCTac–S12. Plasmid pUCTac–S12 was viable in a *lacI*^Q strain, which overproduces Lac repressor but could not be propagated, even in the absence of streptomycin, in the presence of IPTG or in strains producing low levels of Lac repressor. Presumably, the combination of the strong *tac* promoter, the natural leader sequences of the S12 gene, and expression from a multicopy plasmid results in a lethal level of S12 expression.

To allow the *tac*–S12 fusion to be maintained in a single copy per genome, it was cloned into a λ imm²¹ phage. A phage bearing the S12 fusion was constructed from the left and right arms of λ 112,¹⁴ as shown in Fig. 4. This construction also places the *lacZ* gene under control of the *tac* promoter, which allows the presence of a promoter insert to be detected by the formation of blue plaques on X-gal (5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside) plates. Following ligation of the gel-purified fragments, the resulting phage were packaged and infected into GW5100, a streptomycin-resistant, *lacZ*[–] strain (see Table I for genotype). Phage forming turbid blue plaques on X-gal plates were purified by two rounds of replating. Lysogens—designated GW5100(λ tac–S12)—were then isolated by streaking cells from the centers of the turbid plaques onto LB plates containing X-gal, and restreaking candidates until all progeny displayed a homogeneous blue color. To ensure that single lysogens were selected, β -galactosidase assays⁵ were performed on lysogens obtained from four independent plaques and the strain with the lowest level was selected. At most, one or two of the lysogenic strains showed levels of activity that were multiples of the most common single lysogen level.

¹³ E. Amann, J. Brosius, and M. Ptashne, *Gene* **25**, 167 (1983).

¹⁴ R. Maurer, B. J. Meyer, and M. Ptashne, *J. Mol. Biol.* **139**, 147 (1980).

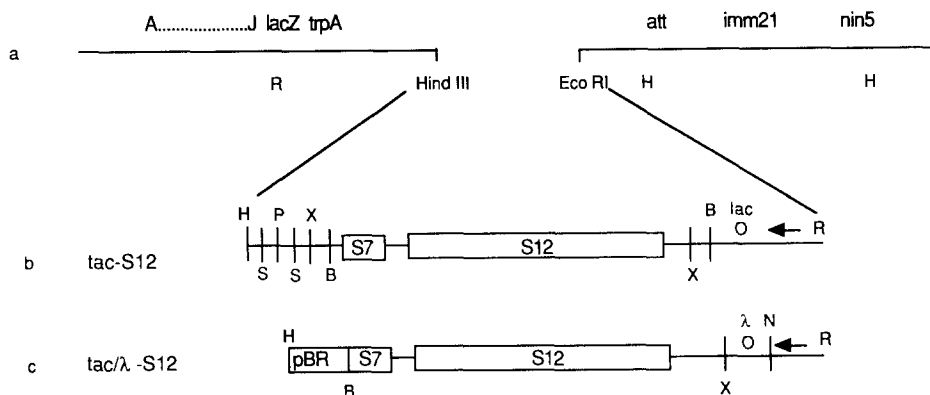
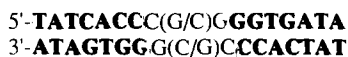


FIG. 4. Prophage maps. (a) The left arm of $\lambda 112^{14}$ contains λ genes A to J, the *lacZ* gene, and the *trpA* gene. Transcription of *lacZ* and *trpA* is dependent on a leftward-directed promoter cloned between the *Hind*III and *Eco*RI sites. The right arm of $\lambda 112$ contains the phage attachment site, the immunity region from phage 21, and the *nin5* deletion. (b) A phage containing the *tac*-S12 fusion was constructed in a three-part ligation containing the left and right arms of $\lambda 112$ and the *Eco*RI-*Hind*III fragment from plasmid pUC*tac*-S12 (see Methods). (c) Phages containing the *tac*/ λ -S12 fusions were constructed in a four-part ligation of the *Eco*RI arm of $\lambda 112$, an *Eco*RI-*Xba*I fragment containing the *tac* promoter and λ operator, the *Xba*I-*Hind*III fragment of pUCS12, and the left arm of $\lambda 112$. Restriction enzyme sites are indicated by single letters: R, *Eco*RI; H, *Hind*III; B, *Bam*HI; X, *Xba*I; N, *Nsi*I; S, *Sac*I; P, *Pst*I; S, *Sma*I.

Similar lysogenic strains were constructed to select for λ Cro DNA-binding activity. In this case, the *tac* promoter was again used to drive wild-type S12 gene expression and a λ operator site was positioned several bases downstream of the -10 region of the *tac* promoter (see Fig. 3C). This hybrid *tac*/ λ promoter-operator was constructed from an *Eco*RI-*Cla*I fragment (containing the -35 region of the promoter) from plasmid pEA300¹³ and a synthetic oligonucleotide that provided the -10 promoter sequence and an operator site for the λ Cro protein. The 17-bp operator sequence synthesized was



The outer 7 bp in each operator half-site correspond to a consensus λ operator site.¹⁵ The central base was synthesized using an equal mixture of C and G. This yielded two different operators with central *Sma*I restriction sites (CCCGGG) shifted by 1 bp. We refer to the operator with a

¹⁵ M. Ptashne, "A Genetic Switch: Gene Control and Phage λ ." Cell Press and Blackwell Scientific Publ., Cambridge, Massachusetts and Palo Alto, California, 1986.

central G on the top strand as $\lambda 1$ and the operator with a central C on the top strand as $\lambda 2$. Additional operators in which a single base pair had been inserted ($\lambda 3$) or deleted ($\lambda 4$) at the center of the site were constructed by cleaving the $\lambda 1$ and $\lambda 2$ operators and swapping their right halves (*SmaI*–*XbaI* fragments). The fusion of the *tac*/ $\lambda 1$ promoter–operator region to the S12 gene was made by ligation of the *EcoRI*–*XbaI* fragment containing the promoter–operator to the *XbaI*–*HindIII* fragment from pUCS12 containing the S12 gene. As described in the legend to Fig. 4, this occurred as part of a four-part ligation used to construct a λimm^{21} phage bearing the *tac*/ $\lambda 1$ –S12 fusion. Phage DNA was packaged *in vitro* and transfected into GW5100. Single lysogens—designated GW5100($\lambda tac/\lambda 1$ -S12)—were isolated as described above. Strains GW5100($\lambda tac/\lambda 2$ -S12), GW5100($\lambda tac/\lambda 3$ -S12), and GW5100($\lambda tac/\lambda 4$ -S12) were constructed in a similar manner.

Lysogens containing the streptomycin-sensitive S12 gene under Lac repressor or λ Cro repressor control were repressible and thus streptomycin resistant in the presence of a wild-type repressor gene. Strain GW5100(λtac -S12) expresses elevated levels of Lac repressor (from the *lacI^Q* gene on the F episome) and was found to be as resistant to streptomycin ($>600 \mu\text{g/ml}$) as GW5100, the nonlysogenic parent strain. However, when the Lac repressor was inactivated by addition of the inducer IPTG, the GW5100(λtac -S12) cells were killed in the presence of $50 \mu\text{g/ml}$ of streptomycin. Similarly, strain GW5100($\lambda tac/\lambda 1$ -S12) was transformed with the Cro-overproducing plasmid pAP119¹⁶ to determine if λ Cro could repress expression of S12. Even at low levels of expression, Cro was able to express S12 synthesis and render the cells resistant to $50 \mu\text{g/ml}$ streptomycin.

To correlate streptomycin resistance with the level of S12-transcription, Lac repressor activity was varied by growth of strain GW5100(λtac -S12) F^- containing the plasmid pI^Q in media containing different amounts of IPTG^{16a} and cells were assayed for growth on LB-amp plates in the presence of 50, 200, and $600 \mu\text{g/ml}$ streptomycin. In parallel experiments, lysates of the same cells grown in LB broth were assayed for β -galactosidase activity to monitor expression from the *tac* promoter. As shown in Table II, cells in which β -gal expression and thus *tac* promoter activity is low are resistant to streptomycin and vice versa.

¹⁶ A. A. Pakula and R. T. Sauer, *Proteins: Struct. Funct. Genet.* **5**, 202 (1989).

^{16a} It is important when attempting to vary Lac repressor activity smoothly over a wide range with IPTG that the host strain be *lacY⁻*, as the Y gene product is both induced by and is a transporter of IPTG. This makes the internal concentration of IPTG in a *lacY⁺* cell increase very rapidly once the amount of IPTG in the media exceeds a critical threshold where passive diffusion into the cell leads to induction.

TABLE II
CORRELATION BETWEEN β -GALACTOSIDASE (β -gal) ACTIVITY AND
STREPTOMYCIN RESISTANCE^a

IPTG (μ M)	β -gal units	Growth on streptomycin		
		50 μ g/ml	200 μ g/ml	600 μ g/ml
0	300	+	+	+
1	200	+	+	+
3	300	+	+	+
10	350	+	+	-
30	800	+	-	-
60	1100	+	-	-
100	1600	-	-	-
300	1800	-	-	-
Control	2000	-	-	-

^a Lac repressor activity was varied by titration with IPTG in strain GW5100(λ tac-S12) F⁻ containing plasmid pI^Q. β -Galactosidase activity of liquid cultures grown in LB broth was assayed by the method of Miller.⁵ Growth on LB plates containing 200 μ g/ml of streptomycin and 100 μ g/ml ampicillin and 15 μ g/ml tetracycline is indicated by a +. The control strain is GW5100(λ tac-S12) F⁻, which contains no Lac repressor.

In a selection experiment, there are several ways other than acquisition of repressor activity that a cell could become streptomycin resistant. These include alteration of the streptomycin-sensitive gene by prophage loss, mutation, or recombination. To assess the probability of these events, we measured the frequency at which colonies lacking a functional Lac repressor acquired a streptomycin-resistant phenotype spontaneously. In either the RecA⁺ strain GW5100(λ tac-S12) F⁻ or the RecA⁻ strain GW5180(λ tac-S12) F⁻, streptomycin-resistant colonies arose at frequencies of approximately 10⁻⁵. Roughly 80% of these resistant colonies were immune to λ imm²¹ and were lacZ⁺, indicating that the cells are still lysogenic and the *tac* promoter is still functional. As a result, resistance appears to arise mainly via mutation of the streptomycin-sensitive S12 gene. We presume that most of these S12 mutations simply result in inactivation of the gene. It is straightforward, however, to distinguish between bona fide candidates that have acquired DNA-binding activity and those with spontaneous mutations to streptomycin resistance because plasmids isolated from the latter cells would not yield streptomycin-resistant colonies at high frequency following retransformation of a naive strain.

Another potential problem in using the streptomycin selection is that

of phenotypic lag. Imagine that the promoter-operator-S12 fusion is present in a cell devoid of endogenous repressor. This cell is then transformed with a population of plasmids, only some of which encode functional protein, and transformants are selected for survival in the presence of streptomycin. A freshly transformed cell will contain a large population of streptomycin-sensitive ribosomes, and the cell might remain sensitive to killing for some time even if the synthesis of the sensitive S12 gene were repressed immediately. To test this possibility, we varied the outgrowth time between transformation of strain GW5100(λ tac/ λ 1-S12) with the Cro-producing plasmid pAP119 and plating of the transformants on plates containing ampicillin (100 μ g/ml) and streptomycin (200 μ g/ml). The ratio of ampicillin-resistant to streptomycin-resistant transformants was found to remain constant from 30 min to 4 hr after transformation, indicating that once the cells stop synthesis of the streptomycin-sensitive S12 protein, they rapidly acquire resistance to the antibiotic.

Applications

Isolation of Neutral Mutations. Selections for activity can be combined with random mutagenesis to investigate the tolerance of different residue position to amino acid substitutions. If a given residue position tolerates chemically diverse side-chain substitutions without affecting protein function, then this position cannot be a critical structural or functional determinant. On the other hand, if a residue position is intolerant of substitutions or accepts only conservative substitutions, then it is likely that the side chain plays an important role in the structure, function, or intracellular stability of the protein.

The Arc repressor of phage P22 is a 53-residue DNA-binding protein (for review, see Ref. 17). To examine which Arc residues were tolerant of substitutions, the *arc* gene in plasmid pUS405 was subjected to "biased" cassette mutagenesis procedure in blocks corresponding to approximately 10 residues (Ref. 18; for description of technique, see Reidhaar-Olson *et al.* [27], this volume). Following transformation of strain UA2F, active *arc* genes were isolated using the streptomycin selection and sequenced.¹⁸ Sixty-one different, functionally neutral amino acid substitutions were identified at 24 residue positions. At many of these positions, nonconservative residue substitutions were allowed. Moreover, at many of the positions where functionally neutral substitutions were not recovered, it was

¹⁷ K. L. Knight, J. U. Bowie, A. K. Vershon, R. D. Kelley, and R. T. Sauer, *J. Biol. Chem.* **264**, 3639 (1989).

¹⁸ J. U. Bowie and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 2152 (1989).

shown that residue substitutions did not disrupt the folded structure of Arc. By comparing the sets of functionally neutral and structurally neutral substitutions, it was determined that 15 Arc residues had the properties expected for functionally important positions.¹⁸

A second example of the use of the streptomycin selection in probing structure–function relationships is illustrated by studies of λ Cro residues 52–58, which form the dimer interface in the crystal structure of the protein.¹⁹ This region of the *cro* gene was mutagenized using a “biased” cassette procedure and streptomycin-resistant colonies were selected.²⁰ When the selection was performed under conditions of low Cro expression (transcription of the *cro* gene repressed by Lac repressor), non-conservative residue changes were tolerated at position 53, but the remaining positions tolerated no changes or only conservative changes, such as Lys-56→Arg and Phe-58→Tyr (see Fig. 5, left side). When the stringency of the selection was reduced by higher-level Cro expression, a greater degree of residue tolerance was evident (see Fig. 5, right side). Now, only position 58 is seen to be strictly conserved, whereas the remaining positions accept a variety of amino acid substitutions. These results show how hierarchies of residue importance can be established. Phe-58 appears to be the most important side chain in this region of λ Cro; Pro-57, Lys-56, Val-55, Glu-54, and Ala-52 seem to play lesser roles in activity; and Glu-53 appears to be relatively unimportant for Cro dimerization.

Isolation of Revertants. Intragenic, second site revertants can provide information about interactions that correct mutant defects or improve protein function in some other way. In DNA-binding proteins, for example, reverting mutations have included sequence changes that enhance DNA binding (Ref. 21; Oxender and Gibson [31], this volume) and improve thermal stability.¹⁶

In P22 Arc, a large number of missense mutations result in loss of Arc function.⁴ Arc gene fragments containing one of three single mutations (Met-4→Ile, Val-18→Gly, and Asp-20→Asn) were subjected to hydroxylamine mutagenesis or random primer mutagenesis *in vitro*, the fragments were recloned into plasmid pUS405, and the streptomycin selection for Arc activity was imposed following transformation of strain UA2F.¹⁰ The *arc* genes from colonies that were both streptomycin resistant and chloramphenicol sensitive (see Methods) were sequenced. Second site revertants were isolated for each of the primary mutations. One revertant

¹⁹ W. F. Anderson, D. H. Ohlendorf, Y. Takeda, and B. W. Matthews, *Nature (London)* **290**, 754 (1981).

²⁰ M. C. Mossing and R. T. Sauer, *Science* **250**, 1712 (1990).

²¹ H. C. M. Nelson and R. T. Sauer *Cell (Cambridge, Mass.)* **42**, 549 (1985).

Near Wild-Type Activity									Moderate Activity								
WT	→	52	53	54	55	56	57	58	WT	→	52	53	54	55	56	57	58
		A	E	E	V	K	P	F			A	E	E	V	K	P	F
13		A	A	E	V	K	P	F	1		A	E	E	V	T	P	F
15		A	E	E	V	K	P	F	2		G	A	E	V	K	P	F
18		A	A	E	V	K	P	F	3		G	D	E	V	K	H	F
19		A	E	E	V	K	P	F	4		A	E	E	V	A	P	F
21		A	D	E	V	K	P	F	5		A	K	E	V	K	L	F
23		A	E	E	V	R	L	F	6		A	V	E	V	R	P	F
24		A	E	E	F	K	P	Y	7		A	P	E	S	R	L	F
31		A	E	E	V	K	P	F	8		G	A	E	V	K	L	F
32		A	S	E	V	K	P	F	9		A	G	E	V	K	P	F
		A	E	E	V	K	P	F	11		A	R	Y	I	N	T	F
			A		F	R	L	Y	25		A	E	V	V	K	L	F
			D						26		G	V	D	V	K	P	F
			S						27		D	E	E	V	K	A	F
									28		T	E	E	V	K	P	F
									29		P	E	A	V	K	P	F
									30		G	Y	E	V	K	L	F
									35		A	I	E	V	K	L	F
									36		A	K	E	V	K	P	F
									38		A	E	E	V	K	T	F
									39		A	D	E	F	K	S	F
									40		G	V	E	V	K	S	F
									41		V	E	G	S	K	T	F
									44		G	V	E	V	R	T	F
									45		A	E	E	V	M	T	F
									46		A	E	D	V	K	P	F
									37		A	E	E	V	K	P	F
											A	E	E	V	K	P	F
											G	V	D	S	R	L	T
											D	A	A	I	T	T	S
											T	D	G	Y	A	S	H
											P	K	V		N		
											V	P	Y		M		
											G						
											R						
											Y						
											I						

FIG. 5. Allowed substitutions in the dimerization region of λ Cro. The wild-type Cro residue is indicated in the one-letter code at the top of each column. Below are listed the sequences from active *cro* genes, isolated following cassette mutagenesis. Sequences listed on the left side were isolated in strain GW5100(λ tac/ λ I-S12), in which Lac repressor limits Cro expression. Sequences listed on the right side were isolated in strain GW5100(λ tac/ λ I-S12) F⁻, where Cro expression is unrepressed. With the pUCroRS vector used for these studies (see below), unrepressed expression of wild-type Cro is lethal; as a result genes with wild-type or near wild-type activity are probably not represented among the moderate activity class. To mutagenize residues 52–58, the corresponding codons of one DNA strand of a cassette were synthesized using a mixture of 62.5% of the wild-type base and 12.5% of each of the other three bases. The double-stranded DNA cassettes for both the randomization and monomer construction were generated by the method of A. K. Oliphant, A. L. Nussbaum, and K. Struhl [*Gene* 44, 177 (1986)] and ligated into pUCroRS. This plasmid consists of the *Eco*RI to *Sma*I fragment of pAP119¹⁶ cloned between the *Eco*RI and *Hinc*II sites of pUC18 [C. Yanisch-Perron, J. Vieira, and J. Messing, *Gene* 33, 103 (1985)].

contained a secondary Gly-52→Asp mutation in addition to the primary mutant. Surprisingly, however, the remaining revertants contained either -1 or +1 frameshift mutations near the 3' end of the *arc* gene. These mutations resulted in bypass of the normal termination codon and resulted in C-terminal additions of 26 residues or 8 residues to the wild-type Arc sequence. These C-terminal extension or "tail" sequences act by slowing the intracellular degradation of the mutant proteins.¹⁰ This occurs because the tails alter the susceptibility of the proteins to an *E. coli* protease that recognizes C-terminal sequences.²² This case demonstrates how the isolation of revertants can reveal unexpected ways of compensating for mutant defects.

The 26-residue long tail sequence described above did not interfere with the DNA-binding activity or gene regulatory properties of the wild-type Arc protein.¹⁰ We wished to make similar C-terminal fusions to the λ Cro protein, as a means of generating an active Cro molecule that could be easily distinguishable from the wild-type protein in gel shift experiments. However, although the C terminus of Cro is disordered in the crystal structure, these residues had been proposed to contribute to DNA binding.¹⁹ As a result, we worried that addition of the tail might perturb the structure or function of Cro. To maximize our chances of obtaining functional Cro variants; we generated proteins with random fusion endpoints, and then identified active molecules by using the streptomycin selection. A plasmid was constructed containing a tandem arrangement of the *cro* gene with its normal TAA termination codon, a 21-bp sequence containing a unique *Pst*I restriction site, and the coding sequences for a 35-residue tail sequence (see Fig. 6; the first 9 residues of the tail sequence correspond to λ repressor residues 94–102 and the next 26 residues correspond to the long tail C-terminal extension discussed above). The plasmid DNA was cleaved at the *Pst*I site and digested with the *Bal*31 exonuclease. At several times aliquots were removed, the exonuclease reaction was stopped, and the linearized plasmids were religated. Following transformation into GW5100(λ tac/ λ 1-S12), colonies resistant to 200 μ g/ml streptomycin were selected. Cro expression in resistant candidates was induced with IPTG and lysates were electrophoresed on sodium dodecyl sulfate (SDS) polyacrylamide gels to determine the size of the fusion product. Two candidates containing Cro variants of increased size were selected and the corresponding *cro* genes were sequenced to determine the nature of the junction. Figure 6 shows the deletion endpoints. The $\Delta 2$ deletion results in a fusion protein that contains all of the wild-type residues in Cro followed by the sequence SLRSEYERKVEAP-

²² D. A. Parsell, K. R. Silber, and R. T. Sauer, *Genes Dev.* **4**, 277 (1990).

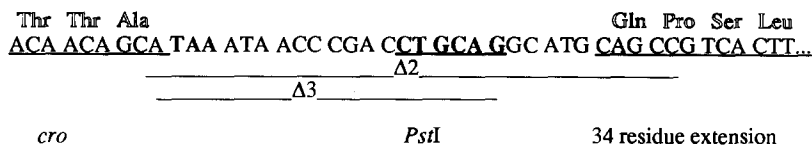


FIG. 6. Strategy for isolating Cro-fusion proteins. The C-terminal codons of the *cro* gene are shown on the left, followed by a 21-bp sequence containing a unique *Pst*I site, followed by the coding sequence for a 34-residue extension. The $\Delta 2$ and $\Delta 3$ deletions were generated by *Bal*31 digestion from the *Pst*I site and result in active Cro variants with C-terminal extensions 33 and 19 residues, respectively (see text).

TAVTVRASVVSKSLEKNQHE. The $\Delta 3$ deletion results in a fusion protein that also contains all of the normal Cro residues, followed by a C-terminal extension with the 19-residue sequence ACSRHLDLSM-SVKLRPQRP.

Attempts to Isolate Cro Variants that Bind +1 or -1 Operators. As described in the Methods section, we constructed variants of the *tac*/ λ -S12 fusion in which the λ operator was altered by insertion or deletion of a single base pair from the center of the operator. Wild-type Cro did not bind specifically to these +1 or -1 operators in gel shift experiments and gave no streptomycin resistant colonies under selection conditions. We then used the library of residue 52-58 randomized sequences to ask if the Cro dimerization interface could be altered by substitution to allow recognition of the +1 or -1 operators. However, no streptomycin-resistant clones were obtained for either mutant operator. Hence, variability in the spacing of operator half-sites cannot be readily accommodated by changes in the sequence of the Cro dimer interface. Recent results, however, suggest that Cro variants with residue insertions in the dimer interface region can repress the +1 operator (M. C. Mossing, unpublished).

Discussion

We have described examples in which the streptomycin selection has been used to probe for DNA-binding activity of three different repressor proteins. The streptomycin selection should, in principle, be applicable to any site-specific DNA-binding protein that can be stably expressed in *E. coli*. The main requirement in modifying the selection for a new DNA-binding protein is that a binding site for the protein be placed in a position where it can repress expression of the wild-type gene for ribosomal protein S12. In the Lac repressor and λ Cro cases, the center to center distances between the -10 region of the *tac* promoter and the operator sites were

20 and 15 bp, respectively (see Fig. 3). Hence, both the *lac* and λ operator sites overlap the startpoint of transcription. To set up the selection for a new DNA-binding protein, it would probably be best to place the DNA-binding site in an analogous position relative to the promoter. In cases where the wild-type gene encoding a DNA-binding protein is available for manipulation, the system can obviously be calibrated with the wild-type protein expressed at various expression levels, and then used to select active variants. The streptomycin selection may also prove useful in cloning genes for DNA-binding proteins whose recognition sites are known, although in this case, it will not always be clear how to position the operator with respect to the promoter or whether bound protein will, in fact, prevent transcription initiation by RNA polymerase.

The efficiency of the streptomycin selection is limited, in some sense, by the frequency at which streptomycin-resistant colonies arise in the absence of the appropriate DNA-binding activity. In the cases where the promoter-operator-S12 fusion is present at the level of a single copy per cell as a lysogen, the background frequency of streptomycin-resistant colonies was about 1 in 100,000. It should, however, be possible to select active proteins that arise less frequently than this by employing a two-step procedure. For example, the selection could be applied, plasmid DNA purified from pooled streptomycin-resistant colonies, and the selection could be reapplied following retransformation of a naive strain. Such two-step procedures would not be effective in cases like that of Arc, where the promoter-operator-S12 fusion is carried on the same plasmid as the gene for the DNA-binding protein. However, in these cases, the effectiveness of the selection could be extended by recloning the gene for the DNA-binding protein between sequential applications of the selection.

Acknowledgment

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[30] Identification of Amino Acid-Base Pair Contacts by Genetic Methods

By RICHARD H. EBRIGHT

Introduction

This chapter discusses the use of genetic methods, i.e., the construction and analysis of single amino acid substitutions, to identify contacts between individual amino acids of a DNA-binding protein and individual base pairs of the DNA site in the protein-DNA complex. The experimental techniques are straightforward and do not require specialized equipment: (1) site-directed mutagenesis of the gene encoding the DNA-binding protein, followed by analysis of the DNA sequence-recognition properties of the substituted protein, and (2) random mutagenesis of the gene encoding the DNA-binding protein, followed by genetic selection and/or screening, DNA sequence determination, and analysis of the DNA sequence-recognition properties of the substituted protein.

The construction and analysis of single amino acid substitutions is an effective method to identify amino acid-base pair contacts. In addition, the construction and analysis of single amino acid substitutions is the sole method to determine the apparent binding free energy contribution and the apparent specificity free energy contribution of an amino acid-base pair contact (see Ref. 1), and the sole method to determine the alternative specificities that arise when an amino acid involved in a specificity-determining contact is replaced by 1 of the 19 other amino acids.

Other methods to identify amino acid-base pair contacts include X-ray diffraction and two-dimensional nuclear magnetic resonance (2D NMR) spectroscopic analysis of the specific protein-DNA complex.²⁻⁵

¹ R. Ebright, A. Kolb, H. Buc, T. Kunkel, J. Krakow, and J. Beckwith, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6083 (1987).

² J. Anderson, M. Ptashne, and S. Harrison, *Nature (London)* **326**, 846 (1987).

³ A. Aggarwal, D. Rodgers, M. Drott, M. Ptashne, and S. Harrison, *Science* **242**, 899 (1988).

⁴ C. Wolberger, Y. Dong, M. Ptashne, and S. Harrison, *Nature (London)* **335**, 789 (1988).

⁵ S. Jordan and C. Pabo, *Science* **242**, 893 (1988).

Loss-of-Contact Approach

The standard genetic method to identify amino acid–base pair contacts is the “loss-of-contact” approach.^{6,7}

Logic

The logic of the loss-of-contact approach is outlined in Fig. 1. The basis of this approach is to eliminate the ability of the amino acid under study to make a contact to DNA, and then to determine at which base pair, if any, specificity is eliminated. To eliminate the ability of the amino acid under study to make a contact to DNA, the amino acid is substituted by an amino acid having a shorter side chain (e.g., Gly or Ala; see below).

A loss-of-contact substitution will result in four identifiable characteristics:

1. The substituted protein will lack the ability to discriminate between the consensus base pair and nonconsensus base pairs at one position, i , within the DNA site, corresponding to the position that in the wild-type protein–DNA complex is contacted by the amino acid under study. (In the case of an inverted-repeat or direct-repeat DNA site, the substituted protein will lack the ability to discriminate between the consensus base pair and nonconsensus base pairs at one position, i , in each repeat.) The substituted protein will exhibit approximately equal affinities for the consensus DNA site and for derivatives of the consensus DNA site having a nonconsensus base pair at position i : $K_{D,\text{noncons},i}/K_{D,\text{cons}} \approx 1$.

2. The substituted protein will retain the ability to discriminate between the consensus base pair and nonconsensus base pairs at each other position, j , within the DNA site. The substituted protein will exhibit a higher affinity for the consensus DNA site than for derivatives of the consensus DNA site having a nonconsensus base pair at position j : $K_{D,\text{noncons},j}/K_{D,\text{cons}} \gg 1$.

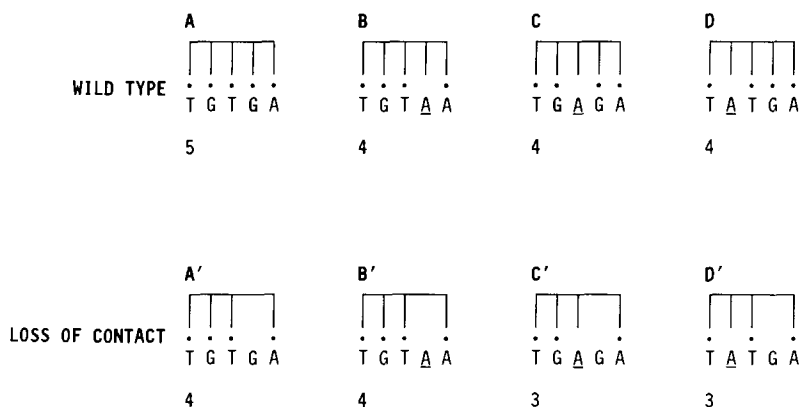
3. As compared to the wild-type protein, the substituted protein will exhibit a decreased affinity for the consensus DNA site. This is due to the presence in the wild-type protein–DNA complex, but absence in the substituted protein–DNA complex, of an energetically favorable interaction between the amino acid under study and the consensus base pair at position i .

4. As compared to the wild-type protein, the substituted protein will exhibit an unchanged affinity (Fig. 1-I) or an increased affinity (Fig. 1-II) for derivatives of the consensus DNA site having a nonconsensus base pair

⁶ R. Ebright, *J. Biomol. Struct. Dyn.* **3**, 281 (1985).

⁷ R. Ebright, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 303 (1986).

I



II

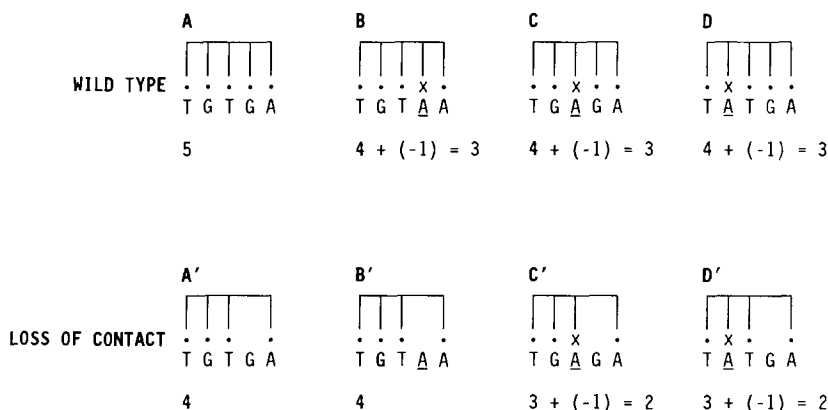


FIG. 1. Loss-of-contact approach: rationale. The basis of the approach is to eliminate the ability of the amino acid under study to make a contact with DNA, and then to determine at which base pair of the DNA site specificity is eliminated. A loss-of-contact substituted DNA binding protein will: (1) lack the ability to discriminate between the consensus base pair and nonconsensus base pairs at one position, i , within the DNA site, corresponding to the position that in the wild-type protein-DNA complex is contacted by the amino acid under study, but (2) retain the ability to discriminate between the consensus base pair and nonconsensus base pairs at each other position, j , within the DNA site. Two cases are presented: case I, which considers only energetically favorable contacts, and case II, which considers both energetically favorable and energetically unfavorable contacts. *Case I.* (A) Interaction of the wild-type DNA-binding protein with five base pairs of the consensus DNA

at position i . This is due to the absence in both the wild-type protein–DNA complex and the substituted protein–DNA complex of an energetically favorable interaction between the amino acid under study and the nonconsensus base pair at position i (Fig. 1-I, 1-II); and to the possible presence in the wild-type protein–DNA complex, but absence in the substituted protein–DNA complex, of an energetically unfavorable interaction between the amino acid under study and the nonconsensus base pair at position i (Fig. 1-I).

Characteristics 1 and 2 in the list above are the critical, defining characteristics of a loss-of-contact substitution. Conclusions regarding the identification of amino acid–base pair contacts can be drawn only if both characteristics 1 and 2 are documented. Characteristics 1 and 2 are specificity characteristics. It is important not to confuse specificity and affinity. Specificity is the factor by which DNA-binding protein X prefers to interact with one DNA site vs another DNA site—for example, to interact with the consensus DNA site vs a nonconsensus DNA site. The ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$ is a measure of specificity.

site (five favorable contacts). The residue of the DNA-binding protein participating in each contact is indicated as a vertical line; the presence of a favorable contact is indicated as a dot. (B–D) Interaction of the wild-type DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one position (four favorable contacts). (A') Interaction of the loss-of-contact substituted DNA-binding protein with the consensus DNA site (four favorable contacts). (B') Interaction of the loss-of-contact substituted DNA-binding protein with the derivative of the consensus DNA site having a nonconsensus base pair at the position, i , that in the wild-type protein–DNA complex is contacted by the amino acid under study (four favorable contacts; identical to the number with the wild-type DNA site). (C' and D') Interaction of the loss-of-contact substituted DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one other position, j (three favorable contacts). *Case II.* (A) Interaction of the wild-type DNA-binding protein with five base pairs of the consensus DNA site (five favorable contacts). The residue of the DNA-binding protein participating in each contact is indicated as a vertical line; the presence of a favorable contact is indicated as a dot. (B–D) Interaction of the wild-type DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one position (four favorable contacts, one unfavorable contact; net three favorable contacts). The presence of an unfavorable contact is indicated as an "x." (A') Interaction of the loss-of-contact substituted DNA-binding protein with the consensus DNA site (four favorable contacts). (B') Interaction of the loss-of-contact substituted DNA-binding protein with the derivative of the consensus DNA site having a nonconsensus base pair at the position, i , that in the wild-type protein–DNA complex is contacted by the amino acid under study (four favorable contacts; identical to the number with the wild-type DNA site). (C' and D') Interaction of the loss-of-contact substituted DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one other position, j (three favorable contacts, one unfavorable contact; net two favorable contacts).

Experimental Design

The optimal experimental design for use of the loss-of-contact approach depends on the amount of information available. Examples are presented here for three cases.

Case One: High-Resolution Structure or Detailed Model of the Protein-DNA Complex Exists. In this case, the structure or model indicates which amino acids are in close proximity to which DNA base pairs in the protein-DNA complex. The objective for use of the loss-of-contact approach is to verify the predicted amino acid-base pair contacts: i.e., to establish that the predicted contacts in fact occur and make energetically significant contributions to DNA binding and DNA sequence specificity.

The optimal experimental design is as follows: Site-directed mutagenesis^{8,9} is used to substitute the amino acid under study by Gly and by Ala. The specificities of the wild-type protein, the Gly-substituted protein, and the Ala-substituted protein are analyzed at the position within the DNA site predicted to be contacted by the amino acid under study (position i), and, as a control, at one other position within the DNA site (position j). The ratio $K_{D,\text{noncons},i}/K_{D,\text{cons}}$ is determined for each of the three possible derivatives of the consensus DNA site having a nonconsensus base pair at position i . (For example, in the case where the consensus base pair at position i is G : C, the ratio $K_{D,\text{noncons},i}/K_{D,\text{cons}}$ is determined for the derivatives of the consensus DNA site having A : T, C : G, and T : A at position i .) The ratio $K_{D,\text{noncons},j}/K_{D,\text{cons}}$ is determined for each of the three possible derivatives of the consensus DNA site having a nonconsensus base pair at position j .

Other amino acid substitutions may be used in this approach^{1,10}; however, in general, the Gly and Ala substitutions are preferred. Gly has no side chain. The Gly substitution has the highest likelihood of eliminating the contact by the amino acid under study, and of not introducing a new energetically significant interaction. Ala has a one-carbon side chain. The Ala substitution has the second highest likelihood of eliminating the contact by the amino acid under study, and of not introducing a new, energetically significant interaction. As compared to the Gly substitution, the Ala substitution has a higher likelihood of not introducing complications due to altered protein conformation.

This experimental design has been used to verify two predicted amino

⁸ T. Kunkel, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 488 (1985).

⁹ J. Wells, M. Vasser, and D. Powers, *Gene* **34**, 315 (1985).

¹⁰ R. Ebright, A. Gunasekera, X. Zhang, T. Kunkel, and J. Krakow, *Nucleic Acids Res.* **18**, 1457 (1990).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
ICAP	A	A	A	T	G	T	G	A	T	C	T	A	G	A	T	C	A	C	A	T	T	T
	T	T	T	A	C	A	C	T	A	G	A	T	C	T	A	G	T	G	T	A	A	A
ICAP-7AS							A									T						
							T									A						
ICAP-7CS							C									G						
							G									C						
ICAP-7TS							T									A						
							A									T						
ICAP-5AS					A															T		
					T															A		
ICAP-5CS					C															G		
					G															C		
ICAP-5TS					T															A		
					A															T		

FIG. 2. DNA sites used to verify a predicted amino acid-base pair contact in the CAP-DNA complex (A. Gunasekera, X. Zhang, M. Smith, T. Kunkel, and R. Ebright, in preparation). ICAP is the consensus DNA site for CAP. The site is 22 bp in length and has perfect twofold sequence symmetry. The twofold axis is located between positions 11 and 12. ICAP-7AS, ICAP-7CS, and ICAP-7TS are derivatives of the consensus DNA site substituted at position 7 of the DNA half-site. ICAP-5AS, ICAP-5CS, and ICAP-5TS are derivatives of the consensus DNA site substituted at position 5 of the DNA half-site.

TABLE I
DNA SEQUENCE-RECOGNITION PROPERTIES OF WILD-TYPE CAP, [Gly-181]CAP, AND [Ala-181]CAP^{a,b}

DNA site ^c	$K_D/K_{D,cons}$		
	Wild-type CAP	[Gly-181]CAP	[Ala-181]CAP
ICAP (consensus site)	[1]	[1]	[1]
ICAP-7AS	>7000	0.6	2
ICAP-7CS	>7000	1	2
ICAP-7TS	3000	2	3
ICAP-5AS	2000	80	60
ICAP-5CS	>7000	300	>800
ICAP-5TS	700	30	50

^a A. Gunasekera, X. Zhang, M. Smith, T. Kunkel, and R. Ebright (in preparation).

^b Data are from nitrocellulose filter-binding assays.

^c Sequences of the DNA sites are in Fig. 2.

acid-base pair contacts in the CAP-DNA complex.^{11,12} Glu-181 of CAP is known to contact position 7 of the DNA site in the CAP-DNA complex^{1,13,14} (see Experimental Design, Case Three, below). In recent work, Glu-181 of CAP has been substituted by Gly and by Ala.¹¹ The data in Fig. 2 and Table I show that [Gly-181]CAP and [Ala-181]CAP lack the ability to discriminate between the consensus base pair G : C and the nonconsensus base pairs A : T, C : G, and T : A at position 7 of the DNA site ($K_{D,\text{noncons},7}/K_{D,\text{cons}} \approx 1$). In contrast, [Gly-181]CAP and [Ala-181]CAP retain the ability to discriminate between the consensus base pair G : C and the nonconsensus base pairs A : T, C : G, and T : A at position 5 of the DNA site ($K_{D,\text{noncons},5}/K_{D,\text{cons}} \gg 1$). These data are consistent with the known contact. The model for the structure of the CAP-DNA complex^{10,14,15} predicts that Arg-180 of CAP contacts position 5 of the DNA site in the CAP-DNA complex. In recent work, Arg-180 of CAP has been substituted by Gly and by Ala.¹² The data show that [Gly-180]CAP and [Ala-180]CAP lack the ability to discriminate between the consensus base pair G : C and the nonconsensus base pairs A : T, C : G, and T : A at position 5 of the DNA site ($K_{D,\text{noncons},5}/K_{D,\text{cons}} \approx 1$). In contrast, [Gly-180]CAP and [Ala-180]CAP retain the ability to discriminate between the consensus base pair G : C and the nonconsensus base pairs A : T, C : G, and T : A at position 7 of the DNA site ($K_{D,\text{noncons},7}/K_{D,\text{cons}} \gg 1$). These data are consistent with the predicted contact. The identification of two amino acid-base pair contacts in the CAP-DNA complex provides information sufficient to define the orientation of the helix-turn-helix DNA-binding motif of CAP with respect to DNA in the CAP-DNA complex.¹²

Case Two: No High-Resolution Structure or Detailed Model of the Protein-DNA Complex Exists; However, Information Suggests Which Amino Acids Make Contacts. In this case, amino acid sequence similarities to a well-characterized DNA-binding protein, or previous biochemical or genetic results, suggest which amino acids of the DNA-binding protein under study are candidates to make amino acid-base pair contacts in the protein-DNA complex. The objectives for use of the loss-of-contact approach are: (1) to determine whether a candidate amino acid makes an amino acid-base pair contact, and, if so, (2) to determine which base pair it contacts.

The optimal experimental design is as follows: Site-directed mutagene-

¹¹ A. Gunasekera, X. Zhang, M. Smith, T. Kunkel, and R. Ebright, in preparation.

¹² X. Zhang and R. Ebright, *Proc. Natl. Acad. Sci. U.S.A.* **87**, 4717 (1990).

¹³ R. Ebright, P. Cossart, B. Gicquel-Sanze, and J. Beckwith, *Nature (London)* **311**, 232 (1984).

¹⁴ R. Ebright, P. Cossart, B. Gicquel-Sanze, and J. Beckwith, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 7274 (1984).

¹⁵ I. Weber and T. Steitz, *Proc. Natl. Acad. Sci. U.S.A.* **81**, 3973 (1984).

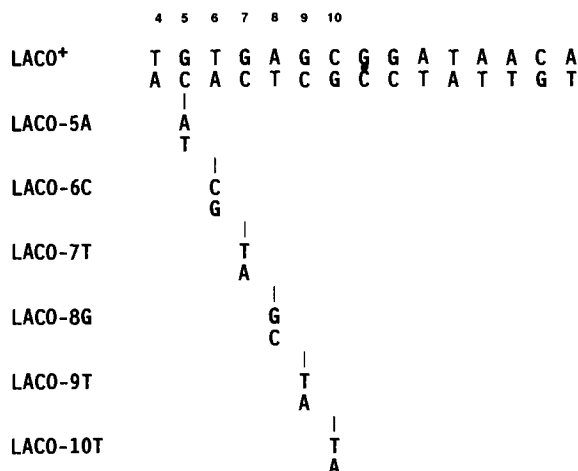


FIG. 3. DNA sites used to identify an amino acid–base pair contact in the Lac repressor–DNA complex [R. Ebright, *J. Biomol. Struct. Dyn.* **3**, 281 (1985); R. Ebright, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 303 (1986); R. Ebright and X. Zhang, unpublished data]. LACO⁺ is the wild-type *lac* operator. The site is 21 bp in length and has approximate twofold sequence symmetry. The twofold axis is located at position 11. (Base positions are numbered with respect to the start point of *lac* transcription; using this convention, the leftmost position in the *lac* operator is designated as position 4.) LACO-5A, LACO-6C, LACO-7T, LACO-8G, LACO-9T, and LACO-10T are derivatives of the *lac* operator substituted at positions 5, 6, 7, 8, 9, and 10 of the DNA half-site.

TABLE II
DNA-SEQUENCE RECOGNITION PROPERTIES OF WILD-TYPE LAC REPRESSOR,
[Gly-18]LAC REPRESSOR, AND [Ala-18]LAC REPRESSOR^{a,b,c}

DNA site ^d	$K_D/K_{D,O^+}$		
	Wild-type Lac repressor	[Gly-18]Lac repressor	[Ala-18]Lac repressor
LACO ⁺ (wild-type <i>lac</i> operator)	[1]	[1]	[1]
LACO-5A	100	20	100
LACO-6C	90	20	60
LACO-7T	50	0.9	0.3
LACO-8G	70	10	50
LACO-9T	20	20	40
LACO-10T	90	20	200

^a R. Ebright, *J. Biomol. Struct. Dyn.* **3**, 281 (1985).

^b R. Ebright, *Proc. Natl. Acad. Sci. U.S.A.* **83**, 303 (1986).

^c R. Ebright and X. Zhang (unpublished data).

^d Sequences of the DNA sites are in Fig. 3.

sis^{8,9} is used to substitute the amino acid under study by Gly and by Ala (see Experimental Design, Case One, above, for the rationale for choice of Gly and Ala substitutions). The specificities of the wild-type protein, the Gly-substituted protein, and the Ala-substituted protein are analyzed at *each* position of the DNA site. For each position of the DNA site, the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$ is determined for one derivative of the consensus DNA site having a nonconsensus base pair at the position. The position that in the wild-type protein-DNA complex is contacted by the amino acid under study is identified as the position, i , within the DNA site at which $K_{D,\text{noncons}}/K_{D,\text{cons}} \approx 1$.

This experimental design has been used to identify an amino acid-base pair contact in the Lac repressor-DNA complex.^{6,7,16} Amino acid sequence similarities to well-characterized helix-turn-helix motif sequence-specific DNA-binding proteins,^{17,18} and previous genetic results,¹⁹ had suggested that Gln-18 of Lac repressor was a strong candidate to make an amino acid-base pair contact in the Lac repressor-DNA complex. Gln-18 of Lac repressor has been substituted by Gly and by Ala.^{6,7,16} The specificities of wild-type Lac repressor and of the two substituted Lac repressor variants have been assessed with respect to positions 5, 6, 7, 8, 9, and 10 of the DNA site.^{6,7,16} The data in Fig. 3 and Table II show that [Gly-18]Lac repressor and [Ala-18]Lac repressor lack the ability to discriminate between the consensus base pair G : C, and the nonconsensus base pair T : A at position 7 of the DNA site ($K_{D,\text{noncons},7}/K_{D,\text{cons}} \approx 1$). In contrast, [Gly-18]Lac repressor and [Ala-18]Lac repressor retain the ability to discriminate between the consensus and nonconsensus base pairs and at each other position of the DNA site ($K_{D,\text{noncons},j}/K_{D,\text{cons}} \gg 1$). These data suggest that Gln-18 of Lac repressor contacts position 7 of the DNA site in the Lac repressor-DNA complex.

This experimental design also has been used to identify amino acid-base pair contacts in the λ repressor-DNA complex,^{20,21} in the λ cro-DNA complex,^{20,22} in the σ^H -DNA complex,²³ and in the AraC-DNA complex.²⁴

¹⁶ R. Ebright and X. Zhang, unpublished data.

¹⁷ C. Pabo and R. Sauer, *Annu. Rev. Biochem.* **53**, 293 (1984).

¹⁸ Y. Takeda, D. Ohlendorf, W. Anderson, and B. Matthews, *Science* **221**, 1020 (1983).

¹⁹ J. Miller, *J. Mol. Biol.* **180**, 205 (1984).

²⁰ A. Hochschild and M. Ptashne, *Cell (Cambridge, Mass.)* **44**, 925 (1986).

²¹ A. Brunelle and R. Schleif, *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6673 (1987).

²² Y. Takeda, A. Sarai, and V. Rivera, *Proc. Natl. Acad. Sci. U.S.A.* **86**, 439 (1989).

²³ P. Zuber, J. Healy, H. L. Carter, S. Cutting, C. Moran, and R. Losick, *J. Mol. Biol.* **206**, 605 (1989).

²⁴ A. Brunelle and R. Schleif, *J. Mol. Biol.* **209**, 607 (1989).

Case Three: No High-Resolution Structure or Detailed Model of the Protein–DNA Complex Exists; No Information Suggests Which Amino Acids Make Contacts. In this case, no predictions are available. The objectives for use of the loss-of-contact approach are: (1) to identify an amino acid making an amino acid–base pair contact, and (2) to determine which base pair it contacts.

The optimal experimental design involves random mutagenesis of the gene encoding the DNA-binding protein, followed by genetic selection and/or screening, followed by DNA sequence determination.

The genetic selection and/or screening procedure takes advantage of characteristics 3 and 4 of a loss-of-contact substitution (decreased affinity for the consensus DNA site, unchanged or increased affinity for derivatives of the consensus DNA site having a nonconsensus base pair at one position, i , within the DNA site; see Logic, above). The genetic selection and/or screening procedure has two steps.¹³ In step 1, a selection or screen is performed in order to isolate a pool of mutant clones, each having a substituted DNA-binding protein with decreased affinity for the consensus DNA site. This is most efficiently done using a selection or screen for decreased fractional occupancy, θ_{cons} , of the consensus DNA site (see Experimental Methods, *In Vivo* Assay, below). In step two, within the resulting pool of mutant clones, a selection or screen is performed in order to isolate rare mutant clones, each having a substituted DNA-binding protein with unchanged or increased affinity for a derivative of the consensus DNA site having a nonconsensus base pair at one position, i , within the DNA site. This is most efficiently done using a selection or screen for unchanged or increased fractional occupancy, $\theta_{\text{noncons},i}$, of a derivative of the DNA site having a nonconsensus base pair at position i . It is possible to reverse the order of steps one and two.^{25,26} In ideal cases, it is possible to perform steps 1 and 2 on a single agar plate.¹³

For each mutant clone isolated in the genetic selection and/or screening procedure, it is imperative to confirm that the substituted DNA-binding protein in the mutant clone exhibits characteristics 1 and 2 of a loss-of-contact substitution (loss of the ability to discriminate between the consensus base pair and nonconsensus base pairs at one position, i , within the DNA site; retention of the ability to discriminate between the consensus base pair and nonconsensus base pairs at each other position, j , within the DNA site; see Logic, above). The specificities of the wild-type DNA-binding protein and of the substituted DNA-binding protein in the mutant

²⁵ D. Siegele, J. Hu, W. Walter, and C. Gross, *J. Mol. Biol.* **206**, 591 (1989).

²⁶ T. Gardella, H. Moyle, and M. Susskind, *J. Mol. Biol.* **206**, 579 (1989).

clone are analyzed at each position of the DNA site. For each position of the DNA site, the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$ is determined for one derivative of the consensus DNA site having a nonconsensus base pair at the position.

For each verified mutant clone, the DNA nucleotide sequence of the gene encoding the DNA-binding protein is then determined. This permits the identification of the amino acid substitution responsible for the loss-of-contact phenotype.

This experimental design has been used to identify an amino acid-base pair contact in the CAP-DNA complex.^{1,13,14} At the time the experiments were initiated (1982), there were conflicting models for the structure of the CAP-DNA complex,²⁷⁻³⁰ and there was no experimental information to suggest which amino acids of CAP make amino acid-base pair contacts. The gene encoding CAP was subjected to random mutagenesis.¹³ A selection was performed in order to isolate mutant clones, each having a substituted CAP variant with decreased affinity for the wild-type *lac* DNA site for CAP; within the resulting pool of mutant clones, a screen was performed in order to isolate rare mutant clones, each having a substituted CAP variant with increased affinity for the derivative of the *lac* DNA site for CAP having A:T at position 7.¹³ The selection and the screen were performed on a single agar plate. Three mutant clones were isolated. DNA nucleotide sequence determination demonstrated that in all three mutant clones the identical amino acid of CAP—amino acid 181—was substituted; the substitutions were Val, Leu, and Lys. It was concluded that Glu-181 contacts base pair 7 of the DNA site in the CAP-DNA complex.^{1,13,14}

This experimental design also has been used to identify amino acid-base pair contacts in the σ^{70} -DNA complex.^{25,26}

Experimental Methods

In Vivo Assay of $K_{D,\text{noncons}}/K_{D,\text{cons}}$. If the DNA-binding protein under study is a transcriptional repressor or a transcriptional activator protein, it is possible to use an *in vivo* assay to determine the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$. This can be efficient and convenient, since it eliminates the requirement to purify the wild-type protein and each substituted protein.

The method employs a set of isogenic "tester" strains. In each tester strain, the identical reporter promoter is placed under the control of the

²⁷ D. McKay and T. Steitz, *Nature (London)* **290**, 744 (1981).

²⁸ F. R. Salemme, *Proc. Natl. Acad. Sci. U.S.A.* **79**, 5263 (1982).

²⁹ R. Ebright, in "Molecular Structure and Biological Activity" (J. Griffen and R. Duax, eds.), p. 91. Elsevier, New York, 1982.

³⁰ C. Pabo and M. Lewis, *Nature (London)* **298**, 443 (1982).

DNA site under study. In one tester strain, the reporter promoter is placed under the control of the consensus DNA site. In the other tester strains, the reporter promoter is placed under the control of derivatives of the consensus DNA site having a nonconsensus base pair at one or more positions within the DNA site.

In cases where the DNA site is an inverted repeat or a direct repeat, equivalent qualitative results are obtained using derivatives of the consensus DNA site substituted in each repeat, or derivatives of the consensus DNA site substituted in only one repeat.¹⁰ Derivatives of the consensus DNA site substituted in each repeat normally are preferred. These derivatives exhibit larger values for the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$, reducing the significance of experimental error. (In most cases, the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$ for a derivative of the consensus DNA site substituted in each of two repeats is the square of the ratio $K_{D,\text{noncons}}/K_{D,\text{cons}}$ for the analogous derivative of the consensus DNA site substituted in only one repeat.)

In order to analyze the profile of specificity of DNA-binding protein X , the plasmid encoding DNA-binding protein X is introduced into each of the tester strains. Note that each of the resulting plasmid-bearing strains is identical except for the sequence of the DNA site. One important implication is that in each of the resulting plasmid-bearing strains the intracellular concentration of DNA-binding protein X is equal. In the data reduction, all calculations are performed using only data from strains having the plasmid encoding the identical DNA-binding protein X . Therefore, the analysis does not incur complications due to either (1) differing rates of synthesis of wild-type vs substituted proteins, or (2) differing stabilities of wild-type vs substituted proteins.

Data collection and data reduction are performed as follows: For each strain, the expression of the reporter promoter is determined. This is accomplished by measurement of the differential rate of synthesis of the reporter mRNA, or by measurement of the differential rate of synthesis of the reporter protein. The fractional occupancy of the reporter promoter, θ , is calculated from the experimental data using Eq. (1) or Eq. (2). Ratios of equilibrium dissociation constants, $K_{D,1}/K_{D,2}$, are calculated from the values for θ using Eq. (3).

In the case of a repressor, the fractional occupancy of the reporter promoter, θ , is calculated as follows^{6,7}:

$$\theta = 1 - (E/E_M) \quad (1)$$

where E denotes the expression of the reporter promoter in the presence of repressor, and E_M denotes the expression of the reporter promoter in the absence of repressor. E_M is determined by measurement of the expres-

sion of the reporter promoter in the presence of a chemical that inactivates the repressor (e.g., isopropyl-thio- β -D-galactoside in the case of Lac repressor).^{6,7} Alternatively, E_M can be determined by measurement of the expression of the reporter promoter in a derivative of the tester strain having a control plasmid that does not encode the repressor.

In the case of an activator protein, the fractional occupancy of the reporter promoter, θ , is calculated as follows¹⁰:

$$\theta = E/E_M \quad (2)$$

where E denotes the expression of the reporter promoter in the presence of the activator protein, and E_M denotes the expression of the reporter promoter in the presence of a saturating concentration of the activator protein. E_M is determined by calculation using comparison *in vitro* data.¹⁰ Alternatively, E_M can be determined by measurement of the expression of the reporter promoter in a derivative of the tester strain having a plasmid construct that results in a saturating concentration of the activator protein. The values used for E and E_M should be corrected for activator-independent expression of the reporter promoter.

Ratios of equilibrium dissociation constants, $K_{D,1}/K_{D,2}$, are calculated as follows^{6,7,10}:

$$K_{D,1}/K_{D,2} = [\theta_2(1 - \theta_1)]/[\theta_1(1 - \theta_2)] \quad (3)$$

Equation (3) applies for data from any two strains 1 and 2 in which the intracellular concentration of unbound DNA-binding protein X in strain 1 is equal to the intracellular concentration of unbound DNA-binding protein X in strain 2 (e.g., any two tester strains having a plasmid encoding the identical DNA-binding protein X ; see above).

In Vitro Assay of $K_{D,noncons}/K_{D,cons}$: Method One. This method is the simplest and most reliable method. In this method, equilibrium DNA binding experiments are performed using purified wild-type DNA binding protein and purified substituted DNA binding protein, and each of a set of double-stranded DNA fragments containing the DNA site. One DNA fragment in the set contains the consensus DNA site. The other DNA fragments in the set contain derivatives of the consensus DNA site having a nonconsensus base pair at one or more positions within the DNA site.

In cases where the DNA site is an inverted repeat or a direct repeat, equivalent qualitative results are obtained using derivatives of the consensus DNA site substituted in each repeat, or derivatives of the consensus DNA site substituted in only one repeat.¹ Derivatives of the consensus DNA site substituted in each repeat normally are preferred (see Experimental Methods, *In Vivo Assay*, above).

Equilibrium dissociation constants, K_D , are measured using the nitrocellulose filter-binding assay,³¹ the gel retardation assay,^{32,33} or the quantitative nuclease footprinting assay.³⁴

In Vitro Assay of $K_{D,noncons}/K_{D,cons}$: Method Two. A chemical modification interference assay is performed as follows^{35,36}: A sample of end-labeled double-stranded DNA fragment containing the DNA site of interest is randomly chemically modified, using reaction conditions such that at most one chemical modification occurs per DNA fragment. The sample of modified DNA fragment then is incubated with excess DNA-binding protein until the establishment of equilibrium. The DNA molecules that remain unbound at equilibrium are separated from the protein-bound DNA molecules. The DNA molecules that remain unbound comprise those having a chemical modification positioned so as to interfere with binding of the protein; the locations of the interfering modifications are identified by thermal- and/or alkali-mediated DNA cleavage at the modified nucleotide, followed by denaturing polyacrylamide gel electrophoresis and autoradiography. There are two relevant types of chemical modification interference assays: methylation interference assays,^{35,36} which probe the effects of methylation of guanine N-7 atoms (and, under certain conditions, also of adenine N-3 atoms), and depurination/depyrimidation interference assays,^{21,35} which probe the effects of removal of purines or pyrimidines.

Hochschild and Ptashne²⁰ and Brunelle and Schleif^{21,24} have shown that a chemical modification interference assay can be a rapid, efficient method to obtain a rough estimate of the ratio $K_{D,noncons}/K_{D,cons}$ at each of several positions within the DNA site. The methylation interference assay permits the estimation of the ratio $K_{D,N-7-methyl-G}/K_{D,cons}$ at each guanine of the DNA site. The depurination/depyrimidation interference assay permits the estimation of the ratio $K_{D,abasic\ site}/K_{D,cons}$ at each base of the DNA site. The procedure is as follows: The methylation interference and/or depurination/depyrimidation interference patterns exhibited by the wild-type protein and a substituted protein are determined and compared. The position that in the wild-type protein-DNA complex is contacted by the amino acid under study is identified as the position, i , at which the wild-

³¹ A. Riggs, H. Suzuki, and S. Bourgeois, *J. Mol. Biol.* **48**, 67 (1970).

³² M. Garner and A. Revzin, *Nucleic Acids Res.* **9**, 3047 (1981).

³³ M. Fried and D. Crothers, *Nucleic Acids Res.* **9**, 6505 (1981).

³⁴ M. Brenowitz, D. F. Seneor, M. A. Shea, and G. K. Ackers, this series, Vol. 130, p. 132.

³⁵ J. Majors, Ph.D. Thesis, Harvard University, Cambridge, Massachusetts (1977).

³⁶ U. Siebenlist, R. Simpson, and W. Gilbert, *Cell (Cambridge, Mass.)* **20**, 269 (1980).

type protein exhibits an interference (i.e., $K_{D,\text{noncons},i}/K_{D,\text{cons}} \gg 1$), but at which the substituted protein does not exhibit an interference (i.e., $K_{D,\text{noncons},i}/K_{D,\text{cons}} \approx 1$).

Using this procedure, all positions within the DNA site susceptible to the chemical modification are assessed *in a single experiment*. This procedure is useful in the case where no high-resolution structure or detailed model of the protein-DNA complex exists, but information suggests which amino acids make contacts (see Experimental Design, Case Two, above).

It is important to note that the methylation interference assay involves the introduction of a methyl group, and the introduction of a positive charge, at the guanine N-7 atom. The depurination/depyrimidation assay involves the introduction of an abasic site. These changes can have significant effects on local DNA conformation, complicating the interpretation of results. Important results obtained using the methylation interference or depurination/depyrimidation assays should be confirmed using one of the two preceding procedures.²⁴

New-Specificity Approach

The logic of the new-specificity approach³⁷ is outlined in Fig. 4. The approach is to substitute the amino acid under study by an amino acid that: (1) makes no energetically favorable interaction with the consensus base pair at the position within the DNA site that in the wild-type protein-DNA complex is contacted by the amino acid under study, but (2) makes an energetically favorable interaction with one nonconsensus base pair at the position within the DNA site that in the wild-type protein-DNA complex is contacted by the amino acid under study. A new-specificity substitution will result in four identifiable characteristics:

1. The substituted protein will prefer a particular nonconsensus base pair to the consensus base pair at one position, i , within the DNA site, corresponding to the position that in the wild-type protein-DNA complex is contacted by the amino acid under study. (In the case of an inverted-repeat or direct-repeat DNA site, the substituted protein will prefer a particular nonconsensus base pair to the consensus base pair at one position, i , in each repeat.) The substituted protein will exhibit a lower affinity for the consensus DNA site than for one derivative of the consensus DNA site having a nonconsensus base pair at position i : $K_{D,\text{noncons},i}/K_{D,\text{cons}} \ll 1$.

2. The substituted protein will retain the wild-type ability to discriminate between the consensus base pair and nonconsensus base pairs at each

³⁷ P. Youderian, A. Vershon, S. Bouvier, R. Sauer, and M. Susskind, *Cell (Cambridge, Mass.)* **35**, 777 (1983).

other position, j , within the DNA site. The substituted protein will exhibit a higher affinity for the consensus DNA site than for derivatives of the consensus DNA site having a nonconsensus base pair at position j : $K_{D, \text{noncons}, j} / K_{d, \text{cons}} \gg 1$.

3. As compared to the wild-type protein, the substituted protein will exhibit a decreased affinity for the consensus DNA site. This is due to the presence in the wild-type protein–DNA complex, but absence in the substituted protein–DNA complex, of an energetically favorable interaction between the amino acid under study and the consensus base pair at position i (Fig. 4-I, 4-II); and to the possible presence in the substituted protein–DNA complex of an energetically unfavorable interaction between the amino acid under study and the consensus base pair at position i (Fig. 4-II).

4. As compared to the wild-type protein, the substituted protein will exhibit an increased affinity for one derivative of the consensus DNA site having a nonconsensus base pair at position i . This is due to the absence in the wild-type protein–DNA complex, but presence in the substituted protein–DNA complex, of an energetically favorable interaction between the amino acid under study and the nonconsensus base pair at position i (Fig. 4-I, 4-II); and to the possible presence in the wild-type protein–DNA complex, but absence in the substituted protein–DNA complex, of an energetically unfavorable interaction between the amino acid under study and the nonconsensus base pair at position i (Fig. 4-II).

Examples of new-specificity substitutions have been obtained in the following ways: (1) in the course of construction of a large number of substitutions (up to all 19 possible substitutions) of an amino acid known or predicted to make an amino acid–base pair contact,^{11,38–40} (2) by interchanging, or “swapping,” one amino acid within a pair of highly homologous DNA-binding proteins having DNA sites that differ at only one base pair⁴¹ (see also Refs. 42–46), and (3) by random mutagenesis of the gene

³⁸ R. Wharton and M. Ptashne, *Nature (London)* **326**, 888 (1987).

³⁹ N. Lehming, J. Sartorius, S. Oehler, B. von Wilcken-Bergmann, and B. Müller-Hill, *Proc. Natl. Acad. Sci. U.S.A.* **85**, 7947 (1988).

⁴⁰ S. Bass, V. Sorrelis, and P. Youderian, *Science* **242**, 240 (1988).

⁴¹ L. Altschmied, R. Baumeister, K. Pfeiderer, and W. Hillen, *EMBO J.* **7**, 4011 (1988).

⁴² N. Lehming, J. Sartorius, M. Niemöller, G. Genenger, B. von Wilcken-Bergmann, and B. Müller-Hill, *EMBO J.* **6**, 3145 (1987).

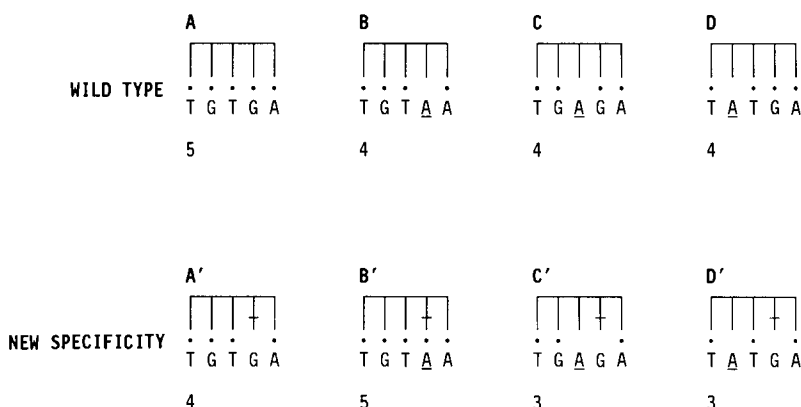
⁴³ S. Hanes and R. Brent, *Cell (Cambridge, Mass.)* **57**, 1275 (1989).

⁴⁴ J. Treisman, P. Gonczy, M. Vashishtha, E. Harris, and C. Desplan, *Cell (Cambridge, Mass.)* **59**, 553 (1989).

⁴⁵ M. Danielson, L. Hinck, and G. Ringold, *Cell (Cambridge, Mass.)* **57**, 1131 (1989).

⁴⁶ K. Umesono and R. Evans, *Cell (Cambridge, Mass.)* **57**, 1139 (1989).

I



II

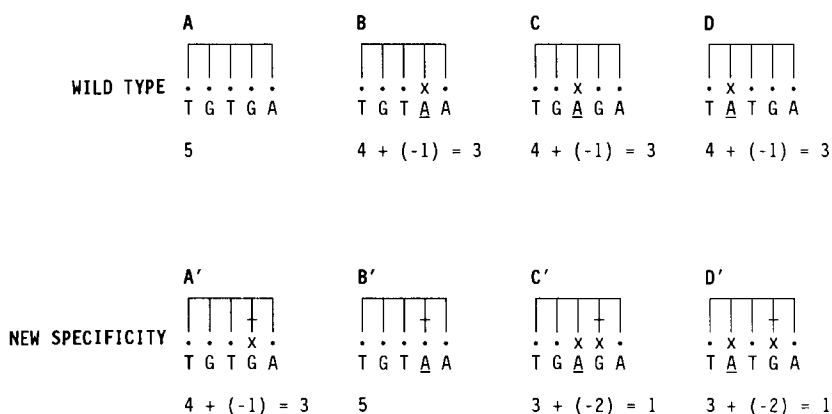


FIG. 4. New-specificity approach: rationale. A new-specificity substituted DNA-binding protein will: (1) prefer a particular nonconsensus base pair to the consensus base pair at one position, i , within the DNA site, corresponding to the position that in the wild-type protein-DNA complex is contacted by the amino acid under study, but (2) retain the wild-type ability to discriminate between the consensus base pair and nonconsensus base pairs at each other position, j , within the DNA site. The figure presents two cases: case I, which considers only energetically favorable contacts, and case II, which considers both energetically favorable and energetically unfavorable contacts. *Case I.* (A) Interaction of the wild-type DNA-binding protein with five base pairs of the consensus DNA site (five favorable contacts). The residue of the DNA-binding protein participating in each contact is indicated as a vertical line; the presence of a favorable contact is indicated as a dot. (B-D) Interaction of the wild-type DNA-binding protein with derivatives of the consensus DNA site, each

encoding the DNA-binding protein, followed by genetic selection and/or screening for decreased affinity for the consensus DNA site and for increased affinity for one nonconsensus derivative of the consensus DNA site.^{23,37}

Empirically, new-specificity substitutions are relatively rare. This is not surprising, since a new-specificity substitution must not only eliminate an interaction, but also introduce a new interaction. Glu-181 of CAP is known to contact position 7 of the DNA site in the CAP-DNA complex^{1,13,14} (see Experimental Design, Case One and Experimental Design, Case Three, above). In recent work, all 19 possible amino acid substitutions at residue 181 of CAP have been constructed; all 19 substituted proteins have been purified; and the DNA sequence-recognition properties of all 19 substituted proteins have been assessed with respect to positions 5, 6, and 7 of the DNA site.¹¹ Fully 15 of 19 amino acid substitutions at residue 181 of CAP result in the loss-of-contact phenotype (Gly, Ala, Val, Leu, Ile, Pro, Ser, Thr, Cys, Met, Asn, Gln, His, Lys, and Arg). Only 1 of 19 amino acid substitutions at residue 181 of CAP results in the new-specificity phenotype (Asp). Empirically, new-specificity substitutions also are difficult to predict. The sole substitution at residue 181 of CAP that results in the new-specificity phenotype was not predicted.¹¹

having a nonconsensus base pair at one position (four favorable contacts). (A') Interaction of the new-specificity substituted DNA-binding protein with the consensus DNA site (four favorable contacts). (B') Interaction of the new-specificity substituted DNA-binding protein with one derivative of the consensus DNA site having a nonconsensus base pair at the position, *i*, that in the wild-type protein-DNA complex is contacted by the amino acid under study (five favorable contacts). (C' and D'). Interaction of the new-specificity substituted DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one other position, *j* (three favorable contacts). *Case II.* (A) Interaction of the wild-type DNA-binding protein with five base pairs of the consensus DNA site (five favorable contacts). The residue of the DNA-binding protein participating in each contact is indicated as a vertical line; the presence of a favorable contact is indicated as a dot. (B-D) Interaction of the wild-type DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one position (four favorable contacts, one unfavorable contact; net three favorable contacts). The presence of an unfavorable contact is indicated as an "x." (A') Interaction of the new-specificity substituted DNA-binding protein with the consensus DNA site (four favorable contacts, one unfavorable contact; net three favorable contacts). (B') Interaction of the new-specificity substituted DNA-binding protein with one derivative of the consensus DNA site having a nonconsensus base pair at the position, *i*, that in the wild-type protein-DNA complex is contacted by the amino acid under study (five favorable contacts). (C' and D') Interaction of the new-specificity substituted DNA-binding protein with derivatives of the consensus DNA site, each having a nonconsensus base pair at one other position, *j* (three favorable contacts, two unfavorable contacts; net one favorable contact).

New-specificity substitutions, when obtained, can permit the identification of amino acid–base pair contacts (cf. Refs. 2, 3, and 38). However, since new-specificity substitutions are rare and difficult to predict, in most instances it is not efficient to seek new-specificity substitutions in order to identify an amino acid–base pair contact.

Altered Methylation Protection Approach

Methylation protection experiments identify guanine N-7 and adenine N-3 atoms of the DNA site that exhibit reduced reactivities to dimethylsulfate on formation of the protein–DNA complex. Methylation protection conventionally is interpreted as indicating that amino acids of the protein are close to, or in contact with, the protected guanine N-7 or adenine N-3 atoms. Pabo *et al.*⁴⁷ and Knight and Sauer⁴⁸ have shown that deletion of, or substitution of, amino acids of the DNA-binding protein can, at least in certain cases, produce position-specific alterations in the methylation protection pattern. Such position-specific alterations in the methylation protection pattern can permit the identification of amino acid–base pair contacts.

Pabo *et al.*⁴⁷ have used the altered methylation protection approach to identify amino acid–base pair contacts in the λ repressor–DNA complex. Wild-type λ repressor protects positions 4, 6, 7, 8, and 9 of the consensus DNA site; in contrast, a derivative of λ repressor, lacking amino acids 1 to 3, protects only positions 4, 6, and 7 of the consensus DNA site. These results suggest that amino acids 1 to 3 of λ repressor are near to, or in contact with, positions 8 and 9 of the consensus DNA site in the λ repressor–DNA complex. This suggestion has been confirmed in two ways: (1) using an approach related to the loss-of-contact approach,⁴⁹ and (2) by determination of the three-dimensional structure of the specific λ repressor–DNA complex.⁵ Knight and Sauer⁴⁸ have used the altered methylation protection approach to identify amino acid–base pair contacts in the Mnt repressor–DNA complex. Wild-type Mnt repressor protects positions 4, 5, 7, 8, 10, and 11 of the consensus DNA site; in contrast, a derivative of Mnt repressor having the Arg $\rightarrow$ Lys substitution at amino acid 2 protects only positions 4, 5, 7, and 8 of the DNA site. These results suggest that Arg-2 of Mnt repressor is near to, or in contact with, positions 10 and 11 of the consensus DNA site in the Mnt repressor–DNA complex.

In the two examples cited, the altered methylation protection approach

⁴⁷ C. Pabo, W. Krovatin, A. Jeffrey, and R. Sauer, *Nature (London)* **298**, 441 (1982).

⁴⁸ K. Knight and R. Sauer, *J. Biol. Chem.* **264**, 13706 (1989).

⁴⁹ J. Eliason, M. Weiss, and M. Ptashne, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 2339 (1985).

has been applied to an extended N-terminal “arm” segment.^{47,48} The success of the approach may have been a function of the relative flexibility⁵⁰ of the arm segment. It is not known whether an amino acid substitution not in an arm segment would be sufficient to permit access of dimethyl sulfate to a previously protected guanine N-7 or adenine N-3 atom.

Caveats

Protein Conformation

When an amino acid substitution is constructed in a protein, the substitution can result in changes in protein conformation. This complicates the interpretation of results. It is necessary to distinguish between results that are due to first-order effects (i.e., the removal and/or addition of amino acid side-chain atoms), and results that are due to second-order effects (i.e., the alteration of protein conformation).

In light of these considerations, to identify an amino acid–base pair contact using the approaches discussed in this chapter, the evidence must meet the following standards:

1. The effects must have large magnitudes. To demonstrate the retention of specificity at a position within the DNA site, or to demonstrate a new specificity at a position within the DNA site, the ratio $K_{D,1}/K_{D,2}$ must be ≥ 5 (≈ 1 kcal/mol). In the case of an inverted-repeat or direct-repeat DNA site altered in each repeat, the ratio $K_{D,1}/K_{D,2}$ must be ≥ 20 (≈ 2 kcal/mol).
2. The effects must be position specific. Substitution of the amino acid under study must affect specificity or methylation protection only at one position, or at two adjacent positions, within the DNA site.
3. At least two different substitutions of the amino acid under study must result in the loss-of-contact phenotype, the new-specificity phenotype, or the altered methylation protection phenotype.
4. The inferred amino acid–base pair contact must be consistent with the structures, if known, of the protein and of the specific protein–DNA complex.

Protein Concentration: In Vivo Assays

When the wild-type DNA-binding protein and a substituted DNA-binding protein are expressed *in vivo* under identical conditions, the intracellular concentration of the unbound wild-type DNA-binding protein can

⁵⁰ M. Weiss, R. Sauer, D. Patel, and M. Karplus, *Biochemistry* **23**, 5090 (1984).

be different from the intracellular concentration of the unbound substituted DNA-binding protein. This is due to several factors: i.e., differences in rates of synthesis, differences in stability, and differences in nonspecific DNA binding affinity.

The section Experimental Methods (see above) describes quantitative *in vivo* assays in which a reporter promoter is placed under the control of the DNA site of interest. In interpreting the results of such assays, one must not compare the levels of expression of the reporter promoter for two different DNA-binding proteins (e.g., for the wild-type DNA-binding protein vs a substituted DNA-binding protein). This comparison is not controlled for differences in the *in vivo* concentrations of the unbound DNA-binding proteins.

It is imperative to restrict all comparisons and calculations to data for a single DNA-binding protein. Failure to observe this requirement has resulted in the misinterpretation of results.⁵¹⁻⁵⁴

Acknowledgments

I thank Dr. Jon Beckwith for helpful discussions. Preparation of this chapter was supported by National Institutes of Health Grant GM41376.

⁵¹ In Refs. 13 and 14, the Val-181, Leu-181, and Lys-181 substitutions of CAP were misidentified as new-specificity substitutions. In fact, these are loss-of-contact substitutions.¹

⁵² In Ref. 40, the Phe-79, Thr-80, and Ser-81 substitutions of Trp repressor are misidentified as new-specificity substitutions. In fact, these are neither new-specificity nor loss-of-contact substitutions; these substitutions result in essentially unchanged, wild-type specificity (Table I of Ref. 40).

⁵³ J. Sartorius, N. Lehming, B. Kisters, B. von Wilcken-Bergmann, and B. Müller-Hill, *EMBO J.* **8**, 1265 (1989).

⁵⁴ In Ref. 53, the [Ser-17;Ala-18], [Ala-17;Ala-18], [Ser-17;Thr-18], [Pro-17;Thr-18], [Ile-17;Thr-18], [His-17;Val-18], and [His-17;Ile-18] substitutions are misidentified as new-specificity substitutions. In fact, these are loss-of-contact substitutions (Table 1 of Ref. 53).

[31] Second-Site Reversion as Means of Enhancing DNA-Binding Affinity

By DALE L. OXENDER and AMY L. GIBSON

Introduction

As a first approach to the study of protein structure, random mutagenesis techniques are useful for identifying key residues and domains of the protein that play important roles in structure or function.^{1,2} Second-site reversion studies of key mutations can be a powerful genetic tool to identify additional important residues and to elucidate complex interactions between specific amino acid residues important for the structure and function of a protein.³⁻⁶ A specific application for the selection of second-site revertants illustrated in this chapter is the production of an altered DNA-binding protein that shows increased binding activity.

Studies by Yanofsky and colleagues demonstrated the range of information available from reversion analysis in classic genetic studies of the α subunit of tryptophan synthase from *Escherichia coli*.^{7,8} Using random mutagenesis techniques, the investigators found that 8 of the 268 residues of the enzyme were essential for function. A missense mutation resulting in replacement of one of these eight amino acid residues inactivated the enzyme. Subsequent mutagenesis and selection for revertants showed that four of these positions could be mutated to one or, in some cases, several new amino acid residues, which restored significant activity. These results are summarized in Fig. 1. For example, replacement of Thr-183 by isoleucine resulted in reduced enzymatic activity. Activity was restored by the substitution of either aspartic acid or serine at position 183. At position 211, seven different primary mutations were reversed by one of eight substitutions.

Table I summarizes the effects on α -tryptophan synthase activity of a variety of reversions, second-site reversions, and double mutations produced by molecular splicing methods. An interesting set of results involves

¹ C. Auerbach and J. M. Robson, *Nature (London)* **157**, 302 (1946).

² R. L. Kelley and C. Yanofsky, *Proc. Natl. Acad. Sci. U.S.A.* **82**, 483 (1985).

³ D. R. Helinski and C. Yanofsky, *J. Biol. Chem.* **238**, 1043 (1963).

⁴ M. H. Hecht and R. T. Sauer, *J. Mol. Biol.* **186**, 53 (1985).

⁵ H. C. M. Nelson and R. T. Sauer, *Cell (Cambridge, Mass.)* **42**, 549 (1985).

⁶ L. S. Klig, D. L. Oxender, and C. Yanofsky, *Genetics* **120**, 651 (1988).

⁷ E. J. Murgola and C. Yanofsky, *J. Mol. Biol.* **86**, 775 (1974).

⁸ C. Yanofsky, *JAMA* **218**, 1026 (1971).

Residue	22	49	175	177	183	211	234	235
Wild type	Phe	Glu	Tyr	Leu	Thr	Gly	Gly	Ser
Primary mutations (activity reduced)	Leu	Gln Met Val	Cys	Arg	Ile	Arg Asp Glu Lys	Asp Cys	Leu
Revertants (activity restored)					Asp Ser Thr	Ala Gly Ser Thr Val	Ala Gly	

FIG. 1. Summary of mutations in the α subunit of tryptophan synthase. Amino acid positions are given, with the original residue, primary mutations, and revertants listed. [Data are from E. J. Murgola and C. Yanofsky, *J. Mol. Biol.* **86**, 775 (1974).]

two missense mutations: substitution of glutamic acid for Gly-211 and replacement of Tyr-175 with cysteine. Either mutation alone results in inactive enzyme, but a double mutant with both changes is active. Yanofsky suggested that these two residues must be close together in the folded protein.

The three-dimensional structure of the $\alpha_2\beta_2$ -tryptophan synthase complex with a substrate analog bound in the α subunit active site was deter-

TABLE I
TRYPTOPHAN SYNTHASE α SUBUNIT: REVERSION AND SECOND-SITE REVERSION^a

Gene source	Amino acid sequence				Activity
	175	177	211	213	
Wild type	-Tyr-Leu-Leu-...	-Gly-Phe-Gly-			+
Primary mutation	-Tyr-Leu-Leu-...	-Glu-Phe-Gly-			-
Phenotypic revertant	-Tyr-Leu-Leu-...	-Val-Phe-Gly-			+
Double mutant	-Tyr-Leu-Leu-...	-Val-Phe-Val-			-
Phenotypic revertant	-Tyr-Leu-Leu-...	-Val-Phe-Ala-			+
Second-site revertant	-Tyr-Leu-Leu-...	-Gly-Phe-Val-			+
Second-site revertant	-Tyr-Leu-Leu-...	-Ala-Phe-Val-			+
Second-site revertant	-Tyr-Leu-Arg-...	-Val-Phe-Val-			+
Recombinant	-Tyr-Leu-Arg-...	-Gly-Phe-Gly-			-
Second-site revertant	-Cys-Leu-Leu-...	-Glu-Phe-Gly-			+
Recombinant	-Cys-Leu-Leu-...	-Gly-Phe-Gly-			-

^a Results of sequential mutagenesis of tryptophan synthase α subunit. The wild-type amino acid sequence at positions 175 to 177 and 211 to 213 are shown. Sequential amino acid replacements are underlined. Effects on enzymatic activity are given by + and -. Two mutations obtained by second-site reversion were placed in an otherwise wild-type protein by gene splicing methods. Data are from C. Yanofsky, *JAMA* **218**, 1026 (1971).

mined.⁹ The structural model for the enzyme confirms predictions made by Yanofsky based on results of the reversion studies. Based on the crystallographic results, at least one of the eight key residues identified in Fig. 1 (Glu-49) is directly involved in catalysis. Six of the remaining seven (absenting Thr-183) participate in the substrate-binding site. As predicted from the study cited in Table I, Tyr-175 and Gly-211 lie in close spatial proximity in the substrate-binding site. Computer graphics model building¹⁰ suggests that either cysteine at position 175 or glutamic acid at position 211 would disrupt the geometry of the binding site, but that the paired changes described in Yanofsky's studies could restore the proper geometry. The close correspondence of results obtained by the two methods illustrates the power of fine-structure genetic analysis involving second-site reversion.

Tryptophan Repressor

Reversion analysis of the tryptophan repressor, TrpR, demonstrated that the method can also be used to obtain second-site mutations that enhance DNA binding by the repressor.⁶ This result may prove to be broadly applicable to many DNA-binding proteins and could provide an additional degree of freedom in applications involving "engineered" regulation of gene expression. Second-site reversion provides an explicit search strategy for finding amino acid substitutions that (introduced singly into wild-type proteins) cause increased DNA binding.

Expression of the tryptophan biosynthetic genes of *E. coli* is regulated by binding of a tryptophan-activated (K_D 10^{-5} M^{11,12}) specific repressor at the promoter/operator region of the operon, preventing initiation of transcription.^{13,14} The repressor is encoded by the *trpR* gene and is subject to autoregulation. TrpR also regulates expression of a third operon, *aroH*, involved in biosynthesis of aromatic amino acids.

The repressor has been purified¹⁵ and crystallographic studies have

⁹ C. C. Hyde, S. A. Ahmed, E. A. Padlan, E. W. Miles, and D. R. Davies, *J. Biol. Chem.* **263**, 17857 (1988).

¹⁰ S. Nagata, C. C. Hyde, and E. W. Miles, *J. Biol. Chem.* **264**, 6288 (1989).

¹¹ D. N. Arvidson, C. Bruce, and R. P. Gunsalus, *J. Biol. Chem.* **261**, 238 (1986).

¹² R. Q. Marmorstein, A. Joachimiak, M. Sprinzl, and P. B. Sigler, *J. Biol. Chem.* **262**, 4922 (1987).

¹³ J. K. Rose, C. L. Squires, C. Yanofsky, H.-L. Yang, and G. Zubay, *Nature (London) New Biol.* **245**, 133 (1973).

¹⁴ C. L. Squires, F. D. Lee, and C. Yanofsky, *J. Mol. Biol.* **92**, 93 (1975).

¹⁵ A. Joachimiak, R. L. Kelley, R. P. Gunsalus, C. Yanofsky, and P. B. Sigler, *Proc. Natl. Acad. Sci. U.S.A.* **80**, 668 (1983).

revealed the structures of the inactive aporepressor,¹⁶ the active holorepressor,¹⁷ and a repressor/operator complex containing an 18-mer synthetic oligonucleotide duplex.¹⁸ The current high-resolution model for tryptophan repressor shows a dimer of unusual architecture. The majority (72%) of the residues participate in one of six helices (designated A-F). There is a solid "central core" formed by the interlocked amino-terminal halves of the two subunits (helices A, B, and C). The central core is symmetrically flanked by two flexible "DNA-reading heads," each formed from the carboxyl-terminal half of one monomer. The D and E helices (residues 68-92) of each monomer form a classic bihelical DNA-binding feature, related by sequence homology to portions of the repressors from λ and coliphage 434, and catabolite repressor protein and P22 Cro protein.²

Surface pockets that bind corepressor are wedged between the solid core and the flexible DNA-reading heads. Side chains of two arginine residues at positions 54 and 84 form a sandwich around the indole moiety of the corepressor, tryptophan. The hydroxyl of Thr-44 and the guanidino group of Arg-84 interact with the carboxyl group of the ligand.

An unusually large DNA contact surface was identified and operator recognition appears to rely on sequence-determined DNA backbone conformation. TrpR functional regions were mapped by random chemical mutagenesis methods.² Mutations were distributed throughout the *trpR* gene, although a cluster of mutations was identified in the helix-turn-helix region of the protein. Primary mutations chosen for second-site reversion analysis were drawn from both classes.

Methods

Choice of Primary Mutations and Mutagens

The pool of TrpR mutants included proteins with altered tryptophan-binding sites (TM44), altered DNA-binding features (GS78) or single mutations affecting both regions (TM81, RH84, and GR85).¹⁹ These primary

¹⁶ R. G. Zhang, A. Joachimiak, C. L. Lawson, R. W. Schevitz, Z. Otwinowski, and P. B. Sigler, *Nature (London)* **327**, 591 (1987).

¹⁷ R. W. Schevitz, Z. Otwinowski, A. Joachimiak, C. L. Lawson, and P. B. Sigler, *Nature (London)* **317**, 782 (1985).

¹⁸ Z. Otwinowski, R. W. Schevitz, R.-G. Zhang, C. L. Lawson, A. Joachimiak, R. Q. Marmorstein, B. F. Luisi, and P. B. Sigler, *Nature (London)* **335**, 321 (1988).

¹⁹ Specific missense mutations are designated by the one-letter amino acid codes of the original residue, followed by the replacement residue, and the number of the amino acid that has been changed. T, Threonine; M, methionine; G, glycine; S, serine; R, arginine; H, histidine; E, glutamic acid; K, lysine; P, proline; L, leucine; D, aspartic acid; N, asparagine; A, alanine; V, valine.

mutations were produced by *in vitro* exposure to hydroxylamine, which causes primarily C-T transitions. The same unidirectional mutagen was chosen for obtaining second-site revertants to minimize the appearance of true revertants. A second advantage of the use of hydroxylamine is that it is possible to obtain a level of mutagenesis that approaches one mutation/ 10^3 bases. Second-site mutation candidate genes should always be completely sequenced to verify that only one new change has been incorporated.

Hydroxylamine Mutagenesis

Mutagenesis was performed on plasmids derived from pRLK13.² Plasmid DNA was purified by CsCl banding and mutagenized by exposure to hydroxylamine according to the protocol of Davis *et al.*²⁰ A solution of 1 M hydroxylamine (pH 6.0) was made immediately before use by mixing 0.56 ml 4 M NaOH and 0.35 g NH_2OH in sterile water to a final volume of 5.0 ml. Four volumes of hydroxylamine solution was added to 1 vol of DNA (0.8 M final hydroxylamine concentration) and the mixture was incubated for 36 hr at 37°. The mutagen was removed by overnight dialysis against TE buffer (10 mM Tris-HCl, pH 7.4, 1 mM EDTA). DNA was ethanol precipitated and resuspended in TE buffer.

Isolation of Revertants

Mutagenized plasmid DNA was introduced into *E. coli* strain CY15075, a λ lysogen carrying a *lacZ* gene fusion producing β -galactosidase under control of the *trp* promoter/operator region.²¹ The strain has a frameshift mutation in the chromosomal *trpR* gene and produces no endogenous repressor. Plasmid-containing colonies were selected for growth on chloramphenicol-containing plates. Cells were replicated to media with and without tryptophan to compare repressor activation. Inclusion of X-Gal permitted screening by color for colonies with reduced β -galactosidase activity. Up to 100,000 transformants were screened from each starting plasmid.

Characterization of Second-Site Mutations

DNA Sequencing. Plasmid DNA was isolated from second-site mutation candidates and the *trpR* genes subcloned to M13 replicative-form DNA for Sanger sequencing.²²

²⁰ R. W. Davis, D. Botstein, and J. R. Roth, eds., "Advances in Bacterial Genetics, a Manual for Genetic Engineering," p. 94. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1980.

²¹ C. Yanofsky and V. Horn, *J. Bacteriol.* **145**, 1334 (1981).

²² F. Sanger, S. Nicklen, and A. R. Coulson, *Proc. Natl. Acad. Sci. U.S.A.* **74**, 5463 (1977).

Repressor Activity Measured by β -Galactosidase Assay. The relative activities of mutant and wild-type repressor molecules were assayed *in vivo* by determining β -galactosidase levels²³ in the expression indicator strain with and without added tryptophan. Overnight cultures of strains bearing double-mutant plasmids were diluted with 5 ml minimal medium containing 1 mM isopropyl- β -D-thiogalactoside (IPTG), and subcultured at 37° to OD₆₀₀ of 0.3 to 0.7. Cultures were chilled on ice and 10 to 100 μ l was mixed with 0.8 ml Z buffer (see below). Cells were permeabilized by adding 2 drops chloroform and 1 drop 0.1% (w/v) sodium dodecyl sulfate (SDS), then vortexed 15 sec. Extracts were warmed to 30° in a water bath and 200 μ l substrate [*o*-nitrophenyl- β -D-galactoside (β -ONPG)] was added. The reaction was timed, stopped by adding 0.5 ml 1 M Na₂CO₃, and left on ice. Tubes were centrifuged 5 min to remove cell debris and the OD₄₂₀ recorded. Activity was calculated according to the formula:

$$\text{Units} = (1000)(\text{OD}_{420})/(\text{time in minutes})(\text{culture volume})(\text{OD}_{600})$$

Z-buffer salts: Na₂HPO₄ · 7H₂O (8.0 g), NaH₂PO₄ · H₂O (2.8 g),
KCl (0.38 g), MgSO₄ · 7H₂O (0.13 g)

Salts were dissolved in water to a volume of 480 ml and the pH was adjusted to 7.0; 13.5 ml 2-mercaptoethanol was added and the final volume adjusted to 500 ml.

Results

Activity levels of the wild-type repressor and single and double mutants are shown in Table II. As expected, compensatory mutations increased the activity of repressors with primary mutations. In the case of TrpR, no double mutant had activity exceeding that of wild-type repressor or of the corresponding superrepressor mutation alone. Second-site reversions that restored activity to four of the five primary mutations are summarized in Table III. A single compensatory mutation was found for TM44 and GS78. Three different second-site reversions were found for both RH84 and GR85. Most of these second-site mutations were represented by several independent isolates. Three additional double mutants, including one that restored activity to TM81, were constructed by subcloning (see Discussion, below). The majority of second-site reversions (18 of 27 isolates) consisted of previously known superrepressor mutations. The remaining nine mutations involved changes at positions 45 or 46. Many of the same second-site changes were obtained regardless of which primary mutation

²³ T. Platt, B. Müller-Hill, and J. Miller, in "Experiments in Molecular Genetics" (J. Miller, ed.), p. 352. Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, 1972.

TABLE II
REPRESSOR ACTIVITY OF *trpR* MUTANTS WITH SECOND-SITE CHANGES AND
SUPERREPRESSOR MUTANTS^a

Mutant plasmid ^b	Second-site change	Units of β -galactosidase activity ^c		Superrepressor plasmid ^b	Units of β -galactosidase activity ^c	
		– Trp	+ Trp		– Trp	+ Trp
TM44		9,000	5,900	(Wild type ^b)	500	10
TM44	EK18	3,700	1,600	No plasmid ^d	12,000	12,000
TM44	EK49	2,100	400	EK18	179	4
GS78		6,000	2,300	EK49	68	4
GS78	EK18	3,500	500	DN46	167	12
TM81		8,300	4,000	AV77	188	7
TM81	EK18	7,800	2,400			
RH84		10,000	5,300			
RH84	EK18	7,500	1,200			
RH84	PL45	7,100	1,000			
RH84	DN46	5,300	400			
RH84	EK49	2,100	500			
GR85		600	2,000			
GR85	EK18	1,800	500			
GR85	EK49	400	400			
GR85	AV77		200			

^a Repressor activity of TrpR proteins containing single and double mutations are compared to activity of wild-type TrpR. Units of β -galactosidase activity (with and without added tryptophan) are shown for mutants with impaired activity, second site revertants of those mutants, and TrpR superrepressors. Reprinted by permission of the Genetics Society of America.

^b Each *trpR* gene was present in pACYC184, in the repression indicator strain, CY15075. This strain contains the *trpR2* allele, a frameshift allele with no repressor activity.

^c Cultures were grown in the presence or absence of exogenous tryptophan and assayed for β -galactosidase activity.

^d Recipient strain. When this strain contained the parental plasmid pACYC184, identical values were obtained.

was involved. No compensatory changes were observed at the codon responsible for the primary mutation.

Construction of Specific Double Mutations

One mutation (EK18), known to compensate for two distinct primary mutations (GS78 and GR85), was introduced by subcloning into plasmids containing distant mutations (TM44, TM81, and RH84). This procedure produced a new class of double mutants with moderate superrepressor

TABLE III
AMINO ACID CHANGES IN PRIMARY MUTANTS AND SECOND-SITE
REVERTANTS AND CONSTRUCTS^a

Name	Primary mutation	Compensatory change	Number of isolates
TM44	Thr-44→Met		
TM44-EK18		Glu-18→Lys	C ^b
TM44-EK49		Glu-49→Lys	3
GS78	Gly-78→Ser		
GS78-EK18		Glu-18→Lys	2
TM81	Thr-81→Met		
TM81-EK18		Glu-18→Lys	C
RH84	Arg-84→His		
RH84-EK18		Glu-18→Lys	C
RH84-PL45		Pro-45→Leu	1
RH84-DN46		Asp-46→Asn	8
RH84-EK49		Glu-49→Lys	6
GR85	Gly-85→Arg		
GR85-EK18		Glu-18→Lys	3
GR85-EK49		Glu-49→Lys	2
GR85-AV77		Ala-77→Val	2

^a Summary of amino acid changes in TrpR primary mutations used in this study, second-site reversions obtained, and selected double mutants constructed by gene splicing. Data are from L. Klig, D. L. Oxender, and C. Yanofsky, *Genetics* **120**, 651 (1988).

^b Double mutant constructed as described in the text.

activity that would not have been detected by the colony-color screening method involving X-Gal plates.

Search for Additional Superrepressors

Superrepressor mutations were originally isolated by screening for the appearance of tryptophan auxotrophy in a tryptophan bradytroph strain transformed with mutagenized pRLK13 plasmid containing the wild-type *trpR* gene. When second-site mutations occurred at positions 45 and 46, a second search for superrepressors was carried out. One of the two new second-site mutations (DN46, isolated as a compensatory mutation for RH84) was recovered, demonstrating that this mutation was also of the superrepressor type.

Discussion

Second-site reversion mutations that restored DNA-binding activity to TrpR were either clustered in the helix-turn-helix region (residues 45, 46, and 49) or scattered elsewhere in the linear amino acid sequence (residues

18 and 77) at positions near the DNA-binding region of the folded protein. The TrpR superrepressors obtained by second-site reversion were essentially equivalent in activity to the wild-type molecule in the presence of tryptophan, but were also active in the absence of corepressor. Three of these mutations involve the loss of a negatively charged residue (glutamic acid or aspartic acid). In the case of DN46, the charged residue is replaced with neutral asparagine. In two superrepressors (EK18 and EK49) glutamic acid was replaced by a positively charged lysine residue. This recurring theme of alterations in surface charges suggests that electrostatic repulsion plays a role in the suboptimal binding activity characteristic of wild-type TrpR. The most common superrepressors (EK18 and EK49) restored activity to a variety of mutant repressors, suggesting that this class of mutations (net charge change of four per dimer) act globally, rather than in a site-specific fashion. EK49 has the highest repressor activity in the absence of tryptophan seen to date. This protein has been purified and its activity has been examined in an *in vitro* transcription assay.²⁴ The superrepressor was shown to have a 10-fold higher affinity for operator DNA due to a decrease in its rate of dissociation from DNA. It is likely that mutations of the EK class may increase activation by providing more favorable electrostatic interactions between the repressor and operator DNA.

The effects of the PL45 compensatory mutation in an otherwise wild-type background have not been observed. This substitution does not affect the charged surface of the protein, but it has been suggested that loss of the proline at that position could result in recruitment of additional residues into helix C, bringing new portions of the solvent-exposed protein surface into contact with DNA.⁶

An alanine residue at position 77 is found in the turn portion of the bihelical DNA-binding motif. The side chain of the residue at position 77 must be small, like that of alanine, to permit a conformational change necessary to go from the active form (with bound tryptophan) to the inactive aporepressor. Substitution of valine at this position (AV77), with its more bulky side chain, keeps the *trp* repressor locked in its active form whether the corepressor is present or absent. Interestingly, several other DNA-binding proteins that employ the bihelical motif have a valine at the position analogous to position 77 in the *trp* repressor. It is interesting that one of the second-site revertants found for GR85 was replacement of alanine with valine (AV77), even though GR85 is unable to bind tryptophan.

The classes of TrpR superrepressors described illustrate three possible molecular mechanisms for enhancing operator binding: (1) Changes in

²⁴ L. S. Klig and C. Yanofsky, *J. Biol. Chem.* **263**, 243 (1988).

surface electrostatic forces (from negative to neutral or positively charged: EK18, DN46, EK49) may act globally to slow dissociation from DNA; (2) changes in the size of the DNA contact surface (e.g., PL45) may alter the hydrophobic effect or intermolecular hydrogen bonding to increase repressor-operator affinity; (3) Steric constraints preventing conformational change (e.g., AV77) may circumvent allosteric control mechanisms.

Using a combination of second-site reversion and subcloning approaches, it was possible to successfully explore the repertoire of changes available to enhance DNA binding by TrpR, and to classify the effects of those changes based on the structure of the protein.

Generalization of Method

Direct genetic selection is the preferred method for detecting mutant proteins with enhanced DNA-binding affinity whenever possible, but for a variety of reasons direct selection may not be available for a given protein. The preferred strategy usually requires establishment of bacterial growth conditions where tighter DNA binding would prevent accumulation or utilization of a toxic metabolite, providing a selective advantage to cells containing a superrepressor-type mutation. An easy to use media-based direct selection strategy may not be available for a particular DNA-binding protein. Development of a direct selection method may also require extensive strain construction and gene splicing. Finally, a wild-type DNA-binding protein may already exhibit near-optimal binding affinity, making the task of detecting incremental improvements even by direct selection more difficult. Direct selection has been used successfully, however, to isolate TrpR superrepressor mutations.² Contributing to the success of the selection strategy were (1) comparatively low operator-repressor binding affinity, (2) the existence of an extensive collection of required background strains, and (3) existence of a substrate analog to permit superrepressor selection (5-methyl anthranilate, which can be converted to a toxic analog, 5-methyltryptophan, in cells in which the *trpEBCD* operon is expressed). TrpR superrepressors obtained using this strategy included EK18 and EK49.

In cases where direct genetic selection methods are difficult or not possible, the second-site reversion approach described above provides a technique to amplify the differential in binding affinities between wild-type and mutant repressors. Mutagenesis of a primary mutant with impaired DNA binding allows screening to detect restored function. When second-site mutations are identified, the specific amino acid changes can be introduced into otherwise wild-type proteins and their effects assayed. The approach efficiently permits the selection of mutations that cause global superrepressor activities.

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